



Command Reference Guide

Release 8.X.0

July 2024

Disclaimer

Copyright © 2022 Exaware Ltd. All rights reserved

The software contains proprietary information of Exaware Ltd.; it is provided under a license agreement containing restrictions on use and disclosure and is also protected by copyright law. Reverse engineering of the software is prohibited.

Due to continued product development this information may change without notice. The information and intellectual property contained herein is confidential between Exaware Ltd. and the client and remains the exclusive property of Exaware Ltd.

If you find any problems in the documentation, please report them to us in writing.

Exaware Ltd. does not warrant that this document is error-free.

No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise without the prior written permission of Exaware Ltd.

Exaware Ltd. 5 Hamelacha St Netanya 42505 ISRAEL

Phone: +972-73-212-4500 Fax: +972-73-212-4600

Email: info@exaware.com

Website: <http://www.exaware.com>

Table of Contents

1. About the CLI Command Reference Manual	15
1.1. Documentation Conventions	15
1.2. Intended Audience	16
1.3. Technical Support	16
2. ARP Commands	17
2.1. ARP Configuration Commands	17
2.1.1. arp interface	17
2.1.2. ndp interface	18
2.2. ARP Operational Commands	19
2.2.1. clear arp	19
2.2.2. clear ndp	20
2.2.3. show arp/ndp	21
3. ETH-OAM Commands	24
3.1. EBN Commands	24
3.1.1. EBN Configuration Commands	24
3.1.2. EBN Operational Commands	31
3.1.3. EBN Show Commands	32
4. Flow Commands	37
4.1. Flowspec Commands	37
4.1.1. Flowspec Operational Commands	38
4.1.2. Flowspec Show Commands	39
4.2. SPAN/RSPAN Commands	42
4.2.1. SPAN/RSPAN Configuration Commands	42
4.2.2. SPAN/RSPAN Show Commands	50
5. Interface Commands	52
5.1. Interface Configuration Commands	52
5.1.1. admin-state	52
5.1.2. aggregator-id	53
5.1.3. arp ageing-timeout	54

5.1.4.	description	55
5.1.5.	ether-type-egress.....	56
5.1.6.	fec.....	57
5.1.7.	flow-control	58
5.1.8.	subinterface-queueing	59
5.1.9.	icmpv4	60
5.1.10.	icmpv6	61
5.1.11.	interface agg-eth.....	62
5.1.12.	interface loopback.....	63
5.1.13.	interface mgmt.....	64
5.1.14.	interface x-eth.....	65
5.1.15.	ipv4-address.....	66
5.1.16.	ipv4-address-secondary.....	67
5.1.17.	ipv6-address.....	68
5.1.18.	l2-transport.....	71
5.1.19.	lACP.....	72
5.1.20.	lACP-interval	73
5.1.21.	lACP-rx-interval	74
5.1.22.	lACP-tx-interval.....	75
5.1.23.	lldp	76
5.1.24.	mac-address	77
5.1.25.	min-links	78
5.1.26.	mpls	79
5.1.27.	mtu.....	80
5.1.28.	mtu-l3 [not supported in release 8.0.0]	81
5.1.29.	ndp ageing-timeout	82
5.1.30.	outer-ether-type	83
5.1.31.	policy	84
5.1.32.	queues-statistics-mode.....	87
5.1.33.	reuse [not supported in release 8.0.0]	88
5.1.34.	routing-policy-accounting [not supported in release 8.0.0]	89
5.1.35.	speed.....	92
5.1.36.	subinterface-queueing	93

5.1.37.	vlan-id	94
5.1.38.	vlan-op-ingress	95
5.1.39.	vlan-tags	96
5.1.40.	vrf	97
5.2.	Interface Operational Commands	98
5.2.1.	clear interface counters	98
5.2.2.	clear interface queues-statistics	99
5.2.3.	clear interface routing-policy-accounting [not supported in release 8.0.0] 100	
5.2.4.	clear lacp counters	101
5.3.	Interface Show Commands	102
5.3.1.	show interface	102
5.3.2.	show interface average-rate	106
5.3.3.	show interface brief	108
5.3.4.	show interface detail	109
5.3.5.	show interface description	116
5.3.6.	show interface ip	117
5.3.7.	show interface optical-monitoring	120
5.3.8.	show interface queues-statistics brief	122
5.3.9.	show interface queues-statistics detail	123
5.3.10.	show interface routing-policy-accounting [not supported in release 8.0.0] 125	
5.3.11.	show interface speed-groups	128
6.	L2 Services Commands	129
6.1.	L2 Services Configuration Commands	129
6.1.1.	VPWS	129
6.1.2.	VPWS Profile	136
6.1.3.	XCONNECT	142
7.	Management Commands	148
7.1.	AAA Commands	148
7.1.1.	AAA Configuration Commands	148
7.1.2.	AAA Operational Commands	184
7.1.3.	AAA Show Commands	185

7.2.	App-VM Commands	187
7.2.1.	App-VM Operational Commands	187
7.2.2.	App-VM Show Commands	200
7.3.	File Commands.....	202
7.3.1.	File Operational Commands.....	202
7.4.	Logging and Debugging Commands.....	219
7.4.1.	Logging and Debugging Configuration Commands	219
7.4.2.	Logging Operational Commands.....	242
7.5.	NTP Commands.....	248
7.5.1.	NTP Configuration Commands	248
7.5.2.	NTP Operational Commands	251
7.6.	Router Administration Commands.....	252
7.6.1.	Configuration Mode.....	252
7.6.2.	Operational Mode.....	292
7.7.	SNMP Commands.....	333
7.7.1.	SNMP Configuration Commands	333
7.7.2.	SNMP Show Commands.....	358
7.8.	Software Download Commands.....	361
7.8.1.	Software Download Operational Commands.....	361
7.8.2.	Software Download Show Commands.....	369
7.9.	System Commands.....	371
7.9.1.	System Configuration Commands	371
7.9.2.	System Operational Commands	377
7.9.3.	System Show Commands.....	386
7.9.4.	System Module Commands	404
8.	MPLS Commands.....	406
8.1.	LDP Commands.....	406
8.1.1.	LDP Configuration Commands.....	406
8.1.2.	LDP Operational Commands.....	439
8.1.3.	LDP Show Commands.....	441
8.2.	RSVP Commands.....	457
8.2.1.	RSVP Configuration Commands	457
8.2.2.	RSVP Operational Commands.....	530

8.2.3.	RSVP Show Commands.....	542
9.	Policy Commands	562
9.1.	Policy Overview	562
9.1.1.	Policy Conditions	568
9.1.2.	Policy Actions	569
9.1.3.	Policy Groups	571
9.1.4.	ACL Policies.....	572
9.1.5.	QoS Policies	575
9.1.6.	Routing Policies.....	581
9.2.	Policy Configuration Commands.....	585
9.2.1.	exec-policy	585
9.2.2.	policy named condition	588
9.2.3.	policy global-parameters.....	589
9.2.4.	policy group acl-forwarding.....	590
9.2.5.	policy group bgp-as-path.....	592
9.2.6.	policy group bgp-community.....	595
9.2.7.	policy group bgp-extcommunity.....	598
9.2.8.	policy group ip-prefix	602
9.2.9.	policy group mac-address	606
9.2.10.	policy (attach QoS and ACL).....	607
9.2.11.	rule-default (QoS and ACL).....	610
9.3.	Policy Show Commands	611
9.3.1.	show configuration policy condition.....	611
9.3.2.	show configuration policy group	612
9.3.3.	show policy	613
9.4.	ACL Policy Configuration Commands	624
9.4.1.	Traffic Conditions	624
9.4.2.	Actions	625
9.4.3.	ctrl-plane policy acl.....	633
9.5.	ACL Policy Operational Commands	634
9.5.1.	clear acl counters.....	634
9.5.2.	clear acl forwarding counters.....	635
9.5.3.	clear ctrl-plane acl counters	636

9.6. ACL Policy Show Commands	637
9.6.1. show acl forwarding interface.....	637
9.6.2. show acl forwarding policy	639
9.6.3. show acl policy	640
9.6.4. show configuration policy acl	641
9.6.5. show ctrl-plane acl	643
9.7. General Routing Policy Actions.....	644
9.7.1. deny	644
9.7.2. default-deny	645
9.7.3. default-permit.....	646
9.7.4. permit.....	647
9.8. BGP Policy Configuration Commands: Routing Conditions.....	648
9.8.1. as-path.....	648
9.8.2. as-path-length	651
9.8.3. as-path-unique-length	652
9.8.4. community	653
9.8.5. extcommunity	656
9.8.6. local-preference	659
9.8.7. med.....	660
9.8.8. next-hop.....	661
9.8.9. orf-prefix	664
9.8.10. origin	667
9.8.11. peer.....	668
9.8.12. prefix.....	671
9.8.13. rib-route.....	673
9.8.14. rpki-origin-as.....	676
9.8.15. tag	677
9.9. BGP Policy Configuration Commands: Actions.....	678
9.9.1. as-path.....	678
9.9.2. community	683
9.9.3. damping.....	687
9.9.4. extcommunity	690
9.9.5. local-preference	694

9.9.6.	med.....	695
9.9.7.	next-hop.....	697
9.9.8.	origin.....	698
9.9.9.	suppress-specific-route.....	699
9.9.10.	tag.....	700
9.9.11.	traffic-counter.....	701
9.9.12.	unsuppress-specific-routes.....	702
9.9.13.	weight.....	703
9.10.	BGP Policy Configuration Commands: Attachment Points.....	704
9.10.1.	accounting policy [not supported in release 8.0.0].....	705
9.10.2.	aggregate-address.....	706
9.10.3.	default-route-originate.....	709
9.10.4.	export-policy.....	710
9.10.5.	flap-damping.....	711
9.10.6.	import-policy.....	712
9.10.7.	neighbor-in.....	713
9.10.8.	neighbor-out.....	714
9.10.9.	network.....	717
9.10.10.	orf.....	718
9.10.11.	redistribute.....	719
9.11.	ISIS Policy Configuration Commands: Routing Conditions.....	720
9.11.1.	level.....	720
9.11.2.	prefix.....	721
9.11.3.	rib-metric.....	724
9.11.4.	rib-route.....	725
9.11.5.	route-has-mpls-label.....	728
9.11.6.	route-has-no-mpls-label.....	729
9.12.	ISIS Policy Configuration Commands: Actions.....	730
9.12.1.	isis-metric.....	730
9.12.2.	metric-type.....	731
9.12.3.	tag.....	732
9.13.	ISIS Policy Configuration Commands: Attachment Points.....	733
9.13.1.	default-route-originate.....	734

9.13.2. redistribute.....	735
9.13.3. route-leak	736
9.14. LDP Policy Configuration Commands: Routing Conditions	737
9.14.1. prefix.....	737
9.15. LDP Policy Configuration Commands: Attachment Points.....	740
9.15.1. label-allocation.....	741
9.16. LDP Policy Operational Commands	742
9.16.1. clear ldp policy label-allocation	742
9.17. LDP Policy Show Commands	743
9.17.1. show ldp policy label-allocation	743
9.18. OSPF Policy Configuration Commands.....	744
9.18.1. inter-area policy	744
9.18.2. rib-filter policy	746
9.19. OSPF Policy Operational Commands.....	747
9.19.1. clear ospf policy default-route-originate	747
9.19.2. clear ospf policy inter-area	748
9.19.3. clear ospf policy redistribute.....	749
9.19.4. clear ospf policy rib-filter.....	750
9.20. OSPF Policy Show Commands.....	751
9.20.1. show ospf policy	751
9.21. OSPFv3 Policy Configuration Commands	752
9.21.1. inter-area policy	752
9.21.2. rib-filter policy	754
9.22. OSPFv3 Policy Operational Commands	755
9.22.1. clear ospfv3 policy inter-area.....	755
9.22.2. clear ospfv3 policy redistribute.....	756
9.22.3. clear ospfv3 policy rib-filter.....	757
9.23. OSPFv3 Policy Show Commands.....	758
9.23.1. show ospfv3 policy.....	758
9.24. QoS Policy Configuration Commands: Actions.....	759
9.24.1. dscp.....	759
9.24.2. header-compensate	760
9.24.3. minimum bandwidth.....	761

9.24.4. mpls exp.....	764
9.24.5. police.....	767
9.24.6. qos-mapping-type.....	780
9.24.7. queue	781
9.24.8. port-level-rate.....	784
9.24.9. precedence.....	785
9.24.10. queue max size.....	786
9.24.11. shape.....	787
9.24.12. weight	789
9.25. QoS Policy Configuration Commands: Traffic Conditions	791
9.25.1. destination ip	791
9.25.2. destination port.....	794
9.25.3. dscp.....	802
9.25.4. ether-type.....	804
9.25.5. frag-non-initial	805
9.25.6. icmpv4-type.....	807
9.25.7. icmpv4-code.....	810
9.25.8. icmpv6-type.....	815
9.25.9. icmpv6-code.....	817
9.25.10. ieee-802.1p	820
9.25.11. mpls exp topmost.....	821
9.25.12. packet-type.....	823
9.25.13. precedence.....	824
9.25.14. protocol-ip	825
9.25.15. queue	828
9.25.16. source ip.....	829
9.25.17. source port.....	832
9.25.18. tcp-flags	839
9.25.19. ttl.....	842
9.26. QoS Policy Operational Commands.....	843
9.26.1. clear qos counters	843
9.27. QoS Policy Show Commands.....	844
9.27.1. show configuration policy qos	844

9.27.2. show qos interface agg-verbose	847
9.27.3. show qos interface brief.....	850
9.27.4. show qos interface detail.....	857
9.27.5. show qos policy.....	864
10. Routing Commands	865
10.1. BFD Routing Commands	865
10.1.1. Single Hop BFD Session	865
10.1.2. Multi Hop BFD Session.....	867
10.1.3. BFD Configuration Commands.....	868
10.1.4. BFD Operational Commands.....	883
10.1.5. BFD Show Commands	886
10.2. BGP Routing Commands.....	890
10.2.1. BGP Configuration Commands	890
10.2.2. BGP Operational Commands	980
10.2.3. BGP Show Commands	1004
10.3. Common Routing Commands.....	1062
10.3.1. Common Routing Configuration Commands.....	1062
10.4. DHCP Snooping Commands.....	1068
10.4.1. DHCP Snooping Configuration Commands.....	1068
10.4.2. DHCP Snooping Operational Commands.....	1077
10.4.3. DHCP Snooping Show Commands.....	1078
10.5. IGMP Routing Commands	1092
10.5.1. IGMP Routing Configuration Commands	1092
10.5.2. IGMP Routing Operational Commands	1100
10.5.3. IGMP Routing Show Commands	1101
10.6. IGMP Snooping Commands (Entel).....	1110
10.6.1. IGMP Snooping Configuration Commands.....	1110
10.6.2. IGMP Snooping Operational Commands.....	1119
10.6.3. IGMP Snooping Show Commands.....	1121
10.7. IS-IS Routing Commands	1125
10.7.1. IS-IS Routing Configuration Commands	1125
10.7.2. IS-IS Routing Operational Commands	1181
10.7.3. IS-IS Routing Show Commands	1189

10.8. Multicast Routing Commands.....	1209
10.8.1. Multicast Routing Configuration Commands.....	1209
10.8.2. Multicast Routing Operational Commands.....	1216
10.8.3. Multicast Routing Show Commands.....	1218
10.9. OSPF Routing Commands.....	1223
10.9.1. OSPF Routing Configuration Commands.....	1223
10.9.2. OSPF Operational Commands.....	1271
10.9.3. OSPF Show Commands.....	1275
10.10. OSPFv3 Routing Commands	1305
10.10.1. OSPFv3 Routing Configuration Commands.....	1305
10.10.2. OSPFv3 Routing Operational Commands.....	1342
10.10.3. OSPFv3 Routing Show Commands.....	1346
10.11. PIM Commands	1369
10.11.1. PIM Configuration Commands.....	1369
10.11.2. PIM Operational Commands.....	1396
10.11.3. PIM Show Commands.....	1398
10.12. Routing and Forwarding Information Base Commands.....	1410
10.12.1. L2 Services Configuration Commands.....	1410
10.12.2. L2 Services Operational Commands.....	1428
10.12.3. L2 Services Show Commands	1431
10.12.4. L3 Services Operational Commands.....	1451
10.12.5. L3 Services Show Commands	1453
10.13. Routing Leak Commands.....	1479
10.13.1. Leak Configuration Commands.....	1479
10.14. Static Routing Commands.....	1484
10.14.1. Static Configuration Commands.....	1484
10.15. VRF Commands	1491
10.15.1. VRF Configuration Commands.....	1491
10.15.2. VRF Show Commands.....	1495
10.16. VRRP Commands	1496
10.16.1. VRRP Configuration Commands.....	1496
10.16.2. VRRP Operational Commands.....	1510
10.16.3. VRRP Show Commands.....	1513

11. Timing Commands	1521
11.1. PTP Commands	1521
11.1.1. PTP Configuration Commands	1521
11.1.2. PTP Operational Commands	1535
11.1.3. PTP Show Commands	1536
11.2. Synchronization Commands	1542
11.2.1. Synchronization Configuration Commands	1542
11.2.2. Synchronization Operational Commands	1553
11.2.3. Synchronization Show Commands	1555
Index.....	1565

1. About the CLI Command Reference Manual

This manual contains the CLI commands for configuring the Exaware System router. Each command description includes the following subsections:

Description	Description of what the command does. Usage guidelines/information about when to invoke the command and additional details.
Command Syntax	Complete syntax of the command.
Command Mode	Mode where the command is executed (ie. configuration).
Parameters Table	Description of command parameters and options.
Examples	Samples of the command syntax.
Notes	Any additional information regarding command usage.

1.1. Documentation Conventions

The following types of formatting in the text identify different components of the commands.

Formatting convention	Type of information
<i>Italic`</i>	Variable expressions such as parameters.
<code>monospace</code>	Command keywords.
{ ... }	Command parameters are mandatory.
[...]	Command parameters are optional.

Note:

Free-formed strings should be surrounded by quotation marks (i.e. ""), when entered into the cli, if containing one or more of the following characters:
\\ ? ! | ; "

1.2. Intended Audience

This guide is designed for technicians in charge of managing and configuring the Exaware System router.

1.3. Technical Support

Contact Exaware Technical Support at: support@exaware.com.

When sending an e-mail, please supply the following information:

- Your name, company name, and customer ID.
- Your contact information (telephone number and e-mail).
- Product name and version number.
- Problem description and any documentation that may help in resolving the problem.
- Hardware model, software version, and memory usage of the router on which the product is installed.

2. ARP Commands

2.1. ARP Configuration Commands

2.1.1. arp interface

Description	Configures a static ARP entry on an interface.		
Command Syntax	<code>arp interface <i>interface-name</i> <i>ip-address</i> <i>mac-address</i></code>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>interface-name</i>	Interface name of type x-eth/agg-eth/mgmt	Interface type and index.
	<i>ip-address</i>	IPv4 address	IPv4 address assigned to the interface.
	<i>mac-address</i>	xx:xx:xx:xx:xx:xx	One byte of MAC address in hexadecimal. 1 is the MSB byte. 00:00:00:00:00:00 is not valid.
Examples	The following example shows how to use this command: exaware# config exaware(config)# arp interface x-eth 0/0/0 1.1.1.1 00:11:aa:bb:cc:dd		

2.1.2. ndp interface

Description	Configures a static IPv6 neighbor discovery entry on an interface.		
Command Syntax	<code>ndp interface <i>interface-name</i> <i>ip-address</i> <i>mac-address</i></code>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>interface-name</i>	Interface name of type x-eth/agg-eth	Interface type and index.
	<i>ip-address</i>	IPv6 address	IPv6 address assigned to the interface.
	<i>mac-address</i>	xx:xx:xx:xx:xx:xx	One byte of MAC address in hexadecimal. 1 is the MSB byte. 00:00:00:00:00:00 is not valid.
Examples	The following example shows how to use this command: <pre>exaware# config exaware(config)# ndp interface x-eth 0/0/0 2001::1 00:11:aa:bb:cc:dd</pre>		

2.2. ARP Operational Commands

2.2.1. clear arp

Description	Clears dynamic ARP entries, either globally or only for the specified interface.		
Command Syntax	<pre>clear arp table all clear arp table interface <i>interface-name</i> [address <i>ipv4-address</i>]</pre>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	all		Clears all dynamic ARP entries.
	<i>Interface-name</i>	x-eth/agg-eth/mgmt interface name	Specific interface to clear of the ARP entries.
	<i>Ipv4-address</i>	IPv4 Address	Specific host IPv4 address to clear of the ARP entries.
Examples	The following example shows how to clear the ARP table for the specified interface: <pre>exaware# clear arp table interface x-eth 0/0/1 exaware# clear arp table interface x-eth 0/0/2 address 134.1.1.1</pre>		

2.2.2. clear ndp

Description	Clears dynamic NDP entries, either globally or only for the specified interface.		
Command Syntax	<pre>clear ndp table all clear ndp table interface <i>interface-name</i> [address <i>ipv6-address</i>]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	all		Clears all NDP entries.
	<i>Interface-name</i>	x-eth/agg-eth interface name	Specific interface to clear of the ARP entries.
	<i>IPv6-address</i>	IPv6 Address	Specific host IPv6 address to clear of the NDP entries.
Examples	The following example shows how to use this command: <pre>exaware# clear ndp table interface x-eth 0/0/1 exaware# clear ndp table interface x-eth 0/0/2 address fe80::200:3bff:fe1c:5cb6</pre>		

2.2.3. show arp/ndp

Description	Displays the arp and ndp entries.		
Command Syntax	<pre>show {arp ndp} all show {arp ndp} [interface interface-name [ip- address] {address ip-address}]</pre>		
Command Mode	operationa		
Parameters Table	Name	Value	Description
	arp ndp	arp or ndp	arp is for IPv4 neighbors and ndp for IPv6.
	all		Display all ARP/NDP entries.
	<i>interface-name</i>	x-eth/agg-eth interface name	Interface name.
	<i>ip-address</i>	x.x.x.x x:::x	Any IPv4 address for arp. Any IPv6 address for ndp.
	L2 address	XX:XX:XX:XX:XX:XX	The hardware address of the entry.

	state		PERMANENT – Static entry. REACHABLE – The entry received slow path traffic. DELAY – The entry is valid, however there was a delay reaching the entry.																								
Examples																											
The following examples show how to use this command: <pre>exaware# show arp address 20.0.0.1</pre> <table> <tr> <th>INSTANCE</th><th>IP</th><th>L2 ADDRESS</th><th>STATE</th></tr> <tr> <td colspan="4">-----</td></tr> <tr> <td>x-eth0/1/0</td><td>20.0.0.1</td><td>00:00:3b:1c:5c:b6</td><td>DELAY</td></tr> </table> <pre>exaware# show ndp interface x-eth0/1/0 fe80::200:3bff:fe1c:5cb6</pre> <table> <tr> <th>INSTANCE</th><th>IP</th><th>L2 ADDRESS</th><th>STATE</th></tr> <tr> <td colspan="4">-----</td></tr> <tr> <td>x-eth0/1/0</td><td>fe80::200:3bff:fe1c:5cb6</td><td>00:00:3b:1c:5c:b6</td><td>DELAY</td></tr> </table>				INSTANCE	IP	L2 ADDRESS	STATE	-----				x-eth0/1/0	20.0.0.1	00:00:3b:1c:5c:b6	DELAY	INSTANCE	IP	L2 ADDRESS	STATE	-----				x-eth0/1/0	fe80::200:3bff:fe1c:5cb6	00:00:3b:1c:5c:b6	DELAY
INSTANCE	IP	L2 ADDRESS	STATE																								

x-eth0/1/0	20.0.0.1	00:00:3b:1c:5c:b6	DELAY																								
INSTANCE	IP	L2 ADDRESS	STATE																								

x-eth0/1/0	fe80::200:3bff:fe1c:5cb6	00:00:3b:1c:5c:b6	DELAY																								

3. ETH-OAM Commands

3.1. EBN Commands

3.1.1. EBN Configuration Commands

3.1.1.1. admin-status

Description	Use the admin-status command to enable/disable the EBN function on the given Ethernet interface. Use the no form of the command to restore the default configuration (disabled).		
Command Syntax	admin-status enable disable no admin-status		
Command Mode	configuration interface eth-oam ebn		
Parameters Table	Name	Value	Description
	enable		Enables EBN on the given interface.
	disable		Disables EBN on the given interface.
Examples	The following example shows how to use this command: exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# eth-oam exaware(config-interface-x-eth-0/0/1-eth-oam)# ebn exaware(config-interface-x-eth-0/0/1-eth-oam-ebn)# admin-status enable		

3.1.1.2. ebn

Description	Use the ebn command to start configuration of the EBN function of the Ethernet OAM on the given Ethernet interface.
Command Syntax	ebn
Command Mode	configuration interface eth-oam
Examples	The following example shows how to use this command: <pre>exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# eth-oam exaware(config-interface-x-eth-0/0/1-eth-oam)# ebn exaware(config-interface-x-eth-0/0/1-eth-oam-ebn)#</pre>

3.1.1.3. eth-oam

Description	Use the eth-oam command to start configuration of the Ethernet OAM on the given Ethernet interface.
Command Syntax	eth-oam
Command Mode	configuration interface x-eth
Examples	The following example shows how to use this command: <pre>exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# eth-oam exaware(config-interface-x-eth-0/0/1-eth-oam)#</pre>

3.1.1.4. level

Description	Use the level command to configure the EBN level on the given Ethernet interface. Use the no form of the command to delete this configuration.		
Command Syntax	<pre>level value no level</pre>		
Command Mode	configuration interface eth-oam ebn		
Parameters Table	Name	Value	Description
	<i>value</i>	0-7	EBN level.
Examples	The following example shows how to use this command: <pre>exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# eth-oam exaware(config-interface-x-eth-0/0/1-eth-oam)# ebn exaware(config-interface-x-eth-0/0/1-eth-oam-ebn)# level 3</pre>		

3.1.1.5. threshold

Description	Use the wait-to-restore-timeout command to configure the wait-to-restore timeout on the given Ethernet interface. Use the no form of the command to restore the default configuration.		
Command Syntax	<pre>threshold value unit no threshold</pre>		
Command Mode	configuration interface eth-oam ebn		
Parameters Table	Name	Value	Description
	<i>value</i>		Specifies the bandwidth in units defined by the next parameter. The default value is 0 – threshold is disabled..
	<i>unit</i>	bps, kbps, mpbs, or gbps	Specifies the unit in which the bandwidth is configured. The default value is bps.
Examples	The following example shows how to use this command: <pre>exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# eth-oam exaware(config-interface-x-eth-0/0/1-eth-oam)# ebn exaware(config-interface-x-eth-0/0/1-eth-oam-ebn)# threshold 1 mbps</pre>		

3.1.1.6. wait-to-restore-control

Description	Use the wait-to-restore-control command to enable/disable the wait-to-restore control on the given Ethernet interface. Use the no form of the command to restore the default configuration (disabled).		
Command Syntax	<pre>wait-to-restore-control enable disable no wait-to-restore-control</pre>		
Command Mode	configuration interface eth-oam ebn		
Parameters Table	Name	Value	Description
	enable		Enables the EBN wait-to-restore control on the given interface.
	disable		Disables the EBN wait-to-restore control on the given interface.
Examples	The following example shows how to use this command: <pre>exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# eth-oam exaware(config-interface-x-eth-0/0/1-eth-oam)# ebn exaware(config-interface-x-eth-0/0/1-eth-oam-ebn)# wait-to-restore-control enable</pre>		

3.1.1.7. wait-to-restore-timeout

Description	Use the wait-to-restore-timeout command to configure the wait-to-restore timeout on the given Ethernet interface. Use the no form of the command to restore the default configuration).		
Command Syntax	<pre>wait-to-restore-timeout seconds no wait-to-restore-timeout</pre>		
Command Mode	configuration interface eth-oam ebn		
Parameters Table	Name	Value	Description
	<i>seconds</i>	0-86400	Specifies the wait-to-restore timeout in seconds. The default value is 300 seconds.
Examples	The following example shows how to use this command: <pre>exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# eth-oam exaware(config-interface-x-eth-0/0/1-eth-oam)# ebn exaware(config-interface-x-eth-0/0/1-eth-oam-ebn)# wait-to-restore-timeout 200</pre>		

3.1.2. EBN Operational Commands

3.1.2.1. clear eth-oam ebn counters

Description	Use the clear eth-oam ebn counters command to clear the EBN counters on the given Ethernet interface.		
Command Syntax	<code>clear eth-oam ebn counters [interface <i>interface</i>]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>interface</i>	x-eth 0/0/port	Specifies the interface. If the parameter is not configured the counters are cleared on all interfaces.
Examples	The following example shows how to use this command: <pre>exaware# clear eth-oam ebn counters interface x-eth 0/0/1</pre>		

3.1.3. EBN Show Commands

3.1.3.1. show eth-oam ebn counters

Description	Use the show eth-oam ebn counters command to display the EBN counters on the given Ethernet interface.										
Command Syntax	show eth-oam ebn counters [interface <i>interface</i>]										
Command Mode	operational										
Parameters Table	Name	Value	Description								
	<i>interface</i>	x-eth 0/0/port	Specifies the interface. If the parameter is not configured the counters are displayed on all interfaces.								
Examples	The following example shows how to use this command:										
	<pre>exaware# show eth-oam ebn counters interface x-eth 0/0/1</pre> <table><tr><td>INTERFACE</td><td>Input Valid Frames</td><td>Input Invalid Frames</td></tr><tr><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>x-eth 0/0/1</td><td>67</td><td>1</td></tr></table>			INTERFACE	Input Valid Frames	Input Invalid Frames	-----	-----	-----	x-eth 0/0/1	67
INTERFACE	Input Valid Frames	Input Invalid Frames									
-----	-----	-----									
x-eth 0/0/1	67	1									

3.1.3.2. show eth-oam ebn status

Description	Use the show eth-oam ebn status command to display the EBN status on the given Ethernet interface.		
Command Syntax	<code>show eth-oam ebn status [interface <i>interface</i>]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>interface</i>	x-eth 0/0/port	Specifies the interface. If the parameter is not configured the counters are displayed on all interfaces.

Examples

The following example shows how to use this command:

```
exaware# show eth-oam ebn status
```

```
* - Current bandwidth < threshold
```

```
Interface: x-eth 0/0/1
```

```
Level: 3 (default)
```

```
Admin-status: Enabled
```

```
Wait-to-restore Control: On
```

```
Wait-to-restore Timeout: 100 seconds
```

```
Threshold Bandwidth: 2000 Mbps
```

```
Period Value: 10 seconds
```

```
Period Value Left: 2 seconds
```

```
Maximum Bandwidth: 10000 Mbps
```

```
Normal Bandwidth: 5000 Mbps
```

```
*Current Bandwidth: 1000 Mbps
```

```
Real Interface Shaper Bandwidth: 2000 Mbps
```

```
Wait-to-restore Time Left: 10 seconds
```

```
Interface: x-eth 0/0/2
```

```
Level: 2
```

```
Admin-status: Disabled
```

```
Wait-to-restore Control: Off (default)
```

```
Wait-to-restore Timeout: 300 seconds (default)
```

```
Threshold Bandwidth: 0 Mbps (default)
```


4. Flow Commands

4.1. Flowspec Commands

In addition to the Flowspec Commands in this section, the following commands are needed for the implementation of Flowspec:

af-ipv4	Enters the routing configuration sub-mode for BGP routing.
show bgp neighbors advertised/received routes	Displays BGP routes advertised to/received from peer.
show bgp table brief/detail	Displays BGP routing table.

4.1.1. Flowspec Operational Commands

4.1.1.1. clear flow-table counters flowspec

Description	Clears flow-table counters.		
Command Syntax	<pre>clear flow-table counters flowspec all clear flow-table counters flowspec dst-prefix <i>ip-prefix</i> clear flow-table counters flowspec src-prefix <i>ip-prefix</i> clear flow-table counters flowspec dst-prefix <i>ip-prefix</i> src-prefix <i>ip-prefix</i></pre>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>ip-prefix</i>	IPv4 address / length	dst/src-prefix from the flowspec rule.
Examples	The following examples show how to use this command: <pre>exaware# clear flow-table counters flowspec all exaware# clear flow-table counters flowspec src-prefix 4.0.0.0/8 exaware# clear flow-table counters flowspec dst-prefix 8.0.0.0/8 exaware# clear flow-table counters flowspec dst-prefix 8.0.0.0/8 src-prefix 4.0.0.0/8</pre>		

4.1.2. Flowspec Show Commands

4.1.2.1. **show flow-table flowspec**

Description	Displays flow-table information.		
Command Syntax	show flow-table flowspec [dst-prefix <i>ip-prefix</i>] [src-prefix <i>ip-prefix</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>ip-prefix</i>	IPv4 address / length	dst/src-prefix from the flowspec rule.
Examples	The following examples show how to use this command:		

```
exaware# show flow-table flowspec
```

FLOW	ACTIONS	PASS PKTS	PASS BYTES	DROP PKTS	DROP BYTES
4.0.0.0/8,7.0.0.0/8,dst-port=1000,src-port=...	permit	10	20k	10	20k
4.0.0.0/8,8.0.0.0/8,dst-port=1002,src-port=...	drop	20	40k	20	40k
6.0.0.0/8,8.0.0.0/8,dst-port=1003,src-port=...	rate=20kbps	30	60k	30	60k

```
exaware# show flow-table flowspec dst-prefix 4.0.0.0/8
```

FLOW	ACTIONS	PASS PKTS	PASS BYTES	DROP PKTS	DROP BYTES
4.0.0.0/8,7.0.0.0/8,dst-port=1000,src-port=...	permit	10	20k	10	20k
4.0.0.0/8,8.0.0.0/8,dst-port=1002,src-port=...	drop	20	40k	20	40k

```
exaware# show flow-table flowspec src-prefix 8.0.0.0/8
```

FLOW	ACTIONS	PASS PKTS	PASS BYTES	DROP PKTS	DROP BYTES
4.0.0.0/8,8.0.0.0/8,dst-port=1002,src-port=...	drop	20	40k	20	40k
6.0.0.0/8,8.0.0.0/8,dst-port=1003,src-port=...	rate=20kbps	30	60k	30	60k

```
exaware# show flow-table flowspec dst-prefix 8.0.0.0/8 src-prefix 4.0.0.0/8
```

FLOW	ACTIONS	PASS PKTS	PASS BYTES	DROP PKTS	DROP BYTES
4.0.0.0/8,8.0.0.0/8,dst-port=1002,src-port=...	deny	20	40k	20	40k

4.2. SPAN/RSPAN Commands

4.2.1. SPAN/RSPAN Configuration Commands

4.2.1.1. description

Description	Sets a mirror session description. Use the no form to restore the default.		
Command Syntax	description <i>text</i> no description		
Command Mode	configuration mirror-session		
Parameters Table	Name	Value	Description
	<i>text</i>	String Max: 255 chars	A text describing the mirror session. The default value – the empty string
Examples	This example configures mirror sessions: <pre> exaware# config exaware(config)# mirror-session mirses exaware(config-mirror-session-mirses)# description "mirror-session 2" exaware(config-mirror-session-mirses)# destination interface x-eth0/0/3 exaware(config-mirror-session-mirses)# sample-rate 400 exaware(config-mirror-session-mirses)# sample- length truncated exaware(config-mirror-session-mirses)# source interface x-eth0/0/1 exaware(config-mirror-session-mirses)# source interface x-eth0/0/2 direction rx exaware(config-mirror-session-mirses)# source interface x-eth0/0/3 direction rx </pre>		

4.2.1.2. destination

Description	The command configures the destination interface for the given SPAN mirror session.		
Command Syntax	destination interface x-eth0/0/ <i>port</i>		
Command Mode	configuration mirror-session		
Parameters Table	Name	Value	Description
	<i>port</i>	Integer number	Specifies an Ethernet port number.
Examples	This example configures mirror sessions: <pre> exaware# config exaware(config)# mirror-session mirses exaware(config-mirror-session-mirses)# description "mirror-session 2" exaware(config-mirror-session-mirses)# destination interface x-eth0/0/3 exaware(config-mirror-session-mirses)# sample-rate 400 exaware(config-mirror-session-mirses)# sample-length truncated exaware(config-mirror-session-mirses)# source interface x-eth0/0/1 exaware(config-mirror-session-mirses)# source interface x-eth0/0/2 direction tx exaware(config-mirror-session-mirses)# source interface x-eth0/0/3 direction both </pre>		
Note	Only one destination interface can be configured into a SPAN session.		

4.2.1.3. mirror-session

Description	Creates a mirror session if it does not exist and enters mirror session configuration mode. Use the no form to remove the given mirror session or all sessions.		
Command Syntax	<pre>mirror-session session-name no mirror-session [session-name]</pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>session-name</i>	Non-empty string up to 16 characters.	Specifies the mirror session name.
Examples	This example shows how to enter mirror session configuration mode: <pre>exaware# config exaware(config)# mirror-session mirses exaware(config-mirror-session-mirses)#</pre>		
Note	• Up to 10 SPAN and RSPAN sessions can be configured. • A mirror session must contain only one destination or reflector interface and at least one source interface. • The same interface can be configured as source only in one mirror session. • A physical port cannot be configured as source if it is defined as destination or reflector in a mirror session. • A sub-interface cannot be configured as source if its physical port is defined as destination or reflector in a mirror session.		

4.2.1.4. reflector

Description The command configures the reflector interface for the given RSPAN mirror session.

Command Syntax	reflector interface x-eth0/0/ <i>port</i> vlan-id <i>vlan_id</i>		
Command Mode	configuration mirror-session		
Parameters Table	Name	Value	Description
	<i>port</i>	Integer number	Specifies an Ethernet port number.
	<i>vlan_id</i>	Integer in range 1..4094	Specifies the VLAN-ID which the mirrored MAC frames are added.
Examples	This example configures mirror sessions: <pre> exaware# config exaware(config)# mirror-session mirses exaware(config-mirror-session-mirses)# description "mirror-session 2" exaware(config-mirror-session-mirses)# reflector interface x-eth0/0/3 vlan-id 100 exaware(config-mirror-session-mirses)# sample-rate 400 exaware(config-mirror-session-mirses)# sample-length truncated exaware(config-mirror-session-mirses)# source interface x-eth0/0/1 exaware(config-mirror-session-mirses)# source interface x-eth0/0/2 direction tx exaware(config-mirror-session-mirses)# source interface x-eth0/0/3 direction both </pre>		
Note	- Only one reflector interface can be configured into a RSPAN session. - A physical port cannot be configured as reflector if it or its sub-interface is defined as source in a mirror session.		

4.2.1.5. sample-length

Description	The command configures the sample length. Use the no form to restore the default configuration.		
Command Syntax	sample-length truncated full no sample-length		
Command Mode	configuration mirror-session		
Parameters Table	Name	Value	Description
	Truncated		The length of sample is truncated until 102 or 231 octets depending on the hardware.
	Full		Samples are not truncated. The default value is full .
Examples	This example configures mirror sessions: <pre> exaware# config exaware(config)# mirror-session mirses exaware(config-mirror-session-mirses)# description "mirror-session 2" exaware(config-mirror-session-mirses)# destination interface x-eth0/0/3 exaware(config-mirror-session-mirses)# sample-rate 400 exaware(config-mirror-session-mirses)# sample-length truncated exaware(config-mirror-session-mirses)# source interface x-eth0/0/1 direction tx exaware(config-mirror-session-mirses)# source interface x-eth0/0/2 direction rx </pre>		

4.2.1.6. sample-rate

Description	The command configures the 1:N sample rate. Use the no form to remove the rate.		
Command Syntax	<pre>sample-rate N no sample-rate</pre>		
Command Mode	configuration mirror-session		
Parameters Table	Name	Value	Description
	N	Integer in range 1..1,048,575	Specifies the 1:N sample rate. The default is 1:1.
Examples	This example configures mirror sessions: <pre>exaware# config exaware(config)# mirror-session mirses exaware(config-mirror-session-mirses)# description "mirror-session 2" exaware(config-mirror-session-mirses)# destination interface x-eth0/0/3 exaware(config-mirror-session-mirses)# sample-rate 400 exaware(config-mirror-session-mirses)# sample-length truncated exaware(config-mirror-session-mirses)# source interface x-eth0/0/1 direction tx exaware(config-mirror-session-mirses)# source interface x-eth0/0/2 direction tx</pre>		

4.2.1.7. source

Description	The command configures a source interface for a given mirror session. Use the no form of the command to remove the source from the session.		
Command Syntax	<pre>source interface x-eth0/0/port[.subinterface-id] [direction {rx tx both}] no source interface x-eth0/0/port[.subinterface-id]</pre>		
Command Mode	configuration mirror-session		
Parameters Table	Name	Value	Description
	<i>port</i>	Integer number	Specifies an inband Ethernet port as source.
	<i>subinterface-id</i>	Integer number into range 1..4094	The optional parameter specifies a sub-interface.
	rx		Specifies that only RX traffic will be mirrored. The default value is rx .
	tx		Specifies that only TX traffic will be mirrored.
	both		Specifies that RX and TX traffic will be mirrored.

Examples	This example configures mirror sessions: <pre>exaware# config exaware(config)# mirror-session mirses exaware(config-mirror-session-mirses)# description "mirror-session 2" exaware(config-mirror-session-mirses)# destination interface x-eth0/0/3 exaware(config-mirror-session-mirses)# sample-rate 400 exaware(config-mirror-session-mirses)# sample-length truncated exaware(config-mirror-session-mirses)# source interface x-eth0/0/1 direction tx exaware(config-mirror-session-mirses)# source interface x-eth0/0/2 direction both</pre>
Note	• Up to 6 subinterfaces per ethernet port are supported. • Up to 50 source interfaces can be configured into a mirror session. • The same interface can be configured as source only in one mirror session. • A physical port cannot be configured as source if it is defined as destination or reflector in a mirror session. • A sub-interface cannot be configured as source if its physical port is defined as destination or reflector in a mirror session.

4.2.2. SPAN/RSPAN Show Commands

4.2.2.1. show mirror-session

Description	Displays the configuration of a given mirror session or all mirror sessions.		
Command Syntax	show mirror-session [<i>session-name</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>session-name</i>	Non-empty string up to 16 characters.	Specifies the session name. If the parameter is omitted, then information about all mirror sessions is displayed.
Examples	This example defines a session:		
<pre>exaware# show mirror-session mirror-session 1 description: session 1 destination: x-eth0/0/1 sample-rate: 1:400 sample-length: truncated sources interface: x-eth0/0/2 direction: tx interface: x-eth0/0/3 direction both mirror-session 2 description: destination: x-eth0/0/3 sample -rate: 1:100 sample-length: full sources interface: x-eth0/0/4 direction: tx mirror-session 4 description: reflector: x-eth0/0/11 vlan-id: 140 sample -rate: 1:1 sample-length: truncated sources</pre>			

```
interface: x-eth0/0/8 direction: tx
```

5. Interface Commands

5.1. Interface Configuration Commands

5.1.1. admin-state

Description	This command sets the administrative state of an interface. Use the no form of the command to change the interface state to its default value. Default values: • loopback: up • physical interfaces (x-eth or mgmt): down • sub-interfaces (x-eth or mgmt): up • aggregation interfaces (agg-eth): up • physical interfaces (x-eth) aggregated with aggregation-id: up		
Command Syntax	<pre>admin-state state no admin-state</pre>		
Command Mode	configuration interface <interface>		
Parameters Table	Name	Value	Description
	state	up	administrative: enabled, traffic: on
		down	administrative: enabled, traffic: down
		inactive	administrative: down
Examples	<pre>exaware# config exaware(config)# interface mgmt 0/0/0 exaware(config-interface-mgmt-0/0/0)# admin-state inactive</pre>		
Notes			

5.1.2. aggregator-id

Description	Sets the aggregator interface of an interface. Use the no form of the command to change the interface state to its default value.		
Command Syntax	<pre>aggregator-id [<i>interface agg id</i>]</pre> <pre>no aggregator-id</pre>		
Command Mode	configuration interface x-eth < <i>interface</i> >		
Parameters Table	Name	Value	Description
	<i>interface agg id</i>	Numeric	The aggregate interface value of the interface.
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# aggregator-id 5</pre>		
Notes	This command is optional. The default is no aggregate. Assigning an aggregation-id to an interface, removes all configuration from the interface, other than admin-state, mac-address and L1 parameters.		

5.1.3. arp ageing-timeout

Description	Configures the Address Resolution Protocol Ageing Timeout on a selected interface, in seconds. Use the no form of the command to set the default value (14400).		
Command Syntax	<pre>arp ageing-timeout [seconds] no arp ageing-timeout</pre>		
Command Mode	configuration interface <interface>		
Parameters Table	Name	Value	Description
	seconds	120-2147483	The time in seconds for which an entry remains in the ARP cache. (Default = 14400)
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/0/0 exaware(config-interface-x-eth-0/0/0)# arp ageing- timeout 1000 exaware(config-interface-x-eth-0/0/0)# no arp ageing-timeout</pre>		
Notes	The ageing timer is randomized +/- 20 secs to prevent arp flooding.		

5.1.4. description

Description	Configures the description associated with the interface.		
Command Syntax	<pre>description <i>description</i> no description</pre>		
Command Mode	configuration interface <interface>		
Parameters Table	Name	Value	Description
	<i>description</i>	String: maximum 256 bytes	User-defined description of the specified interface. Up to 255 characters, including spaces within quotation marks "".
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/3/0 exaware(config-interface-x-eth-0/3/0)# description "Local channel 2" exaware# config exaware(config)# interface x-eth 0/3/0.1 exaware(config-interface-x-eth-0/3/0.1)# description "Local sub Channel 0"</pre>		

5.1.5. ether-type-egress

Description	Sets outer vlan ether-type for push and swap operation. Use the no form of the command to set the QinQ ether-type value to the default (0x8100).		
Command Syntax	<pre>ether-type-egress <i>type</i> no ether-type-egress</pre>		
Command Mode	configuration interface <interface>		
Parameters Table	Name	Value	Description
	<i>type</i>	0x8100 0x88a8 0x9100 custom	Valid vlan ether-type (Default = 0x8100)
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/1/0.53 exaware(config-interface-x-eth-0/1/0.53)# ether-type-egress 0x8100</pre>		
Notes	This command is available only on sub-interfaces.		

5.1.6. fec

Description	The command configures the port FEC coding capability. To restore the default configuration ("none"), use the no form of this command.		
Command Syntax	fec none cl74 cl91 no fec		
Command Mode	configuration x-eth interface		
Parameters Table	Name	Value	Description
	Enable		Enables the FEC coding capability.
	disable		Disables the FEC coding capability.
Examples	exaware# config exaware(config)# interface x-eth 0/0/2 exaware(config-interface-x-eth-0/0/2)# fec cl91		

5.1.7. flow-control

Description	The command enables the Flow Control on a given ethernet interface. To reset the default configuration, use the no form of this command. By default, the Flow Control is disabled.		
Command Syntax	flow-control {disabled rx} no flow-control		
Command Mode	configuration x-eth interface		
Parameters Table	Name	Value	Description
	disabled		Disables the Flow Control.
	rx		Enables the Flow Control and specifies that only the Flow Control input direction (receiving the PAUSE packets) is enabled.
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/0/2 exaware(config-interface-x-eth-0/0/2)# flow-control rx</pre>		

5.1.8. subinterface-queueing

Description	Configures whether TM queues are allocated for the sub interface or is it using the physical port queues. Use the no form, to restore the default configuration (disabled).
Command Syntax	subinterface-queueing {disabled all-queues} no subinterface-queueing
Command Mode	configuration interface x-eth.index
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/0/3.100 exaware(config-interface-x-eth-0/0/3.100)# subinterface-queueing all-queues exaware(config-interface-x-eth-0/0/3.100)# no subinterface-queueing</pre>

5.1.9. icmpv4

Description	Enables the ICMPV4 mode on the interface. (Ethernet interfaces only.) Use the no form of the command to configure the default setting (disable).		
Command Syntax	icmpv4 [enable disable]		
Command Mode	configuration interface		
Parameters Table	Name	Value	Description
		enable disable	Enables/disables ICMPV4 mode on an interface for traffic exceptions like path MTU discovery and TTL expiration. (Default = disable)
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/0/0 exaware(config-interface-x-eth-0/0/0)# icmpv4 enable</pre>		

5.1.10. icmpv6

Description	Enables the ICMPV6 mode on the interface. (Ethernet interfaces only.) Enables the ICMPV6 mode on the interface. (Ethernet intUse the no form of the command to configure the default setting (disable).		
Command Syntax	icmpv6 [enable disable]		
Command Mode	configuration interface		
Parameters Table	Name	Value	Description
		enable disable	Enables/disables ICMPV6 mode on an interface for traffic exceptions like path MTU discovery and TTL expiration. (Default = disable)
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/0/0 exaware(config-interface-x-eth-0/0/0)# icmpv6 enable</pre>		

5.1.11. interface agg-eth

Description	Enters the configuration sub mode for agg-eth interfaces. Use the no form of the command to delete a specific agg-eth interface. (The command cannot delete this interface when other interfaces are aggregated to it.)		
Command Syntax	interface agg-eth <instance> <vlan-id>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	instance	Integer in range: [0..53]	The instance id of the interface.
	vlan-id	Integer in range [1..4094]	The Virtual LAN Identifier.
Examples	<pre>exaware# config exaware(config)# interface agg-eth 0 exaware(config)# interface agg-eth 50 exaware(config)# no interface agg-eth 9</pre>		

5.1.12. interface loopback

Description	Enters the configuration sub-mode for loopback interfaces. Use the no form of the command to delete a specific interface.		
Command Syntax	interface loopback <i>instance</i> no interface loopback <i>instance</i>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>instance</i>	0-65534	Instance ID of the interface.
Examples	exaware# config exaware(config)# interface loopback 10 exaware(config-interface-loopback-10)#		

5.1.13. interface mgmt

Description	Enters the sub configuration mode for the management interface.		
Command Syntax	interface mgmt <i>instance</i>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>instance</i>	0/0/0	Location of the interface.
Examples	exaware# config exaware(config)# interface mgmt 0/0/0 exaware(config-interface-mgmt-0/0/0)#		

5.1.14. interface x-eth

Description	Enters the configuration sub mode for a 10G ethernet interface/sub-interface and creates the sub-interface, if it does not exist. Use the no form of the command to delete a specific sub-interface.		
Command Syntax	<pre>interface x-eth <i>instance</i> no interface x-eth <i>instance</i></pre>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>instance</i>	0/0/port[sub-interface-id]	Location of the interface.
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/0/1.347 exaware(config-interface-x-eth-0/0/1.347)# exaware(config)# no interface x-eth 0/0/1.347</pre>		

5.1.15. ipv4-address

Description	This command assigns a primary IPV4 address to an interface. Use the no form of the command to remove the IP address assignment.		
Command Syntax	<pre>ipv4-address address/prefix-length no ipv4-address</pre>		
Command Mode	configuration interface <interface>		
Parameters Table	Name	Value	Description
	address	IPV4 address	Primary IPV4 address assigned to an interface.
	prefix-length	Integer into range 8..32	IPv4 prefix length.
Examples	<pre>exaware# config exaware(config)# interface mgmt 0/0/0 exaware(config-interface-mgmt-0/0/0)# ipv4-address 10.10.10.10/24 exaware(config-interface-management-0/0/0)# no ipv4- address</pre>		
Notes	An IPv4 address can be configured only on L3 interface not belonging to a LAG.		

5.1.16. ipv4-address-secondary

Description	This command assigns a secondary IPV4 address to an interface. Use the no form of the command to remove the IP address assignment.		
Command Syntax	<pre>ipv4-address-secondary <i>address/prefix-length</i> no ipv4-address-secondary <i>address</i></pre>		
Command Mode	configuration interface < <i>interface</i> >		
Parameters Table	Name	Value	Description
	<i>address</i>	IPV4 address	Secondary IPV4 address assigned to an interface.
	<i>prefix-length</i>	Integer into range 8..32	IPv4 prefix length.
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# ipv4-address- secondary 10.10.10.10/24</pre>		
Notes	An IPv4 address can be configured only on L3 interface not belonging to a LAG.		

5.1.17. ipv6-address

Description	This command assigns an IPV6 address to an interface. Use the no form of the command to remove the specified IP address assignment. Removing all IPv6 addresses disables IPv6 on the interface. Supported on all interfaces types except for mgmt interfaces.
Command Syntax	<pre>ipv6-address link-local [<i>address</i> auto-generated] ipv6-address eui-64 <i>address/prefix-length</i> ipv6-address <i>address/prefix-length</i> no ipv6-address <i>address/prefix-length</i></pre>
Command Mode	configuration interface < <i>interface</i> >
Notes	An IPv6 address can be configured only on L3 interface not belonging to a LAG.

Parameters Table	Name	Value	Description
		auto-generated	Automatically derive the link-local address from the MAC address. Default option for <code>link-local</code> type. When other IPv6 addresses are configured on the interface, configuring "link-local auto-generated" has no effect. When no other IPv6 addresses are configured on the interface, configuring "link-local auto-generated" enables IPv6 with only the default link-local address.
	<i>address/ prefix-length</i>	A:B:C:D:E:F:G:H A-H: each 0000-ffff	IPv6 address assigned to an interface, followed by the prefix length. See Note for more information about IPv6 address format.

Examples	<pre>exaware# config exaware(config)# interface x-eth 0/1/1 exaware(config-interface-x-eth-0/1/1)# ipv6-address link- local fe80::260:3eff:fe11:6770 exaware(config-interface-x-eth-0/1/2)# ipv6-address link- local auto-generated exaware(config-interface-x-eth-0/1/3)# ipv6-address eui- 64 2001:0db8:1234::/64 exaware(config-interface-x-eth-0/1/4)# ipv6-address 2001:0db8:1234::1/48 exaware(config-interface-x-eth-0/1/4)# no ipv6-address 2001:0db8:1234::1/48</pre>
Notes	Up to two unicast IPv6-addresses (either plain or eui-64) can be defined per interface. This command is not supported on 'interface mgmt'. Link-local is not supported on 'interface loopback'. IPv6 addresses are represented as a series of 16-bit hexadecimal fields separated by colons (:) in the format: x:x:x:x:x:x:x. Following are two examples of IPv6 addresses: <pre>2001:0db8:7654:3210:fedc:ba98:7654:3210 2001:0db8::8:800:200c:417a</pre> The address is followed by the prefix length, a decimal value indicating how many of the high-order contiguous bits of the address comprise the prefix (the network portion of the address). A slash (/) must precede the prefix length.

5.1.18. l2-transport

Description	Configures an interface or sub-interface as an L2 service attachment circuit.
Command Syntax	<code>l2-transport {enable disable}</code>
Command Mode	configuration interface
Parameters Table	N/A
Examples	<pre>exaware(config-interface-x-eth-0/1/0)# l2-transport disable exaware(config-Interface-x-eth-0/1/0.10)# l2-transport enable exaware(config-Interface-agg-eth-1)# l2-transport enable exaware(config-Interface-agg-eth-1.10)# no l2-transport</pre>
Notes	Default for all interfaces is disable. Once an interface is set with L2-transport, all L3 configurations are removed. Once an interface is set with L2-transport, all QoS policy attachment points are removed. L2-transport option cannot be set on a LAG member. Only x-eth and agg-eth interfaces can be configured as an L2-transport.

5.1.19. lacp

Description	Sets the LACP protocol status of the interface. Use the no form of the command to set to the default (inactive).		
Command Syntax	<pre>lacp [active inactive] no lacp</pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>lacp protocol status</i>	active / inactive	Defines whether or not LACP will be used. (Default = inactive)
Examples	<pre>exaware# config exaware(config)# interface agg-eth 50 exaware(config-interface-agg-eth-50)# lacp active</pre>		

5.1.20. lacp-interval

Description	Sets the interval used by LACP to check connectivity on the interface. Use the no form of the command to set to the default (short).		
Command Syntax	<pre>lacp-interval [short long] no lacp-interval</pre>		
Command Mode	configuration interface agg-eth		
Parameters Table	Name	Value	Description
	<i>lacp-interval</i>	short / long	The interval used by LACP to check connectivity. (Default = short)
Examples	<pre>exaware# config exaware(config)# interface agg-eth 50 exaware(config-interface-agg-eth-50)# lacp-interval long</pre>		

5.1.21. lacp-rx-interval

Description	Sets the short receive interval used by LACP. Use the no form of the command to set to the default (1000).		
Command Syntax	<code>lacp-rx-interval lacp-rx-interval</code> <code>no lacp-rx-interval</code>		
Command Mode	configuration interface agg-eth		
Parameters Table	Name	Value	Description
	<i>lacp-rx-interval</i>	100-1000	The short receive interval used by LACP. (Default = 1000)
Examples	<pre>exaware# config exaware(config)# interface agg-eth 11 exaware(config-interface-agg-eth-11)# lacp-rx-interval 200 exaware(config-interface-agg-eth-11)# no lacp-rx-interval</pre>		
Notes	The lacp-rx-interval will be available for configuration only if the agg-eth interface is LACP and is active, and the lacp-interval is "short".		

5.1.22. lacp-tx-interval

Description	Sets the short transmit interval used by LACP. Use the no form of the command to set to the default (1000).		
Command Syntax	<code>lacp-tx-interval lacp-tx-interval</code> <code>no lacp-tx-interval</code>		
Command Mode	configuration interface agg-eth		
Parameters Table	Name	Value	Description
	<i>lacp-tx-interval</i>	100-1000	The short transmit interval used by LACP. (Default = 1000)
Examples	<pre>exaware# config exaware(config)# interface agg-eth 11 exaware(config-interface-agg-eth-11)# lacp-tx-interval 200 exaware(config-interface-agg-eth-11)# no lacp-tx-interval</pre>		
Notes	The lacp-tx-interval will be available for configuration only if the agg-eth interface is LACP and is active, and the lacp-interval is "short".		

5.1.23. lldp

Description	To enable the LLDP protocol on an Ethernet port, use the lldp command. To disable LLDP on the Ethernet port, use the no form of this command.		
Command Syntax	lldp tx-only no lldp		
Command Mode	configuration interface x-eth		
Parameters Table	Name	Value	Description
	tx-only		The LLDP messages are sent, received LLDFP messages are dropped.
Examples	The following example shows how to use the command: <pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# lldp tx-only</pre>		

5.1.24. mac-address

Description	Configures the MAC layer address. (Command is not applicable for sub-interfaces.) Address should be unicast address. Use the no form of the command to set the MAC address to its default value (burned hardware value).		
Command Syntax	<pre>mac-address value1:value2:value3:value4:value5:value6</pre> <pre>no mac-address</pre>		
Command Mode	configuration interface		
Parameters Table	Name	Value	Description
	value1...6	00-FF	One byte of MAC address in hexadecimal, 1 is the MSB byte.
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# mac-address 10.22.33.44.AA.1E</pre>		

5.1.25. min-links

Description	Sets the minimum number of links needed for an aggregation interface to be considered up. Use the no form of the command to set to the default.		
Command Syntax	<pre>min-links <i>links</i> no min-links</pre>		
Command Mode	configuration interface agg-eth		
Parameters Table	Name	Value	Description
	<i>links</i>	Numeric	The minimal number of links that must be available for the aggregated interface to be considered up. (Default = 1)
Examples	<pre>exaware# config exaware(config)# interface agg-eth 50 exaware(config-interface-agg-eth-50)# min-links 10</pre>		

5.1.26. mpls

Description	Enables MPLS mode on the interface. By default, MPLS is disabled. Use the no form of the command to disable MPLS on the interface. Command is not applicable for loopback and mgmt interfaces.
Command Syntax	<code>mpls [enable disable]</code> <code>no mpls</code>
Command Mode	configuration interface <interface>
Parameters Table	N/A
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/0/0 exaware(config-interface-x-eth-0/0/0)# mpls enable</pre>

5.1.27. mtu

Description	Sets the MTU of an interface. Use the no form of the command to set the MTU to the default value. Command is not applicable for loopback and agg-eth interfaces.		
Command Syntax	<pre>mtu mtu</pre> <pre>no mtu</pre>		
Command Mode	configuration interface <interface>		
Parameters Table	Name	Value	Description
	mtu	594-9216	Maximum Transmission Unit of the interface L2 in bytes (Default = 0, meaning no MTU is applied)
Examples	<pre>exaware# config</pre> <pre>exaware(config)# interface mgmt 0/0/0</pre> <pre>exaware(config-interface-mgmt-0/0/0)# mtu 1000</pre>		
Notes	The command fails if the mtu shg selector is shg.		

5.1.28. mtu-l3 [not supported in release 8.0.0]

Description	Sets the maximum allowed size of an IP/mpls packet on an interface (including IP header). Use the no form of the command to set the maximum allowed value based on the MTU configuration on the interface. Command is not applicable for loopback and mgmt interfaces.		
Command Syntax	<pre>mtu-l3 mtu no mtu-l3</pre>		
Command Mode	configuration interface <interface>		
Parameters Table	Name	Value	Description
	mtu (Ethernet)	Physical interface: 576-9202 Sub-interface: 576-9198	Maximum allowed size of IPv4/IPv6/mpls packet in bytes.
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/0/0 exaware(config-interface-x-eth-0/0/0)# mtu-l3 4096</pre>		

5.1.29. ndp ageing-timeout

Description	Configures the IPv6 Neighbor Discovery Ageing Timeout on a selected interface, in seconds. Use the no form of the command to set the default value (14400).		
Command Syntax	<pre>ndp ageing-timeout [seconds] no ndp ageing-timeout</pre>		
Command Mode	configuration interface <interface>		
Parameters Table	Name	Value	Description
	seconds	120-2147483	The time in seconds for which an ipv6 neighbor discovery entry remains in the cache. (Default = 14400)
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/0/0 exaware(config-interface-x-eth-0/0/0)# ndp ageing- timeout 1000 exaware(config-interface-x-eth-0/0/0)# no ndp ageing-timeout</pre>		

5.1.30. outer-ether-type

Description	Set outer VLAN ether-type. Use the no form of the command to restore the default configuration (0x8100).		
Command Syntax	outer-ether-type <i>type</i> no outer-ether-type		
Command Mode	configuration interface <i><interface></i>		
Parameters Table	Name	Value	Description
	<i>type</i>	0x8100	Valid vlan ether-type
		0x88a8	
		0x9100	
		custom	
Examples	exaware# config exaware(config)# interface x-eth 0/1/0 exaware(config-interface-x-eth-0/1/0)# outer-ether-type 0x88a8		
Notes	This command is available only on physical interfaces.		

5.1.31. policy

Description	Sets the traffic policy for the interface.		
Command Syntax	policy {acl qos} {in in-port-classification out out-port-pcp-marking} <i>policy-name</i>		
Command Mode	configuration interface <interface>		
Parameters Table	Name	Value	Description
	acl qos		Attach either an ACL or a QoS policy.
	in out in-port-classification out-port-pcp-marking		Attach either an ingress or an egress policy.
	<i>policy-name</i>	name of existing policy	Name of the policy to attach.
Examples	<pre> exaware# config exaware(config)# interface x-eth 0/0/0 exaware(config-interface-x-eth-0/0/0)# policy acl out acl_policy1 exaware(config-interface-x-eth-0/0/0)# policy acl in acl_policy2 exaware(config-interface-x-eth-0/0/0)# policy qos out qos_policy1 exaware(config-interface-x-eth-0/0/0)# policy qos in qos_policy2 </pre>		

Note

- policy acl is not applicable for the loopback interface.
- policy acl can use only the in or out attributes.
- policy qos is not applicable for the mgmt and loopback interfaces.
- policy qos in select queues and policer based on flexible packet fields, and uses internal PMF engine resources.
- policy qos in-port-classification rules are per physical port or aggregate port settings. The selection rules are DSCP for IP traffic and PCP for non-IP traffic, and the result can select the queues.
- policy qos out defines only queues and shaping parameters.
- policy qos out-port-pcp-marking sets the outgoing PCP for VLAN marked packets based on the packet internal queue the packet used.

5.1.32. queues-statistics-mode

Description	Configures the qos statistics mode of the queues assigned to the interface. Use the no form of the command to set to the default (none).		
Command Syntax	queues-statistics-mode {none all ef-queues af-queues be-queue} no queues-statistics-mode		
Command Mode	configuration interface x-eth. <i>n</i>		
Parameters Table	Name	Value	Description
	none		No queue is counted.
	all		Affects all 8 queues.
	ef-queues		Affects EF1 to EF2 queues. Relevant to J2 only.
	af-queues		Affects AF1 to AF5 queues. Relevant to J2 only.
	be-queues		Affects BE queue. Relevant to J2 only.
Examples	<pre>exaware(config-interface-x-eth-0/0/3.100)# queues-statistics-mode af-queues exaware(config-interface-x-eth-0/0/3.100)# no queues-statistics-mode</pre>		

5.1.33. reuse [not supported in release 8.0.0]

Description	Configures the penalty threshold below which the link suppression will be removed. Use the no form of the command to set the reuse threshold value to default.		
Command Syntax	<pre>reuse <reuse-threshold> no reuse</pre>		
Command Mode	configuration interface <x-eth agg-eth> link-damping		
Parameters Table	Name	Value	Description
	<i>reuse-threshold</i>	1-20000	Accumulated penalty amount at which point the link suppression is removed. (Default = 1000)
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/1/4 exaware(config-interface-x-eth-0/1/4)# link-damping exaware(config-interface-x-eth-0/1/4-link-damping)# reuse 5000</pre>		

5.1.34. routing-policy-accounting [not supported in
release 8.0.0]

Description	Enables/disables routing policy accounting on a specific interface. Policy accounting can be enabled on each physical interface with the following options: • Input destination-accounting • Input source-accounting • Output destination-accounting • Output source-accounting The default for 'routing-policy-accounting in' is destination. The default for 'routing-policy-accounting out' is source. Use the no form of the command to disable routing-policy-accounting on the interface.		
Command Syntax	<pre>routing-policy-accounting {in out} [destination source] no routing-policy-accounting [in out]</pre>		
Command Mode	configuration interface <x-eth agg-eth>		
Parameters Table	Name	Value	Description
		in	Ingress direction only.
		out	Egress direction only.
		destination	Set accounting based on destination IP address. (Default for 'in')
		source	Set accounting based on source IP address. (Default for 'out')

Examples	<pre>exaware(config)# interface x-eth 0/0/0 exaware(config-interface-x-eth-0/0/0)# routing-policy- accounting in destination exaware(config-interface-x-eth-0/0/0)# routing-policy- accounting out source exaware(config-interface-x-eth-0/0/0)# no routing- policy-accounting in exaware(config-interface-x-eth-0/0/0)# no routing- policy-accounting</pre>
Notes	When setting routing-policy-accounting on an interface, each packet will be counted on either a specific traffic-counter or a default traffic-counter ("default"). When applying 'routing-policy-accounting out source' on any interface, traffic performance (pps) degradation is expected. 'Mac-accounting' and 'routing-policy-accounting' cannot be configured on the same interface on egress direction.

5.1.35. speed

Description	Configures the port speed.
Command Syntax	speed 1G 10G 25G 40G 100G
Command Mode	configuration interface x-eth
Parameters Table	N/A
Examples	<pre>exaware(config)# interface x-eth-0/0/1 exaware(config-Interface-x-eth-0/0/1)# speed 25G</pre>
Notes	The concrete white box specifies the port default speed and the set of speed values that can be configured.

5.1.36. subinterface-queueing

Description	Configures whether TM queues are allocated for the sub interface or is it using the physical port queues.
Command Syntax	<code>subinterface-queueing {disabled all-queues}</code> <code>no subinterface-queueing</code>
Command Mode	configuration x-eth subinterface
Examples	<pre>exaware# config exaware(config)# interface x-eth 0/0/3.100 exaware(config-interface-x-eth-0/0/3.100)# subinterface-queueing all-queues exaware(config-interface-x-eth-0/0/3.100)# no subinterface-queueing</pre>

5.1.37. vlan-id

Description	Sets the QinQ VLAN values.		
Command Syntax	vlan-id <i>range</i>		
Command Mode	configuration interface <interface>		
Parameters Table	Name	Value	Description
	<i>range</i>	Range of <1-4094>-<1-4094>	Valid VLAN range.
Examples	<pre>exaware(config-interface-x-eth-0/1/0.53)# vlan-id 53 exaware(config-interface-x-eth-0/1/0.53)# vlan-id 71-74 exaware(config-interface-x-eth-0/1/0.53)# vlan-id (71-74, 100-104, 500, 645-701)</pre>		
Notes	This command is available only on x-eth and agg-eth sub-interfaces. By default, vlan-id will be the current interface sub-interface number. The commands "vlan-id" and "vlan-tags" are mutually exclusive, and the new configuration will override old one.		

5.1.38. vlan-op-ingress

Description	Sets the QinQ VLAN operation on ingress.		
Command Syntax	vlan-op-ingress {pop pop-pop pop-swap <i>number</i> swap <i>number</i> }		
Command Mode	configuration interface <interface>		
Parameters Table	Name	Value	Description
	<i>number</i>	1-4094	Valid VLAN number.
Examples	exaware(config-interface-x-eth-0/1/0.53)# vlan-op-ingress pop		
Notes	This command is available only on x-eth and agg-eth sub-interfaces.		
	This command is available only when l2-transport is enabled.		

5.1.39. vlan-tags

Description	Sets the QinQ outer and inner VLAN values.		
Command Syntax	vlan-tags [outer <i>number</i>] [inner <i>range</i>]		
Command Mode	configuration interface <interface>		
Parameters Table	Name	Value	Description
	<i>number</i>	1-4094	Valid VLAN number.
	<i>range</i>	Range of <1-4094>-<1-4094>	Valid VLAN range.
Examples	<pre>exaware(config-interface-x-eth-0/1/0.53)# vlan-tags outer 188 inner 71 exaware(config-interface-x-eth-0/1/0.53)# vlan-tags outer 100 exaware(config-interface-x-eth-0/1/0.53)# vlan-tags inner 30-50 exaware(config-interface-x-eth-0/1/0.53)# vlan-tags outer 188 inner 71-74 exaware(config-interface-x-eth-0/1/0.53)# vlan-tags outer 188 inner (71-74) exaware(config-interface-x-eth-0/1/0.53)# vlan-tags outer 188 inner (71-74, 100-104, 500, 645-701)</pre>		
Notes	This command is available only on x-eth and agg-eth sub-interfaces. The inner range configuration is available only when the outer number configuration exists. The commands "vlan-id" and "vlan-tags" are mutually exclusive, and the new configuration will override old one.		

5.1.40. vrf

Description	This command binds an interface to a VRF. If not configured, the interface is associated with the default VRF which is generated automatically by the system, and cannot be deleted.		
Command Syntax	<pre>vrf vrf-name</pre> <pre>no vrf</pre>		
Command Mode	configuration interface <interface>		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	existing VRF tag	VRF in which this interface operates. (Default = default)
Examples	<pre>exaware# config</pre> <pre>exaware(config)# interface x-eth 0/0/0</pre> <pre>exaware(config-interface-x-eth-0/0/0)# vrf my_vrf</pre>		

5.2. Interface Operational Commands

5.2.1. clear interface counters

Description	Clears counters of all interfaces or a specific interface.		
Command Syntax	<code>clear interface counters [all x-eth <i>eth-if</i> mgmt <i>mgmt-if</i> agg-eth <i>agg-id</i>]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>eth-if</i>	0/0/port[:sub-interface]	Ethernet interface on which to clear counters.
	<i>mgmt-if</i>	0/0/0 0/0/1	Management interface on which to clear counters.
	<i>agg-id</i>	0-79	Aggregation interface ID.
Examples	The following examples show how to use this command: <pre>exaware# clear interface counters mgmt 0/0/0 exaware# clear interface counters x-eth 0/1/10 exaware# clear interface counters agg-eth 7 exaware# clear interface counters all</pre>		

5.2.2. clear interface queues-statistics

Description	Clears the queues statistics either for a specified interface or for all interfaces.
Command Syntax	<code>clear interface queues-statistics {<i>interface-name</i> all}</code>
Command Mode	Operational
Examples	The following examples show how to use this command: <pre>exaware# clear interface queues-statistics x-eth/0/0/0.100</pre>

5.2.3. clear interface routing-policy-accounting [not supported in release 8.0.0]

Description	Clears per interface policy accounting counters. Can be performed on specific interface or all interfaces. Can also be performed for one direction (in/out) or both.		
Command Syntax	clear interface routing-policy-accounting [in out] [all x-eth <i>eth-if</i> agg-eth <i>agg-id</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	direction	in out	Direction of counters to clear. (Default = both in and out).
	interface	all x-eth <i>eth-if</i> agg-eth <i>agg-id</i>	Interface for which to clear counters (or all).
Examples	The following examples show how to use this command: <pre>exaware# clear interface routing-policy-accounting x-eth0/0/0</pre> <pre>exaware# clear interface routing-policy-accounting in all</pre> <pre>exaware# clear interface routing-policy-accounting in x-eth 0/0/0</pre> <pre>exaware# clear interface routing-policy-accounting out all</pre>		

5.2.4. clear lacp counters

Description	Clears LACP counters of all or a specific agg-eth interface.		
Command Syntax	<code>clear lacp counters [all agg-eth <i>eth-if</i>]</code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>eth-if</i>	0/0/port[:channel]	Aggregated ethernet interface on which to clear LACP counters.
Examples	The following example shows how to use this command: <code>exaware# clear lacp counters agg-eth 0/0/1</code>		

5.3. Interface Show Commands

5.3.1. show interface

Description	Shows interface statistics. If no interface is specified, all interfaces except loopback ones are displayed. The Loopback interfaces are not displayed.
Command Syntax	show interface [<i>interface-name</i>]
Command Mode	operational
Parameters Table	N/A
Examples	The following examples show sample output from this command: <pre> exaware# show interface x-eth 0/0/1 interface x-eth 0/0/1 Oper-state: up Link up delay: disabled Link down delay: disabled Link damping: disabled SNMP interface index: 1073741824 Layer 1 Parameters Link capabilities: 10G-Base Link Speed: 100G Flow control: rx Layer 2 Parameters MAC address: 80:a2:35:04:40:0e Encapsulation: Ethernet II Expected Ether-type combinations: [0x8100] [0x8100, 0x8100] [0x88a8] [0x88a8, 0x8100] LLDP: disable L2 MTU: 1500 Layer 3 Parameters VRF name: default VRF ID: 0 L3 MTU: 0 Bytes IPv4 primary address: 27.11.0.1/21 IPv4 unicast RPF: disabled ICMPv4: disabled Handling of IP options: forward IPv6 link local address: fe80::82a2:35ff:fe04:400e </pre>

```
IPv6 unicast addresses: 3027:11::1/64
IPv6 mc solicited addresses: ff02::1:ff04:400e
                             ff02::1:ff00:1

IPv6 unicast RPF: disabled
ICMPv6: disabled
Handling of hop by hop option extension header: forward
MPLS: enabled
HQ: disabled
ARP timeout: 120sec
NDP timeout: 120sec
```

QoS

```
Number of queues: 8
Queues statistics Mode: af-queues
QoS Out Policy: poll
```

Statistics

```
Last clear counters: 1hour 14min 54sec
```

Rx statistics

```
Total bytes: 0
Total packets: 0
Average rate: 0 bps
Average packet rate: 0 pps
```

Tx statistics

```
Total bytes: 0
Total packets: 0
Average rate: 0 bps
Average packet rate: 0 pps
```

Physical port statistics

Rx statistics

```
Total bytes: 3465706791872
Total packets: 13537916414
Average rate: 8.03 gbps
Average packet rate: 3.92 mpps
Received packets: 0
  Ucast: 13537916260
  Mcast: 0
  Bcast: 154
  Control: 0
  Pause: 0
Error packets: 0
  CRC errors: 0
  Runts: 0
  Giants: 0
  Ignored: 0
```

Tx statistics

```
Total bytes: 3625176
Total packets: 29938
Average rate: 8.22 kbps
Average packet rate: 2 pps
Transmitted packets: 0
  Ucast: 0
  Mcast: 29800
  Bcast: 138
  Control: 0
  Pause: 0
Error packets: 0
```

Discard packets:	0		
MAC filtered:	0		
FIFO drop:	0		
exaware# show interface x-eth 0/1/9.1			
interface x-eth 0/1/9.1			
Oper-state: up			
SNMP interface index: 1073922049			
Layer 2 Parameters			
MAC address: 80:a2:35:04:40:0e			
Encapsulation: 802.1Q Virtual LAN, VLAN ID 1			
Added Ether-type: 0x8100			
Layer 3 Parameters			
VRF name: default VRF ID: 0			
L3 MTU: 0 Bytes			
IPv4 primary address: 27.11.8.1/21			
IPv4 unicast RPF: disabled			
ICMPv4: disabled			
Handling of IP options: forward			
IPv6 link local address: fe80::82a2:35ff:fe04:400e			
IPv6 unicast addresses: 3027:11:1::1/64			
IPv6 mc solicited addresses: ff02::1:ff04:400e			
ff02::1:ff00:1			
IPv6 unicast RPF: disabled			
ICMPv6: disabled			
Handling of hop by hop option extension header: forward			
MPLS: enabled			
ARP timeout: 180sec			
NDP timeout: 180sec			
QoS			
Num of queues: 8			
Queues statistics Mode: af-queues			
QoS Out Policy: poll			
Statistics			
Last clear counters: 58min			
Rx statistics		Tx statistics	
Total bytes:	256	Total bytes:	0
Total packets:	4	Total packets:	0

Average rate: 0 bps
Average packet rate: 0 pps

Average rate: 0 bps
Average packet rate: 0 pps

5.3.2. show interface average-rate

Description	Shows the interface average rate (bps) and average packet rate (pps) for both RX/TX directions.																																																																																																																								
Command Syntax	show interface average-rate [<interface>]																																																																																																																								
Command Mode	operational																																																																																																																								
Parameters Table	Name	Value			Description																																																																																																																				
	interface	type 0/0/port			Specifies the interface to be displayed.																																																																																																																				
Examples	The following examples show sample output from this command:																																																																																																																								
<pre>exaware# show interface average-rate x-eth0/1/0</pre> <table><thead><tr><th>INSTANCE</th><th>STATUS</th><th>TX PPS</th><th>TX BPS</th><th>RX PPS</th><th>RX BPS</th></tr></thead><tbody><tr><td>x-eth0/1/0</td><td>up</td><td>0</td><td>0</td><td>21.56 k</td><td>44.16 M</td></tr></tbody></table>								INSTANCE	STATUS	TX PPS	TX BPS	RX PPS	RX BPS	x-eth0/1/0	up	0	0	21.56 k	44.16 M																																																																																																						
INSTANCE	STATUS	TX PPS	TX BPS	RX PPS	RX BPS																																																																																																																				
x-eth0/1/0	up	0	0	21.56 k	44.16 M																																																																																																																				
<pre>exaware# show interface average-rate</pre> <table><thead><tr><th>INSTANCE</th><th>STATUS</th><th>TX PPS</th><th>TX BPS</th><th>RX PPS</th><th>RX BPS</th></tr></thead><tbody><tr><td>x-eth0/1/0</td><td>up</td><td>0</td><td>0</td><td>21.56 k</td><td>44.16 M</td></tr><tr><td>x-eth0/1/1</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/2</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/3</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/4</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/5</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/6</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/7</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/8</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/9</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/10</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/11</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/12</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/13</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/14</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/15</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/16</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>x-eth0/1/17</td><td>admin-down</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></tbody></table>								INSTANCE	STATUS	TX PPS	TX BPS	RX PPS	RX BPS	x-eth0/1/0	up	0	0	21.56 k	44.16 M	x-eth0/1/1	admin-down	0	0	0	0	x-eth0/1/2	admin-down	0	0	0	0	x-eth0/1/3	admin-down	0	0	0	0	x-eth0/1/4	admin-down	0	0	0	0	x-eth0/1/5	admin-down	0	0	0	0	x-eth0/1/6	admin-down	0	0	0	0	x-eth0/1/7	admin-down	0	0	0	0	x-eth0/1/8	admin-down	0	0	0	0	x-eth0/1/9	admin-down	0	0	0	0	x-eth0/1/10	admin-down	0	0	0	0	x-eth0/1/11	admin-down	0	0	0	0	x-eth0/1/12	admin-down	0	0	0	0	x-eth0/1/13	admin-down	0	0	0	0	x-eth0/1/14	admin-down	0	0	0	0	x-eth0/1/15	admin-down	0	0	0	0	x-eth0/1/16	admin-down	0	0	0	0	x-eth0/1/17	admin-down	0	0	0	0
INSTANCE	STATUS	TX PPS	TX BPS	RX PPS	RX BPS																																																																																																																				
x-eth0/1/0	up	0	0	21.56 k	44.16 M																																																																																																																				
x-eth0/1/1	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/2	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/3	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/4	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/5	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/6	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/7	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/8	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/9	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/10	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/11	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/12	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/13	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/14	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/15	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/16	admin-down	0	0	0	0																																																																																																																				
x-eth0/1/17	admin-down	0	0	0	0																																																																																																																				

```
x-eth0/1/18  admin-down  0      0      0      0
x-eth0/1/19  admin-down  0      0      0      0
```

Name	Value	Range or Possible Values
INSTANCE	Interface name	string
STATUS	Interface operational state	string
TX PPS	Average transmit packet rate	Numeric 64bit value.
TX BPS	Average transmit rate	Numeric 64bit value.
RX PPS	Average receive packet rate	Numeric 64bit value.
RX BPS	Average receive rate	Numeric 64bit value.

5.3.3. show interface brief

Description	Shows common interface statistics. If no interface is specified, all interfaces are displayed. The loopback interfaces are not displayed.									
Command Syntax	show interface brief									
Command Mode	operational									
Parameters Table	N/A									
Examples	The following examples show sample output from this command:									
exaware# show interface brief										
INTERFACE	STATUS	SPEED	RX PKTS	RX BYTES	RX ERRS	TX PKTS	TX BYTES	TX ERRS	DESCRIPTION	
x-eth0/0/0	admin-down	10G	0	0	0	0	0	0	Eth0-0-0	
x-eth0/0/0.3	down	10G	0	0	0	0	0	0	Eth0-0-0/3	
x-eth0/0/1	up	100G	428790	0	4	44331	2818612	0		
mgmt0/0/0	down	1G	435	4	0	0	24	2		
mgmt0/0/1	up	1G	422800	0	6	68373	0	0		

5.3.4. show interface detail

Description	Shows detailed operational data for all interfaces or for a specific interface. If no parameter is selected, the full information is displayed for all interfaces.		
Command Syntax	show interface [loopback <i>loop-if</i> x-eth <i>eth-if</i> mgmt <i>mgmt-if</i> agg-eth <i>agg-id</i>] detail		
Command Mode	operational		
Parameters Table	Name	Value	Description
		interface type and instance: loopback <i>loop-if</i> x-eth <i>eth-if</i> mgmt <i>mgmt-if</i> agg-eth <i>agg-id</i>	Specifies the interface to be displayed.

Examples	The following examples show typical output for different types of interfaces:
This example shows a physical Ethernet interface:	
<pre>exaware# show interface x-eth 0/0/1 detail interface x-eth 0/0/1 Oper-state: up MPLS: disabled Multicast: enabled Layer 1 Parameters Link capabilities: 10G 100G Link Speed: 100G Fec mode: cl91 Flow control: disabled Layer 2 Parameters MAC address: 00:0c:29:a2:bd:be L2 MTU: 1500 bytes Encapsulation: Ethernet II LLDP: disable Layer 3 Parameters L3 MTU: 1482 bytes VRF name: default VRF ID: 0 Handling of IP options: forward IPV4 primary address: f:fea2:bdbe IPV4 MTU: 1500 IPV4 unicast RPF: disabled IPV6 link local address: fe80::4212:e4ff:fe00:2c4c IPV6 unicast addresses: 2001::1/64 IPV6 mc solicited addresses: ff02::1:ff00:2c4c ff02::1:ff00:1 IPV6 MTU: 1500 IPV6 unicast RPF: disabled MPLS MTU: 1500 ARP timeout: 14400sec NDP timeout: 14400sec Statistics Last clear counters: 19min 29sec Rx statistics Total bytes: 470649924756 Total packets: 1188509911 IPv4: 1188509911 IPv6: 0 MPLS: 0 Other: 0 Tx statistics Total bytes: 468326579292 Total packets: 1182642877 IPv4: 1182642877 IPv6: 0 MPLS: 0 Other: 0 Physical port statistics Rx statistics Total bytes: 475404199200 Total packets: 1188510498 Average rate: 9.52 gbps Tx statistics Total bytes: 475404203600 Total packets: 1188510509 Average rate: 9.52 gbps</pre>	

Average packet rate:	2.97 mpps	Average packet rate:	2.97 mpps
Received packets:	1188510498	Transmitted packets:	1188510509
Ucast:	1188510498	Ucast:	1188510509
Mcast:	0	Mcast:	0
Bcast:	0	Bcast:	0
Control:	0	Control:	0
Pause:	0	Pause:	0
Error packets:	0	Error packets:	0
CRC errors:	0		
Runts:	0		
Giants:	0		
Ignored:	0		
Discard packets:	0		
MAC filtered:	0		
FIFO drop:	0		

This example shows a physical Ethernet interface with LACP enabled on the aggregator:

```
exaware# show interface x-eth 0/0/2 detail
interface x-eth 0/0/2
  Oper-state: up
  Aggregated in agg-eth 5, LACP-state: up
  Link up delay: disabled
  Link down delay: disabled
  MPLS: disabled
  Interface index: 0x2000fff  SNMP ifIndex: 0x2000000
  Layer 1 Parameters
    Link capabilities: 10G 100G
    Link Speed: 100G
    Fec mode: cl91
    Flow control: disabled
  Layer 2 Parameters
    MAC address: 00:0c:29:89:09:fe
    L2 MTU: 1518 bytes      Encapsulation: Ethernet II
    Link capabilities: 10G-Base
    LLDP: disable
```

This example shows a sub-interface:

```
exaware# show interface x-eth 0/1/5.6 detail
interface x-eth 0/1/5.6
  Oper-state: up
  Link up delay: disabled
  Link down delay: disabled
  Interface index: 0x2045fff   SNMP ifIndex: 0x6045000
  Layer 2 Parameters
    MAC address: 40:12:e4:00:00:1d
    L2 MTU: 9200 Bytes           Encapsulation: Ethernet II
    Link capabilities: 10G-Base
  Layer 3 Parameters
    VRF name: default   VRF ID: 0
    IPv4 primary address: 50.50.50.3/24
    IPv4 MTU: 9200 Bytes
    IPv4 unicast RPF: disabled
    ICMPv4: disabled
    Handling of IP options: forward
    IPv6 link local address: fe80::4212:e4ff:fe00:2c4c
    IPv6 unicast addresses: 2001::1/64
    IPv6 mc solicited addresses: ff02::1:ff00:2c4c
                                   ff02::1:ff00:1

    IPv6 MTU: 9200 Bytes
    IPv6 unicast RPF: disabled
    ICMPv6: disabled
    Handling of hop by hop option extension header: forward
    MPLS: disabled
    MPLS MTU: 9200 Bytes
    ARP timeout: 14400sec
    NDP timeout: 14400sec

  Statistics
  Last clear counters: 19min 29sec
  Rx statistics
    Total bytes:      470649924756
    Total packets:    1188509911
    IPv4:              1188509911
    IPv6:              0
    MPLS:              0
    Other:             0
  Tx statistics
    Total bytes:      468326579292
    Total packets:    1182642877
    IPv4:              1182642877
    IPv6:              0
    MPLS:              0
    Other:             0
```

This example shows a physical management interface:

```
exaware# show interface mgmt 0/0/0 detail
interface mgmt 0/0/0
  Description: mgmt
  Oper-state: up
  Full-duplex 1000Mbit/sec  Auto-negotiation: enabled
  Layer 2 Parameters
    MAC address: 00:0c:29:c9:42:f4
    L2 MTU: 1389 bytes      Encapsulation: Ethernet II
    Link capabilities: 1G 10G
    Link Speed: 10G
  Layer 3 Parameters
    L3 MTU: 1371 bytes
    VRF name: management  VRF ID: 8191
    IPV4 primary address is not set
    IPV4 MTU: 1385
    IPV6: disabled
    IPV6 MTU: 1389
    ARP timeout: 14400sec

  Statistics
    Last clear counters:      19min 37sec
    Rx packets:      118521      Tx packets:      0
    Rx bytes:      2772720      Tx bytes:      0
    Rx drops:      0      Tx drops:      0
    Rx IPV4 packets: 118521      Tx IPV4 packets: 0
    Rx IPV6 packets: 0      Tx IPV6 packets: 0

  L2 Statistics
    Rx packets:      21349      Tx packets:      0
    Rx bytes:      3308471      Tx bytes:      0
    Rx errors:      0      Tx errors:      0
    Rx drops:      0      Tx drops:      0
    Rx ucast packets: 21349      Tx ucast packets: 0
    Rx mcast packets: 0      Tx mcast packets: 0
    Rx bcast packets: 0      Tx bcast packets: 0
    Rx CRC errors:   0      Tx collisions:   0
    Rx runts:      0
    Rx giants:     0
```

This example shows a management VLAN interface:

```
exaware# show interface mgmt 0/0/0.1 detail
interface mgmt 0/0/0.1
  Description: man
  Oper-state: up
  MPLS: disabled
  Multicast: enabled
  Layer 2 Parameters
    MAC address: 00:0c:29:89:09:fe
    L2 MTU: 1500 bytes   Encapsulation: 802.1Q Virtual LAN, VLAN Id 1
  Layer 3 Parameters
    L3 MTU: 1478 bytes
    VRF name: default   VRF ID: 0
    Handling of IP options: forward
    IPv4 primary address is not set
    IPv4 MTU: 1500
    IPv4 unicast RPF: disabled
    IPv6: disabled
    IPv6 MTU: 1500
    IPv6 unicast RPF: disabled
    MPLS MTU: 1500
    ARP timeout: 14400sec

Statistics
Last clear counters:      39sec
Rx packets:      18446744073709551568   Tx packets:      48
Rx bytes:        432345564227567568     Tx bytes:        48
Rx drops:        0                       Tx drops:        0
Rx IPv4 packets: 18446744073709551600   Tx IPv4 packets: 16
Rx IPv6 packets: 0                       Tx IPv6 packets: 0
Rx MPLS packets: 0                       Tx MPLS packets: 0
```

This example shows a loopback interface:

```
exaware# show interface loopback 0 detail
interface loopback 0
  Oper-state: up
  Layer 3 Parameters
    VRF name: default   VRF ID: 0
    IPv4 primary address is not set
    IPv6: disabled
```

This example shows an agg-eth interface with LACP active and where the interval is short:

```
exaware# show interface agg-eth 5
interface agg-eth 5
  Oper-state: up
  Aggregated interfaces: x-eth0/1/5, oper-state: up, member-state: up
                        x-eth0/1/6, oper-state: up, member-state: up
  LACP: active, short-interval (rx-interval: 350ms, tx-interval: 500ms)
  Configured-members: 2, currently-up: 2
  Min-links: 1
  Link up delay: disabled
  Link down delay: disabled
  SNMP interface index: 1610633216
  Layer 1 Parameters
...
```

5.3.5. show interface description

Description	Shows status and description for all interfaces.		
Command Syntax	show interface description <i>[interface]</i>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>interface</i>	type 0/0/port	Specifies the interface to be displayed.
Examples	The following example shows sample output from this command:		
exaware# show interface description			
INTERFACE STATUS DESCRIPTION			

x-eth0/0/0 admin-down Ethernet 0-0-0			
x-eth0/0/0.3 down Ethernet 0-0-0/3			
x-eth0/0/1 up			
mgmt0/0/0 down			
mgmt0/0/1 up			

5.3.6. show interface ip

Description	Shows all IP addresses assigned to each interface, sorted by VRF. You can filter by any of the following parameters: • Specific interface • Specific VRF • IP version (IPv4 or IPv6) The status is also displayed for each interface.		
Command Syntax	<pre>show interface ip [afi-v4 afi-v6] [vrf-tag {interface}]</pre>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vrf-tag</i>	Name of existing VRF	The specific VRF for which to show the IP addresses by interface.
Examples	The following examples show sample output from this command:		

```
exaware# show interface ip
```

VRF NAME	INTERFACE	STATUS	ADDRESS
default	x-eth0/0/0	admin-down	6.6.6.6/24 (v4)
default	x-eth0/0/0.3	down	
default	loopback109	up	109.109.109.2/32 (v4)
management	mgmt0/0/0	up	
management	mgmt0/0/1	admin-down	5.5.5.5/24 (v4)
zzzz	loopback18	up	

```
exaware# show interface ip afi-v4 vrf default
```

VRF NAME	INTERFACE	STATUS	ADDRESS
default	x-eth0/0/0	admin-down	6.6.6.6/24
default	x-eth0/0/0.3	down	
default	loopback109	up	109.109.109.2/32

```
exaware# show interface ip afi-v6 vrf zzz
```

VRF NAME	INTERFACE	STATUS	ADDRESS
zzzz	loopback18	up	

5.3.7. show interface optical-monitoring

Description	Shows operational parameters of an optical transceiver. This helps to measure the optical RX/TX signal power and determine the attenuation and power budget. Command is applicable for physical interfaces only.		
Command Syntax	show interface optical-monitoring <i>[interface]</i>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>interface</i>	interface name	Specifies the interface to be displayed.
Examples	The following examples show sample output from this command:		

```
exaware# show interface optical-monitoring x-eth0/1/6
interface optical-monitoring x-eth0/1/6
  Type: Tunable SFP  10G Base-LR
  Status: OK
  Exaware-SKU: LC-10G-OC-SFP-10K
  Link-Length:      10 Km. SM
  Temperature:      33.37 C
  Temperature-range: -13.00 to 78.00 C
  Supply voltage:    3.00 V
  Voltage range:     2.90 to 3.70 V
  Bias current range:15.00 to 85.00 mA
  TX power range:    -8.00 to 2.00 dBm
  RX power range:    -20.00 to 2.50 dBm
  Wave-Length 0 (1610.92nm):
    Laser bias current: 41.08 mA
    Tx optical power:   -1.39 dBm (0.73 mW)
    Rx optical power:   -1.26 dBm (0.75 mW)
```

For SFP/TSFP where its status is not ok (error/alarm/warning occurred):

In case of the interface optical component is not ok, an error line description will be displayed in the 'status' field in the following priority:

- TSFP error : the error message itself
- Alarm : “Alarm”
- Warning: “Warning”

For example:

```
interface optical-monitoring x-eth0/1/6
  Type: SFP/SFP+ 10G Base-LR
  Status: HW config failure
  Exaware-SKU: LC-10G-OC-SFP-10K
  Link-Length:      10 Km. SM
  Temperature:      33.37 C
  Temperature-range: -13.00 to 78.00 C
  Supply voltage:    3.00 V
  Voltage range:     2.90 to 3.70 V
  Bias current range:15.00 to 85.00 mA
  TX power range:    -8.00 to 2.00 dBm
  RX power range:    -20.00 to 2.50 dBm
  Wave-Length 0 (1610.92nm):
    Laser bias current: 41.08 mA
    Tx optical power:   --- dBm (0 mW)
    Rx optical power:   -1.26 dBm (0.75 mW)
```

5.3.8. show interface queues-statistics brief

Description	Shows a summary of queues statistics collected from all ingress queues towards an egress interface per TC.																																																																																											
Command Syntax	show interface queues-statistics brief { <i>interface-name</i> all}																																																																																											
Command Mode	operational																																																																																											
Parameters Table	Name	Value	Description																																																																																									
	<i>interface</i>	interface name	Specifies the interface to be displayed.																																																																																									
Examples	The following examples show sample output from this command:																																																																																											
exaware# show interface queues-statistics brief x-eth/0/0/0.100																																																																																												
INTERFACE x-eth 0/0/0.100 <table><thead><tr><th rowspan="2">TC</th><th colspan="4">FORWARD</th><th colspan="4">DROP</th></tr><tr><th>BYTES</th><th>AVERAGE RATE[bps]</th><th>PKTS</th><th>AVERAGE RATE [pps]</th><th>BYTES</th><th>AVERAGE RATE[bps]</th><th>PKTS</th><th>AVERAGE RATE[pps]</th></tr></thead><tbody><tr><td>EF1</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td></tr><tr><td>EF2</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td></tr><tr><td>AF1</td><td>1666252800</td><td>536.75 k</td><td>20828160</td><td>5.65 k</td><td>1666925100</td><td>332.77 k</td><td>17546580</td><td>3587</td></tr><tr><td>AF2</td><td>3764480000</td><td>1.01 M</td><td>470560000</td><td>11.56 k</td><td>1666652348</td><td>57.6 k</td><td>17595860</td><td>656</td></tr><tr><td>AF3</td><td>43825029120</td><td>11.9 G</td><td>461316096</td><td>124.14 M</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>AF4</td><td>1000000</td><td>9000</td><td>16000</td><td>200</td><td>400</td><td>80</td><td>5</td><td>1</td></tr><tr><td>AF5</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>BE</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td><td>NC</td></tr></tbody></table> Legend: NC - Not Counted Time since last clear statistics: 1hour 14min 54sec				TC	FORWARD				DROP				BYTES	AVERAGE RATE[bps]	PKTS	AVERAGE RATE [pps]	BYTES	AVERAGE RATE[bps]	PKTS	AVERAGE RATE[pps]	EF1	NC	NC	NC	NC	NC	NC	NC	NC	EF2	NC	NC	NC	NC	NC	NC	NC	NC	AF1	1666252800	536.75 k	20828160	5.65 k	1666925100	332.77 k	17546580	3587	AF2	3764480000	1.01 M	470560000	11.56 k	1666652348	57.6 k	17595860	656	AF3	43825029120	11.9 G	461316096	124.14 M	0	0	0	0	AF4	1000000	9000	16000	200	400	80	5	1	AF5	0	0	0	0	0	0	0	0	BE	NC	NC	NC	NC	NC	NC	NC	NC
TC	FORWARD				DROP																																																																																							
	BYTES	AVERAGE RATE[bps]	PKTS	AVERAGE RATE [pps]	BYTES	AVERAGE RATE[bps]	PKTS	AVERAGE RATE[pps]																																																																																				
EF1	NC	NC	NC	NC	NC	NC	NC	NC																																																																																				
EF2	NC	NC	NC	NC	NC	NC	NC	NC																																																																																				
AF1	1666252800	536.75 k	20828160	5.65 k	1666925100	332.77 k	17546580	3587																																																																																				
AF2	3764480000	1.01 M	470560000	11.56 k	1666652348	57.6 k	17595860	656																																																																																				
AF3	43825029120	11.9 G	461316096	124.14 M	0	0	0	0																																																																																				
AF4	1000000	9000	16000	200	400	80	5	1																																																																																				
AF5	0	0	0	0	0	0	0	0																																																																																				
BE	NC	NC	NC	NC	NC	NC	NC	NC																																																																																				
Notes	<i>Time since last clear statistics</i> is the time passed since the interface statistics were cleared last. <i>Average rate</i> is a moving window average of the last 5 seconds. - LAG interface displays a summary of all the LAG members.																																																																																											

5.3.9. show interface queues-statistics detail

Description	Shows the queues statistics collected for each of ingress queues connected to the egress interface.									
Command Syntax	show interface queues-statistics detail tc {interface-name all}									
Command Mode	operational									
Parameters Table	Name	Value		Description						
	all			Displays all queues.						
	tc	TC (EF1 to BE)		Displays the TC queues.						
Examples	The following examples show sample output from this command:									
exaware# show interface queues-statistics detail tc all agg-eth12										
Legend:										
NC - Not Counted.										
Time since last clear statistics: 1hour 14min 54sec										
INTERFACE agg-eth 12										
#DLC	SOURCE	TC			FORWARD		DROP			
ASIC		BYTES	AVERAGE	PKTS	AVERAGE	BYTES	AVERAGE	PKTS	AVERAGE	
			RATE[bps]		RATE[pps]		RATE[bps]		RATE[pps]	

0	0	EF1	NC	NC	NC	NC	NC	NC	NC	NC
0	0	EF2	NC	NC	NC	NC	NC	NC	NC	NC
0	0	AF1	1666252800	536.75 k	20828160	5.65 k	15100	332.77 k	17540	2535
0	0	AF2	3764480000	1.01 M	4705600	11.56 k	152348	57.6 k	17860	656
0	0	AF3	435029120	11.9 G	4613166	124.14 M	0	0	0	0
0	0	AF4	1000000	9000	16000	200	400	80	5	1
0	0	AF5	0	0	0	0	0	0	0	0
0	0	BE	NC	NC	NC	NC	NC	NC	NC	NC

Notes	• DLC is the unit number (for DDC). • <i>Time since last clear statistics</i> is the time passed since the interface statistics were cleared last. • <i>Average rate</i> is a moving window average of the last 5 seconds. • LAG interface displays a list of all the LAG members. In case the interface is a LAG member the LAG ID should be displayed.
--------------	--

5.3.10. show interface routing-policy-accounting [not supported in release 8.0.0]

Description	Shows routing-policy-accounting statistics per interface and direction (in / out). Only non-zero traffic-counters are displayed.		
Command Syntax	show interface routing-policy-accounting [<i>interface</i> [in out]]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>interface</i>	interface name	Specifies the interface for which routing policy-accounting statistics will be displayed.
	direction	in out	Ingress direction only. Egress direction only.
Examples	The following examples show sample output from this command:		

```
exaware# show interface routing-policy-accounting x-eth0/0/0
```

```
Interface x-eth0/0/0 in destination:
```

Counter-ID	Description	Packets	Bytes
0	Default	10	1000
1	EXAWARE	10	1000
2	Netanya	50	5000
3		10	1000

```
Interface x-eth0/0/0 out source:
```

Counter-ID	Description	Packets	Bytes
0	Default	10	1000
1	EXAWARE	10	1000
2	Netanya	50	5000
3		10	1000

```
exaware# show interface routing-policy-accounting x-eth0/0/0 in
```

```
Interface x-eth0/0/0 in destination:
```

Counter-ID	Description	Packets	Bytes
0	Default	10	1000
1	EXAWARE	10	1000
2	Netanya	50	5000
3		10	1000

Name	Value	Range or possible values
COUNTER-ID	Traffic-counter id	Counter id (0-63)
DESCRIPTION	Traffic-counter description (if configured)	Text
PACKETS	Packets on this traffic-counter	Number 64bit value
BYTES	Bytes on this traffic-counter	Number 64bit value

5.3.11. show interface speed-groups

Description	Displays port groups having the following limitation: if one port is configured speed of 25G, then the other ports of the same group must be configured of this speed too. There is not this limitation for other speeds.
Command Syntax	<code>show interface speed-groups</code>
Command Mode	operational
Examples	The following examples show sample output from this command: <code>exaware# show interface speed-groups</code> <code>x-eth0/0/16-19; x-eth0/0/20-24</code>

6. L2 Services Commands

6.1. L2 Services Configuration Commands

6.1.1. VPWS

6.1.1.1. interface (vpws)

Description	Configures interface (attachment circuit) under the VPWS service. Use the no form of the command to remove the interface configuration under the VPWS service.		
Command Syntax	<code>interface if-name</code>		
Command Mode	configuration l2-services vpws		
Parameters Table	Name	Value	Description
	<i>if-name</i>	Interface	Set VPWS service interface (attachment circuit).
Examples	The following examples show how to configure an interface under the VPWS service: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpws Service1 exaware(config-l2-services-vpws-Service1)# interface x-eth 0/0/1.10</pre>		

6.1.1.2. neighbor (vpws)

Description	Configures the VPWS service neighbor and pw-id.		
Command Syntax	<code>neighbor neighbor-ip-address pw-id pw-id</code>		
Command Mode	configuration l2-services vpws		
Parameters Table	Name	Value	Description
	<code>neighbor-ip-address</code>	IP Address	Remote pseudo-wire neighbor IP address.
	<code>pw-id</code>	1-4294967295	Pseudo-wire service ID.
Examples	This example shows how to configure a pseudo-wire service name: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpws Service1 exaware(config-l2-services-vpws-Service1)# neighbor 1.1.1.1 pw-id 100</pre>		
Notes	VPWS service must be configured with neighbor & pw-id in order to pass commit validation.		

6.1.1.3. profile (vpws)

Description	Configures a pw-profile name under the VPWS service.		
Command Syntax	<code>profile pw-profile-name</code>		
Command Mode	configuration l2-services vpws		
Parameters Table	Name	Value	Description
	<code>pw-profile-name</code>	String	Pseudo-wire profile name.
Examples	These examples show how to configure a pseudo-wire profile name under the VPWS service: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpws Service1 exaware(config-l2-services-vpws-Service1)# profile Profile1</pre>		
Notes	VPWS service must be configured with profile to pass commit validation.		

6.1.1.4. secondary (vpws)

Description	Configures a secondary pseudo-wire in the VPWS service.		
Command Syntax	<code>secondary [neighbor profile revert-mode revert-delay reroute-mode reroute-delay]</code>		
Command Mode	configuration l2-services vpws secondary		
Parameters Table	Name	Value	Description
	<code>neighbor</code>	See neighbor (vpws)	
	<code>profile</code>	See pw-profile (vpws)	
	<code>revert-mode</code>	true / false	
	<code>revert-delay</code>	0-3600	Delay time in seconds
	<code>reroute-mode</code>	immediate	Immediate reroute
		delayed	Delayed reroute
		never	Never reroute
	<code>reroute-delay</code>	0-3600	Delay time in seconds

Examples	These examples show how to configure a secondary pseudo-wire under the VPWS service: <pre> exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpws VPWS1 exaware(config-l2-services-vpws-VPWS1)# interface x-eth 0/0/19.1000 exaware(config-l2-services-vpws-VPWS1)# neighbor 2.2.2.2 pw-id 1 exaware(config-l2-services-vpws-VPWS1)# profile TAG exaware(config-l2-services-vpws-VPWS1)# secondary default exaware(config-vpws-VPWS1-secondary-default)# neighbor 3.3.3.3 pw-id 123 exaware(config-vpws-VPWS1-secondary-default)# profile TAG exaware(config-vpws-VPWS1-secondary-default)# revert- mode true exaware(config-vpws-VPWS1-secondary-default)# revert- delay 11 exaware(config-vpws-VPWS1-secondary-default)# reroute- mode delayed exaware(config-vpws-VPWS1-secondary-default)# reroute- delay 0 </pre>
Notes	VPWS secondary service must be configured with pw-profile in order to pass commit validation.

6.1.1.5. vpws

Description	Configures the VPWS service name. Use the no form of the command to remove the pseudo-wire service name configuration.		
Command Syntax	<code>vpws vpws-name</code>		
Command Mode	configuration l2-services		
Parameters Table	Name	Value	Description
	<code>vpws-name</code>	String	Pseudo-wire service name.
Examples	This examples show how to configure a pseudo-wire service name: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpws Service1 exaware(config-l2-services)# no vpws Service1</pre>		
Notes	VPWS service must be configured with neighbor, pw-id, and pw-profile in order to pass commit validation.		

6.1.2. VPWS Profile

6.1.2.1. control-word (vpws profile)

Description	To configure whether a Control Word must or must not be present in VPWS PW packets, use the control-word CLI command. Use the no form of this command, to restore the default configuration (disable).		
Command Syntax	control-word enable preferr disable no control-word		
Command Mode	configuration l2-services pw-profile		
Parameters Table	Name	Value	Description
	<i>enable</i>		A Control Word must be present in VPWS PW packets,
	<i>preferred</i>		Although the local side prefers Control Word, but it agrees without Control Word, if the remote side cannot support it.
	<i>disable</i>		A Control Word must not be present in VPWS PW packets,
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# pw-profile Profile1 exaware(config-l2-services-pw-profile-Profile1)# control-word enable</pre>		

Notes

This following table specifies the result depended on configuration on both sides:

local side remote side	Enable	prefer	disable
enable	3	3	1
prefer	3	3	2
disable	1	2	2

1: PW is not set

2: PW is set without Control Word.

3: PW is set with Control Word.

6.1.2.2. mtu (pw-profile)

Description	Configures a pseudo-wire profile mtu. Use the no form of the command to remove the pseudo-wire profile mtu configuration.		
Command Syntax	<code>mtu mtu</code>		
Command Mode	configuration l2-services pw-profile		
Parameters Table	Name	Value	Description
	<code>mtu</code>	594-9216	MTU L2 size in bytes, for both physical and sub interface.
Examples	These examples show how to configure a pseudo-wire profile mtu: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# pw-profile Profile1 exaware(config-l2-services-pw-profile-Profile1)# mtu 1400 exaware(config-l2-services-pw-profile-Profile1)# no mtu</pre>		

6.1.2.3. pw-profile

Description	Configures a pseudo-wire profile name. Use the no form of the command to remove the pseudo-wire profile name configuration.		
Command Syntax	<code>pw-profile <i>pw-profile-name</i></code>		
Command Mode	configuration l2-services		
Parameters Table	Name	Value	Description
	<code><i>pw-profile-name</i></code>	String	Pseudo-wire profile name.
Examples	These examples show how to configure a pseudo-wire profile name: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# pw-profile Profile1 exaware(config-l2-services)# no pw-profile Profile1</pre>		

6.1.2.4. type (pw-profile)

Description	Configures the pseudo-wire profile type. Use the no form of the command to remove the pseudo-wire profile type configuration.		
Command Syntax	<code>type type-name</code>		
Command Mode	configuration l2-services		
Parameters Table	Name	Value	Description
	<i>type-name</i>	raw	Ethernet - In raw mode, a frame MAY contain a VLAN tag and if it does, the tag is not a service delimiting tag, and it passes through the PE transparently.
		tagged	VLAN - In tagged mode, each Ethernet frame MUST contain a VLAN tag and the tag must be a service-delimiting tag
Examples	These examples show how to configure a pseudo-wire profile type: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# pw-profile Profile1 exaware(config-l2-services-pw-profile-Profile1)# type tagged exaware(config-l2-services-pw-profile-Profile1)# no type</pre>		
Notes	pw-profile must be configured with type in order to pass commit validation.		

6.1.3. XCONNECT

6.1.3.1. admin-state (xconnect)

Description	Configures the xconnect service admin state. Use the no form of the command to set the xconnect service admin state to the default setting (up).		
Command Syntax	<code>admin-state state</code>		
Command Mode	configuration l2-services xconnect		
Parameters Table	Name	Value	Description
	<code>state</code>	up	Enable xconnect service. (Default)
		down	Disable xconnect service.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# xconnect service1 exaware(config-l2-services-xconnect-service1)# admin-state up exaware(config-l2-services-xconnect-service1)# admin-state down</pre>		

6.1.3.2. interface1 (xconnect)

Description	Configures the interface1 (attachment circuit) under the xconnect service. Use the no form of the command to remove the interface1 configuration under the xconnect service.		
Command Syntax	<code>interface1 if-name</code>		
Command Mode	configuration l2-services xconnect		
Parameters Table	Name	Value	Description
	<i>if-name</i>	Interface	Xconnect service first interface (attachment circuit).
Examples	These examples show how to use this command: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# xconnect service1 exaware(config-l2-services-xconnect-service1)# interface1 xeth0/0/1 exaware(config-l2-services-xconnect-service1)# interface1 x-eth0/0/1.10 exaware(config-l2-services-xconnect-service1)# interface1 agg-eth 1 exaware(config-l2-services-xconnect-service1)# interface1 agg-eth 1.10 exaware(config-l2-services-xconnect-service1)# no interface1</pre>		

6.1.3.3. interface2 (xconnect)

Description	Configures the interface2 (attachment circuit) under the xconnect service. Use the no form of the command to remove the interface2 configuration under the xconnect service.		
Command Syntax	<code>interface2 if-name</code>		
Command Mode	configuration l2-services xconnect		
Parameters Table	Name	Value	Description
	<i>if-name</i>	Interface	Xconnect service second interface (attachment circuit).
Examples	These examples show how to use this command: <pre> exaware# config exaware(config)# l2-services exaware(config-l2-services)# xconnect service1 exaware(config-l2-services-xconnect-service1)# interface2 xeth0/0/1 exaware(config-l2-services-xconnect-service1)# interface2 x-eth0/0/1.10 exaware(config-l2-services-xconnect-service1)# interface2 agg-eth 1 exaware(config-l2-services-xconnect-service1)# interface2 agg-eth 1.10 exaware(config-l2-services-xconnect-service1)# no interface2 </pre>		

6.1.3.4. xconnect

Description	Configures the xconnect service. Use the no form of the command to remove the xconnect service configuration.		
Command Syntax	<code>xconnect xc-name</code>		
Command Mode	configuration l2-services		
Parameters Table	Name	Value	Description
	<code>xc-name</code>	String	Xconnect service name
Examples	These examples show how to configure xconnect service: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# xconnect XcService1 exaware(config-l2-services-pw-profile-Profile1)# no xconnect</pre>		
Notes	Xconnect service must be configured with interface1 and interface2 in order to pass commit validation.		

7. Management Commands

7.1. AAA Commands

7.1.1. AAA Configuration Commands

7.1.1.1. aaa accounting event/destination

Description	Configures an accounting group including types of events for notification, and the list of destinations (external servers or locally).		
Command Syntax	<pre>aaa accounting <i>group-name</i> event [<i>login</i> <i>config</i> <i>show</i> <i>action</i>]</pre> <pre>aaa accounting <i>group-name</i> destination [<i>tacacs</i> <i>radius</i> <i>local</i>]</pre>		
Command Mode	configuration aaa accounting		
Parameters Table	Name	Value	Description
	<i>group-name</i>	string	Name of the accounting group.
	event	<i>login</i> <i>config</i> <i>show</i> <i>action</i>	Type of event for notification – multiple values can be set. login - all login operations attempted on the system. config - all configuration commands issued. show - all show operational commands issued. action - all other commands and actions.

	destination	<i>tacacs <server-ip></i> <i>radius <server-ip></i> <i>local</i>	Type of destination for notification. tacacs – remote server IP address which should be configured under /aaa/tacacs. radius – remote server IP address which should be configured under /aaa/radius. local – local notification located in /var/syslog/accounting/accounting.log.
Examples	This example shows how to configure the acct_grp1 accounting group to send login events locally, and to the TACACS servers at 192.168.0.200 and 3.3.3.3: <pre>exaware# config exaware(config)# aaa accounting acct_grp1 event login exaware(config)# aaa accounting acct_grp1 destination tacacs 192.168.0.200 tacacs 3.3.3.3</pre>		

7.1.1.2. aaa authentication-order

Description	Configures the order in which different types of authentication or different servers with the same type of authentication are to be performed. Local authentication is the default until this command is executed.		
Command Syntax	<pre>aaa authentication-order <i>index</i> authentication-type {local tacacs radius} [server-ip {<i>ip-address</i>} all] no aaa authentication-order</pre>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>index</i>	Integer	Orders the types of authentication.
		local tacacs radius	Type of authentication to perform.
Examples	This example shows how to configure the AAA authentication order: <pre>exaware# config exaware(config)# aaa authentication-order 1 radius all</pre>		

Notes	Authentication failure can be local or remote, and occurs when incorrect credentials are used. Authentication fallback occurs when the external server cannot be reached. When no authentication order is configured, only local authentication is performed. When this command is executed for remote authentication, you must explicitly configure local authentication as one of the methods. It should be configured as the last authentication; after which it will no longer be the default behavior. You must execute this command for the external server to work. Only the authentication method is required. Once a connection with the user is established, both authentication and authorization information is received from the server and operation continues according to this information. Authorization is performed once, at the start of the session. Configuring 'all' before configuring a specific server IP will cause attempts to connect more than once to the same server.
--------------	--

7.1.1.3. aaa authorization-order

Description	Sets the authorization fall back order.		
Command Syntax	aaa authorization-order <i>index</i> authorization-type {local tacacs radius} [server-ip { <i>ip-address</i> } all]		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>index</i>	Integer	Orders the types of authorization.
		local tacacs radius	Type of authorization to perform.
Examples	This example shows how to configure the AAA authorization order: <pre>exaware# config exaware(config)# authorization-order 1 radius exaware(config)# authorization-order 5 tacacs all exaware(config)# authorization-order 10 local</pre>		

Notes

- If radius is configured in authorization order, then no server IP address will be added, and the same radius server from authentication will be used.
- If the authorization order list is empty, then the method that was used for the authentication will be used.
- Configuring authorization order without authentication order is not valid:

Example1:

Authentication – nothing configured

Authorization – nothing configured

Result: use local for both authentication/authorization

Example2:

Authentication – nothing configured

Authorization – some order is configured

Result: not valid

Example3:

Authentication – some order is configured

Authorization – nothing configured

Result: use configured order for authentication and the same method will be used to authorization

- If authentication succeeds with TACACS+ or local server, and the first available method in the authorization order list is a Radius server, then it will be skipped and the next priority will be processed. In that case a notice log will be written.
- If the first priority is configured with an unsupported combination, a warning log will be written.
- Authentication / Authorization combination support table:

	Radius	TACACS	Local
	Authorization	Authorization	Authorization

	Radius Authentication	Valid	Valid	Valid
	TACACS Authentication	Not Supported	Valid	Valid
	Local Authorization	Not Supported	Valid	Valid

7.1.1.3.1. **aaa radius**

Description	Configures a RADIUS server, including IP address and related parameters. Use the no form of the command to remove the server from the configuration.		
Command Syntax	<pre>aaa radius <i>ip-address</i> [<i>vrf vrf-name</i>] authentication-key <i>password</i> timeout <i>timeout</i> source [<i>ip source-ip</i> <i>type instance</i>] account-port <i>acct- port</i> auth-port <i>auth-port</i> reconnect-interval <i>time</i> no aaa radius <i>ip-address</i></pre>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 or IPv6 address	IP address of the RADIUS authentication server.
	<i>vrf-name</i>	Existing VRF name	VRF routing table to be used when sending a request to the RADIUS server.
	<i>password</i>	Clear text DES3-len255 encrypted password	Authentication key ("shared secret") shared between the local agent and the RADIUS server. The RADIUS packets are encrypted using this key. This key must match the key used by the RADIUS daemon.
	<i>timeout</i>	Integer between 1 and 1000	The length of time in seconds that the local router waits to receive a response from the RADIUS server. (Default = 5)

	<i>source-ip</i> OR <i>type instance</i>	IPv4 or IPv6 address OR agg-eth x-eth mgmt loopback 0/0/port	Valid IP address on one of the router interfaces for all outgoing RADIUS packets. This address is used as long as the interface or sub-interface is in the UP state. You can specify a valid source interface instead of an IP address. The IP address of the interface will be used as the source IP. Useful in cases in which the router has many interfaces or sub-interfaces, and you want to ensure that all RADIUS packets from a particular router have the same IP address.
--	--	--	---

	<i>acct-port</i>	Integer between 1024 and 65535	Accounting port where Radius server listens for accounting requests. (Default = 1813)
	<i>auth-port</i>	Integer between 1024 and 65535	Authentication/authorization port where the Radius server listens for authentication and authorization requests. (Default = 1812)
	<i>time</i>	Integer between 1 and 1000	Reconnect-interval value in seconds in which accounting requests to the Radius server will not be sent since the previous unsuccessful attempt. (Default = 30)
Examples	This example shows how to configure a RADIUS server. <pre>exaware# config exaware(config)# aaa radius 10.10.10.10 exaware(config-aaa-radius-10.10.10.10)# authentication-key cl3ark3y exaware(config-aaa-radius-10.10.10.10)# timeout 10 exaware(config-aaa-radius-10.10.10.10)# source ip 10.10.10.20 exaware(config-aaa-radius-10.10.10.10)# account-port 10123 exaware(config-aaa-radius-10.10.10.10)# auth-port 10122 exaware(config-aaa-radius-10.10.10.10)# reconnect-interval 60</pre>		
Notes	If configuring more than one RADIUS server, use the <i>aaa authentication-order</i> (on page 151) command to define the order of connection attempts.		

7.1.1.4. aaa radius-retry

Description	Configures the number of retries when a RADIUS response is not received within the given timeout interval.		
Command Syntax	aaa radius-retry <i>retries</i>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>retries</i>	Integer <1-15>	Number of retries before the RADIUS server is declared unreachable.
Examples	This example shows how to configure the AAA authentication type: exaware# config exaware(config)# aaa radius-retry 3		

7.1.1.5. aaa role

Description	Adds a role to the list of roles that can be defined for a user. You can also define the privileges for the role, as well as exceptions to those privileges. Use the no form of the command to delete a role.		
Command Syntax	<pre>[no] aaa role <i>role-name</i> [privilege <i>privileges</i>] [exception <i>index</i> {command configuration} <i>command-name</i> {action {accept reject}}]</pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>role-name</i>		Pre-defined roles are: super_admin priv_admin
	<i>privilege</i>	Select from the predefined privileges (see list below)	The command level the role is permitted to access, for example, routing or view.
	<i>index</i>	Integer	Exception index will determine the order that the rules are applied. The first match will be executed.
		command / configuration	Specifies whether the exception is a command at the operational level or under the configuration menu.

	<i>command-name</i>	Command name - Enclose in "" if there are spaces in the command name (ie. "system restart").	The command or configuration command from the CLI.
		accept / reject	The action to perform if there is a match on the command.
Examples	This example shows how to add a new role and assign a privilege level: <pre>exaware# config exaware(config)# aaa role guest view</pre> This example shows how to configure exceptions to the privileges for the guest role: <pre>exaware# config exaware(config)# aaa role guest exception 10 command ping action accept exaware(config)# aaa role guest exception 20 command "show configuration aaa" action reject</pre> This example shows how to delete a role: <pre>exaware# config exaware(config)# no aaa role guest</pre>		
Notes	Use the show config aaa role (on page 185) command to see all currently configured roles.		

Defining AAA Authorization

This section will elaborate on the specific commands that a user may perform with a given privilege, not taking exceptions into account. For commands that are available for a privilege, all of their sub-commands are also available, unless stated otherwise.

Default commands for all users

Each user is able to perform the following default commands regardless of his defined privileges (this includes a user with no defined role or with a role empty of privileges):

"clear history"

"show history"

"show session"

"show id"

"help"

"session"

"exit"

Default commands for privileges with configuration permissions

Each user with a role that has privileges with configuration permissions is able to perform the following default commands.

Configuration mode:

"alias"

"abort"

"annotate <sub_menu>"

"cancel"

"commit"

"copy <sub_menu>"

"edit <sub_menu>"

"end"

"exit"

"load merge"

"load replace"

"no"

"pwd"

"resolved"

"rollback"

"run"

"save"

"tag <sub_menu>"

"top"

"save <file_name> alias"

"show candidate"

"show history"

"show configuration running"

Note: The commands "tag", "copy", "edit", and "annotate" all have sub-commands which are privilege specific, and each privilege with configuration abilities will see only its relevant sub-commands.

E.g. aaa privilege will be able to see "tag add aaa", "tag clear aaa", "tag del aaa", "edit aaa", "copy aaa" and "annotate aaa".

Operational mode:

"configure"

"file show"

"file list"

"show configuration alias"

"show configuration details"

Note: The "show configuration details" command is not available for privileges without the "show configuration" ability. The sub-commands under "show configuration details" directly correspond to the sub-commands under "show configuration".

Command definitions for various privileges

The following table lists the commands available for each of the privilege types

Privilege Type	Description	Available commands
all	All commands	
view	Perform all show operations	"show"
system-view	Show system information (who)	"show system" "show configuration system" – excluding "root-password" "show configuration ntp" "show configuration dns" "show configuration tcp" "show configuration telnet-server" "show configuration session" "show who" "show version" "show ntp"

system	Configure and show System information and OS access	system-view commands: "system" "configure system" – excluding root password "configure session" "configure telnet-server" "configure ntp" "configure dns" "configure tcp" "save <file_name> system" (or xml system) "save <file_name> dns" (or xml dns) "save <file_name> ntp" (or xml ntp) "save <file_name> tcp"(or xml tcp) "save <file_name> session" (or xml session) "save <file_name> telnet-server"(or xml telnet-server) "clear system optical-interconnect"
--------	---	--

os-shell	Access to the OS shell and operations	"os-shell" "os-command" (under pipe)
util	File operations	"file" "send"
aaa-view	Show aaa information	"show configuration aaa" "show who" "show configuration system root-password"
aaa	Configure and show login information including users, roles etc.	aaa-view commands: "configure aaa" "clear aaa" "logout" "configure system root-password" "configure save <file_name> aaa" (or xml aaa)
network	Network commands	"ping" "tracert" "ssh"
debug	Debug configuration	"configure debug" "show configuration debug" "configure save <file_name> debug" (or xml debug)

log	Log commands and configuration	"log" "configure log" "clear log" "show configuration log" "show log" "configure save <file_name> log" (or xml log)
interface-view	View interface and vrf information	"show interface" "show fib" "show policy" "show flow-monitor" "show configuration flow-profile" "show configuration interface" "show configuration arp" "show configuration policy" "show configuration vrf" "show configuration ndp" "show lacp"
interface	Configure and view interface and vrf information	interface-view commands: "clear arp" "clear forwarding"

		"clear interface" "clear ndp" "configure interface" "configure flow-profile" "configure vrf" "configure ndp" "configure arp" "configure save <file_name> interface" (or xml interface) "configure save <file_name> vrf" (or xml vrf) "configure save <file_name> ndp" (or xml ndp) "configure save <file_name> arp" (or xml arp)
routing-view	View routing information including route policy	"show dhcp" "show policy" "show bfd" "show route" "show configuration policy" Excluding: "show configuration policy acl"

		"show configuration policy qos" "show configuration routing" "show configuration mpls" "show isis" "show bgp" "show ldp" "show multicast" "show pim" "show rsvp" "show ospfv3" "show ospf"
routing	Configure and view routing information	"config dhcp" "clear dhcp counters" "configure routing" "configure mpls" "configure policy"- Excluding: "configure policy acl" "configure policy qos" "configure save <file_name> routing" (or xml routing)

		"configure save <file_name> mpls" (or xml mpls) "configure save <file_name> policy" (or xml policy) Excluding: "configure save <file_name> policy acl" (and xml acl) "configure save <file_name> policy qos" (and xml qos) "clear bgp" "clear ldp" "clear multicast" "clear pim" "clear rsvp" "clear isis" "clear ospf" "clear ospfv3"
qos-vie PRIV_UTILw	View QoS information including QoS policy	"show qos" "show policy" "show configuration policy" Excluding: "show configuration policy acl"

		"show configuration policy route"
qos	Configure and view QoS information	qos-view commands interface commands "clear qos" "configure save <file_name> policy" (or xml policy) Excluding: "configure save <file_name> policy acl" (and xml policy acl) "configure save <file_name> policy route" (and xml policy route) "configure policy" - Excluding: "configure policy acl" "configure policy route"
acl-view	View ACL information including ACL policy	"show acl" "show ctrl-plane" "show configuration ctrl-plane" "show policy"

		"show configuration policy" – Excluding: "show configuration policy route" "show configuration policy qos"
acl	Configure and view ACL information	acl-view commands interface commands "clear acl" "clear ctrl-plane" "configure save <file_name> ctrl-plane" (or xml ctrl-plane) "configure save <file_name> policy" excluding: "configure save <file_name> policy route" (and xml policy route) "configure save <file_name> policy qos" (and xml policy qos) "configure ctrl-plane" "configure policy" - Excluding:

		"configure policy qos" "configure policy route"
snmp	Configure and view SNMP information	"configure snmp" "configure save <file_name> snmp" (or xml snmp) "show configuration snmp"
internal	Internal functionality for HW access	"internal" "unhide"
config-view	Showing all configurations	"show configuration" "configure save" "compare" "configure show configuration" "configure run show configuration"

7.1.1.6. aaa tacacs

Description	Configures a TACACS+ server, including IP address and related parameters. Use the no form of the command to remove the server from the configuration.		
Command Syntax	<pre>aaa tacacs <i>server-ip</i> authentication-key <i>password</i> timeout <i>timeout</i> source [ip <i>source-ip</i> {<i>type interface</i>}] vrf <i>vrf</i> reconnect- interval <i>interval</i></pre>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>server-ip</i>	IPv4 or IPv6 address	IP address of the TACACS+ authentication server.
	<i>password</i>	HMAC-MD5 encrypted string. Enclose in "" if includes spaces.	Authentication key shared between the local agent and the TACACS+ server. The TACACS+ packets are encrypted using this key. This key must match the key used by the TACACS+ server.
	<i>timeout</i>	1-1000 seconds Default = 5	The length of time in seconds that the local router waits to receive a response from a TACACS+ server.

	<i>ip source-ip</i> OR <i>type interface</i>	IPv4 or IPv6 address OR x-eth mgmt loopback agg-eth 0/0/port	Valid IP address on one of the router interfaces for all outgoing TACACS+ packets. This address is used as long as the interface or sub-interface is in the UP state. You can specify a valid source interface instead of an IP address. The IP address of the interface will be used as the source IP. Useful in cases in which the router has many interfaces or sub-interfaces, and you want to ensure that all TACACS+ packets from a particular router have the same IP address.
--	---	---	--

	<i>vrf</i>	• management (default) • default existing user-defined VRF	VRF that leads to TACACS+ server.
	<i>interval</i>	1-1000 seconds	Time to wait before retrying to connect to the server after a connection failure. (Default = 30)
Examples	This example shows how to configure a TACACS+ server: <pre>exaware# config exaware(config)# aaa tacacs 10.10.10.10 exaware(config-aaa-tacacs-10.10.10.10)# authentication-key "password string" exaware(config-aaa-tacacs-10.10.10.10)# timeout 10 exaware(config-aaa-tacacs-10.10.10.10)# source ip 10.10.10.20 exaware(config-aaa-tacacs-10.10.10.10)# vrf default</pre>		
Notes	Use the aaa authentication-order (on page 151) command to define the order of connection attempts.		

7.1.1.7. aaa user

Description	Creates, edits, and deletes users. You can define the following for each user: - password (may be MD5 encrypted) - role		
Command Syntax	<pre>[no] aaa user <i>user-name</i> password <i>pw-string</i> role <i>role-name</i></pre>		
Command Mode	configuration configuration aaa user <i><user-name></i>		
Parameters Table	Name	Value	Description
	<i>user-name</i>	String	Login name of the user.
	<i>pw-string</i>	Text, maximum 255 characters Use '\$1\$' to indicate an encrypted password	User password which can be saved in MD5 encrypted form.
	<i>role-name</i>	super_admin priv_admin user-defined role	You can specify either one of the pre-defined roles or a user-defined role (added using aaa role (on page 162) command).

Examples	This example shows how to create a new user: <pre>exaware# config exaware(config)# aaa user newuser1 password newpw123 exaware(config-aaa-user-newuser1)#</pre> This example shows how to change the role for an existing user. <pre>exaware# config exaware(config)# aaa user johndoe exaware(config-aaa-user-johndoe)# role network</pre> This example shows how to delete the specified user. <pre>exaware# config exaware(config)# no aaa user johndoe</pre>
Notes	Use the <code>show config aaa user</code> (on page 185) command to see all currently configured users.

7.1.2. AAA Operational Commands

7.1.2.1. clear aaa

Description	Deletes the home directory and user data for external users.		
Command Syntax	<code>clear aaa external-users all user <i>username</i></code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
		all	Clears home folders of all external users.
	<i>username</i>	existing user name	Clears home folders of specified user.
Examples	The following example shows how to clear the user 'johndoe': <code>exaware# clear aaa external-users user johndoe</code>		

7.1.3. AAA Show Commands

7.1.3.1. show configuration aaa

Description	Displays the specified AAA information or configuration: • authentication-order: displays the currently configured authentication order. • role: lists all currently configured roles (user-defined and pre-defined), with their assigned privileges and exceptions. • tacacs: displays the TACACS+ server configuration. • user: lists all currently configured users (user-defined and pre-defined), including passwords and assigned roles. • no option specified: all AAA information is displayed.
Command Syntax	<pre>show configuration aaa [authentication-order role tacacs user]</pre>
Command Mode	operational
Examples	This example shows how to display all currently configured roles: <pre>exaware# show configuration aaa role aaa role priv_admin privilege all exception 1 command os-shell reject ! ! aaa role super_admin privilege all !</pre>

7.1.3.2. show who

Description	Displays a listing of all currently logged on users.					
Command Syntax	show who					
Command Mode	operational					
Examples	This example shows how to display all currently logged on users:					
exaware# show who						
Session	User	Context	From	Proto	Date	Mode
13	root	cli	10.1.1.3	ssh	15:12:16	config-private
11	root	cli	127.0.0.1	console	2010-11-04	operational

7.2. App-VM Commands

7.2.1. App-VM Operational Commands

7.2.1.1. **system app-vm add**

Description	Adds the VM and the resources allocated to it.		
Command Syntax	system app-vm add <i>name</i> image-file <i>filename</i> [cores <i>num-of-cores</i>] [memory <i>MBs</i>] [description <i>description</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>name</i>		The VM name.
	<i>filename</i>		The file attached to the VM.
	<i>num-of-cores</i>	An integer in range 1..4	The number of cores the VM may use according to the platform limitation. Default: 1.
	<i>MBs</i>	An integer in range 0..4000	The amount of memory (in MB) the VM may use.. Default: 2000.
	<i>description</i>		The VM description

Examples	The following examples show how to add a VM: <pre>exaware# system app-vm add TbtVM image-file appvm1.img cores 2 memory 1000 description "Tibbit VM"</pre>
Notes	1. The command fails if the name is used by another VM. 2. The command fails if there are already maximumnumber of VMs added. 3. The command fails if the VM is started already. 4. The command fails if the image file is not of QCOW type. 5. The command fails if the image file does not exist in /opt/compass/system-images/VMs directory. 6. The command fails if the number of cores exceeds the platform limitation. 7. The command fails if the memory exceeds the platform limitation.

7.2.1.2. system app-vm delete

Description	Deletes the VM.		
Command Syntax	<code>system app-vm delete <i>name</i></code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>name</i>		The VM name.
Examples	The following examples show how to delete a VM: <pre>exaware# system app-vm delete TbtVM</pre>		

7.2.1.3. system app-vm file backup

Description	Copies an application VM image file into to a remote directory.		
Command Syntax	<pre>system app-vm file backup local-image-filename [vrf vrf-name] source-address ip-address [source-interface interface-type interface-path] protocol://[user[:password]@]remote-host[:port]/target-image-path</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>local-image-filename</i>		Local file name.
	<i>ip-address</i>	IP address	IPv4 or IPv6 address
	<i>interface-type</i>	agg-eth x-eth loopback mgmt	Type of source interface, e.g x-eth
	<i>interface-path</i>	X/Y/Z	Interface path, e.g. 0/0/2, 0/0/3.100
	<i>protocol</i>	tftp ftp http scp	Backup protocol.
	<i>user</i>		URL user name.
	<i>password</i>		User password.
	<i>remote-host</i>		IPv4 or IPv6 address of the remote host.
	<i>port</i>		Host port number. (Optional if using non-protocol default value)

	<i>image-path</i>		The absolute path where the image will be located on the remote host.
	<i>target-image-filename</i>		The absolute path where the image will be located on the remote host and the file name.
Examples	The following examples show how to copy the image file to a remote directory: <pre>Exaware# system app-vm file backup tbtVm.imgvrf management scp://cronaldo:cr7@10.1.1.2/auto/automation/tftpboot/releases/exaware/VMs/appvm.img</pre>		
Notes	1. If the argument of <code>vrf</code>, <code>source-address</code>, or <code>source-interface</code> is not configured, the vrf "default" is applied and the source address will be the address of the physical interface the traffic is going out through. 2. If the <code>vrf</code> argument is configured, the source address will be the address of the physical interface the traffic is going out through. 3. If the <code>source-address</code> argument is configured, the VRF is specified by the interface which the ip address is configured on. 4. If <code>source-interface</code> is configured, the VRF and source address are taken from the interface configuration. 5. For TFTP protocol the image-path is started from the root TFTP server folder. For example, if folder <code>/a/tftp</code> is the root folder of the TFTP server and an image is <code>/1/tftp/b/c/image.tar.gz</code> then the image-path is <code>"/b/c/image.tar.gz"</code>.		

7.2.1.4. system app-vm file delete

Description	Deletes an application VM image file.		
Command Syntax	<code>system app-vm file delete local-image-file</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<code>local-image-file</code>		Name of the file to be deleted.
Examples	The following examples show how to delete a file: Exaware# system app-vm file delete tbtVm.img		
Notes	The command fails if the file is attached to a VM.		

7.2.1.5. system app-vm file download

Description	Downloads an application VM image file into the local VM images directory.		
Command Syntax	<pre>system app-vm file download [vrf vrf-name] source-address ip-address [source-interface interface-type interface- path] protocol://[user[:password]@]remote- host[:port]/image-path local-image-filename</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>		Name of the vrf to use.
	<i>ip-address</i>	IP address	IPv4 or IPv6 address
	<i>interface-type</i>	agg-eth x-eth loopback mgmt	Type of source interface, e.g x-eth
	<i>interface-path</i>	X/Y/Z	Interface path, e.g. 0/0/2, 0/0/3.100
	<i>protocol</i>	tftp ftp http scp	Download protocol.
	<i>user</i>		URL user name.
	<i>password</i>		User password.
	<i>remote-host</i>		IPv4 or IPv6 address of the remote host.
	<i>port</i>		Host port number. (Optional if using non-protocol default value)

	<i>image-path</i>		The absolute path where the image will be located on the remote host.
	<i>target-image-filename</i>	Relative local path	Local file name.
Examples	The following examples show how to download a new image file: <pre>Exaware# system app-vm file download vrf management scp://cronaldo:cr7@10.1.1.2/auto/automation/tftpboot/releases/exaware/VMs/appvm.img tbtVm.img</pre>		
Notes	1. If the argument of vrf, source-address, or source-interface is not configured, the vrf "default" is applied and the source address will be the address of the physical interface the traffic is going out through. 2. If the vrf argument is configured, the source address will be the address of the physical interface the traffic is going out through. 3. If the source-address argument is configured, the VRF is specified by the interface which the ip address is configured on. 4. If source-interface is configured, the VRF and source address are taken from the interface configuration. 5. For TFTP protocol the image-path is started from the root TFTP server folder. For example, if folder /a/tftp is the root folder of the TFTP server and the image-path in the command is defined as /c/image.tar.gz then the image path on the TFTP server is "/a/tftp/c/image.tar.gz".		

7.2.1.6. system app-vm file rename

Description	Renames an application VM image file.		
Command Syntax	<code>system app-vm file rename <i>local-image-file</i> <i>new-local-image-file</i></code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>local-image-file</i>		Name of the file to be renamed.
	<i>new-local-image-file</i>		New file name.
Examples	The following examples show how to rename a file: Exaware# system app-vm file rename appvm.img appTbt.img		
Notes	The command fails if the file is attached to a VM.		

7.2.1.7. system app-vm start

Description	Starts the VM.		
Command Syntax	<code>system app-vm start <i>name</i></code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>name</i>		The VM name.
Examples	The following examples show how to start a VM: <pre>exaware# system app-vm start TbtVM</pre>		
Notes	1. The command fails if there is no VM by that name. 2. The command fails if the VM is started already.		

7.2.1.8. system app-vm stop

Description	Stops the VM.		
Command Syntax	<code>system app-vm stop <i>name</i></code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>name</i>		The VM name.
Examples	The following examples show how to stop a VM: <pre>exaware# system app-vm stop TbtVM</pre>		
Notes	1. The command fails if there is no VM by that name. 2. A warning message should appear before executing the command.		

7.2.2. App-VM Show Commands

7.2.2.1. show system app-vm

Description	Displays the application VMs.																																				
Command Syntax	show system app-vm																																				
Command Mode	operational																																				
Examples	The following example shows how to display the app VMs: <pre>exaware# show system app-vm</pre> <table><tr><th>VM Name</th><th>STATUS</th><th>File</th><th>Cores</th><th>Memory</th><th>Desc</th></tr><tr><td colspan="6">-----</td></tr><tr><td>1 TBtVM</td><td>Startted</td><td>applvm1.img</td><td>2</td><td>3000MB</td><td>Tibbit Appl</td></tr><tr><td>50:35</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>2 EHVM</td><td>NotStarted</td><td>applvm2.img</td><td>1</td><td>1000MB</td><td>-</td></tr><tr><td>12:10</td><td></td><td></td><td></td><td></td><td></td></tr></table>	VM Name	STATUS	File	Cores	Memory	Desc	-----						1 TBtVM	Startted	applvm1.img	2	3000MB	Tibbit Appl	50:35						2 EHVM	NotStarted	applvm2.img	1	1000MB	-	12:10					
VM Name	STATUS	File	Cores	Memory	Desc																																

1 TBtVM	Startted	applvm1.img	2	3000MB	Tibbit Appl																																
50:35																																					
2 EHVM	NotStarted	applvm2.img	1	1000MB	-																																
12:10																																					

7.2.2.2. show system app-vm file list

Description	Displays the application VM image files.
Command Syntax	<code>show system app-vm list</code>
Command Mode	operational
Examples	The following example shows how to display the contents of the VM image file directory: <pre>exaware# show system app-vm list</pre>

7.3. File Commands

Note: File commands are supported only in the home directory.

7.3.1. File Operational Commands

7.3.1.1. **file copy**

Description	Copy the contents of the specified file from one location in the file system to a different location. Either the source or destination can be remote files. An FTP or TFTP URL is a valid source or destination. Source and destination files can be designated as follows: - local file designation: name and full path of a local file within the user home directory, i.e. /home/<username> - external file designation: <i>external-source</i> where <i>external source</i> is defined as follows: <code>{protocol}://[user[:password]@]{remote_host}/{file_path}</code> (see parameter descriptions in the Parameters Table below)		
Command Syntax	<pre>file copy [vrf <vrf-name> [source-address <IP-address>] source-interface <interface-type> <interface-path>] source destination</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	vrf-name	string	Name of the vrf to use.
	<i>IP-address</i>	IP address	IPv4 or IPv6 source address.
	<i>interface-type</i>	enum	Type of source interface, e.g x-eth.
	<i>interface-path</i>	X/Y/Z	Source interface path, e.g. 0/1/1.
<i>source</i>		Source file to be copied.	

	<i>destination</i>		Destination to which the file should be copied. User should have write permissions on the file and directory.
--	--------------------	--	--

	<i>external source</i>	string	An FTP or TFTP URL for the file containing the configuration.
--	------------------------	--------	---

	<i>protocol</i>	HTTP(S), FTP(S), SFTP, TFTP, SCP	Download protocol.
--	-----------------	-------------------------------------	--------------------

	<i>user</i>	string	URL user name.
--	-------------	--------	----------------

	<i>password</i>	string	User password (Optional, if not given will be queried when executing).
--	-----------------	--------	---

	<i>remote-host</i>	IP address	IP address of the remote host.
	<i>file-path</i>	path	The absolute path where the file is located on the remote host.
Examples	This example shows how to copy a file to a new name: The contents of the user home directory are displayed before and after the copy operation to verify that the file has indeed been successfully copied. <pre>exaware# file list total 4 -rw-r--r-- 1 admin admin 0 Oct 24 06:43 image1.tar.gz -rw-r--r-- 1 admin admin 3594 Nov 4 06:06 testfile exaware# file copy testfile testfile2 Copying file done. exaware# file list total 8 -rw-r--r-- 1 admin admin 0 Oct 24 06:43 image1.tar.gz -rw-r--r-- 1 admin admin 3594 Nov 4 06:06 testfile -rw-r--r-- 1 admin admin 3594 Nov 4 08:57 testfile2 exaware#</pre>		
Notes	VRF (and optional source-address) are mutually exclusive with source interface. If neither is specified, the vrf and source address are taken from the default-management-source configuration. If only VRF is specified, the source address will be the address of the physical interface the traffic is going out through. If source-interface is specified, the VRF and source address are taken from the interface configuration.		

7.3.1.2. file delete

Description	Deletes the specified file from the file system.		
Command Syntax	file delete <i>filename</i>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>filename</i>		Name and full path of the file to be deleted. User should have write permissions on the file and directory.
Examples	This example shows how to delete a file: The contents of the directory are displayed before and after the delete operation to verify that the file has indeed been successfully deleted. <pre>exaware# file list /test2/ total 8 drwx----- 2 test2 test2 4096 Oct 14 13:53 . -rw-r--r-- 1 test2 test2 191 Oct 14 13:34 .bash_profile -rw-r--r-- 1 test2 test2 612 Oct 26 13:08 10_27_config exaware# file delete /test2/10_27_config exaware# file list /test2/ total 6 drwx----- 2 test2 test2 4096 Oct 26 13:08 . -rw-r--r-- 1 test2 test2 191 Oct 14 13:34 .bash_profile</pre>		

7.3.1.3. file edit

Description	Opens the specified file in the vi file editor.		
Command Syntax	<code>file edit <i>filename</i></code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>filename</i>		Name of the file to edit, including full path. If the file does not exist, this command will create a new file with the specified filename and path. User should have write permissions on the file and directory.
Examples	The following example shows a file opened in the Nano editor for editing: Upon exiting the editor, you are prompted to save the file and to assign a new filename, if desired. <pre>exaware# file edit /test/run_config interface management 0/0/0 admin-state up config-state active ipv4 address 10.0.0.2/24 ! interface management 0/0/0 admin-state down !</pre>		

7.3.1.4. file list

Description	Displays the contents of the specified directory.		
Command Syntax	<code>file list [path]</code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>path</i>		Full path of the directory.
Examples	This example shows how to display the contents of the directory <i>/home/test2/</i>: <pre>exaware# file list /test2/ total 18 drwx----- 2 test2 test2 4096 Oct 14 13:53 . drwxr-xr-x 6 root root 4096 Oct 14 16:41 .. -rw-r--r-- 1 test2 test2 191 Oct 14 13:34 .bash_profile</pre>		

7.3.1.5. file monitor start

Description	Starts real time file monitoring.		
Command Syntax	file monitor start <i>filename</i>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>filename</i>		Name and full path of the file to be monitored. Default path is the user home directory. User should have read permissions on the file and directory.
Examples	This example shows how to monitor a file: <pre>exaware# file monitor start /admin/messages</pre>		

7.3.1.6. file monitor stop

Description	Stops real time file monitoring.		
Command Syntax	<code>file monitor stop <i>filename</i></code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>filename</i>		Name and full path of the file to stop monitoring. Default path is the user home directory. User should have read permissions on the file and directory.
Examples	This example shows how to stop monitoring a file: <code>exaware# file monitor stop /admin/messages</code>		

7.3.1.7. file rename

Description	Renames the specified file, and may move the file if the new path is different from the current path.		
Command Syntax	<code>file rename <i>current-filename</i> <i>new-filename</i></code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>current-filename</i>		Current filename and full path of the file.
	<i>new-filename</i>		New filename and path of the file. If the path is different from that specified in the <i>current-filename</i>, the file will be renamed and moved to the new location. User should have permissions on source and destination directory and file.

Examples

This example shows how to rename a file:

```
exaware# file list
total 4
-rw-r--r-- 1 admin admin    0 Oct 24 06:43 image1.tar.gz
-rw-r--r-- 1 admin admin 3533 Nov  4 09:11 testfile
exaware# file rename testfile testfile2
exaware# file list
total 4
-rw-r--r-- 1 admin admin    0 Oct 24 06:43 image1.tar.gz
-rw-r--r-- 1 admin admin 3533 Nov  4 09:11 testfile2
```

This example shows how to use this command to move a file:

```
exaware# file list
total 1
-rw-r--r-- 1 admin admin    0 Oct 24 06:43 image1.tar.gz
drwxr-xr-x 2 admin admin 1024 Nov  4 09:26 test
exaware# file list test
total 4
-rw-r--r-- 1 admin admin 3533 Nov  4 09:11 testfile
exaware# file rename test/testfile testfile2
exaware# file list
total 5
-rw-r--r-- 1 admin admin    0 Oct 24 06:43 image1.tar.gz
drwxr-xr-x 2 admin admin 1024 Nov  4 09:30 test
-rw-r--r-- 1 admin admin 3533 Nov  4 09:11 testfile2
```

7.3.1.8. file show

Description	Displays the contents of the specified file.		
Command Syntax	file show <i>filename</i>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>filename</i>		Name and full path of the file to be displayed. User should have read permissions on the file. Default path is the user home directory.
Examples	This example shows how to display the contents of a file: <pre>exaware# file show testfile interface management 0/0/0 admin-state up config-state active ipv4 address 10.0.0.2/24 ! interface management 0/0/1 admin-state down ! system hostname exaware ! ! exaware#</pre>		

7.4. Logging and Debugging Commands

7.4.1. Logging and Debugging Configuration Commands

7.4.1.1. debug

Description	Directs debug messages from the specified category to the specified file. If the file does not exist, it is created. Use the no form of the command to disable debugging.		
Command Syntax	debug [file <i>filename</i>] <i>process category</i>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>filename</i>	name of debug file	Can be either an existing file or a new file.
	<i>process</i>	query to obtain list of processes	
	<i>category</i>		Available values are dependent on process selected.

Examples

This example shows how to send messages to a debug file from BGP and ISIS:

```
exaware# config
exaware(config)# debug file bgp.log bgp fsm
exaware(config-debug-file-bgp.log)# isis spf
```

This example shows how to configure rotate parameters for a debug file:

```
exaware(config)# debug file debug.log rotate files 5
```

This example shows how to disable debugging for a specific category:

```
exaware(config)# no debug file bgp.log
```

This example shows how to disable debugging globally:

```
exaware(config)# no debug
```

7.4.1.2. facility-override (log remote-server)

Description	Configures facility to be listed as the source facility on all system log messages sent to the remote server, regardless of which facility actually sent the syslog message. Use the no form of the command to disable the facility override.		
Command Syntax	[no] facility-override <i>facility</i>		
Command Mode	configuration log output remote-server		
Parameters Table	Name	Value	Description
	<i>facility</i>	0-kern	Facility that will be shown in specified remote log-server for all logs from the configured Exaware router.
		1-user	
		2-mail	
		3-daemon	
		4-auth	
		5-syslog	
		6-lpr	
		7-news	
		8-uucp	
		9-cron	
		10-security	
		11-ftp	
		12-ntp	
		13-logaudit	

	14-logalert 15-clock 16-local0 17-local1 18-local2 19-local3 20-local4 21-local5 22-local6 23-local7	
Examples	This example shows how to configure a facility override for the specified remote server. <pre>exaware# config exaware(config)# log output remote-server 10.10.10.10 exaware(config-log-output-remote-server-10.10.10.10)# facility-override 19-local3</pre> This example shows how to disable facility override for the specified remote server. <pre>exaware# config exaware(config)# log output remote-server 10.10.10.10 exaware(config-log-output-remote-server-10.10.10.10)# no facility-override</pre>	
Notes	The facility-override configuration is per a specific remote log-server. A facility must be specified to use as the override. There is no default facility value.	

7.4.1.3. filter

Description	Filters the log output by either: • facility and severity of the message • including / excluding given strings You can filter output to the following log devices: • console (log-output-console) • log file (log-file-<i><filename></i>) • log user (log-file-<i><username></i>) • log remote-server (log-remote-server-<i><ip></i> or log-remote-server-<i><hostname></i>) Use the no form of the command to disable filtering.		
Command Syntax	<pre>[no] filter facility <i>facility</i> severity <i>severity</i></pre> <pre>[no] filter include exclude "<i>string1</i>"</pre>		
Command Mode	<pre>configuration log output console</pre> <pre>configuration log output file <i><filename></i></pre> <pre>configuration log output user file <i><username></i></pre> <pre>configuration log output-remote-server <i><ip></i> <i><vrf></i></pre> <pre>configuration log output remote-server <i><hostname></i> <i><vrf></i></pre>		
Parameters Table	Name	Value	Description
	<i>facility</i>	any (default) acl auth authpriv config-change cron	Facility that should send messages to the selected output device. "Any" allows all facilities.

		daemon	
		ftp	
		infra	
		infra-utils	
		interactive-cmd	
		kernel	
		lpr	
		mail	
		news	
		syslog	
		user	
		uucp	

	<i>severity</i>	0-emergency 1-alert 2-critical 3-error 4-warning 5-notice 6-info 7-debug any none (default)	Minimum severity of messages to send to the selected output device. "Any" allows all levels. 'None' excludes the specified facility.
--	-----------------	--	--

	<i>string</i>	String enclosed in "".	Include: Send only messages containing the indicated string or strings. Exclude: Do not send messages containing the indicated string or strings.
--	---------------	------------------------	---

Examples	This example shows how to filter log output by facility and severity: <pre>exaware# config exaware(config)# log output console filter facility user severity 2</pre> This example shows how to exclude the daemon facility, while allowing all others: <pre>exaware# config exaware(config)# log output file log_file filter facility any severity any exaware(config-log-output-file-log_file-filter-facility- any)# log output file log_file filter facility daemon severity none</pre> This example shows how to filter log output for admin messages containing the string "configure": <pre>exaware# config exaware(config)# log output user admin filter include "configure"</pre> This example shows how to filter log output to exclude all messages containing the string "isis": <pre>exaware# config exaware(config)# log output file log_file filter exclude isis</pre> This example shows how to disable filtering: <pre>exaware# config exaware(config)# no log output file log_file filter facility any severity any exaware(config)# no log output file log_file filter exclude isis</pre>
Notes	A command with facility any is applied to all facilities that do not have explicit configuration (see the examples above).

7.4.1.4. log output console

Description	Directs log messages to the console. Use the no form of command to disable logging.
Command Syntax	[no] log output console
Command Mode	configuration
Parameters Table	N/A
Examples	This example shows how to configure logging to the console: <pre>exaware# config exaware(config)# log output console exaware(config-log-output-console)#</pre> This example shows how to disable logging to the console: <pre>exaware# config exaware(config)# no log output console</pre>

7.4.1.5. log output file

Description	Directs log messages to the specified output file under /var/log directory. Use the no form of the command to disable logging to a specific file.		
Command Syntax	[no] log output file <i>filename</i>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>filename</i>	String	Name of file under /var/log directory. May be new file or existing file.
Examples	This example shows how to configure logging to the specified file: <pre>exaware# config exaware(config)# log output file new_log exaware(config-log-output-file-new_log)#</pre> This example shows how to disable logging to the specified file: <pre>exaware# config exaware(config)# no log output file new_log</pre>		

7.4.1.6. log output remote-server

Description	Directs log messages to the specified output remote server. Also enters configuration mode to configure the following server parameters. Use the no form of the command to disable logging.		
Command Syntax	[no] log output remote-server <i>ip-address</i> [vrf <i>vrf-name</i> [source <i>ip-address</i>]]		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4/IPv6 IP address	IP address of the remote server.
	<i>vrf-name</i>	default management	VRF name is where the remote server is located. (Default = management)

	source ip-address	IPv4/IPv6 IP address	Source IP address of the sent log messages. This IP address must be one of the IP addresses configured on an interface belonging to the same VRF where the remote server is located. The default value depends on VRF: • default: IP address of the outgoing interface. • management: IP address of the management (OOB) interface.
--	-------------------	----------------------	---

Examples

This example shows how to configure logging to the specified remote server (2.2.2.2). Source IP address that will be used for these logs is 3.3.3.3:

```
exaware# config
exaware(config)# log output remote-server 2.2.2.2 vrf
default source 3.3.3.3
```

This example shows how to disable vrf and source IP for logs to the specified remote server:

```
exaware# config
exaware(config)# no log output remote-server
10.10.10.10 vrf default
```

This example shows how to disable logging to all remote servers:

```
exaware# config
exaware(config)# no log output remote-server
```


7.4.1.7. log output user

Description	Directs log messages to the specified user. Use the no form of command to disable logging.		
Command Syntax	[no] log output user { <i>username</i> <i>all</i> }		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>username</i>	Existing username	
		all	Log to all users.
Examples	This example shows how to configure logging to the specified user: <pre>exaware# config exaware(config)# log output user marysmith exaware(config-log-output-user-marysmith)#</pre> This example shows how to disable logging to the specified user: <pre>exaware# config exaware(config)# no log output user marysmith</pre> This example shows how to disable logging to all users: <pre>exaware# config exaware(config)# no log output user</pre>		

7.4.1.8. rotate (log file)

Description	Configures the archive parameters for the log file. Use the no form of the command to restore the default archive parameters.		
Command Syntax	[no] rotate {size <i>max-size</i> files <i>max-#-files</i> frequency <daily weekly>}		
Command Mode	configuration log-file <filename>		
Parameters Table	Name	Value	Description
	<i>max-size</i>	1-10 (Default = 5 MB)	The maximum file size (in MB). When the file reaches the specified size, it is archived automatically.
	<i>max-#-files</i>	1-20 (Default = 7)	Maximum number of files that can be stored. When this number is reached, oldest archived file is deleted.
		daily (default) weekly	Frequency that log files are archived.
Examples	This example shows how to filter log output by facility and severity: <pre>exaware# config exaware(config)# log output file log_file rotate size 10 files 10 frequency weekly</pre>		

Notes

Archiving is performed automatically when either of these conditions are met:

- File reaches the configured maximum size, or
- At the end of the configured time interval (day or week)

7.4.1.9. transport (log remote-server)

Description	Specifies the transport type and port number to communicate with remote server.		
Command Syntax	<code>transport {tcp udp} port <i>port</i></code>		
Command Mode	configuration log remote-server < <i>ip-address</i> or <i>hostname</i> >		
Parameters Table	Name	Value	Description
		TCP UDP	Transport type for communicating with the remote server. (Default)
	<i>port</i>	Port number	Port that communicates with remote server for the specified transport type. (Default = 514 for both transport types)
Examples	This example shows how to configure the transport type and port number: <pre>exaware# config exaware(config)# log output remote-server 10.10.10.10 default exaware(config-log-output-remote-server-10.10.10.10/default)# transport udp port 212</pre>		

7.4.2. Logging Operational Commands

7.4.2.1. log monitor start

Description	Starts real time log monitoring.		
Command Syntax	<code>log monitor start <i>log</i></code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>log</i>		Name and full path of the log to monitor in the /var/log/ directory. User must have read permissions on the file and directory.
Examples	This example shows how to monitor a file: <code>exaware# monitor log start /var/log/messages</code>		

7.4.2.2. log monitor stop

Description	Stops real time log monitoring.		
Command Syntax	log monitor stop <i>log</i>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>log</i>		Name and full path of the file to stop monitoring in the /var/log/ directory. User should have read permissions on the file and directory.
Examples	This example shows how to stop monitoring a log. <pre>exaware# log monitor stop var/log/messages</pre>		

7.4.2.3. log copy

Description	Copy the contents of the specified log file to the specified destination file.		
Command Syntax	<code>log copy <i>source-log destination</i></code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>source-log</i>	Path and filename of the source file.	Name of the log to copy from the /var/log/ directory.
	<i>destination</i>	Path and filename of the destination file.	Destination to which the file should be copied. Destination directory should be the user home directory. Default path is the user home directory.
Examples	This example shows how to copy a log file: <pre>exaware# log copy ? Possible completions: FILE Source log file name apt/ audit.log boot bootstrap.log snmp.log exaware# log copy syslog ? Possible completions: FILE Destination log file name .bash_logout .bash_profile .bashrc .profile .vimrc exaware# log copy syslog syslog-copy</pre>		

7.4.2.4. log delete

Description	Deletes the specified log file from the file system.		
Command Syntax	<code>log delete <i>filename</i></code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>filename</i>	Path and filename to delete.	Name of the log to delete from /var/log/ directory.
Examples	This example shows how to delete a log file: <code>exaware# log delete wtmp.3</code>		

7.4.2.5. show log

Description	Displays contents of the specified log file.		
Command Syntax	<code>show log <i>filename</i></code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>filename</i>	Existing log file	File to display.
Examples	This example shows how to display the specified log file: <code>exaware# show log messages</code>		

7.5. NTP Commands

7.5.1. NTP Configuration Commands

7.5.1.1. ntp

Description	Configures NTP options including: Add/Remove NTP servers to/from the list of available time source servers. Specify the source interface/IP address from which NTP requests will be sent.		
Command Syntax	<pre>ntp vrf {vrf-name} [source {interface-type} {interface-path} source {source-ip}] server {ip-address} ...</pre>		
Command Mode	configuration ntp		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	string	The VRF name to use for NTP server communications.
	<i>interface-type</i>	enum	loopback, x-eth, mgmt, agg-eth.
	<i>interface-path</i>	##/## or #	0/0/port convention for eth, or number for loopback.
	<i>source-ip</i>	IP address	IPv4 / IPv6 address to be used as the source address.
	<i>ip-address</i>	IP address	IPv4 / IPv6 address of the time server.

Examples	The following examples show how to configure the NTP server: <pre>exaware# config exaware(config)# ntp exaware(config-ntp)# vrf default exaware(config-ntp-vrf-default)# source x-eth 0/1/2 exaware(config-ntp-vrf-default)# server 10.1.1.2 exaware(config-ntp-vrf-default)# source loopback 2</pre>
Notes	Only one VRF is supported.

7.5.2. NTP Operational Commands

7.5.2.1. show ntp associations

Description	Displays a list of the currently configured NTP time servers.																
Command Syntax	show ntp associations																
Command Mode	operational																
Examples	This example shows how to display the current NTP status. <pre>exaware# show ntp associations</pre> <table><tr><th>NTP Server</th><th>VRF Name</th><th>Stratum</th><th>Next update</th></tr><tr><td>122.168.2.2</td><td>management</td><td>0</td><td>09:35:40</td></tr><tr><td>122.168.2.3</td><td>management</td><td>0</td><td>09:40:44</td></tr><tr><td>122.168.2.3</td><td>management3</td><td>2</td><td>09:34:30</td></tr></table>	NTP Server	VRF Name	Stratum	Next update	122.168.2.2	management	0	09:35:40	122.168.2.3	management	0	09:40:44	122.168.2.3	management3	2	09:34:30
NTP Server	VRF Name	Stratum	Next update														
122.168.2.2	management	0	09:35:40														
122.168.2.3	management	0	09:40:44														
122.168.2.3	management3	2	09:34:30														

7.6. Router Administration Commands

7.6.1. Configuration Mode

7.6.1.1. **abort**

Description	Aborts the configuration session.
Command Syntax	abort
Command Mode	configuration
Parameters Table	N/A
Examples	<pre>exaware# config exaware(config)# interface agg-eth 11 exaware(config-interface-agg-eth-11)# lacp-rx-interval 200 exaware(config-interface-agg-eth-11)# abort Exaware#</pre>

7.6.1.2. alias

Description	Can be used anywhere on the command line. After a command with an alias has been entered, the expanded command line appears, enabling you to verify the alias value.		
Command Syntax	<code>alias <i>alias-name</i> expansion <i>alias-expansion</i></code>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>alias-name</i>	String	Alias input.
	<i>alias-expansion</i>	String	Alias output. (Default = none)
Examples	<pre>exaware(config)# alias www expansion "run who" exaware(config)# commit Commit complete. exaware(config)# www exaware(config)# run who</pre>		

7.6.1.3. annotate

Description	Associates an annotation with a given configuration statement. To remove an annotation leave the text empty		
Command Syntax	<code>annotate <i>statement</i> "<i>text</i>"</code>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>statement</i>	You can display a listing of options	The command or command/parameter where to add a comment.
	<i>"text"</i>	string enclosed in " "	
Examples	<code>exaware(config)# annotate system hostname "put your comment here in quotes"</code>		

7.6.1.4. cancel

Description	Removes all uncommitted configuration changes.
Command Syntax	cancel
Command Mode	configuration
Parameters Table	N/A
Examples	<pre>exaware(config)# cancel All configuration changes will be lost. Proceed? [yes,no] yes exaware(config)#</pre>

7.6.1.5. `commit`

Description

Commits the current set of changes. When a configuration is committed, the CLI first validates the configuration and if there is a problem, then the problem is displayed.

The configuration lock needs to be used for confirmed commit functionality to work appropriately. Therefore, the confirmed commit command is not supported in 'private' mode.

The confirming commit will be rolled back if the CLI session is terminated before confirming the commit.

You can also:

- Associate a comment with the commit. The comment can later be seen when examining rollback files.
- Associate a label with the commit. The label can later be seen when examining rollback files.
- Assign a 'persist' ID. If 'persist' is specified, then the CLI session can be terminated and a later session can confirm the pending commit by supplying the assigned persist token as an argument to the commit command. If a prior confirming commit operation has been performed with the persist argument, then to modify the ongoing confirming commit process, the persist-id argument needs to be supplied with the same persist token. This makes it possible, for example, to abort an ongoing persist commit.

Command Syntax

```
commit [check | and-quit | [persist token]] [comment  
text] [label text] [persist-id id]
```

Command Mode Configuration

Parameters Table	Name	Value	Description
		and-quit	Commit current set of changes and exit configuration mode.
		check	Validate current configuration.

	<i>token</i>		Assign an ID by which these commit changes can be identified in future commit commands.
	<i>text</i>	string	Text of comment or label to display in rollback files.
	<i>id</i>	existing persist token	ID assigned to previous committed configuration using the <i>persist token</i> option.
Examples	The following example illustrates some configuration changes that are committed: <pre>exaware# config exaware(config)# aaa authentication-order 1 authentication-type tacacs all exaware(config)# commit</pre>		

7.6.1.6. dns

Description	Set the DNS domain and name servers.		
Command Syntax	dns vrf { <i>vrf-name</i> } [source { <i>interface-type</i> } { <i>interface-path</i> }] [source { <i>source-ip</i> }] domain-name { <i>domain-name</i> } name-server { <i>ip-address</i> } ...		
Command Mode	configuration		
Parameters Table	Name	Values	Description
	<i>vrf-name</i>	string	Name of VRF to use for DNS server communications.
	<i>interface-type</i>	enum	loopback, x-eth, mgmt, agg-eth, address.
	<i>interface-path</i>	##/# or #	0/0/port convention for eth, or number for loopback.
	<i>source-ip</i>	IP address	IPv4 / IPv6 address to be used as the source address.
	<i>domain-name</i>	string	DNS domain name of the router.
	<i>ip-address</i>	IP address	IPv4 / IPv6 address of the DNS server.
Examples	<pre> exaware(config-dns)# vrf default exaware(config-dns-vrf-default)# source loopback 2 exaware(config-dns-vrf-default)# source 10.1.1.2 exaware(config-dns-vrf-default)# domain-name cmpsys.com exaware(config-dns-vrf-default)# name-server 192.168.2.2 exaware(config-dns-vrf-default)# name-server 192.168.3.3 </pre>		
Notes	Only one VRF is supported.		

7.6.1.7. edit

Description	Opens a configuration statement in a text editor for editing.		
Command Syntax	<code>edit option</code>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>option</i>	You can display a listing of options	The command or command/parameter to edit.
Examples	The following example shows how to edit an AAA role: <pre>exaware(config)# edit aaa role priv_admin exception 1 Loading editor... aaa role priv_admin exception 1 command os-shell action reject !</pre>		
Notes	The type of editor to be used can be defined using the command <i>session editor</i> (on page 286) emacs/vi.		

7.6.1.8. end

Description	Terminates the configuration session.
Command Syntax	end
Command Mode	configuration
Parameters Table	N/A
Examples	<pre>exaware(config-routing-bgp-999-vrf-default)# end exaware#</pre>

7.6.1.9. exit

Description	Exits from the current configuration mode.
Command Syntax	exit
Command Mode	configuration
Parameters Table	N/A
Examples	<pre>exaware(config-routing-bgp-999-vrf-default)# exit exaware(config-routing-bgp-999)# exit exaware(config)#</pre>

7.6.1.10. load

Description	Loads the running configuration from an ASCII file. Choose from the following load options: • Override the complete current configuration. The entire configuration is deleted before loading the new configuration. • Merge with the current configuration. • Replace the delta in the current configuration with corresponding sections from the file.		
Command Syntax	<pre>load [(merge override replace) {<i>local-filename</i> [vrf <i>vrf-name</i>] [source-address <<i>ip-address</i>>] source-interface <<i>interface-type</i>> <<i>interface-path</i>>] external-source</pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
		merge	Merge the loaded configuration with the existing running configuration.
		override	Completely overwrite the existing configuration.
		replace	Replace changed sections in the current configuration.
	<i>local-filename</i>	string	Name and path of a local file.
	vrf-name	string	Name of the vrf to use.
	<i>ip-address</i>	IP address	IPv4 or IPv6 source address.

	<i>interface- type</i>	enum	Type of source interface, e.g x-eth.
--	----------------------------	------	--------------------------------------

	<i>interface- path</i>	X/Y/Z	Source interface path, e.g. 0/1/1.
--	----------------------------	-------	------------------------------------

	external-source	string	An FTP or TFTP URL for the file containing the configuration.
--	-----------------	--------	---

	Protocol	ftp tftp	Download protocol.
--	----------	-------------	--------------------

	User	string	URL user name.
--	------	--------	----------------

	Password	string	User password (Optional, if not provided, then it will be queried when executing).
--	----------	--------	--

	remote-host	IP address	IP address of the remote host.
	file-path	path	The absolute path where the file is located on the remote host.
Examples	This example shows how to load a configuration from a local file, merging with the current configuration: <pre> exaware# config exaware(config)# load merge ? Possible completions: FILE Source file name, local or ftp/tftp URL format - {protocol}://{user[:password]@{remote_host}}/{file_path} .bash_logout .bash_profile .bashrc .profile .vimrc a.txt b.txt c.txt vrf Virtual Routing and Forwarding name exaware(config)# load merge ftp://ftpuser:ftppassword@10.1.70.200//pc13/base- cmp1.initial.trunk.cfg Loading... Copying file done. Operation completed successfully in 1.98 seconds </pre>		
Notes	VRF (and optional source-address) are mutually exclusive with source interface. If neither is specified, the vrf and source address are taken from the default-management-source configuration. If only VRF is specified, the source address will be the address of the physical interface the traffic is going out through. If source-interface is specified, the VRF and source address are taken from the interface configuration.		

7.6.1.11. pwd

Description	Displays the current mode path.
Command Syntax	pwd
Command Mode	configuration
Parameters Table	N/A
Examples	<pre>exaware(config)# vrf jjj description j exaware(config-vrf-jjj)# pwd Current submode path: vrf jjj exaware(config-vrf-jjj)#</pre>

7.6.1.12. resolved

Description	Resolves conflicts. When two different sessions edit the same part of the configuration, a conflict occurs. When a second session tries to commit its modifications, a "conflict" message appears and cannot commit until the "resolved" command has been issued.
Command Syntax	resolved
Command Mode	configuration
Parameters Table	N/A
Examples	The following message appears when the <code>resolve</code> command needs to be executed: <pre>Aborted: there are conflicts. ----- Resolve needed before configuration can be committed. View conflicts with the command 'show candidate' and execute the command 'resolved' when done, or exit configuration mode to abort. Conflicting configuration items are indicated with a leading '!' Conflicting users: test -----</pre>

7.6.1.13. rollback

Description	Rolls back the database to the specified version. It returns the configuration to the previously committed version. The system stores a limited number of old configurations. If this amount is exceeded, the system automatically removes the oldest configuration before creating a new one.		
Command Syntax	<code>rollback [number]</code>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>number</i>		The version number of previously committed configuration to rollback to. (Default = last committed version -no number indicated)
Examples	This example rolls back the configuration to the specified version: <pre>exaware(config)# rollback 2</pre> This example rolls back the configuration to the last committed version: <pre>exaware(config)# rollback</pre>		

7.6.1.14. run

Description	Runs an Operational mode command from configuration mode.
Command Syntax	run
Command Mode	configuration
Parameters Table	N/A
Examples	<pre>exaware(config)# run show session autowizard enable complete-on-space enable history 100 idle-timeout 20 output-file terminal paginate enable screen-length 24 screen-width 80 show-defaults disable terminal-type xterm exaware(config)#</pre>

7.6.1.15. save

Description	Saves the contents of the running configuration to an ASCII file.		
Command Syntax	<code>save filename</code>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>filename</i>		Name and full path of the file to which the running configuration should be saved.
Examples	This example shows how to save and the display the running configuration. <pre> exaware# config exaware(config)# save running_config exaware(config)# run file show running_config interface management 0/0/0 admin-state up config-state active ipv4 address 10.0.0.2/24 ! interface management 0/0/0 admin-state down system hostname exaware ! ! exaware(config)# </pre>		

7.6.1.16. show candidate

Description	Displays current configuration changes that have not been committed.		
Command Syntax	<code>show candidate [<i>configuration option</i>] [all diff merge]</code>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>configuration option</i>		Specific command group, such as 'interface' or 'vrf'. Query for available options.
		all	Show the complete candidate configuration tree.
		diff	Show differences between running and merged configuration.
		merge	Show contents of merged configuration.
Examples	<pre>exaware(config)# show candidate session complete-on-space enable session autowizard enable session display-level 10 !end-of-config</pre>		

7.6.1.17. show configuration commit

Description	Displays configuration changes. You can display changes for a specific commit action or for all commit actions.																																																													
Command Syntax	show configuration commit [list]																																																													
Command Mode	operational																																																													
Parameters Table	N/A																																																													
Examples	<pre>exaware# show configuration commit list</pre> <table> <thead> <tr> <th>SNo.</th><th>ID</th><th>User</th><th>Client</th><th>Time Stamp</th><th>Label</th><th>Comment</th></tr> <tr> <th>----</th><th>-----</th><th>-----</th><th>-----</th><th>-----</th><th>-----</th><th>-----</th></tr> </thead> <tbody> <tr> <td>0</td><td>10269</td><td>admin</td><td>cli</td><td>2021-08-02 02:44:35</td><td></td><td></td></tr> <tr> <td>1</td><td>10268</td><td>admin</td><td>cli</td><td>2021-08-02 02:44:12</td><td></td><td></td></tr> <tr> <td>2</td><td>10267</td><td>admin</td><td>cli</td><td>2021-08-02 02:44:10</td><td></td><td></td></tr> <tr> <td>3</td><td>10266</td><td>admin</td><td>cli</td><td>2021-08-01 23:44:09</td><td></td><td></td></tr> <tr> <td>4</td><td>10265</td><td>admin</td><td>cli</td><td>2021-08-01 23:43:56</td><td></td><td></td></tr> <tr> <td>5</td><td>10264</td><td>admin</td><td>cli</td><td>2021-08-02 02:43:48</td><td></td><td></td></tr> </tbody> </table>						SNo.	ID	User	Client	Time Stamp	Label	Comment	----	-----	-----	-----	-----	-----	-----	0	10269	admin	cli	2021-08-02 02:44:35			1	10268	admin	cli	2021-08-02 02:44:12			2	10267	admin	cli	2021-08-02 02:44:10			3	10266	admin	cli	2021-08-01 23:44:09			4	10265	admin	cli	2021-08-01 23:43:56			5	10264	admin	cli	2021-08-02 02:43:48		
SNo.	ID	User	Client	Time Stamp	Label	Comment																																																								
----	-----	-----	-----	-----	-----	-----																																																								
0	10269	admin	cli	2021-08-02 02:44:35																																																										
1	10268	admin	cli	2021-08-02 02:44:12																																																										
2	10267	admin	cli	2021-08-02 02:44:10																																																										
3	10266	admin	cli	2021-08-01 23:44:09																																																										
4	10265	admin	cli	2021-08-01 23:43:56																																																										
5	10264	admin	cli	2021-08-02 02:43:48																																																										

7.6.1.18. show configuration rollback

Description	Displays rollback changes. You can display changes for a specific rollback action, for all rollback actions, or for rolling back a specified number of commit actions.		
Command Syntax	show config rollback change [<i>commit-id</i>] [diff <i>number</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>commit-id</i>	integer	ID number of the commit action. If no number is specified, all changes are listed.
	<i>number</i>	integer	Changes for rolling back the specified number of commits.
Examples	<pre>exaware# show configuration rollback changes 3 routing bgp 999 vrf default ! !</pre>		

7.6.1.19. show configuration running

Description	Displays the current running configuration. By default, the whole configuration is displayed. It is possible to limit what is shown by supplying a pathfilter. A pathfilter may be either a path pointing to a specific instance, or if an instance id is omitted, the part following the omitted instance is treated as a filter.		
Command Syntax	show config running [<i>pathfilter</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>pathfilter</i>		A path pointing to a specific instance, or if an instance ID is omitted, the part following omitted instance is treated as a filter. (Default = no filter)
Examples	<pre> exaware# show config running aaa aaa user admin password \$1\$U4\$uyksWEZhyup0h5Jj5126H0 role super_admin ! aaa user private password \$1\$UD\$PqCdoyKFNQDoHqrMo750n. role super_admin ! aaa user public password \$1\$L1\$wSdI9o9tB3CeSFFNBKSiS0 role super_admin ! aaa role priv_admin privilege all exception 1 command os-shell action reject ! ! aaa role super_admin privilege all </pre>		

7.6.1.19.1. **show history (configuration)**

See ***show history*** (on page 323).

7.6.1.20. session

Many of the operational session commands are also available in configuration mode. Commands in the operational mode only affect the current session, while commands executed in configuration mode affect all sessions.

See:

- **session autowizard** (*on page 304*)
- **session complete-on-space** (*on page 305*)
- **session history** (*on page 307*)
- **session idle-timeout** (*on page 308*)
- **session ignore-leading-space** (*on page 309*)
- **session paginate** (*on page 310*)
- **session show-defaults** (*on page 314*)

The following session commands are available only in configuration mode:

- **session display-level** (*on page 285*)
- **session editor** (*on page 286*)

7.6.1.21. session display-level

Description	Maximum number of levels to show when displaying the configuration.		
Command Syntax	<code>session display-level <i>level</i></code>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>level</i>	1-64	Maximum number of levels to display.
Examples	<pre>exaware# config exaware(config)# session display-level 10</pre>		

7.6.1.22. session editor

Description	Specifies the editor to be used for editing the candidate configuration. Choose either: • emacs editor • vi editor (default)
Command Syntax	<code>session editor {emacs vi}</code>
Command Mode	configuration
Parameters Table	N/A
Examples	<pre>exaware# config exaware(config)# session editor emacs</pre>

7.6.1.23. session prompt-timestamp

Description	Sets the timestamp in the prompt.		
Command Syntax	session prompt-timestamp [enable disable]		
Command Mode	configuration		
Parameters Table	Name	Value	Description
		enable	Displays the timestamp in the prompt.
		disable	Does not display the timestamp in the prompt. (Default)
Examples	exaware# config exaware(config)# session prompt-timestamp enable		

7.6.1.24. tag

Description	Adds, clears, or deletes a statement tag.		
Command Syntax	tag {add clear del} <i>option</i> "tag"		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>option</i>	You can display a listing of options	The command or command/parameter where to add the tag.
	"tag"	string enclosed in " "	
Examples	exaware(config)# tag add aaa "put your tag here in quotes"		

7.6.1.25. tcp

Description	Sets the TCP configuration of the system.		
Command Syntax	<pre>tcp max-segment-size {MSS} tcp window-size {window size}</pre>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>MSS</i>	68-9100	Maximum segment size for a TCP session. (Default = 1460)
	<i>window size</i>	2048-1000000	The send and receive window size of a TCP session. (Default = 65535)
Examples	<pre>exaware(config)# tcp max-segment-size 576 exaware(config)# tcp window-size 32768</pre>		

7.6.1.26. telnet-server

Description	Enables or Disables the Telnet server.
Command Syntax	<code>telnet-server</code> <i>enable/disable</i>
Command Mode	configuration
Notes	The Telnet server is disabled by default.

7.6.1.27. top

Description	Exits to the top level and optionally runs the command. Equivalent to executing exit commands until the (config)# prompt level is reached.
Command Syntax	top
Command Mode	configuration
Parameters Table	N/A
Examples	<pre>exaware(config-routing-bgp-999-vrf-default) # top exaware(config)#</pre>

7.6.2. Operational Mode

7.6.2.1. clear history

Description	Clears the command history.
Command Syntax	<code>clear history</code>
Command Mode	operational
Parameters Table	N/A
Examples	<code>exaware# clear history</code>

7.6.2.2. compare

Description	Compares the running configuration to another configuration or to the contents of a specified file line-by-line, and displays any differences that are found.		
Command Syntax	<code>compare filename</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>filename</i>		Name and full path of the file to be compared to the running configuration.
Examples	This example shows the results of the compare operation: <pre>exaware# compare file /home/test/10_26_config < admin-state down --- > admin-state up !</pre>		

7.6.2.3. configure

Description	Enters the Configuration mode to edit a private copy of the candidate configuration.
Command Syntax	<code>configure [private]</code>
Command Mode	operational
Examples	<pre>exaware# config Entering configuration mode private exaware(config) #</pre>

7.6.2.4. exit

Description	Exits the current mode. If performed on the top level, this command exits the Configuration mode. To close an active terminal session and log off the router, use the exit command in Operational mode. To return the router to the next higher Configuration mode, use the exit command in the Configuration mode.
Command Syntax	<code>exit</code>
Command Mode	operational and configuration
Parameters Table	N/A
Examples	<code>exaware# exit</code>

7.6.2.5. help

Description	Shows the help text for the commands.		
Command Syntax	<code>help <i>command</i></code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>command</i>		Command for which to show the help text.
Examples	<pre>exaware# help clear interface Help for command: clear interface Clear interface counters</pre>		

7.6.2.6. logout

Description	Logs out a user from all CLI sessions.		
Command Syntax	logout [user <i>username</i> session <i>sessionid</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>username</i>		Logout a specific user. (Default = user issuing the command)
	<i>sessionid</i>		Terminate a specific session. (Default = current session)
Examples	exaware# logout user user1		

7.6.2.7. os-shell

Description	Logs in to the Linux operational system.
Command Syntax	os-shell
Command Mode	operational
Parameters Table	N/A
Examples	<pre>exaware# os-shell admin@exaware[CPM-0-0_A]:> ls -ltr total 4 -rwxr-xr-x 1 admin admin 200 Jun 17 12:50 open. admin@exaware[CPM-0-0_A]:> pwd /home/admin admin@exaware[CPM-0-0_A]:> exit exit exaware#</pre>

7.6.2.8. ping

Description	Sends ping (ICMP ECHO) request packets and gathers responses.		
Command Syntax	<code>ping [vrf vrf-name] host [<option1 option2 option3....>]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>vrf</i>	vrf-name	VRF instance to use.
	<i>host</i>	ip-address	IP address to ping.
	<i>count</i>	Integer, >0	Number of ping requests to send. (Default = 5)
	<i>DF</i> (Supported only for IPv4)		Turn on the 'don't fragment' flag. (Default = No DF)
	<i>random-pattern</i>		Use a different random pattern for each packet. (Default = No)
	<i>pattern</i>	Hexi-decimal data (max 16 char)	Use a custom pattern for the packet payload. (Default = 0x1234)
	<i>interval</i>	3600 or less	Interval between ping requests, in seconds. (Default = 1)
	<i>resolve</i>		Enable reverse DNS lookup. (Default = No)

	<i>source-interface</i>	Interface	Use the specified interface for outgoing requests.
	<i>size</i>	≥ 28	Ping packet size in bytes (L3 size). (Default = 64)
	<i>sweep-size</i>	≥ 28	Minimal packet size for sweep.
		≥ 28	Maximum packet size for sweep. Must be larger than the specified minimal size.
		Integer, >0	Size increment for sweep. (Default: 1)
	<i>source-address</i>	IP address	Source address.
	<i>timeout</i>	Integer, >0	Maximum wait time for response in seconds. (Default = 10)
	<i>tos</i>	0-255	Type Of Service field. (Default = 0)
	<i>ttl</i>	0-255	Time To Live field. (Default = 64)
	<i>verbose</i>		Verbose output on errors. (Default = No

	<i>RR</i> (Supported only for IPv4)		Enable the 'Record Route' IPv4 option. (Default = No)
	<i>TS</i> (Supported only for IPv4)		Enable the 'Time Stamp' IPv4 option. (Default = No)
	<i>loose-source</i>	Array of IP addresses	Enable the 'Loose Source Routing' IP option.
	<i>strict-source</i>	Array of IP addresses	Enable the 'Strict Source Routing' IP option.
	<i>override-routing</i>		Override the routing nexthop (send directly on interface). (Default = No)
	<i>ipv4</i>		Force IPv4 destination.
	<i>ipv6</i>		Force IPv6 destination.
	<i>egress-interface</i>		Set the egress interface.
Examples	<pre>exaware# ping 10.1.1.1 count 7 random-pattern exaware# ping vrf default 10.1.1.1 count 1 sweep-size 30-40-2 source-interface mgmt 0/0/0 interval 0.3 pattern abcdef</pre>		
Notes	ping ip-address is equivalent to: <pre>ping vrf default ip-address count 5 timeout 10 interval 1 size 64</pre>		

7.6.2.9. send

Description	Sends a message to the terminal of one or all users.		
Command Syntax	send { <i>username</i> all} <i>message</i>		
Command Mode	operational		
Parameters Table	Name	Value	Description
		<i>username</i>	Sends message only to this user name.
		<i>All</i>	Sends message to all users.
Examples	<pre>exaware# send all "Good Day !!!" Message from admin@10.10.10.10 at 18:23:58... Good Day !!! exaware#</pre>		

7.6.2.10. session autowizard

Description	Controls the default value for the autowizard setting in the CLI environment. If the autowizard is enabled, then the CLI prompts the user for the required values when a new configuration instance is created in Configuration mode. By default, the autowizard is disabled and is automatically disabled.		
Command Syntax	<code>session autowizard [enable disable]</code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
		enable/ disable	Toggles the state between enabled and disabled. (Default = disable)
Examples	<code>exaware# session autowizard enable</code>		

7.6.2.11. session complete-on-space

Description	Enables or disables space completion in the CLI environment. When space completion is enabled, a CLI command is completed automatically when a space is entered.		
Command Syntax	session complete-on-space {enable disable}		
Command Mode	operational		
Parameters Table	Name	Value	Description
		enable	Enable space completion in the CLI environment.
		disable	Disable space completion in the CLI environment. (Default)
Examples	exaware# session complete-on-space enable		

7.6.2.12. session configure-prompt

Description	Enables or disables the command line prompt in configuration mode.		
Command Syntax	session configure-prompt {enable disable}		
Command Mode	operational		
Parameters Table	Name	Value	Description
		enable	Prompt appears in configuration mode. (Default)
		disable	Prompt does not appear in configuration mode.
Examples	<pre> exaware# session configure-prompt enable exaware# config Entering configuration mode private exaware(config)# exit exaware# session configure-prompt disable exaware# config Entering configuration mode private exit exaware# </pre>		

7.6.2.13. session history

Description	Configures the history size (relevant only for the current session context). The size of the history list is configured using the history CLI setting. If you specify a history limit, only the last number of commands up to that limit is shown.		
Command Syntax	session history <i>size</i>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>size</i>		Size of history list configured using history CLI setting. (Default = 100)
Examples	<pre> exaware# session history 7 exaware# show session autowizard enable complete-on-space enable history 7 idle-timeout 1800 output-file terminal paginate enable screen-length 24 screen-width 80 show-defaults disable terminal linux exaware# show history 06:08:14 -- session history 7 06:08:14 -- session history 7 06:08:14 -- session history 7 06:08:14 -- session history 7 06:08:14 -- session history 7 06:08:25 -- show session 06:08:28 -- show history </pre>		

7.6.2.14. session idle-timeout

Description	Configures the length of time, in minutes, after which the current session will timeout if it has been idle.		
Command Syntax	<code>session idle-timeout <i>interval</i></code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>interval</i>	Integer	Idle timeout value for the current session in minutes. (Default = 30) Use '0' to prevent the session from timing out.
Examples	<code>exaware# session idle-timeout 15</code>		

7.6.2.15. session ignore-leading-spaces

Description	Enables or disables ignoring leading white space in the command line.		
Command Syntax	session ignore-leading-spaces {enable disable}		
Command Mode	operational		
Parameters Table	Name	Value	Description
		enable	Ignores leading white space in the command line.
		disable	Does not ignore leading white space in the command line.
Examples	exaware# session ignore-leading-spaces enable		

7.6.2.16. session paginate

Description	Paginates the output from CLI commands (relevant only for the current session context). It enables a filter for paging through the CLI command output text one screen at a time as opposed to continuous scrolling.		
Command Syntax	session paginate {enable disable}		
Command Mode	Operational		
Parameters Table	Name	Value	Description
		enable/ disable	Toggles the state between enabled and disabled. (Default = enable)
Examples	exaware# session paginate disable		

7.6.2.17. session record start / stop

Description	Starts or stops recording the terminal outputs.		
Command Syntax	session record [start <i>filename</i> stop]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>filename</i>		Name and full path of the saved file record.
Examples	<pre> exaware# session record start test_record exaware# show configuration snmp community snmp community public ! exaware# file show test_record exaware# show configuration snmp community snmp community public ! exaware# file show test_record exaware# session record stop </pre>		

7.6.2.18. session screen-length

Description	Configures the screen length (relevant only for the current session context).		
Command Syntax	session screen-length <i>length</i>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>length</i>		Screen length. (Default = current screen length)
Examples	exaware# session screen-length 777		

7.6.2.19. session screen-width

Description	Configures the current screen width (relevant only for the current session context).		
Command Syntax	<code>session screen-width <i>width</i></code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>width</i>	integer	Screen width. (Default = current screen width)
Examples	<code>exaware#session screen-width 777</code>		

7.6.2.20. session show-defaults

Description	Shows the default values when showing the configuration (relevant only for the current session context).		
Command Syntax	session show-defaults [enable disable]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
		enable / disable	Toggles the state between enabled and disabled. (Default = disable)
Examples	<pre>exaware# session show-defaults enable exaware# show configuration interface mgmt system module 0/cpm0 admin-state active ! active ! exaware#</pre>		

7.6.2.21. session terminal-type

Description	Sets the terminal type for the current session..		
Command Syntax	session terminal-type [generic vt100 xterm]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
		ansi	
		generic	
		linux	
		vt100	
		xterm	Default
Examples	exaware# session terminal-type vt100		

7.6.2.22. show (output filters)

Description	Filters the output display information about the system configuration or operational state. Use show commands in either Operational or Configuration mode with the relevant output filter.		
Command Syntax	show <command> [begin count exclude include linnum include more nomore save]		
Command Mode	operational and configuration		
Parameters Table	Name	Value	Description
		append	Append output text to a file.
		begin	Begin with the line that matches.
		count	Count the number of lines in the output.
		exclude	Exclude lines that match.
		include	Include lines that match.
		linnum	Enumerate lines in the output.
		more	Paginate output.
		nomore	Suppress pagination.
		os-command	Execute operating system commands.
		save	Save output text to a file.
		until	End with the line that matches.

Examples	<pre>exaware# show session lnum 1: autowizard disable 2: complete-on-space disable 3: history 100 4: idle-timeout 1800 5: output-file terminal 6: paginate enable 7: screen-length 27 8: screen-width 80 9: show-defaults disable 10: terminal xterm</pre>
-----------------	---

7.6.2.23. show configuration (operational)

Description	Shows the current user-configured parameters for the specified sub-mode option. This is to be distinguished from <code>show sub-mode-option</code> commands (such as <code>show config session</code> as opposed to <code>show session</code>), which display the complete current configuration, both user-configured and default. One notable exception to this general rule is the <code>show config aaa</code> command. The <code>show config aaa user</code> and the <code>show config aaa role</code> commands display both user-configured and default pre-configured users/roles. The entire configuration is displayed by default. It is possible to limit what is shown by supplying a pathfilter. The pathfilter may be either a path pointing to a specific instance, or if an instance id is omitted, the portion following the omitted instance is treated as a filter.		
Command Syntax	<code>show configuration [option]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description

	<i>option</i>	aaa	Set the AAA parameters
		alias	Create command alias
		arp	Arp commands
		ctrl-plane	Control-plane options
		debug	Debugging options
		details	By default only elements that have been explicitly set to a value are shown. This makes it easier to handle large configurations. However, it is possible to force the show command to display all elements using 'details'.
		[<pathfilter>]	
		dns	DNS related configuration
		interface	Interface commands
		log	Logging configuration
		mpls	MPLS Commands
		ndp	Neighbor discovery commands
		ntp	NTP related configuration
		policy	Policy commands
		routing	Routing protocols commands
		session	Global default session parameters
		snmp	Set the SNMP Agent parameters
		system	Global system commands
		tcp	Set tcp parameters
		telnet-server	Enable/disable the Telnet server
		vrf	

		VRF routing table
Examples	This example shows the difference between the outputs for <code>show</code> and <code>show config</code> commands. <pre>exaware# show configuration session session idle-timeout 35 exaware# show session autowizard disable complete-on-space disable history 100 idle-timeout 35 output-file terminal paginate enable screen-length 24 screen-width 80 show-defaults disable terminal vt100</pre>	

7.6.2.24. show history

Description	Shows the command history of the current mode. When changing to Configuration mode, you cannot access command history from the Operational mode session. When exiting back into Operational mode, you can access the command history from the preceding Operational mode session.		
Command Syntax	show history [<i>limit</i>]		
Command Mode	operational and configuration		
Parameters Table	Name	Value	Description
	<i>limit</i>		Number of items to show (Default = no limit, show complete history)
Examples	<pre>exaware# config exaware(config)# show history 2 10:44:54 -- config 10:45:25 -- show history 2</pre>		

7.6.2.25. show id

Description	Shows the user ID information; uid, gid, and groups.
Command Syntax	<code>show id</code>
Command Mode	operational
Parameters Table	N/A
Examples	<code>exaware# show id</code> <code>user = admin(517), gid=500, groups=admin, gids=500</code>

7.6.2.26. show session

Description	Displays the CLI settings.
Command Syntax	show session
Command Mode	operational
Parameters Table	N/A
Examples	<pre>exaware# show session autowizard disable complete-on-space disable history 100 idle-timeout 1800 output-file terminal paginate enable screen-length 27 screen-width 80 show-defaults disable terminal xterm</pre>

7.6.2.27. ssh

Description	Opens a secure shell connection on a remote host (based on the Linux SSH command).		
Command Syntax	ssh [vrf < <i>vrf-name</i> > [source-address < <i>ip-address</i> >] source-interface < <i>interface-type</i> > < <i>interface-path</i> >] [< <i>username</i> >@] < <i>host</i> > [ipv4 ipv6]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	string	Name of the VRF on which to open the secure shell.
	<i>ip-address</i>	IPv4 / IPv6 address	IP source address.
	<i>interface-type</i>	enum	Type of source interface, e.g x-eth.
	<i>interface-path</i>	x/y/z	Source interface path, e.g. 0/1/1.
	<i>username</i>	string	User name to use for login.
	<i>host</i>	string or IP address	Name or IP address of a remote host.
	ipv4		Force IPv4 destination.
	ipv6		Force IPv6 destination.
Examples	<pre>exaware# ssh 192.168.2.1 exaware# ssh vrf management joe@192.168.2.1 exaware# ssh source-interface loopback 0 joe@foo.Exaware.com ipv6 exaware# ssh source-address fc00::ffff:0a01:d602 fc00::ffff:0a01:0103</pre>		

Notes	VRF (and optional source-address) are mutually exclusive with source interface. If neither is specified, the vrf and source address are taken from the default-management-source configuration. If only VRF is specified, the source address will be the address of the physical interface through which the traffic flows. If source-interface is specified, the VRF and source address are taken from the interface configuration. If the address family is not specified, the default is IPv4.
--------------	--

7.6.2.28. traceroute

Description	Traces the path to a host. It finds all the routers on the path to a specific host by sending probe packets. It sends probe packets with rising TTL (Time-To-Live) and gathers ICMP responses, producing a route description with round-trip times.		
Command Syntax	traceroute [vrf <i>vrf-name</i>] <i>host</i> [<i>option1 option2 option3...</i>]		
Command Mode	Operational		
Parameters Table	Name	Values	Description
	<i>vrf</i>	vrf-name	VRF instance to use.
	<i>host</i>	ip-address	The target IP address.
	<i>count</i>	Integer >0	Number of probes to send per hop. (Default = 3)
	<i>interval</i>	Float, ≥0.2	Delay between packets. ≤9 in seconds, >9 in ms. (Default = 1)
	<i>resolve</i>		Enable reverse DNS and AS lookup. (Default = No)
	<i>source-interface</i>	Interface	Use the specified interface for outgoing requests.
	<i>source-address</i>	IP address	Source address.
	<i>timeout</i>	Integer >0	Maximum wait time for response in seconds. (Default = 30)

	<i>tos</i>	0-255	Type Of Service field. (Default = 0)
	<i>minttl</i>	1-255	Minimum Time To Live field. (Default = 1)
	<i>maxttl</i>	1-255	Maximum Time To Live field. (Default = 30)
	<i>override- routing</i>		Override the routing next hop (send directly on interface). (Default = No)
	<i>simultaneous- packets</i>	Integer >0	Maximum number of outstanding packets. (Default = 1)
	<i>seq</i>	1-65534	'Sequence' number: Port number for UDP. (Default = 33433) Seq number for ICMP. (Default = 1)
	<i>ipv4</i>		Force IPv4 destination.
	<i>ipv6</i>		Force IPv6 destination.
	<i>gateway</i>	IP address	Address of router gateway to route through.
Examples	<pre>exaware# traceroute 10.1.1.1 count 7 traceroute vrf default 10.1.1.1 count 1 source-interface management 0/0/0 interval 0.3</pre>		

Notes

```
traceroute 'host'  
is equivalent to:  
traceroute vrf default 'host' count 5 timeout 5 interval  
1 size 64
```


7.7. SNMP Commands

7.7.1. SNMP Configuration Commands

7.7.1.1. agent

Description	Enters SNMP agent configuration mode. Use the no format to remove the SNMP agent configuration.		
Command Syntax	agent network platform no agent network platform		
Command Mode	configuration snmp		
Parameters Table	Name	Value	Description
	platform		Specifies the platform SNMP agent. This agent is responsible for mibs describing the White Box itself (for example, entity mib).
	network		Specifies the network SNMP agent. This agent is responsible for network mibs (for example, ifTable, mib2).
Examples	This example shows how to enter SNMP agent configuration mode: <pre>exaware# config exaware(config)# snmp exaware(config-snmp)# agent platform exaware(config-snmp-agent-platform)#</pre>		

7.7.1.2. authentication-key

Description	Configures authentication key in the case of version v3usm.		
Command Syntax	authentication-key <i>key</i>		
Command Mode	configuration snmp agent version v3usm configuration snmp trap target version v3usm		
Parameters Table	Name	Value	Description
	key	String Min: 8 chars Max: 20 chars	User Password.
Examples	This example shows how to configure an authentication key: <pre>exaware# config exaware(config)# snmp exaware(config-snmp)# agent network exaware(config-snmp-agent-network)# version v3usm exaware(config-snmp-agent-network-version-v3usm)# authentication-key AS7AeV63</pre>		
Notes	The parameter is mandatory if a value of security-level is AuthNoPriv or AuthPriv. A new CLI command overrides the previous one entering in the same configuration mode.		

7.7.1.3. authentication-protocol

Description	Configures authentication key protocol in the case of version v3usm.		
Command Syntax	authentication-protocol md5 sha		
Command Mode	configuration snmp agent version v3usm configuration snmp trap target version v3usm		
Parameters Table	Name	Value	Description
	md5		The md5 authentication protocol is applied.
	sha		The sha authentication protocol is applied.
Examples	This example shows how to configure an authentication protocol:		
	<pre>exaware# config exaware(config)# snmp exaware(config-snmp)# agent network exaware(config-snmp-agent-network)# version v3usm exaware(config-snmp-agent-network-version-v3usm)# authentication-protocol sha</pre>		
Notes	The parameter is mandatory if a value of security-level is AuthNoPriv or AuthPriv. A new CLI command overrides the previous one entering in the same configuration mode.		

7.7.1.4. community

Description	Configures a user defined SNMP community name in the case of version v2c.		
Command Syntax	community <i>community-name</i>		
Command Mode	configuration snmp agent version v2c configuration snmp trap target version v2c		
Parameters Table	Name	Value	Description
	<i>community-name</i>	string Min: 1 char Max: 20 chars	A unique name for the community.
Examples	This example shows how to configure a community:		
<pre>exaware# config exaware(config)# snmp exaware(config-snmp)# trap exaware(config-snmp-trap)# target nsml exaware(config-snmp-trap-target-nsmlt)# version v2c exaware(config-snmp-trap-target-nsml-version-v2c)# community NewCommunityName</pre>			
Notes	A new CLI command overrides the previous one entering in the same configuration mode.		

7.7.1.5. contact

Description	Sets a value of the MIB-II contact parameter. Use the no form to restore the default.		
Command Syntax	contact <i>contact</i> no contact		
Command Mode	configuration mib2		
Parameters Table	Name	Value	Description
	<i>contact</i>	String Max: 255 chars	A value of the MIB-II contact parameter. The default value – the empty string
Examples	This example shows how to configure the MIB-II contact parameter: <pre>exaware# config exaware(config)# mib2 exaware(config-mib2)# contact bob@company.com</pre>		

7.7.1.6. description

Description	Sets a value of the MIB-II description parameter. Use the no form to restore the default.		
Command Syntax	description <i>description</i> no description		
Command Mode	configuration mib2		
Parameters Table	Name	Value	Description
	<i>description</i>	String	A value of the MIB-II description parameter.
		Max: 255 chars	The default value – the empty string
Examples	This example shows how to configure the MIB-II name parameter: <pre>exaware# config exaware(config)# mib2 exaware(config-mib2)# description "core router"</pre>		
Notes	A new CLI command overrides the previous one entering in the same configuration mode.		

7.7.1.7. engine-id

Description	Configures security level in the case of version v3usm.		
Command Syntax	engine-id {free-text <i>string</i> mac-address}		
Command Mode	configuration snmp agent version v3usm configuration snmp trap target version v3usm		
Parameters Table	Name	Value	Description
	free-text <i>string</i>	ascii string of 1-27 characters length	Text Address schema engineID format: • First 4 octets: 0x8060 (enterprise id). • Fifth octet: Set to 4 to indicate the text that follows. • Last up to 27 octets: <i>string</i>.
	mac-address		Text Address schema engineID format: • First 4 octets: 0x8060 (enterprise id). • Fifth octet: Set to 3 to indicate the MAC address that follows. • Last 6 octets: the MAC address of interface mgmt. 0/0/0.
Examples	This example shows how to configure an engine identifier: <pre>exaware# config exaware(config)# snmp exaware(config-snmp)# agent network exaware(config-snmp-agent-network)# version v3usm exaware(config-snmp-agent-network-version-v3usm)# engine-id free-text alpha</pre>		

Notes	A new CLI command overrides the previous one entering in the same configuration mode.
--------------	---

7.7.1.8. ip-address

Description	Configures an IPv4 address that a SNMP agent should listen to for queries from NMS's or should send traps to.		
Command Syntax	<code>ip-address ip-address</code>		
Command Mode	configuration snmp agent configuration snmp trap target		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 address in format nnn.nnn.nnn.nnn	Agent: The NSM IPv4 to which the SNMP agent should listen. "0.0.0.0" means that the SNMP agent listens to the snmp port for all IPv4 addresses on the machine. (Default = 0.0.0.0) Target: The NSM IPv4 which the SNMP traps will be sent to.

Examples	This example shows how to configure the IP address that the SNMP agent should listen to: <pre>exaware# config exaware(config)# snmp exaware(config-snmp)# agent network exaware(config-snmp-agent-network)# ip-address 1.1.1.1 exaware(config-snmp-agent-network)# trap exaware(config-snmp-trap)# target nsm1 exaware(config-snmp-trap-target-nsm1)# ip-address 1.1.1.1 exaware(config-snmp-trap-target-nsm1)# target nsm2 exaware(config-snmp-trap-target-nsm2)# ip-address 2.2.2.2</pre>
Notes	A new CLI command overrides the previous one entering in the same configuration mode.

7.7.1.9. location

Description	Sets a value of the MIB-II location parameter. Use the no form to restore the default.		
Command Syntax	location <i>location</i> no location		
Command Mode	configuration mib2		
Parameters Table	Name	Value	Description
	<i>location</i>	String Max 255 chars	A value of the MIB-II location parameter. The default value – the empty string
Examples	This example shows how to configure the MIB-II location parameter: <pre>exaware# config exaware(config)# mib2 exaware(config-mib2)# location "New York"</pre>		
Notes	A new CLI command overrides the previous one entering in the same configuration mode.		

7.7.1.10. mib2

Description	Enters MIB-II configuration mode.
Command Syntax	mib2
Command Mode	configuration
Examples	This example shows how to enter MIB-II configuration mode: <pre>exaware# config exaware(config)# mib2 exaware(config-mib2)#</pre>

7.7.1.11. name

Description	Sets a value of the MIB-II name parameter. Use the no form to restore the default.		
Command Syntax	name <i>name</i> no name		
Command Mode	configuration mib2		
Parameters Table	Name	Value	Description
	<i>name</i>	String Max 255 chars	A value of the MIB-II name parameter. The default value – the empty string
Examples	This example shows how to configure the IP address that the SNMP agent should listen to: exaware# config exaware(config)# mib2 exaware(config-mib2)# name "Router AA"		
Notes	A new CLI command overrides the previous one entering in the same configuration mode.		

7.7.1.12. privilege-key

Description	Configures privilege key in the case of version v3usm.		
Command Syntax	privilege-key <i>key</i>		
Command Mode	configuration snmp agent version v3usm configuration snmp trap target version v3usm		
Parameters Table	Name	Value	Description
	<i>key</i>	String, Min: 12 Max: 20.	privilege key
Examples	This example shows how to configure a privilege:		
<pre>exaware# config exaware(config)# snmp exaware(config-snmp)# agent network exaware(config-snmp-agent-network)# version v3usm exaware(config-snmp-agent-network-version-v3usm)# privilege-key A&SAeV63</pre>			
Notes	The parameter is mandatory if a value of security-level is AuthPriv. A new CLI command overrides the previous one entering in the same configuration mode.		

7.7.1.13. privilege-key-protocol

Description	Configures privilege key protocol in the case of version v3usm.		
Command Syntax	privilege-key-protocol aes128 des		
Command Mode	configuration snmp agent version v3usm configuration snmp trap target version v3usm		
Parameters Table	Name	Value	Description
	aes128		The md5 authentication protocol is applied.
	des		The sha authentication protocol is applied.
Examples	This example shows how to configure a privilege key protocol:		
	<pre>exaware# config exaware(config)# snmp exaware(config-snmp-network)# agent network exaware(config-snmp-agent-network)# version v3usm exaware(config-snmp-agent-network-version-v3usm)# privilege-key-protocol des</pre>		
Notes	The parameter is mandatory if a value of security-level is AuthPriv. A new CLI command overrides the previous one entering in the same configuration mode.		

7.7.1.14. security-level

Description	Configures security level in the case of version v3usm.		
Command Syntax	Security-level noAuthNoPriv AuthNoPriv AuthPriv		
Command Mode	configuration snmp agent version v3usm configuration snmp trap target version v3usm		
Parameters Table	Name	Value	Description
	noAuthNoPriv		Without authentication and without encryption.
	AuthNoPriv		With authentication and without encryption.
	AuthPriv		With authentication and with encryption.
Examples	This example shows how to configure a privilege key protocol: <pre> exaware# config exaware(config)# snmp exaware(config-snmp)# agent network exaware(config-snmp-agent-network)# version v3usm exaware(config-snmp-agent-network-version-v3usm)# security-level AuthPriv </pre>		
Notes	A new CLI command overrides the previous one entering in the same configuration mode.		

7.7.1.15. snmp

Description	Enters SNMP configuration mode. Use the no form to remove the SNMP configuration of the all agents and trap targets.
Command Syntax	<code>snmp</code> <code>no snmp</code>
Command Mode	configuration
Examples	This example shows how to enter SNMP agent configuration mode: <code>exaware# config</code> <code>exaware(config)# snmp</code> <code>exaware(config-snmp)#</code>
Note	The Router supports two SNMP agents: SNMP Network agent and SNMP Platform agent. The first one provides access to the network information including network environment information and the second one provides access to non-network environment information. To get access to the SNMP Network agent, use the standard SNMP UDP port 161. To get access to the SNMP Platform agent, use the UDP port 1161.

7.7.1.16. status

Description	Enables or disables the SNMP agents. Use the no form to restore the default.		
Command Syntax	<pre>status {enabled disabled} no status</pre>		
Command Mode	configuration snmp		
Parameters Table	Name	Value	Description
		enabled disabled	Enables/disables the SNMP agents. Default: disabled
Examples	This example shows how to enable the SNMP agents: <pre>exaware# config exaware(config)# snmp exaware(config-snmp)# status enabled</pre>		

7.7.1.17. target

Description	Enters trap target configuration mode. Use the no form the command to remove the target configuration.		
Command Syntax	target <i>name</i> no target <i>name</i>		
Command Mode	configuration trap		
Parameters Table	Name	Value	Description
	<i>name</i>	String Max: 32 char	A unique name for the target NSM.
Examples	This example shows how to configure the IP address that the SNMP agent should listen to: <pre>exaware# config exaware# snmp exaware(config-snmp)# trap exaware(config-snmp -trap)# target nsm1 exaware(config-snmp -trap-target-nsm1)#</pre>		
Notes	Up to 32 target NSMs are supported.		

7.7.1.18. trap

Description	Enters SNMP trap configuration mode. Use the no form to remove all trap targets.
Command Syntax	trap no trap
Command Mode	configuration snmp
Examples	This example shows how to enter SNMP network trap configuration mode: <pre>exaware# config exaware(config)# snmp exaware(config-snmp)# trap exaware(config-snmp-trap)#</pre>

7.7.1.19. version

Description	Sets SNMP agent/trap version and enters SNMP agent/trap target version configuration mode		
Command Syntax	version v2c v3usm		
Command Mode	configuration snmp agent configuration snmp trap target		
Parameters Table	Name	Value	Description
	v2c		Specifies that the SNMP agent will support the v2c version.
	V3usm		Specifies that the SNMP agent will support the v3usm version.
Examples	This example shows how to configure the version of a SNMP agent: <pre>exaware# config exaware(config)# snmp exaware(config-snmp)# agent network exaware(config-snmp-agent-network)# version v2c exaware(config-snmp-agent-network-version-v2c)#</pre>		
Notes	A new CLI command overrides the previous one entering in the same configuration mode and removes the previous the version configuration.		

7.7.1.20. vrf

Description	Sets the vrf that where the SNMP agent will listen to for queries from NMS's or traps will be sent to.		
Command Syntax	<code>vrf vrf-name</code>		
Command Mode	configuration snmp agent configuration snmp trap target		
Parameters Table	Name	Value	Description
	<code>vrf-name</code>	String	vrf name from given vrf list. (Default = default)
Examples	This example shows how to configure the vrf that the SNMP agent should listen to: <pre>exaware# config exaware(config)# snmp exaware(config-snmp)# agent network exaware(config-snmp-agent-network)# vrf manag</pre>		
Notes	A new CLI command overrides the previous one entering in the same configuration mode.		

7.7.1.21. udp-port

Description	Sets a udp port which traps will be sent to.		
Command Syntax	udp-port <i>port</i>		
Command Mode	configuration snmp trap target		
Parameters Table	Name	Value	Description
	<i>port</i>	Integer in rang 1-65535	Destination UDP port in sent traps. (Default = 162)
Examples	This example shows how to configure the UDP port which traps will be sent to: <pre>exaware# config exaware(config)# snmp exaware(config-snmp)# trap exaware(config-snmp-trap)# target nsml exaware(config-snmp-trap-nsml)# udp-port 24564</pre>		
Notes	A new CLI command overrides the previous one entering in the same configuration mode.		

7.7.1.22. username

Description	Configures a username in the case of version v3usm.		
Command Syntax	username <i>name</i>		
Command Mode	configuration snmp agent version v3usm configuration snmp trap target version v3usm		
Parameters Table	Name	Value	Description
	<i>name</i>	string Max 255 characters.	Username
Examples	This example shows how to configure a username: <pre>exaware# config exaware(config)# snmp exaware(config-snmp-network)# agent network exaware(config-snmp-agent-network)# version v3usm exaware(config-snmp-agent-network-version-v3usm)# username Bob</pre>		
Notes	The parameter is mandatory if a value of security-level is AuthNoPriv or AuthPriv. A new CLI command overrides the previous one entering in the same configuration mode.		

7.7.2. SNMP Show Commands

7.7.2.1. **show snmp**

Description	Displays snmp configured parameters
Command Syntax	show snmp
Command Mode	operational
Examples	The following example shows sample output from this command:

```
exaware# show snmp
agent network
  IP address: 1.1.1.1
  VRF: default
  Version: v3usm
    Security-level: AuthPriv
    Username: bob
    Authentication-key: ASr543###
    Authentication-key-protocol: sha
    Privilege-key: AS56trFG
    Privilege-key-protocol: aes128
    Engine-id: free-text: SA54v& (0x806004534105047626)
Agent platform
  IP address: 1.1.1.1
  VRF: default
  Version: v3usm
    Security-level: AuthNoPriv
    Username: Bob
    Authentication-key: ASr543###
    Authentication-key-protocol: sha
    Engine-id: free-text: SA54v& (0x806004534105047626)
Trap
  Target nsm1
    NMS IP address: 1.1.1.1
    VRF: default
    Version: v3usm
      Security-level: AuthPriv
      Username: bob
      Authentication-key: ASr543###
      Authentication-key-protocol: sha
      Privilege-key: AS56trFG
      Privilege-key-protocol: aes128
      Engine-id: MAC address: 00:11:22:33:44:55 (0x806003001122334455)
  Target nsm2
    IP address: 2.2.2.2
    VRF: default
    UDP Port: 12544
    Version: v2c
    Community: comm
```

7.7.2.2. show mib2

Description	Displays MIB-II configured parametersr
Command Syntax	Shoe snmp
Command Mode	operational
Examples	The following example shows sample output from this command: <pre>exaware# show mib2 contact: bob@company.com description: core router location: Boston name: Router AA</pre>

7.8. Software Download Commands

7.8.1. Software Download Operational Commands

7.8.1.1. system software download

Description	Downloads a new software version into the user's home directory.		
Command Syntax	<pre>system software download [vrf vrf-name] [source-interface interface-type interface-path] protocol://[user[:password]@] remote-host[:port]//image-path target-image-filename</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>		Name of the vrf to use.
	<i>interface-type</i>	enum	Type of source interface, e.g x-eth
	<i>interface-path</i>	X/Y/Z	Interface path, e.g. 0/0/2
	<i>protocol</i>	tftp ftp http scp	Download protocol.
	<i>user</i>		URL user name. (Optional)
	<i>password</i>		User password. (Optional)
	<i>remote-host</i>		IPv4 or IPv6 address of the remote host.
	<i>port</i>		Host port number. (Optional if using non-protocol default value)

	<i>image-path</i>		The absolute path where the image will be located on the remote host.
	<i>target-image-filename</i>	Relative local path	Local file path and name.

Examples

The following examples show how to download a new software version:

```
EC-3017[2019-05-30-09:03:50](config)# run system software download vrf
management
```

```
scp://eyalo:welc0me@10.1.1.2://auto/automation/tftpboot/releases/exaware/d
evelop/b635-vbox-vdevelop.5-2019-05-30-g62d1b3ef-635.tar.gz
QoSThursday.tar.gz
```

```

** info: Start Download new image flow
** info: Extracted host from given url 10.1.1.2
** info: Validate host 10.1.1.2
** info: Host 10.1.1.2 Validated
** info: Download destination: /opt/Exawaew/images/QoSThursday.tar.gz
** info: MD5(bsc) file name QoSThursday.tar.gz.bsc
** info: MD5(bsc) url:
scp://eyalo:welc0me@10.1.1.2://auto/automation/tftpboot/releases/exaware/develop/b635-
vbox-vdevelop.5-2019-05-30-g62d1b3ef-635.tar.gz.bsc
** info: MD5(bsc) destination path /opt/Exaware/images/QoSThursday.tar.gz.bsc
** info: File is allowed to be downloaded, does not exceed 4 images in the system
** info: Download MD5(bsc) VRF: -v management SA: URL:
scp://eyalo:welc0me@10.1.1.2://auto/automation/tftpboot/releases/exaware/develop/b635-
vbox-vdevelop.5-2019-05-30-g62d1b3ef-635.tar.gz.bsc FILE_PATH:
/opt/Exaware/images/QoSThursday.tar.gz.bsc % Total % Received % Xferd Average Speed
Time Time Time Current
                                Dload Upload Total Spent Left Speed
100 33 100 33 0 0 19 0 0:00:01 0:00:01 --:--:-- 19

100 33 100 33 0 0 19 0 0:00:01 0:00:01 --:--:-- 19
** info: MD5(bsc) file download done
** info: Download image VRF: -v management SA: URL:
scp://eyalo:welc0me@10.1.1.2://auto/automation/tftpboot/releases/exaware/develop/b635-
vbox-vdevelop.5-2019-05-30-g62d1b3ef-635.tar.gz FILE_PATH:
/opt/Exaware/images/QoSThursday.tar.gz
% Total % Received % Xferd Average Speed Time Time Time Current
                                Dload Upload Total Spent Left Speed
100 688M 100 688M 0 0 6255k 0 0:01:52 0:01:52 --:--:-- 6507k
100 688M 100 688M 0 0 6255k 0 0:01:52 0:01:52 --:--:-- 6255k
** info: Image file download done
** info: Download new software image flow succeed

```


Notes

1. In addition to the requested image, the command downloads a ".bsc" file, which should reside along the required image with the same name. In the case this file is not found, the download command will fail.

The ".bsc" file contains md5 signature used for image verification.

2. If neither argument of `vrf`, `source-address`, or `source-interface` is not configured, the vrf "default" is applied and the source address will be the address of the physical interface the traffic is going out through.
3. If the `vrf` argument is configured, the source address will be the address of the physical interface the traffic is going out through.
4. If the `source-address` argument is configured, the VRF is specified by the interface which the ip address is configured on.
5. If `source-interface` is configured, the VRF and source address are taken from the interface configuration.
6. For TFTP protocol the image-path is started from the root TFTP server folder. For example, if folder `/a/tftp` is the root folder of the TFTP server and the image-path in the command is defined as `/c/image.tar.gz` then the image path on the TFTP server is `"/a/tftp/c/image.tar.gz"`.

7.8.1.2. system software install

Description	Installs a new software version on the router. In case of system image installation – reboots the system.		
Command Syntax	<code>system software install {image-filename}</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>image-filename</i>		Image name on the router, including its relative path.
Examples	The following example shows how to install a new software version: <pre>exaware# system software install image1.tar.gz</pre>		
Notes	The image is installed on the non-active flash device. In the case of installing software image (vbox-*.tar.gz) - The image will be extracted into the alternate directory structure. In case of system-image (onie-installer-*) - The install operation installs both the application software and a new ONL host image. Since this operation is carries using the ONIE boot loader, the system reboots during this command.		

7.8.1.3. system software switch

Description	Switches to the alternate software image and restarts the machine. By default, the alternate image will be loaded with the current configuration. By default, the command will auto-revert to the previous image in case of an unsuccessful CDB upgrade.		
Command Syntax	<pre>system software switch [<i>current-config</i> <i>alternate-image-config</i> <i>factory-default-config</i>]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
		<i>current-config</i>	Switch to the alternate image, restart the machine, and load the configuration which is currently stored with the current image.
		<i>alternate-image-config</i>	Switch to the alternate image, restart the machine, and load the configuration which is currently stored with the alternate image.
		<i>factory-default-config</i>	Switch to the alternate image, restart the machine, and load the factory-default configuration.

7.8.1.4. system software verify

Description	Checks integrity of the downloaded image before installation.		
Command Syntax	<code>system software verify {<i>image-filename</i>}</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>image-filename</i>		The downloaded image to be verified.
Examples	The following examples show the messages for both a corrupted and a usable image: <pre>exaware# system software verify images/broken.tar.gz The /home/images/broken.tar.gz image is corrupted!</pre> <pre>exaware# system software verify images/auto.tar.gz The /home/images/auto.tar.gz image can be installed</pre>		

7.8.2. Software Download Show Commands

7.8.2.1. show system software images

Description	Lists all the software images currently installed on the system.
Command Syntax	<code>show system software images</code>
Command Mode	operational
Examples	The following example shows how to list the software images: <pre>exaware# show system software images Images on 0/cpm0: 2.0.0 DTH011_b71 102812_2251 (front) (default, active) 2.0.0 DTH010.1_b67 102212_0133 (back) Images on 0/cpm1: 2.0.0 DTH011_b71 102812_2251 (front) (default, active) 2.0.0 DTH010.1_b67 102212_0133 (back)</pre>

7.8.2.2. show system software version

Description	Displays the software version.
Command Syntax	<code>show system software version</code>
Command Mode	operational
Examples	The following example shows how to display the software version: <pre>exaware# show system software version 2.0.0 DTH010.1_b67 102212_0133</pre>
Notes	The command can be shortened to "show version".

7.9. System Commands

7.9.1. System Configuration Commands

7.9.1.1. system banner announcement

Description	Configures system announcement message that appears after login.		
Command Syntax	<code>system banner announcement <i>message</i></code>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>message</i>	text must be in quotes	Up to 2048 characters, including spaces, horizontal tabs (\t) and newlines (\n).
Examples	<code>exaware(config)# system banner announcement "This is the announcement message"</code>		

7.9.1.2. system banner login

Description	Configures system banner login message that appears before login.		
Command Syntax	<code>system banner login <i>message</i></code>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>message</i>	text must be in quotes	Up to 2048 characters, including spaces, horizontal tabs (\t) and newlines (\n).
Examples	<code>exaware(config)# system banner login "This is login message"</code>		

7.9.1.3. system default-management-source

Description	Sets the default vrf or source address for locally generated IP traffic. This configuration is used by default and it is overridden by protocol-specific configurations and operational commands. When a vrf is configured, the source address of outgoing traffic is determined by the routing table of the particular vrf. When an interface is specified, the vrf and source address are derived from the interface configuration. The user can configure either a source interface or vrf, but not both.		
Command Syntax	<pre>system default-management-source {vrf <i>vrf-name</i> interface}</pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	string	Name of VRF to use for locally generated traffic.
	<i>interface</i>	enum	Type of source interface, currently only loopback is supported.
	<i>interface-path</i>	number	Source interface path, e.g. 0
Examples	<pre>exaware(config)# system default-management-source vrf management</pre> <pre>exaware(config)# system default-management-source loopback 0</pre>		

7.9.1.4. system hostname

Description	Specifies or modifies the hostname for the router. This command is used in system configuration sub-mode, in prompts, and default configuration filenames.		
Command Syntax	system hostname <i>name</i>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>name</i>	String: max 63 characters, restricted to alphanumeric, dot and dash characters	Used in prompts and default configuration filenames.
Examples	<pre>exaware(config)# system hostname Box101 exaware(config)# commit Commit complete.</pre>		

7.9.1.5. system run-app-vm

Description	Run the VM file.		
Command Syntax	<code>system run-app-vm <i>filename</i></code> <code>no system run-app-vm</code>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>filename</i>	Non-empty string of up to 128 characters	The name of file containing the 3rd party VM image.
Examples	This example shows how to configure the time zone. <pre>exaware# config exaware(config)# system run-app-vm vm1.im</pre>		

7.9.1.6. system time-zone

Description	Configures the time zone.		
Command Syntax	<code>system time-zone <i>zone</i></code>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>zone</i>	time zone	Time zone, as defined by GMT or geographic location. (Default = GMT)
Examples	This example shows how to configure the time zone. <pre>exaware# config exaware(config)# system time-zone Asia/Jerusalem</pre> or <pre>exaware(config)# system time-zone GMT+2</pre>		

7.9.2. System Operational Commands

7.9.2.1. clear system alarm history

Description	Clears the history of alarms in the system.
Command Syntax	<code>clear system alarm history</code>
Command Mode	operational
Parameters	N/A
Examples	<code>exaware# clear system alarm history</code>

7.9.2.2. **clear system to-us-traffic queues-statistics**

Description	Clears the queues-statistics for the CPU ports at the BCM.
Command Syntax	<code>Clear system to-us-traffic queues-statistics</code>
Command Mode	Operational
Examples	The following examples show how to use this command: <pre>exaware# clear systemto-us-traffic statistics</pre>

7.9.2.3. system date

Description	Sets the current date and time.		
Command Syntax	<code>system date <i>date time</i></code>		
Command Mode	Operational		
Parameters Table	Name	Values	Description
	<i>date</i>	string	Date string in format of YYYY-MM-DD.
	<i>time</i>	string	Time string in the format of HH:MM:SS.
Examples	<code>exaware# system date 2012-04-04 10:30:00</code>		

7.9.2.4. system mtu-shg-selector

Description	Configures whether the system supports MTU or 1:N Split Horizon Groups.
Command Syntax	<code>system mtu-shg-selector mtu shg</code>
Command Mode	operational
Examples	<code>exaware# system mtu=shg-selector mtu</code>
Notes	• The action requires system reboot. • Setting the value to mtu is not allowed if there is SHG configured. • Setting the value to shg is not allowed if there is an interfaces with mtu configured. • The default value is shg.

7.9.2.5. system recovery reset factory-default

Description	Performs a system restart after resetting the configuration to system default. All rollback configurations will be deleted as well.
Command Syntax	<code>system recovery reset factory-default</code>
Command Mode	operational
Parameters Table	N/A
Examples	<pre>exaware# system recovery reset factory-default</pre> The command will permanently remove configuration rollbacks and revert to factory-default configuration, followed by system restart. Are you sure?[yes NO]
Notes	Useful in cases where the system cannot be initialized correctly because of a faulty configuration.

7.9.2.6. system recovery reset snapshot

Description	Applies an existing snapshot. A configuration snapshot can be used to reset the system to a well-known state; after the reset, system is rebooted.		
Command Syntax	system recovery reset snapshot <type> <location> <number>		
Command Mode	Operational		
Parameters Table	Name	Values	Description
	type	enum {configuration}	Snapshot type ("configuration" or "all").
	location	enum {local}	Snapshot location (only "local").
	number	1-5	Snapshot index.
Examples	<pre>exaware# system recovery reset snapshot configuration local 3</pre> The command will permanently remove configuration rollbacks and revert to the configuration snapshot, followed by system restart. Are you sure? [yes,NO]		

7.9.2.7. system restart

Description	Restarts the system. (Any pending configuration is lost upon restart.)		
Command Syntax	system restart [hard soft]		
Command Mode	operational		
Parameters Table	Name	Values	Description
	hard		Performs HW reset including NP.
	soft		Performs SW restart only without HW reset (Default).
Parameters Table	N/A		
Examples	<pre>exaware# system restart hard Restart the system? [yes,no] yes Broadcast message from root (pts/1) (Wed Jun 12 11:37:52 2002): The system is going down for reboot NOW! exaware# Connection to 10.10.10.10 closed by remote host. Connection to 10.10.10.10 closed.</pre>		

7.9.2.8. system snapshot create

Description	Saves a new configuration snapshot to the local directory. A configuration snapshot consists of the complete set of CDB files, which can be later used for system recovery.		
Command Syntax	system snapshot create [<type>] [<location>] <description>		
Command Mode	operational		
Parameters Table	Name	Values	Description
	type	enum {configuration}	Snapshot type ("configuration" or "all"). (Default = all)
	location	enum {local}	Snapshot location (only "local"). (Default = local)
	description	text	Snapshot label.
Examples	exaware# system snapshot create configuration local "Snapshot description" Snapshot saved.		

7.9.2.9. system tech-support

Description	Generates a TAR file for technical support. You can specify either detailed output or brief output (default).		
Command Syntax	<code>system tech-support <i>file-prefix</i> [basic brief detailed]</code>		
Command Mode	Operational		
Parameters Table	Name	Values	Description
	<i>file-prefix</i>	string	Identifying string to prefix to the TAR filename.
Examples	The following example shows how to generate a TAR file for technical support: <pre>exaware# system tech-support tech_sprt</pre>		

7.9.3. System Show Commands

7.9.3.1. show system alarm

Description	Displays the active alarms in the system.
Command Syntax	show system alarm
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command:
exaware# show system alarm	
TIMESTAMP SEVERITY DESCRIPTION	
2012-08-07 00:46:06 Major Egress HW resource leakage due to 0/cpm0	
2012-08-06 18:24:36 Major 0/fan1 Removed	

7.9.3.2. show system alarm configuration-mismatch

Description	Displays only the active configuration mismatch alarms in the system.
Command Syntax	show system alarm configuration-mismatch
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command: exaware# show system alarm configuration-mismatch TIMESTAMP SEVERITY DESCRIPTION ----- 2015-01-13 11:41:50 Major Misconfiguration - '...', conflicts with ...

7.9.3.3. show system alarm history

Description	Displays the history of alarms in the system.
Command Syntax	show system alarm history
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command: exaware# show system alarm history TIME STATUS SEVERITY DESCRIPTION 2012-08-07 00:46:06 Cleared Minor x-eth 0/1/1 Interface down 2012-08-07 00:46:06 Raised Major Egress HW resource leakage due to 0/cpm0 2012-08-06 20:30:36 Raised Major x-eth 0/1/1 Interface down 2012-08-06 18:24:36 Raised Major 0/fan1 Removed 2012-08-06 07:41:04 Cleared Minor System initialization in progress 2012-08-06 07:00:04 Raised Minor System initialization in progress

7.9.3.4. show system build-parameters

Description	Displays the system build-parameters.
Command Syntax	show system build-parameters
Command Mode	operational
Parameters Table	N/A
Examples	<pre>exaware# show system build-parameters Platform build id = develop/b60 Application build id = release/2.0.0@710c8f8 Zebos release = 7.8.1 Confd release = 3.9.3.3 User name = builds slave Date = 101612_1826 Build = DTH09.2_b62</pre>

7.9.3.5. show system date

Description	Displays the current date and time.
Command Syntax	<code>show system date</code>
Command Mode	operational
Parameters Table	N/A
Examples	<code>exaware# show system date</code> 2012-04-04 10:30:00

7.9.3.6. show system description

Description	Displays the system description.
Command Syntax	<code>show system description</code>
Command Mode	operational
Parameters Table	N/A
Examples	<code>exaware# show system description</code> "Exaware router"

7.9.3.7. show system hardware [not supported in release 8.0.0]

Description	Shows the system hardware inventory in the router.		
Command Syntax	show system hardware [brief detail]		
Command Mode	Operational		
Parameters Table	Name	Values	Description
		brief	Shows a brief description of the hardware modules. (Default)
		detail	Shows a detailed description of the hardware modules.
Examples	The following example shows sample output from this command.		
<pre>exaware# show system hardware detail System Information: Product Name: 5916-54XK-O-AC-F Part Number: FP1ZZ56540A9A Serial Number: 591654XK1828004 MAC: 3c:2c:99:f3:a0:40 MAC Range: 256 Manufacturer: Accton Manufacture Date: 07/17/2018 12:52:44 Vendor: Edgecore Platform Name: x86_64-accton_as5916_54xk-r1 Label Revision: R02A Country Code: TW Diag Version: b0.0.0.9 ONIE Version: 2017.08.00.03 Fans: Name: Fan 1 Description: Chassis Fan - 1 State: Present Status: Running RPM: 8700 Speed: 34% Airflow: Front-to-Back Name: Fan 2 Description: Chassis Fan - 2 State: Present Status: Running</pre>			

```
RPM: 8700
Speed: 34%
Airflow: Front-to-Back
Name: Fan 3
Description: Chassis Fan - 3
State: Present
Status: Running
RPM: 8700
Speed: 34%
Airflow: Front-to-Back
Name: Fan 4
Description: Chassis Fan - 4
State: Present
Status: Running
RPM: 8600
Speed: 33%
Airflow: Front-to-Back
Name: Fan 5
Description: Chassis Fan - 5
State: Present
Status: Running
RPM: 8700
Speed: 34%
Airflow: Front-to-Back
Name: Fan 6
Description: Chassis Fan - 6
State: Present
Status: Running
RPM: 8800
Speed: 34%
Airflow: Front-to-Back
Thermals:
Name: Thermal 1
Description: CPU Core
Status: Functional
Temperature: 39.0 C
Name: Thermal 2
Description: LM75-1-48
Status: Functional
Temperature: 28.0 C
Name: Thermal 3
Description: LM75-2-49
Status: Functional
Temperature: 25.0 C
Name: Thermal 4
Description: LM75-3-4A
Status: Functional
Temperature: 24.0 C
Name: Thermal 5
Description: LM75-3-4B
Status: Functional
Temperature: 23.0 C
PSUs:
```

```
Name: PSU 1
Description: PSU-1
State: Present
Status: Running
Model: YM-2651Y?
Type: AC
Vout: 12.1
Iout: 11.5
Pout: 139.0
Fans:
    Name: Fan 7
    Description: PSU 1 - Fan 1
    State: Present
    Status: Running
    RPM: 6800
    Speed: 26%
    Airflow: Front-to-Back
Thermals:
    Name: Thermal 6
    Description: PSU-1 Thermal Sensor 1
    Status: Functional
    Temperature: 25.0 C
Name: PSU 2
Description: PSU-2
State: Present
Status: Unplugged
LEDs:
    Name: LED 1
    Description: Chassis LED 1 (LOC LED)
    State: Present
    Mode: OFF
    Name: LED 2
    Description: Chassis LED 2 (DIAG LED)
    State: Present
    Mode: OFF
    Name: LED 3
    Description: Chassis LED 3 (PSU1 LED)
    State: Present
    Mode: AUTO
    Name: LED 4
    Description: Chassis LED 4 (PSU2 LED)
    State: Present
    Mode: AUTO
    Name: LED 5
    Description: Chassis LED 5 (FAN LED)
    State: Present
    Mode: AUTO
```

7.9.3.8. show system memory

Description	Displays the system memory usage.
Command Syntax	show system memory
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command:

```

exaware# show system memory
Processes on 0/cpm0
-----
            total      used      free      shared      buffers      cached
Mem:      2056276      690024      1366252          0          0      210864
-/+ buffers/cache:  479160      1577116

Processes on 0/cpm1
-----
            total      used      free      shared      buffers      cached
Mem:      2056276      690024      1366252          0          0      210864
-/+ buffers/cache:  479160      1577116

Processes on 0/lc1
-----
            total      used      free      shared      buffers      cached
Mem:      2056276      690024      1366252          0          0      210864
-/+ buffers/cache:  479160      1577116

```

7.9.3.9. show system mtu-shg-selector

Description	Displays the mtu-shg-selector current mode.
Command Syntax	Show system mtu-shg-selector
Command Mode	operational
Examples	exaware# show system mtu-shg-selector

7.9.3.10. show system processes

Description	Displays the status of running system processes.										
Command Syntax	show system processes [top] [<i>number</i> all]										
Command Mode	Operational										
Parameters Table	Name	Values				Description					
	<i>number</i>	1-1000000				Number of processes to show. (Default = 25)					
Examples	The following examples show sample output from this command:										
<pre>exaware# show system processes Processes on 0/cpm0 ----- : USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND root 2969 0.0 0.0 5496 1084 ? Ss Jun17 0:00 /usr/sbin/sshd root 2991 0.0 0.0 3808 876 ? Ss Jun17 0:00 /usr/sbin/inetd root 3021 0.0 0.0 4008 1116 ? Ss Jun17 0:03 /usr/sbin/cron root 3161 0.0 0.1 7696 2892 ? S Jun17 0:00 /exaware/sw_a/bin/sysi root 3176 0.0 0.0 7696 1992 ? S Jun17 0:00 /exaware/sw_a/bin/sysi root 3177 0.0 0.2 76464 6076 ? Sl Jun17 1:54 /exaware/sw_a/bin/sysa : Processes on 0/cpm1 ----- : USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND root 2969 0.0 0.0 5496 1084 ? Ss Jun17 0:00 /usr/sbin/sshd root 2991 0.0 0.0 3808 876 ? Ss Jun17 0:00 /usr/sbin/inetd root 3021 0.0 0.0 4008 1116 ? Ss Jun17 0:03 /usr/sbin/cron root 3161 0.0 0.1 7696 2892 ? S Jun17 0:00 /exaware/sw_a/bin/sysi root 3176 0.0 0.0 7696 1992 ? S Jun17 0:00 /exaware/sw_a/bin/sysi</pre>											

```

root      3177  0.0  0.2  76464  6076 ?          Sl   Jun17   1:54
/exaware/sw_a/bin/sysa
:
Processes on 0/lc0
-----
:
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root      2969  0.0  0.0   5496   1084 ?        Ss   Jun17    0:00
/usr/sbin/sshd
root      2991  0.0  0.0   3808    876 ?        Ss   Jun17    0:00
/usr/sbin/inetd
root      3021  0.0  0.0   4008   1116 ?        Ss   Jun17    0:03
/usr/sbin/cron
root      3161  0.0  0.1   7696   2892 ?        S    Jun17    0:00
/exaware/sw_a/bin/sysi
root      3176  0.0  0.0   7696   1992 ?        S    Jun17    0:00
/exaware/sw_a/bin/sysi
root      3177  0.0  0.2  76464  6076 ?          Sl   Jun17    1:54
/exaware/sw_a/bin/sysa
:

exaware# show system processes top
Top processes on 0/lc2
-----
top - 01:16:21 up 2 days, 1:13, 1 user, load average: 0.05, 0.02, 0.00
Tasks: 133 total,  3 running, 130 sleeping,  0 stopped,  0 zombie
Cpu(s): 1.9%us, 0.4%sy, 0.0%ni, 97.2%id, 0.3%wa, 0.0%hi, 0.1%si, 0.0%st
Mem:   2056276k total,  690024k used, 1366252k free,      0k buffers
Swap:      0k total,      0k used,      0k free,  210568k cached

PID USER      PR  NI  VIRT  RES  SHR S  %CPU  %MEM    TIME+  COMMAND
  1 root        20   0   1984   692   592 S   0.0   0.0    0:01.27 init
  2 root        20   0     0     0     0 S   0.0   0.0    0:00.00 kthreadd
  3 root        RT   0     0     0     0 S   0.0   0.0    0:00.00 migration/0
  4 root        20   0     0     0     0 S   0.0   0.0    0:00.00 ksoftirqd/0
  5 root        RT   0     0     0     0 S   0.0   0.0    0:00.00 watchdog/0
  6 root        20   0     0     0     0 S   0.0   0.0    0:00.01 events/0
  7 root        20   0     0     0     0 S   0.0   0.0    0:00.00 khelper
 12 root        20   0     0     0     0 S   0.0   0.0    0:00.00 async/mgr
MORE...
```

7.9.3.11. show system snapshot

Description	Shows the list of existing snapshots, filtered by type, if requested.		
Command Syntax	show system snapshot [<type>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	type	enum {configuration}	Snapshot type ("configuration" or "all") (Default = all)
Examples	The following example shows sample output from this command:		
<pre>exaware# show system snapshot TYPE INDEX LOCATION USER TIMESTAMP DESCRIPTION ----- config 1 local admin 2013-05-10::23:59:59 desc</pre>			

7.9.3.12. show system to-us-traffic queues-statistics brief

Description	Shows a summary of queues statistics collected from all ingress VOQs towards the CPU ports.																																																																																																							
Command Syntax	show system to-us-traffic queues-statistics brief																																																																																																							
Command Mode	Operational																																																																																																							
Examples	The following example shows sample output from this command:																																																																																																							
exaware# show system to-us-traffic queues-statistics brief																																																																																																								
Time since last clear statistics: 1hour 14min 54sec																																																																																																								
<table><thead><tr><th rowspan="2">TC</th><th rowspan="2">BYTES</th><th colspan="2">FORWARD</th><th colspan="2">CPU</th><th colspan="2">DROP</th><th rowspan="2">PKTS</th><th rowspan="2">AVERAGE RATE[pps]</th></tr><tr><th>AVERAGE RATE[bps]</th><th>PKTS</th><th>AVERAGE RATE[pps]</th><th>BYTES</th><th>AVERAGE RATE[bps]</th></tr></thead><tbody><tr><td>EF1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>EF2</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>AF1</td><td>1652800</td><td>536.75 k</td><td>208281</td><td>5.65 k</td><td>15100</td><td>332.77 k</td><td>17546</td><td>3587</td><td></td></tr><tr><td>AF2</td><td>376448</td><td>1.01 M</td><td>470560</td><td>11.56 k</td><td>1662348</td><td>57.6 k</td><td>17590</td><td>656</td><td></td></tr><tr><td>AF3</td><td>43825020</td><td>11.9 G</td><td>461396</td><td>124.14 M</td><td>0</td><td>0</td><td>0</td><td>0</td><td></td></tr><tr><td>AF4</td><td>1000000</td><td>9000</td><td>16000</td><td>200</td><td>400</td><td>80</td><td>5</td><td>1</td><td></td></tr><tr><td>AF5</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td></td></tr><tr><td>BE</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td></td></tr></tbody></table>										TC	BYTES	FORWARD		CPU		DROP		PKTS	AVERAGE RATE[pps]	AVERAGE RATE[bps]	PKTS	AVERAGE RATE[pps]	BYTES	AVERAGE RATE[bps]	EF1	0	0	0	0	0	0	0	0	0	EF2	0	0	0	0	0	0	0	0	0	AF1	1652800	536.75 k	208281	5.65 k	15100	332.77 k	17546	3587		AF2	376448	1.01 M	470560	11.56 k	1662348	57.6 k	17590	656		AF3	43825020	11.9 G	461396	124.14 M	0	0	0	0		AF4	1000000	9000	16000	200	400	80	5	1		AF5	0	0	0	0	0	0	0	0		BE	0	0	0	0	0	0	0	0	
TC	BYTES	FORWARD		CPU		DROP		PKTS	AVERAGE RATE[pps]																																																																																															
		AVERAGE RATE[bps]	PKTS	AVERAGE RATE[pps]	BYTES	AVERAGE RATE[bps]																																																																																																		
EF1	0	0	0	0	0	0	0	0	0																																																																																															
EF2	0	0	0	0	0	0	0	0	0																																																																																															
AF1	1652800	536.75 k	208281	5.65 k	15100	332.77 k	17546	3587																																																																																																
AF2	376448	1.01 M	470560	11.56 k	1662348	57.6 k	17590	656																																																																																																
AF3	43825020	11.9 G	461396	124.14 M	0	0	0	0																																																																																																
AF4	1000000	9000	16000	200	400	80	5	1																																																																																																
AF5	0	0	0	0	0	0	0	0																																																																																																
BE	0	0	0	0	0	0	0	0																																																																																																
Notes	<i>Time since last clear statistics</i> is the time passed since the interface statistics were cleared last. <i>Average rate</i> is a moving window average of the last 5 seconds.																																																																																																							

7.9.3.13. show system to-us-traffic queues-statistics detail

Description	Shows the statistics of queues statistics collected from all ingress VOQs towards the CPU ports.								
Command Syntax	show system to-us-traffic queues-statistics detail								
Command Mode	Operational								
Examples	The following example shows sample output from this command:								
exaware# show system to-us-traffic queues-statistics detail									
Time since last clear statistics: 1hour 14min 54sec									
CPU 1									
TC	BYTES	FORWARD AVERAGE RATE[bps]	PKTS	AVERAGE RATE[pps]	BYTES	DROP AVERAGE RATE[bps]	PKTS	AVERAGE RATE[pps]	

EF1	0	0	0	0	0	0	0		
EF2	0	0	0	0	0	0	0		
AF1	1652800	536.75 k	208281	5.65 k	15100	332.77 k	17546	3587	
AF2	376448	1.01 M	470560	11.56 k	1662348	57.6 k	17590	656	
AF3	43825020	11.9 G	461396	124.14 M	0	0	0	0	
AF4	1000000	9000	16000	200	400	80	5	1	
AF5	0	0	0	0	0	0	0	0	
BE	0	0	0	0	0	0	0	0	
CPU 2									
TC	BYTES	FORWARD AVERAGE RATE[bps]	PKTS	AVERAGE RATE[pps]	BYTES	DROP AVERAGE RATE[bps]	PKTS	AVERAGE RATE[pps]	

EF1	0	0	0	0	0	0	0	0	
EF2	0	0	0	0	0	0	0	0	
AF1	1652800	536.75 k	208281	5.65 k	15100	332.77 k	17546	3587	
AF2	376448	1.01 M	470560	11.56 k	1662348	57.6 k	17590	656	
AF3	43825020	11.9 G	461396	124.14 M	0	0	0	0	
AF4	1000000	9000	16000	200	400	80	5	1	
AF5	0	0	0	0	0	0	0	0	
BE	0	0	0	0	0	0	0	0	

Notes	• <i>Time since last clear statistics</i> is the time passed since the interface statistics were cleared last. • <i>Average rate</i> is a moving window average of the last 5 seconds.
--------------	--

7.9.3.14. show system uptime

Description	Displays the system uptime.
Command Syntax	<code>show system uptime</code>
Command Mode	operational
Parameters Table	N/A
Examples	<code>exaware# show system uptime</code> System uptime is 2 days, 20 hours, 19 minutes, 0 seconds

7.9.4. System Module Commands

7.9.4.1. System Module Show Commands

7.9.4.1.1.1. show system module

Description	Shows the system module status.				
Command Syntax	show system module				
Command Mode	operational				
Parameters Table	N/A				
Examples	The following example shows sample output from this command:				
exaware# show system module					
System Date: 2012-08-09 12:59:05					
HA Role: A - active, S - standby					
SLOT INDEX	TYPE	ADMIN STATE	OPER STATE	STATE DURATION	ADDITIONAL INFO

0/lc1	20X10Gigab	Active	Card-Ready	3d5h28m12s	
0/lc2	20X10Gigab	Active	Calibrating	2h04m27s	
0/lc3	20X10Gigab	Active	Not-Calibrated	2h04m27s	
0/lc4	20X10Gigab	Active	Partially-Calibrated	2h04m27s	
0/cpm0	N/A	Active	A Card-Ready	3d5h32m29s	
0/cpm1	N/A	Active	S Card-Ready	3d5h32m29s	

Notes	Operating States: 1. Calibrating - LC configured successfully and waiting for optical-interconnect calibration to finish. 2. Not-calibrated - A problem in optical-interconnect connectivity causing all LC interfaces to be down. 3. Partially-calibrated - A problem in optical-interconnect connectivity causing some interfaces in the LC to be down (one engine). In the event that there are active alerts in the system, then the following message will be displayed: Note: There are active alerts. Run 'show system alerts' for more info.
--------------	---

8. MPLS Commands

8.1. LDP Commands

8.1.1. LDP Configuration Commands

8.1.1.1. admin-state

Description	Administratively enable/disable communication with a specific targeted peer by enabling/disabling targeted hello exchange.		
Command Syntax	admin-state [<i>up</i> <i>down</i>] no admin-state		
Command Mode	configuration mpls ldp targeted-peer		
Parameters Table	Name	Value	Description
	<i>status</i>	<i>up</i>	Enable communication with targeted peer. (Default)
		<i>down</i>	Disable communication with targeted peer.
Examples	This example shows how to disable communication with a targeted peer: exaware# config exaware(config)# mpls ldp exaware(config-mpls-ldp-default)# targeted-peer 10.10.10.20 exaware(config-mpls-ldp-default-targeted-peer-10.10.10.20)# admin-state down		

8.1.1.2. af-ipv4

Description	Enters the address-family ipv4 mode. Use the no form of the command to remove all af-ipv4 configurations. All relevant parameters will revert to their default values.
Command Syntax	<pre>af-ipv4 no af-ipv4</pre>
Command Mode	configuration mpls ldp <instance> interface-<interface-id>
Parameters	N/A
Examples	This example shows how to enter the address-family ipv4 mode: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# interface x-eth 0/0/2 exaware(config-mpls-ldp-interface-x-eth-0/0/2)# af-ipv4 exaware(config-interface-x-eth-0/0/2-af-ipv4)#</pre>
Notes	The IPv4 address family should be enabled under each LDP interface. IPv4 is the only address family currently supported.

8.1.1.3. discovery-hello-holdtime (timers)

Description	Configures the discovery hello hold time for LDP protocol globally or for a specific interface. Use the no form of the command to revert to the default discovery hello hold time (15 seconds).		
Command Syntax	<pre>discovery-hello-holdtime <i>seconds</i> no discovery-hello-holdtime</pre>		
Command Mode	<pre>configuration mpls ldp <instance> timers configuration mpls ldp <instance> interface <interface-id> timers</pre>		
Parameters Table	Name	Value	Description
	<i>seconds</i>	3-65535	Discovery hello hold time in seconds. This is the length of time an LSR maintains its record of hellos from a peer on not receiving another hello from that peer. (Default = 15 seconds)
Examples	This example shows how to set the discovery hello hold time to 300 seconds for a specific interface: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# interface x-eth 0/0/0 exaware(config-mpls-ldp-default-interface-x-eth-0/0/0)# timers exaware(config-interface-x-eth-0/0/0-timers)# discovery- hello-holdtime 300</pre> This example shows how to set the discovery hello hold time to 300 seconds globally: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# timers exaware(config-mpls-ldp-default-timers)# discovery- hello-holdtime 300</pre>		

Notes

Discovery-hello-holdtime for an interface should be at least three times greater than ***discovery-hello-interval*** (on page 411).

If the discovery-hello-interval is not configured for the specific interface, the discovery-hello-holdtime on the interface should be at least three times greater than the global discovery-hello-interval.

8.1.1.4. discovery-hello-interval (timers)

Description	Configures the interval after which the multicast discovery hello messages are sent, either for LDP protocol globally or for a specific interface. Use the no form of the command to revert to the default discovery hello interval (5 seconds).		
Command Syntax	<pre>discovery-hello-interval seconds no discovery-hello-interval</pre>		
Command Mode	<pre>configuration mpls ldp <instance> timers configuration mpls ldp <instance> interface <interface-id> timers</pre>		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-21845	Discovery hello interval timer in seconds. (Default = 5 seconds)
Examples	This example shows how to set the discovery hello interval timer to 120 seconds for a specific interface: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp)# interface x-eth 0/0/2 exaware(config-interface-x-eth-0/0/2)# timers exaware(config-interface-x-eth-0/0/2-timers)# discovery-hello-interval 120</pre> This example shows how to set the discovery hello interval timer to 120 seconds globally: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# timers exaware(config-mpls-ldp-default-timers)# discovery-hello-interval 120</pre>		

Notes	Discovery-hello-interval should not be more than 1/3 of the <i>discovery-hello-holdtime</i> (on page 408). If the discovery-hello-holdtime is not configured for the specific interface, the discovery-hello-interval on the interface should not be more than 1/3 the global discovery-hello-holdtime.
--------------	--

8.1.1.5. discovery-hellos

Description	Configures discovery hello exchange for LDP protocol globally or for a specific interface. Use the no form of the command to return to the default (enabled) discovery hello exchange.		
Command Syntax	<pre>discovery-hellos [<i>enable</i> <i>disable</i>]</pre> <pre>no discovery-hellos</pre>		
Command Mode	<pre>configuration mpls ldp <instance></pre> <pre>configuration mpls ldp <instance> interface <interface-id></pre>		
Parameters Table	Name	Value	Description
		<i>enable</i>	Enable multicast hello exchange. (Default)
		<i>disable</i>	Disable multicast hello exchange.
Examples	This example shows how to disable multicast hello exchange for a specific interface: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# interface x-eth 0/0/2 exaware(config-interface-x-eth-0/0/2)# discovery-hellos disable</pre> This example shows how to disable multicast hello exchange globally: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# discovery-hellos disable</pre>		

8.1.1.6. entropy-label (ldp)

Description	To enable Entropy label on the egress LDP LSP router, use entropy-label CLI command. Use the no form of this command, to disable Entropy label (default).		
Command Syntax	entropy-label enable disable no entropy-label		
Command Mode	configuration mpls ldp <instance>		
Parameters Table	Name	Value	Description
	enable		The egress LDP LSP router supports the Entropy label and informs upstream LDP LSP routers about it.
	disable		The egress LDP LSP router does not support the Entropy label.
Examples	This example shows how to use this command: <pre>exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# entropy-label enable</pre>		

8.1.1.7. explicit-null

Description	Configures LSR to advertise explicit null labels instead of implicit null labels. Use the no form of the command or the explicit-null disable to revert to the default (implicit null labels).		
Command Syntax	explicit-null [<i>enable</i> <i>disable</i>] no explicit-null		
Command Mode	configuration mpls ldp <instance>		
Parameters Table	Name	Value	Description
		<i>enable</i>	Set LSR to advertise explicit null labels.
		<i>disable</i>	Set LSR to advertise implicit null labels. (Default)
Examples	This example shows how to enable advertising explicit null labels: exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# explicit-null enable		

8.1.1.8. hello-holdtime

Description	Configures timeout before closing an adjacency with a targeted peer. The command configures the hold time for a specific targeted peer. Use the no form of the command to revert to the globally configured or default value.		
Command Syntax	<pre>hello-holdtime seconds</pre> <pre>no hello-holdtime</pre>		
Command Mode	configuration mpls ldp targeted-peer timers		
Parameters Table	Name	Value	Description
	seconds	1-65535	Targeted hello hold time value. (Default = 45)
Examples	This example shows how to set the hello hold time for a specific targeted peer to 180 seconds: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# targeted-peer 10.10.10.20 exaware(config-mpls-ldp-default-targeted-peer-10.10.10.20)# timers hello-holdtime 180</pre>		
Notes	When a targeted peer configuration exists, it overrides the global configuration. The hello-holdtime value should be at least 3 times greater than the hello-interval value.		

8.1.1.9. hello-interval

Description	Configures the interval for sending unicast hello messages to targeted peers. This command configures the hello interval for a specific targeted peer. Use the no form of the command to revert to the globally configured value.		
Command Syntax	<pre>hello-interval seconds</pre> <pre>no hello-interval</pre>		
Command Mode	configuration mpls ldp targeted-peer timers		
Parameters Table	Name	Value	Description
	<i>seconds</i>	3-21845	Targeted hello interval value. (Default = 15 seconds)
Examples	This example shows how to set the hello interval to 120 seconds for a specific peer: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# targeted-peer 1.1.1.1 exaware(config-mpls-ldp-default-targeted-peer-1.1.1.1)# timers exaware(config-targeted-peer-1.1.1.1-timers)# hello- interval 120</pre>		
Notes	When a targeted peer configuration exists, it overrides the global configuration. The hello-interval value should not be more than 1/3 of the hello-holdtime value.		

8.1.1.10. igp-synchronization-delay

Description	Configures the time LDP waits upon LDP convergence completion before notifying the IGP. Use the no form of the command to remove the LDP-related configuration.		
Command Syntax	<pre>igp-synchronization-delay <i>seconds</i> no igp-synchronization-delay</pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>seconds</i>	5-300	IGP-synchronization delay. (Default = 10)
Examples	This example shows how to enter the LDP configuration sub-mode: <pre>exaware# config exaware(config)# igp-synchronization-delay 50</pre>		

8.1.1.11. interface

Description	Enters the configuration mode for the specified interface in LDP protocol sub-mode. Use the no form of the command to remove the interface from LDP protocol and to remove all interface-related configuration. All relevant parameters will revert to their default values.		
Command Syntax	<pre>interface <i>type instance</i> no interface <i>type instance</i></pre>		
Command Mode	configuration mpls-ldp		
Parameters Table	Name	Value	Description
	<i>type</i>	Interface type	
	<i>instance</i>	0/0/port	Interface index
Examples	This example shows how to enter the interface configuration sub-mode for the specified interface: <pre>exaware# config exaware(config)# mpls ldp exaware(config-mpls-ldp-default)# interface x-eth 0/0/2 exaware(config-mpls-ldp-default-interface-x-eth-0/0/2)# af-ipv4 exaware(config-interface-x-eth-0/0/2-af-ipv4)#</pre>		
Notes	The IPv4 address family should be enabled under each LDP interface. IPv4 is the only address family currently supported.		

8.1.1.12. label-allocation

Description	Sets the policy to specify the IGP routes which are allocated labels by the router (instead of the default MPLS LDP behavior of allocation of labels for all IGP routes). Use the no form of the command to remove the label-allocation configuration.		
Command Syntax	<pre>label-allocation policy <i>policy-name</i> [<i>additional-policy-name</i>]</pre> <pre>no label-allocation</pre>		
Command Mode	configuration mpls ldp <instance>		
Parameters Table	Name	Value	Description
	<i>policy-name</i>	policy-name	Apply policy to filter labels allocated locally.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# label-allocation policy LDP_filter1</pre>		
Notes	Supported conditions: prefix. Supported actions: permit, deny, default-permit, default-deny. There is an implicit deny at the end of the policy. For LDP Policy information, see LDP Policy Configuration Commands: Routing Conditions.		

8.1.1.13. ldp-synchronization

Description	Enables LDP synchronization with IGP protocol by temporarily setting the link metric to the highest value. The command enables the LDP-IGP synchronization globally for IGP protocol (OSPF/ISIS) or per OSPF area or per OSPF interface or per level in ISIS interface. Use the no form of the command to disable LDP synchronization.		
Command Syntax	<pre>ldp-synchronization [enable disable] no ldp-synchronization</pre>		
Command Mode	configuration routing isis configuration routing isis interface level configuration routing ospf vrf configuration routing ospf vrf area configuration routing ospf vrf area interface		
Parameters Table	Name	Value	Description
		<i>enable</i>	Enable LDP synchronization.
		<i>disable</i>	Disable LDP synchronization. (Default)
Examples	This example shows how to enable LDP synchronization: <pre>exaware# config exaware(config)# routing ospf ospf1 vrf default exaware(config-routing-ospf-ospf1-vrf-default)# ldp-synchronization enable</pre>		

8.1.1.14. ldp-synchronization-holdtime

Description	Configures the time the Exaware router waits to allow LDP to synchronize with IGP before advertising the real link metric, unless the notification about convergence completion is accepted first. The command configures the holdtime globally for IGP protocol (OSPF/ISIS) or per OSPF area or per OSPF interface or per level in ISIS interface. A holdtime value of 0 means infinity. Use the no form of the command to set the LDP synchronization holdtime to the default value of 0 seconds.		
Command Syntax	<pre>ldp-synchronization-holdtime <i>seconds</i> no ldp-synchronization-holdtime</pre>		
Command Mode	<pre>configuration routing isis configuration routing isis interface level configuration routing ospf vrf configuration routing ospf vrf area configuration routing ospf vrf area interface</pre>		
Parameters Table	Name	Value	Description
	<i>seconds</i>	0-65535	LDP synchronization holdtime. (Default = 0)
Examples	This example shows how to set the LDP synchronization holdtime: <pre>exaware# config exaware(config)# routing ospf ospf1 vrf default exaware(config-routing-ospf-ospf1-vrf-default)# ldp-synchronization-holdtime 100</pre>		
Notes	A holdtime value of 0 means infinity.		

8.1.1.15. local-address

Description	Configures IPv4 local address for the session. Use the no form of the command to remove the local address configuration. Local address is the source address for the ldp session (usually a loopback address).		
Command Syntax	<pre>local-address ipv4 ip-address no local-address</pre>		
Command Mode	configuration mpls ldp <instance>		
Parameters Table	Name	Value	Description
	ip-address	IP address	Local IP address for the session.
Examples	This example shows how to configure the local IP address: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# local-address ipv4 10.10.10.10</pre>		
Notes	It is recommended that the LDP local address entry is explicitly configured in order to avoid inconsistency in implicit selection.		

8.1.1.16. mpls ldp

Description	Enters the LDP configuration sub-mode.		
	Use the no form of the command to remove the LDP-related configuration.		
Command Syntax	<code>mpls ldp <i>instance</i></code>		
	<code>no mpls ldp <i>instance</i></code>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>instance</i>	existing LDP instance	Currently only a value of 'default' is supported.
Examples	This example shows how to enter the LDP configuration sub-mode: <code>exaware# config</code> <code>exaware(config)# mpls ldp default</code> <code>exaware(config-mpls-ldp-default)#</code>		

8.1.1.17. password

Description	Configures the MD5 password for the specified peer. Configuring a password enables the TCP MD5 Signature option. LSR applies the MD5 algorithm as specified in RFC2385 (http://www.rfc-editor.org/rfc/rfc2385.txt) to compute the MD5 digest for a TCP segment to be sent to a peer. This computation makes use of the peer password as well as the TCP segment. Use the no form of the command to disable the TCP MD5 Signature option for the specified peer. The IP address is the session source (local address) of the peer, and not the address of the interface from which the announcements emanate.		
Command Syntax	<pre>password ip-address password no password ip-address</pre>		
Command Mode	configuration mpls ldp <instance>		
Parameters Table	Name	Value	Description
	ip-address	IP address	IP address of the peer.
	password	String, maximum 79 characters	Use either clear-text or encrypted format string. When using encrypted format, a '\$\$\$' string should be prepended to the encrypted key. (Default = no password)

Examples

This example shows how to configure a password for a specified peer:

```
exaware# config
exaware(config)# mpls ldp default
exaware(config-mpls-ldp-default)# password 10.10.10.10
my123pw
```

8.1.1.18. router-id

Description	Manually configures the router ID. An LDP identifier is a six-octet quantity. The router ID uses an ipv4-format IP address (first four bytes), which must be a globally unique value. The last two octets identify a specific label space within the LSR. This command manually sets the router ID to the specified value. Use the no form of the command to revert to the default behavior: the router ID is automatically selected from the highest loopback ipv4 address. If the loopback interface does not exist, the highest configured IP address is selected.		
Command Syntax	<pre>router-id ip-address no router-id</pre>		
Command Mode	configuration mpls ldp <instance>		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IP address	IP address of the router.
Examples	This example shows how to configure the local IP address: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# router-id 10.10.10.10</pre>		
Notes	It is recommended that the LDP router id entry is explicitly configured in order to avoid inconsistency in implicit selection.		

8.1.1.19. session-keepalive-interval (timers)

Description	Configures the interval after which session keep-alive messages are sent out. Use the no form of the command to revert to the globally configured value. If no global value is configured, then the interval will revert to the default.		
Command Syntax	<pre>session-keepalive-interval <i>seconds</i> no session-keepalive-interval</pre>		
Command Mode	configuration mpls ldp targeted-peer timers		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-21845	Keep-alive interval in seconds. (Default = 10 seconds)
Examples	This example shows how to set the keep-alive interval to 120 seconds globally: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# timers exaware(config-mpls-ldp-default-timers)# session-keepalive-interval 120</pre> This example shows how to set the keep-alive interval to 120 seconds for a specific interface. <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# interface x-eth 0/0/4 exaware(config-mpls-ldp-default-interface-x-eth-0/0/4)# timers exaware(config-interface-x-eth-0/0/4-timers)# session-keepalive-interval 120</pre>		
Notes	Session-keepalive-interval should not be more than 1/3 of the session-keepalive-timeout. If the session-keepalive-timeout is not configured for the specific interface, the session-keepalive-interval on the interface should not be more than 1/3 of the global session-keepalive-timeout. When targeted peer configuration exists, it overrides the global configuration.		

8.1.1.20. session-keepalive-timeout (timers)

Description	Configures the keep-alive timeout interval after which a session with a peer is closed. Use the no form of the command to revert to the globally configured value. If no global value is configured, then the interval will revert to the default.		
Command Syntax	<pre>session-keepalive-timeout <i>seconds</i> no session-keepalive-timeout</pre>		
Command Mode	configuration mpls ldp targeted-peer timers		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	Keep-alive timeout in seconds. (Default = 30 seconds)
Examples	This example shows how to set the keep-alive timeout to 120 seconds globally: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# timers exaware(config-mpls-ldp-default-timers)# session-keepalive-timeout 120</pre> This example shows how to set the keep-alive timeout to 120 seconds for a specific interface: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# interface x-eth 0/0/4 exaware(config-mpls-ldp-default-interface-x-eth-0/0/4)# timers exaware(config-interface-x-eth-0/0/4-timers)# session-keepalive-timeout 120</pre>		

Notes	Session-keepalive-timeout should be at least three times the session-keepalive-interval. If the session-keepalive-interval is not configured for the specific interface, the session-keepalive-timeout on the interface should be at least three times the global session-keepalive-interval.
--------------	---

8.1.1.21. targeted-hello-holdtime

Description	Configures the global timeout before closing an adjacency with targeted peers. The command configures the hold time globally for the LDP protocol.		
Command Syntax	targeted-hello-holdtime <i>seconds</i> no targeted-hello-holdtime		
Command Mode	configuration mpls ldp timers		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	Targeted hello hold time value. (Default = 45)
Examples	This example shows how to set the targeted hello hold time to 180 seconds: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# timers targeted-hello-holdtime 180</pre>		
Notes	Targeted-hello-holdtime should be at least 3 times higher than targeted-hello-interval.		

8.1.1.22. targeted-hello-interval

Description	Configures the global interval for sending unicast hello messages to targeted peers. The command configures the hello interval globally for the LDP protocol.		
Command Syntax	targeted-hello-interval <i>seconds</i> no targeted-hello-interval		
Command Mode	configuration mpls ldp timers		
Parameters Table	Name	Value	Description
	<i>seconds</i>	3-21845	Targeted hello interval value. (Default = 15)
Examples	This example shows how to set the targeted hello interval to 60 seconds: <pre>exaware# config exaware(config)# mpls ldp exaware(config-mpls-ldp-default)# timers targeted-hello-interval 60</pre>		
Notes	The targeted-hello-interval should be at most 1/3 of the targeted-hello-holdtime.		

8.1.1.23. targeted-hellos

Description	Sets the status of hello exchange for all targeted peers. This global configuration can be overridden by configuring hello exchange for specific targeted peer.		
Command Syntax	<pre>targeted-hellos [enable disable strict] no targeted-hellos</pre>		
Command Mode	configuration mpls ldp		
Parameters Table	Name	Value	Description
		<i>enable</i>	Enable targeted hello exchange between peers, including peers not configured locally in LDP or in RSVP (LDP-tunneling enabled). (Default)
		<i>disable</i>	Disable targeted hello exchange between targeted peers.
		<i>strict</i>	Allow targeted hello exchange only with peers configured locally (see note below).
Examples	This example shows how to enable hello exchange for all targeted peers: <pre>exaware# config exaware(config)# mpls ldp exaware(config-mpls-ldp-default)# targeted-hellos enable</pre>		
Notes	Currently, "configured locally" means that ldp-tunneling is enabled for the RSVP-TE tunnel with the same IP destination.		

8.1.1.24. targeted-peer

Description	Enter the targeted peer configuration sub-mode.		
Command Syntax	targeted-peer <i>ip-address</i> no targeted-peer <i>ip-address</i>		
Command Mode	configuration mpls ldp		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 address	IPv4 address.
Examples	This example shows how to enter the targeted peer sub-mode: <pre> exaware# config exaware(config)# mpls ldp exaware(config-mpls-ldp-default)# targeted-peer 10.10.10.20 exaware(config-mpls-ldp-default-target-peer-10.10.10.20)# </pre>		

8.1.1.25. timers

Description	Enters the timers configuration mode for LDP protocol globally, for a specific interface, or for a specific targeted peer. You can configure the following timers: • discovery-hello-holdtime (on page 408) • discovery-hello-interval (on page 411) • session-keepalive-interval (on page 429) • session-keepalive-timeout (on page 431) Use the no form of the command to remove timers-related configuration. • For global configuration: All relevant parameters will revert to their default values. • For a specific interface: All relevant parameters will revert the globally configured values.
Command Syntax	timers no timers
Command Mode	configuration mpls ldp configuration mpls ldp targeted-peer
Parameters Table	N/A
Examples	This example shows how to enter timers configuration mode for a specific interface: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# interface x-eth 0/0/2 exaware(config-mpls-ldp-default-interface-x-eth-0/0/2)# timers exaware(config-interface-x-eth-0/0/2-timers)#</pre> This example shows how to enter timers configuration mode globally: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# timers exaware(config-mpls-ldp-default-timers)#</pre>

8.1.2. LDP Operational Commands

8.1.2.1. clear ldp

Description	Restarts an LDP session with specific peer or restarts all LDP sessions. This command resets only TCP sessions, and not UDP adjacencies.		
Command Syntax	<code>clear ldp [all neighbors ipv4 {<i>ip-address</i>}]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>ip address</i>	IP address	IP address of session peer.
		all	Clear all TCP-based sessions.
Examples	The following example shows how to restart all TCP-based LDP sessions: exaware# clear ldp all		

8.1.2.2. clear ldp statistics

Description	Clears global LDP messages counters.
Command Syntax	<code>clear ldp statistics</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows how to clear LDP counters: <pre>exaware# clear ldp statistics</pre>

8.1.3. LDP Show Commands

8.1.3.1. **show ldp**

Description	Displays basic LDP attributes defined for current LSR. The output is the same with or without the 'summary' keyword.
Command Syntax	show ldp [summary]
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows a sample output from this command: <pre>exaware# show ldp Global Parameters Router ID: 192.168.0.200 IPv4 Local Address: 192.168.0.200 Table Version: 1 Label Advertisement Mode: Downstream Unsolicited Label Retention Mode: Liberal Label Control Mode: Independent Propagate Release: Disabled Discovery Hellos: Enabled Targeted Peer Hellos: Enabled Explicit Null: Disabled Import BGP: Disabled Entropy label: Disabled Configured Timers Session Keepalive Interval: 10sec Session Keepalive Timeout: 30sec Discovery Hello Interval: 5sec Discovery Hello Holdtime: 15sec Targeted Hello Interval: 15sec Targeted Hello Holdtime: 45sec IGP Synchronization delay: 10sec Graceful Restart: Recovery Time: 120sec Reconnect Time: 120sec</pre>

8.1.3.2. show ldp adjacency

Description	Displays a listing of all adjacencies discovered by the current LSR in table format.																																									
Command Syntax	show ldp adjacency [brief] [<i>ip-address</i>]																																									
Command Mode	operational																																									
Parameters Table	Name	Value	Description																																							
	<i>ip-address</i>	IP address	Adjacency IP address.																																							
Examples	The following example shows a sample of the output from this command: exaware# show ldp adjacency																																									
	<table><tr><td>ADJACENCY</td><td>TYPE</td><td>LDP ID</td><td>INTERFACE</td><td>TIMEOUT</td></tr><tr><td colspan="5">-----</td></tr><tr><td>11.10.10.2</td><td>Auto</td><td>192.168.0.11</td><td>x-eth0/0/1</td><td>25s</td></tr><tr><td>192.168.0.22</td><td>Tunnel</td><td>192.168.0.22</td><td></td><td>37s</td></tr><tr><td>192.168.0.23</td><td>Tunnel</td><td>192.168.0.23</td><td></td><td>37s</td></tr><tr><td>192.168.0.24</td><td>Targeted</td><td>192.168.0.24</td><td></td><td>37s</td></tr><tr><td>192.168.0.25</td><td>VPWS</td><td>192.168.0.25</td><td></td><td>37s</td></tr><tr><td>192.168.0.26</td><td>Tunnel, VPWS</td><td>192.168.0.26</td><td></td><td>37s</td></tr></table>			ADJACENCY	TYPE	LDP ID	INTERFACE	TIMEOUT	-----					11.10.10.2	Auto	192.168.0.11	x-eth0/0/1	25s	192.168.0.22	Tunnel	192.168.0.22		37s	192.168.0.23	Tunnel	192.168.0.23		37s	192.168.0.24	Targeted	192.168.0.24		37s	192.168.0.25	VPWS	192.168.0.25		37s	192.168.0.26	Tunnel, VPWS	192.168.0.26	
ADJACENCY	TYPE	LDP ID	INTERFACE	TIMEOUT																																						

11.10.10.2	Auto	192.168.0.11	x-eth0/0/1	25s																																						
192.168.0.22	Tunnel	192.168.0.22		37s																																						
192.168.0.23	Tunnel	192.168.0.23		37s																																						
192.168.0.24	Targeted	192.168.0.24		37s																																						
192.168.0.25	VPWS	192.168.0.25		37s																																						
192.168.0.26	Tunnel, VPWS	192.168.0.26		37s																																						
Notes	Type : Auto, Tunnel, Target, Tunnel+VPWS Interface: Interface name in case of auto-discovered peer or empty in case of targeted-peer. "show ldp adjacency ip-address" is equivalent to "show ldp adjacency brief ip-address"																																									

8.1.3.3. show ldp interfaces

Description	Displays a listing of all interfaces for which LDP is enabled for at least one address family in table format. The output is the same with or without the 'brief' keyword.																	
Command Syntax	show ldp interfaces [brief { <i>interface</i> }]																	
Command Mode	Operational																	
Parameters Table	Name	Value	Description															
	<i>interface</i>	Interface type and index	Name of interface to display.															
Examples	The following example shows a sample output from this command:																	
	exaware# show ldp interfaces																	
	<table><tr><td>INTERFACE</td><td>AFI</td><td>MPLS STATE</td></tr><tr><td colspan="3">-----</td></tr><tr><td>x-eth0/0/0</td><td>IPV4</td><td>Down (interface down)</td></tr><tr><td>x-eth0/0/2</td><td>IPV4</td><td>Up</td></tr><tr><td>x-eth0/0/3</td><td>IPV4</td><td>Up</td></tr></table>			INTERFACE	AFI	MPLS STATE	-----			x-eth0/0/0	IPV4	Down (interface down)	x-eth0/0/2	IPV4	Up	x-eth0/0/3	IPV4	Up
	INTERFACE	AFI	MPLS STATE															

x-eth0/0/0	IPV4	Down (interface down)																
x-eth0/0/2	IPV4	Up																
x-eth0/0/3	IPV4	Up																

8.1.3.4. show ldp local-bindings

Description	Displays a table of LDP local bindings.																																																							
Command Syntax	show ldp local-bindings [brief detail] [<i>prefix</i>]																																																							
Command Mode	Operational																																																							
Parameters Table	Name	Value	Description																																																					
	<i>prefix</i>	IP address/length	IP address prefix of FEC to display.																																																					
Examples	The following examples show sample output from this command:																																																							
	<pre>exaware# show ldp local-bindings</pre> <table><thead><tr><th>FEC</th><th>LOCAL LABEL</th><th>UPSTREAM NEIGHBOR COUNT</th></tr></thead><tbody><tr><td>1.0.0.0/32</td><td>53127</td><td>1</td></tr><tr><td>20.200.0.0/24</td><td>impl-null</td><td>4</td></tr><tr><td>20.210.0.0/24</td><td>53120</td><td>3</td></tr><tr><td>22.1.1.0/24</td><td>impl-null</td><td>4</td></tr></tbody></table> <pre>exaware# show ldp local-bindings detail</pre> <table><thead><tr><th>FEC</th><th>LOCAL LABEL</th><th>NEIGHBOR</th></tr></thead><tbody><tr><td>1.0.0.0/32</td><td>53127</td><td>192.168.0.20</td></tr><tr><td>20.200.0.0/24</td><td>impl-null</td><td>192.168.0.20</td></tr><tr><td></td><td></td><td>22.1.1.101</td></tr><tr><td></td><td></td><td>22.1.2.101</td></tr><tr><td></td><td></td><td>22.1.3.101</td></tr><tr><td>20.210.0.0/24</td><td>53120</td><td>22.1.1.101</td></tr><tr><td></td><td></td><td>22.1.2.101</td></tr><tr><td></td><td></td><td>22.1.3.101</td></tr><tr><td>22.1.1.0/24</td><td>impl-null</td><td>192.168.0.20</td></tr><tr><td></td><td></td><td>22.1.1.101</td></tr><tr><td></td><td></td><td>22.1.2.101</td></tr><tr><td></td><td></td><td>22.1.3.101</td></tr></tbody></table>			FEC	LOCAL LABEL	UPSTREAM NEIGHBOR COUNT	1.0.0.0/32	53127	1	20.200.0.0/24	impl-null	4	20.210.0.0/24	53120	3	22.1.1.0/24	impl-null	4	FEC	LOCAL LABEL	NEIGHBOR	1.0.0.0/32	53127	192.168.0.20	20.200.0.0/24	impl-null	192.168.0.20			22.1.1.101			22.1.2.101			22.1.3.101	20.210.0.0/24	53120	22.1.1.101			22.1.2.101			22.1.3.101	22.1.1.0/24	impl-null	192.168.0.20			22.1.1.101			22.1.2.101		
FEC	LOCAL LABEL	UPSTREAM NEIGHBOR COUNT																																																						
1.0.0.0/32	53127	1																																																						
20.200.0.0/24	impl-null	4																																																						
20.210.0.0/24	53120	3																																																						
22.1.1.0/24	impl-null	4																																																						
FEC	LOCAL LABEL	NEIGHBOR																																																						
1.0.0.0/32	53127	192.168.0.20																																																						
20.200.0.0/24	impl-null	192.168.0.20																																																						
		22.1.1.101																																																						
		22.1.2.101																																																						
		22.1.3.101																																																						
20.210.0.0/24	53120	22.1.1.101																																																						
		22.1.2.101																																																						
		22.1.3.101																																																						
22.1.1.0/24	impl-null	192.168.0.20																																																						
		22.1.1.101																																																						
		22.1.2.101																																																						
		22.1.3.101																																																						
Notes	“show ldp local-bindings” is equivalent to “show ldp local-bindings brief” “show ldp local-bindings prefix” is equivalent to “show ldp local-bindings brief prefix”																																																							

8.1.3.5. show ldp local-bindings vpws

Description	Displays VPWS service related LDP information.		
Command Syntax	show ldp local-bindings vpws [brief] [<i>neighbor</i> [<i>pw-id</i>]]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>neighbor</i>	IP address	Neighbor IP address.
	<i>pw-id</i>	1-4294967295	Pseudo-wire id
Examples	The following example shows a sample of the output from this command: <pre>exaware# show ldp local-bindings vpws NEIGHBOR PW-ID LOCAL LABEL ----- 10.2.255.12 64 1000</pre>		

8.1.3.6. show ldp neighbors

Description	Displays a listing of all sessions or particular session established between current LSR and other LSRs in brief or detail format.			
Command Syntax	show ldp neighbors [brief detail] [<i>ip-address</i>]			
Command Mode	operational			
Parameters Table	Name	Value	Description	
	<i>ip-address</i>	IP address	IP address of the peer to display.	
Examples	The following examples show sample output from this command:			
exaware# show ldp neighbors				
NEIGHBOR	TYPE	SESSION MODE	SESSION STATE	TIMEOUT EXPIRATION

172.30.117.101	Auto	Active	Operational	13s
172.30.117.102	Auto	Active	Non_existent (GR)	0s
192.168.0.22	Tunnel	Active	Operational	35s
192.168.0.23	Tunnel	Active	Operational	35s
192.168.0.24	Target	Active	Operational	35s
192.168.0.25	Auto, Target	Active	Operational	35s
192.168.0.26	Auto, Tunnel	Active	Operational	35s
192.168.0.26	Auto, VPWS	Active	Operational	35s
192.168.0.26	Tunnel, VPWS	Active	Operational	35s
192.168.0.26	Auto, Tunnel, VPWS	Active	Operational	35s
exaware# show ldp neighbors detail 192.168.0.20				
LDP Neighbor: 192.168.0.20				
LDP Session parameters				
Session state:		Operational		
Session mode:		Active		
Session type:		Auto, VPWS		
Advertisement mode:		Downstream Unsolicited		
Label retention mode:		Liberal		
Neighbor Password:		N/A		
TCP Connection parameters				
Connection state:		Established		
Local Router ID:		192.168.0.82	Remote Router ID:	192.168.0.20
Local Host:		192.168.0.82	Remote Host:	192.168.0.20
Local Port:		41055	Remote Port:	646

Timers

Negotiated Session Timeout: 30sec
 Timeout Expiration: 25s
 Reconnect Interval: 15sec
 Next Reconnect: N/A

Graceful Restart: Enabled

Recovery Timeout: 60sec
 Reconnect Timeout: 60sec
 Recovery Expiration: N/A
 Reconnect Expiration: N/A

Adjacencies:

20.200.0.20, 40.200.0.20, 192.168.0.20

Addresses Bound to Neighbor:

20.200.0.20
 20.210.0.20
 40.200.0.20
 192.168.0.20

Received Labels:

FEC	OUT LABEL	ENTROPY LABEL
0.0.0.0/0	352160	False
1.0.0.0/32	352192	True
1.0.0.1/32	352208	True
1.0.0.2/32	352224	True
1.0.0.3/32	352240	True
1.0.0.4/32	352256	True
2.0.0.0/32	352272	True
3.0.0.0/32	308704	True
22.1.1.0/24	352176	False
22.1.2.0/24	352176	False
22.1.3.0/24	352176	False
22.1.21.0/24	352176	False
22.1.22.0/24	352176	False
22.1.31.0/24	352176	False
22.1.32.0/24	352176	False
22.1.33.0/24	352176	False
22.3.1.21/32	352288	False
22.3.1.22/32	352304	False
30.200.0.0/24	352176	False
40.201.0.0/24	352176	False
40.202.0.0/24	352176	False
40.203.0.0/24	352176	False
40.204.0.0/24	352176	False

40.205.0.0/24	352176	False
50.200.0.0/24	352176	False
50.201.0.0/24	352176	False
50.202.0.0/24	352176	False
50.203.0.0/24	352176	False
50.204.0.0/24	352176	False
192.168.0.20/32	impl-null	False
192.168.0.23/32	308688	False
192.168.0.82/32	352176	False
PW-ID:101	351696	False
PW-ID:102	351712	False
PW-ID:103	351728	False

Sent Labels:

FEC	IN LABEL	ENTROPY LABEL
0.0.0.0/0	65536	False
1.0.0.0/32	65539	True
1.0.0.1/32	65540	True
1.0.0.2/32	65541	True
1.0.0.3/32	65542	True
1.0.0.4/32	65543	True
2.0.0.0/32	65544	True
20.200.0.0/24	impl-null	False
22.1.1.0/24	impl-null	False
22.1.2.0/24	impl-null	False
22.1.3.0/24	impl-null	False
22.1.21.0/24	impl-null	False
22.1.22.0/24	impl-null	False
22.1.31.0/24	impl-null	False
22.1.32.0/24	impl-null	False
22.1.33.0/24	impl-null	False
22.3.1.21/32	65546	False
22.3.1.22/32	65547	False
30.200.0.0/24	impl-null	False
40.200.0.0/24	impl-null	False
40.201.0.0/24	impl-null	False
40.202.0.0/24	impl-null	False
40.203.0.0/24	impl-null	False
40.204.0.0/24	impl-null	False
40.205.0.0/24	impl-null	False
50.200.0.0/24	impl-null	False
50.201.0.0/24	impl-null	False
50.202.0.0/24	impl-null	False
50.203.0.0/24	impl-null	False
50.204.0.0/24	impl-null	False

```

60.1.1.0/24          impl-null      False
192.168.0.82/32      impl-null      False
PW-ID:101            65552          False
PW-ID:102            65553          False
PW-ID:103            65554          False

exaware# show ldp neighbors detail 172.30.117.102
LDP Neighbor: 172.30.117.102
LDP Session parameters
  Session state:      Non_existent (GR)
  Session mode:       Active
  Session type:       Auto
  Advertisement mode: Downstream Unsolicited
  Label retention mode: Liberal
  Neighbor Password:  N/A

TCP Connection parameters
  Connection state: Not Established
  Local Router ID:   192.168.0.82          Remote Router ID:
172.30.117.102
  Local Host:        N/A                  Remote Host:      N/A
  Local Port:        N/A                  Remote Port:      N/A

Timers
  Negotiated Session Timeout:  30sec
  Timeout Expiration:          0s
  Reconnect Interval:          15sec
  Next Reconnect:              N/A

Graceful Restart:      Enabled
  Recovery Timeout:      60sec
  Reconnect Timeout:     60sec
  Recovery Expiration:   N/A
  Reconnect Expiration:  52s

Adjacencies:
  22.1.2.101
  22.1.3.101

Addresses Bound to Neighbor:
  22.1.2.101 (Stale)
  22.1.3.101 (Stale)

Received Labels:
  FEC                OUT LABEL          ENTROPY LABEL
  -----
  1.0.0.0/32         21000 (Stale)    False

```

1.0.0.1/32	21001 (Stale)	False
1.0.0.2/32	21002 (Stale)	False
1.0.0.3/32	21003 (Stale)	False
1.0.0.4/32	21004 (Stale)	False Received Labels:
Notes	Type: Auto, Tunnel, Target, VPWS or following combinations: Auto+Target, Auto+Tunnel, Auto+VPWS, Tunnel+VPWS, Auto+Tunnel+VPWS Session Mode: Active, Passive Session state: Non_existent, Initialized, OpenSent, OpenRec, Operational, Non_existent (GR) "show ldp neighbors" is equivalent to "show ldp neighbors brief" "show ldp neighbors ip-address" is equivalent to "show ldp neighbors detail ip-address"	

8.1.3.7. show ldp statistics messages

Description	Displays LDP packet statistics.
Command Syntax	show ldp statistics messages
Command Mode	operational
Examples	The following example shows a sample of the output from this command: <pre>exaware# show ldp statistics messages TYPE SENT RECEIVED ----- Notification 0 0 Hello 3980 0 Initialization 1 1 Keepalive 1932 1932 Address 1 1 Address Withdraw 0 0 Label Mapping 7 7 Label Request 0 0 Label Withdraw 1 0 Label Release 0 1 Request Abort 0 0</pre>

8.1.3.8. show ldp table

Description	Displays the LDP table entries.																																																																								
Command Syntax	show ldp table [brief] [<i>prefix</i>]																																																																								
Command Mode	operational																																																																								
Parameters Table	Name	Value	Description																																																																						
	<i>prefix</i>	IP address/length	IP address prefix of FEC to display.																																																																						
Examples	The following example shows a sample output from this command:																																																																								
<pre>exaware# show ldp table Flags : > - Selected, S - Stale, A - Alternate path</pre> <table><tr><th>FEC</th><th>IN LABEL</th><th>FLAGS</th><th>NEIGHBOR</th><th>OUT LABEL</th><th>NEXTHOP</th><th>IF/TUNNEL</th></tr><tr><td>1.0.0.1/32</td><td>65537</td><td>> Ax</td><td>1.0.0.1</td><td>impl-null</td><td>22.1.2.106</td><td>x-th0/0/1.1002</td></tr><tr><td></td><td></td><td>></td><td>1.0.0.1</td><td>impl-null</td><td>22.1.1.106</td><td>x-eth0/0/1.1001</td></tr><tr><td>2.2.2.2/32</td><td>impl-null</td><td>></td><td></td><td></td><td></td><td>connected</td></tr><tr><td>22.1.1.0/24</td><td>impl-null</td><td>></td><td>1.0.0.1</td><td>impl-null</td><td></td><td>connected</td></tr><tr><td>22.1.2.0/24</td><td>impl-null</td><td>></td><td>1.0.0.1</td><td>impl-null</td><td></td><td>connected</td></tr><tr><td>22.1.3.0/24</td><td>impl-null</td><td>></td><td></td><td></td><td></td><td>connected</td></tr><tr><td>192.168.0.82/32</td><td>impl-null</td><td>></td><td></td><td></td><td></td><td>connected</td></tr><tr><td>192.168.0.106/32</td><td>65538</td><td>> A</td><td>1.0.0.1</td><td>impl-null</td><td>22.1.2.106</td><td>x-eth0/0/1.1002</td></tr><tr><td></td><td></td><td>></td><td>1.0.0.1</td><td>impl-null</td><td>22.1.1.106</td><td>x-eth0/0/1.1001</td></tr></table>				FEC	IN LABEL	FLAGS	NEIGHBOR	OUT LABEL	NEXTHOP	IF/TUNNEL	1.0.0.1/32	65537	> Ax	1.0.0.1	impl-null	22.1.2.106	x-th0/0/1.1002			>	1.0.0.1	impl-null	22.1.1.106	x-eth0/0/1.1001	2.2.2.2/32	impl-null	>				connected	22.1.1.0/24	impl-null	>	1.0.0.1	impl-null		connected	22.1.2.0/24	impl-null	>	1.0.0.1	impl-null		connected	22.1.3.0/24	impl-null	>				connected	192.168.0.82/32	impl-null	>				connected	192.168.0.106/32	65538	> A	1.0.0.1	impl-null	22.1.2.106	x-eth0/0/1.1002			>	1.0.0.1	impl-null	22.1.1.106	x-eth0/0/1.1001
FEC	IN LABEL	FLAGS	NEIGHBOR	OUT LABEL	NEXTHOP	IF/TUNNEL																																																																			
1.0.0.1/32	65537	> Ax	1.0.0.1	impl-null	22.1.2.106	x-th0/0/1.1002																																																																			
		>	1.0.0.1	impl-null	22.1.1.106	x-eth0/0/1.1001																																																																			
2.2.2.2/32	impl-null	>				connected																																																																			
22.1.1.0/24	impl-null	>	1.0.0.1	impl-null		connected																																																																			
22.1.2.0/24	impl-null	>	1.0.0.1	impl-null		connected																																																																			
22.1.3.0/24	impl-null	>				connected																																																																			
192.168.0.82/32	impl-null	>				connected																																																																			
192.168.0.106/32	65538	> A	1.0.0.1	impl-null	22.1.2.106	x-eth0/0/1.1002																																																																			
		>	1.0.0.1	impl-null	22.1.1.106	x-eth0/0/1.1001																																																																			
Notes	"show ldp table brief" is equivalent to "show ldp table"																																																																								
	"show ldp table brief prefix" is equivalent to "show ldp table <i>prefix</i> "																																																																								

8.1.3.9. show ldp table vpws

Description	Displays VPWS service related LDP information.																	
Command Syntax	show ldp table vpws [brief] [<i>neighbor</i> [<i>pw-id</i>]]																	
Command Mode	operational																	
Parameters Table	Name	Value	Description															
	<i>neighbor</i>	IP address	Neighbor IP address.															
	<i>pw-id</i>	1-4294967295	Pseudo-wire id															
Examples	The following example shows a sample of the output from this command: <pre>exaware# show ldp table vpws Flags : S - Stale</pre> <table><tr><td>NEIGHBOR</td><td>PW ID</td><td>LOCAL LABEL</td><td>FLAGS</td><td>REMOTE LABEL</td></tr><tr><td colspan="5">-----</td></tr><tr><td>10.2.255.12</td><td>64</td><td>1000</td><td>S</td><td>2000</td></tr></table>			NEIGHBOR	PW ID	LOCAL LABEL	FLAGS	REMOTE LABEL	-----					10.2.255.12	64	1000	S	2000
NEIGHBOR	PW ID	LOCAL LABEL	FLAGS	REMOTE LABEL														

10.2.255.12	64	1000	S	2000														
Notes	"show ldp table vpws neighbor [pw-id]" is equivalent to "show ldp table vpws brief neighbor [pw-id] "																	

8.1.3.10. show ldp tunnels

Description	Displays a list of all RSVP-TE tunnels enabled for LDP tunneling. The output is the same with or without the 'brief' keyword.		
Command Syntax	show ldp tunnels [brief]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>tunnel</i>		Tunnel name.
Examples	The following example shows a sample output from this command: <pre>exaware# show ldp tunnels TUNNEL NAME TUNNEL ID DEST IP ----- TunnelToSanD..12345 106 192.168.3.101 TunnelToSanF..o_225 105 192.168.0.24 t1 101 192.168.0.22 t2 102 192.168.0.22 t3 103 192.168.2.101</pre>		

8.2. RSVP Commands

8.2.1. RSVP Configuration Commands

8.2.1.1. admin-state

Description	Sets the admin-state of the tunnel. Use the no form of the command to return to default value (up).		
Command Syntax	admin-state <i>state</i> no admin-state		
Command Mode	configuration mpls te rsvp tunnel		
Parameters Table	Name	Value	Description
	<i>state</i>	down	Administratively shutdown a tunnel.
		up	Administratively activate a tunnel. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te 1 exaware(config-mpls-te-rsvp-default-tunnel-te-1)# admin-state down exaware(config-mpls-te-rsvp-default-tunnel-te-1)# no admin-state</pre>		

8.2.1.2. affinity (Interface)

Description	Adds the specified administrative group to an interface. Use the no form of the command to remove the administrative group from the interface.		
Command Syntax	<pre>mpls te rsvp interface affinity <i>group-name</i> no mpls te rsvp interface affinity <i>group-name</i></pre>		
Command Mode	configuration mpls te rsvp interface		
Parameters Table	Name	Value	Description
	<i>group-name</i>	Maximum 15 characters, no spaces	Name of the administrative group.
Examples	The following example shows how to add an administrative group to an interface: <pre>exaware(config-mpls-te-rsvp-default-interface-x-eth-0/0/0.1)# affinity [group1 group2]</pre>		

8.2.1.3. affinity (Tunnel)

Description	Adds the specified administrative group to an LSP (Primary/Secondary) include/exclude list. Use the no form of the command to remove the administrative group from the include/exclude list.		
Command Syntax	<pre>mpls te rsvp tunnel-te <i>tunnel-name</i> [<i>secondary</i>] affinity <i>list-type</i> <i>group-name</i> no mpls te rsvp tunnel-te <i>tunnel-name</i> [<i>secondary</i>] affinity <i>list-type</i> <i>group-name</i></pre>		
Command Mode	configuration mpls te rsvp tunnel-te configuration mpls te rsvp tunnel-te secondary		
Parameters Table	Name	Value	Description
	<i>group-name</i>	Maximum 15 characters, no spaces	Name of the administrative group.
	<i>list-type</i>	{include-any exclude-any}	Set the list match type.
Examples	The following examples show how to add an administrative group to an LSP include/exclude list: <pre>exaware(config-mpls-te-rsvp-default-tunnel-te-1)# affinity include-any group1 exaware(config-mpls-te-rsvp-default-tunnel-te-1)# affinity exclude-any group2</pre>		
Notes	If an include list is configured, all chosen links must belong to at least one of the administrative groups enumerated in the include list. If an exclude list is specified, any link that includes even one of the administrative groups specified in the exclude list will not be chosen.		

8.2.1.4. affinity-map

Description	Defines the specified administrative group name and assigns a numeric value. Use the no form of the command to delete the specified administrative group and value.		
Command Syntax	<pre>mpls te affinity-map <i>group-name</i> <i>value</i> no mpls te affinity-map <i>group-name</i></pre>		
Command Mode	configuration mpls te		
Parameters Table	Name	Value	Description
	<i>group-name</i>	Maximum 15 characters, no spaces	Name of the administrative group.
	<i>value</i>	0-31	Number assigned to the group.
Examples	This example shows how to define a new group and assign to it the number '30': <pre>exaware# config exaware(config)# mpls te affinity-map group30 30</pre> This example shows how to delete 'group30'. The associated index (30) is then available to be assigned to a different group: <pre>exaware#config exaware(config)# no mpls te affinity-map group30</pre>		
Notes	The link group definition should be consistent throughout the network.		

8.2.1.5. auto-bandwidth adjust-interval

Description	Configures the bandwidth adjust interval. Use the no form of the command to revert to the default interval configuration.		
Command Syntax	<pre>auto-bandwidth adjust-interval <i>number period</i> no auto-bandwidth adjust-interval</pre>		
Command Mode	<pre>configuration mpls te rsvp <instance> tunnel-te <tunnel-name></pre>		
Parameters Table	Name	Value	Description
	<i>number</i>	depends on the sample-interval	The adjust interval can be between one minute and one year, expressed in terms of a number of either seconds, minutes, hours, days, or weeks. The adjust-interval should be at least 3 times higher than the sample-interval.
	<i>period</i>	day hour min sec week	

Examples

This example shows how to configure bandwidth adjust interval of 1 day..

```
exaware# config
exaware(config)# mpls te rsvp default
exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1
exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)#
auto-bandwidth adjust-interval 1 day
```

This example shows how to revert to the default bandwidth adjust interval (86400):

```
exaware# config
exaware(config)# mpls te rsvp default
exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1
exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)#
no auto-bandwidth adjust-interval
```

8.2.1.6. auto-bandwidth adjust-threshold

Description	Configures the threshold at which bandwidth is automatically adjusted. Use the no form of the command to revert to the default bandwidth adjustment threshold.		
Command Syntax	<pre>auto-bandwidth adjust-threshold percent no auto-bandwidth adjust-threshold</pre>		
Command Mode	configuration mpls te rsvp <instance> tunnel-te <tunnel-		
Parameters Table	Name	Value	Description
	<i>percent</i>	1-100	Bandwidth differential percentage at which point adjustment occurs. (Default = 15)
Examples	This example shows how to configure a bandwidth adjustment threshold of 20%: <pre>exaware# config exaware(config)# mpls te exaware(config-mpls-te)# rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnell exaware(config-mpls-te-rsvp-default-tunnel-te- tunnell)# auto-bandwidth adjust-threshold 20</pre> This example shows how to revert to the default bandwidth adjustment threshold (15%): <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnell exaware(config-mpls-te-rsvp-default-tunnel-te- tunnell)# no auto-bandwidth adjust-threshold</pre>		

8.2.1.7. auto-bandwidth max-bandwidth

Description	Configures the maximum bandwidth allowed for the LSP. Use the no form of the command to remove the minimum bandwidth configuration.		
Command Syntax	<pre>auto-bandwidth max-bandwidth <i>value</i> [<i>unit</i>]</pre> <pre>no auto-bandwidth max-bandwidth</pre>		
Command Mode	configuration mpls te rsvp tunnel-te		
Parameters Table	Name	Value	Description
	<i>value</i>	1-42000000000	The maximum amount of bandwidth.
	<i>unit</i>		Units in which the bandwidth is measured.
		bps	Bits per second
		kbps	Kilo bps (Default)
		mbps	Mega bps
		gbps	Giga bps
Examples	This example shows how to configure a maximum bandwidth of 1 Gbps: <pre>exaware# config exaware(config)# mpls te exaware(config-mpls-te)# rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# auto-bandwidth max-bandwidth 1 gbps</pre> This example shows how to revert to the default maximum allowable bandwidth: <pre>exaware# config exaware(config)# mpls te exaware(config-mpls-te)# rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# no auto-bandwidth max-bandwidth</pre>		

8.2.1.8. auto-bandwidth min-bandwidth

Description	Configures the minimum bandwidth allowed for the LSP. Use the no form of the command to remove the minimum bandwidth configuration.		
Command Syntax	<pre>auto-bandwidth min-bandwidth <i>value</i> [<i>unit</i>]</pre> <pre>no auto-bandwidth min-bandwidth</pre>		
Command Mode	configuration mpls te rsvp tunnel-te		
Parameters Table	Name	Value	Description
	<i>value</i>	1-42000000000	The minimum amount of bandwidth.
	<i>unit</i>		Units in which the bandwidth is measured.
		bps	Bits per second
		kbps	Kilo bps (Default)
		mbps	Mega bps
		gbps	Giga bps
Examples	This example shows how to configure a minimum bandwidth of 100000: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# auto-bandwidth min-bandwidth 100000</pre> This example shows how to revert to the default minimum allowable bandwidth: <pre>exaware# config exaware(config)# mpls te exaware(config-mpls-te)# rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# no auto-bandwidth min-bandwidth</pre>		

8.2.1.9. auto-bandwidth monitor

Description	Starts bandwidth monitoring. Use the no form of the command to stop bandwidth monitoring.
Command Syntax	<pre>auto-bandwidth monitor [<i>enable / disable</i>]</pre> <pre>no auto-bandwidth monitor</pre>
Command Mode	configuration mpls te rsvp <instance> tunnel-te < <i>tunnel-name</i> >
Parameters	N/A
Examples	This example shows how to start bandwidth monitoring: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# auto-bandwidth monitor enable</pre> This example shows how to stop bandwidth monitoring: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# no auto-bandwidth monitor</pre>
Notes	In order for bandwidth monitoring to work 'auto-bandwidth adjust-interval' must be configured.

8.2.1.10. auto-bandwidth overflow-limit

Description	Configures the number of consecutive bandwidth overflow samples that triggers automatic bandwidth adjustment. Use the no form of the command to revert to the default setting.		
Command Syntax	<pre>auto-bandwidth overflow-limit <i>number</i> no auto-bandwidth overflow-limit</pre>		
Command Mode	configuration mpls te rsvp <instance> tunnel-te < <i>tunnel-name</i> >		
Parameters Table	Name	Value	Description
	<i>number</i>	1-65535	Number of consecutive bandwidth overflow samples that triggers automatic bandwidth adjustment. (Default = 0 - disabled)
Examples	This example shows how to configure a bandwidth overflow limit of 10: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# auto-bandwidth overflow-limit 10</pre> This example shows how to revert to the default bandwidth overflow limit (0): <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# no auto-bandwidth overflow-limit</pre>		

8.2.1.11. auto-bandwidth sample-interval

Description	Configures the time interval between sampling bandwidth utilization statistics. Use the no form of the command to stop the sampling.		
Command Syntax	<pre>auto-bandwidth sample-interval <i>number period</i> no auto-bandwidth sample-interval</pre>		
Command Mode	configuration mpls te rsvp <instance>		
Parameters Table	Name	Value	Description
	<i>number</i>	depends on the period	The sample interval can be between one minute and one year, expressed in terms of a number of either seconds, minutes, hours, days, or weeks. The sample interval should be at least 3 times lower than any adjust interval.
	<i>period</i>	day hour min sec week	

Examples

This example shows how to configure a sample interval of 600 seconds:

```
exaware# config
exaware(config)# mpls te rsvp default
exaware(config-mpls-te-rsvp-default)# auto-bandwidth
sample-interval 600 sec
```

This example shows how to configure a sample interval of one year (365 days):

```
exaware# config
exaware(config)# mpls te rsvp default
exaware(config-mpls-te-rsvp-default)# auto-bandwidth
sample-interval 365 day
```

8.2.1.12. bandwidth

Description	Reserves the bandwidth for the current tunnel. Use the no form of the command to remove the configured bandwidth value.		
Command Syntax	<code>bandwidth value [unit]</code> <code>no bandwidth</code>		
Command Mode	configuration mpls te rsvp tunnel-te configuration mpls te rsvp tunnel-te secondary configuration mpls te rsvp interface tunnel-bypass (possible interfaces: agg-eth, x-eth and sub-interfaces of the same)		
Parameters Table	Name	Value	Description
	<i>value</i>	1-42000000000 for primary LSP 0-42000000000 for secondary LSP	Bandwidth (Default = 0)
	<i>unit</i>	bps kbits mbps gbps	Bits per second Kilo bps (Default) Mega bps Giga bps

Examples	These examples show how to use this command: <pre>exaware# config exaware(config)# mpls te rsvp exaware(config-mpls-te-rsvp-default)# interface x-eth 0/0/1 exaware(config-mpls-te-rsvp-default-interface-x-eth-0/0/1)# tunnel-bypass MY_BACKUP_TUNNEL exaware(config-mpls-te-rsvp-default-interface-x-eth-0/0/1- tunnel-bypass-MY_BACKUP_TUNNEL)# bandwidth 5 gbps exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# bandwidth 10 mbps exaware(config-mpls-te-rsvp-default-interface-x-eth-0/0/1- tunnel-bypass-MY_BACKUP_TUNNEL)# no bandwidth</pre>
Notes	Each LSP has an associated bandwidth attribute. The bandwidth value is included in the sender's RSVP Path message, and specifies the bandwidth to be reserved for the LSP. It is specified in bits per second. A zero bandwidth reserves no resources, although exchanges labels. The user can explicitly set zero bandwidth on the secondary LSP.

8.2.1.13. cspf

Description	Disables / Re-enables the use of Constrained Shortest Path First (CSPF). Use the no form of the command to restore CSPF to the default (enabled).
Command Syntax	cspf {enable disable} no cspf
Command Mode	configuration mpls te rsvp tunnel-te configuration mpls te rsvp tunnel-te secondary configuration mpls te rsvp interface tunnel-bypass (possible interfaces: agg-eth, x-eth and sub-interfaces of the same)
Parameters Table	N/A
Examples	This example shows how to disable CSPF: exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# cspf disable
Notes	Disabling CSPF will disable the ability to dynamically calculate paths (ERO) to the tunnel-destination.

8.2.1.14. entropy-label (rsvp)

Description	To enable Entropy label on the egress RSVP LSP router, use entropy-label CLI command. Use the no form of this command, to disable Entropy label (default).		
Command Syntax	entropy-label enable disable no entropy-label		
Command Mode	configuration mpls te rsvp <instance>		
Parameters Table	Name	Value	Description
	enable		The egress RSVP LSP router supports the Entropy label and informs upstream RSVP LSP routers about it.
	disable		The egress RSVP LSP router does not support the Entropy label.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# entropy-label enable</pre>		

8.2.1.15. explicit-null

Description	Sends explicit-null label instead of implicit-null label to the upstream LSR (Label Switch Router). Use the no form of the command to revert to the default setting (implicit-null labels).		
Command Syntax	<pre>explicit-null {enable disable} no explicit-null</pre>		
Command Mode	configuration mpls te rsvp <instance>		
Parameters Table	Name	Value	Description
		enable	Send explicit-null labels.
		disable	Send implicit-null labels. (Default)
Examples	This example shows how to enable explicit-null labels: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# explicit-null enable</pre>		
Notes	In order for the change to take place, it's required to clear (re-signal) the tunnel(s).		

8.2.1.16. hello-messages interval

Description	Configures the time, in seconds, of the interval between hello message packets. Use the no form of the command to revert to the default interval (2 seconds).		
Command Syntax	<pre>hello-messages interval <i>seconds</i> no hello-messages interval</pre>		
Command Mode	<pre>configuration mpls te rsvp <instance> configuration mpls te rsvp <instance> interface [x-eth <i>eth-id</i> <i>agg-eth eth-id</i>]</pre>		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	Length of the interval between hello packets, in seconds. (Default = 2 seconds)

Examples	This example shows how to set the time interval between hello messages to 60 seconds: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# hello-messages interval 60</pre> This example shows how to revert to the default interval between hello messages (2 seconds): <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# no hello- messages interval</pre> This example shows how to set the hello messages interval for the specified Ethernet interface to 60 seconds: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# interface x-eth 0/0/2 exaware(config-mpls-te-rsvp-default-interface-x-eth- 0/0/2)# hello-messages interval 60</pre>
Notes	For optimum performance, this value should not be more than 1/3 of the hello timeout (on page 481) value.

8.2.1.17. hello-messages neighbor

Description	Explicitly specifies a neighbor with which to exchange hello messages. Use the no form of the command to remove the neighbor.		
Command Syntax	<pre>hello-messages neighbor {ip-address} no hello-messages neighbor</pre>		
Command Mode	configuration mpls te rsvp <instance>		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 address	Neighbor IPv4 address.
Examples	This example shows how to configure the node at 10.10.10.10 as a neighbor for exchanging hello messages for RSVP globally: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# hello-messages neighbor 10.10.10.10</pre> This example shows how to remove the neighbor: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# no hello-messages neighbor</pre>		
Notes	Hello messages from neighbors that are not explicitly or implicitly learned will be rejected.		

8.2.1.18. hello-messages receipt

Description	Enables the receipt of hello messages from peers. Use the no form of the command to disable the exchange of hello messages.		
Command Syntax	<pre>hello-messages receipt {enable disable} no hello-messages receipt</pre>		
Command Mode	<pre>configuration mpls te rsvp <instance> configuration mpls te rsvp <instance> interface [x-eth <i>eth-id</i> <i>agg-eth eth-id</i>]</pre>		
Parameters Table	Name	Value	Description
		enable	Enable receiving hello messages.
		disable	Disable receiving hello message. (Default)
Examples	This example shows how to enable receiving hello messages from peers: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# hello-messages receipt enable</pre> This example shows how to enable receiving hello messages for the specified Ethernet interface: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# interface x-eth 0/0/2 exaware(config-mpls-te-rsvp-default-interface-x-eth-0/0/2)# hello-messages receipt enable</pre>		

8.2.1.19. hello-messages timeout

Description	Configures the time, in seconds, to wait before declaring a neighbor to be down. Use the no form of the command to revert to the default hello timeout value (10 seconds).		
Command Syntax	<pre>hello-messages timeout <i>seconds</i> no hello-messages timeout</pre>		
Command Mode	<pre>configuration mpls te rsvp <instance> configuration mpls te rsvp <instance> interface [x-eth <i>eth-id</i> <i>agg-eth eth-id</i>]</pre>		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	Time to wait before declaring a neighbor to be down, in seconds. (Default = 10 seconds)

Examples	This example shows how to configure a timeout period of 300 seconds for RSVP globally: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)#hello-messages timeout 300</pre> This example shows how to revert to the default timeout period (10 seconds): <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# no hello-messages timeout</pre> This example shows how to configure a timeout period of 300 seconds for the specified Ethernet interface: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# interface x-eth 0/0/2 exaware(config-mpls-te-rsvp-default-interface-x-eth- 0/0/2)# hello-messages timeout 300</pre>
Notes	If an LSR has not received a Hello message from a peer within the number of seconds specified by this command, all sessions shared with this peer will be reset.

8.2.1.20. hop-limit

Description	Configures the hop limit for the RSVP tunnel. Use the no form of the command to revert to the default hop limit (255).		
Command Syntax	<pre>hop-limit limit no hop-limit</pre>		
Command Mode	<pre>configuration mpls te rsvp <instance> tunnel-te-<tunnel-name> configuration mpls te rsvp <instance> tunnel-te-<tunnel-name> secondary</pre>		
Parameters Table	Name	Value	Description
	<i>limit</i>	1-255	Maximum number of hops for the RSVP tunnel. (Default = 255)
Examples	This example shows how to set a hop limit of 100: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# hop-limit 100</pre> This example shows how to remove the hop limit: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# no hop-limit</pre> This example shows how to set a hop limit of 100 for the RSVP tunnel secondary configuration path: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# secondary default exaware(config-tunnel-te-tunnel1-secondary-default)# hop-limit 100</pre>		

Notes	Upon configuring a hop-limit number, the hop-limit is compared with calculated/configured path (ERO). If the calculated/configured path (ERO) will contain more hops than the hop-limit number, the tunnel will not be signaled due to "CSPF path not found" or "Hop limit conflict".
--------------	---

8.2.1.21. igp-metric

Description	Configures the LSP metric value for use by Interior Gateway Protocol (IGP). Use the no form of the command to use the IGP-assigned metric (default).		
Command Syntax	<pre> igp-metric <i>value</i> no igp-metric </pre>		
Command Mode	configuration mpls te rsvp <instance> tunnel-te < <i>tunnel-name</i> >		
Parameters Table	Name	Value	Description
	<i>value</i>	1-65535	The LSP metric value to be used by IGP.
Examples	This example shows how to configure an LSP metric value of 5000: <pre> exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# igp-metric 5000 </pre> This example shows how to revert to the IGP-assigned metric: <pre> exaware# config exaware(config)# mpls te exaware(config-mpls-te)# rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# no igp-metric </pre>		

8.2.1.22. igp-shortcut

Description	Enables Interior Gateway Protocol (IGP) shortcuts. Use the no form of the command to disable IGP shortcuts.		
Command Syntax	<pre>igp-shortcut {enable disable} no igp-shortcut</pre>		
Command Mode	configuration mpls te rsvp <instance> tunnel-te <tunnel-name>		
Parameters Table	Name	Value	Description
		enable	Enable IGP shortcuts.
		disable	Disable IGP shortcuts.
Examples	This example shows how to enable IGP shortcuts: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# igp-shortcut enable</pre> This example shows how to disable IGP shortcuts: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# no igp-shortcut</pre>		
Notes	When IGP shortcuts are enabled, link-state IGP adds the tunnel-te to their topology.		

8.2.1.23. interface

Description	Configures a specific interface with RSVP and enters interface command sub-mode. Use the no form of the command to remove the interface from RSVP and remove all RSVP-related configuration on the interface. The following options can be configured for a specific RSVP interface: • hello-messages options • keep-multiplier (on page 491) • max-reservable-bw (on page 498) • refresh-time (on page 514)		
Command Syntax	<pre>interface type instance no interface type instance</pre>		
Command Mode	configuration mpls te rsvp		
Parameters Table	Name	Value	Description
	type	ETH	Type of interface.
	instance	0/0/port	
Examples	This example shows how to configure RSVP on the specified Ethernet interface. It also enters the interface configuration sub-mode for the specified interface: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# interface x-eth 0/0/2 exaware(config-mpls-te-rsvp-default-interface-x-eth- 0/0/2)#</pre>		

8.2.1.24. ip-source

Description	Sets the IPv4 source address to be used as tunnel-source for all RSVP tunnels. Use the no form of the command to revert to the default setting.		
Command Syntax	<pre>ip-source ip-address no ip-source</pre>		
Command Mode	configuration mpls te rsvp <instance>		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 address	IPv4 address to set as the default RSVP source address.
Examples	This example shows how to configure the source IP address to 10.10.10.10: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# ip-source 10.10.10.10</pre>		
Notes	RSVP tunnel "tunnel-source <IPv4 address>" configuration will override the globally configured IP source address.		

8.2.1.25. keep-multiplier

Description	Configures the constant to calculate a valid reservation lifetime for an LSP. Use the no form of the command to revert to the default keep multiplier (3).		
Command Syntax	<pre>keep-multiplier value no keep-multiplier</pre>		
Command Mode	<pre>configuration mpls te rsvp <instance> configuration mpls te rsvp <instance> interface [x-eth eth-id agg-eth eth-id]</pre>		
Parameters Table	Name	Value	Description
	<i>value</i>	1-255	Keep multiplier value. (Default = 2)
Examples	This example shows how to configure the keep multiplier value to 50: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# keep-multiplier 50</pre> This example shows how to configure the keep multiplier value to 50 for a specific Ethernet interface: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# interface x-eth 0/0/2 exaware(config-mpls-te-rsvp-default-interface-x-eth-0/0/2)# keep-multiplier 50</pre>		

Notes

The **refresh time** (on page 514) and keep multiplier are two interrelated timing parameters used to calculate the valid Reservation Lifetime for an LSP.

Use the following formula to calculate the reservation lifetime for an LSP:

$$\text{Life-time} \geq (K + 0.5) * 1.5 * R$$

- K = keep-multiplier
- R = refresh timer

The router sends refresh messages periodically so that the neighbors do not timeout.

8.2.1.26. ldp-tunneling

Description	Enable/disable ldp-tunneling for a specific tunnel.
Command Syntax	<pre>ldp-tunneling {<i>enable</i> <i>disable</i>} no ldp-tunneling</pre>
Command Mode	configuration mpls te rsvp tunnel-te
Examples	This example shows how to enable ldp tunneling for a specific rsvp tunnel: <pre>exaware# config exaware(config)# mpls te rsvp exaware(config-mpls-te-rsvp-default)# tunnel-te rsvptunnel ldp-tunneling enable</pre>

8.2.1.27. local-protection (interface)

Description	Specifies that LSP going through the specified interface will be protected by a bypass tunnel in either link mode or node-link mode. Use the no form of the command to remove the protection from the interface configuration.		
Command Syntax	<pre>local-protection protection-type no local-protection</pre>		
Command Mode	configuration mpls te rsvp interface (possible interfaces: agg-eth, x-eth and sub-interfaces of the same)		
Parameters Table	Name	Value	Description
	protection-type	link-only	Link protection (NHOP).
		node-link	Node and link protection (NNHOP).
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# mpls te rsvp exaware(config-mpls-te-rsvp-default)# interface x-eth 0/0/0 exaware(config-mpls-te-rsvp-default-interface-x-eth-0/0/0)# local-protection link-only exaware(config-mpls-te-rsvp-default-interface-x-eth-0/0/0)# no local-protection</pre>		

8.2.1.28. local-protection (tunnel)

Description	Specifies that a tunnel will request a link or node-link protection. Use the no form of the command to remove the protection from the tunnel configuration.		
Command Syntax	<pre>local-protection protection-type no local-protection</pre>		
Command Mode	configuration mpls te rsvp tunnel		
Parameters Table	Name	Value	Description
	protection-type	link-only node-link	Link protection (NHOP). Node and link protection (NNHOP).
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# mpls te rsvp exaware(config-mpls-te-rsvp-default)# tunnel-te cmp_to_j2 exaware(config-mpls-te-rsvp-default-tunnel-te-cmp_to_j2)# local-protection link-only exaware(config-mpls-te-rsvp-default-tunnel-te-cmp_to_j2)# no local-protection</pre>		

8.2.1.29. lsp-style

Description	Configures the filter for the LSP reservation style. Use the no form of the command to revert to the default reservation style (shared-explicit).		
Command Syntax	<pre>lsp-style style no lsp-style</pre>		
Command Mode	configuration mpls te rsvp <instance> tunnel-te <tunnel-name>		
Parameters Table	Name	Value	Description
	style	fixed-filter	Specifies the fixed-filter reservation style.
		shared-explicit	Specifies shared-explicit reservation style. (Default)
Examples	This example shows how to configure the fixed-filter reservation style: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# lsp-style fixed-filter</pre>		

8.2.1.30. max-reservable-bw

Description	Configures the maximum amount of bandwidth that can be reserved on the interface. Use the no form of the command to restore the default value.		
Command Syntax	max-reservable-bw <i>value</i> [<i>unit</i>]		
Command Mode	configuration /mpls/te/rsvp/interface		
Parameters Table	Name	Value	Description
	<i>value</i>	1-42000000000	Amount of bandwidth to reserve. The default value is set according to interface type (10Gbps or 100Gbps).
	<i>unit</i>		Units in which the bandwidth is measured.
		bps	Bits per second
		kbps	Kilo bps (Default)
		mbps	Mega bps
		gbps	Giga bps
		percent	Percentage of the interface bandwidth

Examples

This example shows how to set a maximum reserved bandwidth of 500 mbps in bandwidth for the specified interface:

```
exaware# config
exaware(config)# mpls te rsvp default
exaware(config-mpls-te-rsvp-default)# interface x-eth
0/0/2
exaware(config-mpls-te-rsvp-default-interface-x-eth-
0/0/2)# max-reservable-bw 500 mbps
```

This example shows how to set a maximum reserved bandwidth of 5 times the current bandwidth for the specified interface:

```
exaware# config
exaware(config)# mpls te rsvp default
exaware(config-mpls-te-rsvp-default)# interface x-eth
0/0/2
exaware(config-mpls-te-rsvp-default-interface-x-eth-
0/0/2)# max-reservable-bw 500 percent
```


8.2.1.31. nexthop

Description	Adds a nexthop IP to the path. Use the no form of the command to remove the IP from the path.		
Command Syntax	nexthop <i>nexthop-ip</i> [<i>path-action</i>] no nexthop		
Command Mode	configuration mpls te rsvp path		
Parameters Table	Name	Value	Description
	<i>nexthop-ip</i>	IPv4 address	nexthop IPv4 address.
	<i>path-action</i>	strict	Specifies whether the sub-object is strict,
		loose	loose, or exclude.
exclude		(Default = loose)	
Examples	This example shows how to define an explicit route sub-object exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# path path1 exaware(config-mpls-te-rsvp-default-path-path1)# nexthop 10.10.10.10 strict		
Notes	strict: The network node and its preceding node in the ERO must be adjacent and directly connected. This ensures that routing is enforced on the basis of each link. loose: The network node specified in the ERO must be in the path, but is not required to be directly connected to its preceding node. exclude: Exclude a nexthop IP from being used along the path.		

8.2.1.32. optimization-timer

Description	Specifies the interval between consecutive optimization processes. Use the no form of the command to disable the optimization timer.		
Command Syntax	<pre>optimization-timer <value> <units> no optimization-timer</pre>		
Command Mode	<pre>configuration /mpls/te configuration /mpls/te/rsvp/tunnel-te</pre>		
Parameters Table	Name	Value	Description
	<i>value</i>	0, 1–6550	1-6550: The interval between two consecutive optimization processes. 0: Optimization timer is disabled. (Default)
	<i>units</i>	minutes, hours, days, weeks	Units in which the optimization interval is measured. (Default = minutes)
Examples	This example shows how to set the optimization interval to 30 minutes: <pre>exaware# config exaware(config)# mpls te optimization-timer 30 min</pre> This example shows how to disable the optimization timer: <pre>exaware# config exaware(config)# no mpls te optimization-timer</pre>		

8.2.1.33. path (command mode)

Description	Creates a new RSVP path and enters path command mode. The nexthop option defines an explicit route sub-object as being either loose or strict. A list of sub-objects specifies an explicit LSP route to the egress router. Use the no form of the command to delete the specified path.		
Command Syntax	<pre>path path-name [nexthop ip-address {strict loose}] no path path-name</pre>		
Command Mode	configuration mpls te rsvp <instance>		
Parameters Table	Name	Value	Description
	<i>path-name</i>	Maximum 128 characters, no spaces	Name of the path.
	<i>ip-address</i>	IPv4 address	Next sub-object IPv4 address.
		<i>strict</i> <i>loose</i> (default)	Specifies whether the sub-object is strict or loose.
Examples	This example shows how to create a new RSVP path named <i>path1</i> and enter the path command mode: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# path path1 exaware(config-mpls-te-rsvp-default-path-path1)#</pre> This example shows defining an explicit route sub-object: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# path path1 nexthop 10.10.10.10 strict</pre>		

Notes	If path-name already exists, then this command simply enters the command mode for that path (no new path is created). strict: The network node and its preceding node in the ERO must be adjacent and directly connected. This ensures that routing is enforced on the basis of each link. loose: The network node specified in the ERO must be in the path, but is not required to be directly connected to its preceding node.
--------------	---

8.2.1.34. path

Description	Specifies an existing path (defined under rsvp) to be used within the tunnel. Use the no form of the command to remove the path specification.		
Command Syntax	<pre>path <i>path-name</i> no path <i>path-name</i></pre>		
Command Mode	<pre>configuration mpls te rsvp tunnel-te configuration mpls te rsvp tunnel-te secondary configuration mpls te rsvp interface tunnel-bypass (possible interfaces: agg-eth, x-eth and sub-interfaces of the same)</pre>		
Parameters Table	Name	Value	Description
	<i>path-name</i>	String	RSVP path to be used.
Examples	This example shows how to specify an RSVP path to be used: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# path path1</pre> This example shows how to specify an RSVP path to be used in the MPLS tunnel secondary configuration path: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# secondary exaware(config-tunnel-te-tunnel1-secondary-default)# path path1</pre> This example shows how to specify an RSVP path to be used on an interface: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# tunnel-bypass MY_BACKUP_TUNNEL exaware(config-interface-x-eth-0/0/1-tunnel-bypass- MY_BACKUP_TUNNEL)# path path1</pre>		

Notes	The path specified in this command must be an existing path created using the <i>path (command mode)</i> (on page 503) command.
-------	--

8.2.1.35. **priority**

Description	Configures the setup priority and the hold priority for the specified tunnel. Use the no form of the command to revert to the default priority values. Note: For a tunnel, the hold priority must not be higher than the setup priority.		
Command Syntax	<pre>priority setup <i>setup-priority</i> hold <i>hold-priority</i> no priority no priority setup no priority hold</pre>		
Command Mode	<pre>configuration mpls te rsvp tunnel-te configuration mpls te rsvp tunnel-te secondary configuration mpls te rsvp interface tunnel-bypass (possible interfaces: agg-eth, x-eth and sub-interfaces of the same)</pre>		
Parameters Table	Name	Value	Description
	<i>setup-priority</i>	0-7	Value for the setup priority. (Default = 7)
	<i>hold-priority</i>	0-7	Value for the hold priority. (Default = 0)

Examples

This example shows how to set the setup priority to 5 and the hold priority to 1:

```
exaware# config.  
exaware(config)# mpls te rsvp default  
exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1  
exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)#  
priority setup 5 hold 1
```

This example shows how to revert to the default priority values:

```
exaware# config  
exaware(config)# mpls te rsvp default  
exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1  
exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)#  
no priority
```

This example shows how to set the setup priority to 5 and the hold priority to 1 for the RSVP tunnel secondary configuration path:

```
exaware# config.  
exaware(config)# mpls te rsvp default  
exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1  
exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)#  
secondary default  
exaware(config-tunnel-te-tunnel1-secondary-default)#  
priority setup 5 hold 1
```

Notes	In case of insufficient bandwidth, remove less important existing LSPs to free up the bandwidth. This can be done by preempting one or more of the signaled LSPs. Setup priority determines whether a new LSP that preempts an existing LSP may be established. The setup priority of the new LSP must be higher than the hold priority of an existing LSP for the existing LSP to be preempted. Hold priority determines the degree to which an LSP holds onto its reservation for a session after the LSP has been set up successfully. When the hold priority is high, the existing LSP is less likely to give up its reservation.
--------------	---

8.2.1.36. record

Description	Enables recording of the route taken by path and reservation request (RSVP) messages. Use the no form of the command to disable recording.		
Command Syntax	<pre>record {enable disable} no record</pre>		
Command Mode	configuration mpls te rsvp <instance> tunnel-te <tunnel-name>		
Parameters Table	Name	Value	Description
		enable	Enable recording. (Default)
		disable	Disable recording.
Examples	This example shows how to disable recording: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnell exaware(config-mpls-te-rsvp-default-tunnel-te-tunnell)# no record</pre>		

8.2.1.37. refresh-time

Description	Configures the RSVP refresh interval timer. The timer specifies the interval after which PATH and/ or RESV messages will be sent out. Use the no form of the command to revert to the default and/ or RESV messages will be sent out.		
Command Syntax	<pre>refresh-time seconds no refresh-time</pre>		
Command Mode	<pre>configuration mpls te rsvp <instance> configuration mpls te rsvp <instance> interface [x-eth eth-id agg-eth eth-id]</pre>		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	Number of seconds between refresh (PATH/RESV) messages. (Default = 30 seconds)
Examples	This example shows how to set the RSVP refresh interval timer to 60 seconds: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# refresh-time 60</pre> This example shows how to set the RSVP refresh interval timer to 60 seconds for a specific Ethernet interface: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# interface x-eth 0/0/2 exaware(config-mpls-te-rsvp-default-interface-x-eth- 0/0/2)# refresh-time 60</pre>		

Notes

The refresh time and ***keep multiplier*** (on page 491) are two interrelated timing parameters used to calculate the valid reservation lifetime for an LSP. Refresh time regulates the interval between refresh messages, which include path and reservation request (RESV) messages. Refresh messages are sent periodically so that reservation does not timeout in the neighboring nodes. Each sender and receiver host sends Path and RESV messages, downstream and upstream respectively, along the paths.

8.2.1.38. retry-limit

Description	Configures the maximum number of retries allowed for this RSVP tunnel Use the no form of the command to revert to the default retry limit (0).		
Command Syntax	<pre>retry-limit <i>number</i> no retry-limit</pre>		
Command Mode	<pre>configuration mpls te rsvp <instance> tunnel-te-<<i>tunnel-name</i>> configuration mpls te rsvp <instance> tunnel-te-<<i>tunnel-name</i>> secondary</pre>		
Parameters Table	Name	Value	Description
	<i>number</i>	1-65535	The maximum number of retries permitted. (Default = 0)
Examples	This example shows how to configure 5 retries: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# retry-limit 5</pre> This example shows how to configure five retries for the RSVP tunnel secondary configuration path: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# secondary default exaware(config-tunnel-te-tunnel1-secondary-default)# retry-limit 5</pre>		
Notes	If the retry limit is 0, the system tries to create the tunnel indefinitely.		

8.2.1.39. retry-limit-cspf

Description	Configures the maximum number of CSPF calculation retries allowed for this RSVP tunnel Use the no form of the command to revert to the default retry limit (0).		
Command Syntax	<pre>retry-limit-cspf <i>number</i> no retry-limit-cspf</pre>		
Command Mode	configuration mpls te rsvp <instance> tunnel-te-< <i>tunnel-name</i> > secondary		
Parameters Table	Name	Value	Description
	<i>number</i>	1-65535	The maximum number of retries permitted. (Default = 0)
Examples	This example shows how to configure 5 retries: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# secondary default exaware(config--tunnel-te-tunnel1-secondary-default)# retry-limit-cspf 5</pre>		
Notes	If the retry limit is 0, the system tries to create the tunnel indefinitely.		

8.2.1.40. retry-timer

Description	Configures the interval between retries for this RSVP tunnel. Use the no form of the command to revert to the default retry interval (30 seconds).		
Command Syntax	<pre>retry-timer seconds no retry-timer</pre>		
Command Mode	<pre>configuration mpls te rsvp <instance> tunnel-te-<tunnel-name> configuration mpls te rsvp <instance> tunnel-te-<tunnel-name> secondary</pre>		
Parameters Table	Name	Value	Description
	seconds	1-600	The number of seconds between retries. (Default = 30 seconds)
Examples	This example shows how to configure the retry interval to 90 seconds: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# retry-timer 90</pre> This example shows how to configure the retry interval to 90 seconds for the RSVP tunnel secondary configuration path: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# secondary default exaware(config-tunnel-te-tunnel1-secondary-default)# retry-timer 90</pre>		
Notes	When the ingress tries to set up an LSP and the setup fails due to the receipt of a Path Error message, the system waits for the time specified by this command before retrying the LSP setup process.		

8.2.1.41. revert-timer

Description	Sets the auto-revert interval to perform a reversion back to the primary. Use the no form of the command to return to the default behavior – revert-timer set to 0 seconds (disabled).		
Command Syntax	<pre>revert-timer <i>time-value</i> [<i>time-unit</i>]</pre> <pre>no revert-timer</pre>		
Command Mode	<pre>configuration mpls te rsvp</pre> <pre>configuration mpls te rsvp tunnel-te</pre>		
Parameters Table	Name	Value	Description
	<i>time-value</i>	0 - 65535	Interval value (0 = disable auto-revert). (Default = 0)
	<i>time-unit</i>	[sec min]	The time unit.(Default = sec)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te 1 exaware(config-mpls-te-rsvp-default-tunnel-te-1)# revert-timer 90 sec</pre> <pre>exaware(config-mpls-te-rsvp-default-tunnel-te-1)# no revert-timer</pre>		
Notes	Setting time-value to 0 will disable auto-revert. Auto-revert for a specific tunnel will behave according to the global rsvp revert-timer configuration, unless specifically configured on that tunnel.		

8.2.1.42. rsvp

Description	Enters RSVP sub-mode from MPLS-TE mode and enables the RSVP daemon, if it is not already enabled. Use the no form of the command to disable RSVP and to remove all RSVP configuration, reverting all values to the default settings.		
Command Syntax	<pre>rsvp [instance] no rsvp</pre>		
Command Mode	configuration mpls te		
Parameters Table	Name	Value	Description
	instance	existing RSVP instance	If the instance is not specified, the default RSVP instance is assumed.
Examples	This example shows how to enable RSVP and enter RSVP sub-mode: <pre>exaware# config exaware(config)# mpls te rsvp exaware(config-mpls-te-rsvp-default)#</pre>		
Notes	This is the main sub-mode for configuring the RSVP. Configure RSVP, ISIS-TE, and enable MPLS on at least one interface to start RSVP.		

8.2.1.43. secondary

Description	Enters the secondary configuration path for the current tunnel. Use the no form of the command to delete all secondary configuration.
Command Syntax	secondary no secondary
Command Mode	configuration mpls te rsvp <instance> tunnel-te <tunnel-name>
Parameters	N/A
Examples	This example shows how to configure a bypass tunnel: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# secondary exaware(config-tunnel-te-tunnel1-secondary-default)#</pre>
Notes	If for any reason the primary LSP is down (after it was up), traffic will be routed into the secondary LSP. If the instance is not specified, the default instance is assumed.

8.2.1.44. standby

Description	Enables or disables standby state in the secondary tunnel (default).
Command Syntax	standby {enable disable}
Command Mode	configuration mpls te rsvp <instance> tunnel-te < <i>tunnel-name</i> > secondary
Parameters	N/A
Examples	This example shows how to disable standby state in the secondary tunnel: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# secondary default exaware(config-tunnel-te-tunnel1-secondary-default)# standby disable</pre>

8.2.1.45. tunnel-bypass

Description	Creates a manual bypass tunnel object or enters the configuration of a bypass tunnel. Use the no form of the command to remove the bypass tunnel from the configuration.		
Command Syntax	<pre>tunnel-bypass <i>tunnel-name</i> no tunnel-bypass <i>tunnel-name</i></pre>		
Command Mode	configuration mpls te rsvp interface (possible interfaces: agg-eth, x-eth and sub-interfaces of the same)		
Parameters Table	Name	Value	Description
	tunnel-name	String	A string uniquely identifying the bypass tunnel.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# mpls te rsvp exaware(config-mpls-te-rsvp-default)# interface x-eth 0/0/1 exaware(config-mpls-te-rsvp-default-interface-x-eth-0/0/1)# tunnel-bypass MY_BACKUP_TUNNEL exaware(config-mpls-te-rsvp-default-interface-x-eth-0/0/1)# no local-protection</pre>		
Notes	The tunnel-name must be unique between the tunnel-bypass and tunnel-te names (i.e. a bypass tunnel cannot have the same name as an existing te tunnel and vice-versa). The tunnel name may not begin with the string "_Bypass" – this validation will also be added to regular "tunnel-te" names. Unless a path explicitly asked for the next interface, the manual bypass will exclude it from the bypass path being created to ensure that it is link protected.		

8.2.1.46. tunnel-destination

Description	Specifies the egress address for an LSP. Use the no form of the command to remove the egress address configuration.		
Command Syntax	tunnel-destination <i>destination-ip</i> no tunnel-destination		
Command Mode	configuration mpls te rsvp tunnel-te configuration mpls te rsvp interface tunnel-bypass (possible interfaces: agg-eth, x-eth and sub-interfaces of the same)		
Parameters Table	Name	Value	Description
	<i>destination-ip</i>	IPv4 address	Destination egress IPv4 address.
Examples	This example shows how to configure the egress IP address to 10.10.10.10: exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# tunnel-destination 10.10.10.10		
Notes	When configuring an LSP, you must specify the address of the egress router by including this to command in the tunnel node. The egress definition is a mandatory attribute, and an RSVP session will not be created if an egress is not defined.		

8.2.1.47. tunnel-source

Description	Specifies the sender IP address in the sender template object in Path messages. Use the no form of the command to revert to the default setting.		
Command Syntax	<pre>tunnel-source source-ip no tunnel-source</pre>		
Command Mode	<pre>configuration mpls te rsvp tunnel-te configuration mpls te rsvp interface tunnel-bypass</pre> (possible interfaces: agg-eth, x-eth and sub-interfaces of the same)		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 address	Sender IPv4 address.
Examples	This example shows how to configure the sender IP address to 10.10.10.10: <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te tunnel1 exaware(config-mpls-te-rsvp-default-tunnel-te-tunnel1)# tunnel-source 10.10.10.10</pre>		
Notes	The source IP must belong to the router. If not, the validation error will be: "Aborted: ' tunnel-source': source IP address should belong to this router". If no IP is specified, the source-ip for the tunnel will be taken from the mple/te/rsvp/default/ip-source entry (if exists). If ip-source doesn't exist, the source address will be 0.0.0.0.		

8.2.1.48. tunnel-te

Description	Creates a new RSVP tunnel and enters the tunnel configuration sub-mode. Use the no form of the command to remove the tunnel.		
Command Syntax	<code>tunnel-te <i>tunnel-name</i></code> <code>no tunnel-te <i>tunnel-name</i></code>		
Command Mode	configuration mpls te rsvp <instance>		
Parameters Table	Name	Value	Description
	<i>tunnel-name</i>	Maximum 128 characters, no spaces	RSVP tunnel name.
Examples	This example shows how to create a tunnel named 'new-tunnel'. <pre>exaware# config exaware(config)# mpls te rsvp default exaware(config-mpls-te-rsvp-default)# tunnel-te new-tunnel exaware(config-mpls-te-rsvp-default-tunnel-te-new-tunnel)#</pre>		

8.2.2. RSVP Operational Commands

8.2.2.1. clear rsvp auto-bandwidth

Description	Clears the RSVP auto-bandwidth statistics.
Command Syntax	<code>clear rsvp auto-bandwidth</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows how to use this command: <code>exaware# clear rsvp auto-bandwidth</code>

8.2.2.1.1. clear rsvp bypass-tunnel

Description	Clears a bypass tunnel. When the optimize option is used the command triggers tunnel optimization.		
Command Syntax	<code>clear rsvp bypass-tunnel [optimize] <i>bypass-tunnel</i></code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>bypass-tunnel</i>	String	Bypass tunnel to be cleared.
Examples	The following example shows how to use this command: <code>exaware# clear rsvp bypass-tunnel MY_BAKCUP_TUNNEL</code> <code>exaware# clear rsvp bypass-tunnel optimize MY_BACKUP_TUNNEL</code>		

8.2.2.1.2. clear rsvp bypass-tunnels

Description	Clears all bypass tunnels. When the optimize option is used the command triggers tunnel optimization.
Command Syntax	<pre>clear rsvp bypass-tunnels [optimize]</pre>
Command Mode	operational
Parameters Table	N/A
Examples	The following examples show how to use this command: <pre>exaware# clear rsvp bypass-tunnels optimize</pre>

8.2.2.1.3. clear rsvp statistics

Description	Resets all RSVP statistics counters to zero.
Command Syntax	<code>clear rsvp statistics</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following examples show how to use this command: <code>exaware# clear rsvp statistics</code>
Notes	This command sets the path receipt and sent counters to zero. Once this command is executed, all counters read zero, and if a new message is sent or received it shows up in the statistics.

8.2.2.1.4. clear rsvp statistics neighbor

Description	Clears counters for a specific neighbor.		
Command Syntax	<code>clear rsvp statistics neighbor {<i>neighbor-address</i>}</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>neighbor-address</i>	Specific neighbor address	Neighbor address to be cleared.
Examples	The following example shows how to use this command: <code>exaware# clear rsvp statistics neighbor 120.0.0.2</code>		

8.2.2.1.5. clear rsvp statistics neighbors

Description	Clears all neighbor counters.
Command Syntax	<code>clear rsvp statistics neighbors</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following examples show how to use this command: <pre>exaware# clear rsvp statistics neighbors</pre>

8.2.2.1.6. clear rsvp tunnel

Description	Resets a specific tunnel.		
Command Syntax	clear rsvp tunnel name <i>tunnel-name</i>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>tunnel-name</i>	existing tunnel	Name of the RSVP tunnel.
Examples	The following example shows how to use this command: exaware# clear rsvp tunnel tunnel3		

8.2.2.1.7. clear rsvp tunnel optimize

Description	Perform optimization for a specific tunnel now.		
Command Syntax	<code>clear rsvp tunnel name <i>tunnel-name</i> optimize</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>tunnel-name</i>	existing tunnel	Name of the RSVP tunnel to optimize.
Examples	The following example shows how to use this command: <pre>exaware# clear rsvp tunnel name exaware-to-cisco optimize</pre>		

8.2.2.1.8. clear rsvp tunnel revert-primary

Description	Reverts to primary on a specific tunnel.		
Command Syntax	<code>clear rsvp tunnel name <i>tunnel-name</i> revert-primary</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>tunnel-name</i>	existing tunnel	Name of the RSVP tunnel to revert to primary.
Examples	The following example shows how to use this command: <pre>exaware# clear rsvp tunnel name exaware-to-cisco revert-primary</pre>		

8.2.2.1.9. clear rsvp tunnels

Description	Resets all RSVP configured tunnels.
Command Syntax	<code>clear rsvp tunnels</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following examples show how to use this command: <code>exaware# clear rsvp tunnels</code>

8.2.2.1.10. clear rsvp tunnels optimize

Description	Perform optimization on all existing RSVP tunnels now.
Command Syntax	<code>clear rsvp tunnels optimize</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following examples show how to use this command: <pre>exaware# clear rsvp tunnels optimize</pre>

8.2.2.1.11. clear rsvp tunnels revert-primary

Description	Reverts to the primary LSP on all tunnels.
Command Syntax	<code>clear rsvp tunnels revert-primary</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following examples show how to use this command: <code>exaware# clear rsvp tunnels revert-primary</code>

8.2.3. RSVP Show Commands

8.2.3.1. show rsvp bypass

Description	Displays LSP-related information for RSVP te-tunnels (non-backup).		
Command Syntax	show rsvp bypass [<i>filter-type</i>] [name <i>bypass-name</i>] [brief detail summary]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>filter-type</i>	Operational	Shows only operational (up) sessions.
		Non-operational	Shows only non-operational (down) sessions.
	<i>tunnel-name</i>	Tunnel name	Tunnel name to be filtered.
	level of detail option (Default = brief)	Brief	Shows a summary of information for all LSP-related information in table form.
		Detail	Shows a list of detailed LSP-related information.
		Summary	Shows a summary of LSP-related information.

Examples	The following examples show sample output from this command:						
<pre>exaware# show rsvp bypass brief</pre>							
TUNNEL	TUNNEL	LSP					LABEL
NAME	ID	ID	Type	SOURCE	DESTINATION	STATE	OUT

-							
_Bypass->1.1.1.1	2	18740	Auto	129.250.0.59	1.1.1.1	Down	345
Man-Bypass	5	18742	Manual	129.250.0.59	1.1.1.1	UP	1111
Man-Bypass	5	18742	Manual-Sib	129.250.0.59	1.1.1.1	Down	32323
<pre>exaware# show rsvp bypass detail</pre>							
<pre>_Bypass->1.1.1.1: Up (Operational), Type: Auto-Bypass/Man-Bypass (in-use/standby), Uptime: 40s Tunnel-id: 3, Destination: 11.86.255.1 Source: 129.250.0.59, Extended-id: 129.250.0.59 Style: shared-explicit, Metric: 33 Optimization: Not Scheduled Protected Link: x-eth0/1/0, 12.1.1.2 Protected Sessions: 3 LSP-id: 18789, Priority: Setup: 7, Hold: 0 Bandwidth: 0 bps, CSPF-Metric: 50 Downstream: 13.1.1.1, agg-eth13 Label-Out: 53766, Label Recording Enable CSPF: Enable, Configured Path: PATH-R1 (in-use) Retry: count: 0, interval: 30 seconds Path MTU: 9182 Explicit Route: 12.1.1.1 11.81.0.81 11.180.46.6 Record route:</pre>							
ip-address	is-node	in-label	protection	in-use			

<self>	no	--	none	no			
13.1.1.1	no	53766	none	no			
11.180.34.4	no	300240	none	no			
11.180.34.1	yes	300240	none	no			
11.180.46.6	no	3	none	no			
none	no						
<pre>Protected sessions: Tunnel-id: 1, Lsp-id 5, source 1.2.3.54 destination 2.34.5.6 Tunnel-id: 2, Lsp-id 6, source 1.2.3.54 destination 2.34.5.6 Tunnel-id: 3, Lsp-id 7, source 1.2.3.54 destination 2.34.5.6</pre>							

8.2.3.2. show rsvp interface

Description	Displays RSVP-specific information for interfaces.																																						
Command Syntax	show rsvp interface [<i>interface-name</i>] [brief detail]																																						
Command Mode	operational																																						
Parameters Table	Name	Value			Description																																		
	<i>interface-name</i>	Specific interface name			Interface name to be filtered.																																		
Examples	The following examples show sample output from this command:																																						
<pre>exaware# show rsvp interface brief</pre> <table><thead><tr><th>INTERFACE</th><th>STATE</th><th>MAX-RESV BW</th><th>AVAILABLE BW</th><th>RESERVED BW</th><th>ACTIVE SESSIONS</th><th>CONFIGURED PROTECTION</th><th>LOCAL AFFINITIES</th></tr></thead><tbody><tr><td colspan="8">-----</td></tr><tr><td>agg-eth1</td><td>UP</td><td>9.5Gbps</td><td>8.655Gbps</td><td>845Mbps</td><td>121</td><td>Node-link</td><td>Gold(10) Silver(11) Bronze(12)</td></tr><tr><td>agg-eth2</td><td>UP</td><td>9.5Gbps</td><td>8.655Gbps</td><td>845Mbps</td><td>121</td><td>Link-only</td><td>Gold(10) Silver(11) Bronze(12)</td></tr></tbody></table> <pre>exaware# show rsvp interface detail x-eth0/0/1.120 x-eth0/0/1.120, 120.0.0.10 Hello Messages: Receipt: Enable, Interval: 2, Timeout: 10 Refresh-Time: 30, Keep Multiplier: 2 Refresh Reduction: Disable Message Acknowledgment: Disable, Timeout: 10 Affinities: aff1 Bandwidth: 10 Gbps Max Reserved Bandwidth: 10 Gbps Available Bandwidth at Priority 0: 10 Gbps Available Bandwidth at Priority 1: 10 Gbps Available Bandwidth at Priority 2: 10 Gbps Available Bandwidth at Priority 3: 10 Gbps Available Bandwidth at Priority 4: 10 Gbps Available Bandwidth at Priority 5: 10 Gbps Available Bandwidth at Priority 6: 10 Gbps Available Bandwidth at Priority 7: 10 Gbps Protection: node-link (NNHOP) _Bypass->1.1.1.1 (node 1.1.1.1) _Bypass->2.2.2.2 (node 2.2.2.2)</pre>								INTERFACE	STATE	MAX-RESV BW	AVAILABLE BW	RESERVED BW	ACTIVE SESSIONS	CONFIGURED PROTECTION	LOCAL AFFINITIES	-----								agg-eth1	UP	9.5Gbps	8.655Gbps	845Mbps	121	Node-link	Gold(10) Silver(11) Bronze(12)	agg-eth2	UP	9.5Gbps	8.655Gbps	845Mbps	121	Link-only	Gold(10) Silver(11) Bronze(12)
INTERFACE	STATE	MAX-RESV BW	AVAILABLE BW	RESERVED BW	ACTIVE SESSIONS	CONFIGURED PROTECTION	LOCAL AFFINITIES																																

agg-eth1	UP	9.5Gbps	8.655Gbps	845Mbps	121	Node-link	Gold(10) Silver(11) Bronze(12)																																
agg-eth2	UP	9.5Gbps	8.655Gbps	845Mbps	121	Link-only	Gold(10) Silver(11) Bronze(12)																																

MY_MANUAL_BACKUP (manual)

Signaled LSPs:

Name	Role

MY_TUNNEL	ingress
_Bypass->3.3.3.3	ingress
Tunnel-to-cmp2	Transit

8.2.3.3. show rsvp lsp

Description	Displays LSP-related information for RSVP te-tunnels (non-backup).		
Command Syntax	show rsvp lsp [<i>filter-type</i>] [name <i>tunnel-name</i>] [brief detail summary]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>filter-type</i> (Default = all)	Ingress	Shows LSP-related information that originated from this router.
		Transit	Shows LSP-related information that pass through this router.
		Egress	Shows LSP-related information whose end point is this router.
		Operational	Shows LSP-related information of the specified type that are currently operational (up).

		Non-operational	Shows LSP-related information of the specified type that are currently not operational (down).
	<i>tunnel-name</i>	Tunnel name	Tunnel name to be filtered.
	level of detail option (Default = brief)	Brief	Shows a summary of information for all LSP-related information in table form.
		Detail	Shows a list of detailed LSP-related information.
		Summary	Shows a summary of LSP-related information.
Examples	The following examples show sample outputs from this command:		
<pre>exaware# show rsvp lsp Flags: A - tunnel is running with FRR TUNNEL TUNNEL LSP NAME ID ID TYPE ROLE SOURCE DESTINATION STATE IN OUT ----- - CMP.PC136-1 1 2815 --- Transit 30.136.136.1 30.107.1.1 Up 5504 453764 PC143-to-PC133 1 18780 Pri Ingress 129.250.0.59 40.254.254.13 Up - 12345 PC143-to-PC133 1 18781 Sec Ingress 129.250.0.59 40.254.254.13 Down - - PC143-to-PC133 1 18782 Pri-Sib Ingress 129.250.0.59 40.254.254.13 Down - - exaware# show rsvp lsp ingress detail Test Test: Up (Operational), Role: Ingress, Type: Primary (in-use), Uptime: 4s Tunnel-id: 3, Extended-id: 129.250.0.59 Source: 129.250.0.59, Destination: 11.86.255.1 LSP-id: 18789, Priority: Setup: 7, Hold: 0 Bandwidth: 0 bps, CSPF-Metric: 50 Downstream: 13.1.1.1, agg-eth13</pre>			


```

Label-Out: 53766, Label Recording Enable
CSPF: Enable, Configured Path: PATH-R1 (in-use)
Retry: count: 0, interval: 30 seconds
Path MTU: 9182
Affinity:
  exclude-any: Ex4500 (11)
  exclude-any: PC100 (23)
Protection available: Link / Node / None
Locally protected by: Bypass name / Secondary (lsp-id) / None
Explicit Route:
  13.1.1.1 strict
  11.81.0.81 strict
  11.180.46.6 strict

```

Record route:

ip-address	is-node-id	in-label	protection	in-use

<self>	no	--	link-only	no
13.1.1.1	no	53766	link-only	no
11.180.34.4	no	300240	node-link	no
11.180.34.1	yes	300240	node-link	no
11.180.46.6	no	3	none	no

Secondary

Test: Up (Operational), Role: Ingress, Type: Secondary (Standby), Uptime: 10s // time from session establishment)

```

Tunnel-id: 3, Extended-id: 129.250.0.59
Source: 129.250.0.59, Destination: 11.86.255.1
LSP-id: 90873, Priority: Setup: 7, Hold: 0
Bandwidth: 0 bps, CSPF-Metric: 50
Downstream: 12.1.11.1, agg-eth12
Label-Out: 53733, Label Recording Enable
CSPF: Enable, Configured Path: PATH-R1 (in-use)
Retry: count: 0, interval: 30 seconds
Path MTU: 9182

```

Affinity:

```

  exclude-any: Ex4500 (11)
  exclude-any: PC100 (23)

```

Protection available: None

Explicit Route:

```

  13.1.1.1 strict
  11.81.0.81 strict
  11.180.46.6 strict

```

Record route:

ip-address	is-node-id	in-label	protection	in-use

<self>	no	--	none	no
13.1.1.1	no	53766	none	no
11.180.34.4	no	300240	none	no
11.180.34.1	yes	300240	none	no
11.180.46.6	no	3	none	no

```
Transit LSP

CMP.PC136-TEQ.Ixial07-1: Up (Operational), Role: Transit, Uptime: 40s
  Tunnel-id: 1, Extended-id: 30.136.136.136
  Source: 30.136.136.136, Destination: 30.107.1.1
  LSP-id: 2815, Priority: Setup: 7, Hold: 0
  Bandwidth: 13.23 Kbps, CSPF-Metric: 50
  Downstream: 11.54.0.0, x-eth0/1/5.1153, Upstream: 13.1.1.3, agg-eth13
  Label-In: 53764, Label-Out: 3, Label Recording Enabled
  Retry: count: 0, interval: 30 seconds
  Path MTU: 9182
Protection available: Node
Locally protected by: Bypass>30.19.19.136->11.81.0.81
  Explicit Route:
    11.54.0.0 strict
  Record route:
ip-address      is-node-id    in-label      protection    in-use
-----
30.19.19.136    no            --            none          no
13.1.1.1 <self>  no            53766         link-only     no
11.180.34.4     no            300240        node-link     no
11.180.34.1     yes           300240        node-link     no
11.180.46.6     no            3             none          no

Egress LSP
FR2: Up (Operational), Role: Egress, Uptime: 1d23h43m28s
  Tunnel-id: 37910, Extended-id: 11.64.255.
  Source: 11.64.255.1, Destination: 129.250.0.59
  LSP-id: 3, Priority: Setup: 7, Hold: 6
  Bandwidth: 0 bps
  Upstream: 30.19.19.136, x-eth0/1/19
  Label-In: 3, Label Recording Enable
  Explicit Route:
    13.1.1.1 strict
    11.81.0.81 strict
    11.180.46.6 strict

exaware# show rsvp lsp summary
  Ingress : 103, Up 0, Down 103
  Transit : 0, Up 0, Down 0
  Egress  : 0, Up 0, Down 0
  Total configured : 103, Up 0, Down 103

exaware# show rsvp lsp egress summary
  Egress  : 0, Up 0, Down 0

exaware# show rsvp lsp transit summary
  Transit : 0, Up 0, Down 0

exaware# show rsvp lsp ingress summary
  Ingress : 103, Up 0, Down 103
```

```
exaware# show rsvp lsp operational summary
  Total configured   : 103, Up 0, Down 103

exaware# show rsvp lsp non-operational summary
  Total configured   : 103, Up 0, Down 103
```

8.2.3.4. show rsvp neighbors

Description	Displays the list of RSVP neighbors.
Command Syntax	show rsvp neighbors
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command:
exaware# show rsvp neighbors	
NEIGHBOR	UP STRM DN STRM REFRESH REDUCTION REFRESH IN EXPLICIT

29.29.29.2	1 1 Disable - Implicit

8.2.3.5. show rsvp paths

Description	Displays the configured RSVP paths and their configured hops.		
Command Syntax	show rsvp paths [<i>path-name</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>path-name</i>	string	RSVP path name.
Examples	The following example shows sample output from this command: exaware# show rsvp paths PATH ID HOPS ADDRESS TYPE ----- p 1 2 1.1.1.1 loose 3.3.3.3 strict		

8.2.3.6. show rsvp statistics

Description	Displays the counts for various messages exchanged by the daemon. This displays the list of packet types, the number of sent packets and the number of received packets.																																																																								
Command Syntax	show rsvp statistics [brief detail]																																																																								
Command Mode	operational																																																																								
Parameters Table	N/A																																																																								
Examples	The following examples show sample output from this command: <pre>exaware# show rsvp statistics</pre> <table><tr><th>MSG</th><th>SEND</th><th>RCV</th></tr><tr><td colspan="3">-----</td></tr><tr><td>Path</td><td>0</td><td>0</td></tr><tr><td>PathErr</td><td>0</td><td>0</td></tr><tr><td>PathTear</td><td>0</td><td>0</td></tr><tr><td>Resv</td><td>0</td><td>0</td></tr><tr><td>ResvErr</td><td>0</td><td>0</td></tr><tr><td>ResvTear</td><td>0</td><td>0</td></tr><tr><td>Hello</td><td>0</td><td>0</td></tr></table> <pre>exaware# show rsvp statistics detail</pre> <table><tr><th>MSG</th><th>SEND</th><th>RCV</th></tr><tr><td colspan="3">-----</td></tr><tr><td>Path</td><td>0</td><td>0</td></tr><tr><td>PathErr</td><td>0</td><td>0</td></tr><tr><td>PathTear</td><td>0</td><td>0</td></tr><tr><td>ResvFf</td><td>0</td><td>0</td></tr><tr><td>ResvWf</td><td>0</td><td>0</td></tr><tr><td>ResvSe</td><td>0</td><td>0</td></tr><tr><td>ResvErr</td><td>0</td><td>0</td></tr><tr><td>ResvTear</td><td>0</td><td>0</td></tr><tr><td>ResvConf</td><td>0</td><td>0</td></tr><tr><td>Hello</td><td>0</td><td>0</td></tr><tr><td>Bundle</td><td>0</td><td>0</td></tr><tr><td>ACK</td><td>0</td><td>0</td></tr><tr><td>SRefresh</td><td>0</td><td>0</td></tr></table>	MSG	SEND	RCV	-----			Path	0	0	PathErr	0	0	PathTear	0	0	Resv	0	0	ResvErr	0	0	ResvTear	0	0	Hello	0	0	MSG	SEND	RCV	-----			Path	0	0	PathErr	0	0	PathTear	0	0	ResvFf	0	0	ResvWf	0	0	ResvSe	0	0	ResvErr	0	0	ResvTear	0	0	ResvConf	0	0	Hello	0	0	Bundle	0	0	ACK	0	0	SRefresh	0	0
MSG	SEND	RCV																																																																							

Path	0	0																																																																							
PathErr	0	0																																																																							
PathTear	0	0																																																																							
Resv	0	0																																																																							
ResvErr	0	0																																																																							
ResvTear	0	0																																																																							
Hello	0	0																																																																							
MSG	SEND	RCV																																																																							

Path	0	0																																																																							
PathErr	0	0																																																																							
PathTear	0	0																																																																							
ResvFf	0	0																																																																							
ResvWf	0	0																																																																							
ResvSe	0	0																																																																							
ResvErr	0	0																																																																							
ResvTear	0	0																																																																							
ResvConf	0	0																																																																							
Hello	0	0																																																																							
Bundle	0	0																																																																							
ACK	0	0																																																																							
SRefresh	0	0																																																																							

8.2.3.7. show rsvp statistics neighbor

Description	Displays control message statistics for an RSVP neighbor.		
Command Syntax	show rsvp statistics neighbor [<i>neighbor-address</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>neighbor-address</i>	Specific neighbor address	Neighbor address filtered.
Examples	The following example shows sample output from this command:		
<pre>exaware# show rsvp statistics neighbor 104.0.0.4 Neighbor 104.0.0.4 PACKET SENT RECEIVED ----- PATH 642 219 PATH ERR 0 0 PATH TEAR 0 0 RESV 326 432 RESV ERR 0 0 RESV TEAR 0 0 HELLO 0 344 Neighbor 120.0.0.2 PACKET SENT RECEIVED ----- PATH 642 219 PATH ERR 0 0 PATH TEAR 0 0 RESV 326 432 RESV ERR 0 0 RESV TEAR 0 0 HELLO 0 344</pre>			

8.2.3.8. show rsvp summary

Description	Displays data about the RSVP daemon.
Command Syntax	<code>show rsvp [summary]</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows a sample output from this command: <pre>exaware# show rsvp summary RSVP Summary: Version : 1 Entropy label : Disabled Uptime : 1h8m Refresh timer : 30 Keep multiplier : 2 Graceful restart : Disable</pre>

8.2.3.9. show rsvp tunnels

Description	Displays tunnel-related information for configured LSPs.		
Command Syntax	show rsvp tunnels [brief detail summary] [<i>filter-type</i>] [name <i>tunnel-name</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>tunnel-name</i>	Tunnel name	Tunnel name to be filtered.
	<i>filter-type</i>	Ingress	Shows tunnels that originated from this router.
		Transit	Shows tunnels passing through this router.
		Egress	Shows tunnels ending at this router.
		Operational	Shows tunnels that are currently operational.
		Non-operational	Shows tunnels that are currently non-operational.
	level of detail option (Default = brief)	Brief	Shows a summary of information for all tunnels in table form.
		Detail	Shows a list of detailed information.
		Summary	Shows only the number of tunnels.

Examples	The following examples show sample outputs from this command:
<pre> exaware# show rsvp tunnels Flags: ER - end-to-end protection is ready EA - end-to-end protection is active FR - FRR bypass (local-protection) is ready FA - FRR bypass (local-protection) is active TUNNEL TUNNEL LSP NAME ID ID ROLE SOURCE DESTINATION STATE LABEL LABEL ----- R7_to_R3_NNH 3 19935 Ingress 7.7.7.7 3.3.3.3 Up - 301040 R7_to_R2 2 19934 Ingress 7.7.7.7 2.2.2.2 Up(ER) - 3 R7_R3_R2_R1_R5 1 - Ingress 7.7.7.7 5.5.5.5 Up(EA) - - R3_to_R7_NNH 43753 15 Egress 3.3.3.3 7.7.7.7 Up(FR) 3 - Bypass->11.2.7.2 20194 1 Egress 2.2.2.2 7.7.7.7 Up(FA) 3 - Bypass->11.2.3.1 43779 1 Egress 3.3.3.3 7.7.7.7 Down 3 - exaware# show rsvp tunnels detail 152_TO_138A_1 152_TO_138A_1: Up (Operational), Role: Ingress, LSP: Primary, Uptime: 2h27m15s Tunnel-id: 1, Destination: 11.138.255.1 Source: 11.1.255.1, Extended-id: 11.1.255.1 Style: shared-explicit, Metric: 11 IGP-Shortcut: Enable, LDP-Tunneling: Enable Optimization: Due-in 18s End-to-end protection: Available (lsp-id 42203) Protection desired: none Auto-Bandwidth: Monitoring Max-Avg-BW: 1.584 Kbps Sample-Interval: 60 sec, Total bits: 57309221804310528 Adjust-Threshold: 15%, Adjust-Interval: 180 sec Overflow-Limit: 0, Overflow-Count: 147 Bandwidth-Adjust-In: 78 sec, Counter-ID: 1 Currently active LSP information: LSP-id: 42167, Priority: setup 0, hold 0 Bandwidth: 0 bps, CSPF-Metric: 11 Downstream: 11.238.152.138, agg-eth52 Label-Out: 73740, Label Recording: Enable CSPF: Enable Retry: count: 0, interval: 30 seconds Path MTU: 4470 Protection available: none Locally protected by: Secondary (42203) Affinity: exclude-any: LEFT (25) exclude-any: PC140 (8) exclude-any: RIGHT (26) exclude-any: T-POS (29) Explicit Route: 11.238.152.138 strict 11.138.238.0 strict Record Route: </pre>	

ip-address	is-node-id	in-label	protection	in-use
<self>	no	-	none	no
11.238.152.138	no	73740	none	no
11.138.238.0	no	3	none	no
exaware# show rsvp tunnels summary				
Ingress : 103, Up 0, Down 103				
Transit : 0, Up 0, Down 0				
Egress : 0, Up 0, Down 0				
Total configured : 103, Up 0, Down 103				
exaware# show rsvp tunnels egress summary				
Egress : 0, Up 0, Down 0				
exaware# show rsvp tunnels transit summary				
Transit : 0, Up 0, Down 0				
exaware# show rsvp tunnels ingress summary				
Ingress : 103, Up 0, Down 103				
exaware# show rsvp tunnels operational summary				
Total configured : 103, Up 0, Down 103				
exaware# show rsvp tunnels non-operational summary				
Total configured : 103, Up 0, Down 103				

Notes	Following the introduction of local protection (FRR) since version 4.1, new potential tunnel states can occur. The states are identified using a two character flag inside of brackets "()" next to the tunnel state. The state of the tunnel is "up" if signaling is valid, and the tunnel is capable of forwarding traffic by the means of a primary lsp, secondary lsp or via a bypass. A tunnel is "down" if it is not capable of forwarding traffic. Tunnel is "up" with no flags attached to if it is not locally protected on the current network element. These are the flags that reflect the various protection options: ER - end-to-end protection is ready – The tunnel is protected by secondary LSP end to end protection. This state is only possible on the headend of the tunnel. EA - end-to-end protection is active - The tunnel is under protection by secondary LSP end to end protection (i.e. protection is active). This state is only possible on the headend of the tunnel. FR - FRR bypass (local-protection) is ready - The tunnel is protected by a bypass LSP. This state is possible on a potential PLR (which might or might not be the headend). FA - FRR bypass (local-protection) is active - The tunnel is under protection by a bypass LSP (i.e. protection is active). This state is possible on a PLR.
--------------	--

9. Policy Commands

9.1. Policy Overview

A policy is a set of rules. A rule is a policy composed of match conditions and list of actions to perform if conditions are met. Each rule is structured "if condition(s), then action(s)". There are two general types of policies:

- data-plane policies: QoS, Ethernet-Services Access Control List, and Access Control List policies
 - control-plane policies: route policies
 - A QoS policy defines actions such marking, setting a priority or policing, or shaping of the selected traffic.
 - An Access Control List policy defines one or more access control entries (ACE) which collectively define the network traffic profile.
 - A Routing policy defines actions such as setting a route attribute (for example, local-preference), or defining the route (by setting the next-hop or as-path).

Each rule consists of conditions and actions. Conditions and actions can be created separately and referenced by name (named condition/action), or defined inline in the policy itself.

You can also define parameters to be used in the rules. This allows you to use the same policy on different attach points, but to assign different values to the parameters for each attach point as necessary.

Here is the general structure of a policy:

```
policy [qos | acl | route] policy-name
  rule rule-name
    if [match-any | match-all]
      condition-1
      ...
      condition-n
    then
      action-1
      ...
      action-n
  end-policy
```

For QoS and ACL policies, the last rule in a policy is the default rule and has no condition applied on it. A default rule is automatically created when a policy is attached on an interface. A QoS policy with the default rule looks like this:

```
policy qos policy-name
  rule rule-name
    if [match-any | match-all]
      condition-1
      ...
      condition-n
    then
      action-1
      ...
      action-n
  rule rule-default
    action-1
    ...
    action-n
  end-policy
```

A policy is attached to a target entity and is executed in the context of that target entity. For example, a QoS policy is attached to an interface, and a BGP policy can be attached to a BGP neighbor.

Note. A global QoS policy is attached to the ctrl-plane attachment point.

To attach a QoS or ACL policy on an interface (under config hierarchy):

```
interface [interface-name] policy [qos | acl] [in | out] policy-name
```

A routing policy is attached by specifying the policy in a supported routing CLI command.

To apply a routing policy:

```
routing bgp 2
vrf default
neighbor 10.0.0.1
remote-as 3
af-ipv4 unicast
default-route-originate policy policy-name
```

Summary of steps for defining a policy:

1. Define global parameters and named actions for use with routing policies to be used in the rule definitions. (Optional).

```
policy global-parameters (on page 589) $parameter1 $parameter2...
policy condition (on page 588) [traffic|route] condition-name
```

2. Define policy group. (Optional)

Groups can be IP prefix groups or BGP groups.

```
policy group ip-prefix (on page 602) policy-group-name
```

3. Define the actual policy.

```
policy [qos | acl | route] policy-name
rule rule-name1
  if [match-any | match-all]
    condition-1
    ...
    condition-n
  then
    action-1
    ...
    action-n
end-policy
rule rule-name2
  if [match-any | match-all]
    condition-1
    ...
    condition-n
  then
    action-1
    ...
```



```
    action-n
rule rule-name...

[rule rule-default
    action-1
    ...
    action-n]
end-policy
```

Conditions may be either inline or named

Attach the policy to an interface or a routing attach point or ctrl-plane.

```
interface interface-name policy (on page 607) [qos | acl] [in | out] qos-policy-name
```

OR

```
routing <instance> <command> routing-policy-name
```

Rules

Each rule in the policy consists of <condition(s), action(s)> tuples.

The conditions come after "*if*". Conditions define the traffic flows by filters such as source/destination address or source/destination ports. Each traffic flow is associated with an action such as:

- ACL: *permit* or *deny*
- QoS: *police*, *shape*, or *marking (set)*
- BGP: *permit*, *deny*, or set a BGP route attribute value such as *as-path* or *damping*
- ISIS: *permit*, *deny*, or set an ISIS route attribute value such as *metric-type* or *tag*
- RSVP: *permit* or *deny*

The actions come after "*then*". Actions can be either a single action or a list of actions. Rules can be either conditional-rule (defined within an if/then clause) or non-conditional-rule (defined as a list of actions).

A non-conditional rule can be used only as the last (default) rule.

Default rule (QoS and ACL only)

A policy will automatically generate a default rule with default actions for the policy if a default rule is not explicitly defined. You can override the default rule actions by using the reserved default rule name **rule default** (on page 610). If a default rule is explicitly defined, it must be the last rule and there are no conditions attached, only action(s).

Parameters (not in QoS and ACL policies)

You can either define parameters in the policy or define global parameters in a separate command. These global parameters may then be used inside any policy definition. A parameter name is indicated by a '\$' at the beginning of the name: '\$parameter-name'.

Global Parameters

To define global parameters, use the **policy global-parameter** (on page 589) command:

```
policy global-parameters
  parameter GLOBAL_DSCP_VALUE 6
end-global-parameters
!
```

Using parameters in a policy

When a policy uses parameters they are used inside the policy definition as part of a rule definition (condition or action). This is an example of a global parameter, using the parameter defined from the example above:

```
policy acl my_policy
  rule r1
    if dscp eq $GLOBAL_DSCP_VALUE
    then permit
  end-policy
```

Nesting or hierarchical policies

You can create hierarchical policies by nesting one policy within another. To do this, define the child policy and then create a parent policy that uses the `exec-policy` (on page 585) command to call the child policy.

Examples

Example of a QoS policy specifying all conditions and actions inline:

```
policy qos QOS_POLICY
  rule r1
    if dscp eq 3
    then set dscp 5
  rule r2
    if queue eq 3
    then minimum-bandwidth 5 mbps
```

```
end-policy
```

Example of an ACL policy specifying all conditions and actions inline:

```
policy acl ACL_POLICY
rule RULE1
  if protocol-ip exist-in (46)
  then deny
end-policy
```

Example of a policy using a named action:

```
policy route MY_POLICY
rule Rule1
  if source-ip eq 20.20.20.20
  then action MY_ACTION (5)
end-policy
```

Example of a policy using parameters:

```
policy qos QOS_PLICER_PLCY_IN
rule RULE_1
  if precedence eq $precedence
  then
    police cir 500 kbps 5000 bytes
end-policy
```

Example of a policy that explicitly defines the default rule:

```
policy qos MY_POLICY
rule RULE_1
  if precedence ipv4 exist-in (4,5)
  then
    dscp set 3
    police cir 1 mbps burst 300 kbps
rule rule-default
  dscp set 0
end-policy
```

9.1.1. Policy Conditions

Conditions are used to filter the traffic based on different criteria such as source address, destination address, and protocol-specific parameters.

Conditions may be specified inline or named in a separate `policy condition` command.

Inline conditions:

Example of an inline condition:

```
policy qos MY_POLICY
  rule MY_RULE
    if source-ip exist-in (1.1.1.1, 2.2.2.2)
    then...
  end-policy
```

Named conditions:

Example of a named condition:

```
policy condition traffic MY_TRAFFIC_CONDITION
  source-ip exist-in (3.3.3.3/32)
  ttl eq 40
end-condition

policy qos MY_POLICY
  rule RULE1
    if destination-ip exist-in (10.0.0.0/8)
    then ...
  rule RULE2
    if condition MY_TRAFFIC_CONDITION
    then ...
  end-policy
```

9.1.2. Policy Actions

Actions are operations defined on a rule to be executed if a packet matches the rule condition. The default rule (supported for QoS and ACL only) is an exception. It has no condition, since all packets that did not match any of the preceding rules will match the default rule.

Actions are defined after the 'then' part of the if/then statement:

Example of actions in a policy:

```
policy qos MY_POLICY
  rule MY_RULE1
    if ...
    then
      minimum-bandwidth 5000 kbps
      set precedence 5
end-policy
```

Actions may be specified inline or named in a separate `policy condition` command.

Inline actions:

Example of an inline actions:

```
policy qos MY_POLICY
  rule MY_RULE1
    if ...
    then
      minimum-bandwidth 5000 kbps
      set precedence 5
end-policy
```

Named actions:

Example of a named condition:

```
policy action MY_ACTION
  minimum-bandwidth 5000 kbps
  set precedence 5
end-action

policy qos MY_POLICY
  rule MY_RULE1
    if ...
    then
      action MY_ACTION
    end-policy
```

9.1.3. Policy Groups

Policy groups

You can create policy groups. There are three types of policy groups:

- ip-prefix groups (all policy types): contains a group of IP addresses that may be referenced by any policy.
- Use the `policy group ip-prefix` (on page 602) command
- BGP groups (routing only): defines a group based on one of the following BGP route attributes:
 - AS path: use the `policy group bgp-as-path` (on page 592) command
 - BGP community: use the `policy group bgp-community` (on page 595) command
 - Extended community: use the `policy group bgp-extcommunity` (on page 598) command
 - Route Distinguisher: use the `policy group bgp-rd` command

Create a policy group:

```
policy group ip-prefix MY_IP_PREFIX_LIST
  10.10.0.0/16
  20.20.0.0/16
  3333:1:2:3::/64
end-group
```

How to use the group in a policy:

```
policy acl MY_ACL_POLICY
  rule BLACK_LIST
    if source-ip exist-in MY_IP_PREFIX_LIST
    then deny
  end-policy
```

9.1.4. ACL Policies

9.1.4.1. ACL Groups

The following types of groups are supported for ACL policies:

- ACL-based Forwarding (ABF) groups [not supported in release 8.0.0] - ABF groups hold the next-hops for the ACL 'forward-to' action. After the group has been created, any ABF rule (within an ACL policy) can use this group.
- IP prefix groups - IP-prefix groups contain a group of IP addresses that may be referenced by any policy. The group can contain any combination of IPv4 and IPv6 addresses.

9.1.4.2. ACL Conditions

Traffic conditions allow filtering through a variety of packet fields. Each of the match options in the traffic condition can be used inline (or via named condition) in a rule of the policy.

The following traffic condition options are supported for ACL policies:

Match any or match-all

- destination-ip
- destination-port
- dscp
- frag-non-initial
- icmpv4-code
- icmpv4-type
- icmpv6-code
- icmpv6-type
- precedence
- protocol-ip
- source-ip
- source-port
- tcp-flags
- ttl

Attach points for conditions

Condition \ interface	Ingress	Egress
destination-ip	+	+
destination-port	+	+
dscp	+	+
ether-type	-	-
frag-non-initial	+	+
icmpv4-code	+	+
icmpv4-type	+	+
icmpv6-code	+	+
icmpv6-type	+	+
ieee-802.1p	-	-
local-gen-pkt	-	-
mpls-exp imposition	-	-
mpls-exp topmost	-	-
precedence	+	+
protocol-ip	+	+
source-ip	+	+
source-port	+	+
tcp-flags	+	+
ttl	+	+

9.1.4.3. ACL Actions

ACL actions:

- permit
- deny
- forward-to [not supported in release 8.0.0]
- default-forward-to [not supported in release 8.0.0]

General notes regarding ACL actions:

- Default rule is 'deny', so the access list should contain at least one permit statement or all packets are denied.
- 'permit', 'deny', 'forward-to', and 'default-forward-to' actions are mutually exclusive. At least one of them must be defined in a rule.
- 'forward-to' and 'default-forward-to' actions can be attached only to ingress.

9.1.5. QoS Policies

Hierarchical service policy allows enforcement of SLA agreements while applying classification for different QoS classes on the traffic (inbound or outbound).

Ingress hierarchical policy:

Up to two levels of hierarchy are supported. The highest level corresponds to the aggregated interface (or VLAN) traffic, and the only supported actions for this level are queue assignment and police.

The policy is configured on two levels (on ingress only):

- Parent level – consists of rule-default only and has optional police with transmit/drop actions that are applied to the whole traffic of the attachment point.
- Child level – consists of a flat policy that can be configured with any condition/action supported on the ingress attach point and map the traffic under the rule for specific egress queues and optionally police on ingress this traffic.

This allows the user to police the whole ingress interface (parent level) while applying different classification of the traffic to different rule through the child policy as well as additional policing for this traffic,

Egress rate control hierarchical policy:

Up to two levels are supported. The highest level defines the shaper rate for outbound traffic, while allowing a different rate allocation to different QoS classes.

The shaping is done on one level – parent level (on egress only):

- Parent level – consists of rule-default with only shape action.
- Child level – consists of a policy that control the queue's parameters (guaranteed rate, maximum rate, and weight for excess traffic). Total minimum-bandwidth allocated in child policy can not exceed parent shape rate.

The system supports physical level shaping. For ports assigned with hierarchical shaping, per VLAN shaping on interfaces set for hierarchical queueing. VLAN shaping configuration includes minimum rate, maximum rate, and weight for excess traffic.

9.1.5.1. QoS Groups

The following types of groups are supported for QoS policies:

- IP prefix groups

9.1.5.2. QoS/Traffic Conditions

Traffic conditions allow filtering through a variety of packet fields. Each of the match options in the traffic condition can be used inline in a rule of the policy.

The following traffic condition options are supported for QoS policies:

Match any or match-all

- destination-ip
- destination-port
- dscp
- ether-type
- frag-non-initial
- icmpv4-code
- icmpv4-type
- icmpv6-code
- icmpv6-type
- ieee-802.1p
- local-gen-pkt
- mpls-exp
- precedence
- protocol-ip
- source-ip
- source-port
- tcp-flags
- ttl

9.1.5.3. Attach points for conditions

Condition \ Interface	Ingress	Egress
destination-ip	+	
destination-port	+	
dscp	+	
ether-type	+	
frag-non-initial	+	

Condition \ Interface	Ingress	Egress
icmpv4-code	+	
icmpv4-type	+	
icmpv6-code	+	
icmpv6-type	+	
ieee-802.1p (x-eth interface)	+	
local-gen-pkt	-	
mpls-exp imposition	+	
mpls-exp topmost	+	
precedence	+	
protocol-ip	+	
source-ip	+	
source-port	+	
tcp-flags	+	
ttl	+	
queue		+

9.1.5.4. QoS Actions

Ingress Policy Queuing Actions (QoS only):

- All ingress actions except marking and policing map traffic to a queue. A policy can have up to eight queues.

- A default queue is automatically allocated on any physical interface when created (regardless of any policy attached to it) and for any VLAN for hierarchical qos enabled ports.

Ingress QoS Actions

- queue
- police
- set (packet marking)

Egress Policy Queuing actions (QoS only):

- All egress actions except marking controls the queue parameters.
- A default queue exists in the HW and used if a policy is not configured to the queue.

Egress QoS actions

- header-compensate
- minimum-bandwidth
- queue-max-size
- set (packet marking)
- shape
- weight

9.1.5.5. Attach Point for QoS Actions

Action \ interface	Ingress	Egress
header-compensate		+
Queue	+	
Marking	+*	+**
Policing	+	
minimum-bandwidth		+
queue-max-size		+
Shape		+

Action \ interface	Ingress	Egress
Weight (for excess traffic)		+
exec-policy	+(police)	+(shape)

*) Marking on the ingress can be any of the following:

dscp

**) Marking on the egress can be any of the following:

- dscp
- precedence
- mpls-exp topmost

9.1.5.6. Policing

The following policing actions are possible:

- cir (conform rate)
- pir (peak rate)
- conform action
- exceed action
- violate action

The cir subcommand is mandatory, however the rest of the actions have default values so you do not have to explicitly configure these.

9.1.5.7. QoS Marking

The Marking action set the qos marking for a packet for future use. The marking action can be the result of classification of policing.

Unless stated otherwise, the marking can be part of either ingress or egress policy.

The following marking actions are possible:

- dscp
- precedence
- mpls-exp

9.1.6. Routing Policies

9.1.6.1. Supported Routing GroupsAS path

- BGP community
- BGP extended community
- IP prefix

9.1.6.2. Supported BGP Routing Conditionsas-path

- as-path-length
- as-path-unique-length
- community
- extcommunity
- local-preference
- med
- next-hop
- orf-prefix
- origin
- peer
- prefix
- rib-route
- rpki-origin-as
- tag

9.1.6.3. Supported BGP Routing Actions

- as-path
- community
- damping
- default-deny
- default-permit
- deny
- extcommunity
- local-preference
- log
- med
- next-hop
- origin
- permit
- suppress-specific-route
- tag
- traffic-counter
- unsuppress-specific-route
- weight

9.1.6.4. BGP Attach Points

The following BGP attach points, which are BGP routing commands with a 'policy' option, are supported:

- accounting policy
- aggregate-address
- default-route-originate
- export-policy
- flap-damping
- import-policy
- neighbor-in
- neighbor-out
- network
- orf
- redistribute

9.1.6.5. ISIS Routing Conditions

The following ISIS routing conditions are supported:

- level
- prefix
- rib-metric
- rib-route
- route-has-mpls-label
- route-has-no-mpls-label
- tag

9.1.6.6. ISIS Routing Actions

The following ISIS routing actions are supported:

- isis-metric
- metric-type
- tag

9.1.6.7. ISIS Attach Points

The following ISIS attach points, which are ISIS routing commands with a 'policy' option, are supported:

- default-route-originate
- redistribute
- route-leak

9.1.6.8. RSVP Routing Conditions

The following RSVP routing conditions are supported:

- destination-ip
- source-ip

9.1.6.9. RSVP Routing Actions

The following RSVP routing actions are supported:

- deny
- igp-forward
- permit

9.1.6.10. RSVP Attach Points

The following RSVP attach points, which are RSVP routing commands with a 'policy' option, are supported:

rsvp

9.2. Policy Configuration Commands

9.2.1. exec-policy

Description	Creates hierarchical service policies. It defines a policy action that calls a sub-policy. The policy that has the action of exec-policy is usually known as the parent-policy whereas the policy referenced by the exec-policy command is known as child-policy. The policy-name referenced by the exec-policy (child-policy) bases its services on the parent service. Up to two levels of hierarchy are supported. Only a default rule can be configured on the parent level.		
Command Syntax	<code>exec-policy <i>policy-name</i></code>		
Command Mode	configuration policy [qos acl route]		
Parameters Table	Name	Value	Description
	<i>policy-name</i>	Name of existing policy	Name of sub-policy to execute.

Examples	Example of hierarchal policies applied on the egress: <pre>policy qos PARENT rule rule-default shape 1000000 bps burst 1000 bytes exec-policy CHILD end-policy policy qos CHILD rule gold if queue eq af1 then minimum-bandwidth percent 40 rule silver if queue eq af2 then minimum-bandwidth percent 40 rule bronze if queue eq af3 then minimum-bandwidth percent 20 end-policy policy qos CHILD rule GOLD if precedence eq 1 then police cir 1 mbps burst 250000 byte conform-action set dscp cs3 exceed-action drop priority-queue high rule SILVER if precedence exist-in (2,3) then minimum-bandwidth 60 mbps remaining-bandwidth percent 40 rule BRONZE if precedence exist-in (4,5) then minimum-bandwidth 30 mbps remaining-bandwidth percent 20 end-policy</pre>
-----------------	--

9.2.2. policy named condition

Description	Configures a traffic or routing condition that can then be used in defining a policy.		
Command Syntax	<code>policy condition [traffic route] <i>named-condition</i></code>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>named-condition</i>		Name of the condition.
Examples	Define the condition: <pre>policy condition traffic MY_TRAFFIC_CONDITION destination-ip exist-in (3.3.3.3/32) ttl eq 40 end-condition</pre> Use the condition in a policy: <pre>policy qos MY_POLICY rule RULE1 if condition MY_TRAFFIC_CONDITION then ... end-policy</pre>		

9.2.3. policy global-parameters

Description	Defines parameters that can then be used for defining rules (conditions or actions) inside a policy.		
Command Syntax	<pre>policy global-parameters parameter <i>name</i> <i>value</i> policy global-parameters end-global-parameters</pre>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>name</i>		Name of the parameter.
	<i>value</i>		Value of the parameter.
Examples	Define the parameters: <pre>policy global-parameters parameter precedence1 1000 end-global-parameters !</pre> Use the parameters in a policy: <pre>policy qos QOS_PLICER_PLCY_IN rule RULE_1 if precedence eq \$precedence1 then police cir 500 kbps burst 5000 bytes end-policy</pre>		

9.2.4. policy group acl-forwarding

Description	The acl-forwarding group holds the next-hops for the ACL 'forward-to' or 'default-forward-to' actions. After it has been created, any ABF rule (within the ACL policy) can use this group.		
Command Syntax	<pre> policy group acl-forwarding <group-name> next-hop1 ip <ip-addr> [vrf <vrf-name>] next-hop2 ip <ip-addr> [vrf <vrf-name>] next-hop3 ip <ip-addr> [vrf <vrf-name>] </pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>group-name</i>		Name of the policy group.
	next-hop1	IP address or interface name	First priority next-hop IP/interface.
	next-hop2	IP address or interface name	Second priority next-hop IP/interface.
	next-hop3	IP address or interface name	Third priority next-hop IP/interface.
	<i>ip-addr</i>		IP address.
	<i>vrf-name</i>		VRF name to which the IP address is attached.

Examples	<pre>policy group acl-forwarding VIP_USERS next-hop1 ip 10.0.10.0 next-hop2 ip 20.20.20.20 vrf vip_vrf next-hop3 ip 30.30.30.5 ! policy acl a2 rule r1 if dscp eq 2 then forward-to group VIP_USERS rule r2 if dscp eq 3 then default-forward-to group VIP_USERS end-policy ! interface x-eth 0/0/0 policy acl in a2</pre>
Notes	At least one of the next-hop options must be configured.

9.2.5. policy group bgp-as-path

Description	Configures a policy group based on the BGP AS-path value. You can also specify one or more parameters to use in the group.		
Command Syntax	<pre>policy group bgp-as-path <i>group-name</i> [<i>\$parameter</i> <i>element1</i> [<i>element2</i> <i>element3</i>....] end-group</pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>group-name</i>		Name of the policy group.
	<i>\$parameter</i>	String starting with \$ example: \$parameter-name	Parameter name.

	<i>element1...</i>		Element in a group could be one of the following types: regex : regular expression enclosed in single quote (POSIX 1003.2) AS number (including in 4-byte format) range of AS numbers (range 0 - 4,294,967,295)
Examples	<pre>policy group bgp-as-path MY_BGP_AS_PATH_GROUP 213 450-567 888 1234-5678 end-group</pre>		

9.2.6. policy group bgp-community

Description	Configures a policy group based on the BGP community group value. You can also specify one or more parameters to use in the group.		
Command Syntax	<pre>policy group bgp-community <i>group-name</i> [<i>\$parameter</i>] <i>element1</i> (<i>element2</i> <i>element3</i>...) end-group</pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>group-name</i>		Name of the policy group.
	<i>\$parameter</i>	String starting with \$ example: \$parameter-name	Parameter name.

	<i>element1...</i>		Values in community group can be expressed in the following ways: numerical value in the format: <number>:<number> well-known value : no-export, no-advertise, no-export-subconfed, internet wildcard for any of the decimal values : * range for any of the decimal values: [low_val-high_val] parameter (starts with \$) for any decimal value regex : regular expression enclosed in single quotes (POSIX 1003.2)
Examples	<pre>policy group bgp-community MY_COMMUNITY_GROUP 12:34 12:* 200-1400:400 500:\$arg1 '1(2 3)':*</pre>		

9.2.7. policy group bgp-extcommunity

Description	Configures a policy group based on BGP extended community values. You can also specify one or more parameters to use in the group. Two types of extended communities are supported in bgp-extcommunity groups: • target - (route-target extcommunity) • origin - (soo extcommunity) The extended community format is: type:administrator:tag where type can be either origin (soo extcommunity) or target (route-target extcommunity). Administrator:tag pair can be: <2-byte number>:<4-byte number> (2-byte ASN notation) <4-byte number>L:<2-byte number> (4-byte ASN notation) <IPv4 address>:<2-byte number> (IPv4 address notation)		
Command Syntax	<pre>policy group bgp-extcommunity <i>group-name</i> [<i>\$parameter</i>] <i>element1</i> (<i>element2</i> <i>element3</i>...) end-group</pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>group-name</i>		Name of the policy group.
	<i>\$parameter</i>	String starting with \$ example: \$parameter-name	Parameter name.

	<i>element1...</i>		route-target format: <target:administrator:tag> soo format: <origin:administrator:tag> where: administrator = IP address, ASN (1-65535), or ASN4 (1-4,294,967,295). tag = site-specific number. The range of the tag is dependent on the administrator type. For ASN4 and IP addresses: range = 1-65535. For ASN: range = 1- 4,294,967,295. Each value can be number, range, regexp, parameter, or wildcard.
--	--------------------	--	---

Examples

```
policy group bgp-extcommunity MY_EXTCOMY_GROUP1
target:1.2.3.4:666
target:1234:666
target:1234:1000000
target:1234L:65535
target:1.2.3.4:*
target:1234:5
target:`1234(1-9)`:*
target:1200-4000:100-200
target:100.1.1.0/24:200
target:$ip_addr:100
target:12.1.1.1:$tag
origin:1.2.3.4:666
origin:1234:666
origin:1234L:666
origin:1.2.3.4:*
origin:1234:5
origin:`1234(1-9)`:*
origin:1200-4000:100-200
origin:100.1.1.0/24:200
origin:334324L:132
origin:$ip_addr:100
origin:12.1.1.1:$tag
```

9.2.8. policy group ip-prefix

Description	Configures an ip-prefix group. The ip-prefix group contains a group of IP addresses. The same ip-prefix group may be referenced by any policy type such as acl, route, or qos. A prefix is identified by ip address/mask and length. The ip-prefix group can contain both IPv4 and IPv6 formats.		
Command Syntax	<pre>policy group ip-prefix <i>group-name</i> <i>prefix1</i> <i>prefix2</i> <i>prefix3.....</i> end-group</pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>group-name</i>		Name of the policy group.

	<i>prefix1....</i>	IPv4 or IPv6 prefix in format: <code><address[/mask-len]</code> <code>[matching-len n1[-n2]]></code> examples: 2.2.0.0 3.0.0.0/8 matching-len 16 3001:AF:1000::/36 matching-len 48-96	
--	--------------------	--	--

Examples	<pre> policy group ip-prefix MY_BLACK_PFX_GROUP 60.60.0.0/16 212.33.44.55 2001:0db8:85a3:0000:0000:8a2e:0370:7334 2001::0DB8:800:200C:417A/128 3001::0DB8:800:200C:417A/128 3.0.0.0/8 matching-len 16 end-group </pre>
Notes	Prefix value indicates the number of masking bits of the IP address. The conditions are performed only on the mask value. IP-prefix group can contain a list of elements of the same type; IPv4/IPv6 addresses or a mixed list of IPv4 and IPv6 addresses. The IP address could be sole address with no length argument i.e.: 1.1.1.1 means host 1.1.1.1, mask length = 32. Matching Length matching-len allows you to compare a route prefix against a range of subnet-masks rather than just a single subnet-mask. The default minimum matching-len is the mask length. If a minimum matching-len is specified, then the default maximum matching-len must be less than 32 for IPv4 prefixes or 128 for IPv6 prefixes. If neither a minimum nor maximum length is specified, the default maximum length is the mask length. (192.168.2.0/24) – compares against one possible value 192.168.2.0/24. (192.168.3.0/24 matching-len 28-32) – compares against a range of prefix values, from 192.168.3.0/28 to 192.168.3.255/32. (192.168.3.0/24 matching-len 28) – compares against a range of prefix values, from 192.168.3.0/28 to 192.168.3.240/28.

9.2.9. policy group mac-address

Description	Configures a mac-address group. The mac-address group contains a group of Ethernet mac-addresses.		
Command Syntax	<code>policy group mac-address <group-name></code>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>group-name</i>		Name of the policy group.
Examples	<pre>policy group mac-address WhiteListGr 00:00:00:00:11:11 00:00:00:00:22:22/24 00:00:00:00:33:33 end-group</pre>		
Notes	The conditions are performed only on the mask value.		

9.2.10. policy (attach QoS and ACL)

Description	Attaches a QoS or ACL policy to an interface. In order for a policy to take effect, it must be attached to an interface in a specific direction. Once a policy is attached to an interface in the ingress and/or egress direction and committed successfully, the QoS or ACL rules apply on the traffic of that interface. Use the no form of the command to remove the policy from the interface.		
Command Syntax	<pre>policy [qos acl] [in out] qos-policy-name</pre> <pre>no policy [qos acl] qos [in out] qos-policy-name</pre>		
Command Mode	configuration interface <interface-name> policy-qos		
Parameters Table	Name	Value	Description
	qos-policy-name		Name of the QoS policy to attach.
Examples	Attach policy to interface: <pre>exaware(config-interface-x-eth-0/0/1)# policy qos in QOS_PLCY_IN</pre> <pre>exaware(config-interface-x-eth-0/0/1)# policy qos out QOS_PLCY_OUT</pre> <pre>exaware(config-interface-x-eth-0/0/1)# policy acl in ACL_PLCY_IN</pre> <pre>exaware(config-interface-x-eth-0/0/1)# policy acl out ACL_PLCY_OUT</pre> Detach policy from interface: <pre>exaware(config-interface-x-eth-0/0/1)# no policy qos in</pre> The following example will detach acl policies from ingress and egress (since direction was not specified): <pre>exaware(config-interface-x-eth-0/0/1)# no policy acl</pre>		

Notes	All semantic validation checks will be done on attachment point. Egress ACL can be applied on physical interface as a whole and when applied affects all the sub interfaces traffic on this port.
--------------	---

9.2.11. rule-default (QoS and ACL)

Description	Overrides default action in a QoS or ACL rule. Once a QoS or ACL policy is attached to an interface, it will automatically generate a default rule with default actions for that interface if you do not explicitly define a rule. You can override the default rule actions by using the reserved default rule name 'rule-default'.
Command Syntax	rule-default
Command Mode	configuration policy qos/acl
Parameters Table	N/A
Examples	In the following example default queue is changed by the user: <pre> policy qos MY_POLICY rule RULE_1 if precedence exist-in (4,5) then set dscp 3 police cir 1 mbps burst 300 kbytes rule rule-default set dscp 5 end-policy </pre>
Notes	The default rule is always the last one. No rules can be inserted after the default. The rule-default action in acl is 'deny'.

9.3. Policy Show Commands

9.3.1. show configuration policy condition

Description	Displays all named conditions of the specified type.
Command Syntax	<code>show configuration policy condition [traffic route]</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following examples show sample outputs from this command: <pre>exaware# show configuration policy condition traffic policy condition traffic MY_ACL_CONDITION_1 match-all ttl eq \$TTL_DISCARD_VAL source-ip exist-in (10.8.0.0/16, 10.7.0.0/16) destination-ip exist-in (10.2.0.0/16, 10.3.0.0/16) end-condition policy condition traffic MY_ACL_CONDITION_2 match-any destination-ip eq 10.6.0.0/32 end-condition</pre>

9.3.2. show configuration policy group

Description	Displays the policy groups of the specified type configured on the router. If no type is specified, all groups are displayed.
Command Syntax	show configuration policy group [ip-prefix bgp-as-path bgp-community bgp-extcommunity bgp-rd]
Command Mode	operational
Examples	The following examples show sample outputs from this command. exaware# show configuration policy group ip-prefix policy group ip-prefix MY_IP_PREFIX_LIST 10.10.0.0/16 20.20.0.0/16 3333:1:2:3::/64 end-group !

9.3.3. show policy

Description	Displays all the active policy elements; those policy elements that are attached directly or indirectly. The brief option displays only the names of the policy elements. The detail option displays the content of the policy elements (as seen in <code>show running-configuration</code>).		
Command Syntax	<pre>show policy [attached unattached] [acl qos route] [brief detail] show policy unused [brief detail]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
		attached	Show all attached policies and all elements referenced (directly or indirectly) by this policy.
		unattached	Show all unattached policies or elements referenced (directly or indirectly) only by unattached elements. For example, an ip-prefix that is referenced by an attached ACL policy will not be displayed as an unattached QoS.
		route	Show all route related elements or elements referenced (directly or indirectly) by route elements.

	acl	Show all acl related elements or elements referenced (directly or indirectly) by acl elements.
--	-----	--

	qos	Show all QoS related elements or elements referenced (directly or indirectly) by QoS elements.
--	-----	--

	unused	Show all elements not referenced (directly or indirectly) and not attached.
--	--------	--

		brief	Show only element names. (Default)
--	--	-------	------------------------------------

		detail	Show full content elements.
--	--	--------	-----------------------------

Examples

The following examples show sample outputs from this command.

Example 1:

```
#show configuration policy qos
policy group ip-prefix P_GROUP
    56.56.56.0/24
    40.40.40.0/24
end-group
!
policy condition traffic MY_SRC_COND match-all
    precedence eq 5
    source-ip eq 10.10.10.0/24
end-condition
!
policy qos policy1
    rule rule1
        if condition MY_SRC_COND
            then set dscp cs3
!

R10004# show policy unused
policy group ip-prefix P_GROUP

R10004# show policy unattached
policy condition traffic MY_SRC_COND
policy qos policy1
```

Example 2:

```
#show configuration
policy group ip-prefix DUMMY_GROUP
    56.56.56.0/24
    40.40.40.0/24
end-group
!
policy acl DUMMY_ACL_POLICY
rule R1
    if precedence eq 2
        source-ip exist-in DUMMY_GROUP
        then deny
end-policy
!
```



```
policy qos DUMMY_QOS_POLICY
rule RULE1
  if queue eq afl
  then set mpls-exp imposition 4
rule rule-default
  queue-max-size 100 ms
  remaining-bandwidth percent 1
end-policy
!

R10004# show policy unattached qos
policy qos DUMMY_QOS_POLICY

R10004# show policy unattached acl
policy acl DUMMY_ACL_POLICY
policy group ip-prefix DUMMY_GROUP

R10004# show policy unattached
policy qos DUMMY_QOS_POLICY
policy acl DUMMY_ACL_POLICY
policy group ip-prefix DUMMY_GROUP
```

Example 3:

```
#show configuration
policy group ip-prefix CLUSTER_GROUP
  56.56.56.0/24
  40.40.40.0/24
end-group
!
policy condition traffic FROM_CLUSTER_COND
  source-ip exist-in CLUSTER_GROUP
end-condition
!
policy qos CHILD_POLICY
rule RULE1
  if condition FROM_CLUSTER_COND
  then
    police
      cir 10 mbps burst 10000 bytes
      conform-action set dscp af11
      exceed-action set dscp af13
      violate-action drop
```

```
rule rule-default
    set precedence 0
    set mpls-exp topmost 0
end-policy
!
policy qos PARENT_POLICER
rule default match-all
police rate cir 20 mbps  cir-burst 200000 bytes
exec-policy CHILD_POLICY
end-policy
!
policy qos MY_QOS_POLICY
rule RULE1
if queue eq afl
then
mpls-exp set 2
    queue-max-size 100000 bytes
rule default
    dscp set 1
    queue-max-size 100 ms
remaining-bandwidth percent 1
end-policy
!
policy acl MY_ACL_POLICY
rule Rule1
    if match-all
        source-ip exist-in CLUSTER_GROUP
    then permit
end-policy
!
interface x-eth 0/0/0.1
policy acl in MY_ACL_POLICY
!
interface x-eth0/0/1
policy qos in PARENT_POLICER
!
interface x-eth0/0/1
policy qos out MY_QOS_POLICY
!

R10004# show policy attached qos
policy group ip-prefix CLUSTER_GROUP
policy condition traffic FROM_CLUSTER_COND
```

```
policy qos CHILD_POLICY
policy qos MY_QOS_POLICY
policy qos PARENT_POLICER
```

```
R10004# show policy attached acl
policy group ip-prefix CLUSTER_GROUP
policy acl MY_ACL_POLICY
```

```
R10004# show policy attached
policy group ip-prefix CLUSTER_GROUP
policy condition traffic FROM_CLUSTER_COND
policy qos CHILD_POLICY
policy qos MY_QOS_POLICY
policy qos PARENT_POLICER
policy acl MY_ACL_POLICY
```

9.4. ACL Policy Configuration Commands

9.4.1. Traffic Conditions

The following configuration commands for traffic conditions are supported for ACL policies. See the referenced command descriptions in the **QoS Policy Configuration Commands: Traffic Conditions** section.

- **destination ip** (on page 791)
- **destination port** (on page 794)
- **dscp** (on page 794)
- **ether-type** (on page 804)
- **frag-non-initial** (on page 805)
- **icmpv4-code** (on page 810)
- **icmpv4-type** (on page 807)
- **icmpv6-code** (on page 817)
- **icmpv6-type** (on page 815)
- **ieee-802.1p** (on page 820)
- **local-gen-pkt**
- **mpls-exp**
- **ip=packet-length**
- **precedence** (on page 823)
- **protocol-ip** (on page 825)
- **source ip** (on page 828)
- **source port** (on page 832)
- **tcp-flags** (on page 832)
- **ttl** (on page 842)

9.4.2. Actions

9.4.2.1. **default-forward-to [not supported in release 8.0.0]**

Description	Determines the next-hop of a filtered packet. The 'default-forward-to' action can set up to three next-hops. The packet will be forwarded according to normal route lookup results if it exists in the FIB. The ABF is applied only to traffic for which, in addition to the ACL match, the fib-lookup on the packet destination-IP had no result, or which hits the default route (0.0.0.0/0).		
Command Syntax	<pre>default-forward-to next-hop1 ip <ip-addr> [vrf <vrf-name>] next-hop2 ip <ip-addr> [vrf <vrf-name>] next-hop3 ip <ip-addr> [vrf <vrf-name>] group <acl-forwarding group name></pre>		
Command Mode	configuration policy acl		
Parameters Table	Name	Value	Description
	next-hop1	IP address or interface name	First priority next-hop IP/interface.
	next-hop2	IP address or interface name	Second priority next-hop IP/interface.
	next-hop3	IP address or interface name	Third priority next-hop IP/interface.
	<i>ip-addr</i>		IP address.
	<i>vrf-name</i>		VRF name to which the IP address is attached.
	<i>acl-forwarding group name</i>		Pre-defined ABF set of next-hops.

Examples	<pre>policy acl MY_ACL_POLICY rule RULE_2 if source-ip eq 101.1.1.6 then default-forward-to group VIP_USERS rule RULE_1 if destination-ip eq 10.10.10.10 then default-forward-to next-hop1 ip 1.1.1.1 end-policy</pre>
Notes	'permit', 'deny', 'forward-to', and 'default-forward-to' actions are mutually exclusive. At least one of them must be defined in a rule. A policy with the default-forward-to action can be attached only to ingress. At least one of the next-hop options or the acl-forwarding group need to be configured. next-hop configuration and group configuration are mutually exclusive.

9.4.2.2. deny

Description	Defines an ACL policy action that drops any packet that matches the rule conditions.
Command Syntax	deny
Command Mode	configuration policy acl
Parameters Table	N/A
Examples	<pre>policy acl MY_ACL_POLICY rule RULE_1 if destination-ip eq 10.10.10.10 then deny end-policy</pre>
Notes	'permit', 'deny', 'forward-to', and 'default-forward-to' actions are mutually exclusive. At least one of them must be defined in a rule.

9.4.2.3. forward-to [not supported in release 8.0.0]

Description	Determines the next-hop of a filtered packet. The 'forward-to' action can set up to three next-hops. The packet will be forwarded to the first reachable next hop. If none of the next-hops is reachable, the forwarding will be based on the normal route lookup decision.		
Command Syntax	<pre>forward-to next-hop1 ip <ip-addr> [vrf <vrf-name>] next-hop2 ip <ip-addr> [vrf <vrf-name>] next-hop3 ip <ip-addr> [vrf <vrf-name>] group <acl-forwarding group name></pre>		
Command Mode	configuration policy acl		
Parameters Table	Name	Value	Description
	next-hop1	IP address or interface name	First priority next-hop IP/interface.
	next-hop2	IP address or interface name	Second priority next-hop IP/interface.
	next-hop3	IP address or interface name	Third priority next-hop IP/interface.
	<i>ip-addr</i>		IP address.
	<i>vrf-name</i>		VRF name to which the IP address is attached.
	<i>acl-forwarding group name</i>		Pre-defined ABF set of next-hops.

Examples	<pre>policy acl MY_ACL_POLICY rule RULE_2 if source-ip eq 101.1.1.6 then forward-to group VIP_USERS rule RULE_1 if destination-ip eq 10.10.10.10 then forward-to next-hop1 ip 1.1.1.1 end-policy</pre>
Notes	'permit', 'deny', 'forward-to', and 'default-forward-to' actions are mutually exclusive. At least one of them must be defined in a rule. A policy with the forward-to action can be attached only to ingress. At least one of the next-hop options or the acl-forwarding group need to be configured. next-hop configuration and group configuration are mutually exclusive.

9.4.2.4. permit

Description	Defines an ACL policy action that allows further processing for packets that match the rule conditions.
Command Syntax	<code>permit</code>
Command Mode	configuration policy acl
Examples	<pre>policy acl MY_ACL_POLICY rule RULE_1 if source-ip eq 10.2.2.2 then permit end-policy</pre>
Notes	'permit', 'deny', 'forward-to', and 'default-forward-to' actions are mutually exclusive. At least one of them must be defined in a rule.

9.4.3. ctrl-plane policy acl

Description	Applies an ACL to the ctrl-plane interface.		
Command Syntax	ctrl-plane policy acl {egress nni-ingress} <i>policy-acl-name</i> [nni-ingress]		
Command Mode	configuration policy acl		
Parameters Table	Name	Value	Description
	<i>policy-acl-name</i>		Reference to ACL policy.
Examples	<pre>exaware(config)# ctrl-plane policy acl nni-ingress MY_MGMT_ACL_POLICY exaware(config)# ctrl-plane policy acl egress MY_MGMT_ACL_POLICY</pre>		
Notes	Egress mode means the ACL is applied on the egress direction of the ctrl-plane interface, meaning traffic directed to the CPU. Nni-ingress mode means the ACL will be applied on the ingress direction of all network interfaces. In this case the default rule of the policy action is set to NA. The ACL attached to the control-plane does not apply to the management interface.		

9.5. ACL Policy Operational Commands

9.5.1. clear acl counters

Description	Resets ACL policy counters.		
Command Syntax	clear acl counters {all interface <interface-name> [in out]}		
Command Mode	Operational		
Parameters Table	Name	Value	Description
		all	Clears all counters.
	interface-name		Name of interface to which the ACL is attached.
		in out	Policy direction on interface; ingress or egress.
Examples	To clear all ACL counters: <pre>exaware# clear acl counters all</pre> To clear only the ingress counters on a specific interface: <pre>exaware# clear acl counters interface x-eth 0/0/1 in</pre>		

9.5.2. clear acl forwarding counters

Description	Resets ACL-based Forwarding policy counters.		
Command Syntax	<pre>clear acl forwarding counters {all interface <interface-name> [in]}</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
		all	Clears all counters.
	<i>interface-name</i>		Name of interface to which the ACL is attached.
	direction	in	Ingress direction on the interface.
Examples	To clear all ABF counters: <pre>exaware# clear acl forwarding counters all</pre> To clear only the ingress counters on a specific interface: <pre>exaware# clear acl forwarding counters interface x-eth 0/0/0 in</pre>		

9.5.3. clear ctrl-plane acl counters

Description	Resets management ACL policy counters.
Command Syntax	<code>clear ctrl-plane acl counters</code>
Command Mode	operational
Notes	The command 'clear ctrl-plane acl counters' clears control plain ACL counters. The existing command 'clear acl counters all' does not clear control plain ACL counters.

9.6. ACL Policy Show Commands

9.6.1. show acl forwarding interface

Description	Displays ABF counters per ACL rule where ABF is configured.																																											
Command Syntax	show acl forwarding interface [brief detail[interface-instance]]																																											
Command Mode	operational																																											
Parameters Table	N/A																																											
Examples	The following examples show sample output from this command:																																											
exaware# show acl forwarding interface detail																																												
<table><tr><td></td><td></td><td></td><td>PKT</td><td>BYTE</td></tr><tr><td>INTERFACE</td><td>POLICY</td><td>RULE</td><td>COUNT</td><td>COUNT</td></tr><tr><td colspan="5">-----</td></tr><tr><td>x-eth0/1/1</td><td>VIP_ABF</td><td>RULE_1</td><td>100</td><td>4200</td></tr><tr><td></td><td></td><td>RULE_2</td><td>1550</td><td>24959</td></tr><tr><td>x-eth0/1/1</td><td>VIP2_ABF</td><td>RULE_1</td><td>234</td><td>2334</td></tr><tr><td>x-eth0/2/0</td><td>MY_ABF</td><td>RULE_1</td><td>165</td><td>8240</td></tr><tr><td></td><td></td><td>RULE_2</td><td>8790</td><td>65449</td></tr></table>								PKT	BYTE	INTERFACE	POLICY	RULE	COUNT	COUNT	-----					x-eth0/1/1	VIP_ABF	RULE_1	100	4200			RULE_2	1550	24959	x-eth0/1/1	VIP2_ABF	RULE_1	234	2334	x-eth0/2/0	MY_ABF	RULE_1	165	8240			RULE_2	8790	65449
			PKT	BYTE																																								
INTERFACE	POLICY	RULE	COUNT	COUNT																																								

x-eth0/1/1	VIP_ABF	RULE_1	100	4200																																								
		RULE_2	1550	24959																																								
x-eth0/1/1	VIP2_ABF	RULE_1	234	2334																																								
x-eth0/2/0	MY_ABF	RULE_1	165	8240																																								
		RULE_2	8790	65449																																								
exaware# show acl forwarding interface detail x-eth0/1/1																																												
<table><tr><td></td><td></td><td></td><td>PKT</td><td>BYTE</td></tr><tr><td>INTERFACE</td><td>POLICY</td><td>RULE</td><td>COUNT</td><td>COUNT</td></tr><tr><td colspan="5">-----</td></tr><tr><td>x-eth0/1/1</td><td>VIP_ABF</td><td>RULE_1</td><td>100</td><td>4200</td></tr><tr><td></td><td></td><td>RULE_2</td><td>1550</td><td>24959</td></tr><tr><td>x-eth0/1/1</td><td>VIP2_ABF</td><td>RULE_1</td><td>234</td><td>2334</td></tr></table>								PKT	BYTE	INTERFACE	POLICY	RULE	COUNT	COUNT	-----					x-eth0/1/1	VIP_ABF	RULE_1	100	4200			RULE_2	1550	24959	x-eth0/1/1	VIP2_ABF	RULE_1	234	2334										
			PKT	BYTE																																								
INTERFACE	POLICY	RULE	COUNT	COUNT																																								

x-eth0/1/1	VIP_ABF	RULE_1	100	4200																																								
		RULE_2	1550	24959																																								
x-eth0/1/1	VIP2_ABF	RULE_1	234	2334																																								
exaware# show acl forwarding interface brief																																												
<table><tr><td>INTERFACE</td><td>POLICY</td><td>FORWARD-TO</td></tr><tr><td></td><td></td><td>COUNT</td></tr><tr><td colspan="3">-----</td></tr><tr><td>x-eth0/0/0</td><td>pol1</td><td>88446744</td></tr><tr><td>x-eth0/0/1</td><td>pol1</td><td>356344</td></tr><tr><td>x-eth0/2/0</td><td>pol1</td><td>4907023</td></tr></table>					INTERFACE	POLICY	FORWARD-TO			COUNT	-----			x-eth0/0/0	pol1	88446744	x-eth0/0/1	pol1	356344	x-eth0/2/0	pol1	4907023																						
INTERFACE	POLICY	FORWARD-TO																																										
		COUNT																																										

x-eth0/0/0	pol1	88446744																																										
x-eth0/0/1	pol1	356344																																										
x-eth0/2/0	pol1	4907023																																										

```
exaware# show acl forwarding interface brief x-eth0/2/0
```

INTERFACE	POLICY	FORWARD-TO COUNT

x-eth0/2/0	pol1	18446744

9.6.2. show acl forwarding policy

Description	Displays the active next-hop per policy per ACL rule where ABF is configured.					
Command Syntax	show acl forwarding policy [<i>policy-name</i>]					
Command Mode	operational					
Parameters Table	N/A					
Examples	The following examples show sample output from this command:					
exaware# show acl forwarding policy						
POLICY	RULE	NEXTHOP	VRF	VIA	INTERFACE	LABEL

POL_1	RULE_1	1.1.1.1	my_vrf_1	3.3.3.3	x-eth0/1/1	50
				4.4.4.4	x-eth0/1/1	60
				5.5.5.5	x-eth0/1/2	
POL_2	RULE_2	2001:0DB8:AC10:FE01::	my_vrf_2	fe80::123:122	x-eth0/1/3	
	RULE_1	-	-	FIB-FORWARDING		
	RULE_2	11.55.33.1	default	11.55.33.1	x-eth0/1/4	
exaware# show acl forwarding policy POL_1						
POLICY	RULE	NEXTHOP	VRF	VIA	INTERFACE	LABEL

POL_1	RULE_1	1.1.1.1	my_vrf_1	3.3.3.3	x-eth0/1/1	50
				4.4.4.4	x-eth0/1/1	60
				5.5.5.5	x-eth0/1/2	
	RULE_2	2001:0DB8:AC10:FE01::	my_vrf_2	fe80::123:122	x-eth0/1/3	

9.6.3. show acl policy

Description	Displays information about the ACL policies in table format.				
Command Syntax	show acl policy [brief]				
Command Mode	operational				
Parameters Table	N/A				
Examples	The following example shows sample output from this command:				
exaware# show acl policy					
INTERFACE	DIR	POLICY	RULE	ACTION	HIT-COUNT

x-eth0/0/1	in	MY_ACL_POLICY_IN	RULE_1	deny	900
	RULE_2	deny			120
	RULE_3	deny			5
	default	permit			5520
x-eth0/0/1	out	MY_ACL_POLICY_OUT	RULE_1	deny	100
	RULE_2	deny			105
	RULE_3	next-hop			2000
	default	permit			4007
x-eth0/0/2	out	MY_ACL_POLICY	MY_WORST	permit	50
	default	deny			1000

9.6.4. show configuration policy acl

Description	Displays all ACL configured policies. You can copy the output and use it for new configurations.		
Command Syntax	<code>show configuration policy acl</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>interface-name</i>		Name of interface to which the ACL is attached
		in out	Policy direction on interface; ingress or egress

Examples	The following example shows sample output from this command:
<pre>exaware# show configuration policy acl policy acl MY_ACL_POLICY_IN rule RULE_1 if source-ip exist-in \$MY_IP_PREFIX_LIST then deny rule RULE_2 if condition MY_ACL_CONDITION_1 then deny rule RULE_3 if condition MY_ACL_LONG_CONDITION then deny rule rule-default permit end-policy ! policy acl MY_ACL_POLICY_OUT rule RULE_1 if destination-ip exist-in (10.8.0.0/32) then deny rule RULE_2 if condition MY_ACL_CONDITION_2 then deny rule RULE_3 if source-ip exist-in (10.1.0.0/16, 10.2.0.0/16) then deny rule rule-default permit end-policy !</pre>	

9.6.5. show ctrl-plane acl

Description	Displays all control-plane ACL configured policies.			
Command Syntax	show ctrl-plane acl [brief detail]			
Command Mode	operational			
Parameters Table	N/A			
Examples	The following examples show sample output from this command:			
exaware# show ctrl-plane acl detail				
mode	POL	RULE	ACTION	HIT-COUNT

nni-ingress	MY_NNI_INGRESS_POLICY	RULE_1	deny	950
		default	NA	
egress	MY_MGMT_ACL_POLICY	RULE_1	deny	900
		RULE_2	permit	120
		RULE_3	permit	5
		default	deny	5520
exaware# show ctrl-plane acl brief				
mode	POL	PERMIT	DENY	

nni-ingress	MY_NNI_INGRESS_POLICY	0	950	
egress	MY_MGMT_ACL_POLICY	125	6420	

9.7. General Routing Policy Actions

9.7.1. deny

Description	Defines a route policy action that drops any route that matches the rule conditions.
Command Syntax	deny
Command Mode	configuration policy route
Parameters Table	N/A
Examples	<pre>policy route MY_ROUTE_POLICY rule RULE_1 if prefix exist-in \$MY_IP_PREFIX_LIST then deny end-policy</pre>
Notes	'deny' action and 'permit' action are mutually exclusive.

9.7.2. default-deny

Description	Defines a route policy action that changes the default action to deny, and continues evaluation.
Command Syntax	default-deny
Command Mode	configuration policy route
Parameters Table	N/A
Examples	<pre>policy route MY_ROUTE_POLICY rule RULE_1 if prefix exist-in \$MY_IP_PREFIX_LIST then default-deny end-policy</pre>

9.7.3. default-permit

Description	Defines a route policy action that changes the default action to permit, and continues evaluation.
Command Syntax	default-permit
Command Mode	configuration policy route
Parameters Table	N/A
Examples	<pre>policy route MY_ROUTE_POLICY rule RULE_1 if prefix exist-in \$MY_IP_PREFIX_LIST then default-permit end-policy</pre>

9.7.4. permit

Description	Defines a route policy action that allows further processing for routes that match the rule conditions.
Command Syntax	<code>permit</code>
Command Mode	configuration policy route
Parameters Table	N/A
Examples	<pre>policy route MY_BGP_POLICY rule RULE_1 if prefix exist-in \$MY_IP_PREFIX_LIST then permit end-policy</pre>
Notes	'deny' action and 'permit' action are mutually exclusive.

9.8. BGP Policy Configuration Commands: Routing Conditions

9.8.1. as-path

Description	Matches on the AS number of a route.		
Command Syntax	as-path [not] [include-any include-all learned-from originated-in empty] [<i>inline-as-path-group</i> <i>named-as-path-group</i> <i>\$parameter</i>]		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>inline-as-path-group</i>	list of AS numbers	AS number, or as-path expression, keyword, or list separated by spaces.
	<i>named-as-path-group</i>	as-path-group	User-defined as-path group or list of as-path groups separated by spaces.
	<i>parameter</i>		Parameter containing an AS number or name of an as-path group.

Examples	If the elements in the group are numeric values, then a match occurs if at least one AS number matches one of the values on the list: <pre>rule RULE_1 if as-path include-all (1200, 1400, 1600) then...</pre> If the element in the group is a range of as numbers, then a match occurs if at least one AS number falls in the given range: <pre>rule RULE_2 if as-path include-any (64512-65535) then...</pre> This example illustrates using a named as-path group: <pre>rule RULE_3 if include-any MY_AS_PATH_GROUP_NAME then...</pre> There are several keywords that can be used in place of regular as-path expressions: <pre>as-path empty =as-path ^\$ as-path originated-in 100 =as-path _100\$ as-path learned-from 100= as-path ^100_</pre> Multiple as-path match items can be defined in one rule: <pre>Rule ONE if match-any as-path include-any MY_AS_PATH_GROUP_NAME1 as-path include-any MY_AS_PATH_GROUP_NAME2 then ...</pre>						
Expressions	Commonly used regular as-path expressions:						
	<table> <tr> <th data-bbox="409 1627 638 1675">Expression</th><th data-bbox="638 1627 1396 1675">Meaning</th></tr> <tr> <td data-bbox="409 1675 638 1751">.*</td><td data-bbox="638 1675 1396 1751">Anything</td></tr> <tr> <td data-bbox="409 1751 638 1820">^\$</td><td data-bbox="638 1751 1396 1820">Locally originated routes (empty)</td></tr> </table>	Expression	Meaning	.*	Anything	^\$	Locally originated routes (empty)
Expression	Meaning						
.*	Anything						
^\$	Locally originated routes (empty)						

	<code>^100_</code>	Learned from AS 100
	<code>_100\$</code>	Originated in AS 100
	<code>_100_</code>	Passes through AS 100
	<code>^[0-9]+\$</code>	Directly connected AS's
Notes	The named as-path group is defined using the <i>policy group bgp-as-path</i> (on page 592) command. Each of the inline as-path group elements can use local/global parameters. Learned-from: The leftmost as-path value matches an as-path element. Originated-in: The rightmost as-path value matches an as-path element.	

9.8.2. as-path-length

Description	Matches on the number of ASNs in the AS path attribute.		
Command Syntax	as-path-length [not] [eq ge gt le lt] [<i>number</i> <i>\$parameter</i>]		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>number</i>	integer	AS path length.
	<i>parameter</i>		Parameter containing an AS path length.
Examples	Multiple as-path lengths can be defined in one rule: <pre>policy route MY_BGP_POLICY Rule ONE if match-any as-path-length eq 20 as-path-length eq 40 then ...</pre>		

9.8.3. as-path-unique-length

Description	Matches on the number of unique ASNs in the AS path attribute.		
Command Syntax	as-path-unique-length [not] [eq ge gt le lt] [<i>number</i> <i>\$parameter</i>]		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>number</i>	integer	AS path length.
	<i>parameter</i>		Parameter containing an AS path length.
Examples	Multiple as-path lengths can be defined in one rule: <pre> policy route MY_BGP_POLICY Rule ONE if match-any as-path-unique-length eq 20 as-path-unique-length eq 40 then ... </pre>		

9.8.4. community

Description	Matches on the community attribute of a route. A community is a 4-byte value, expressed as two unsigned decimal integers in the range 0 to 65535 separated by a colon. Examples: 100:200, 1400:1000		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>inline-community-group</i>	list of community numbers	Community number or list separated by spaces or well-known values: internet, no-advertise, no-export, no-export-subconfed.
	<i>named-community-group</i>	community group	User-defined community group or list of community groups separated by spaces.
	<i>parameter</i>		Parameter containing a community number or name of a BGP community group.
Examples	<pre>policy route MY_BGP_POLICY(\$as_num, \$tag) rule ONE if community include-any (\$as_num:\$tag, 100:1200, 200:1400) then ... rule TWO if community empty then</pre>		

Notes	The named community group is defined using the <i>policy group bgp-community</i> (on page 595) command. Each of the inline community group elements can use local/global parameters.
--------------	---

9.8.5. extcommunity

Description	Matches on the extended community attribute of a route. An extended community is similar to a regular BGP community, but contains more data (eight bytes vs four bytes on the normal community), and has a different structure for encoding types and values. Two types of extended communities are supported: • target - (route-target extcommunity) • origin - (soo extcommunity) The extended community format is: type:administrator:tag where type can be either origin (soo extcommunity) or target (route-target extcommunity). Administrator:tag pair can be: <2-byte number>:<4-byte number> (2-byte ASN notation) <4-byte number>L:<2-byte number> (4-byte ASN notation) <IPv4 address>:<2-byte number> (IPv4 address notation)		
Command Syntax	<pre>extcommunity [not] [include-any include-all empty empty-origin empty-target] [inline-extcommunity-group named-extcommunity-group \$parameter]</pre>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>inline-extcommunity-group</i>	list of extended community numbers	Extended community number or list separated by spaces.

	<i>named-extcommunity-group</i>	extended community group	User-defined extended community group or list of extended community groups separated by spaces.
	<i>parameter</i>		Parameter containing an extended community number or name of a BGP extended community group.
Examples	<pre> policy route MY_BGP_POLICY(\$as_num, \$tag) rule ONE if if extcommunity include-any (origin:12000L:*, target:3333L:*) then .. </pre>		
Notes	The named extended community group is defined using the <i>policy group bgp-extcommunity</i> (on page 598) command. Each of the inline community group elements can use local/global parameters. Using the EMPTY options: • extcommunity empty evaluates to true if no extended community attribute is set for the route. • extcommunity empty-target evaluates to true if no target extended community attribute is set for the route. • extcommunity empty-origin evaluates to true if no origin extended community attribute is set for the route.		

9.8.6. local-preference

Description	Matches on the local preference attribute of a BGP route.		
Command Syntax	local-preference [eq ge gt le lt] [<i>number</i> <i>\$parameter</i>]		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>number</i>	0 - 4294967295	Local preference value.
	<i>parameter</i>		Parameter containing local preference value.
Examples	<pre> policy route MY_BGP_POLICY rule ONE if match-any local-preference eq 40 local-preference eq 50 then ... </pre>		

9.8.7. med

Description	Matches on the Multi Exit Discriminator (MED) attribute of the route.		
Command Syntax	med [not] [eq ge gt le lt] [<i>number</i> <i>\$parameter</i>]		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>number</i>	0 - 4294967295	MED number.
	<i>parameter</i>		Parameter containing MED number.
Examples	Multiple MED match items can be defined in one rule: <pre> policy route MY_BGP_POLICY Rule ONE if match-any med eq 20 med eq 40 then ... </pre>		

9.8.8. next-hop

Description	Matches on the next hop associated with the route.		
Command Syntax	<pre>next-hop [not] exist-in [<i>inline-ip-prefix-group</i> <i>named-ip-prefix-group</i> <i>\$parameter</i>]</pre> <pre>next-hop [not] eq <i>ip-address</i></pre>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>inline-ip-prefix-group</i>	IP prefix or list of IP prefixes	Prefix (IP address/mask length) or list of IP prefixes separated by spaces.
	<i>named-ip-prefix-group</i>	ip-prefix-group	User-defined ip-prefix group or list of ip-prefix groups separated by spaces.
	<i>parameter</i>		Parameter containing an IPv4 or IPv6 prefix or name of an ip-prefix group.
	<i>ip-address</i>	IP address	Specific IPv4 or IPv6 IP address to match.
Examples	<pre>policy route MY_BGP_POLICY rule RULE_1 if match-all next-hop exist-in (10.0.0.0/8) next-hop exist-in MY_PREFIX_GROUP then...</pre>		

Notes

The named ip-prefix group is defined using the ***policy group ip-prefix*** (on page 602) command.

The 'exist-in' can be followed by a list of groups. Each group in the list can be either inline or named ip-prefix group. The group names in the list must be separated by spaces.

Each of the inline prefix group elements can use local/global parameters.

9.8.9. orf-prefix

Description	Matches on the IP prefix in the context of the ORF policy attachment point.		
Command Syntax	orf-prefix exist-in [<i>inline-ip-prefix-group</i> <i>named-ip-prefix-group</i> <i>\$parameter</i>]		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>inline-ip-prefix-group</i>	IP prefix or list of IP prefixes	Prefix (IP address/mask length) or list of IP prefixes separated by spaces.
	<i>named-ip-prefix-group</i>	ip-prefix-group	User-defined ip-prefix group or list of ip-prefix groups separated by spaces.
	<i>parameter</i>		Parameter containing an IP prefix or name of an ip-prefix group.

Examples	<pre> policy route MY_BGP_POLICY rule ONE if match-any orf-prefix exist-in (10.10.10.1/32, 22.22.22.0/24 matching-len 24-32) then permit rule TWO if match-any orf-prefix exist-in (\$prefix, 10.10.0.0/16) then deny rule THREE if match-any orf-prefix exist-in MY_PREFIX_GROUP then deny rule FOUR if match-any orf-prefix exist-in \$MY_PREFIX_GROUP_NAME then permit </pre>
Notes	• orf-prefix cannot be used in the context of the logical operator "match-all". • orf-prefix cannot be used in the context of a compound condition. • The named ip-prefix group is defined using the policy group ip-prefix (on page 602) command. • The 'exist-in' can be followed by a list of groups. Each group in the list can be either inline or named ip-prefix group. The group names in the list must be separated by spaces. See configuration example. • Each of the inline prefix group elements can use local/global parameters.

9.8.10. origin

Description	Matches on the origin attribute of the route.
Command Syntax	<code>origin [not] eq [egp igp incomplete]</code>
Command Mode	configuration policy route
Parameters Table	N/A
Examples	Multiple origin match items can be defined in one rule: <pre>policy route MY_BGP_POLICY Rule ONE if match-any origin eq egp origin eq incomplete then ...</pre>

9.8.11. peer

Description	Matches on neighbor IP address.		
Command Syntax	<pre>peer exist-in [<i>inline-ip-prefix group</i> <i>named-ip-prefix-group</i> \$parameter] peer eq [<i>ip-address</i> <i>\$parameter</i>]</pre>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>inline-ip-prefix-group</i>	prefix or list of prefixes	Prefix (IP address/mask length) or list of prefixes separated by spaces.
	<i>named-ip-prefix-group</i>	ip-prefix-group	User-defined ip-prefix group or list of ip-prefix groups separated by spaces.
	<i>parameter</i>		Parameter containing a prefix or name of an ip-prefix group.
	<i>ip-address</i>	IP address	Specific IP address to match.

Examples	<pre>policy route MY_BGP_POLICY rule ONE if ... peer eq 10.10.1.1 then ... rule TWO if ... peer exist-in (10.10.1.0/24) then ... rule THREE if ... peer exist-in \$MY_PEER_GROUP then ...</pre>
Notes	The named ip-prefix group is defined using the <i>policy group ip-prefix</i> (on page 602) command. The 'exist-in' can be followed by a list of groups. Each group in the list can be either inline or named ip-prefix group. The group names in the list must be separated by spaces. See configuration example. Each of the inline prefix group elements can use local/global parameters. The <code>matching-len</code> keyword is not supported.

9.8.12. prefix

Description	Matches on the route prefix, known as the network-layer reachability information (NLRI). The prefix comprises an IP address and a mask length.		
Command Syntax	<pre>prefix [not] exist-in [<i>inline-ip-prefix-group</i> <i>named-ip-prefix-group</i> <i>\$parameter</i>]</pre>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>inline-ip-prefix-group</i>	IP prefix or list of IP prefixes	Prefix (IP address/mask length) or list of IP prefixes separated by spaces.
	<i>named-ip-prefix-group</i>	ip-prefix-group	User-defined ip-prefix group or list of ip-prefix groups separated by spaces.
	<i>parameter</i>		Parameter containing an IPv4 or IPv6 prefix or name of an ip-prefix group.

Examples	<pre> policy group ip-prefix IPV4_PFX_GRP1 10.20.0.0/16 30.40.0.0/16 end-group policy group ip-prefix IPV6_PFX_GRP2 2001:0db8:85a3:0:0:8a2e:0370:7334 1050:0:0:0:5:600:300c:326b end-group policy route MY_BGP_POLICY rule RULE_1 if match-all prefix exist-in (20.20.20.0/24, 30.30.30.0/24) (IPV4_PFX_GRP1 IPV6_PFX_GRP2) then... rule RULE_2 prefix exist-in (1050::5:600:0:0/32, 1000::1:1::0:0/32) , 30.30.30.0/24) then... </pre>
Notes	The named ip-prefix group is defined using the <i>policy group ip-prefix</i> (on page 602) command. The 'exist-in' can be followed by a list of groups. Each group in the list can be either inline or named ip-prefix group. The group names in the list must be separated by spaces. See configuration example. Each of the inline prefix group elements can use local/global parameters.

9.8.13. rib-route

Description	Matches on IPv4 or IPv6 prefixes in the RIB. This condition is supported for both BGP and ISIS routes.		
Command Syntax	<pre>rib-route [not] [protocol {isis bgp connected ospf ospfv3 rsvp static}] [include-any include-all] [inline-ip-prefix group named-ip-prefix-group \$parameter]</pre>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>inline-ip-prefix-group</i>	IP prefix or list of IP prefixes	Prefix (IP address/mask length) or list of IP prefixes separated by spaces.
	<i>named-ip-prefix-group</i>	ip-prefix-group	User-defined ip-prefix group or list of ip-prefix groups separated by spaces.
	<i>parameter</i>		Parameter containing an IP prefix or name of an ip-prefix group.
Examples	Multiple rib-route conditions can be defined in a rule: <pre>policy route MY_BGP_POLICY rule ONE if match-any rib-route include-any (10.0.0.0/8, 20.0.0.0/8) rib-route include-any MY_PREFIX_GROUP rib-route protocol bgp include-any (30.0.0.0/8) rib-route include-all (100.0.0.0/8, 200.10.0.0/16) then ...</pre>		

Notes	The named ip-prefix group is defined using the <i>policy group ip-prefix</i> (on page 602) command. Each of the inline prefix group elements can use local/global parameters. If no protocol is specified, the condition matches on any route source.
--------------	---

9.8.14. rpki-origin-as

Description	Matches on the RPKI Origin AS number of a route.		
Command Syntax	<code>rpki-origin-as eq {not-found invalid}</code>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	not-found		The rule is applied only to routes that RPKI does not contain.
	invalid		The rule is applied only to routes rejected by RPKI.
Examples	This example shows how to use this command: <pre>rule RULE_1 if rpki-origin-as eq not-found then deny</pre>		
Note	This rule is applied only when it is attached to point 'neighbor in'.		

9.8.15. tag

Description	Matches on the tag attribute of the RIB route.		
Command Syntax	tag [eq ge gt le lt] [<i>number</i> <i>\$parameter</i>]		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>number</i>	0 - 4294967295	Tag value.
	<i>parameter</i>		Parameter containing tag value.
Examples	Multiple tag match items can be defined in one rule: <pre> policy route MY_BGP_POLICY rule ONE if match-any tag eq 40 tag eq 30 then ... rule TWO if tag ge 40 tag le 60 ... then ... </pre>		

9.9. BGP Policy Configuration Commands: Actions

9.9.1. as-path

Description	Manipulates the AS path attribute of a route. Two actions are supported: • prepend: Prepends the specified additional AS numbers or as-path to the existing as-path. • replace: Replaces the sequence of AS path segments in the existing as-path with the local AS number. Can be defined more than once in a rule.		
Command Syntax	<pre>as-path prepend [<i>as-number</i> <i>\$parameter1</i> last-as] [count {<i>number</i> <i>\$parameter2</i>}]</pre> <pre>as-path prepend path [<i>as-path-string</i>]</pre> <pre>as-path replace [<i>as-path-string</i> private-as]</pre>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>as-path-string</i>	as-path segments	Sequence of one or more as-path segments to be replaced.
	<i>as-number</i>	AS number	Specific AS number to prepend.
	<i>parameter1</i>		Parameter containing an AS number.
		last-as	Prepend the last AS to the as-path before the local AS is added.
		count	Specifies the number of times to prepend the AS number.
	<i>number</i>	Integer	Number of times to prepend the AS number.
	<i>parameter1</i>		Parameter containing the number of times to prepend the AS number.

		private-as	Replace private AS numbers with the local AS number.
--	--	------------	--

Examples	Prepend the AS number <i>1880</i> before local as number is added <pre>rule RULE_1 if ... then as-path prepend 1880</pre> Prepend as number <i>\$as_num</i> <i>n</i> times. (where <i>as_num</i> and <i>n</i> are parameters): <pre>rule RULE_2 if ... then as-path prepend \$as_num count \$n</pre> Prepend the last AS 400 times before local AS is added. <pre>rule RULE_3 if ... then as-path prepend last-as count 400</pre> Replace the sequence of '23 1220 1000' with local AS number. <pre>rule RULE_4 if ... then as-path replace "23 1220 1000"</pre> Replace any sequence of private AS numbers with local AS number. <pre>rule RULE_4 if ... then as-path replace private-as</pre> Replace the sequence of two as-path segments: "3 1220" and "1000" with local as number. <pre>rule RULE_4 if ... Then as-path replace "(23 1220) 1000" "(23 1220) 1000"</pre>
------------------------	--

Expression	Commonly used regular as-path expressions:	
	Expression	Meaning
	. *	Anything
	^\$	Locally originated routes (empty)
	^100_	Learned from AS 100
	_100\$	Originated in AS 100
	100	Passes through AS 100
	^[0-9]+\$	Directly connected AS's
Notes	The named as-path group is defined using the policy group bgp-as-path (on page 592) command. Each of the inline as-path group elements can use local/global parameters. Learned-from: The leftmost as-path value matches an as-path element. Originated-in: The rightmost as-path value matches an as-path element.	

9.9.2. community

Description	Sets the community attribute of a route. A community is a 32-bit value carried in BGP routes. Three actions are supported: • set: Replaces the current community values with specified values. Can be defined only one time per rule. • add: Adds community values to the existing attribute values. Can be defined more than one time per rule. • delete: Deletes values from the current community attribute all: Removes all communities values associated with the route. Can be defined only one time per rule. group: Removes any community that matches any element in the community group. Can be defined more than one time per rule. not-in-group: Removes all communities that don't match any of the elements in the community group. Can be defined more than one time per rule.
Command Syntax	<pre>community set [<i>inline-community-group</i> <i>named-community-group</i> \$parameter] community add [<i>inline-community-group</i> <i>named-community-group</i> \$parameter] community delete [all group {<i>inline-community-group</i> <i>named- community-group</i> \$parameter} not-in-group {<i>inline-community-group</i> <i>named-community-group</i> \$parameter}]</pre>
Command Mode	configuration policy route

Parameters Table	Name	Value	Description
	<i>inline-community-group</i>	list of community numbers	Community number or list separated by spaces.
	<i>named-community-group</i>	community group	User-defined community group or list of community groups separated by spaces.
	<i>parameter</i>		Parameter containing a community value or name of a BGP community group.
Examples	Delete any communities listed in named community group. <pre>policy route MY_BGP_POLICY rule ONE if ... then community delete group MY_COMMUNITY_GROUP</pre> Delete any communities listed in inline community group. <pre>policy route MY_BGP_POLICY rule ONE if ... then community delete group (10:*, 1000:2000, '[100-200]':1000, no-export, '123[4 5 6]':*)</pre> Add community values to the existing community attribute of a route. <pre>policy route MY_BGP_POLICY rule ONE if ... then community add (100:200, 3000:5000)</pre>		

Notes	The named community group is defined using the <i>policy group bgp-community</i> (on page 595) command.
-------	--

9.9.3. damping

Description	Specifies the the route dampening behavior.		
Command Syntax	damping set [half-time {minutes1 \$parameter1} max-suppress {minutes2 \$parameter2} reuse {penalty-units1 \$parameter3} suppress {penalty-units2 \$parameter4}]		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
		half-time	Sets the time after which the penalty value is halved if a route stays stable.
		max-suppress	Sets the maximum time in minutes that a route can be suppressed no matter how unstable it has been prior to this period of stability.
		reuse	Penalty threshold, below which a suppressed route is re-advertised.
		suppress	Threshold above which a route advertisement will be suppressed.
	minutes1	integer	Half-time value in minutes. (Default = 15 min)
	\$parameter1		Parameter containing half-time value.
	minutes2	integer	Max-suppress time in minutes. (Default = 60 min)

	<i>\$parameter</i> 2		Parameter containing max-suppress time value.
	<i>penalty-units1</i>	integer	Reuse penalty threshold. (Default = 750)
	<i>\$parameter</i> 3		Parameter containing reuse penalty threshold value.
	<i>penalty-units2</i>	integer	Suppress penalty threshold. (Default = 3000)
	<i>\$parameter</i> 4		Parameter containing suppress penalty threshold value.
Examples	<pre>rule RULE_1 if ... then damping set half-time 10 max-suppress 45 reuse 600 suppress 3500</pre>		

9.9.4. extcommunity

Description	Sets the extended community attribute of a route. An extended community is similar to a regular BGP community, but contains more data (eight bytes vs four bytes on the normal community), and has a different structure for encoding types and values. Two types of extended communities are supported: • route target - (route-target extcommunity) • route origin - (soo extcommunity) Three actions are supported: • set: Replaces the current extended community values with specified values. Can be defined only one time per rule. • add: Adds extended community values to the existing attribute values. Can be defined more than one time per rule. • delete: Deletes values from the current extended community attribute - all: Removes all extended communities values associated with the route. Can be defined only one time per rule. - group: Removes any extended community that matches any element in the extended community group. Can be defined more than one time per rule. - not-in-group: Removes all extended communities that don't match any of the elements in the extended community group. Can be defined more than one time per rule. • The extended community format is: type:administrator:tag where type can be either origin (soo extcommunity) or target (route-target extcommunity). • Administrator:tag pair can be: <2-byte number>:<4-byte number> (2-byte ASN notation) <4-byte number>L:<2-byte number> (4-byte ASN notation) <IPv4 address>:<2-byte number> (IPv4 address notation)
--------------------	--

Command Syntax	<pre> extcommunity set [<i>inline-extcommunity-group</i> <i>named-extcommunity-group</i> <i>\$parameter</i>] extcommunity add [<i>inline-extcommunity-group</i> <i>named-extcommunity-group</i> <i>\$parameter</i>] extcommunity delete [all group {<i>inline-extcommunity-group</i> <i>named-extcommunity-group</i> <i>\$parameter</i>} not-in-group {<i>inline-extcommunity-group</i> <i>named-extcommunity-group</i> <i>\$parameter</i>}] </pre>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>inline-extcommunity-group</i>	list of community values	Extended community value or list separated by spaces.
	<i>named-extcommunity-group</i>	extended community group	User-defined extended community group.
	<i>parameter</i>		Parameter containing all or part of (ASN, tag or IP) a community value or name of a BGP extended community group.

Examples	Delete RT extended community values defined in inline group, including parameters. <pre>policy route MY_BGP_POLICY rule ONE if ... then extcommunity delete group (target:67:29, target:\$ip_addr:\$tag)</pre> Add extended community values using named group. <pre>policy group bgp-extcommunity MY_EXTCOMMUNITY_GROUP origin:1000:2000 origin:12:65 origin:4.3.2.1:100 end-group</pre> <pre>policy route MY_BGP_POLICY rule ONE if ... then extcommunity add group MY_EXTCOMMUNITY_GROUP</pre>
Notes	The named extended community group is defined using the <i>policy group bgp-extcommunity</i> (on page 598) command.

9.9.5. local-preference

Description	Sets the local preference attribute of a BGP route.		
Command Syntax	<code>local-preference set [<i>number</i> <i>\$parameter</i>]</code>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>number</i>	0 - 4294967295	Local preference value.
	<i>parameter</i>		Parameter containing local preference value.
Examples	<pre> policy route MY_BGP_POLICY rule ONE if ... then local-preference set 500 </pre>		

9.9.6. med

Description	Sets the Multi Exit Discriminator (MED) attribute of the route. Three actions are supported: • set: set the MED to the specified value. • increase: set the MED to the current MED value plus the specified value (offset). • decrease: set the MED to the current MED value minus the specified value (offset).		
Command Syntax	<pre>med set [number \$parameter igp-cost best worst] med increase [number \$parameter] med decrease [number \$parameter]</pre>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>number</i>	0 - 4294967295	MED number.
	<i>parameter</i>		Parameter containing MED number.
		igp-cost	Set the MED number to the igp cost of the bgp next hop.
		best	Set the MED number to the best value cost = 0.
		worst	Set the MED number to the worst value.
Examples	<pre>policy route MY_BGP_POLICY rule ONE if ... then med set best</pre>		

9.9.7. next-hop

Description	Replace the existing next hop associated with the route with the specified IP address.		
Command Syntax	next-hop set [ipv4-address { <i>ipv4 address</i> <i>\$parameter</i> } ipv6-address { <i>ipv6 address</i> <i>\$parameter</i> } peer-address self]		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>ipv4/6-address</i>	IP address	IP address of next hop.
		peer-address	Set the next hop to the IP address of the remote BGP peer.
		self	Set local ip address to be next hop.
	<i>parameter</i>		Parameter containing an IP address.
Examples	<pre> policy route MY_BGP_POLICY rule RULE_1 if ip-prefix exist-in MY_PREFIFX_GROUP then next-hop set peer-address policy route MY_BGP_POLICY2 rule RULE_1 if ip-prefix exist-in MY_PREFIFX_GROUP then next-hop set ipv4-address 2.2.2.2 </pre>		

9.9.8. origin

Description	Sets the origin attribute of the route.
Command Syntax	<code>origin set [egp igp incomplete]</code>
Command Mode	configuration policy route
Parameters Table	N/A
Examples	<pre>policy route MY_BGP_POLICY rule ONE if ... then origin set igp</pre>

9.9.9. suppress-specific-route

Description	Suppresses the advertisement of more specific routes. This action applies only to aggregate attach-point.
Command Syntax	suppress-specific-routes
Command Mode	configuration policy route
Parameters Table	N/A
Examples	<pre>policy route MY_BGP_POLICY rule ONE if prefix exist-in (10.0.0.0/8) then suppress-specific-routes</pre>

9.9.10. tag

Description	Sets the tag attribute associated with a given route in the RIB.		
Command Syntax	tag set [<i>number</i> <i>\$parameter</i>]		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>number</i>	0 - 4294967295	Tag value.
	<i>parameter</i>		Parameter containing tag value.
Examples	<pre> policy route MY_BGP_POLICY rule ONE if prefix exist-in (10.0.0.0/8) then tag set 40 </pre>		

9.9.11. traffic-counter

Description	Sets traffic counters indexes to routes matching the policy rule in order to track the number of packets/bytes matching these routes. All the packets matched to the rules conditions will count on this traffic-counter. Each traffic-counter will be assigned per interface for which routing-policy-accounting is enabled.		
Command Syntax	<code>traffic-counter set <counter-id></code>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>counter-id</i>	integer (0-63)	Counter bucket number for this rule.
Examples	<pre> policy route my_policy rule my_rule if community include-all (internet) then traffic-counter set 1 rule my_2nd_rule if community include-all (1000:20) then traffic-counter set 2 !</pre>		
Notes	The 'traffic-counter set' action can be used only with a routing policy attached to an "account" attach point.		

9.9.12. unsuppress-specific-routes

Description	This action affects routes that have been suppressed in the generation of BGP route aggregation. When it is attached at a neighbor-out attach point, it guarantees that the routes that it affects are advertised to that neighbor even if the route was suppressed either by using the suppress-route command in a policy at the aggregation attach point or due to "advertise summary" option being activated on the aggregate route.
Command Syntax	<code>unsuppress-specific-routes</code>
Command Mode	configuration policy route
Parameters Table	N/A
Examples	<pre>policy route MY_BGP_POLICY rule ONE if prefix exist-in (10.0.0.0/8) then unsuppress-specific-routes</pre>

9.9.13. weight

Description	Sets the weight for a route.		
Command Syntax	weight set [<i>number</i> <i>\$parameter</i>]		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>number</i>	0 - 65535	Weight.
	<i>parameter</i>		Parameter containing weight value.
Examples	<pre> policy route MY_BGP_POLICY rule ONE if prefix exist-in (10.0.0.0/8) then weight set 40 </pre>		

9.10. BGP Policy Configuration Commands: Attachment Points

A BGP policy can be applied in an attachment point or as a parameter of a CLI command.

The following CLI commands define the attachment points and BGP policies that are applied in these points:

- accounting policy
- aggregate-address
- default-route-originate
- export-policy
- flap-damping
- import-policy
- neighbor in (import)
- neighbor out (export)
- network
- orf
- redistribute

There is no limit on number of policies attached to one entity.

The following CLI commands can specify a policy as their optional parameter:

- import

9.10.1. accounting policy [not supported in release 8.0.0]

Description	The account policy attach point allows the user to assign traffic counters for specific prefixes that are installed in the BGP table (for the routing-policy-accounting feature).
Examples	<pre> policy route my_policy rule my_rule if community include-all (internet) then traffic-counter set 1 rule my_2nd_rule if community include-all (1000:20) then traffic-counter set 2 routing bgp 100 vrf default af-ipv4 unicast accounting policy BGP_ACCOUNT </pre>
Notes	Any action besides 'traffic-counter set' and 'permit/deny/default-permit/default-deny' cannot be used in a routing policy attached to "account" attach point. The permit/deny actions in account policy are used to either accept or reject the traffic counter set by the policy. They will not affect the BGP best path selection in any way. The 'traffic-counter set' action can be used only in a routing policy attached to an "account" attach point.

9.10.2. aggregate-address

Description

The aggregate-address attach point generates an aggregate route to be advertised based on the conditional presence of more specific routes of that aggregate address. Policies attached at this attach point can also set any of the BGP attributes on the aggregate route itself (med, community, etc).

The specified aggregate address is generated if any routes evaluated by the policy are permitted.

Any 'set attribute' action taken on rule match will affect the attributes of the aggregate route. If the '*full*' flag is set on the aggregate-address (it is configured to send both aggregated and more specific routes), it is possible to filter the advertisement of some of the more specific routes by using the 'suppress-specific-route' action.

Examples

In the following example, the aggregate address 10.0.0.0/8 is generated if there are any routes in the range 10.0.0.0/8 matching-len 8-25 with the exception of 10.2.0.0/24.

Since the aggregate-address is set to full, all specific routes of the aggregate address are advertised. However, the specific route 10.1.0.0 is suppressed.

```
policy route AGGREGATE
  rule SET-AGG-COMMUNITY
    if
      prefix exist-in (10.0.0.0/8 matching-len 8-25)
      prefix not exist-in (10.2.0.0/24)
    then community set(10:33)
  rule SUPPRESS
    if prefix exist-in (10.1.0.0/24)
    then suppress-specific-route
  end-policy

routing bgp 1
  vrf default
    af-ipv4 unicast
      aggregate-address 10.0.0.0/8 advertise full policy
      AGGREGATE
```


9.10.3. default-route-originate

Description	The default-route-originate attach point allows the default route (0.0.0.0/0) to be conditionally generated and advertised to a peer, based on the presence of other routes in the RIB. If any route in the policy is permitted, then the default route is generated.
Examples	The following policy generates and sends a default-route to the BGP neighbor 10.0.0.1 if any ISIS routes that match 10.0.0.0/8 matching-len 8-32 are present in RIB. <pre>policy route DEFAULT-ORIGINATE rule CHECK-RIB if rib-route protocol isis include any (10.0.0.0/8 matching-len 8-32) then permit end-policy routing bgp 2 vrf default neighbor 10.0.0.1 remote-as 3 af-ipv4 unicast default-route-originate policy DEFAULT-ORIGINATE</pre>

9.10.4. export-policy

Description	The export policy attach point controls the policy applied to routes prior to being handed over to the export-rt mechanism, and whether they are handed over (permitted) or not (denied).
Examples	<pre>routing bgp 123 vrf VRF1 af-ipv4 unicast export-policy EXPORT-POLICY !</pre>
Notes	Export-policy is not available under VRF default. Routes will go through export-policy, then route-targets will be added according to export-rt statements to the routes permitted by the policy. At the end of these two processes, any route with a route-target attached to it will be exported.

9.10.5. flap-damping

Description	The damping attach point configures the damping behavior. Damping behavior is controlled through following set of parameters: • suppress : penalty threshold, above which a route advertisement will be suppressed. (Default: 3000) • reuse : penalty threshold, below which a suppressed route is re-advertised. (Default: 750) • half-time: Number of minutes after which the penalty value is halved if a route stays stable. (Default: 15) • max-suppress: Maximum time in minutes that a route can be suppressed no matter how unstable it has been prior to this period of stability. (Default: 60)
Examples	The following policy sets damping values so that routes more specific than /25 will take more time to get out of suppressed state than routes that are less specific than /25. The permit action in addition to the damping action will stop the evaluation of the policy. Deny action will not apply any damping on the matched routes. <pre> policy route DAMPING rule SPECIFIC_ROUTES if prefix exist-in (0.0.0.0/0 matching-len 25-32) then damping set half-time 30 permit rule DEFAULT damping set half-time 20 end-policy routing bgp 1 vrf default af-ipv4 unicast flap-damping policy DAMPING </pre>

9.10.6. import-policy

Description	The import policy attach point controls the policy applied to routes imported according to import-rt statements.
Examples	<pre>routing bgp 123 vrf VRF1 af-ipv4 unicast import-policy IMPORT-POLICY !</pre>
Notes	Import-policy is not available under VRF default. Routes will be imported according to import-rt statements, then go through import-policy.

9.10.7. neighbor-in

Description	The neighbor-in (import) attach point controls the reception of routes from a specific peer. All routes that are received by a peer are run through the attached policy. Any route that is permitted will be installed in the BGP routing table (in-RIB), and becomes a candidate to be selected as best BGP path.
Examples	This example shows two import polices attached to peer 10.0.0.1: • The first one sets local preference and med values based on received <i>community</i> attribute. • The second policy drops any prefix that matches 20.0.0.0/8. In addition this policy also adds community (200:1000) to prefixes that match 30.0.0.0/8. <pre> policy route IMPORT_POLICY_COMMON rule COMM_A if community includes-any (30:1000) then local-preference set 200 med set 40 rule DEFAULT local-preference 300 med set 60 end-policy router bgp 1 neighbor 10.0.0.1 remote-as 2 af-ipv4 unicast policy in IMPORT_POLICY_COMMON IMPORT_POLICY_CUSTA policy route IMPORT_POLICY_CUSTA rule DENY_PREF if prefix exist-in (20.0.0.0/8) then deny rule SET_CUST_COMM if prefix exist-in (30.0.0.0/8) then community add (200:1000) permit end-policy </pre>

9.10.8. neighbor-out

Description	The neighbor-out (export) attach point selects the BGP routes to send to a given neighbor. The routes are selected by running the BGP routes through the policy. Any routes that are permitted will be sent as updates to the neighbor. The route attributes can be altered before been sent. Denied routes will not be advertised to the neighbor.
--------------------	---

Examples	In this example, some neighbors are allowed to be advertised with server prefixes and some are not: • A common policy (IPV4-PEER-OUT-COMMON) is applied to all neighbors. It checks mandatory community values, and also sets the default permit action to the server prefixes. • The second policy, which is attached only to neighbors with server prefixes that are not allowed, denies these prefixes, thus preventing them from being advertised to the neighbor. <pre> policy group ip-prefix SERVER_PREFIX 12.1.1.0/24 12.1.2.0/24 end-group policy route IPV4-PEER-OUT-COMMON rule MANDATORY_COMMUNITIES if community not include-any (3:100, *:2000) then deny rule DEF_PERMIT_SERVER_PREF if prefix exist-in SERVER_PREFIX then default-permit end-policy policy route IPV4-PEER_DENY_OUT rule DENY_SERVER_PREF if prefix exist-in SERVER_PREFIX then deny end-policy routing bgp 1 vrf default neighbor 10.0.0.5 remote-as 3 af-ipv4 unicast policy out IPV4-PEER-OUT-COMMON IPV4-PEER_DENY_OUT neighbor 10.0.0.6 remote-as 4 af-ipv4 unicast policy out IPV4-PEER-OUT </pre>
-----------------	---

9.10.9. network

Description	The network attach point controls the injection of routes from the RIB into BGP. A policy attached at this point is able to set any of the valid BGP attributes on the routes that are being injected.
Examples	The following example shows a policy that sets the community to <i>no-export</i> for any routes more specific than /24: <pre>policy route NETWORK_POLICY rule SET_COMMUNITY if prefix exist-in (0.0.0.0/0 matching-len 25-32) then community add (no-export) end-policy routing bgp 2 vrf default af-ipv4 unicast network 180.20.0.32/27 policy NETWORK_POLICY</pre>

9.10.10. orf

Description	The orf attach point provides filtering of incoming prefixes via BGP update. In addition to filtering, the prefixes are sent to upstream neighbors as an Outbound Route Filter (ORF) to allow them to perform filtering before sending downstream.
Examples	This example shows a policy applied to the neighbor ORF attach point: • The prefix of the route is dropped if it matches any prefix in (100.1.1.0/24, 200.1.1.0/24). • In addition to the inbound filtering, each prefix entry is sent to the upstream neighbor with a permit or deny indication, so that the neighbor can filter updates before sending them on to their destination. (RFC 5292) <pre>policy route ORF rule DENY_BLACK_LIST if orf-prefix exist-in (100.1.1.0/24, 200.1.1.0/24) then deny end-policy end-policy routing bgp 1 vrf default neighbor 10.0.0.5 remote-as 3 af-ipv4 unicast orf policy ORF</pre>

9.10.11. redistribute

Description	The redistribute attach point allows routes from other protocols to be advertised by BGP. A policy attached at this point is able to set any of the valid BGP attributes on the routes that are being redistributed. In addition, this policy allows the control and filtering of routes sourced by other protocols.
Examples	The following example: • Redistributes all ISIS inst 1 routes except for default route. • Routes with tag 10 are also set with local pref 300 and community (2:200). • Sets all other routes (matching the default rule) with local-pref 200 and community (2:100). <pre>policy route BGP_REDIST rule DROP_DEFAULT if prefix exist-in (0.0.0.0/0) then deny rule TAG_10 if tag eq 10 then local-preference set 300 community set (2:200) permit rule DEFAULT_RULE local-preference set 200 community set (2:100) end-policy routing bgp 2 af-ipv4 unicast redistribute isis policy BGP_REDIST</pre>

9.11. ISIS Policy Configuration Commands: Routing Conditions

9.11.1. level

Description	Matches on the level attribute of the route.
Command Syntax	<code>level eq [level-1 level-2]</code>
Command Mode	configuration policy route
Examples	<pre>policy route MY_ISIS_POLICY rule ONE if match-all rib-metric eq 100 level eq level-1 then ...</pre>

9.11.2. prefix

Description	Matches on the route prefix, known as the network-layer reachability information (NLRI). The prefix comprises an IP address, a mask length and an optional matching-len specification.		
Command Syntax	<pre>prefix [not] exist-in [<i>inline-ip-prefix-group</i> <i>named-ip-prefix-group</i> <i>\$parameter</i>]</pre>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>inline-ip-prefix-group</i>	IP prefix or list of IP prefixes	Prefix (IP address/mask length) or list of IP prefixes separated by spaces.
	<i>named-ip-prefix-group</i>	ip-prefix-group	User-defined ip-prefix group or list of ip-prefix groups separated by spaces.
	<i>parameter</i>		Parameter containing an IPv4 or IPv6 prefix or name of an ip-prefix group.

Examples	<pre> policy group ip-prefix IPV4_PFX_GRP1 10.20.0.0/16 30.40.0.0/16 end-group policy group ip-prefix IPV6_PFX_GRP2 2001:0db8:85a3:0:0:8a2e:0370:7334 1050:0:0:0:5:600:300c:326b end-group policy route MY_ISIS_POLICY rule RULE_1 if match-all prefix exist-in (20.20.20.0/24, 30.30.30.0/24) (IPV4_PFX_GRP1 IPV6_PFX_GRP2) then... rule RULE_2 prefix exist-in (1050::5:600:0:0/32, 1000::1:1::0:0/32) , 30.30.30.0/24) then... </pre>
Notes	The named ip-prefix group is defined using the <i>policy group ip-prefix</i> (on page 602) command. The 'exist-in' can be followed by a list of groups. Each group in the list can be either inline or named ip-prefix group. The group names in the list must be separated by spaces. See configuration example. Each of the inline prefix group elements can use local/global parameters.

9.11.3. rib-metric

Description	Matches on the RIB metric associated with the route.		
Command Syntax	rib-metric [not] [eq ge gt le lt] [<i>number</i> <i>\$parameter</i>]		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>number</i>	0-4294967295	RIB metric value.
	<i>parameter</i>		Parameter containing a RIB metric value.
Examples	<pre> policy route MY_ISIS_POLICY rule ONE if match-all rib-metric eq 40 rib-metric eq 30 then ... rule TWO if rib-metric ge 40 rib-metric le 60 then ... </pre>		

9.11.4. rib-route

Description	Matches on IPv4 or IPv6 prefixes in the ISIS. This condition is supported for all protocols mentioned.		
Command Syntax	<pre>rib-route [not] [protocol {isis bgp connected ospf ospfv3 rsvp static}] [include-any include-all] [inline-ip-prefix group named-ip-prefix-group \$parameter]</pre>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>inline-ip-prefix-group</i>	IP prefix or list of IP prefixes	Prefix (IP address/mask length) or list of IP prefixes separated by spaces.
	<i>named-ip-prefix-group</i>	ip-prefix-group	User-defined ip-prefix group or list of ip-prefix groups separated by spaces.
	<i>parameter</i>		Parameter containing an IP prefix or name of an ip-prefix group.
Examples	Multiple rib-route conditions can be defined in a rule: <pre>policy route MY_ISIS_POLICY rule ONE if match-any rib-route include-any (10.0.0.0/8, 20.0.0.0/8) rib-route include-any MY_PREFIX_GROUP rib-route protocol isis include-any (30.0.0.0/8) rib-route include-all (100.0.0.0/8, 200.10.0.0/16) then ...</pre>		

Notes	The named ip-prefix group is defined using the <i>policy group ip-prefix</i> (on page 602) command. Each of the inline prefix group elements can use local/global parameters. If no protocol is specified, the condition matches on any route source.
--------------	---

9.11.5. route-has-mpls-label

Description	Matches on existence of an MPLS label on the route.
Command Syntax	route-has-mpls-label
Command Mode	configuration policy route
Parameters Table	N/A
Examples	<pre>policy route MY_ISIS_POLICY rule ONE if match-all rib-metric eq 100 route-has-mpls-label then...</pre>

9.11.6. route-has-no-mpls-label

Description	Matches on absence of an MPLS label on the route.
Command Syntax	route-has-no-mpls-label
Command Mode	configuration policy route
Parameters Table	N/A
Examples	<pre>policy route MY_ISIS_POLICY rule ONE if match-all rib-metric eq 100 route-has-no-mpls-label then...</pre>

9.12. ISIS Policy Configuration Commands: Actions

9.12.1. isis-metric

Description	Sets the ISIS metric attribute for the route. Three actions are supported: • set: set the ISIS metric to the specified value. • increase: set the ISIS metric to the current ISIS metric value plus the specified value (offset). • decrease: set the ISIS metric to the current ISIS metric value minus the specified value (offset).		
Command Syntax	<pre>isis-metric set [number \$parameter] isis-metric increase [number \$parameter] isis-metric decrease [number \$parameter]</pre>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>number</i>	0 - 63	ISIS metric value.
	<i>parameter</i>		Parameter containing ISIS metric value.
Examples	<pre>policy route MY_ISIS_POLICY Rule MY_RULE1 if ... then isis-metric set 4 rule MY_RULE2 if ... then isis-metric set \$LEVEL_1_METRIC rule MY_RULE3 if ... then isis-metric increase 10 rule MY_RULE4 if ... then isis-metric decrease 20</pre>		

9.12.2. metric-type

Description	Sets the metric type for the route.
Command Syntax	<code>metric-type set [internal external]</code>
Command Mode	configuration policy route
Parameters Table	N/A
Examples	<pre>policy route MY_ISIS_POLICY Rule MY_RULE1 if ... then then metric-type set internal rule MY_RULE2 if ... then metric-type set external</pre>

9.12.3. tag

Description	Sets the tag attribute associated with a given route in the RIB.		
Command Syntax	tag set [<i>number</i> <i>\$parameter</i>]		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>number</i>	0 - 4294967295	Tag value.
	<i>parameter</i>		Parameter containing tag value.
Examples	<pre> policy route MY_ISIS_POLICY Rule ONE if prefix exist-in (10.0.0.0/8) then tag set 40 </pre>		

9.13. ISIS Policy Configuration Commands: Attachment Points

An ISIS policy can be attached to the following attachment points:

- default-route-originate
- redistribute
- route-leak

There is no limit on the number of policies attached to one entity.

9.13.1. default-route-originate

Description	The default-route-originate attach point allows the default route (0.0.0.0/0) to be conditionally generated and advertised to other IS-IS routers, based on the presence of other routes in the RIB. If any route in the policy is permitted, then the default route is generated.
Examples	The following policy generates and sends a default-route to the ISIS neighbor 10.0.0.1 if any BGP routes that match 10.0.0.0/8 matching-len 8-32 are present in RIB. <pre>policy route ISIS-DEFAULT-ORIGINATE rule CHECK-RIB if rib-route bgp exist-in (10.0.0.0/8 matching-len 8-32) then permit end-policy routing isis 2 is-type level-1-2 net 49.0001.7777.1111.1111.00 af-ipv4 unicast default-route-originate policy ISIS-DEFAULT-ORIGINATE</pre>

9.13.2. redistribute

Description	The redistribute attach point allows routes from other protocols to be advertised by ISIS. A policy attached at this point is able to set any of the valid ISIS attributes on the routes that are being redistributed. In addition, this policy allows the control and filtering of routes sourced by other protocols.
Examples	The following example: • Redistributes all ISIS routes except for default route. • Sets the following values for the distributed route: isis metric = 200 and metric type = internal. <pre>policy route ISIS_REDIST rule DROP_DEFAULT if prefix exist-in (0.0.0.0/0) then deny rule DEFAULT_RULE isis-metric set 200 metric-type set internal end-policy policy route ISIS_LEVEL1_REDIST rule DROP_DEFAULT if prefix exist-in (100.10.0.0/16) then deny rule DEFAULT_RULE isis-metric set 100 metric-type set internal end-policy routing isis test is-type level-1-2 net 49.0001.7777.1111.1111.00 af-ipv4 unicast redistribute bgp level-1-2 ISIS_REDIST redistribute connect level-1-only policy ISIS_REDIST</pre>

9.13.3. route-leak

Description	The route-leak attach point within IS-IS allows the prefixes to be conditionally propagated from one level to another level within the same IS-IS instance.
Examples	The following example shows how to allow prefixes to be leaked from the level 1 LSP into the level 2 LSP if any of the prefixes match 10.0.0.0/24, greater than 8 and less than 32. <pre>policy route ISIS-LEAK rule CHECK-METRIC if prefix exist-in (10.0.0.0/24 matching-len 8-32) then permit end-policy routing isis test is-type level-1-2 net 49.0001.7777.1111.1111.00 af-ipv4 unicast level-1 route-leak policy ISIS-LEAK</pre>

9.14. LDP Policy Configuration Commands: Routing Conditions

9.14.1. prefix

Description	Matches on neighbor IP address.		
Command Syntax	<pre>prefix exist-in [<i>inline-ip-prefix group</i> <i>named-ip-prefix-group</i> \$parameter]</pre>		
Command Mode	configuration policy route		
Parameters Table	Name	Value	Description
	<i>inline-ip-prefix-group</i>	prefix or list of prefixes	Prefix (IP address/mask length) or list of prefixes separated by spaces.
	<i>named-ip-prefix-group</i>	ip-prefix-group	User-defined ip-prefix group or list of ip-prefix groups separated by spaces.
	<i>parameter</i>		Parameter containing a prefix or name of an ip-prefix group.
	<i>ip-address</i>	IP address	Specific IP address to match.

Examples	<pre>policy route MY_LDP_POLICY rule ONE if ... prefix exist in (10.10.1.1/32) then ... rule TWO if ... prefix exist-in (10.10.1.0/24) then ... rule THREE if ... prefix exist-in \$MY_PEER_GROUP then ...</pre>
Notes	The named ip-prefix group is defined using the <i>policy group ip-prefix</i> (on page 602) command. The 'exist-in' can be followed by a list of groups. Each group in the list can be either inline or named ip-prefix group. The group names in the list must be separated by spaces. See configuration example. Each of the inline prefix group elements can use local/global parameters.

9.15. LDP Policy Configuration Commands: Attachment Points

An LDP policy can be attached to the following attachment points:

- label-allocation

There is no limit on the number of policies attached to one entity.

9.15.1. label-allocation

Description	Sets the policy to specify the IGP routes which are allocated labels by the router (instead of the default MPLS LDP behavior of allocation of labels for all IGP routes). Use the no form of the command to remove the label-allocation configuration.		
Command Syntax	<pre>label-allocation policy <i>policy-name</i> [<i>additional-policy-name</i>]</pre> <pre>no label-allocation</pre>		
Command Mode	configuration mpls ldp <instance>		
Parameters Table	Name	Value	Description
	<i>policy-name</i>	policy-name	Apply policy to filter labels allocated locally.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# mpls ldp default exaware(config-mpls-ldp-default)# label-allocation policy LDP_filter1</pre>		
Notes	Supported conditions: prefix. Supported actions: permit, deny, default-permit, default-deny. There is an implicit deny at the end of the policy.		

9.16. LDP Policy Operational Commands

9.16.1. clear ldp policy label-allocation

Description	Clears the LDP label-allocation policy counters.
Command Syntax	<code>clear ldp policy label-allocation</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows how to clear LDP label-allocation policy counters: <pre>exaware# clear ldp policy label-allocation</pre>

9.17. LDP Policy Show Commands

9.17.1. show ldp policy label-allocation

Description	Displays LDP label-allocation policy information.
Command Syntax	show ldp policy label-allocation
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows a sample of the output from this command: <pre>exaware# show ldp policy label-allocation POLICY RULE HIT COUNT ----- P1 r1 50 r2 10000 P2 alloc1 10 alloc2 0 alloc3 6</pre>

9.18. OSPF Policy Configuration Commands

9.18.1. inter-area policy

Description	The policy command allows filtering of OSPF type-3 summary LSAs. In - Allow to filter inbound OSPF type-3 summary LSAs. Out - Allow to filter Outbound OSPF type-3 summary LSAs.		
Command Syntax	<code>inter-area policy {in out} <i>policy-names</i></code>		
Command Mode	configuration ospf vrf area		
Parameters Table	Name	Value	Description
	<i>policy-names</i>	string	Policy name.
Examples	Policy configuration: <pre>policy route OSPF-area rule k1 if prefix exist-in (20.20.20.0/24) then deny rule k2 if prefix exist-in (21.21.21.0/24) then permit</pre> Policy application: <pre>routing ospf 11 vrf default area 1 inter-area policy in OSPF-area routing ospf 11 vrf default area 1 inter-area policy out OSPF-area</pre>		

Policy Operations:

Command	Tag	prefix	next-hop	rib-metric	rib-route	route-has-mpls-label	Area-id	route-type
Inter-area Policy in/out	n/a	+	+	+	n/a	n/a	+	n/a

Supported actions per command:

Command	metric	metric-type	tag
Inter-area Policy in/out	n/a	n/a	n/a

9.18.2. rib-filter policy

Description	The rib-filter command filters routes from entering the routing table. The LSAs will exist in the LSDB, despite the filtering, since all routers in the OSPF area must be synchronized and have the same LSDB.		
Command Syntax	<code>rib-filter policy <i>policy-names</i></code>		
Command Mode	configuration ospf vrf		
Parameters Table	Name	Value	Description
	<i>policy-names</i>	string	Policy name.
Examples	Policy configuration: <pre>policy route OSPF-rib-filter rule k1 if prefix exist-in (20.20.20.0/24) then deny</pre> Policy application: <pre>routing ospf 11 vrf default rib-filter policy OSPF-rib-filter</pre>		

Policy Operations:

Command	Tag	prefix	next-hop	rib-metric	rib-route	route-has-mpls-label	Area-id	route-type
rib-filter	+	+	+	+	n/a	n/a	n/a	+

Supported actions per command:

Command	metric	metric-type	tag
rib-filter	n/a	n/a	n/a

9.19. OSPF Policy Operational Commands

9.19.1. clear ospf policy default-route-originate

Description	Clears the OSPFv2 default-route-originate policy counters.
Command Syntax	<code>clear ospf policy default-route-originate</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows how to use this command: <pre>exaware# clear ospf policy default-route-originate</pre>

9.19.2. clear ospf policy inter-area

Description	Clears the OSPFv2 inter-area policy counters.		
Command Syntax	<code>clear ospf policy inter-area <i>area-id</i> {in out}</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>area-id</i>	decimal	<0-4294967295> Area ID as a decimal value.
		A.B.C.D	Area ID in IP address format.
Examples	The following example shows how to use this command: <code>exaware# clear ospf policy inter-area 0.0.0.100 in</code>		

9.19.3. clear ospf policy redistribute

Description	Clears the OSPFv2 redistribute policy counters.
Command Syntax	<code>clear ospf policy redistribute</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows how to use this command: <code>exaware# clear ospf policy redistribute</code>

9.19.4. clear ospf policy rib-filter

Description	Clears the OSPFv2 rib-filter policy counters.
Command Syntax	<code>clear ospf policy rib-filter</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows how to use this command: <pre>exaware# clear ospf policy rib-filter</pre>

9.20. OSPF Policy Show Commands

9.20.1. show ospf policy

Description	Displays OSPF policy information.
Command Syntax	show ospf policy {default-route-originate redistribute inter-area rib-filter}
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command: <pre> exaware# show ospf policy default-route-originate HIT POLICY RULE COUNT ----- default r1_filter 0 exaware# show ospf policy inter-area HIT AREA-ID DIRECTION POLICY RULE COUNT ----- 0.0.0.100 in OSPF_area_in1 k1 0 0.0.0.200 out OSPF_area_in1 k1 2 0.0.0.300 out OSPF_area_in1 k1 6 0.0.0.400 out OSPF_area_in1 k1 5 exaware# show ospf policy redistribute HIT PROTOCOL POLICY RULE COUNT ----- connected p1 r1 1 static p2 r2 4 exaware# show ospf policy rib-filter HIT POLICY RULE COUNT ----- p1 r1 1 p2 r2 4 </pre>

9.21. OSPFv3 Policy Configuration Commands

9.21.1. inter-area policy

Description	The policy command allows filtering of OSPF type-3 summary LSAs. In - Allow to filter inbound OSPF type-3 summary LSAs. Out - Allow to filter Outbound OSPF type-3 summary LSAs.		
Command Syntax	<code>inter-area policy {in out} <i>policy-names</i></code>		
Command Mode	configuration ospf vrf area		
Parameters Table	Name	Value	Description
	<i>policy-names</i>	string	Policy name.
Examples	Policy configuration: <pre> policy route OSPFv3-area rule k1 if prefix exist-in (2001:564:23::/48) then deny rule k2 if prefix exist-in (2001:1000:58:5633::/64) then permit </pre> Policy application: <pre> routing ospfv3 11 vrf default area 1 inter-area policy in OSPFv3-area routing ospfv3 11 vrf default area 1 inter-area policy out OSPFv3-area </pre>		

Policy Operations:

Command	Tag	prefix	next-hop	rib-metric	rib-route	route-has-mpls-label	Area-id
Inter-area Policy in/out	n/a	+	+	+	n/a	n/a	+

Supported actions per command:

Command	metric	metric-type	tag
Inter-area Policy in/out	n/a	n/a	n/a

9.21.2. rib-filter policy

Description	The rib-filter command filters routes from entering the routing table. The LSAs will exist in the LSDB, despite the filtering, since all routers in the OSPF area must be synchronized and have the same LSDB.		
Command Syntax	<code>rib-filter policy <i>policy-names</i></code>		
Command Mode	configuration ospfv3 vrf		
Parameters Table	Name	Value	Description
	<i>policy-names</i>	string	Policy name.
Examples	Policy configuration: <pre>policy route OSPFv3-rib-filter rule k1 if prefix exist-in (2001:564:23::/48) then deny</pre> Policy application: <pre>routing ospfv3 11 vrf default rib-filter policy OSPFv3-rib-filter</pre>		

Policy Operations:

Command	Tag	prefix	next-hop	rib-metric	rib-route	route-has-mpls-label	Area-id	route-type
rib-filter	+	+	+	+	n/a	n/a	n/a	+

Supported actions per command:

Command	metric	metric-type	tag
rib-filter	n/a	n/a	n/a

9.22. OSPFv3 Policy Operational Commands

9.22.1. clear ospfv3 policy inter-area

Description	Clears the OSPFv3 inter-area policy counters.		
Command Syntax	clear ospfv3 policy inter-area <i>area-id</i> {in out}		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>area-id</i>	decimal	<0-4294967295> Area ID as a decimal value.
		A.B.C.D	Area ID in IP address format.
Examples	The following example shows how to use this command: exaware# clear ospfv3 policy inter-area 0.0.0.100 in		

9.22.2. clear ospfv3 policy redistribute

Description	Clears the OSPFv3 redistribute policy counters.
Command Syntax	<code>clear ospfv3 policy redistribute</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows how to use this command: <pre>exaware# clear ospfv3 policy redistribute</pre>

9.22.3. clear ospfv3 policy rib-filter

Description	Clears the OSPFv3 rib-filter policy counters.
Command Syntax	<code>clear ospfv3 policy rib-filter</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows how to use this command: <pre>exaware# clear ospfv3 policy rib-filter</pre>

9.23. OSPFv3 Policy Show Commands

9.23.1. show ospfv3 policy

Description	Displays OSPFv3 policy information.																																																		
Command Syntax	show ospfv3 policy {redistribute inter-area rib-filter}																																																		
Command Mode	operational																																																		
Parameters Table	N/A																																																		
Examples	The following example shows sample output from this command: exaware# show ospfv3 policy inter-area <table><thead><tr><th>AREA-ID</th><th>DIRECTION</th><th>POLICY</th><th>RULE</th><th>HIT COUNT</th></tr></thead><tbody><tr><td>0.0.0.100</td><td>in</td><td>OSPF_area_in1</td><td>k1</td><td>0</td></tr><tr><td>0.0.0.200</td><td>out</td><td>OSPF_area_in1</td><td>k1</td><td>2</td></tr><tr><td>0.0.0.300</td><td>out</td><td>OSPF_area_in1</td><td>k1</td><td>6</td></tr><tr><td>0.0.0.400</td><td>out</td><td>OSPF_area_in1</td><td>k1</td><td>5</td></tr></tbody></table> exaware# show ospfv3 policy redistribute <table><thead><tr><th>PROTOCOL</th><th>POLICY</th><th>RULE</th><th>HIT COUNT</th></tr></thead><tbody><tr><td>connected</td><td>p1</td><td>r1</td><td>1</td></tr><tr><td>static</td><td>p2</td><td>r2</td><td>4</td></tr></tbody></table> exaware# show ospfv3 policy rib-filter <table><thead><tr><th>POLICY</th><th>RULE</th><th>HIT COUNT</th></tr></thead><tbody><tr><td>p1</td><td>r1</td><td>1</td></tr><tr><td>p2</td><td>r2</td><td>4</td></tr></tbody></table>					AREA-ID	DIRECTION	POLICY	RULE	HIT COUNT	0.0.0.100	in	OSPF_area_in1	k1	0	0.0.0.200	out	OSPF_area_in1	k1	2	0.0.0.300	out	OSPF_area_in1	k1	6	0.0.0.400	out	OSPF_area_in1	k1	5	PROTOCOL	POLICY	RULE	HIT COUNT	connected	p1	r1	1	static	p2	r2	4	POLICY	RULE	HIT COUNT	p1	r1	1	p2	r2	4
AREA-ID	DIRECTION	POLICY	RULE	HIT COUNT																																															
0.0.0.100	in	OSPF_area_in1	k1	0																																															
0.0.0.200	out	OSPF_area_in1	k1	2																																															
0.0.0.300	out	OSPF_area_in1	k1	6																																															
0.0.0.400	out	OSPF_area_in1	k1	5																																															
PROTOCOL	POLICY	RULE	HIT COUNT																																																
connected	p1	r1	1																																																
static	p2	r2	4																																																
POLICY	RULE	HIT COUNT																																																	
p1	r1	1																																																	
p2	r2	4																																																	

9.24. QoS Policy Configuration Commands: Actions

9.24.1. dscp

Description	Defines a QoS policy action that sets the DSCP value (available at Ingress Qos Policy). See additional option to set DSCP by qos-mapping-type command.		
Command Syntax	dscp [<i>value</i> <i>\$parameter</i>]		
Command Mode	configuration policy qos		
Parameters Table	Name	Value	Description
	<i>value</i>	0-63	DSCP value.
	<i>\$parameter</i>		Parameter that contains DSCP value.
Examples	This example shows how to set 'af13' value on all traffic (IPv4 and IPv6). <pre>rule rule_one if dscp eq cs1 then set dscp af13</pre>		
Notes	Precedence and dscp values are mutually exclusive.		

9.24.2. header-compensate

Description	The command specifies the difference between input and output frame sizes (available on egress only). By default, the difference is 0.		
Command Syntax	<code>header-compensate value</code>		
Command Mode	configuration policy qos		
Parameters Table	Name	Value	Description
	<code>value</code>	An integer value	Specifies the difference between input and output frame sizes.
Examples	<pre> policy qos PARENT rule-default shape 1000000 kbps header-compensate -4 exec-policy CHILD end-policy policy qos CHILD rule GOLD if source-ip eq 2.2.2.2 then minimum-bandwidth percent 40 rule SILVER if source-ip eq 4.4.4.4 then minimum-bandwidth percent 20 end-policy !</pre>		
Notes	The command is not supported on child policy.		

9.24.3. minimum bandwidth

Description	Defines a QoS policy action that configures the guaranteed minimum amount of bandwidth for traffic.		
Command Syntax	minimum-bandwidth { <i>value</i> [<i>units</i>] percent <i>percent-val</i> }		
Command Mode	configuration policy qos		
Parameters Table	Name	Value	Description
	<i>value</i>	1- 294967295	Minimum bandwidth.
	<i>units</i>	bps kbps (default) mbps gbps	Minimum bandwidth units.
	<i>percent-val</i>	1-100	Percent is based on physical interface rate unless shaper is defined.

Examples	Example 1: <pre> policy qos GOLD_POLICY rule MY_RULE if queue eq af1 then minimum-bandwidth 1000000 kbps </pre> Example 2: <pre> policy qos PARENT rule rule-default shape 1000000 kbps exec-policy CHILD end-policy ! policy qos CHILD rule GOLD if queue eq af1 then minimum-bandwidth percent 40 rule silver if queue eq af2 then minimum-bandwidth percent 40 rule bronze if queue eq af3 then minimum-bandwidth percent 20 end-policy </pre>
Notes	Total guaranteed bandwidth on sub-interfaces cannot exceed the shape rate of the sub-interface. This minimum-bandwidth action is valid only on egress policy. The actual rate might be different than the configured value due to hardware granularity.

9.24.4. mpls exp

Description	Defines a QoS policy action that sets the MPLS EXP field value (available on ingress only).		
Command Syntax	<code>set mpls-exp [imposition topmost] [<i>value</i> <i>\$parameter</i>]</code>		
Command Mode	configuration policy qos		
Parameters Table	Name	Value	Description
	<i>value</i>	0-7	MPLS EXP field value.
	<i>\$parameter</i>		Parameter that contains MPLS EXP field value.
		imposition	Preserves the original IP precedence value in a packet when forwarding it from an IP network to an MPLS network. [not supported in release 8.0.0]

Examples	Example 1: <pre> policy condition traffic PREC_IS_DEFAULT_IN_IP4 match-all precedence eq 0 protocol eq ipv4 end-condition policy condition traffic PREC_IS_NC_IN_IP4 match-all precedence eq 1 protocol eq ipv4 end-condition policy qos MPLS_NHB rule MY_RULE1 if condition PREC_IS_DEFAULT_IN_IP4 then set mpls-exp imposition 0 rule MY_RULE2 if condition PREC_IS_NC_IN_IP4 then set mpls-exp imposition \$MY_NC_VAL end-policy </pre> Example 2: <pre> rule INC_MPLS_VAL if mpls-exp topmost eq 4 then set mpls-exp topmost \$MY_TOPMOST_VAL </pre>
Notes	The <code>mpls exp</code> condition is mutually exclusive with the following conditions: • source-ip / destination-ip • dscp • precedence • tcp-flags • source-port / destination-port • icmpv4-code / icmpv4-type • icmpv6-code / icmpv6-type • frag-non-initial • ipv4 / ipv6

9.24.5. police

Description	Configure police actions. The police action is composed of five sub-commands: • <code>cir</code> (conform rate) • <code>pir</code> (peak rate) The <code>cir</code> subcommand is mandatory, however the other commands have default values so it is not mandatory to configure them explicitly. The police rate limits traffic based on the Two Rates Three Color Marking (trTCM) algorithm (RFC 2698). trTCM is useful when the peak rate needs to be enforced separately from the committed rate. In this policy configuration, two rates are set: Committed Information Rate (CIR) and Peak Information Rate (PIR). Each packet is metered and assigned accordingly with one of the following operations: • conform • exceed • violate Defaults: • Default value of PIR value is CIR value		
Command Syntax	<pre>police cir [rate-value rate-units percent percent-value] [burst burst-value burst-units] [pir rate-value rate-units percent percent-value] [burst burst-value burst-units]</pre>		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	<code>rate-value</code>	1- 4294967295	traffic rate.

	<i>rate-units</i>	bps kbps (default) mbps gbps	Units for rate values.
--	-------------------	---------------------------------------	------------------------

	<i>percent-value</i>	1-100	
--	----------------------	-------	--

	<i>burst-value</i>	Ingress 1- 4294967295 Egress 1-8355840	
--	--------------------	---	--

	<i>burst-units</i>	bytes (default) kbytes mbytes gbytes msec usec	
--	--------------------	---	--

	<i>set-value</i>		See 'Set Actions' table below.
--	------------------	--	--------------------------------

		set	Set a value according to police rates and transmit. When <code>set</code> is configured, it implicitly includes <code>transmit</code> action as well. Note: more than one <code>set</code> command can be defined for an action.
--	--	-----	---

		transmit	Transmit the packet.
--	--	----------	----------------------

		drop	Drop the packet.
--	--	------	------------------

Examples	<pre>policy qos MY_QOS_POLICY rule MY_RULE1 if dscp eq ef then police cir 64000 kbps burst 10000 bytes pir 128000 kbps burst 20000 bytes conform-action set red-tag 1 exceed-action set red-tag 3 violate-action drop end-policy policy qos MY_QOS_POLICY rule MY_RULE1 if mpls-exp topmost eq 0 then police cir 250 kbps burst 500 bytes conform-action set mpls-exp topmost 4 exceed-action drop end-policy</pre>		
Notes	Set Actions		
	Field	Value	
	dscp	DSCP value - range (0-63) - One of dscp values (on page 802)	
	mpls-exp topmost	value range 0-7	
	mpls-exp imposition	value range 0-7	
	red-tag	Red-tag value [0-7]	
	precedence	Precedence value [0-7]	
	Actions allowed on policer according to interface:		
	Set command	Ingress policy	Egress policy
	dscp value	+	-
	Precedence	+	+

mpls-exp topmost [	+	+
mpls-exp imposition [not supported in release 8.0.0]	+	-
red-tag	+	-

- Policer configured on the egress can support only one set action per each rate (dscp or precedence).
- Police CIR rate/percent value is mandatory configuration (burst value is optional).
- Multiple set actions are allowed. For example:

```
if ...
then
  police
    cir 64000 kbps
    conform-action set red-tag 1
    You cannot set different values for same type . For
example, the following sequence of commands:
    police
      conform action set red-tag 1
      conform action set red-tag 2

will result in :

police
  conform action set red-tag 2
```
- Drop, transmit, and set are mutually exclusive within a specific action. For example, the following sequence of commands:

```
police
  conform action transmit
  conform action drop

will result in :

police
  conform action set drop
```
- If PIR is not configured, it will be set with the same values as the CIR. In this case, you can configure conform, exceed, and violate. Since CIR equals PIR, no packets will match the exceed criteria.
- Conform action cannot be set to *drop*.
- Police-actions DSCP and precedence marking support filtering for ipv4 only or ipv6 only:

```
conform-action set dscp [ipv4|ipv6] set {value |
$parameter}
conform-action set precedence {value | $parameter}
exceed-action set dscp {value | $parameter}
exceed-action set precedence {value | $parameter}
violate-action set dscp {value | $parameter}
violate-action set precedence {value | $parameter}
```

9.24.6. qos-mapping-type

Description	Defines QoS mapping type.		
Command Syntax	qos-mapping-type QosTagToExp ExpIngress ExpEgress		
Command Mode	configuration policy qos		
Parameters Table	Name	Value	Description
	QosTagToExp		qos-tag to EXP mapping. In this release the qos-tag is equal to the dscp which was set by the system by the ingress policies.
Examples	Example 1: <pre> policy qos MplsEgress2 qos-mapping-type QosTagToExp rule r1 if qos-tag eq 18 then set mpls-exp topmost 6 rule r2 if qos-tag eq 26 then set mpls-exp topmost 7 rule rule-default end-policy !</pre>		
Notes	This command will be replaced by another type of attachment point in future releases. This policy type can be only attached to the ctrl-plane attachment point and is global to the node.		

9.24.7. queue

Description	Defines the queue for the packets conforming to the rules under the policy. EF1 is high priority queue that bypass the scheduling and sub-interface shaping. It should be used with care for user traffic as it is the same queue as for control traffic. EF2 is high priority queue which is served first before all other queues until it is empty. AF1-AF5 queues are queues with optional minimum bandwidth configuration. BE is best effort queue, which is served only if all other queues in the hierarchy are empty. Use the no form of the command to revert to the default queue parameters.		
Command Syntax	<pre>set queue [queue-name] no queue</pre>		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
		ef1	Map traffic to EF1 queue
		ef2	Map traffic to EF2 queue
		af1	Map traffic to AF1 queue
		af2	Map traffic to AF2 queue
		af3	Map traffic to AF3 queue
		af4	Map traffic to AF4 queue

		af5	Map traffic to AF5 queue
		be	Map traffic to BE queue
Examples	<pre> policy qos InChild rule r1 if dscp eq ef then set queue af1 set dscp af42 rule r2 if source-ip eq 11.1.1.1 then set queue ef1 rule rule-default end-policy !</pre>		

9.24.8. port-level-rate

Description	Defines a QoS policy action that defines rate and burst size for broadcast and unicast unknown packets. It can be attached only as ingress policy.		
Command Syntax	<code>port-level-rate rate pps burst size packets</code>		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	<code>rate</code>	43-6,718,863	Port Level rate in packet per second. Default value is 148,800.
	<code>size</code>	2-65,020	Burst size in packets. Default value is 100.
Examples	<pre> policy qos POL rule RULE_1 if packet-type eq broadcast then port-level-rate 50000 pps burst 10 packets rule rule-default end-policy </pre>		

9.24.9. precedence

Description	Defines a QoS policy action that sets the precedence value.		
Command Syntax	<code>precedence [value \$parameter]</code>		
Command Mode	configuration policy qos		
Parameters Table	Name	Value	Description
	<i>value</i>	0-7	Precedence value.
	<i>\$parameter</i>		Parameter that contains precedence value.
Examples	<pre>rule r1 if precedence eq 3 then set precedence 4 rule r2 if precedence eq 5 then set precedence \$MY_PRECEDENCE_VAL rule r3 if precedence eq 6 then set precedence 4</pre>		
Notes	Precedence and dscp values are mutually exclusive. Precedence of ipv4 and ipv6 are mutually exclusive.		

9.24.10. queue max size

Description	Configures maximum queue size (available on egress only). When a specific class is assigned to a queue, all packets of that class are directed to that queue and wait for the scheduler to service them. On excess traffic overflow packets are dropped (tail drop). By default, a queue is allocated for best effort with size = 100msec unless shaper is defined.		
Command Syntax	queue-max-size <i>value</i> [<i>units</i>]		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	<i>value</i>	1- 1300000000	Maximum queue size.
	<i>units</i>	bytes (default) kbytes mbytes gbytes msec usec	Queue size units.
Examples	<pre>rule TAIL_DROP_RULE if dscp eq 3 then queue-max-size 1000000 bytes</pre>		
Notes	All queues share buffers from the same pool. Default value of queue-max-size is 100 ms of its parent bandwidth. Restrictions: This action is valid only for egress policy.		

9.24.11. shape

Description	Defines a QoS policy action that shapes traffic to a certain average rate (available on egress only). Shaping can be configured for physical interface level, and if hierarchical queueing is enabled, on sub-interfaces as well.		
Command Syntax	<pre>shape {rate [rate-units] percent percent-val} [burst burst [burst_units]]</pre>		
Command Mode	configuration policy qos		
Parameters Table	Name	Value	Description
	<i>rate</i>	1- 4294967295	
	<i>rate-units</i>	bps kbps (default) mbps gbps	
	<i>percent-val</i>	1-100	Percent of physical rate.
	<i>burst</i>	1- 4294967295	
	<i>burst_units</i>	bytes (default) kbytes mbytes gbytes msec usec	

Examples	<pre>policy qos PARENT rule-default shape 1000000 kbps exec-policy CHILD end-policy policy qos CHILD rule GOLD if source-ip eq 2.2.2.2 then minimum-bandwidth percent 40 rule SILVER if source-ip eq 4.4.4.4 then minimum-bandwidth percent 20 end-policy !</pre>
-----------------	--

9.24.12. weight

Description	Defines a sub-interface weight.		
Command Syntax	<code>weight value</code>		
Command Mode	configuration policy qos		
Parameters Table	Name	Value	Description
	<i>value</i>	Parent policy: 1- 1024 Child policy: 1-512	
Examples	<pre> policy qos PARENT rule-default shape 1000000 kbps weight 100 exec-policy CHILD end-policy policy qos CHILD rule GOLD if source-ip eq 2.2.2.2 then minimum-bandwidth percent 40 rule SILVER if source-ip eq 4.4.4.4 then minimum-bandwidth percent 20 end-policy </pre>		
Note	A policy containing this action can be attached only on sub-interface. The weight determines the relative weight of an active flow EIR compared to other flows. For a VLAN EIR, the lower the weight the higher proportion of the traffic - traffic for VLAN j is proportional to $(1/W_j)/\text{SUM}(1/W_i)$, W_i is considered only for active flows. For a queue EIR, the higher the weight the higher proportion of the traffic - traffic for queue j is proportional to $W_j/\text{SUM}(W_i)$, W_i is considered only for active flows.		

9.25. QoS Policy Configuration Commands: Traffic Conditions

9.25.1. destination ip

Description	Enables classifying based on destination IP address.		
Command Syntax	<code>destination-ip [eq {ip-address \$parameter} exist-in ip-prefix inline-ip-prefix-group named-ip-prefix-group]</code>		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IP address	IP address.
	<i>parameter</i>		Parameter containing one IP address.
	<i>ip-prefix</i>	ip-prefix	
	<i>inline-ip-prefix-group</i>	list of ip prefixes	IP address range or list of IP addresses.
	<i>named-ip-prefix-group</i>	ip-prefix-group	User-defined ip-prefix group.

Examples	<pre> policy group ip-prefix VPN_CLUSTER 10.20.0.0/16 30.40.0.0/16 2001:0db8:85a3:0:0:8a2e:0370:7334 1050:0:0:0:5:600:300c:326b end-group policy qos MY_QOS_POLICY rule RULE_1 if match-any destination-ip exist-in VPN_CLUSTER then set dscp 1 rule RULE_2 if destination-ip exist-in (20.20.20.0/24,30.30.30.0) then set dscp 2 rule RULE_3 if destination-ip eq 20.20.20.0 then set dscp 3 end-policy !</pre>
Notes	The named ip-prefix group is defined using the <i>policy group ip-prefix</i> (on page 602) command. The 'exist-in' can be followed by list of groups. Each group in the list can be either inline or named ip-prefix group. All groups in the list will be on the same line, separated by spaces. See configuration example. Each of the inline prefix group elements can use local/global parameters.

9.25.2. destination port

Description	Enables classifying based on destination port.		
Command Syntax	destination-port [eq {value \$parameter} exist-in {inline-port-group} gt {value \$parameter} lt {value \$parameter}]		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	value	0-65535 Predefined values: according to TCP and UDP tables	Destination port number or destination-ports.
	parameter		Parameter that contains the destination port number or range.
	inline-port-group		List of destination port numbers or ranges used for filtering. Range is used only in exist-in list. Range is set by using "-" between two numbers.

Examples	<pre>policy global-parameters parameter PORT_MAX 42000 parameter PORT_BOTTOM 40000 end-global-parameters ! policy qos MY_ACL_POLICY rule RULE_1 if destination-port eq 2042 then set dscp 1 rule RULE_2 if destination-port exist-in (\$PORT_BOTTOM-\$PORT_MAX) then set dscp 2 rule RULE_2 if destination-port exist-in (38000-39000) then set dscp 3 rule RULE_3 if destination-port gt 38000 then set dscp 4 rule default end-policy !</pre>
Notes	Range is not supported in release 8.0.0

TCP-port predefined-numbers:

Port number	Description	Predefined value
179	Boarder Gateway Protocol	bgp
21	Domain Name System	ftp
20	File Transfer Protocol	ftp-data
80	Hypertext Transfer Protocol	http
443	Secure HTTP	https
220	Internet Message Access Protocol	imap
389	Lightweight Directory Access Protocol	ldap
646	Label Distribution Protocol	ldp
445	Microsoft DS (AD)	ms-ds
496	PIM Auto-RP	pim-auto-rp
110	Post Office Protocol v3	pop3
995	Secure Post Office Protocol v3	pop3s
1723	Point-to-Point Tunneling protocol	pptp
25	Simple Mail Transport Protocol	smtp
22	Secure shell	ssh

514	Syslog	syslog
49	TAC Access Control System	tacacs

UDP-port predefined-numbers:

Port number	Description	Predefined value
68	Bootstrap Protocol (BOOTP) client	bootpc
67	Bootstrap Protocol (BOOTP) server	bootps
53	Domain Name Service (DNS)	domain
500	Internet Security Association and Key Management Protocol	isakmp
389	Lightweight Directory Access Protocol	ldap
646	Label Distribution Protocol	ldp
445	Microsoft DS (SMB)	ms-ds
138	NetBios datagram service	netbios-dgm
137	NetBios name service	netbios-ns
139	NetBios session service	netbios-ss

Port number	Description	Predefined value
4500	Internet Security Association and Key Management Protocol	non500-isakmp
123	Network Time Protocol	ntp
496	PIM Auto-RP	pim-auto-rp
1646, 1813	RADIUS accounting	radacct
1645, 1812	RADIUS authentication	radius
161	Simple Network Management Protocol	snmp
162	SNMP Traps	snmptrap
514	System Logger	syslog
49	TAC Access Control System	tacacs
69	Trivial File Transfer Protocol	tftp
68	Bootstrap Protocol (BOOTP) client	bootpc

Port number	Description	Predefined value
67	Bootstrap Protocol (BOOTP) server	bootps
53	Domain Name Service (DNS)	domain
500	Internet Security Association and Key Management Protocol	isakmp
389	Lightweight Directory Access Protocol	ldap
646	Label Distribution Protocol	ldp
445	Microsoft DS (SMB)	ms- ds
138	NetBios datagram service	netbios-dgm
137	NetBios name service	netbios-ns
139	NetBios session service	netbios- ss
4500	Internet Security Association and Key Management Protocol	non500-isakmp
123	Network Time Protocol	ntp
496	PIM Auto-RP	pim-auto-rp

9.25.3. dscp

Description	Enables classifying based on DSCP.		
Command Syntax	<code>dscp [not] [eq {value \$parameter} exist-in inline-dscp-group]</code>		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	<i>value</i>	0-63	DSCP value in TOS byte in IP header.
	<i>parameter</i>		Parameter that contains a DSCP value.
	<i>inline-group</i>		Group of DSCP values.
Examples	<pre>rule RULE1 if dscp eq ef then remaining-bandwidth percent 20 rule RULE2 if dscp exist-in (cs1, cs2,\$NC_DSCP) then remaining-bandwidth percent 40 rule RULE3 if dscp eq 40 then ...</pre>		

DSCP values

DSCP value	Reserved keyword
0	be
10	af11
12	af12
14	af13
18	af21
20	af22

DSCP value	Reserved keyword
22	af23
26	af31
28	af32
30	af33
34	af41
36	af42
38	af43
46	ef
8	cs1
16	cs2
24	cs3
32	cs4
40	cs5
48	cs6
56	cs7

9.25.4. ether-type

Description	Enables classifying based on source IP address type.
Command Syntax	<code>ether-type eq [ipv4 ipv6 mpls-unicast]</code>
Command Mode	configuration policy qos
Examples	<pre>rule RULE1 if match-all protocol-ip eq tcp ether-type eq ipv4 then ...</pre>
Notes	The <code>ether-type mpls</code> condition is mutually exclusive with the following conditions: • source-ip / destination-ip • dscp • precedence • tcp-flags • source-port / destination-port • icmpv4-code / icmpv4-type • icmpv6-code / icmpv6-type • frag-non-initial • ipv4 / ipv6

9.25.5. frag-non-initial

Description	Enables classifying non-initial fragments only. As an IP packet moves through the Internet, it might need to cross a route that cannot handle the size of the packet. The packet will be divided, or fragmented, into smaller packets and reassembled later.
Command Syntax	frag-non-initial
Command Mode	configuration policy qos/acl
Parameters Table	N/A
Examples	<pre>rule MY_RULE1 if frag-non-initial then ...</pre>

Note: Fragmentation behavior in ACL

Non-initial fragments contain only Layer 3 information. Access-list entries containing only Layer 3 information can be applied to non-initial fragments. The fragment has all the information the system requires to filter, so the entry is applied to the fragments.

By specifying the 'frag-non-initial' keyword in an access-list condition that particular condition applies only to non-initial fragments of packets; the fragment is either permitted, denied or logged.

The behavior of access-list entries regarding the presence or absence of the 'frag-non-initial' keyword can be summarized as follows:

frag-non-initial keyword in a condition	No L4 information in a condition	L3 & L4 information in a condition
No 'frag-non-initial' keyword	Condition is applied to all packets: 1. Initial fragmented packets 2. Non-initial fragmented packets 3. Non-fragmented packets	Condition and action (permit/deny) applies only for: 1. Initial fragmented packets 2. Non-fragmented packets
frag-non-initial keyword	Condition is applied only to non-initial fragmented packets.	Invalid state

9.25.6. icmpv4-type

Description	Enables traffic filtering based on ICMPv4 protocol type fields. The filtering is done according to the first 16 bits of the ICMPv4 header. icmpv4-type can be used without icmpv4-code.		
Command Syntax	<pre>icmpv4-type [eq {value \$parameter} exist-in inline-icmpv4-type-group]</pre>		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	<i>value</i>	0-255 Predefined values: according to ICMPv4 Type predefined values table	Numeric value of ICMPv4 type.
	<i>parameter</i>	Parameter starting with \$	Parameter that contains the ICMPv4 type.
	<i>inline-icmpv4-type-group</i>		List of inline ICMPv4 message types.
Examples	<pre>rule MY_RULE1 if protocol eq icmpv4 icmpv4-type eq echo-reply then ... rule MY_RULE2 if icmpv4-type exist-in (1,3,5) then ...</pre>		
Notes	ICMPv4 Type predefined values:		
	ICMPv4 Type	Description	Predefined value
	0	Echo reply (0)	echo-reply

	3	Unreachable (3)	unreachable
	4	Source quench (4)	source-quench
	5	Redirect (5)	redirect
	8	Echo request (8)	echo-request
	9	Router advertisement (9)	router-advertisement
	10	Router solicit (10)	router-solicit
	11	Time exceeded (11)	time-exceeded
	12	Parameter problem (12)	parameter-problem
	13	Timestamp (13)	timestamp
	14	Timestamp reply (14)	timestamp-reply
	15	Informational request (15)	info-request
	16	Informational reply (16)	info-reply
	17	Mask request (17)	mask-request
	18	Mask reply (18)	mask-reply

9.25.7. icmpv4-code

Description	Enables classifying based on ICMPv4 protocol code fields. The filtering is done according to the first 16 bits of the ICMPv4 header. icmpv4-code must be configured with icmpv4-type. icmpv4-code cannot be used in a match-any condition.		
Command Syntax	<pre>icmpv4-code [eq {value \$parameter} exist-in inline-icmpv4-code]</pre>		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	<i>value</i>	0-255 Predefined values: according to ICMPv4 code tables	Numeric value of ICMPv4 code.
	<i>parameter</i>	Parameter starting with \$	Parameter that contains the ICMPv4 code.
	<i>inline-icmpv4-code</i>		List of inline ICMPv4 message codes.
Examples	<pre>rule MY_RULE1 if protocol eq icmpv4 icmpv4-type eq unreachable icmpv4-code eq dest-net-unknown then ... rule MY_RULE2 if icmpv4_type eq 1 icmpv4-code exist-in (1,3,5) then ...</pre>		

Notes

ICMPv4 predefined-codes (Destination Unreachable (type=3))		
ICMPv4 Code	Description	Predefined value
0	Net Unreachable	net-unreachable
1	Host Unreachable	host-unreachable
2	Protocol Unreachable	protocol-unreachable
3	Port Unreachable	port-unreachable
4	Fragmentation Needed and Don't Fragment was Set	fragmentation-needed
5	Source Route Failed	source-route-failed
6	Destination Network Unknown	dest-net-unknown
7	Destination Host Unknown	dest-host-unknown
8	Source Host Isolated	source-host-isolated
9	Communication with Destination Network is Administratively Prohibited	dest-net-prohibited
10	Communication with Destination Host is Administratively Prohibited	dest-host-prohibited
11	Destination Network Unreachable for Type of Service	net-unreachable-for-tos
12	Destination Host Unreachable for Type of Service	host-unreachable-for-tos

13	Communication Administratively Prohibited	communication-prohibited
14	Host Precedence Violation	host-precedence-violation
15	Precedence cutoff in effect	precedence-cutoff-in-effect
ICMPv4 predefined-codes (Redirect (type=5))		

ICMPv4 Code	Description	Predefined value
0	Redirect Datagram for the Network (or subnet)	redirect-for-net
1	Redirect Datagram for the Host	redirect-for-host
2	Redirect Datagram for the Type of Service and Network	redirect-for-tos-and-net
3	Redirect Datagram for the Type of Service and Host	redirect-for-tos-and-host

ICMPv4 predefined-codes (Time-exceeded (type=11))		
ICMPv4 Code	Description	Predefined value
0	Time to Live exceeded in Transit	ttl-exceeded-during-transit
1	Fragment Reassembly Time Exceeded	ttl-exceeded-during-reassembly

ICMPv4 predefined-codes (Parameter-problem (type=12))		
ICMPv4 Code	Description	Predefined value
0	Pointer indicates the error	pointer-indicate-err
1	Missing a Required Option	required-option-missing

9.25.8. icmpv6-type

Description	Enables traffic filtering based on ICMPv6 protocol type fields. The filtering is done according to the first 16 bits of the ICMPv6 header. icmpv6-type can be used without icmpv6-code.		
Command Syntax	<pre>icmpv6-type [eq {value \$parameter} exist-in inline-icmpv6-type-group]</pre>		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	<i>value</i>	0-255 Predefined values: according to ICMPv6 Type predefined values table	Numeric value of ICMPv6 type.
	<i>parameter</i>	Parameter starting with \$	Parameter that contains the ICMPv6 type.
	<i>inline-icmpv6-type-group</i>		List of inline ICMPv6 message types.
Examples	<pre>rule MY_RULE1 if protocol eq icmpv6 icmpv6-type eq destination-unreachable then ... rule MY_RULE2 if icmpv6-type exist-in (1,3,5) then ...</pre>		

Notes

ICMPv6 Type predefined values:		
ICMPv6 Type	Description	Predefined value
1	Destination Unreachable (3)	destination-unreachable
2	Packet Too Big (2)	packet-too-big
3	Time Exceeded (3)	time-exceeded
4	Parameter Problem (4)	parameter-problem
128	Echo Request (128)	echo-request
129	Echo Reply (129)	echo-reply
133	Router Solicitation (133)	router-solicitation
134	Router Advertisement (134)	router-advertisement
135	Neighbor Solicitation (135)	neighbor-solicitation
136	Neighbor Advertisement (136)	neighbor-advertisement
137	Redirect Message (137)	redirect-message
130	Multicast Listener Query (130)	MLD- Query
131	Multicast Listener Report (131)	MLD-request
132	Multicast Listener Done (132)	MLD-done
143	Version 2 Multicast Listener Report (143)	MLDv2-request

9.25.9. icmpv6-code

Description	Enables classifying based on ICMPv6 protocol code fields. The filtering is done according to the first 16 bits of the ICMPv6 header. icmpv6-code and icmpv6-type must be used together. icmpv6-code must be configured with icmpv6-type. icmpv6-code cannot be used in a match-any condition.		
Command Syntax	<pre>icmpv6-code [eq {<i>value</i> <i>\$parameter</i>} exist-in <i>inline-icmpv6-code</i>]</pre>		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	<i>value</i>	0-255 Predefined values: according to ICMPv6 code tables	Numeric value of ICMPv6 code.
	<i>parameter</i>	Parameter starting with \$	Parameter that contains the ICMPv6 code.
	<i>inline-icmpv6-code</i>		List of inline ICMPv6 message codes.

Examples

```
rule MY_RULE1
if
    protocol eq icmpv6
    icmpv6-type eq unreachable
    icmpv6-code eq no-route-to-destination
then ...

rule MY_RULE2
if
    icmpv6_type eq 1
    icmpv6-code exist-in (1,3,5)
then ...
```

Notes

ICMPv6 predefined-codes (Destination Unreachable (type=1))		
ICMPv6 Code	Description	Predefined value
0	No Route To Destination	no-route-to-destination
1	Administratively Prohibited	administratively-prohibited
2	Beyond Scope of Source Address	beyond-scope-of source address
3	Address Unreachable	address-unreachable
4	Port Unreachable	port-unreachable

ICMPv6 predefined-codes (Time-exceeded (type=3))		
ICMPv6 Code	Description	Predefined value
0	Hop Limit Exceeded in Transit	hop-limit-exceeded-during-transit
1	Fragment Reassembly Time Exceeded	fragment-reassembly-time-exceeded

ICMPv6 predefined-codes (Parameter-problem (type=4))		
ICMPv6 Code	Description	Predefined value
0	Erroneous Header Field Encountered	erroneous-header-field
1	Unrecognized Next Header Type Encountered	unrecognized-next-header type
2	Unrecognized Ipv6 Option Encountered	unrecognized-IPv6-option

9.25.10. ieee-802.1p

Description	Enables classifying based on layer 2 class as defined in the TCI field of the Ethernet frame in IEEE 802.1p. The three bits give eight different classes ranging from 0 to 7.		
Command Syntax	<code>ieee-802.1p [not] [eq {value \$parameter} exist-in inline-ieee-802.1-group]</code>		
Command Mode	configuration policy qos		
Parameters Table	Name	Value	Description
	<i>value</i>	0-7	ieee802.1p numeric value.
	<i>parameter</i>		Parameter that contains an ieee802.1p value.
	<i>inline-group</i>		Group of ieee802.1p values.
Examples	<pre>rule RULE1 if ieee-802.1p exist-in (1, 2, 3, 4) then set precedence 4 rule RULE2 if ieee-802.1p eq 6 then set precedence 6 rule LAN2_RULE if ieee-802.1p eq \$IEEE_VAL then set precedence 2</pre>		
Notes	Match on ieee802.1p is not allowed in egress policy.		

9.25.11. mpls exp topmost

Description	Enables classifying based topmost MPLS EXP field. This value matches against the three bits of the EXP field value in the MPLS protocol. There are eight possible values with a range of 0-7.		
Command Syntax	<code>mpls-exp topmost [not] [eq {value \$parameter} exist-in inline-mpls-exp-group]</code>		
Command Mode	configuration policy qos		
Parameters Table	Name	Value	Description
	<i>value</i>	0-7	exp value in the MPLS protocol.
	<i>parameter</i>		Parameter that contains an mpls exp value.
	<i>inline-group</i>		Group of mpls exp values.
Examples	Example 1: <pre>rule MPLS_TO_IP_RULE if mpls-exp topmost eq 4 then set dscp 4</pre> Example 2: <pre>policy condition traffic HIGH_PRIO_MPLS_COND mpls-exptopmost eq \$MPLS_VAL !</pre> <pre>policy qos MY_QOS_POLICY rule MY_RULE if condition HIGH_PRIO_MPLS_COND then ...</pre> Example 3: <pre>rule MY_RULE3 if mpls-exp topmost exist-in (1, 2) then ...</pre>		

9.25.12. packet-type

Description	Enables classifying based on packet type.		
Command Syntax	packet-type eq broadcast		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	broadcast		Specifies MAC frames with the BROADCAST Destination MAC address.
Examples	<pre> policy qos POL rule RULE_1 if packet-type eq broadcast then port-level-rate 45 pps burst 2 packets rule rule-default end-policy </pre>		

9.25.13. precedence

Description	Enables classifying based on precedence. The precedence value is located in the TOS byte in the IP header. Bits [0-3] of the TOS byte define eight possible values (0-7) that categorize the priority of the packet.		
Command Syntax	<pre>precedence[not] [eq {value \$parameter} exist-in inline- precedence-group]</pre>		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	<i>value</i>	0-7	Precedence value in TOS byte in IP header.
	<i>parameter</i>		Parameter the contains a precedence value.
	<i>inline-group</i>		Group of precedence values.
Examples	<pre>policy qos MY_QOS_POLICY rule Rule1 if precedence eq 5 then set red-tag 1 rule Rule2 if precedence eq \$NC_PRECEDENCE then set red-tag 2 rule Rule3 if precedence not eq 0 then set red-tag 3 rule Rule4 if precedence exist-in (1,3, \$MY_PRECEDENCE_VAL) then set red-tag 4</pre>		

9.25.14. protocol-ip

Description	Enables classifying based on protocol. There are two types of protocols: • Transport protocols: The value will be checked against the protocol field in the IP header. • Network protocol: ipv4/ipv6/mpls. The value will be checked against the protocol field inside the layer 2 header.		
Command Syntax	<code>protocol [eq {value \$parameter} exist-in inline-protocol-group]</code>		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	<i>value</i>	0-255	Can be any number in dec/hex representation.
		protocol name	Possible values for transport protocols: tcp, udp, ahp1, esp, gre, icmpv4, igmp,igrp, ipinip, nos2, ospf, pcp3, pim, sctp Possible values for network protocols: ipv4, ipv6 and mpl.
	<i>parameter</i>		Parameter representing a protocol value.
	<i>inline-group</i>		Group of protocol values.

Examples	Example 1: Will match all tcp based packets originated from 192.0.2.78: <pre>rule MY_RULE1 if match-all protocol-ip eq tcp source-ip eq 192.0.2.78 then set dscp 1</pre> Example 2: <pre>rule MY_RULE2 if match-all protocol-ip eq \$MY_PROTOCOL source-ip eq \$MEDIA_DEVICE_IP then set dscp 2</pre> Example 3: <pre>rule MY_RULE3 if protocol-ip exist-in (icmpv4, igmp, 51, 94) then set dscp 3</pre>
Notes	For the MO (adm subscribers) the values are represented as numerical values only, and not as enums. Each of the inline group elements can use local/global parameters.

9.25.15. queue

Description	Configures the queue parameters.		
Command Syntax	queue eq [<i>queue-type</i>]		
Command Mode	configuration policy qos		
Parameters Table	Name	Value	Description
	<i>queue-type</i>	ef1	The queue this policy relates to.
		ef2	
		af1	
		af2	
		af3	
		af4	
		af5	
		be	
Examples	<pre>policy qos Egress rule r1 if queue eq af1 then minimum-bandwidth 10 mbps queue-max-size 100000 bytes rule rule-default end-policy !</pre>		
Notes	A policy matching on queue can only be used on egress policies. minimum-bandwidth is supported for af1-af5 queue types.		

9.25.16. source ip

Description	Enables classifying based source IP address.		
Command Syntax	<pre>source-ip [eq {ip-address \$parameter} exist-in ip-prefix inline-ip-prefix-group named-ip-prefix-group]</pre>		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IP address	IP address.
	<i>parameter</i>		Parameter containing one IP address (either ipv4 or ipv6 format).
	<i>ip-prefix</i>	ip-prefix	
	<i>inline-ip-prefix-group</i>	ip prefixes	IP address range or list of IP addresses.
	<i>named-ip-prefix-group</i>	ip-prefix-group	User-defined ip-prefix group.

Examples	<pre> policy group ip-prefix VPN_CLUSTER 10.20.0.0/16 30.40.0.0/16 2001:0db8:85a3:0:0:8a2e:0370:7334 1050:0:0:0:5:600:300c:326b end-group ! policy qos MY_QOS_POLICY rule RULE_1 if match-any source-ip exist-in VPN_CLUSTER then set dscp 1 rule RULE_2 if source-ip exist-in (20.20.20.0/24 ,30.30.30.0/24) then set dscp 2 rule RULE_3 if source-ip eq 20.20.20.0 then set dscp 3 </pre>
Notes	The named ip-prefix group is defined using the policy group ip-prefix (on page 602) command. The 'exist-in' can be followed by list of groups. Each group in the list can be either inline or named ip-prefix group. All groups in the list will be in same line, separated by space . See configuration example. Each of the inline prefix group elements can use local/global parameters.

9.25.17. source port

Description	Enables classifying based source port.		
Command Syntax	<code>source-port [eq {value \$parameter} exist-in {inline-port-group} gt {value \$parameter} lt {value \$parameter}]</code>		
Command Mode	configuration interface <interface-name> policy-qos		
Parameters Table	Name	Value	Description
	<i>value</i>	0-65535 Predefined values: according to TCP and UDP tables	Source port number or source-ports.
	<i>parameter</i>		Parameter that contains the source port number or range.
	<i>inline-port-group</i>		List of source port numbers or ranges used for filtering. Range is used only in exist-in list. Range is set by using "-" between two numbers.

Examples	<pre> policy global-parameters parameter PORT_MAX 42000 parameter PORT_BOTTOM 40000 end-global-parameters ! policy qos MY_ACL_POLICY rule RULE_1 if source-port eq 2042 then set dscp 1 rule RULE_2 if source-port exist-in (\$PORT_BOTTOM-\$PORT_MAX) then set dscp 2 rule RULE_3 if source-port exist-in (38000-39000) then set dscp 3 rule RULE_4 if source-port gt 38000 then set dscp 4 rule rule-default end-policy !</pre>
Notes	Range is not supported in release 8.0.0

TCP-port predefined-numbers:

Port number	Description	Predefined value
179	Boarder Gateway Protocol	bgp
21	Domain Name System	ftp
20	File Transfer Protocol	ftp-data
80	Hypertext Transfer Protocol	http
443	Secure HTTP	https
220	Internet Message Access Protocol	imap

Port number	Description	Predefined value
389	Lightweight Directory Access Protocol	ldap
646	Label Distribution Protocol	ldp
445	Microsoft DS (AD)	ms-ds
496	PIM Auto-RP	pim-auto-rp
110	Post Office Protocol v3	pop3
995	Secure Post Office Protocol v3	pop3s
1723	Point-to-Point Tunneling protocol	pptp
25	Simple Mail Transport Protocol	smtp
22	Secure shell	ssh
514	Syslog	syslog
49	TAC Access Control System	tacacs

UDP-port predefined-numbers:

Port number	Description	Predefined value
68	Bootstrap Protocol (BOOTP) client	bootpc
67	Bootstrap Protocol (BOOTP) server	bootps
53	Domain Name Service (DNS)	domain
500	Internet Security Association and Key Management Protocol	isakmp
389	Lightweight Directory Access Protocol	ldap
646	Label Distribution Protocol	ldp
445	Microsoft DS (SMB)	ms-ds
138	NetBios datagram service	netbios-dgm
137	NetBios name service	netbios-ns
139	NetBios session service	netbios-ss

Port number	Description	Predefined value
4500	Internet Security Association and Key Management Protocol	non500-isakmp
123	Network Time Protocol	ntp
496	PIM Auto-RP	pim-auto-rp
1646	RADIUS accounting	radacct
1645	RADIUS authentication	radius
161	Simple Network Management Protocol	snmp
162	SNMP Traps	snmptrap
514	System Logger	syslog
49	TAC Access Control System	tacacs
69	Trivial File Transfer Protocol	tftp
68	Bootstrap Protocol (BOOTP) client	bootpc

Port number	Description	Predefined value
67	Bootstrap Protocol (BOOTP) server	bootps
53	Domain Name Service (DNS)	domain
500	Internet Security Association and Key Management Protocol	isakmp
389	Lightweight Directory Access Protocol	ldap
646	Label Distribution Protocol	ldp
445	Microsoft DS (SMB)	ms- ds
138	NetBios datagram service	netbios-dgm
137	NetBios name service	netbios-ns
139	NetBios session service	netbios- ss
4500	Internet Security Association and Key Management Protocol	non500-isakmp
123	Network Time Protocol	ntp
496	PIM Auto-RP	pim-auto-rp

9.25.18. tcp-flags

Description	Enables classifying based on specified TCP protocol flags. The filtering is done according to the five bits of the TCP header flags.		
Command Syntax	tcp-flags [established initial] [set {[fin] [syn] [rst] [psh] [ack]} unset {[fin] [syn] [rst] [psh] [ack]}]		
Command Mode	configuration policy qos/acl		
Parameters Table	Name	Value	Description
		established	Optional flag to indicate established connection.
		initial	Optional flag to indicate first packet of the connection.
		set/unset <i>predefined flag</i>	Set or unset inline TCP flags for filtering TCP packets.
Examples	<pre>rule MY_RULE1 if protocol eq tcp tcp-flags established then ... rule MY_RULE2 if protocol eq tcp tcp-flags set (ack) tcp-flags unset (syn, push) then ...</pre>		
Notes	'tcp-flags set' and 'tcp-flags unset' commands are not mutually exclusive. They can both appear in the same condition. 'tcp-flags established' is mutually exclusive with any other 'tcp-flags' command. 'tcp-flags initial' is mutually exclusive with any other 'tcp-flags' command.		

TCP supported flags:

TCP flag value	Description	Predefined value
1	No more data from sender	fin
2	Synchronize sequence numbers.	syn
4	Reset the connection	rst
8	Push function. Asks to push the buffered data to the receiving application	psh
16	Indicates that the Acknowledgment field is significant.	ack

9.25.19. ttl

Description	Enables classifying based on the time to live (TTL) field. The filtering is done according to the eight bits of the 'TTL' field, in the IP packet header. TTL is a limit on the period of time or number of iterations or transmissions that a unit of data can experience before it should be discarded.
Command Syntax	<code>ttl [eq <i>value</i> exist-in <i>inline-ttl-group</i> gt <i>value</i> lt <i>value</i>]</code>
Command Mode	configuration policy qos/acl
Examples	<pre>rule MY_RULE1 if ttl eq 40 then set dscp 1 rule MY_RULE2 if ttl lt 4 then set dscp 2 rule MY_RULE3 if ttl exist-in (12,13,155) then set dscp 3</pre>

9.26. QoS Policy Operational Commands

9.26.1. clear qos counters

Description	Resets the qos counters either for a specified interface or for all interfaces.		
Command Syntax	clear qos counters [all interface <i>interface-id</i> {in out}]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>interface-id</i>	valid interface ID	Interface on which to clear the qos counters. Use the all keyword to clear counters on all interfaces.
		in out	Traffic direction.
Examples	The following examples show how to use this command: <pre>exaware# clear qos counters interface x-eth 0/0/1 in exaware# clear qos counters all</pre>		

9.27. QoS Policy Show Commands

9.27.1. `show configuration policy qos`

Description	Displays all QoS policies configured in the router.
Command Syntax	<code>show configuration policy qos</code>
Command Mode	operational

Examples

The following example shows sample output from this command:

```
exaware# show configuration policy qos
policy qos CHILD_POLICY
  rule RULE1
    if
    then
      police
        cir 10 mbps burst 100 kbytes
        conform-action set red-tag 3
        exceed-action set red-tag 6
      rule rule-default
        queue-max-size 100 msec
        remaining-bandwidth percent 1
    end-policy
  !
policy qos EGR_QOS
  rule r1
    if source-ip exist-in (20.20.20.0/24)
    then
      minimum-bandwidth 20 mbps
      queue-max-size 5 mbytes
    rule rule-default
  end-policy
  !
policy qos PARENT_POLICER
  rule rule-default
    police
      cir 20 mbps burst 200 kbytes
      conform-action transmit
      exceed-action transmit
      violate-action drop
  end-policy
  !
policy qos PARENT_POLICY
  rule rule-default
    police
      cir 20 mbps burst 200 kbytes
      conform-action transmit
      exceed-action transmit
      violate-action drop
      exec-policy CHILD_POLICY
  end-policy
  !
policy qos POL
  rule RULE_1
    if packet-type eq broadcast
      then port-level-rate 45 pps burst 2 packets
  end-policy
  !
```

9.27.2. show qos interface agg-verbose

Description	Displays detailed QOS information for the LAG (or LAG sub-interface) and all its members.		
Command Syntax	show qos interface agg-verbose <lag-interface> [in out]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	lag-interface	LAG interface ID	agg-eth[0-79].[1-4094]
		in out	Traffic direction.
Examples	The following example shows sample outputs from this command:		
<pre>exaware# show qos interface agg-verbose agg-eth5.1 interface:agg-eth5.1, direction:out, ref:parent, policy:q1 rule r1 ----- Classification packets/bytes Total match 0/0 Queuing statistics Default-queue is used. rule r2 ----- Classification packets/bytes Total match 0/0 Queuing statistics Default-queue is used. rule rule-default ----- Classification packets/bytes Total match 0/0 Queuing hardware values Priority: Best-effort</pre>			

Min bandwidth:	0 bps (0%)
Min burst:	0 bytes
Remaining bandwidth:	0%
Max bandwidth:	0 bps
Max burst:	0 bytes
Guaranteed queue rate:	0 bps
Max queue size:	0 bytes / 0 usec
Instantaneous queue size	NA
Default-queue queuing statistics	packets/bytes
Total transmitted	0/0
Tail drop	0/0
interface:x-eth0/0/0.1, direction:out, ref:parent, policy:q1	
rule r1	

Classification	packets/bytes
Total match	0/0
Queuing statistics	
Default-queue is used.	
rule r2	

Classification	packets/bytes
Total match	0/0
Queuing statistics	
Default-queue is used.	
rule rule-default	

Classification	packets/bytes
Total match	0/0
Queuing hardware values	
Priority:	Best-effort
Min bandwidth:	0 bps (0%)
Min burst:	0 bytes
Remaining bandwidth:	0%
Max bandwidth:	0 bps
Max burst:	0 bytes
Guaranteed queue rate:	0 bps
Max queue size:	0 bytes / 0 usec
Instantaneous queue size	0 bytes
Default-queue queuing statistics	packets/bytes

Total transmitted	0/0
Tail drop	0/0

9.27.3. show qos interface brief

Description	Displays statistical information on QoS traffic per interface and direction.		
Command Syntax	show qos interface brief [<i>interface-id</i> [in out]]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>interface-id</i>	valid interface ID	Interface for which to display the statistics.
		in out	Traffic direction.
Examples	The following examples show sample outputs from this command:		

Example 1: qos parent and child policies attached to x-eth 0/0/0.100 interface.

Configuration:

```

policy qos in_child
  rule r1
    if dscp eq 1
    then
      police
        cir 1 gbps
        conform-action transmit
        exceed-action transmit
        violate-action drop
  rule rule-default
end-policy
!
policy qos in_parent
  rule rule-default
    exec-policy in_child
  police
    cir 3 gbps
    conform-action transmit
    exceed-action transmit
    violate-action drop
end-policy
!
```

```

interface x-eth 0/0/0.100
  policy qos in in_parent
!

exaware# show qos interface brief x-eth0/0/0.100 in

interface:eth0/0/0.100, direction:in, ref:parent, policy:in_parent

  rule rule-default
  -----
  Classification                                packets/bytes
    Total match                                300000/298800000

  Policer statistics                            packets/bytes
    Conform                                    300000/298800000
    Violate(drop)                             0/0

interface:eth0/0/0.100, direction:in, ref:child, policy:in_child

  rule r1
  -----
  Classification                                packets/bytes
    Total match                                50000/48900000

  Policer statistics                            packets/bytes
    Conform                                    50000/48900000
    Exceed                                     0/0
    Violate(drop)                             0/0

  rule rule-default
  -----
  Classification                                packets/bytes
    Total match                                100000/97800000

```

Example 2: qos parent and child policies attached to agg-eth 22 interface.

agg-eth 22 has one member x-eth/0/2

Configuration:

```

policy group qos-red-curve CURVE1
  0:0
  10:10
  100:100
end-group
!

```

```

policy qos out_child
rule r1
  if queue eq afl
  then set dscp 2
rule r2
  if queue eq af2
  then queue-max-size 200 mbytes
  rule rule-default
end-policy
!
policy qos out_parent
rule rule-default
  shape 10 gbps
  exec-policy out_child
end-policy
!
interface agg-eth 22
  policy qos out out_parent
!
interface x-eth 0/0/2
  aggregator-id 22
!

exaware# show qos interface brief

interface:agg-eth22, direction:out, ref:parent, policy:out_parent

rule rule-default
-----
Classification                                packets/bytes
Total match                                333/323826

interface:agg-eth22, direction:out, ref:child, policy:out_child

rule r1
-----
Classification                                packets/bytes
Total match                                77/75306

Queuing statistics
Default-queue is used.

rule r2
-----
Classification                                packets/bytes
Total match                                48/46944

Queuing statistics                                packets/bytes

```



```

Total transmitted          48/46944
Tail drop                  49952/48853056

rule r3
-----
Classification              packets/bytes
  Total match                48/46944

Queuing statistics          packets/bytes
  Total transmitted          48/46944
  Tail drop                  49952/48853056

Random-detect (precedence based)
Value                       curve
drop(packets/bytes)
  3                           CURVE1              0/0
  default                     default-curve       0/0

rule rule-default
-----
Classification              packets/bytes
  Total match                61/57810

Default-queue queuing statistics
  Total transmitted          138/133116
  Tail drop                  149863/146566014

interface:x-eth0/0/2, direction:out, ref:parent, policy:out_parent

rule rule-default
-----
Classification              packets/bytes
  Total match                333/323826

interface:x-eth0/0/2, direction:out, ref:child, policy:out_child

rule r1
-----
Classification              packets/bytes
  Total match                77/75306

Queuing statistics
  Default-queue is used.

rule r2
-----

```

Classification	packets/bytes		
Total match	48/46944		
Queuing statistics	packets/bytes		
Total transmitted	48/46944		
Tail drop	49952/48853056		
rule r3			

Classification	packets/bytes		
Total match	48/46944		
Queuing statistics	packets/bytes		
Total transmitted	48/46944		
Tail drop	49952/48853056		
Random-detect (precedence based)			
Value	curve		
drop(packets/bytes)			
3	CURVE1		0/0
default	default-curve		0/0
rule rule-default			

Classification	packets/bytes		
Total match	61/57810		
Default-queue queuing statistics	packets/bytes		
Total transmitted	138/133116		
Tail drop	149863/146566014		

Notes	In hierarchical ingress policies, the parent processing precedes the child processing. All packets that are dropped by the parent policer are not counted in the child. The parent policer does the actual packet drops. All dropped packets are counted at the parent level. In hierarchical egress policies, the child processing precedes the parent. Packets dropped at the child level (due to tail-drop, red-drop, or policer-drop) are not counted at the parent level.
--------------	---

Explanation of compound values

Name	Description	Comments	Default
Total Match	Number of packets/bytes that matched the rule	ERRATA : The total match counter does not include the packets that are dropped as a result of queue tail-drop or red-drop.	
Conform	Total number of packets/bytes conformed to the cir and cir-burst values		
Exceed	Total number of packets/bytes conformed to the pir and pir-burst values	If the drop is configured for exceed actions, it will appear in brackets.	

Name	Description	Comments	Default
Violate	Total number of packets/bytes that violate policer's rates	If the drop is configured for exceed actions, it will appear in brackets.	
Total Transmitted	Total packets /bytes transmitted	Total transmitted packets. This field is shown only if queue is allocated for the rule.	
Tail drop	Total number of packets/bytes dropped as a result of queue overflow (tail drop).	When the rule queue is full, packets will be dropped. This counter counts total packets dropped as a result.	

9.27.4. show qos interface detail

Description	Displays the exact configured values used in the hardware after some rounding was made along with statistical data. Aggregated interfaces are not been presented in this show.		
Command Syntax	show qos interface brief [<i>interface-id</i> [in out]]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	interface-id	valid interface ID	Interface for which to display the statistics.
		in out	Traffic direction.
Examples	The following example shows sample output from this command:		

```
# show qos interface detail x-eth0/0/1 out
Exaware(config-policy-qos-in_child-rule-r1)# run show qos interface brief

interface:x-eth0/0/0.100, direction:in, ref:parent, policy:in_parent
rule rule-default
-----
Classification                                packets/bytes
Total match                                300000/298800000

Policer configuration:
CIR rate:          1000000 kbps,   CIR Burst:      2097120 bytes (default)
PIR rate:          1500000 kbps ,   PIR Burst:      2397120 bytes (default)

Policer statistics                                packets/bytes
Conform                                300000/298800000
Violate(drop)                            0/0

interface:x-eth0/0/0.100, direction:in, ref:child, policy:in_child
rule r1
-----
Classification                                packets/bytes
Total match                                50000/48900000
```

```

Policer configuration:
    CIR rate:      1000000 kbps,   CIR Burst:      2097120 bytes (default)
    PIR rate:      1500000 kbps ,   PIR Burst:      2397120 bytes (default)

Policer statistics
Conform           packets/bytes      50000/48900000
Exceed            0/0
Violate(drop)     0/0

rule rule-default
-----
Classification    packets/bytes
Total match       100000/97800000

interface:x-eth0/0/0.200, direction:out, ref:parent, policy:out_parent
rule rule-default
-----
Shaper hardware values
Rate:             0 bps
Burst:            0 bytes

Classification    packets/bytes
Total match       0/0

interface:x-eth0/0/0.200, direction:out, ref:child, policy:out_child
rule r1
-----
Classification    packets/bytes
Total match       77/75306

Queuing statistics
Default-queue is used.

rule r2
-----
Classification    packets/bytes
Total match       48/46944

Queuing configuration
Min bandwidth     2000128 kbps (20 %)
Min burst         100000bps
Remaining bandwidth 10% (default value)
Max bandwidth     10000128 kbps (default)
Max burst         31248 bytes
Guaranteed queue rate 4666666 kbps
Max queue size    12500000 bytes / 100 ms (default)
Instantaneous queue size (bytes) 76749

```

```

Queuing statistics                                packets/bytes
Total transmitted                                48/46944
Tail drop                                         49952/48853056

rule r3
-----
Classification                                packets/bytes
Total match                                     48/46944

Policer configuration:
  CIR rate:          1000000 kbps,   CIR Burst:      2097120 bytes (default)
  PIR rate:          1500000 kbps ,   PIR Burst:      2397120 bytes (default)

Policer statistics                                packets/bytes
Conform                                              48/46944
Exceed                                              0/0
Violate(drop)                                       0/0

Queuing configuration
  Min bandwidth      2000128 kbps (20 %)
  Min burst          1000000bps
  Remaining bandwidth 10% (default value)
  Max bandwidth      10000128 kbps (default)
  Max burst          31248 bytes
  Guaranteed queue rate 4666666 kbps
  Max queue size      12500000 bytes / 100 ms (default)
  Instantaneous queue size (bytes) 76749

Queuing statistics                                packets/bytes
Total transmitted                                48/46944
Tail drop                                         49952/48853056

Random-detect (precedence based)
  Value                                                    curve
drop(packets/bytes)
  3                                                         CURVE1          0/0
  default                                                    default-curve      0/0

rule r4
-----
Classification                                packets/bytes
Total match                                     38403/37558134

Policer configuration:
  CIR rate:          1000000 kbps,   CIR Burst:      2097120 bytes (default)
  PIR rate:          1500000 kbps ,   PIR Burst:      2397120 bytes (default)

Policer statistics                                packets/bytes

```

Conform	99/96822
Exceed	0/0
Violate(drop)	38304/37461312
Queuing configuration	
Min bandwidth	2000128 kbps (20 %)
Min burst	100000bps
Remaining bandwidth	10% (default value)
Max bandwidth	10000128 kbps (default)
Max burst	31248 bytes
Guaranteed queue rate	4666666 kbps
Max queue size	12500000 bytes / 100 ms (default)
Instantaneous queue size (bytes)	76749
Queuing statistics	packets/bytes
Total transmitted	99/96822
Tail drop	11597/11341866
rule rule-default	

Classification	packets/bytes
Total match	61/57810
Queuing configuration	
Min bandwidth	2000128 kbps (20 %)
Min burst	100000bps
Remaining bandwidth	10% (default value)
Max bandwidth	10000128 kbps (default)
Max burst	31248 bytes
Guaranteed queue rate	4666666 kbps
Max queue size	12500000 bytes / 100 ms (default)
Instantaneous queue size (bytes)	76749
Default-queue queuing statistics	packets/bytes
Total transmitted	138/133116
Tail drop	149863/146566014

Notes	In hierarchical ingress policies, the parent processing precedes the child processing. All packets that are dropped by the parent policer are not counted in the child. The parent policer does the actual packet drops. All dropped packets are counted at the parent level. In hierarchical egress policies, the child processing precedes the parent. Packets dropped at the child level (due to tail-drop, red-drop, or policer-drop) are not counted at the parent level.
--------------	---

Explanation of compound values

For an explanation of compound values related to statistical data refer to:

`show qos interface brief`

Name	Description	Comments	Default
Rate	Shaper rate	When the rate is higher than 1000, the values will be shown in kbps units.	
Burst	Shaper burst		
CIR rate	Committed Information Rate value	When the user configures QoS values, there might be something rounded up done by h/w. the value displayed here are the actual values the h/w uses. When the rate is higher than 1000, the values will be shown in kbps units.	
CIR Burst	Committed Information Rate burst rate		

Name	Description	Comments	Default
PIR rate	Peak Information Rate rate values	When the rate is higher than 1000, the values will be shown in kbps units.	
PIR Burst	Peak Information Rate burst		
Min bandwidth	minimum-bandwidth guaranteed for the rule packets	When the rate is higher than 1000, the values will be shown in kbps units.	
Min burst	Burst value allowed		
Remaining bandwidth	Percent of remaining b/w allocated for the rule	When there is leftover b/w, it is allocated between all rules according to user configuration (by bandwidth-remaining is configured). If the user did not configure this value explicitly, the remain b/w will be divided equally.	
Max bandwidth	Maximum amount of traffic allowed to be used by the rule (shaper)		
Max burst	Maximum amount of traffic allowed to be sent in a burst in relation to the value of maximum b/w		

Name	Description	Comments	Default
Guaranteed queue rate	Minimum bandwidth used by the rule plus amount remaining bandwidth	If user configures minimum b/w more than possible capacity (shape or physical rate), the value here is identical to minimum b/w. When the rate is higher than 1000, the values will be shown in kbps units.	
Max-queue size	Max queue size used. Packets exceeding this limit are dropped		

9.27.5. show qos policy

Description	Displays a summary of QoS information.																																																																										
Command Syntax	show qos policy statistics [<i>interface-id</i> [in out]]																																																																										
Command Mode	operational																																																																										
Parameters Table	Name	Value	Description																																																																								
	interface-id	valid interface ID	Interface for which to display the statistics.																																																																								
		in out	Traffic direction.																																																																								
Examples	The following example shows sample output from this command: <pre>exaware# show qos policy statistics</pre> Queue flags: S - Rule is using shared queue N/A - No queue configured <table><thead><tr><th>IF</th><th>DIR</th><th>REF</th><th>POL</th><th>RULE</th><th>HIT COUNT</th><th>POLICE DROP</th><th>QUEUE DROP</th></tr></thead><tbody><tr><td>x-eth0/0/0.1</td><td>in</td><td>parent</td><td>in_p</td><td>r1</td><td>200</td><td>45</td><td>N/A</td></tr><tr><td>x-eth0/0/0.1</td><td>out</td><td>parent</td><td>ch</td><td>r1</td><td>1000</td><td>0</td><td>423</td></tr><tr><td></td><td></td><td></td><td></td><td>r2</td><td>200</td><td>0</td><td>S</td></tr><tr><td></td><td></td><td></td><td></td><td>def</td><td>40088</td><td>0</td><td>423</td></tr><tr><td>x-eth0/0/0.2</td><td>out</td><td>parent</td><td>pa</td><td>def</td><td>1000</td><td>356</td><td>N/A</td></tr><tr><td></td><td></td><td>child</td><td>ch</td><td>r1</td><td>200</td><td>0</td><td>S</td></tr><tr><td></td><td></td><td></td><td></td><td>r2</td><td>40088</td><td>0</td><td>S</td></tr><tr><td></td><td></td><td></td><td></td><td>def</td><td>500</td><td>0</td><td>423</td></tr></tbody></table>			IF	DIR	REF	POL	RULE	HIT COUNT	POLICE DROP	QUEUE DROP	x-eth0/0/0.1	in	parent	in_p	r1	200	45	N/A	x-eth0/0/0.1	out	parent	ch	r1	1000	0	423					r2	200	0	S					def	40088	0	423	x-eth0/0/0.2	out	parent	pa	def	1000	356	N/A			child	ch	r1	200	0	S					r2	40088	0	S					def	500	0	423
IF	DIR	REF	POL	RULE	HIT COUNT	POLICE DROP	QUEUE DROP																																																																				
x-eth0/0/0.1	in	parent	in_p	r1	200	45	N/A																																																																				
x-eth0/0/0.1	out	parent	ch	r1	1000	0	423																																																																				
				r2	200	0	S																																																																				
				def	40088	0	423																																																																				
x-eth0/0/0.2	out	parent	pa	def	1000	356	N/A																																																																				
		child	ch	r1	200	0	S																																																																				
				r2	40088	0	S																																																																				
				def	500	0	423																																																																				

10. Routing Commands

10.1. BFD Routing Commands

10.1.1. Single Hop BFD Session

A Single Hop BFD session is identified by an interface and a remote IP address. The Single Hop BFD is applied to the following BFD clients:

- OSPF
- OSPFv3
- IS-IS

All Single Hop sessions are running on the same interface for a given IP protocol (IPv4 or IPv6) have the same BFD parameters. The following example configures the IPv4 BFD sessions parameters running on interface x-eth 0/0/4:

```
exaware# config
exaware(config)# bfd single-hop af-ipv4
exaware(config-bfd-single-hop-af-ipv4)# interface x-eth 0/0/4
exaware(config-bfd-single-hop-af-ipv4-interface-x-eth-0/0/4)# admin-
state up
exaware(config-bfd-single-hop-af-ipv4-interface-x-eth-0/0/4)#
minimal-tx-interval 100
exaware(config-bfd-single-hop-af-ipv4-interface-x-eth-0/0/4)#
multiplier 4
```

A Single Hop BFD session is established based on the needs of the BFD client that will be making use of it. It is up to the application to determine the need for BFD, and the addresses to use -- there is no discovery mechanism in BFD. For example, an OSPF implementation may request a BFD session to be established to a neighbor discovered using the OSPF Hello protocol.

For example, the following OSPF configuration requires to run the BFD sessions for each its neighbors on all interfaces except x-eth 0/0/4:

```
exaware(config)# routing ospf ospf1
exaware(config-routing-ospf-ospf1)# vrf vrf1
exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10
exaware(config-routing-ospf-ospf1-vrf-vrf1-area-0.0.0.10)# interface
x-eth 0/0/4
exaware(config-routing-ospf-ospf1-vrf-vrf1-area-0.0.0.10-interface-
x-eth- 0/0/4)# bfd disabled
```

10.1.2. Multi Hop BFD Session

A Multi Hop BFD Session is identified by an address family and a neighbor's remote IP address.

The following example configures the default BFD parameters for all Multi Hop IPv4 sessions and private BFD parameters for IPv4 multi-hop session with neighbor 10.1.10.3:

```
exaware# config
exaware(config)# bfd multi-hop af-ipv4
exaware(config-bfd-multi-hop-af-ipv4)# admin-state local-only
exaware(config-bfd-multi-hop-af-ipv4)# minimal-rx-interval 100
exaware(config-bfd-multi-hop-af-ipv4)# minimal-tx-interval 100
exaware(config-bfd-multi-hop-af-ipv4)# multiplier 4
exaware(config-bfd-multi-hop-af-ipv4)# neighbor 10.1.10.3 vrf
default
exaware(config-bfd-multi-hop-af-ipv4-10.1.10.3)# minimal-rx-interval
1000
exaware(config-bfd-multi-hop-af-ipv4-10.1.10.3)# minimal-tx-interval
1000
exaware(config-bfd-multi-hop-af-ipv4-10.1.10.3)# multiplier 3
```

The following example activates BFD for BGP with neighbors 10.1.10.3 and 10.1.10.4:

```
exaware(config)# routing bgp 61
exaware(config-routing-bgp-61)# vrf default
exaware(config-routing-bgp-61-vrf-default)# neighbor 10.1.10.3
exaware(config-routing-bgp-61-vrf-default-neighbor-10.1.10.3)# bfd
enable
exaware(config-routing-bgp-61-vrf-default)# neighbor 10.1.10.4
exaware(config-routing-bgp-61-vrf-default-neighbor-10.1.10.4)# bfd
enable
```

10.1.3. BFD Configuration Commands

10.1.3.1. **admin-state**

Description	The command configures the BFD Administrative State. Use the no form of the command to restore the default configuration. The default state depends on Command Mode: • bfd multi-hop af-ipv4: down. • bfd multi-hop af-ipv6: down. • bfd single-hop af-ipv4 interface: local-only. • bfd single-hop af-ipv6 interface: local-only.		
Command Syntax	admin-state {down local-only up} no admin-state		
Command Mode	configuration bfd multi-hop af-ipv4 configuration bfd multi-hop af-ipv6 configuration bfd single-hop af-ipv4 interface configuration bfd single-hop af-ipv6 interface		
Parameters Table	Name	Value	Description
	down		• Requests from local clients are saved, but the BFD sessions are not created. • Requests from remote peers are rejected.
	local-only		• Requests from local clients are accepted. • Requests from remote peers are rejected.
	up		• Requests from local clients are accepted. • Requests from remote peers are accepted.

Examples

This example shows how to use this command:

Example 1. Single Hop BFD

```
exaware# config
exaware(config)# bfd single-hop af-ipv4
exaware(config-bfd-single-hop-af-ipv4)# interface x-eth 0/0/4
exaware(config-bfd-single-hop-af-ipv4-interface-x-eth-0/0/4)#
admin-state up
```

Example 2. Multi Hop BFD

```
exaware# config
exaware(config)# bfd multi-hop af-ipv4
exaware(config-bfd-af-ipv4)# interface x-eth 0/0/4
exaware(config-bfd-af-ipv4-interface-x-eth-0/0/4)# admin-state up
```

10.1.3.2. bfd

Description	The command enters the interface BFD context allowing configuring the BFD parameters. Use the no form of the command to restore the default BFD interface configuration.		
Command Syntax	bfd {single-hop multi-hop} {af-ipv4 af-ipv6} no {single-hop multi-hop} {af-ipv4 af-ipv6}		
Parameters Table	Name	Value	Description
	single-hop		BFD Single Hop
	multi-hop		BFD Multi Hop
	af-ipv4		The BFD is running over IPv4.
	af-ipv6		The BFD is running over IPv6.
Command Mode	configuration		
Examples	These examples show how to use this command: <pre>exaware# config exaware(config)# bfd multi-hop af-ipv4 exaware(config-bfd-multi-hop-af-ipv4)#</pre>		

10.1.3.3. interface

Description	Enters BFD Single Hop configuration mode for a specific interface. Use the no form of the command to disable BFD for the interface or for the level.		
Command Syntax	<pre>interface <i>interface-name</i> no interface <i>interface-name</i></pre>		
Command Mode	<pre>configuration bfd single-hop af-ipv4 configuration bfd single-hop af-ipv6</pre>		
Parameters Table	Name	Value	Description
	<i>interface-name</i>	<pre>x-eth 0/0/port agg-eth <i>number</i></pre>	Interface name.
Examples	These examples show how to use this command: <pre>exaware# config exaware(config)# bfd single-hop af-ipv4 exaware(config-bfd-single-hop-af-ipv4)# interface x-eth 0/0/4 exaware(config-bfd-single-hop-af-interface-x-eth-0/0/4)#</pre>		

10.1.3.4. minimal-rx-interval

Description	The command configures the BFD Session Minimal Required RX Interval. Use the no form of the command to restore the default configuration.		
Command Syntax	minimal-rx-interval <i>milliseconds</i> no minimal-rx-interval		
Command Mode	configuration bfd multi-hop af-ipv4 configuration bfd multi-hop af-ipv6 configuration bfd multi-hop af-ipv4 neighbor configuration bfd multi-hop af-ipv6 neighbor configuration bfd single-hop af-ipv4 interface configuration bfd single-hop af-ipv6 interface		
Parameters Table	Name	Value	Description
	milliseconds	Value into range: 3..1700	Specifies the minimal interval in milliseconds. The value applied to a BFD session is: • Single hop: ○ A value configured per the interface (highest priority) ○ 50 msec • Multi hop: ○ A value configured per IP the Address (highest priority) ○ A value configured per the Address Family • 500 msec

Examples	These examples show how to use this command: Example 1. Single Hop BFD <pre>exaware# config exaware(config)# bfd single-hop af-ipv4 exaware(config-bfd-single-hop-af-ipv4)# interface x-eth 0/0/4 exaware(config-bfd-single-hop-af-ipv4-interface-x-eth-0/0/4)# minimal-rx-interval 100</pre> Example 2. Multi Hop Hop BFD <pre>exaware# config exaware(config)# bfd multi-hop af-ipv4 exaware(config-bfd-af-ipv4)# minimal-rx-interval 500 exaware(config-bfd-af-ipv4)# neighbor 10.0.0.3 vrf default exaware(config-bfd-af-ipv4-10.0.0.3)# minimal-rx-interval 100</pre>
Notes	• multiplier * minimal-rx-interval must be less or equal 5000

10.1.3.5. minimal-tx-interval

Description	The command configures the BFD Session Minimal Desired TX Interval. Use the no form of the command to restore the default configuration.		
Command Syntax	minimal-tx-interval <i>milliseconds</i> no minimal-tx-interval		
Command Mode	configuration bfd multi-hop af-ipv4 configuration bfd multi-hop af-ipv6 configuration bfd multi-hop af-ipv4 neighbor configuration bfd multi-hop af-ipv6 neighbor configuration bfd single-hop af-ipv4 interface configuration bfd single-hop af-ipv6 interface		
Parameters Table	Name	Value	Description
	milliseconds	Integer from the following range: 3.3, 10, 20, 50, 100, 200, 500, 1000	Specifies the minimal interval in milliseconds. The value applied to a BFD session is: • Single hop: ○ A value configured per the interface (highest priority) ○ 50 msec • Multi hop: ○ A value configured per the IP Address (highest priority) ○ A value configured per the Address Family ○ 500 msec

Examples	These examples show how to use this command: Example 1. Single Hop BFD <pre>exaware# config exaware(config)# bfd single-hop af-ipv4 exaware(config-bfd-single-hop-af-ipv4)# interface x-eth 0/0/4 exaware(config-bfd-single-hop-af-interface-x-eth-0/0/4)# minimal- tx-interval 100</pre> Example 2. Multi Hop Hop BFD <pre>exaware# config exaware(config)# bfd multi-hop af-ipv4 exaware(config-bfd-af-ipv4)# minimal-tx-interval 500 exaware(config-bfd-af-ipv4)# neighbor 10.0.0.3 vrf default exaware(config-bfd-af-ipv4-10.0.0.3)# minimal-tx-interval 100</pre>
Notes	• multiplier * minimal-tx-interval must be less or equal 5000

10.1.3.6. multiplier

Description	The command configures the BFD Session Multiplier. Use the no form of the command to restore the default configuration.		
Command Syntax	multiplier <i>number</i> no multiplier		
Command Mode	configuration bfd multi-hop af-ipv4 configuration bfd multi-hop af-ipv6 configuration bfd multi-hop af-ipv4 neighbor configuration bfd multi-hop af-ipv6 neighbor configuration bfd single-hop af-ipv4 interface configuration bfd single-hop af-ipv6 interface		
Parameters Table	Name	Value	Description
	number	Integer range 1..10.	Specifies the BFD multiplier. The value applied to a BFD session is: • Single hop: ○ A value configured per the interface (highest priority) ○ 3 • Multi hop: ○ A value configured per the IP Address (highest priority) ○ A value configured per the Address Family ○ 3

Examples	These examples show how to use this command: Example 1. Single Hop BFD <pre>exaware# config exaware(config)# bfd single-hop af-ipv4 exaware(config-bfd-single-hop-af-ipv4)# interface x-eth 0/0/4 exaware(config-bfd-single-hop-af-interface-x-eth-0/0/4)# multiplier 4</pre> Example 2. Multi Hop BFD <pre>exaware# config exaware(config)# bfd multi-hop af-ipv4 exaware(config-bfd-af-ipv4)# multiplier 5 exaware(config-bfd-af-ipv4)# neighbor 10.0.0.3 vrf default exaware(config-bfd-af-ipv4-10.0.0.3)# multiplier 4</pre>
Notes	• multiplier * minimal-rx-interval must be less or equal 5000 • multiplier * minimal-tx-interval must be less or equal 5000

10.1.3.7. neighbor

Description	The command enters the Multi Hop BFD multi-hop neighbor command mode allowing configuring the BFD session parameters. Use the no form of the command to restore the default BFD configuration.		
Command Syntax	neighbor ip-address vrf <i>vrf-name</i> no neighbor ip-address vrf <i>vrf-name</i>		
Parameters Table	Name	Value	Description
	ip-address	IPv4 or IPv6 address depending on the command mode.	Specifies the remote address of the Multi Hop BFD session.
	<i>vrf-name</i>	Existing VRF name	Specifies the VRF which the remote address of the Multi Hop BFD session belongs to.
Command Mode	configuration bfd multi-hop af-ipv4 configuration bfd multi-hop af-ipv6		
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# bfd multi-hop af-ipv4 exaware(config-bfd-af-ipv4)# neighbor 10.1.10.3 vrf VRFA exaware(config-bfd-af-ipv4-10.1.10.3)#</pre>		

10.1.4. BFD Operational Commands

10.1.4.1. **clear bfd session**

Description	The command clears a BFD session.		
Command Syntax	clear bfd session {single-hop {af-ipv4 af-ipv6} interface <i>if-name</i> ip-address <i>ip-address</i> } {multi-hop {af-ipv4 af-ipv6} ip-address <i>ip-address</i> vrf <i>vrf-name</i> }		
Parameters Table	Name	Value	Description
	single-hop		Specifies a single-hop session.
	multi-hop		Specifies a multi-hop session.
	af-ipv4		Specifies that the remote IP address is an IPv4 address.
	af-ipv6		Specifies that the remote IP address is an IPv6 address.
	if-name	Interface Name	Specifies the interface which the single-hop BFD session is running on.
	ip-address	IPv4 or IPv6 address depending on the command mode.	Specifies the remote address of the BFD session.
	<i>vrf-name</i>	Existing VRF name	Specifies the VRF which the remote address of the mlti-hops BFD session belongs to.
Command Mode	Operational		
Examples	This example shows how to use this command: <pre>exaware# clear bfd session single-hop af-ipv4 interface x-eth 0/0/3 ip-address 143.10.5.6 exaware# clear bfd session multi-hop af-ipv4 ip-address 10.10.10.10 vrf default</pre>		

Notes	○ If the session is an unsolicited one, then: • If the session state is UP, then a few AdminDown BFD messages are sent to change the remote state. • The session is removed. ○ If the session is a local one, then: • If the session state is UP, then a few AdminDown BFD messages are sent to change the remote state. • The session state is changed to DOWN and the session is reinitialized.
--------------	---

10.1.5. BFD Show Commands

10.1.5.1. show bfd sessions

Description	The command displays the BFD information.					
Command Syntax	show bfd sessions brief detail					
Command Mode	Operational					
Examples	The following examples show sample outputs from this command:					
exaware# show bfd sessions brief						
Single Hop Sessions						
AFS	INTERFACE	ADMIN STATE	REMOTE IP ADDRESS	LOCAL IP ADDRESS	LD/RD	STATE

IPv4	x-eth 0/0/1	Up	192.0.0.3	192.0.0.2	1/2	Up
			192.0.0.13	192.0.0.2	2/3	Up
	x-eth 0/0/2	Up	192.0.1.1	192.0.1.2	2/3	Up
IPv6	x-eth 0/0/1	Up	FE80::3334	FE80::193	4/5	Up
			FE80::4444	FE80::193	5/5	Up
	x-eth 0/0/2	Up	FE80::3334	FE80::223	6/5	Up
Multi Hop Sessions						
AFS	ADMIN STATE	VRF	REMOTE IP ADDRESS	LOCAL IP ADDRESS	LD/RD	STATE

IPv4	Up	default	10.0.0.3	10.0.0.3	1/2	Up
			10.0.1.3	10.0.0.3	2/3	Up
		vpn-a	1.0.1.3	1.0.1.1	3/1	Up
		vpn-b	1.0.1.3	1.0.1.1	4/1	Up
IPv6	Up	aaa	2008::12	4008::986	3/1	Up
			2008::13	4008::986	4/1	Up
exaware# show bfd sessions detail						
Single Hop Sessions						
IPv4						
Interface x-eth 0/0/1						
AdminStatus: Up						
MinTxInt: 1000ms, MinRxInt: 1000ms, Multiplier: 8						
Remote IP address: 192.0.0.3, Local IP Address: 192.0.0.2 LD/RD: 1/24						
Clients: OSPF, IS-IS						
Local						
Diag No Diagnostic, State: Up						
Demand bit: 0, Poll bit: 0, Final bit: 0						
Received:						

```
MinTxInt: 100ms, MinRxInt:100ms, Multiplier: 3, Min Echo interval: 0
Diag: No Diagnostic, State: Up
Demand bit: 0, Poll bit: 0, Final bit: 0
Negotiated: TxInterval: 100ms, Detected Interval: 300ms
Creation: 12-Jan-2019 12:00:01
Last Up: 12-Jan-2019 12:00:04
Last Down:
Number of Downs: 0
Remote IP address: 192.0.0.4, Local IP Address: 192.0.0.2
LD/RD: 2/24
Clients: IS-IS
Local
Diag No Diagnostic, State: Up
Demand bit: 0, Poll bit: 0, Final bit: 0
Received:
MinTxInt: 100ms, MinRxInt:100ms, Multiplier: 3, Min Echo interval: 0
Diag: No Diagnostic, State: Up
Demand bit: 0, Poll bit: 0, Final bit: 0,
Negotiated: TxInterval: 100ms, Detected Interval: 300ms
Creation: 12-Jan-2019 12:00:01
Last Up: 12-Jan-2019 12:00:04
Last Down:
Number of Downs: 0
Interface x-eth 0/0/2
AdminStatus: Up
MinTxInt: 1000ms, MinRxInt: 1000ms, Multiplier: 8
Remote IP address: 192.0.2.3, Local IP Address: 192.0.0.2 LD/RD: 3/24
Clients: OSPF, IS-IS
Local
Diag No Diagnostic, State: Up
Demand bit: 0, Poll bit: 0, Final bit: 0
Received:
MinTxInt: 100ms, MinRxInt:100ms, Multiplier: 3, Min Echo interval: 0
Diag: No Diagnostic, State: Up
Demand bit: 0, Poll bit: 0, Final bit: 0
Negotiated: TxInterval: 100ms, Detected Interval: 300ms
Creation: 12-Jan-2019 12:00:01
Last Up: 12-Jan-2019 12:00:04
Last Down:
Number of Downs: 0
IPv6
Interface x-eth 0/0/1
AdminStatus: Up
MinTxInt: 1000ms, MinRxInt: 1000ms, Multiplier: 8
Remote IP address: FE80::123:23A, Local IP Address: FE80:123:3A1
LD/RD: 4/24
Clients: OSPFv3
Local
Diag No Diagnostic, State: Up
Demand bit: 0, Poll bit: 0, Final bit: 0
Received:
MinTxInt: 100ms, MinRxInt:100ms, Multiplier: 3, Min Echo interval: 0
Diag: No Diagnostic, State: Up
Demand bit: 0, Poll bit: 0, Final bit: 0
Negotiated: TxInterval: 100ms, Detected Interval: 300ms
Creation: 12-Jan-2019 12:00:01
Last Up: 12-Jan-2019 12:00:04
```

Last Down:
Number of Downs: 0

Multihops Sessions

IPv4

AdminStat: Up
MinTxInt: 500ms, MinRxInt: 500ms, Multiplier: 3
VRF: default, Remote IP Address: 10.0.0.3, Local IP Address: 10.0.0.2

LD/RD: 1/0

MinTxInt: 1000ms, MinRxInt: 1000ms, Multiplier: 4
Clients: BGP

State: Down

Diag: No Diagnostic, Demand mode: 0, Poll bit: 0

Creation: 12-Jan-2019 12:00:01

Last Up: 12-Jan-2019 12:00:21

Last Down:

Number of Downs: 0

VRF: default, Remote IP Address: 10.0.0.4, Local IP Address: 10.0.0.2

LD/RD: 1/24

MinTxInt: 700ms, MinRxInt: 300ms, Multiplier: 3

Clients: BGP, Remote

State: Up

Diag: No Diagnostic, Demand mode: 0, Poll bit: 0

Received:

MinTxInt: 400ms, MinRxInt: 400ms, Multiplier: 4, Min Echo interval: 0

Diag: No Diagnostic, State: Up

Demand bit: 0, Poll bit: 0, Final bit: 0

Negotiated: TxInterval: 700ms, Detected Interval: 1600ms

Creation: 12-Jan-2019 12:00:01

Last Up: 12-Jan-2019 13:00:24

Last Down: 12-Jan-2019 13:00:14

Number of Downs: 1

VRF: default, Remote IP Address: 10.0.0.5, Local IP Address: 10.0.0.3

LD/RD: 1/24

MinTxInt: 100ms, MinRxInt: 100ms, Multiplier: 3

Clients: BGP

State: Up

Diag: No Diagnostic, Demand mode: 0, Poll bit: 0

Received:

MinTxInt: 100ms, MinRxInt: 100ms, Multiplier: 3, Min Echo interval: 0

Diag: No Diagnostic, State: Up

Demand bit: 0, Poll bit: 0, Final bit: 0

Negotiated: TxInterval: 1000ms, Detected Interval: 300ms

Creation: 12-Jan-2019 12:00:01

Last Up: 12-Jan-2019 13:00:24

Last Down: 12-Jan-2019 13:00:14

Number of Downs: 1

VRF: default, Remote IP Address: 10.0.0.13, Local IP Address: 10.0.0.2

LD/RD: 11/4

MinTxInt: 100ms, MinRxInt: 100ms, Multiplier: 3

Clients: BGP

State: Up

Diag: No Diagnostic, Demand mode: 0, Poll bit: 0

Received:

MinTxInt: 100ms, MinRxInt: 100ms, Multiplier: 3, Min Echo interval: 0

Diag: No Diagnostic, State: Up

```
Demand bit: 0, Poll bit: 0, Final bit: 0
Negotiated: TxInterval: 100ms, Detected Interval: 300ms
Creation: 12-Jan-2019 12:00:01
Last Up: 12-Jan-2019 12:00:04
Last Down:
Number of Downs: 0
IPv6
AdminStat: Up
MinTxInt: 500ms, MinTxInt: 500ms, Multiplier: 3
VRF: default, Remote IP Address: 2101::76, Local IP Address: 2008::43:0:2 LD/RD: 1/24
MinTxInt: 1000ms, MinRxInt: 1000ms, Multiplier: 3
Clients: BGP
State: Up
Diag: No Diagnostic, Demand mode: 0, Poll bit: 0
Received:
MinTxInt: 100ms, MinRxInt:100ms, Multiplier: 3, Min Echo interval: 0
Diag: No Diagnostic, State: Up
Demand bit: 0, Poll bit: 0, Final bit: 0
Negotiated: TxInterval: 1000ms, Detected Interval: 300ms
Creation: 12-Jan-2019 12:00:01
Last Up: 12-Jan-2019 12:00:04
Last Down:
Number of Downs: 0
```

10.2. BGP Routing Commands

10.2.1. BGP Configuration Commands

10.2.1.1. admin-state (peer)

Description	Administratively shut down the neighbor or neighbor group. Use the no form of the command to reset to the default state (up).		
Command Syntax	admin-state up down no admin-state		
Command Mode	configuration routing bgp vrf neighbor configuration routing bgp vrf neighbor-group		
Parameters Table	Name	Value	Description
		up down	Administrative state. (Default = up)
Examples	This example shows how to use this command to shut down the specified neighbor: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# admin-state down</pre>		

10.2.1.2. admin-state (rpki-server)

Description	To set the RPKI cache-server administrative state, use the shutdown command. To reset the default configuration (enabled), use the no form of this command.		
Command Syntax	admin-state up down no admin-state		
Command Mode	configuration routing bgp vrf rpki server transport		
Parameters Table	Name	Value	Description
	up		Enables the RPKI server.
	down		Disables the RPKI server.
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# rpki exaware(config-routing-bgp-61-vrf-default-rpki)# rpki-server 11.1.1.1 exaware(config-routing-bgp-61-vrf-default-rpki-server- 11.1.1.1)#admin-state down </pre>		

10.2.1.3. af-ipv4

Description	Enters the routing configuration sub-mode for BGP routing, as follows: • IP address family and routing scheme for the specified VRF • IP address family and routing scheme for the specified neighbor • IP address family and routing scheme for the specified neighbor group Use the no form of the command to disable BGP for the IP address family and routing scheme for the specified routing instance.		
Command Syntax	af-ipv4 <i>routing-scheme</i> no af-ipv4 <i>routing-scheme</i>		
Command Mode	configuration routing bgp vrf configuration routing bgp vrf neighbor configuration routing bgp vrf neighbor-group		
Parameters Table	Name	Value	Description
	<i>routing-scheme</i>	labeled-unicast unicast vpn flowspec	The sub-address family (SAFI) to enter for BGP routing configuration.

Examples	This example shows how to enter configuration sub-mode for the IP address family and routing scheme for the specified VRF: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrfl exaware(config-routing-bgp-61-vrf-vrfl)# af-ipv4 unicast exaware(config-vrf-vrfl-af-ipv4-unicast)#</pre> This example shows how to enter configuration sub-mode for the IP address family and routing scheme for the specified neighbor: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrfl exaware(config-routing-bgp-61-vrf-vrfl)# neighbor 10.10.10.10 exaware(config-vrf-vrfl-neighbor-10.10.10.10)# af-ipv4 unicast exaware(config-vrf-vrfl-neighbor-10.10.10.10-af-ipv4-unicast)#</pre>
Notes	The Flowspec routing scheme is applicable only to neighbor and neighbor-group. SAFI labeled-unicast and vpn are only supported in VRF default.

10.2.1.4. af-ipv6

Description	Enters the routing configuration sub-mode for BGP routing, as follows: • IP address family and routing scheme for the specified VRF • IP address family and routing scheme for the specified neighbor • IP address family and routing scheme for the specified neighbor group Use the no form of the command to disable BGP for the IP address family and routing scheme for the specified routing instance.		
Command Syntax	<pre>af-ipv6 <i>routing-scheme</i> no af-ipv6 <i>routing-scheme</i></pre>		
Command Mode	<pre>configuration routing bgp vrf configuration routing bgp vrf neighbor configuration routing bgp <i>vrf</i> neighbor-group</pre>		
Parameters Table	Name	Value	Description
	<i>routing-scheme</i>	label-unicast unicast vpn	The sub-address family (SAFI) to enter for BGP routing configuration.

Examples	This example shows how to enter configuration sub-mode for the IP address family and routing scheme for the specified VRF: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrfl exaware(config-routing-bgp-61-vrf-vrfl)# af-ipv6 unicast exaware(config-vrf-vrfl-af-ipv6-unicast)#</pre> This example shows how to enter configuration sub-mode for the IP address family and routing scheme for the specified neighbor: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrfl exaware(config-routing-bgp-61-vrf-vrfl)# neighbor 10.10.10.10 exaware(config-vrf-vrfl-neighbor-10.10.10.10)# af-ipv6 unicast exaware(config-vrf-vrfl-neighbor-10.10.10.10-af-ipv6-unicast)#</pre>
Notes	SAFI labeled-unicast and vpn are only supported in VRF default.

10.2.1.5. af-l2vpn vpls

Description	Enables the L2VPN VPLS AFI/SAFI in neighbor or neighbor group. Use the no form of the command to disable the L2VPN VPLS AFI/SAFI in neighbor or neighbor group.
Command Syntax	<pre>af-l2vpn vpls no af-l2vpn vpls</pre>
Command Mode	<pre>configuration routing bgp vrf neighbor configuration routing bgp vrf neighbor-group</pre>
Parameters Table	N/A
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 123 exaware(config-routing-bgp-123)# vrf default exaware(config-routing-bgp-123-vrf-default)# neighbor 3.3.3.2 exaware(config-vrf-vrf1-neighbor-3.3.3.2)# af-l2vpn vpls</pre> <pre>exaware# config exaware(config)# routing bgp 123 exaware(config-routing-bgp-123)# vrf default exaware(config-routing-bgp-123-vrf-default)# neighbor-group VPLS-neighbors exaware(config-vrf-vrf1-neighbor-group-VPLS-neighbors)# af- l2vpn vpls</pre>
Notes	The af-l2vpn vpls command is supported only in VRF default.

10.2.1.6. aggregate-address

Description	Configures a summary address to summarize IPv4 and IPv6 reachability information. Use the no form of the command to disable the summary address.		
Command Syntax	<pre>aggregate-address <i>ip-prefix</i> [as-path overwrite] [advertise {full summary}] [policy <i>policy-name</i>]</pre> <pre>no aggregate-address</pre>		
Command Mode	configuration routing bgp vrf afi safi		
Parameters Table	Name	Value	Description
	<i>ip-prefix</i>	IPv4 or IPv6 prefix	Aggregate IP address and mask.
	as-path	overwrite	Overwrite AS path with local AS number. (default)
	advertise	full	Advertise both summarized and specific routes. (default)
		summary	Advertise only summarized routes.
	<i>policy-name</i>	existing policy name	Associated policy.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# af-ipv4 unicast exaware(config-vrf-vrf1-af-ipv4-unicast)# aggregate-address 10.10.10.0/24 as-path overwrite advertise summary</pre>		

10.2.1.7. allocate-label

Description	Enables redistribution of prefixes between unicast and labeled-unicast, optionally applying policy on prefixes redistributed from unicast. Use the no form of the command to disable the allocate-label configuration.		
Command Syntax	<pre>allocate-label {all policy <i>policy-name</i>} no allocate-label</pre>		
Command Mode	configuration routing bgp vrf af-ipv6 unicast		
Parameters Table	Name	Value	Description
	<i>policy-name</i>	Name	Policy Name.
Examples	The following examples show how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# af-ipv6 unicast exaware(config-vrf-default-af-ipv6-unicast)# allocate-label policy policy1 exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# af-ipv6 unicast exaware(config-vrf-default-af-ipv6-unicast)# allocate-label all exaware(config-vrf-default-af-ipv6-unicast)# no allocate-label</pre>		
Notes	Default configuration is no allocate-label. Removal of policy will be done implicitly by setting "all". This command is applicable for VRF default IPv6 unicast only.		

10.2.1.8. allow-as-in

Description	Accepts routes up to the specified number of instances of the local AS number in the AS-Path attribute. Use the no form of the command to disable the allow-as-in configuration.		
Command Syntax	<pre>allow-as-in <i>number</i> no allow-as-in</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi</pre>		
Parameters Table	Name	Value	Description
	<i>number</i>	1-10	Number of instances of the local AS number.
Examples	The following examples show how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 3.3.3.2 exaware(config-vrf-vrf1-neighbor-3.3.3.2)# af-ipv4 unicast exaware(config-neighbor-3.3.3.2-af-ipv4-unicast)# allow-as-in 2</pre> <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor-group CLIENTS exaware(config-vrf-vrf1-neighbor-group-CLIENTS)# af-ipv4 unicast exaware(config-neighbor-group-CLIENTS-af-ipv4-unicast)# no allow-as-in</pre>		
Notes	This command is only available under unicast SAFI, VRF default, and VPN SAFI.		

10.2.1.9. alternate-path

Description	Installs BGP alternate-path. Use the no form of the command to disable BGP alternate-path.
Command Syntax	<pre>alternate-path {enable disable} no alternate-path</pre>
Command Mode	configuration routing bgp vrf afi safi
Parameters Table	N/A
Examples	The following examples show how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# af-ipv4 unicast exaware(config-vrf-default-af-ipv4-unicast)# alternate-path enable exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# af-ipv6 labeled unicast exaware(config-vrf-default-af-ipv6-labeled-unicast)# alternate- path disable exaware(config-vrf-default-af-ipv6-unicast)# no alternate-path</pre>
Notes	Default configuration from version 4.2 is alternate-path enabled for SAFIs supporting multipath. This command is applicable only to unicast, labeled-unicast, and vpn SAFIs.

10.2.1.10. as-override

Description	Overrides each instance of the remote AS number in the AS-Path attribute with the local AS number. Use the no form of the command to revert to the default setting (disable).		
Command Syntax	<pre>as-override {enable disable} no as-override</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi</pre>		
Parameters Table	Name	Value	Description
		enable	Enable as-override.
		disable	Disable as-override. (Default)
Examples	The following examples show how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 3.3.3.2 exaware(config-vrf-vrf1-neighbor-3.3.3.2)# af-ipv4 unicast exaware(config-neighbor-3.3.3.2-af-ipv4-unicast)# as-override enable</pre> <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor-group CLIENTS exaware(config-vrf-vrf1-neighbor-group-CLIENTS)# af-ipv4 unicast</pre> <pre>exaware(config-neighbor-group-CLIENTS-af-ipv4-unicast)# no as-override</pre>		
Notes	This command is only available under non-default VRF and unicast SAFI.		

10.2.1.11. bfd

Description	Enable or disable BFD session to the given remote peer. Use the no form of the command to reset to the default state (disable).		
Command Syntax	bfd enable disable no bfd		
Command Mode	configuration routing bgp vrf neighbor configuration routing bgp vrf neighbor-group		
Parameters Table	Name	Value	Description
	enable		A BFD session is used to check the peer state.
	disable		A BFD session is not used to check the peer state.
Examples	This example shows how to use this command to shut down the specified neighbor: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# bfd enable</pre>		

10.2.1.12. bgp first-as-enforcement

Description	Allows the BGP to receive an update from an eBGP peer when the first AS in the AS path is different from the configured remote AS. Use the no form of the command to revert to the default (disable enforcement).		
Command Syntax	<pre>bgp first-as-enforcement [enable disable] no bgp first-as-enforcement</pre>		
Command Mode	configuration routing bgp		
Parameters Table	Name	Value	Description
		enable	BGP enforces that the first AS path of a route received from an eBGP peer must be the same as the configured remote AS (default).
		disable	The first AS path of a route received from an eBGP peer can be different from the configured remote AS.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# bgp first-as-enforcement enable</pre>		

10.2.1.13. bgp extended-asn-capability

Description	Allows the router to send 4-octet ASN numbers. Use the no form of the command to revert to the default (enable).		
Command Syntax	<pre>bgp extended-asn-capability [enable disable] no bgp extended-asn-capability</pre>		
Command Mode	configuration routing bgp		
Parameters Table	Name	Value	Description
		enable	Enables sending 4-octet ASN numbers. (Default)
		disable	Disables sending 4-octet ASN numbers.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# bgp extended-asn-capability disable</pre>		

10.2.1.14. bgp graceful-restart

Description	Configures the graceful restart parameter settings. You can configure the following parameters: - graceful restart deferral time: maximum time, in seconds, to defer initial route selection - graceful reset: enables a graceful reset if configuration change forces peer reset - graceful restart restart time: maximum time restart time advertised to neighbors, in seconds, - graceful restart stalepath time: maximum time, in seconds, to retain stale paths from a restarting neighbor By default, graceful restart is enabled (<code>graceful-reset = enable</code>)		
Command Syntax	<pre>bgp graceful-restart [deferral-time <i>time1</i>] [graceful-reset (enable disable)] [restart-time <i>time2</i>] [stalepath-time <i>time3</i>]</pre>		
Command Mode	configuration routing bgp		
Parameters Table	Name	Value	Description
	<i>time1</i>	1-3600	Maximum time to defer initial route selection, in seconds. (Default = 120)
	<i>time2</i>	1-3600	Maximum restart time advertised to neighbors, in seconds. (Default = 120)
	<i>time3</i>	1-3600	Maximum amount of time that stale paths from restarting neighbors are retained, in seconds. (Default = 360)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# bgp graceful-restart deferral-time 240 restart-time 240 stalepath-time 720</pre>		

10.2.1.15. bgp local-preference

Description	Configures the default local-preference. Use the no form of the command to revert to the default (100).		
Command Syntax	<code>bgp local-preference <i>value</i></code> <code>no bgp local-preference</code>		
Command Mode	configuration routing bgp		
Parameters Table	Name	Value	Description
	<i>value</i>	0-4294967295	Local preference value. (Default = 100)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# bgp local-preference 12345</pre>		

10.2.1.16. bgp path-selection

Description	Configures changes in BGP default path selection criteria. Use the no form of the command to reset the default values (see Notes).		
Command Syntax	<pre> bgp path-selection compare-as-path [enable disable] compare-router-id [enable disable] always-compare-med [enable disable] missing-med [worst best] compare- med-confed [enable disable] deterministic-med [enable disable] no bgp path-selection </pre>		
Command Mode	configuration routing bgp		
Parameters Table	Name	Value	Description
	compare-as-path	enable	BGP compares as-path length when calculating preferred paths.
		disable	BGP ignores as-path length when calculating preferred paths. (Default = enable)
	compare-router-id	enable	BGP compares router ID when calculating preferred paths.
		disable	BGP ignores router ID when calculating preferred paths. (Default = enable)

	always-compare-med	enable	BGP compares MED even if the update came from the same neighbor AS. BGP compares MED only if the update came from a different neighbor AS. (Default = disable)
		disable	

	missing-med	worst	Add value of 'infinite' to updates received without MED.
		best	Add value of '0' to updates received without MED. (Default = best)

	compare-med-confed	enable	Compare MED on a group of peers belonging to the same AS.
		disable	Do not group peers based on AS before MED comparison. (Default = disable)

	deterministic-med	enable disable	Enable MED comparison among paths learned from confederation peers. BGP does not compare the MED of paths containing only confederation segments. (Default = disable)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# bgp path-selection compare-as-path disable compare-router-id disable always-compare-med enable missing-med best compare-med-confed enable deterministic-med enable</pre>		
Notes	Following is a summary of the default values for this command: BGP path selection algorithm selects the path with the shortest as-path attribute. BGP path selection algorithm selects the path received from the neighbor that has the lowest router ID. BGP path selection algorithm selects the path with the lowest MED value. BGP path selection algorithm compares MED only if both paths came from a different neighbor AS. BGP assigns a MED value of 0 to prefixes received without MED. BGP does not compare updates with only confederation info in their as-path.		

10.2.1.17. bgp timers

Description	Configures the following timers in BGP: - hold time - keep alive interval Use the no form of the command to reset the BGP timer to the default value.		
Command Syntax	<pre>bgp timers [holdtime <i>time1</i> keepalive <i>time2</i>]</pre> <pre>no bgp timers [holdtime keepalive]</pre>		
Command Mode	configuration routing bgp		
Parameters Table	Name	Value	Description
	<i>time1</i>	3-65535	Interval without receiving a keepalive message after which a peer is declared dead, in seconds. (Default = 180)
	<i>time2</i>	1-65535	Frequency of the keepalive messages in seconds. (Default = 60)
Examples	This example shows how to configure the BGP timers: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# bgp timers holdtime 120 keepalive 180</pre>		
Notes	The hold timer (<i>time1</i>) must be greater than the keepalive timer (<i>time2</i>).		

10.2.1.18. capability

Description	Configures the capabilities to be advertised to the peer. The options available for this command depend on the configuration submode from which it is executed: • bgp-vrf-neighbor/neighbor-group: dynamic and route-refresh • bgp-vrf-neighbor/neighbor-group-afi-safi: graceful-restart and orf Use the no form of the command to disable advertising capabilities.		
Command Syntax	<pre>capability [dynamic {enable disable}] [route-refresh {enable disable}] capability [graceful-restart {enable disable}] [orf {disable both receive send}] no capability</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor configuration routing bgp vrf neighbor-group configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi</pre>		
Parameters Table	Name	Value	Description
		dynamic	Advertises dynamic capability to its neighbors. (Default = disable)
		route-refresh	Advertises route-refresh capability to its neighbors. (Default = enable)
		graceful-restart	Advertises graceful-restart capability to its neighbors. (Default = enable)

	orf	both disable receive send	Advertises ORF capability to its neighbors either on transmission or reception, or both directions. (Default = both)
--	-----	------------------------------------	---

Examples

This example shows how to use this command in the bgp-vrf-neighbor/neighbor-group configuration sub-mode:

```
exaware# config
exaware(config)# routing bgp 61
exaware(config-routing-bgp-61)# vrf vrfl
exaware(config-routing-bgp-61-vrf-vrfl)# neighbor 10.10.10.10
exaware(config-vrf-vrfl-neighbor-10.10.10.10)# capability dynamic
enable route-refresh disable
```

This example shows how to use this command in the bgp-vrf-neighbor/neighbor-group-afi-safi configuration sub-mode:

```
exaware# config
exaware(config)# routing bgp 61
exaware(config-routing-bgp-61)# vrf vrfl
exaware(config-routing-bgp-61-vrf-vrfl)# neighbor-group group1
exaware(config-vrf-vrfl-neighbor-group-group1)# af-ipv4 unicast
exaware(config-vrf-vrfl-neighbor-group-group1-af-ipv4-unicast)#
capability graceful-restart disable orf receive
```

10.2.1.19. client-to-client-reflection

Description	Enable or disable client to client route reflection of iBGP client updates. Use the no form of the command to disable the client to client reflections.		
Command Syntax	<code>client-to-client-reflection [enable disable]</code> <code>no client-to-client-reflection</code>		
Command Mode	configuration routing bgp		
Parameters Table	Name	Value	Description
		enable disable	Default = disable
Examples	This example shows how to use this command: exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# client-to-client-reflection enable		

10.2.1.20. cluster-id

Description	Defines a BGP cluster (route-reflector and its clients). Use the no form of the command to remove the cluster ID.		
Command Syntax	<pre>cluster-id [cluster-id ip-address] no cluster-id</pre>		
Command Mode	configuration routing bgp		
Parameters Table	Name	Value	Description
	<i>cluster-id</i>	1-4294967295	Cluster ID as a numerical value.
	<i>ip-address</i>	IPv4 or IPv6 address	Cluster ID as an IP address.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# cluster-id 3456</pre>		
Notes	By default, BGP has no cluster ID.		

10.2.1.21. confederation

Description	Configures a BGP confederation on the specified VRF. Use the no form of the command to disable confederation.		
Command Syntax	<pre>confederation {identifier peers} as-number</pre> <pre>no confederation {identifier peers} as-number</pre>		
Command Mode	configuration routing bgp vrf		
Parameters Table	Name	Value	Description
	as-number	1-65535	Confederation or peer AS number.
Examples	The following example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 123 exaware(config-routing-bgp-123)# vrf default exaware(config-routing-bgp-123-vrf-default)# confederation peers 777 exaware(config-routing-bgp-123-vrf-default)# confederation identifier 888</pre>		
Notes	Confederation configuration is available only under VRF default, but is valid to all VRFs. Use the identifier option to set the AS number for the confederation. Use the peers option to configure the AS number of the peer-AS's in the confederation.		

10.2.1.22. default-route-originate

Description	Causes BGP to advertise the default route to a specific neighbor or neighbor group even without the existence of a default route in the routing table. Use the no form of the command to restore the default (disable).		
Command Syntax	<pre>default-route-originate policy <i>policy-name</i> [enable disable] no default-route-originate</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi</pre>		
Parameters Table	Name	Value	Description
		enable	BGP will send a default route to the selected peer.
		disable	BGP will not send a default route to the selected peer (default).
	<i>policy-name</i>	existing policy name	Associated policy.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# af-ipv4 unicast exaware(config-vrf-vrf1-neighbor-10.10.10.10-af-ipv4-unicast)# default-route-originate policy policyname enable</pre>		

10.2.1.23. description (peer)

Description	Meaningful description of the neighbor. Use the no form of the command to remove the description.		
Command Syntax	description <i>text</i> no description		
Command Mode	configuration routing bgp vrf <i>neighbor</i> configuration routing bgp vrf neighbor-group		
Parameters Table	Name	Value	Description
	<i>text</i>	Maximum 255 characters enclosed in quotation marks, may include spaces	Free text describing or annotating the neighbor. (Default = no description)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# description "router in CO in Anywhere, USA"</pre>		

10.2.1.24. description (rpki-server)

Description	To specify a description for the RPKI cache-server, use the description command. To restore the default configuration, use the no form of this command.		
Command Syntax	<pre>description <i>string</i> no description</pre>		
Command Mode	configuration routing bgp vrf rpki server		
Parameters Table	Name	Value	Description
	string	String: maximum 256 bytes	User-defined description of the specified RPKI cache-server. Up to 255 characters, including spaces within quotation marks "". The default value is an empty string,
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# rpki exaware(config-routing-bgp-61-vrf-default-rpki)# rpki-server 11.1.1.1 exaware(config-routing-bgp-61-vrf-default-rpki-server-11.1.1.1)# description ServerA</pre>		

10.2.1.25. distance

Description	Define an administrative distance.		
Command Syntax	<pre>distance external <i>ext-distance</i> distance internal <i>int-distance</i> distance local <i>local-distance</i></pre>		
Command Mode	configuration routing bgp vrf afi safi		
Parameters Table	Name	Value	Description
	<i>ext-distance</i>	1-255	Default distance for external routes. (Default = 20)
	<i>int-distance</i>	1-255	Default distance for internal routes. (Default = 200)
	<i>local-distance</i>	1-255	Default distance for local routes. (Default = 200)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrfl exaware(config-routing-bgp-61-vrf-vrfl)# af-ipv4 unicast exaware(config-routing-bgp-61-vrf-vrfl-af-ipv4)# distance local 50</pre>		
Notes	SAFI can be only unicast or labeled-unicast.		

10.2.1.26. distance source-ip

Description	Define the source IP address for setting administrative distance.		
Command Syntax	<code>distance source-ip <i>ip-prefix</i></code>		
Command Mode	configuration routing bgp		
Parameters Table	Name	Value	Description
	<i>ip-prefix</i>	IPv4 or IPv6 address / prefix length	IP address and prefix length of the source.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# distance source-ip 10.10.10.0/24</pre>		

10.2.1.27. ebgp-multihop

Description	Enables EBGp multihop and specifies the maximum number of hops. When multihop is enabled, connections to eBGP peers on networks that are not directly connected are allowed. Use the no form of the command to disable multihop.		
Command Syntax	<pre>ebgp-multihop [enable disable] hops #hops no ebgp-multihop</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor configuration routing bgp vrf neighbor-group</pre>		
Parameters Table	Name	Value	Description
		enable disable	Enable/disable multi-hop. (Default = disable)
	#hops	2-255	Maximum number of hops to the remote peer. (Default = 255)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# ebgp-multihop enable hops 100</pre>		

10.2.1.28. explicit-null

Description	Sets null-label allocation to explicit-null. Use the no form of the command to revert to the default setting (disable).		
Command Syntax	<pre>explicit-null {enable disable} no explicit-null</pre>		
Command Mode	configuration routing bgp vrf neighbor afi labeled-unicast configuration routing bgp vrf neighbor-group afi labeled-unicast		
Parameters Table	Name	Value	Description
		enable	Enable explicit-null.
		disable	Disable explicit-null. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 3.3.3.2 exaware(config-vrf-vrf1-neighbor-3.3.3.2)# af-ipv4 labeled-unicast exaware(config-neighbor-3.3.3.2-af-ipv4-labeled-unicast)# explicit-null enable</pre> <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor-group CLIENTS exaware(config-vrf-vrf1-neighbor-group-CLIENTS)# af-ipv4 labeled-unicast exaware(config-neighbor-group-CLIENTS-af-ipv4-labeled-unicast)# no explicit-null</pre>		
Notes	This command is only available under labeled-unicast SAFI.		

10.2.1.29. export-policy

Description	Configures export policy. Use the no form of the command to remove the export-policy configuration.		
Command Syntax	<pre>export-policy <i>policy-name</i> no export-policy</pre>		
Command Mode	configuration routing bgp vrf afi unicast		
Parameters Table	Name	Value	Description
	<i>policy-name</i>	Name	Export policy.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 123 exaware(config-routing-bgp-123)# vrf vrf1 exaware(config-routing-bgp-123-vrf-vrf1)# af-ipv4 unicast exaware(config-vrf-vrf1-af-ipv4-unicast)# export-policy EXPORT-POLICY exaware(config-vrf-vrf1-af-ipv4-unicast)# no export-policy</pre>		
Notes	Export-policy is not available under VRF default. Routes will go through export-policy, then route-targets will be added according to export-rt statements to the routes permitted by the policy. At the end of these two processes, any route with a route-target attached to it will be exported.		

10.2.1.30. export-rt

Description	Configures export route-target. Use the no form of the command to remove the export-rt configuration.		
Command Syntax	<pre>export-rt route-target no export-rt route-target</pre>		
Command Mode	configuration routing bgp vrf afi unicast		
Parameters Table	Name	Value	Description
	<i>route-target</i>	<2-byte ASN>:<4-byte number>	2-byte ASN notation.
		<4-byte ASN>L:<2-byte number>	4-byte ASN notation.
		<IPv4 address>:<2-byte number>	IPv4 address notation.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 123 exaware(config-routing-bgp-123)# vrf vrf1 exaware(config-routing-bgp-123-vrf-vrf1)# af-ipv4 unicast exaware(config-vrf-vrf1-af-ipv4-unicast)# export-rt 100:1 exaware(config-vrf-vrf1-af-ipv4-unicast)# no export-rt 100:1 exaware(config-vrf-vrf1-af-ipv4-unicast)# export-rt 70222L:2 exaware(config-vrf-vrf1-af-ipv4-unicast)# export-rt 1.2.3.4:3</pre>		
Notes	Export-rt is not available under VRF default.		

10.2.1.31. fast-external-turnover

Description	Enables fast external turnover. This option immediately resets the session when a link to a directly-connected external peer goes down. Use the no form of the command to disable fast external turnover.		
Command Syntax	fast-external-turnover [enable disable] no fast-external-turnover		
Command Mode	configuration routing bgp		
Parameters Table	Name	Value	Description
		enable disable	Default = enable
Examples	This example shows how to use this command: exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# fast-external-turnover disable		

10.2.1.32. flap-damping

Description	Configures flap damping parameters. Use the no form of the command to disable flap damping.		
Command Syntax	<pre>flap-damping suppress <i>cutoff-threshold</i> half-time <i>decay-half-life</i> reuse <i>reuse-threshold</i> max-suppress <i>max-hold-down-time</i> [policy <i>policy-name</i>] no flap-damping</pre>		
Command Mode	configuration routing bgp vrf afi safi		
Parameters Table	Name	Value	Description
	<i>cutoff-threshold</i>	1-20000	Accumulated penalty amount at which the route is suppressed. (Default =3000)
	<i>decay-half-life</i>	1-45	Amount of time in minutes after which a penalty is reduced by half. (Default =15)
	<i>reuse-threshold</i>	1-20000	Accumulated penalty amount at which the route is re-advertised. (Default =750)
	<i>max-hold-down-time</i>	1-255	Maximum time in minutes that a route is suppressed. (Default =60)
	<i>policy-name</i>	existing policy number	Associated policy.

Examples	This example shows how to use configure flap damping: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrfl cexaware(config-routing-bgp-61-vrf-vrfl)# af-ipv4 unicast exaware(config-vrf-vrfl-af-ipv4-unicast)# flap-damping supress 5000 half-time 30 reuse 1000 max-supress 75 policy polname oper-state enable</pre>
Notes	By default, route flap damping is disabled.

10.2.1.33. import

Description	To allow import routes to the given VRF from the source VRF with policy, use the import command. Use the no form of the command to remove the configuration.		
Command Syntax	<pre>import source-vrf vrf-name [policy policy-name] no import</pre>		
Command Mode	<pre>configuration routing bgp vrf af-ipv4 unicast configuration routing bgp vrf af-ipv6 unicast</pre>		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>		The name of the source VRF.
	<i>policy-name</i>		The policy name.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-vrf1)# af-ipv4 unicast exaware(config-vrf-vrf1-af-ipv4-unicast)# import source-vrf VRF2 policy POL1</pre>		
Notes	vrf "management" cannot be defined as source-vrf		

10.2.1.34. import-policy

Description	Configures import policy. Use the no form of the command to remove the import-policy configuration.		
Command Syntax	<pre>import-policy <i>policy-name</i> no import-policy</pre>		
Command Mode	configuration routing bgp vrf afi unicast		
Parameters Table	Name	Value	Description
	<i>policy-name</i>	Name	Import policy.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 123 exaware(config-routing-bgp-123)# vrf vrf1 exaware(config-routing-bgp-123-vrf-vrf1)# af-ipv4 unicast exaware(config-vrf-vrf1-af-ipv4-unicast)# import-policy IMPORT-POLICY exaware(config-vrf-vrf1-af-ipv4-unicast)# no import-policy</pre>		
Notes	Import-policy is not available under VRF default. Routes will be imported according to import-rt statements, and will then go through import-policy.		

10.2.1.35. import-rt

Description	Configures import route-target. Use the no form of the command to remove the import-rt configuration.		
Command Syntax	<pre>import-rt route-target</pre> <pre>no import-rt route-target</pre>		
Command Mode	configuration routing bgp vrf afi unicast		
Parameters Table	Name	Value	Description
	<i>route-target</i>	<2-byte ASN>:<4-byte number>	2-byte ASN notation.
		<4-byte ASN>L:<2-byte number>	4-byte ASN notation.
		<IPv4 address>:<2-byte number>	IPv4 address notation.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 123 exaware(config-routing-bgp-123)# vrf vrf1 exaware(config-routing-bgp-123-vrf-vrf1)# af-ipv4 unicast exaware(config-vrf-vrf1-af-ipv4-unicast)# import-rt 100:2 exaware(config-vrf-vrf1-af-ipv4-unicast)# no import-rt 100:2 exaware(config-vrf-vrf1-af-ipv4-unicast)# import-rt 70222L:10 exaware(config-vrf-vrf1-af-ipv4-unicast)# import-rt 1.2.3.4:30</pre>		
Notes	Import-rt is not available under VRF default.		

10.2.1.36. inbound-soft-reconfiguration

Description	Enables local storage of inbound routes and per neighbor soft reconfiguration. Use the no form of the command to restore the default (disable).		
Command Syntax	inbound-soft-reconfiguration [enable disable] no inbound-soft-reconfiguration		
Command Mode	configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi		
Parameters Table	Name	Value	Description
		enable	Enable local storage of inbound routes.
		disable	Disable local storage of inbound routes. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# af-ipv4 unicast exaware(config-vrf-vrf1-neighbor-10.10.10.10-af-ipv4-unicast)# inbound-soft-reconfiguration enable</pre>		

10.2.1.37. local-address

Description	Specifies the address of the local end of a BGP session. Use the no form of the command to revert to the default (see Note).		
Command Syntax	<pre>local-address [<i>ip-address</i> <i>interface-name</i>]</pre> <pre>no local-address</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor</pre> <pre>configuration routing bgp vrf neighbor-group</pre>		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 or IPv6 address	IP address to use as local address.
	<i>interface-name</i>	0/0/port	Interface ID to be used as local address.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# local-address x-eth 0/0/0</pre>		
Notes	By default, BGP uses the outgoing interface toward the neighbor as its source. In order to change the existing local-address, first execute a <code>no local-address</code> command and then configure the new local-address.		

10.2.1.38. log-neighbor-changes

Description	Enables or disables generating a log message each time the neighbor state changes. Use the no form of the command to revert to the default setting (enable).		
Command Syntax	<pre>log-neighbor-changes [enable disable] no log-neighbor-changes</pre>		
Command Mode	configuration routing bgp		
Parameters Table	Name	Value	Description
		enable	Enables logging BGP neighbor status changes (up or down) (Default).
		disable	Disables logging BGP neighbor status changes (up or down).
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# log-neighbor-changes disable</pre>		

10.2.1.39.martians

Description	Sets a Martian policy to validate received IPv4/IPv6 prefixes. Use the no form of the command to remove the policy.		
Command Syntax	<pre>martians policy <i>policy-name</i> no martians</pre>		
Command Mode	<pre>configuration routing bgp vrf af-ipv4 configuration routing bgp vrf af-ipv6</pre>		
Parameters Table	Name	Value	Description
	<i>policy-name</i>	Existing policy name	Associated policy.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# af-ipv4 unicast exaware(config-vrf-vrf1-neighbor-10.10.10.10)# martians policy policy-martians-ipv4</pre>		

Notes

If the command is not configured for an address family the default algorithm is applied.

- The following IPv4 addresses are Martian ones:
 - 0.0.0.0/8
 - 127.0.0.0/8
 - 191.255.0.0/16
 - 192.0.0.9/24
 - 223.255.255.0/24
- The following IPv6 addresses are Martian ones:
 - fe80::/10

10.2.1.40. maximum-prefix

Description	Sets the maximum number of prefix accepted from this peer and configures the action to take if this maximum is exceeded. Use the no form of the command to restore the default maximum.		
Command Syntax	<pre>maximum-prefix number <i>max</i> threshold <i>percent</i> action [warn terminate] no maximum-prefix</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi</pre>		
Parameters Table	Name	Value	Description
	<i>max</i>	0- 4294967295	Maximum number of prefixes. (Default = 2097152)
	<i>percent</i>	1-100	Threshold (percentage of maximum) at which warnings will be generated. (Default = 75)
		warn	Generate a warning when the maximum number of prefixes is exceeded. (Default)
		terminate	Terminate the BGP session when the maximum number of prefixes is exceeded.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# af-ipv4 unicast exaware(config-vrf-vrf1-neighbor-10.10.10.10-af-ipv4-unicast)# maximum-prefix number 456789 threshold 90 action terminate</pre>		

10.2.1.41. minimum-interval

Description	Sets the minimum interval between two consecutive update advertisements. Use the no form of the command to reset to the default interval (10).		
Command Syntax	<pre>minimum-interval <i>interval</i> no minimum-interval</pre>		
Command Mode	<pre>configuration routing bgp vrf <i>neighbor</i> configuration routing bgp vrf neighbor-group</pre>		
Parameters Table	Name	Value	Description
	<i>interval</i>	0-600	Number of seconds BGP waits between two updates. (Default = 10)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# minimum- interval 10</pre>		

10.2.1.42. multipath

Description	Enables or disables the BGP neighbor-group from participating in BGP multipath.		
Command Syntax	multipath {enable disable}		
Command Mode	configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi		
Parameters Table	Name	Value	Description
		enable	Enables the BGP neighbor-group from participating in BGP multipath. (Default)
		disable	Disables the BGP neighbor-group from participating in BGP multipath.
	Name	Value	Description
Examples	This example shows how to use this command: <pre>exaware# routing bgp 100 vrf default neighbor 10.0.0.2 multipath enable</pre>		
Notes	If the multipath is configured on both a neighbor and a neighbor group to which the neighbor belongs, the neighbor configuration is used. Not available under SAFI flowspec.		

10.2.1.43. multipath ebgp | ibgp

Description	Configures the BGP multipath details.		
Command Syntax	<code>multipath {ebgp ibgp} [<i>number</i>] [igp-metric {compare ignore}] [neighbor-as {compare ignore}] [as-length {compare ignore}]</code>		
Command Mode	configuration routing bgp vrf afi safi		
Parameters Table	Name	Value	Description
	<i>number</i>	1-32	Maximum number of paths allowed for BGP multipath feature. (Default = 1)
	igp-metric	compare	Take the IGP metric into consideration. (Default)
		ignore	Ignore the IGP metric.
	neighbor-as	compare	Compare neighboring AS or sub-AS. (Default)
		ignore	Ignore neighboring AS or sub-AS.
	as-length	compare	Take as-length into consideration. (Default)
		ignore	Ignore as-length.
Examples	This example shows how to use this command: <pre>exaware# routing bgp 100 vrf default af-ipv4 unicast multipath ebgp 8 igp-metric compare as-length ignore</pre>		
Notes	Multipath is not available under SAFI flowspec.		

10.2.1.44. neighbor

Description	Enters neighbor configuration sub-mode to configure a neighbor router. Use the no form of the command to remove the neighbor router.		
Command Syntax	<code>neighbor ip-address</code> <code>no neighbor ip-address</code>		
Command Mode	configuration routing bgp vrf		
Parameters Table	Name	Value	Description
	<code>ip-address</code>	IPv4 or global IPv6 address	IP address of the neighbor router.
Examples	This example shows how to enter configuration sub-mode for the specified neighbor for the specified VRF. Note the change in the command line prompt: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)#</pre>		

10.2.1.45. neighbor-group

Description	Enters neighbor-group configuration sub-mode to configure a neighbor group. Use the no form of the command to remove the neighbor group.		
Command Syntax	<pre>neighbor-group group-name no neighbor-group group-name</pre>		
Command Mode	configuration routing bgp vrf		
Parameters Table	Name	Value	Description
	group-name	string	Name to assign to the neighbor group.
Examples	This example shows how to enter configuration sub-mode for the neighbor-group for the specified VRF: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor-group group1 exaware(config-vrf-vrf1-neighbor-group-group1)#</pre>		
Notes	Neighbor group name is unique per vrf. CLI commands under neighbor-group have the same command set as 'neighbor', except 'neighbor-group'. <address-family-routing scheme> (for example, af-ipv4 unicast) must be configured for the group. (Unless the neighbor which this group is bound to already has an address-family-routing-scheme configured. In general address family must be configured either on the neighbor or on the neighbor-group.)		

10.2.1.46. network

Description	Instructs BGP to advertise locally known prefixes to its neighbors. Use the no form of the command to disable advertising.		
Command Syntax	<pre>network <i>ip-prefix</i> [backdoor {disable enable}] [policy <i>policy-name</i>]</pre> <pre>no network</pre>		
Command Mode	configuration routing bgp vrf afi safi		
Parameters Table	Name	Value	Description
	<i>ip-prefix</i>	IPv4 or IPv6 address / prefix length	IP address and prefix length where to broadcast.
	backdoor	enable/ disable	Sets the administrative distance of external routes to be the same as locally sourced routes. (Default = disable)
	<i>policy-name</i>	existing policy number	Associated policy.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# af-ipv4 unicast exaware(config-vrf-vrf1-af-ipv4-unicast)# network 10.10.10.0/24 backdoor disable</pre>		
Notes	The ORIGIN attribute will be set to IGP for prefixes advertised through the network command. SAFI can be only unicast or labeled-unicast.		

10.2.1.47. next-hop-self

Description	Enables the next-hop-self option. This forces the BGP router to report itself as the next hop for advertised routes. Use the no form of the command to revert to the default setting (disable).		
Command Syntax	<pre>next-hop-self [enable disable] no next-hop-self</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi</pre>		
Parameters Table	Name	Value	Description
		enable	Enable next hop self.
		disable	Disable next hop self. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# af-ipv4 unicast exaware(config-vrf-vrf1-neighbor-10.10.10.10-af-ipv4-unicast)# next-hop-self enable</pre>		
Notes	Not available under SAFI flowspec.		

10.2.1.48. orf

Description	Configures a policy for Outbound Route Filtering (ORF) for the neighbor or neighbor group. Use the no form of the command to remove the policy.		
Command Syntax	<pre>orf policy <i>policy-name</i> no orf policy</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor afi unicast configuration routing bgp vrf neighbor-group afi unicast</pre>		
Parameters Table	Name	Value	Description
	<i>policy-name</i>	existing policy name	Policy to be used for ORF.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# af-ipv4 unicast exaware(config-vrf-vrf1-neighbor-10.10.10.10-af-ipv4-unicast)# orf policy polname</pre>		

10.2.1.49.origin-as-invalid-action (rpki)

Description	To specify the action applied to routes marked with an 'invalid' origin-as by RPKI, use the origin-as-allow-invalid command. To return the device to default action (deny), use the no form of this command.		
Command Syntax	<pre>origin-as-invalid-action permit deny no origin-as-invalid-action</pre>		
Command Mode	configuration routing bgp vrf rpki		
Parameters Table	Name	Value	Description
	permit		Permit the received routes marked with an 'invalid' origin-as by RPKI.
	deny		Deny the received routes marked with an 'invalid' origin-as by RPKI.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# rpki exaware(config-routing-bgp-61-vrf-default-rpki)# origin-as- invalid-action permit</pre>		
Note	The action configured by this command can be overridden by the BGP Routing policy (rpki-origin-as invalid).		

10.2.1.50.origin-as-not-found-action (rpki)

Description	To specify the action applied to routes marked with an “not-found” origin-as by RPKI, use the origin-as-allow-invalid command. To return the device to default action (permit), use the no form of this command.		
Command Syntax	<pre>origin-as-not-found-action permit deny no origin-as-not-found-action</pre>		
Command Mode	configuration routing bgp vrf rpki		
Parameters Table	Name	Value	Description
	permit		Permit the received routes marked as “not-found” by RPKI.
	deny		Deny the received routes marked as “not-found” by RPKI.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# rpki exaware(config-routing-bgp-61-vrf-default-rpki)# origin-as-not-found-action deny</pre>		
Note	The action configured by this command can be overridden by the BGP Routing policy (rpki-origin-as not-found).		

10.2.1.51. passive-session

Description	Local BGP does not initiate any TCP open requests and the session can be established only when the request comes from the remote end. Use the no form of the command to allow BGP to initiate sessions.		
Command Syntax	passive-session [enable disable] no passive-session		
Command Mode	configuration routing bgp vrf neighbor configuration routing bgp vrf neighbor-group		
Parameters Table	Name	Value	Description
		enable	BGP does not attempt to initiate the session.
		disable	BGP can both initiate and accept session. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# passive-session enable</pre>		

10.2.1.52. password

Description	Configures a password for the neighbor group. Use the no form of the command to remove the password.		
Command Syntax	password <i>pw-string</i> no password		
Command Mode	configuration routing bgp vrf <i>neighbor</i> configuration routing bgp vrf neighbor-group		
Parameters Table	Name	Value	Description
	<i>pw-string</i>	string	Max 63 characters. Use \$\$ to indicate encrypted string.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrfl exaware(config-routing-bgp-61-vrf-vrfl)# neighbor 10.10.10.10 exaware(config-vrf-vrfl-neighbor-10.10.10.10)# password pw123</pre>		

10.2.1.53. password (rpki-server)

Description	To specify a SSH password for RPKI cache-server, use the password command. To remove the SSH password, use the no form of this command.		
Command Syntax	password <i>password</i> no password		
Command Mode	configuration routing bgp rpki-server transport ssh		
Parameters Table	Name	Value	Description
	<i>pw-string</i>	string	Max 63 characters. Use \$\$ to indicate encrypted string.
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# rpki-server 11.1.1.1 exaware(config-routing-bgp-61-rpki-server-11.1.1.1)# transport ssh exaware(config-routing-bgp-61-rpki-server-11.1.1.1-transport-ssh)# password aqwen12 </pre>		

10.2.1.54. policy

Description	Sets an inbound or outbound policy on routes published or received from a neighbor. Use the no form of the command to remove a specific or all inbound or outbound policies.		
Command Syntax	<pre>policy {in out} <i>policy-name</i> no policy [in out]</pre>		
Command Mode	configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi		
Parameters Table	Name	Value	Description
	<i>policy-name</i>	Existing policy name	Associated policy.
Examples	This example shows how to use this command: <pre>exaware # config exaware(config)# routing bgp 65000 exaware(config-routing-bgp-65000)# vrf default neighbor 1.1.1.1 exaware(config-vrf-default-neighbor-1.1.1.1)# af-ipv4 unicast exaware(config-neighbor-1.1.1.1-af-ipv4-unicast)# policy in poll exaware(config-neighbor-1.1.1.1-af-ipv4-unicast)#</pre>		

10.2.1.55. preference (rpki-server)

Description	To specify a preference value for the RPKI cache-server, use the preference command. To restore the default configuration, use the no form of this command.		
Command Syntax	<pre>preference value no preference</pre>		
Command Mode	configuration routing bgp vrf rpki server		
Parameters Table	Name	Value	Description
	<i>value</i>	1-20	Specifies a RPKI cache preference value. The default value is 1.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# rpki exaware(config-routing-bgp-61-vrf-default-rpki)# rpki-server 11.1.1.1 exaware(config-routing-bgp-61-vrf-default-rpki-server-11.1.1.1)# preference 2</pre>		
Note	Different preference must be configured for different RPKI servers.		

10.2.1.56. preserve-next-hop

Description	Preserves the next-hop attribute of a route when advertising it. Use the no form of the command to revert to the default setting (disable).		
Command Syntax	<pre>preserve-next-hop {enable disable} no preserve-next-hop</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi</pre>		
Parameters Table	Name	Value	Description
		enable	Enable preserve-next-hop.
		disable	Disable preserve-next-hop. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 3.3.3.2 exaware(config-vrf-vrf1-neighbor-3.3.3.2)# af-ipv4 unicast exaware(config-neighbor-3.3.3.2-af-ipv4-unicast)# preserve- next-hop enable exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor-group CLIENTS exaware(config-vrf-vrf1-neighbor-group-CLIENTS)# af-ipv4 unicast exaware(config-neighbor-group-CLIENTS-af-ipv4-unicast)# no preserve-next-hop</pre>		

10.2.1.57. private-as

Description	Configures how to handle private AS numbers. Use the no form of the command to restore the default (private AS numbers are not removed).		
Command Syntax	<pre>private-as [remove leave] no private-as</pre>		
Command Mode	configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi		
Parameters Table	Name	Value	Description
		remove	Remove private AS numbers from outbound updates.
		leave	Do not remove private AS numbers (default).
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# af-ipv4 unicast exaware(config-vrf-vrf1-neighbor-10.10.10.10-af-ipv4-unicast)# private-as remove</pre>		

10.2.1.58. purge-time (rpki)

Description	To configure the time BGP waits to keep routes from RPKI cache-server after the cache session drops, use the purge-time command. To restore the default configuration (7200 seconds), use the no form of this command.		
Command Syntax	purge-time <i>seconds</i> no purge-time		
Command Mode	configuration routing bgp vrf rpki		
Parameters Table	Name	Value	Description
	<i>seconds</i>	600-172800	Specifies the purge-time in seconds.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# rpki exaware(config-routing-bgp-61-vrf-default-rpki)# purge-time 700</pre>		
Note	When a cache session is dropped then a "purge-timer" is started for that cache. If the session re-establishes within the timer interval, then the purge timer is stopped, and no further action is taken. If the cache session does not re-establish within the timer interval, only then does BGP remove all ROAs from the cache.		

10.2.1.59. redistribute

Description	Redistributes reachability information from other routing protocols. Use the no form of the command to disable redistribution.		
Command Syntax	<pre>redistribute <i>routing-type</i> [policy <i>policy-name</i>]</pre> <pre>no redistribute</pre>		
Command Mode	configuration routing bgp vrf afi safi		
Parameters Table	Name	Value	Description
	<i>routing-type</i>	connected static isis ospf rsvp	Specify the type of routing to redistribute: • direct connections • statically configured routes • routes learned via IS-IS • routes learned via OSPF • routes learned via RSVP (Default = no redistribution)
	<i>policy-name</i>	existing policy name	Associated policy.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# af-ipv4 unicast exaware(config-vrf-vrf1-af-ipv4-unicast)# redistribute connected</pre>		
Notes	You can redistribute more than one routing type, but you can only specify one routing type per command. SAFI can be only unicast or labeled-unicast.		

10.2.1.60. refresh-time (rpki)

Description	To configure the time BGP waits in between sending periodic serial queries to the RPKI server, use the refresh-time command. To restore the default configuration (3600 seconds), use the no form of this command.		
Command Syntax	<pre>refresh-time seconds no refresh-time</pre>		
Command Mode	configuration routing bgp vrf rpki		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-86400	Specifies the refresh-time in seconds.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# rpki exaware(config-routing-bgp-61-vrf-default-rpki)# refresh-time 100</pre>		

10.2.1.61. remote-as-number

Description	Specifies the number of the AS to which this neighbor belongs. Use the no form of the command to reset to the default AS (0).		
Command Syntax	<code>remote-as-number <i>as-number</i></code> <code>no remote-as-number</code>		
Command Mode	configuration routing bgp vrf neighbor configuration routing bgp vrf neighbor-group		
Parameters Table	Name	Value	Description
	<i>as-number</i>	1-4294967295	Autonomous system number.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor-group group1 exaware(config-vrf-vrf1-neighbor-group-group1)# remote-as-number 100</pre>		
Notes	<i>as-number</i> is a mandatory field that does not have any default.		

10.2.1.62. route-reflector-client

Description	Configures the neighbor or neighbor group as a route reflector client in the AS. Use the no form of the command to restore the default (disable).		
Command Syntax	<pre>route-reflector-client [enable disable] no route-reflector-client</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi</pre>		
Parameters Table	Name	Value	Description
		enable	Enable the neighbor as a route reflector client.
		disable	Disable the neighbor as a route reflector client. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# af-ipv4 unicast exaware(config-vrf-vrf1-neighbor-10.10.10.10-af-ipv4-unicast)# route-reflector-client disable</pre>		

10.2.1.63. router-id

Description	Configures the router identifier for the VRF. Use the no form of the command to disable the manually-configured router ID.		
Command Syntax	<pre>router-id ip-address no router-id</pre>		
Command Mode	configuration routing bgp vrf		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 address	Manually configured router identifier.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 123 exaware(config-routing-bgp-123)# vrf vrf1 exaware(config-routing-bgp-123-vrf-vrf1)# router-id 1.2.3.4</pre>		
Notes	The router-id configuration also exists under /routing/bgp for backward compatibility reasons. Explicitly configured router-id under routing/bgp/vrf takes precedence over globally configured router-id. If no router-id is configured for a specific VRF, and no router-id is configured globally, then the router-id for that VRF is selected according to the regular algorithm, only from IP addresses configured on interfaces belonging to VRF default.		

10.2.1.64. routing bgp

Description	Enters routing BGP configuration mode. This mode configures and enables BGP routing for a specified AS. Use the no form of the command to disable BGP routing for the specified AS.		
Command Syntax	<pre>routing bgp <i>as-number</i> no routing bgp <i>as-number</i></pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>as-number</i>	1-4294967295	Autonomous system number.
Examples	This example shows how to configure BGP routing for AS 61 Note the change in the command line prompt: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)#</pre>		
Notes	You must execute this command to enable BGP routing for the specified AS.		

10.2.1.65. rpki

Description	To enter resource public key infrastructure (rpki) configuration mode and enable rpki parameters configuration, use the rpki command. To remove the rpki configuration mode and remove the rpki configuration parameters, use the no form of this command. The following options are available in the rpki-server configuration mode: • origin-as-invalid-action • origin-as-not-found-action • purge-time • refresh-time • server
Command Syntax	rpki no rpki
Command Mode	configuration routing bgp vrf
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# rpki exaware(config-routing-bgp-61-vrf-default-rpki)#</pre>
Note	RPKI can be configured only for the "default" vrf.

10.2.1.66. send-community

Description	Allows BGP to send community attributes (standard and/or extended) to its eBGP neighbor. Use the no form of the command to restore the default (disable).		
Command Syntax	<pre>send-community [standard extended all none] no send-community</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi</pre>		
Parameters Table	Name	Value	Description
		standard	BGP will send standard community attribute to the specified neighbor.
		extended	BGP will send extended community attribute to the specified neighbor. (Default for VPN SAFI)
		all	BGP will send both standard and extended community attribute to the specified neighbor.
		none	BGP will not send community attribute to the specified neighbor. (Default for non VPN SAFIs)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# af-ipv4 unicast exaware(config-vrf-vrf1-neighbor-10.10.10.10-af-ipv4-unicast)# send-community all</pre>		

10.2.1.67. server (rpki)

Description	To enter resource public key infrastructure (RPKI) cache-server (rpki-sever) configuration mode and enable rpki casche-server parameters configuration, use the rpki-server command. To remove the rpki-server configuration mode and delink cache-server from the cache list, use the no form of this command. The following options are available in the rpki-server configuration mode: • admin-state • description • preference • transport		
Command Syntax	<pre>server ip-address no server</pre>		
Command Mode	configuration routing bgp vrf rpki		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 or IPv6 address	IP address of the rpki server.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# rpki exaware(config-routing-bgp-61-vrf-default-rpki)# server 11.1.1.1 exaware(config-routing-bgp-61-vrf-default-rpki-server-11.1.1.1)#</pre>		

10.2.1.68. strict-capability-match

Description	Enforce a strict match between local and remote capabilities and reset the session if they do not match. Use the no form of the command to revert to the default (disable).		
Command Syntax	<pre>strict-capability-match [enable disable] no strict-capability-match</pre>		
Command Mode	configuration routing bgp vrf <i>neighbor</i> configuration routing bgp vrf neighbor-group		
Parameters Table	Name	Value	Description
		enable	Accept peering only if capabilities match.
		disable	Accept peering even if there is some mismatch between capabilities. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# strict- capability-match enable</pre>		

10.2.1.69. soo

Description	Configures Site of Origin. Use the no form of the command to configure soo to the default (None).		
Command Syntax	<pre>soo ext-community</pre> <pre>no soo</pre>		
Command Mode	configuration routing bgp vrf neighbor afi unicast configuration routing bgp vrf neighbor-group afi unicast		
Parameters Table	Name	Value	Description
	<i>ext-community</i>	<2-byte ASN>:<4-byte number>	2-byte ASN notation.
		<4-byte ASN>L:<2-byte number>	4-byte ASN notation.
		<IPv4 address>:<2-byte number>	IPv4 address notation.
Examples	The following examples show how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrfl exaware(config-routing-bgp-61-vrf-vrfl)# neighbor 3.3.3.2 exaware(config-vrf-vrfl-neighbor-3.3.3.2)# af-ipv4 unicast exaware(config-neighbor-3.3.3.2-af-ipv4-unicast)# soo 100:200</pre> <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrfl exaware(config-routing-bgp-61-vrf-vrfl)# neighbor-group CLIENTS exaware(config-vrf-vrfl-neighbor-group-CLIENTS)# af-ipv4 unicast exaware(config-neighbor-group-CLIENTS-af-ipv4-unicast)# no soo</pre>		
Notes	This command only available under non-default VRF and unicast SAFI.		

10.2.1.70. timers

Description	Configures the following timers in neighbor sub-mode: - hold time - keep alive interval Use the no form of the command to reset the neighbor timer to the default value.		
Command Syntax	<pre>timers [holdtime <i>time1</i> keepalive <i>time2</i>]</pre> <pre>no timers [holdtime keepalive]</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor</pre> <pre>configuration routing bgp vrf neighbor-group</pre>		
Parameters Table	Name	Value	Description
	<i>time1</i>	3-65535	Interval without receiving a keepalive message after which a peer is declared dead, in seconds. (Default = 180)
	<i>time2</i>	1-65535	Frequency of the keepalive messages in seconds. (Default = 60)
Examples	This example shows how to configure the BGP timers: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 3.3.3.2 exaware(config-vrf-vrf1-neighbor-3.3.3.2)# timers holdtime 120 keepalive 180</pre>		
Notes	The hold timer (<i>time1</i>) must be greater than the keepalive timer (<i>time2</i>).		

10.2.1.71. transport (rpki-server)

Description	To enter resource public key infrastructure (RPKI) cache-server (rpki-sever) configuration mode and enable rpki parameters configuration, use the rpki-server command. To remove the rpki-server configuration mode and delink cache-server from the cache list, use the no form of this command.		
Command Syntax	<pre>transport {ssh [port <i>ssh-port-number</i>]} {tcp port <i>port-number</i>}</pre> <pre>no transport</pre>		
Command Mode	configuration routing bgp vrf rpki server		
Parameters Table	Name	Value	Description
	ssh		Specifies the SSH transport.
	<i>ssh-port-number</i>	1 to 65535	Specifies the SSH transport port. The default is 22.
	tcp		Specifies the TCP transport.
	<i>port-number</i>	1 to 65535	Specifies the TCP transport port.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf default exaware(config-routing-bgp-61-vrf-default)# rpki exaware(config-routing-bgp-61-vrf-default-rpki)# server 11.1.1.1 exaware(config-routing-bgp-61-vrf-default-rpki-server-11.1.1.1)# transport tcp port 1234</pre>		

10.2.1.72. vrf

Description	Enters routing BGP configuration mode for a specified VRF. Use the no form of the command to disable BGP routing for the specified VRF.		
Command Syntax	<pre>vrf vrf-name no vrf vrf-name</pre>		
Command Mode	configuration routing bgp		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Maximum 128 characters, no spaces	Name of the VRF for which to configure BGP routing. A default VRF is pre-configured in the system. To configure a user-defined VRF for BGP routing, you must first define it. See vrf (on page 1494).
Examples	This example shows how to enter configuration mode for the specified VRF: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)#</pre>		
Notes	When using a non-default VRF, the user-defined VRF must have a configured RD value.		

10.2.1.73.username (rpki-server)

Description	To specify a SSH username for RPKI cache-server, use the username command. To remove the SSH username, use the no form of this command.		
Command Syntax	username <i>user-name</i> no password		
Command Mode	configuration routing bgp rpki-server transport ssh		
Parameters Table	Name	Value	Description
	<i>user-name</i>	string	Max 63 characters.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# rpki-server 11.1.1.1 exaware(config-routing-bgp-61-rpki-server-11.1.1.1)# transport ssh exaware(config-routing-bgp-61-rpki-server-11.1.1.1-transport-ssh)# username bob</pre>		

10.2.1.74. weight

Description	Sets the default weight for routes advertised by the neighbor or neighbor group. Use the no form of the command to reset to the default weight (0).		
Command Syntax	<pre>weight weight-value no weight</pre>		
Command Mode	<pre>configuration routing bgp vrf neighbor afi safi configuration routing bgp vrf neighbor-group afi safi</pre>		
Parameters Table	Name	Value	Description
	<i>weight-value</i>	0-65535	Default weight for routes. (Default = 0)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing bgp 61 exaware(config-routing-bgp-61)# vrf vrf1 exaware(config-routing-bgp-61-vrf-vrf1)# neighbor 10.10.10.10 exaware(config-vrf-vrf1-neighbor-10.10.10.10)# af-ipv4 unicast exaware(config-neighbor-10.10.10.10-af-ipv4-unicast)# weight 1800</pre>		

10.2.2. BGP Operational Commands

10.2.2.1. **clear bgp all**

Description	Clears BGP peers. Options: Graceful reset (Default): - Gracefully restarts all BGP sessions if possible. It removes the TCP connection to the neighbor, but all received BGP routes from the neighbor are marked as stale. If a neighbor cannot be restarted gracefully, then the hard reset is performed. Hard reset: - Will reset all BGP sessions. It removes the TCP connection to the neighbor and all BGP routes received from the neighbor, and then re-establishes the session with the neighbor. Soft reset: - When specifying outbound, all advertised routes will be re-advertised. - When specifying inbound, all received routes will be cleared and refresh requests will be sent to the related neighbors (in the case where the neighbor supports route refresh capability). - In case of a soft reconfiguration set under the related peer, use the storage of inbound routes for reevaluation of the received route. - If specifying both, the behavior will be as mentioned above regarding inbound and outbound.		
Command Syntax	<pre>clear bgp all [<i>afi safi</i>] [graceful hard soft {both inbound outbound}]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>afi</i>	ipv4	IP address family (AFI) name. (Default = all)
		ipv6	
		l2vpn	

	<i>safi</i>	labeled-unicast unicast vpn flowspec vpls	SAFI name. (Default = all)
--	-------------	---	-------------------------------

	soft		Clear BGP routing information without tearing the BGP session.
		inbound outbound both	Incoming routes will cleared. Outgoing routes will cleared. Both incoming and outgoing routes will cleared. (Default = both) These are the available options for the 'soft' reconfiguration option.
	hard		Clear bgp routing information by resetting BGP sessions and clearing all routing information received from this neighbor.
	graceful		Clear bgp routing information by resetting BGP sessions gracefully. Mark all routing information received from neighbor as stale.

Examples

The following examples show how to use this command:

```
exaware# clear bgp all
```

```
exaware# clear bgp all soft inbound
```

```
exaware# clear bgp all soft outbound
```

Notes

In 4.0, if AFI is IPv6, only unicast is a valid SAFI completion.

10.2.2.2. clear bgp groups

Description	Clears BGP groups. Options: Graceful reset (Default): - Gracefully restarts all BGP sessions if possible. It removes the TCP connection to the neighbor, but all received BGP routes from the neighbor are marked as stale. If a neighbor cannot be restarted gracefully, then the hard reset is performed. Hard reset: - Will reset the BGP session with specified neighbor(s). It removes the TCP connection to the neighbor and all BGP routes received from the neighbor, and then re-establishes the session with the neighbor. Soft reset: - When specifying outbound, all advertised routes will be re-advertised. - When specifying inbound, all received routes will be cleared and refresh requests will be sent to the related neighbors (in the case where the neighbor supports route refresh capability). - In case of a soft reconfiguration set under the related peer: - use the storage of inbound routes for reevaluation of the received route. If specifying both, the behavior will be as mentioned above regarding inbound and outbound.
Command Syntax	<pre>clear bgp groups [vrf <i>vrf-name</i>] {name <i>group-name</i> all <i>afi safi</i>} [graceful hard soft {both inbound outbound}]</pre>
Command Mode	Operational

Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Name	VRF name. (Default = default)
	<i>group-name</i>	Name	Group name. (Default = all)
	<i>afi</i>	ipv4 ipv6 l2vpn	IP address family (AFI) name. (Default = all)
	<i>safi</i>	labeled-unicast unicast vpn flowspec vpls	SAFI name. (Default = all)
	<i>soft</i>		Clear using a soft reconfiguration.
		inbound outbound both	Incoming routes will cleared. Outgoing routes will cleared. Both incoming and outgoing routes will cleared. (Default = both) These are the available options for the 'soft' reconfiguration option.

		inbound	Incoming routes will cleared.
		outbound	Outgoing routes will cleared.
		both	Both incoming and outgoing routes will cleared. (Default = both) These are the available options for the 'soft' reconfiguration option.
	hard		Clear BGP routing information by resetting BGP sessions and clearing all routing information received from this neighbor.
	graceful		Clear BGP routing information by resetting BGP sessions gracefully. Mark all routing information received from neighbor as stale.
Examples	The following example shows how to use this command: exaware# clear bgp groups vrf-all all		
Notes	If no VRF filter is specified, this command will clear neighbors in VRF default only.		

10.2.2.3. clear bgp neighbors

Description	Clears BGP neighbors. Options: Graceful reset (Default): - Gracefully restarts all BGP sessions if possible. It removes the TCP connection to the neighbor, but all received BGP routes from the neighbor are marked as stale. If a neighbor cannot be restarted gracefully, then the hard reset is performed. Hard reset: - Will reset the BGP session with specified neighbor(s). It removes the TCP connection to the neighbor and all BGP routes received from the neighbor, and then re-establishes the session with the neighbor. Soft reset: - When specifying outbound, all advertised routes will be re-advertised. - When specifying inbound, all received routes will be cleared and refresh requests will be sent to the related neighbors (in the case where the neighbor supports route refresh capability). - In case of a soft reconfiguration set under the related peer: - use the storage of inbound routes for reevaluation of the received route. - If specifying both, the behavior will be as mentioned above regarding inbound and outbound.		
Command Syntax	<pre>clear bgp neighbors [vrf <i>vrf-name</i>] {peer-ip <i>neighbor-ip</i> all <i>afi safi</i>} [graceful hard soft {both inbound outbound}]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Name	VRF name. (Default = default)
	<i>neighbor-ip</i>	IPv4 or IPv6 address	Neighbor's IP address. (Default = all)

	<i>afi</i>	ipv4 ipv6 l2vpn	IP address family (AFI) name. (Default = all)
--	------------	-----------------------	--

	<i>safi</i>	labeled-unicast unicast vpn flowspec vpls	SAFI name. (Default = all)
--	-------------	---	-------------------------------

	soft		Clear using a soft reconfiguration.
		inbound outbound both	Incoming routes will cleared. Outgoing routes will cleared. Both incoming and outgoing routes will cleared. (Default = both) These are the available options for the 'soft' reconfiguration option.
	hard		Clear bgp routing information by resetting BGP sessions and clearing all routing information received from this neighbor.
	graceful		Clear bgp routing information by resetting BGP sessions gracefully. Mark all routing information received from neighbor as stale.
Examples	The following example shows how to use this command: exaware# clear bgp neighbors vrf default 3.3.3.2 hard		
Notes	If no VRF filter is specified, this command will clear neighbors in VRF default only.		

10.2.2.4. clear bgp policy accounting

Description	Clears the BGP accounting policy hit counters.		
Command Syntax	clear bgp policy accounting [vrf <i>vrf-name</i> vrf-all] [all <i>afi safi</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Name	VRF name. (Default = default)
	<i>afi</i>	ipv4 ipv6	IP address family (AFI) name. (Default = all)
	<i>safi</i>	unicast	SAFI name. (Default = all)
Examples	The following example shows how to use this command: <pre>exaware# clear bgp policy accounting vrf RED ipv4 unicast</pre>		
Notes	If no VRF filter is specified, this command will clear accounting policies in VRF default only.		

10.2.2.5. clear bgp policy aggregate-address

Description	Clears the BGP aggregate-address policy hit counters.		
Command Syntax	<code>clear bgp policy aggregate-address [vrf <i>vrf-name</i> vrf-all] [all <i>afi safi</i>]</code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Name	VRF name. (Default = default)
	<i>afi</i>	ipv4 ipv6	IP address family (AFI) name. (Default = all)
	<i>safi</i>	unicast	SAFI name. (Default = all)
Examples	The following example shows how to use this command: <pre>exaware# clear bgp policy aggregate-address vrf-all</pre>		
Notes	If no VRF filter is specified, this command will clear policies in VRF default only.		

10.2.2.6. clear bgp policy allocate-label

Description	Clears the BGP allocate-label policy.		
Command Syntax	<code>clear bgp policy allocate-label [vrf default vrf-all] [all <i>afi safi</i>]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<code>afi</code>	<code>ipv6</code>	AFI name. (Default = all)
	<code>safi</code>	<code>unicast</code>	SAFI name. (Default = all)
Examples	The following example shows how to use this command: <code>exaware# clear bgp policy allocate-label vrf default ipv6 unicast</code>		
Notes	This command is relevant only to default VRF.		

10.2.2.7. clear bgp policy default-route-originate

Description	Clears the BGP neighbor's default-route-originate policy hit counters.		
Command Syntax	<code>clear bgp policy default-route-originate [vrf <i>vrf-name</i>] [peer-ip <i>neighbor-ip</i>] [all <i>afi safi</i>]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Name	VRF name. (Default = default)
	<i>neighbor-ip</i>	IPv4 or IPv6 address	Neighbor's IP address. (Default = all)
	<i>afi</i>	ipv4 ipv6	IP address family (AFI) name. (Default = all)
	<i>safi</i>	unicast	SAFI name. (Default = all)
Examples	The following example shows how to use this command: <pre>exaware# clear bgp policy default-route-originate vrf BLUE ipv4 unicast</pre>		
Notes	If no VRF filter is specified, this command will clear policies in VRF default only.		

10.2.2.8. clear bgp policy import/export/import-from-vrf

Description	Clears the BGP import/export/import-from-vrf policy hit counters.		
Command Syntax	clear bgp policy {import export import-from-vrf} [vrf <i>vrf-name</i> vrf-all] [all <i>afi safi</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Name	VRF name. (Default = default)
	<i>afi</i>	ipv4 ipv6	IP address family (AFI) name. (Default = all)
	<i>safi</i>	unicast	SAFI name. (Default = all)
Examples	The following example shows how to use this command: <pre>exaware# clear bgp policy import vrf vrf1 ipv4 unicast exaware# clear bgp policy export vrf-all ipv4 unicast exaware# clear bgp policy import vrf vrf1 all exaware# clear bgp policy import-from-vrf vrf-all ipv4 unicast</pre>		

10.2.2.9. clear bgp policy neighbors

Description	Clears the BGP neighbor policy hit counters.		
Command Syntax	<code>clear bgp policy neighbors [vrf <i>vrf-name</i>] [peer-ip <i>neighbor-ip</i>] [all <i>afi safi</i>] [in out]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	vrf-name	Name	VRF name. (Default = default)
	neighbor-ip	IPV4 or IPV6 address	Neighbor's IP address. (Default = all)
	afi	ipv4 ipv6 l2vpn	IP address family to be cleared. (Default = all)
	safi	labeled-unicast unicast vpn flowspec vpls	SAFI name. (Default = all)
		in out	Incoming policy counters are cleared. Outgoing policy counters are cleared.
Examples	The following example shows how to use this command: <pre>exaware# clear bgp policy neighbors peer-ip 1.2.3.4 out</pre>		

Notes	If no VRF filter is specified, this command will clear policies in VRF default only. If no direction filter is used, this command will clear policies in both directions.
--------------	---

10.2.2.10. clear bgp policy network

Description	Clears the BGP network policy hit counters.		
Command Syntax	clear bgp policy network [vrf <i>vrf-name</i> vrf-all] [all <i>afi safi</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	vrf-name	Name	VRF name. (Default = default)
	afi	ipv4 ipv6	IP address family to be cleared. (Default = all)
	safi	labeled-unicast unicast	SAFI name. (Default = all)
Examples	The following example shows how to use this command: <pre>exaware# clear bgp policy network vrf GREEN all</pre>		
Notes	If no VRF filter is specified, this command will clear policies in VRF default only.		

10.2.2.11. clear bgp policy redistribute

Description	Clears the BGP redistribute policy hit counters.		
Command Syntax	<pre>clear bgp policy redistribute [vrf <i>vrf-name</i> vrf-all] [all <i>afi safi</i>] [<i>protocol</i>]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Name	VRF name. (Default = default)
	<i>afi</i>	ipv4 ipv6	IP address family to be cleared. (Default = all)
	<i>safi</i>	labeled-unicast unicast	SAFI name. (Default = all)
	<i>protocol</i>	connected static isis ospf	Directly connected routes, e.g. interfaces address. Statically configured routes. Routes learned via isis. Routes learned via ospf.
Examples	The following example shows how to use this command: <pre>exaware# clear bgp policy redistribute vrf-all ipv6 unicast static</pre>		
Notes	If no VRF filter is specified, this command will clear policies in VRF default only. If no protocol filter is specified, this command will clear the redistribute policy counters of all protocols.		

10.2.3. BGP Show Commands

Notes:

- Filtering on any SAFI other than unicast and flowspec can only be done under VRF default. i.e. show bgp {auto-groups | damping | groups | neighbors | table} vrf non-default-vrf: AFI SAFI filter can only be ipv4 unicast or ipv6 unicast.
- Filtering on RD can only be done under VRF default, IPv4, and SAFI VPN. i.e. show bgp {neighbors {advertised-routes | received-routes} | table} [vrf vrf-name | all] afi non-vpn-safi has no RD filter option.
- Label view and filtering can only be done under IPv4 and IPv6 labeled-unicast or IPv4/IPv6 VPN.
- User can use * in place of vrf-name to show all VRF's.
- In any show command which lists several VRF's, the order will be default, then other VRF's ordered alphabetically.
- In any show command which lists several AFI's, the order will be IPv4, IPv6, and then l2vpn.
- In any show command which lists several SAFI's, the order will be unicast, labeled-unicast, VPN, and then flowspec.

10.2.3.1. show bgp auto-groups

Description	Displays information regarding BGP auto-groups (internal logic groups).							
Command Syntax	show bgp auto-groups [vrf <i>vrf-name</i>] [<i>afi safi</i>] [brief [<i>group-index</i>]]							
Command Mode	Operational							
Parameters Table	Name	Value			Description			
	<i>vrf-name</i>	Name, *			Name of VRF instance for BGP in which the group resides. (Default = default)			
	<i>afi</i>	ipv4 ipv6 l2vpn			IP address family (AFI) name. (Default = all)			
	<i>safi</i>	labeled-unicast unicast vpn flowspec vpls			SAFI name. (Default = all)			
	<i>group-index</i>	A			Group index. (Default = all)			
Examples	The following examples show sample output from this command:							
exaware# show bgp auto-groups vrf * VRF default:								
GROUP INDEX	NEIGHBOR	PEER AS	MSG RCVD	MSG SENT	UP/DOWN TIME	AFI/SAFI	NEIGHBOR TABLE VERSION	STATE/ PFX-RCVD

0	3.3.3.4	123	37664	44129	21h59m22s	IPv4u	1191146	1
	3.3.3.5	123	37665	44119	21h59m22s	IPv4u	1191146	1
1	3.3.3.6	123	37664	44129	21h59m22s	IPv4u	1191146	1
	3.3.3.7	123	37665	44119	21h59m22s	IPv4u	1191146	1
VRF BLUE:								
GROUP INDEX	NEIGHBOR	PEER AS	MSG RCVD	MSG SENT	UP/DOWN TIME	AFI/SAFI	NEIGHBOR TABLE VERSION	STATE/ PFX-RCVD
0	7.7.7.2	777	37664	44129	21h59m22s	IPv4u	1191146	1
	7.7.7.3	777	37665	44119	21h59m22s	IPv4u	1191146	1
exaware# show bgp auto-groups vrf * brief 0								
VRF default:								
GROUP INDEX	NEIGHBOR	PEER AS	MSG RCVD	MSG SENT	UP/DOWN TIME	AFI/SAFI	NEIGHBOR TABLE VERSION	STATE/ PFX-RCVD
0	3.3.3.4	123	37664	44129	21h59m22s	IPv4u	1191146	1
	3.3.3.5	123	37665	44119	21h59m22s	IPv4u	1191146	1
VRF BLUE:								
GROUP INDEX	NEIGHBOR	PEER AS	MSG RCVD	MSG SENT	UP/DOWN TIME	AFI/SAFI	NEIGHBOR TABLE VERSION	STATE/ PFX-RCVD
0	7.7.7.2	777	37664	44129	21h59m22s	IPv4u	1191146	1
	7.7.7.3	777	37665	44119	21h59m22s	IPv4u	1191146	1
exaware# show bgp auto-groups ipv4 flowspec								
VRF default:								
GROUP INDEX	NEIGHBOR	PEER AS	MSG RCVD	MSG SENT	UP/DOWN TIME	AFI/SAFI	NEIGHBOR TABLE VERSION	STATE/ PFX-RCVD
0	3.3.3.4	123	37664	44129	21h59m22s	IPv4fs	1191146	1
	3.3.3.5	123	37665	44119	21h59m22s	IPv4fs	1191146	1
exaware# show bgp auto-groups l2vpn vpls								
VRF default:								
GROUP INDEX	NEIGHBOR	PEER AS	MSG RCVD	MSG SENT	UP/DOWN TIME	AFI/SAFI	NEIGHBOR TABLE VERSION	STATE/ PFX-RCVD
0	3.3.3.4	123	37664	44129	21h59m22s	L2VPNvpls	1191146	1
	3.3.3.5	123	37665	44119	21h59m22s	L2VPNvpls	1191146	1

Notes	If a VRF is not specified, data is shown for the default VRF. AFI I2vpn is only available in VRF default, and SAFI vpls is only supported in AFI I2vpn, and is the only option there. SAFI labeled-unicast and vpn are supported only in VRF default.
--------------	--

10.2.3.2. show bgp damping

Description	Displays BGP damping information and statistics.		
Command Syntax	show bgp damping [vrf <i>vrf-name</i>] [<i>afi safi</i>] {{dampened-paths flap-statistics} [<i>ip-prefix</i>] parameters}		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Name, *	Name of VRF instance for BGP. (Default = default)
	<i>afi</i>	ipv4 ipv6	IP address family (AFI) name. (Default = all)
	<i>safi</i>	labeled-unicast unicast vpn	SAFI name. (Default = all)
	<i>ip-prefix</i>	IPv4 or IPv6 address/length	Prefix.
Examples	The following examples show sample outputs from this command:		
<pre>exaware# show bgp damping vrf * dampened-paths VRF default: Flags: s - suppressed, d - damped, h - history, * - valid, > - best i - internal, S - stale Origin: i - IGP, e - EGP, ? - incomplete IPv4 Unicast: Flags: s - suppressed, d - damped, h - history, * - valid, > - best i - internal, S - stale Origin: i - IGP, e - EGP, ? - incomplete IPv4 Labeled-Unicast: Flags: s - suppressed, d - damped, h - history, * - valid, > - best i - internal, S - stale</pre>			

```

Origin:  i - IGP, e - EGP, ? - incomplete
IPv4 VPN:

NETWORK          PEER          FLAGS  REUSE          AS
              PATH  ORIGIN
-----
10.10.10.10/32   30.30.30.1    d      01:25:00   777    i
10.10.10.11/32   30.30.30.1    d      01:25:00   777    i

Flags:  s - suppressed, d - damped, h - history, * - valid, > - best
        i - internal, S - stale
Origin:  i - IGP, e - EGP, ? - incomplete

IPv6 Unicast

VRF BLUE:
Flags:  s - suppressed, d - damped, h - history, * - valid, > - best
        i - internal, S - stale
Origin:  i - IGP, e - EGP, ? - incomplete
IPv4 Unicast:

NETWORK          PEER          FLAGS  REUSE          AS
              PATH  ORIGIN
-----
10.10.10.10/32   40.40.40.1    d      01:25:00   888    i
10.10.10.11/32   40.40.40.1    d      01:25:00   888    i

Flags:  s - suppressed, d - damped, h - history, * - valid, > - best
        i - internal, S - stale
Origin:  i - IGP, e - EGP, ? - incomplete
IPv6 Unicast:

exaware# show bgp damping vrf * ipv4 unicast flap-statistics
VRF default:
Flags:  s - suppressed, d - damped, h - history, * - valid, > - best
        i - internal, S - stale
Origin:  i - IGP, e - EGP, ? - incomplete

NETWORK          PEER          FLAGS  FLAPS  DURATION  REUSE          AS
              PATH  ORIGIN
-----
10.10.10.10/32   30.30.30.1    *d      4      00:00:18  00:36:00   216    i
10.10.10.11/32   30.30.30.1    *d      4      00:00:18  00:36:00   216    i

VRF BLUE:
Flags:  s - suppressed, d - damped, h - history, * - valid, > - best
        i - internal, S - stale
Origin:  i - IGP, e - EGP, ? - incomplete

NETWORK          PEER          FLAGS  FLAPS  DURATION  REUSE          AS
              PATH  ORIGIN
-----
10.10.10.10/32   30.30.30.1    *d      4      00:00:18  00:36:00   216    i
10.10.10.11/32   30.30.30.1    *d      4      00:00:18  00:36:00   216    i

```

```
exaware# show bgp damping vrf * parameters
VRF default:
IPv4 Unicast:

Damping Control Block(s):
  Reachability Half-Life time      : 15 min
  Reuse penalty                    : 750
  Suppress penalty                 : 3000
  Max suppress time                : 60 min
  Max penalty (ceil)              : 11999
  Min penalty (floor)             : 375
  Penalty increment value         : 1000

IPv4 Labeled-Unicast:

Damping Control Block(s):
  Reachability Half-Life time      : 15 min
  Reuse penalty                    : 750
  Suppress penalty                 : 3000
  Max suppress time                : 60 min
  Max penalty (ceil)              : 11999
  Min penalty (floor)             : 375
  Penalty increment value         : 1000

IPv4 VPN:

Damping Control Block(s):
  Reachability Half-Life time      : 15 min
  Reuse penalty                    : 750
  Suppress penalty                 : 3000
  Max suppress time                : 60 min
  Max penalty (ceil)              : 11999
  Min penalty (floor)             : 375
  Penalty increment value         : 1000

IPv4 RT-Constrains:

Damping Control Block(s):
  Reachability Half-Life time      : 15 min
  Reuse penalty                    : 750
  Suppress penalty                 : 3000
  Max suppress time                : 60 min
  Max penalty (ceil)              : 11999
  Min penalty (floor)             : 375
  Penalty increment value         : 1000

IPv6 Unicast

Damping Control Block(s):
  Reachability Half-Life time      : 15 min
  Reuse penalty                    : 750
```

```

Suppress penalty           : 3000
Max suppress time         : 60 min
Max penalty (ceil)        : 11999
Min penalty (floor)       : 375
Penalty increment value   : 1000

```

VRF BLUE:

IPv4 Unicast:

Damping Control Block(s):

```

Reachability Half-Life time : 15 min
Reuse penalty               : 750
Suppress penalty           : 3000
Max suppress time         : 60 min
Max penalty (ceil)        : 11999
Min penalty (floor)       : 375
Penalty increment value   : 1000

```

IPv6 Unicast:

Damping Control Block(s):

```

Reachability Half-Life time : 15 min
Reuse penalty               : 750
Suppress penalty           : 3000
Max suppress time         : 60 min
Max penalty (ceil)        : 11999
Min penalty (floor)       : 375
Penalty increment value   : 1000

```

Notes

If a VRF is not specified, data is shown for the default VRF.

SAFI labeled-unicast and vpn are supported only in VRF default.

10.2.3.3. show bgp groups

Description	Displays information regarding configured BGP neighbor-groups.		
Command Syntax	<code>show bgp groups [vrf <i>vrf-name</i>] [afi <i>safi</i>] [brief [<i>group-name</i>]]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Name, *	Name of VRF instance for BGP. (Default = default)
	<i>afi</i>	ipv4 ipv6 l2vpn	IP address family (AFI) name. (Default = all)
	<i>safi</i>	labeled-unicast unicast vpn flowspec vpls	SAFI name. (Default = all)
	<i>group-name</i>	Name	Group name. (Default = all)
Examples	The following example shows sample outputs from this command:		

exaware# show bgp groups vrf *										
VRF default:										
NEIGHBOR GROUP	NEIGHBOR	STATE	PEER AS	MSG RCVD	MSG SENT	UP/DOWN TIME	AFI/SAFI	NEIGHBOR TABLE VERSION	STATE/ PFX-RCVD	
--										
Clients	3.3.3.4	up	123	39780	46612	1D15h37m34s	IPv4u	1191146		1
	3.3.3.5	up	123	39781	46596	1D15h37m34s	L2VPNvpls	1191146		1
VRF BLUE:										
NEIGHBOR GROUP	NEIGHBOR	STATE	PEER AS	MSG RCVD	MSG SENT	UP/DOWN TIME	AFI/SAFI	NEIGHBOR TABLE VERSION	STATE/ PFX-RCVD	
--										
CE	3.3.3.4	up	123	39780	46612	1D15h37m34s	IPv4u	1191146		1
Notes	If a VRF is not specified, data is shown for the default VRF.									
	AFI l2vpn is only available in VRF default, and SAFI vpls is only supported in AFI l2vpn, and is the only option there.									
	SAFI labeled-unicast and vpn are supported only in VRF default.									

10.2.3.4. show bgp neighbors advertised/received routes

Description	Displays BGP routes advertised to / received from the peer.		
Command Syntax	<pre>show bgp neighbors {advertised-routes received-routes} [vrf <i>vrf-name</i>] [<i>afi safi</i>] [brief detail] [<i>neighbor-ip</i> [<i>ip-prefix</i>]] show bgp neighbors {advertised-routes received-routes} [vrf default] <i>afi</i> vpn rd <i>route-distinguisher</i> [brief detail] [<i>neighbor-ip</i> [<i>ip-prefix</i>]] show bgp neighbors {advertised-routes received-routes} [vrf <i>vrf-name</i>] ipv4 flowspec [brief detail] <i>neighbor-ip</i> [[dst-prefix <i>ip-prefix</i>] [src-prefix <i>ip-prefix</i>] [<i>nlri-abbreviation</i>]] show bgp neighbors {advertised-routes received-routes} [vrf default] l2vpn vpls [rd <i>route-distinguisher</i>] [brief detail] [<i>neighbor-ip</i> [<i>vpls-prefix</i>]]</pre>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Name, *	Name of VRF instance for BGP in which peer resides. (Default = default)
	<i>afi</i>	ipv4 ipv6 l2vpn	IP address family (AFI) name. (Default = all)

	<i>safi</i>	labeled-unicast unicast vpn flowspec vpls	SAFI name. (Default = all)
--	-------------	---	-------------------------------

	<i>neighbor-ip</i>	IPv4 or IPv6 address	IP address of specific neighbor. (Default = all)																																			
	<i>vpls-prefix</i>	<2-byte VE>:<2-byte offset>/32	VPLS prefix.																																			
	<i>ip-prefix</i>	IPv4 or IPv6 address / length	Neighbor-IP Prefix or dst/src-prefix from the flowspec rule.																																			
	<i>nlri-abbreviation</i>	String, optionally in quotation marks, if the character “ ” is part of the string	Start of flowspec rule string (will return all nlri matching the abbreviation).																																			
	<i>route-distinguisher</i>	<2-byte ASN>:<4-byte number>	2-byte ASN notation.																																			
		<4-byte ASN>L:<2-byte number>	4-byte ASN notation.																																			
		<IPv4 address>:<2-byte number>	IPv4 address notation.																																			
Examples	The following examples show sample output from this command:																																					
<pre>exaware# show bgp neighbors received-routes vrf * VRF default: Origin: i - IGP, e - EGP, ? - incomplete IPv4 VPN Route Distinguisher 100:200 (VRF RED):</pre> <table><thead><tr><th>NEIGHBOR</th><th>NETWORK</th><th>NEXT HOP</th><th>METRIC</th><th>LOC PRF</th><th>AS PATH</th><th>ORIGIN</th></tr></thead><tbody><tr><td>3.3.3.2</td><td>100.4.1.0/24</td><td>4.4.4.1</td><td>100</td><td>100</td><td>64401</td><td>i</td></tr><tr><td></td><td>100.5.1.0/24</td><td>4.4.4.2</td><td>155</td><td>55</td><td>64401</td><td>i</td></tr><tr><td>3.3.3.3</td><td>100.4.1.0/24</td><td>4.4.4.1</td><td>100</td><td>100</td><td>64401</td><td>i</td></tr><tr><td></td><td>100.5.1.0/24</td><td>4.4.4.2</td><td>155</td><td>55</td><td>64401</td><td>i</td></tr></tbody></table>				NEIGHBOR	NETWORK	NEXT HOP	METRIC	LOC PRF	AS PATH	ORIGIN	3.3.3.2	100.4.1.0/24	4.4.4.1	100	100	64401	i		100.5.1.0/24	4.4.4.2	155	55	64401	i	3.3.3.3	100.4.1.0/24	4.4.4.1	100	100	64401	i		100.5.1.0/24	4.4.4.2	155	55	64401	i
NEIGHBOR	NETWORK	NEXT HOP	METRIC	LOC PRF	AS PATH	ORIGIN																																
3.3.3.2	100.4.1.0/24	4.4.4.1	100	100	64401	i																																
	100.5.1.0/24	4.4.4.2	155	55	64401	i																																
3.3.3.3	100.4.1.0/24	4.4.4.1	100	100	64401	i																																
	100.5.1.0/24	4.4.4.2	155	55	64401	i																																

```

Origin: i - IGP, e - EGP, ? - incomplete
IPv4 VPN Route Distinguisher 100:300 :

```

NEIGHBOR	NETWORK	NEXT HOP	METRIC	LOC PRF	AS PATH	ORIGIN
3.3.3.2	150.2.1.0/24	5.5.5.1	-	100	64402	i
	150.3.1.0/24	5.5.5.2	-	100	64402	i
3.3.3.3	150.2.1.0/24	5.5.5.1	-	100	64402	i
	150.3.1.0/24	5.5.5.2	-	100	64402	i

```

VRF BLUE:
Origin: i - IGP, e - EGP, ? - incomplete
IPv4 Unicast:

```

NEIGHBOR	NETWORK	NEXT HOP	METRIC	LOC PRF	AS PATH	ORIGIN
7.7.7.2	100.4.1.0/24	7.7.7.2	100	100	64403	i
	100.5.1.0/24	7.7.7.2	155	55	64403	i
7.7.7.3	100.4.1.0/24	7.7.7.3	100	100	64403	i
	100.5.1.0/24	7.7.7.3	155	55	64403	i

```

exaware# show bgp neighbors advertised-routes ipv4 vpn detail 3.3.3.2
Neighbor 3.3.3.2
Route Distinguisher 100:100 (VRF BLUE)
  BGP advertised table for 100.4.1.0/24
    As path: 64401 Next-hop: 4.4.4.1
    Origin: i, metric: 100, localpref: 100, weight: 0
    Extended Community: RT:64401:100 SOO:64401:1
    Label: 16033
  BGP advertised table for 100.5.1.0/24
    As path: 64401 Next-hop: 4.4.4.2
    Origin: i, metric: 155, localpref: 55, weight: 0
    Extended Community: RT:64401:200 SOO:64401:2
    Label: 16034
Route Distinguisher 100:200 (VRF RED)
  BGP advertised table for 150.2.1.0/24
    As path: 64402 Next-hop: 5.5.5.1
    Origin: i, localpref: 100, weight: 0
    Extended Community: RT:64402:100 SOO:64402:1
    Label: 29055
  BGP advertised table for 150.3.1.0/24
    As path: 64402 Next-hop: 5.5.5.2
    Origin: i, localpref: 100, weight: 0
    Extended Community: RT:64402:200 SOO:64402:2
    Label: 29066

exaware# show bgp neighbors advertised-routes ipv4 vpn rd 100:200
Origin: i - IGP, e - EGP, ? - incomplete
IPv4 VPN Route Distinguisher 100:200 (VRF RED):

```

NEIGHBOR	NETWORK	NEXT HOP	METRIC	LOC PRF	AS PATH	ORIGIN
3.3.3.2	100.4.1.0/24	4.4.4.1	100	100	64401	i
	100.5.1.0/24	4.4.4.2	155	55	64401	i

```

3.3.3.3    100.4.1.0/24  4.4.4.1  100      100  64401    i
           100.5.1.0/24  4.4.4.2  155      55   64401    i

exaware# show bgp neighbors received-routes ipv4 flowspec
VRF default:
Origin: i - IGP, e - EGP, ? - incomplete
IPv4 Flowspec:

NEIGHBOR  NETWORK      NEXT      LOC
HOP      METRIC  PRF  AS  PATH  ORIGIN
-----
3.3.3.2   4.0.0.0/8,*  0.0.0.0  100   100  64401    i
           5.0.0.0/8,*  0.0.0.0  155   55   64401    i
3.3.3.3   4.0.0.0/8,*  0.0.0.0  100   100  64401    i
           5.0.0.0/8,*  0.0.0.0  155   55   64401    i

exaware# show bgp neighbors received-routes ipv4 flowspec
VRF default:
Origin: i - IGP, e - EGP, ? - incomplete
IPv4 Flowspec:

NEIGHBOR NETWORK      NEXT      LOC  AS
HOP      METRIC  PRF  PATH  ORIGIN
-----
3.3.3.2   4.0.0.0/8,7.0.0.0/8,dst-port=1000,src-...  0.0.0.0  100   100  64401    i
           5.0.0.0/8,*  0.0.0.0  155   55   64401    i
3.3.3.3   4.0.0.0/8,*  0.0.0.0  100   100  64401    i
           5.0.0.0/8,*  0.0.0.0  155   55   64401    i

exaware# show bgp neighbors advertised-routes ipv4 flowspec detail 3.3.3.2
Neighbor 3.3.3.2
BGP advertised table for 4.0.0.0/8,7.0.0.0/8,dst-port=1000,src-port=2000
  As path: 64401
  Next-hop: 0.0.0.0
  Origin: i, weight: 0
  Extended community: Flowspec-Rate:0:777777
BGP advertised table for 4.0.0.0/8,*
  As path: 64401
  Next-hop: 0.0.0.0
  Origin: i, weight: 0
  Extended community: Flowspec-Rate:0:777777

BGP advertised table for 5.0.0.0/8,*
  As path: 64401
  Next-hop: 0.0.0.0
  Origin: i, weight: 0
  Extended community: Flowspec-Rate:0:discard

exaware# show bgp neighbors received-routes vrf default l2vpn vpls
VRF default:

Origin: i - IGP, e - EGP, ? - incomplete
L2VPN VPLS Route Distinguisher 100:200 :

NEIGHBOR  NETWORK      NEXT      LOC
HOP      METRIC  PRF  AS  PATH  ORIGIN
-----

```



```

-----
3.3.3.2    3:1/32      4.4.4.1  100      100  64401    i
           3:101/32   4.4.4.1  100      100  64401    i

Origin: i - IGP, e - EGP, ? - incomplete
L2VPN VPLS Route Distinguisher 100:300 :
      NEXT      LOC
NEIGHBOR  NETWORK  HOP      METRIC  PRF  AS  PATH  ORIGIN
-----
3.3.3.2    57:1/32      4.4.4.1  100      100  64401    i
           57:101/32  4.4.4.1  100      100  64401    i

exaware# show bgp neighbors received-routes l2vpn vpls rd 100:200 detail 3.3.3.2
Neighbor 3.3.3.2
Route Distinguisher 100:200
  BGP received table for 3:1/32
    Next-hop: 4.4.4.1
    Origin: i, metric: 100, localpref: 100, weight: 0
    Extended Community: RT:64401:200, L2-info:VPLS:0x80:1500
    Label-base: 16033, range: 10
  BGP received table for 3:101/32
    Next-hop: 4.4.4.1
    Origin: i, metric: 100, localpref: 100, weight: 0
    Extended Community: RT:64401:200, L2-info:VPLS:0x0:1500
    Label-base: 16043, range: 10
Route Distinguisher 100:300
  BGP received table for 57:1/32
    Next-hop: 4.4.4.1
    Origin: i, metric: 100, localpref: 100, weight: 0
    Extended Community: RT:64402:30, L2-info:VPLS:0x0:1500
    Label-base: 16053, range: 10
  BGP received table for 57:101/32
    Next-hop: 4.4.4.1
    Origin: i, metric: 100, localpref: 100, weight: 0
    Extended Community: RT:64401:300, L2-info:VPLS:0x0:1500
    Label-base: 16063, range: 10

exaware# show bgp neighbors advertised-routes l2vpn vpls rd 100:400
Origin: i - IGP, e - EGP, ? - incomplete
L2VPN VPLS Route Distinguisher 100:400 :
      NEXT      LOC
NEIGHBOR  NETWORK  HOP      METRIC  PRF  AS  PATH  ORIGIN
-----
3.3.3.2    150:1/32    3.3.3.1  100      100  64401    i
           150:101/32  3.3.3.1  100      100  64401    i

exaware# show bgp neighbors advertised-routes l2vpn vpls rd 100:200 detail 3.3.3.2
Neighbor 3.3.3.2
Route Distinguisher 100:400
  BGP advertised table for 111:1/32
    As path: 64401
    Next-hop: 3.3.3.1
    Origin: i, metric: 100, localpref: 100, weight: 0
    Extended Community: RT:64401:200, L2-info:VPLS:0x80:1500
    Label-base: 8017, range: 16
  BGP advertised table for 111:101/32
    As path: 64401

```

<pre> Next-hop: 3.3.3.1 Origin: i, metric: 100, localpref: 100, weight: 0 Extended Community: RT:64401:200, L2-info:VPLS:0x0:1500 Label-base: 8033, range: 16 Route Distinguisher 100:500 BGP advertised table for 150:1/32 As path: 64401 Next-hop: 3.3.3.1 Origin: i, metric: 100, localpref: 100, weight: 0 Extended Community: RT:64401:300, L2-info:VPLS:0x0:1500 Label-base: 8001, range: 16 BGP advertised table for 150:101/32 As path: 64401 Next-hop: 3.3.3.1 Origin: i, metric: 100, localpref: 100, weight: 0 Extended Community: RT:64401:300, L2-info:VPLS:0x0:1500 Label-base: 8049, range: 16 </pre>	
Notes	If VRF is not specified, data is shown for the default VRF. Default view for this command is brief. When specifying neighbor by itself, the default view is brief. When specifying neighbor and prefix, the default view is detailed. Filtering on RD can only be done under VRF default and SAFI vpn or vpls. AFI I2vpn is only available in VRF default, and SAFI vpls is only supported in AFI I2vpn, and is the only option there. SAFI labeled-unicast and vpn are supported only in VRF default. Filtering on flowspec components (dst/src-prefix or nlri abbreviation) can only be done under SAFI flowspec.

10.2.3.5. show bgp neighbors advertised/received routes label view

Description	Displays label information of BGP routes advertised to / received from the peer.		
Command Syntax	<pre>show bgp neighbors advertised-routes [vrf default] {ipv4 labeled-unicast <i>afi</i> vpn [rd <i>route-distinguisher</i>]} labels [local <i>label-id</i> <i>neighbor-ip</i>] show bgp neighbors received-routes [vrf default] {ipv4 labeled-unicast <i>afi</i> vpn [rd <i>route-distinguisher</i>]} labels [received <i>label-id</i> <i>neighbor-ip</i>]</pre>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>afi</i>	ipv4 ipv6	IP address family (AFI) name. (Default = all)
	<i>route-distinguisher</i>	<2-byte ASN>:<4-byte number>	2-byte ASN notation.
		<4-byte ASN>L:<2-byte number>	4-byte ASN notation.
		<IPv4 address>:<2-byte number>	IPv4 address notation.
	<i>label-id</i>	A	Label number. (Default = all)

	<i>neighbor-ip</i>	IPv4 or IPv6 address	IP address of specific neighbor. (Default = all)
Examples	The following examples show sample output from this command:		
<pre>exaware# show bgp neighbors advertised-routes ipv4 vpn labels IPv4 VPN Route Distinguisher 100:100 (VRF BLUE): NEXT NEIGHBOR NETWORK HOP LABEL ----- 3.3.3.2 100.4.1.0/24 4.4.4.1 16033 100.5.1.0/24 4.4.4.2 16033 3.3.3.3 100.4.1.0/24 4.4.4.1 16033 100.5.1.0/24 4.4.4.2 16033 IPv4 VPN Route Distinguisher 100:200 (VRF RED): NEXT NEIGHBOR NETWORK HOP LABEL ----- 3.3.3.2 150.2.1.0/24 5.5.5.1 29055 150.3.1.0/24 5.5.5.2 29055 3.3.3.3 150.2.1.0/24 5.5.5.1 29055 150.3.1.0/24 5.5.5.2 29055 exaware# show bgp neighbors advertised-routes ipv4 vpn labels local 16033 IPv4 VPN Route Distinguisher 100:100 (VRF BLUE): NEXT NEIGHBOR NETWORK HOP LABEL ----- 3.3.3.2 100.4.1.0/24 4.4.4.1 16033 100.5.1.0/24 4.4.4.2 16033 3.3.3.3 100.4.1.0/24 4.4.4.1 16033 100.5.1.0/24 4.4.4.2 16033 exaware# show bgp neighbors advertised-routes ipv4 labeled-unicast labels 3.3.3.2 NEXT NEIGHBOR NETWORK HOP LABEL ----- 3.3.3.2 100.4.1.0/24 4.4.4.1 55044 100.5.1.0/24 4.4.4.2 55044</pre>			

Notes	If VRF is not specified, data is shown for the default VRF. This command has no detailed view. Filtering on RD can only be done under VRF default and SAFI vpn. SAFI labeled-unicast and vpn are supported only in VRF default.
--------------	---

10.2.3.6. show bgp neighbors brief/detail

Description	Displays BGP neighbor operational data.							
Command Syntax	show bgp neighbors [vrf <i>vrf-name</i>] [<i>afi safi</i>] [brief detail] [<i>neighbor-ip</i>]							
Command Mode	Operational							
Parameters Table	Name	Value				Description		
	<i>vrf-name</i>	Name, *				Name of VRF instance for BGP. (Default = default)		
	<i>afi</i>	ipv4 ipv6 l2vpn				IP address family (AFI) name. (Default = all)		
	<i>safi</i>	labeled-unicast unicast vpn flowspec vpls				SAFI name. (Default = all)		
	<i>neighbor-ip</i>	IPv4/IPv6 address				IPv4 or IPv6 IP address of specific neighbor. (Default = all)		
Examples	The following examples show sample outputs from this command:							
exaware# show bgp neighbors VRF default :								
NEIGHBOR	STATE	PEER AS	MSG RCVD	MSG SENT	UP/DOWN TIME	AFI/SAFI	TABLE VERSION	STATE/ PFX-RCVD
1.1.1.1	down	111	0	0	N/A	L2VPNvpls	0	Connect

```

20.200.0.20    up        65001  3380  3585  1D1h27m40s  IPv4u        30          19
                  L2VPNvpls  3          2

exaware# show bgp neighbors vrf BLUE brief 3.3.3.3

NEIGHBOR  STATE  PEER  MSG  MSG  UP/DOWN  TABLE  STATE/
AS        RCVD  SENT  TIME  AFI/SAFI  VERSION  PFX-RCVD
-----
3.3.3.3   up     64400 19724 23086 4D22h49m9s  IPv4u    1191077  1

exaware# show bgp neighbors detail 20.200.0.20
VRF default :
BGP peer: 20.200.0.20
Admin status: up, BGP state: Established (up for: 1d2h26m14s), link type: external
BFD: disabled
Connection parameters
  Local Router ID:      192.168.0.82      Remote Router ID:      192.168.0.20
  Local host:           20.200.0.82      Remote host:           20.200.0.20
  Local TCP port:       179              Remote TCP port:       63819
  Local AS:             1                Remote AS:             65001
  Min interval:         30
  Keepalive 30sec (configured 60sec)      Hold time 90sec (configured 180sec)
Statistics
  Connections established: 1              Connections dropped: 0
  Rx messages:            3511            Tx messages:          3724
  Rx notifications:       0              Tx notifications:     0
  Rx updates:             16              Tx updates:           5
  Rx route refresh req:   0              Tx route refresh req: 0
  IPv4 Unicast table:
    Table version:        31              Neighbor version:     30
    Rx prefixes:          19              Tx prefixes:          9
  L2VPN VPLS table:
    Table version:        5              Neighbor version:     5
    Rx prefixes:          2              Tx prefixes:          0
Neighbor parameters
  Peer multihop:          disabled
  IPv4 Unicast parameters
    Route reflector client: disabled
    Inbound soft reconfigure: disabled
    Next hop self:        disabled
    Send community:       none
    Weight:                0
    Maximum prefixes:     524288
    Preserve next hop:    disabled
    Allow AS in:          disabled
    Multipath:            disabled
  L2VPN VPLS parameters
    Route reflector client: disabled
    Inbound soft reconfigure: enabled
    Send community:       extended
    Weight:                0
    Maximum prefixes:     1000
Capabilites
  Multiprotocol (1)
    IPv4 Unicast:          advertised and received
    IPv4 Labeled-Unicast:  none

```


	<pre> IPv4 VPN: none IPv4 Flowspec: none IPv6 Unicast: none IPv6 Labeled-Unicast: none IPv6 VPN: none L2VPN VPLS: advertised and received Route refresh (2): advertised and received (old and new) Outbound Route Filter (3) IPv4 Unicast Send mode: advertised Receive mode: advertised Rx prefix filter: none Graceful Restart (64): advertised and received Maximum stale time: 360sec Restart time Advertised: 120sec Received: 120sec IPv4 Unicast: advertised and received L2VPN VPLS: advertised and received Extended asn capability (65): advertised and received Dynamic capabilities (67): none </pre>
Notes	If a VRF is not specified, data is shown for the default VRF. AFI I2vpn is only available in VRF default, and SAFI vpls is only supported in AFI I2vpn, and is the only option there. SAFI labeled-unicast and vpn are supported only in VRF default. Default view is brief. When specifying a neighbor-ip, default view is detailed.

10.2.3.7. show bgp neighbors received-prefix-filter

Description	Displays received ORF prefix-list information.		
Command Syntax	show bgp neighbors received-prefix-filter [vrf <i>vrf-name</i>] [<i>afi</i> unicast] [<i>neighbor-ip</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Name, *	Name of VRF instance for BGP. (Default = default)
	<i>afi</i>	ipv4 ipv6	IP address family (AFI) name.
	<i>neighbor-ip</i>	IPv4 or IPv6 address	IPv4 or IPv6 IP address of specific neighbor. (Default = all)
Examples	The following example shows sample output from this command: exaware# show bgp neighbors received-prefix-filter vrf * VRF default: IPv4 Unicast: NEIGHBOR SEQ ACTION PREFIX GE LE ----- 10.135.140.2 11.54.255.1 20.101.100.1 100.100.100.100 40 permit 0.0.0.0/0 31 0 45 permit 0.0.0.0/0 32 0 IPv6 Unicast:		
Notes	If a VRF is not specified, data is shown for the default VRF. SAFI filter can only be unicast.		

10.2.3.8. show bgp policy

Description	Displays information regarding BGP policies.																		
Command Syntax	show bgp policy {accounting aggregate-address default-route-originate flap-damping neighbors network redistribute import export import-from-vrf} [vrf <i>vrf-name</i>] [<i>afi safi</i>]																		
Command Mode	Operational																		
Parameters Table	Name	Value	Description																
	<i>vrf-name</i>	Name, *	Name of VRF instance for BGP. (Default = default)																
	<i>afi</i>	ipv4 ipv6 l2vpn	IP address family (AFI) name. (Default = all)																
	<i>safi</i>	labeled-unicast unicast vpn flowspec vpls	SAFI name. (Default = all)																
Examples	The following example shows sample output from this command:																		
exaware# show bgp policy redistribute vrf * VRF default: IPv4 Unicast:																			
<table><thead><tr><th>PROTOCOL</th><th>POLICY</th><th>RULE</th><th>HIT COUNT</th></tr></thead><tbody><tr><td>connected</td><td>UNICAST-REDIST-POLICY</td><td>CONNECTED</td><td>3</td></tr><tr><td></td><td></td><td>STATIC</td><td>0</td></tr><tr><td></td><td></td><td>DEFAULT</td><td>15</td></tr></tbody></table>				PROTOCOL	POLICY	RULE	HIT COUNT	connected	UNICAST-REDIST-POLICY	CONNECTED	3			STATIC	0			DEFAULT	15
PROTOCOL	POLICY	RULE	HIT COUNT																
connected	UNICAST-REDIST-POLICY	CONNECTED	3																
		STATIC	0																
		DEFAULT	15																

static	UNICAST-REDIST-POLICY	CONNECTED	0
		STATIC	5
		DEFAULT	1
PROTOCOL	POLICY	RULE	HIT COUNT
-----	-----	-----	-----
static	BLOCK-MARTIANS	MARTIANS	2
		DEFAULT	34
IPv4 Labeled-Unicast:			
IPv4 VPN:			
PROTOCOL	POLICY	RULE	HIT COUNT
-----	-----	-----	-----
connected	VPN-REDIST-POLICY	CONNECTED	19
		STATIC	0
		DEFAULT	0
static	VPN-REDIST-POLICY	CONNECTED	0
		STATIC	2
		DEFAULT	3
IPv6 Unicast:			
PROTOCOL	POLICY	RULE	HIT COUNT
-----	-----	-----	-----
connected	IPV6-POLICY	CONNECTED	0
		STATIC	0
		DEFAULT	23
static	IPV6-POLICY	CONNECTED	0
		STATIC	0
		DEFAULT	50
VRF BLUE:			
IPv4 Unicast:			
PROTOCOL	POLICY	RULE	HIT COUNT
-----	-----	-----	-----
connected	BLUE-REDIST-POLICY	CONNECTED	2
		STATIC	0
		DEFAULT	0
static	BLUE-REDIST-POLICY	CONNECTED	0
		STATIC	5
		DEFAULT	1
PROTOCOL	POLICY	RULE	HIT COUNT
-----	-----	-----	-----
static	BLOCK-MARTIANS	MARTIANS	30
		DEFAULT	24

IPv6 Unicast:

PROTOCOL	POLICY	RULE	HIT COUNT

connected	BLUE-IPV6-POLICY	CONNECTED	0
		STATIC	0
		DEFAULT	23
static	BLUE-IPV6-POLICY	CONNECTED	0
		STATIC	0
		DEFAULT	50

exaware# show bgp policy neighbors vrf default l2vpn vpls

VRF default:

L2VPN VPLS:

NEIGHBOR	DIRECTION	POLICY	RULE	HIT COUNT

3.3.3.2	out	override-peer-MEDs	ATT-HONOR-MEDS	0
			PEER-ROUTES	0
			DEFAULT	2144898
		final-filter	FIX_MED	0
			DONE	0
			ACCEPT	70439

exaware# show bgp policy import vrf vrf1

VRF vrf1:

IPv4 Unicast:

POLICY	RULE	HIT COUNT

p1	r1	1
p2	r2	4

VRF vrf1:

IPv6 Unicast:

POLICY	RULE	HIT COUNT

p1	r1	2

exaware# show bgp policy import vrf vrf1 ipv4 unicast

VRF vrf1:

IPv4 Unicast:

POLICY	RULE	HIT COUNT

p1	r1	1

```
p2          r2          4

exaware# show bgp policy import vrf vrfl ipv6 unicast
VRF vrfl:
IPv6 Unicast:

POLICY      RULE      HIT
COUNT
-----
p1          r1          2

exaware# show bgp policy export vrf vrfl
VRF vrfl:
IPv4 Unicast:

POLICY      RULE      HIT
COUNT
-----
p1          r1          1
p2          r2          4

VRF vrfl:
IPv6 Unicast:

POLICY      RULE      HIT
COUNT
-----
p1          r1          2

exaware# show bgp policy export vrf vrfl ipv4 unicast
VRF vrfl:
IPv4 Unicast:

POLICY      RULE      HIT
COUNT
-----
p1          r1          1
p2          r2          4

exaware# show bgp policy export vrf vrfl ipv6 unicast
VRF vrfl:
IPv6 Unicast:

POLICY      RULE      HIT
COUNT
-----
p1          r1          2

exaware# show bgp policy import-from-vrf vrf vrfl ipv4 unicast
VRF vrfl:
IPv4 Unicast:
```

SOURCE VRF	POLICY	RULE	HIT COUNT
vpn1	p1	r1	1
vpn2	p2	r2	4

Notes

If a VRF is not specified, data is shown for the default VRF.

AFI I2vpn is only available in VRF default, and SAFI vpls is only supported in AFI I2vpn, and is the only option there.

SAFI labeled-unicast and vpn are supported only in VRF default.

Only the neighbor policy exists for the flowspec and vpls SAFI.

Only IPv4/IPv6 unicast are available for the import and export commands.

10.2.3.9. show bgp policy allocate-label

Description	Displays information regarding BGP allocate-label policy.		
Command Syntax	show bgp policy allocate-label [vrf <i>vrf-name</i>] [<i>afi safi</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	default	Name of VRF instance for BGP. (Default = default)
	<i>afi</i>	ipv6	AFI name. (Default = all)
	<i>safi</i>	unicast	SAFI name. (Default = all)
Examples	The following example shows sample output from this command:		
<pre>exaware# show bgp policy allocate-label vrf default ipv6 unicast VRF default: POLICY RULE COUNT ----- permit all policy a 0</pre>			

10.2.3.10. show bgp rpki status

Description	To display the Public Key Infrastructure (RPKI) status, use the show bgp rpki configuration command.
Command Syntax	<code>show bgp rpki status</code>
Command Mode	operational
Examples	This example shows how to use this command: <pre>exaware# show bgp rpki status BGP 61 VRF: default Origin-as not-found action: permit Origin-as invalid action: deny Purge time: 700s Refresh time: 120s</pre>

10.2.3.11. show bgp rpki servers

Description	To display resource public key infrastructure (RPKI) servers, use the show bgp rpki servers command.		
Command Syntax	<code>show bgp rpki servers [<i>ip-address</i>]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 or IPv6 address	Specifies the rpki-server IP address.

Examples

This example shows how to use this command:

```
exaware# show bgp rpki servers
```

```
BGP 61
```

```
  VRF: default
```

```
    Server 10.1.1.5
```

```
      Description: ServerA
```

```
      Preference: 6
```

```
      Transport: tcp
```

```
        Port: 2497
```

```
        Administrative State: Up
```

```
        Operational State: Active
```

```
    Server 10.1.10.3
```

```
      Description: ServerB
```

```
      Preference: 4
```

```
      Transport: tcp
```

```
        Port: 2497
```

```
        Administrative State: Up
```

```
        Operational State: Backup
```

```
    Server 10.1.11.6
```

```
      Description:
```

```
      Preference: 5
```

```
      Transport: ssh
```

```
        Username: Bob
```

```
        Password: *****
```

```
        Port: 2497
```

```
        Administrative State: Down
```

```
        Operational State: Down
```

10.2.3.12. show bgp rpki table

Description	To display resource public key infrastructure (RPKI) cache, use the show bgp rpki table command.																										
Command Syntax	show bgp rpki table [<i>prefix-addr</i>] [[af-ipv4 af-ipv6] [origin-as <i>as-number</i>]]																										
Command Mode	operational																										
Parameters Table	Name	Value	Description																								
	<i>prefix-addr</i>	IPv4 or IPv6 prefix address	Specifies the IPv4 or IPv6 prefix address.																								
	<i>as-number</i>	1-4294967295	Specifies the AS number.																								
Examples	This example shows how to use this command: <pre>exaware# show bgp rpki table</pre> <table> <thead> <tr> <th>Prefix Address</th><th>MaxLen</th><th>MinLen</th><th>Origin-AS</th></tr> <tr> <th>-----</th><th>-----</th><th>-----</th><th>-----</th></tr> </thead> <tbody> <tr> <td>1.1.1.0</td><td>32</td><td>16</td><td>9991</td></tr> <tr> <td>67.64.0.0</td><td>32</td><td>24</td><td>4002</td></tr> <tr> <td>67.128.0.0</td><td>32</td><td>24</td><td>4003</td></tr> <tr> <td>2002:1::10</td><td>72</td><td>64</td><td>4008</td></tr> </tbody> </table>			Prefix Address	MaxLen	MinLen	Origin-AS	-----	-----	-----	-----	1.1.1.0	32	16	9991	67.64.0.0	32	24	4002	67.128.0.0	32	24	4003	2002:1::10	72	64	4008
Prefix Address	MaxLen	MinLen	Origin-AS																								
-----	-----	-----	-----																								
1.1.1.0	32	16	9991																								
67.64.0.0	32	24	4002																								
67.128.0.0	32	24	4003																								
2002:1::10	72	64	4008																								

10.2.3.13. show bgp summary

Description	Displays summary of BGP operational data.		
Command Syntax	show bgp summary [general [<i>specific-info</i>] vrf <i>vrf-name</i>]		
Command Mode	operational		
	Name	Value	Description

Parameters Table	<i>specific-info</i>	as-path-entries as-path-compare attribute-entries bgp-holdtime-timer bgp-keepalive-timer client-to-client-reflection-enabled community-entries default-local-preference deferral-time extended-as-capability fast-external-turnover graceful-reset local-as-number log-neighbor-changes-enabled med-compare-always med-compare-confed med-deterministic med-missing-worst nexthop-tracking-delay nexthop-tracking-enabled nexthop-tracking-threshold restart-time router-id-compare scan-time stalepath-time	Specific parameter. (Default = all)
------------------	----------------------	--	--

	<i>vrf-name</i>	Name, *	Name of VRF instance for BGP. (Default = default)
--	-----------------	---------	--

Examples	The following example shows sample output from this command: <pre> exaware# show bgp summary Local AS number 123 Attribute entries 26 AS-PATH entries 4 Community entries 2 Timers: Scan time 60sec Keepalive 60sec Holdtime 180sec Graceful restart parameters: Graceful reset enabled Restart time 120sec Stalepath time 360sec Deferral time 120sec Path Selection: Compare AS-PATH enabled Compare router ID enabled Always compare MED enabled Missing MED best Compare MED among confeds enabled Deterministic MED enabled Log neighbor changes enabled Client to client reflection enabled Fast external turnover enabled Extended asn capability enabled Default local preference 100 VRF default Router identifier: 11.11.11.11 Table version IPv4 Unicast: 22 IPv4 Labeled-Unicast: 33 IPv4 VPN: 44 IPv4 Flowspec: 55 IPv6 Unicast: 0 IPv6 Labeled-Unicast: 8 IPv6 VPN: 4 L2VPN VPLS: 88 Number of peers: 6 IPv4 Unicast Distances: Local 200 Internal 200 External 20 IPv4 Labeled-Unicast Distances: Local 200 Internal 200 External 20 IPv6 Unicast Distances: Local 200 Internal 200 External 20 VRF BLUE Router identifier: 7.7.7.7 Table version: 36 Number of peers: 6 IPv4 Unicast Distances: Local 200 Internal 200 </pre>
-----------------	--

	External 20 IPv4 Labeled-Unicast Distances: Local 200 Internal 200 External 20 IPv6 Unicast Distances: Local 200 Internal 200 External 20
Notes	If no option is used, displays general and all VRF's sections. Filtering on general displays only the global (non-VRF-specific) parameters. Filtering on a VRF displays only that VRF's portion. The command "show bgp" is equivalent to "show bgp summary".

10.2.3.14. show bgp table brief/detail

Description	Displays the BGP routing table.		
Command Syntax	<pre>show bgp table [vrf <i>vrf-name</i>] [<i>afi safi</i>] [community <i>community-value</i>] [ext-community <i>ext-community-value</i>] [brief detail inconsistent-as] [<i>ip-prefix</i>] show bgp table [vrf default] <i>afi</i> vpn rd <i>route-distinguisher</i> [brief detail inconsistent-as] [<i>ip-prefix</i>] show bgp table [vrf <i>vrf-name</i>] ipv4 flowspec [brief detail inconsistent-as] [[<i>dst-prefix ip-prefix</i>] [<i>src-prefix ip-prefix</i>] [<i>nlri-abbreviation</i>]] show bgp table [vrf default] l2vpn vpls [rd <i>route-distinguisher</i>] [brief detail inconsistent-as] [<i>vpls-prefix</i>]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Name, *	Name of VRF instance for BGP. (Default = default)
	<i>afi</i>	ipv4 ipv6 l2vpn	IP address family (AFI) name. (Default = all)
	<i>safi</i>	labeled-unicast unicast vpn flowspec vpls	SAFI name. (Default = all)

	<i>as-number</i>	1-4294967295	Autonomous system number.
--	------------------	--------------	---------------------------

	<i>community-value</i>	- numerical value in the format: <number>:<number> - well-known value : no-export, no-advertise, no-export-subconfed, internet - wildcard for any of the decimal values : * - range for any of the decimal values: [low_val-high_val] - regex : regular expression enclosed in single quotes (POSIX 1003.2)	Community value
--	------------------------	---	-----------------

	<i>ext-community-value</i>	<2-octet-ASN>:<4-octet number>	2-octet-ASN notation
		<4-octet-ASN>L:<2-octet number>	4-octet-ASN notation
		<IPv4 address> <2-octet number>	IPv4 address notation
	<i>nlri-abbreviation</i>	<4-byte ASN>L:<2-byte number>	4-byte ASN notation.
	<i>vpls-prefix</i>	<IPv4 address>:<2-byte number>	IPv4 address notation.
	<i>route-distinguisher</i>	<2-byte ASN>:<4-byte number>	2-byte ASN notation.
		<4-byte ASN>L:<2-byte number>	4-byte ASN notation.
		<IPv4 address>:<2-byte number>	IPv4 address notation.
Examples	The following examples show sample output from this command:		
<pre>exaware# show bgp table vrf test1 detail VRF test1 : IPv4 Unicast BGP routing table for 77.77.77.0/24 2 paths available Advertised to auto-groups: 0 65004 60001 172.16.11.2 (metric: 0) from 172.16.11.2 (172.16.11.2) Origin: IGP, metric: 0, localpref: 100, weight: 0, valid, external, best Last update: Sun Dec 28 19:30:27 2014 65002 65004 100000 65004 65004 101000 102000 65004 65004 65004 172.16.10.2 (metric: 0) from 172.16.10.2 (172.16.10.2)</pre>			


```

Origin: IGP, metric: 0, localpref: 100, weight: 0, valid, external, alternate
Extended community: SOO:8:8
Last update: Sun Dec 28 19:30:16 2014

exaware# show bgp table vrf test1

VRF test1:
Flags:    s - suppressed, d - damped, h - history, * - valid, > - best
          i - internal, S - stale, = - multipath, a - alternate
Origin:   i - IGP, e - EGP, ? - incomplete

IPv4 Unicast:

NETWORK          PEER IP      FLAGS NEXT HOP      METRIC  LOC
PRF  WEIGHT AS PATH  ORIGIN
-----
-
77.77.77.0/24    172.16.11.2  *>    172.16.11.2  0        -    0        65004 60001 i
              172.16.10.2 *a     172.16.10.2  0        -    0        65002 65004
100000 65004 65004 101000 102000 65004 65004 65004 i

exaware# show bgp table ipv4 flowpsec
VRF default:
Flags:    s - suppressed, d - damped, h - history, * - valid, > - best
          i - internal, S - stale, = - multipath
Origin:   i - IGP, e - EGP, ? - incomplete

NETWORK          PEER IP      FLAGS  NEXT HOP      METRIC  LOC
PRF  WEIGHT AS PATH  ORIGIN
-----
-
4.0.0.0/8,*      4.4.4.1  *>i    0.0.0.0  100      100  0        64401 i
4.0.0.0/8,7.0.0.0/8,dst-... 4.4.4.2  *>i    0.0.0.0  100      100  0        64401 i
5.0.0.0/8,*      4.4.4.2  *>i    0.0.0.0  100      100  0        64401 i

exaware# show bgp table ipv4 flowspec detail
BGP routing table for 4.0.0.0/8,*
1 path available
Advertised to auto-groups:
0
64401
0.0.0.0 (metric: 0) from 4.4.4.1 (4.4.4.1)
Origin: IGP, metric: 0, localpref: 100, weight: 0, valid, external, best
Extended community: Flowspec-Rate:0:discard
Last update: Wed Jan 7 16:57:28 2015
BGP routing table for 4.0.0.0/8,7.0.0.0/8,dst-port=1000,src-port=2000
1 path available
Advertised to auto-groups:
0
64401
0.0.0.0 (metric: 0) from 4.4.4.1 (4.4.4.1)
Origin: IGP, metric: 0, localpref: 100, weight: 0, valid, external, best
Extended community: Flowspec-Rate:0:777777
Last update: Wed Jan 7 16:57:28 2015
BGP routing table for 5.0.0.0/8,*

```

```
1 path available
Advertised to auto-groups:
0
64401
0.0.0.0 (metric: 0) from 4.4.4.2 (4.4.4.2)
  Origin: IGP, metric: 0, localpref: 100, weight: 0, valid, external, best
  Extended community: Flowspec-Rate:0:888888
  Last update: Wed Jan  7 16:57:28 2015
```

exaware# show bgp table l2vpn vpls

VRF default:

Flags: s - suppressed, d - damped, h - history, * - valid, > - best

i - internal, S - stale, = - multipath, a - alternate

Origin: i - IGP, e - EGP, ? - incomplete

L2VPN VPLS Route Distinguisher 100:200 :

NETWORK	PEER IP	FLAGS	NEXT HOP	METRIC	LOC		AS		ORIGIN
					PRF	WEIGHT	PATH		
3:1/32	3.3.3.2	*>i	4.4.4.1	100	100	0	64401	i	
3:101/32	3.3.3.2	*>i	4.4.4.1	100	100	0	64401	i	

Flags: s - suppressed, d - damped, h - history, * - valid, > - best

i - internal, S - stale, = - multipath, a - alternate

Origin: i - IGP, e - EGP, ? - incomplete

L2VPN VPLS Route Distinguisher 100:300 :

NETWORK	PEER IP	FLAGS	NEXT HOP	METRIC	LOC		AS		ORIGIN
					PRF	WEIGHT	PATH		
57:1/32	3.3.3.2	*>i	4.4.4.1	100	100	0	64401	i	
57:101/32	3.3.3.2	*>i	4.4.4.1	100	100	0	64401	i	

Flags: s - suppressed, d - damped, h - history, * - valid, > - best

i - internal, S - stale, = - multipath, a - alternate

Origin: i - IGP, e - EGP, ? - incomplete

L2VPN VPLS Route Distinguisher 100:400 :

NETWORK	PEER IP	FLAGS	NEXT HOP	METRIC	LOC		AS		ORIGIN
					PRF	WEIGHT	PATH		
111:1/32	3.3.3.2	*>	3.3.3.1	100	100	0	64401	i	
111:101/32	3.3.3.2	*>	3.3.3.1	100	100	0	64401	i	

Flags: s - suppressed, d - damped, h - history, * - valid, > - best

i - internal, S - stale, = - multipath, a - alternate

Origin: i - IGP, e - EGP, ? - incomplete

L2VPN VPLS Route Distinguisher 100:500 :

NETWORK	PEER IP	FLAGS	NEXT HOP	METRIC	LOC		AS		ORIGIN
					PRF	WEIGHT	PATH		
150:1/32	3.3.3.2	*>	3.3.3.1	100	100	0	64401	i	
150:101/32	3.3.3.2	*>	3.3.3.1	100	100	0	64401	i	

```
Exaware# show bgp table l2vpn vpls rd 100:200 detail
Route Distinguisher 100:200
=====
L2VPN VPLS
BGP routing table for 3:1/32
1 path available
Not advertised to any peer
  64401
    4.4.4.1 (metric: 10) from 3.3.3.2 (3.3.3.2)
      Origin: IGP, metric: 100, localpref: 100, weight: 0, valid, internal, best
      Extended community: RT:64401:200 L2-info:VPLS:0x80:1500
      Label-base: 16033, range: 10
      Last update: Wed Apr 29 10:28:52 2015

L2VPN VPLS
BGP routing table for 3:101/32
1 path available
Not advertised to any peer
  64402
    4.4.4.2 (metric: 10) from 3.3.3.3 (3.3.3.3)
      Origin: IGP, metric: 100, localpref: 100, weight: 0, valid, internal, best
      Extended community: RT:64401:200 L2-info:VPLS:0x0:1500
      Label-base: 16043, range: 10
      Last update: Wed Apr 29 10:28:52 2015
```

Notes	If a VRF is not specified, data is shown for the default VRF. Default view for this command is brief. When specifying a prefix, default view is detailed. Filtering on RD can only be done under VRF default and SAFI vpn or vpls. Filtering on flowspec components (dst/src-prefix or nlri abbreviation) can only be done under SAFI flowspec. Additional field "(Via VRF vrf-name)" will be hidden when the NH is reachable via the same VRF. SAFI labeled-unicast and vpn are supported only in VRF default.
--------------	---

10.2.3.15. show bgp table import brief/detail

Description	Displays the BGP routing table.		
Command Syntax	show bgp table import [vrf <i>vrf-name</i>] [source-vrf <i>source-vrf-name</i>] [<i>afi safi</i>] [brief detail] [<i>ip-prefix</i>]		
Command Mode	Operational`		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Name, *	Name of VRF instance for BGP. (Default = default)
	<i>source-vrf-name</i>	Name, *	Name of VRF instance from which the route is imported. (Default = default)
	<i>afi</i>	ipv4 ipv6	IP address family (AFI) name. (Default = all)
	<i>safi</i>	unicast	SAFI name. (Default = all)
	<i>ip-prefix</i>	IPv4 or IPv6 address / length	IP prefix of learned network, or dst/src-prefix from the flowspec rule.

Examples	The following examples show sample output from this command:																										
<pre>exaware# show bgp table import vrf default detail VRF default : Flags: s - suppressed, d - damped, h - history, * - valid, > - best i - internal, S - stale, = - multipath, a - alternate IPv4 Unicast BGP routing table for 77.77.77.0/24 2 paths available Advertised to auto-groups: 0 65004 60001 172.16.11.2 (metric: 0) from 172.16.11.2 (172.16.11.2) Source VRF: VRF1, metric: 0, localpref: 100, weight: 0, valid, external, best Last update: Sun Dec 28 19:30:27 2014 65002 65004 100000 65004 65004 101000 102000 65004 65004 65004 172.16.10.2 (metric: 0) from 172.16.10.2 (172.16.10.2) Source VRF: VRF1, metric: 0, localpref: 100, weight: 0, valid, external, alternate Last update: Sun Dec 28 19:30:16 2014 exaware# show bgp table import vrf default VRF default : Flags: s - suppressed, d - damped, h - history, * - valid, > - best i - internal, S - stale, = - multipath, a - alternate IPv4 Unicast:</pre> <table><thead><tr><th>NETWORK</th><th>PEER IP</th><th>FLAGS</th><th>NEXT HOP</th><th>METRIC</th><th>LOC PRF</th><th>WEIGHT</th><th>AS PATH</th><th>SOURCE VRF</th></tr></thead><tbody><tr><td rowspan="2">77.77.77.0/24</td><td>172.16.11.2</td><td>*></td><td>172.16.11.2</td><td>0</td><td>-</td><td>0</td><td>65004 60001</td><td>VRF1</td></tr><tr><td>172.16.10.2</td><td>*a</td><td>172.16.10.2</td><td>0</td><td>-</td><td>0</td><td>65002 65004</td><td>VRF1</td></tr></tbody></table>		NETWORK	PEER IP	FLAGS	NEXT HOP	METRIC	LOC PRF	WEIGHT	AS PATH	SOURCE VRF	77.77.77.0/24	172.16.11.2	*>	172.16.11.2	0	-	0	65004 60001	VRF1	172.16.10.2	*a	172.16.10.2	0	-	0	65002 65004	VRF1
NETWORK	PEER IP	FLAGS	NEXT HOP	METRIC	LOC PRF	WEIGHT	AS PATH	SOURCE VRF																			
77.77.77.0/24	172.16.11.2	*>	172.16.11.2	0	-	0	65004 60001	VRF1																			
	172.16.10.2	*a	172.16.10.2	0	-	0	65002 65004	VRF1																			
Notes	If a VRF is not specified, data is shown for the default VRF. Default view for this command is brief. When specifying a prefix, default view is detailed.																										

10.2.3.16. show bgp table label view

Description	Displays label information of routes in the BGP routing table.		
Command Syntax	show bgp table [vrf default] {ipv4 labeled-unicast afi vpn [rd route-distinguisher]} labels [{local received} label-id ip-prefix]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	afi	ipv4	IP address family (AFI)
		ipv6	name.
		route-distinguisher	<2-byte ASN>:<4-byte number>
		<4-byte ASN>L:<2-byte number>	4-byte ASN notation.
		<IPv4 address>:<2-byte number>	IPv4 address notation.
	label-id	A	Label number. (Default = all)
ip-prefix	IPv4 or IPv6 address / length	IP prefix of learned network.	
Examples	The following examples show sample output from this command:		
exaware# show bgp table ipv4 vpn labels Flags: s - suppressed, d - damped, h - history, * - valid, > - best i - internal, S - stale, = - multipath Origin: i - IGP, e - EGP, ? - incomplete IPv4 VPN Route Distinguisher 100:100 (VRF BLUE): NETWORK PEER IP FLAGS HOP NEXT RCVD LOC LABEL LABEL			

```

-----
100.2.1.0/24  4.4.4.1  *>i    4.4.4.1  nolabel 16033
100.4.1.0/24  4.4.4.2  *>i    4.4.4.2  nolabel 16033

Flags:  s - suppressed, d - damped, h - history, * - valid, > - best
        i - internal, S - stale, = - multipath
Origin:  i - IGP, e - EGP, ? - incomplete
IPv4 VPN Route Distinguisher 100:300 :
      NEXT      RCVD      LOC
NETWORK      PEER IP  FLAGS  HOP      LABEL  LABEL
-----
150.2.1.0/24  5.5.5.1  *>     5.5.5.1  29055   nolabel
150.3.1.0/24  5.5.5.2  *>     5.5.5.2  29055   nolabel

Flags:  s - suppressed, d - damped, h - history, * - valid, > - best
        i - internal, S - stale, = - multipath
Origin:  i - IGP, e - EGP, ? - incomplete
IPv4 VPN Route Distinguisher 100:400 :
      NEXT      RCVD      LOC
NETWORK      PEER IP  FLAGS  HOP      LABEL  LABEL
-----
160.2.1.0/24  4.4.4.1  *>i    4.4.4.1  29055   nolabel
160.4.1.0/24  4.4.4.2  *>i    4.4.4.2  29055   nolabel

exaware# show bgp table ipv4 vpn labels received 29055
Flags:  s - suppressed, d - damped, h - history, * - valid, > - best
        i - internal, S - stale, = - multipath
Origin:  i - IGP, e - EGP, ? - incomplete
IPv4 VPN Route Distinguisher 100:300 :
      NEXT      RCVD      LOC
NETWORK      PEER IP  FLAGS  HOP      LABEL  LABEL
-----
150.2.1.0/24  5.5.5.1  *>     5.5.5.1  29055   nolabel
150.3.1.0/24  5.5.5.2  *>     5.5.5.2  29055   nolabel

Flags:  s - suppressed, d - damped, h - history, * - valid, > - best
        i - internal, S - stale, = - multipath
Origin:  i - IGP, e - EGP, ? - incomplete
IPv4 VPN Route Distinguisher 100:400 :
      NEXT      RCVD      LOC
NETWORK      PEER IP  FLAGS  HOP      LABEL  LABEL
-----
160.2.1.0/24  4.4.4.1  *>i    4.4.4.1  29055   nolabel
160.4.1.0/24  4.4.4.2  *>i    4.4.4.2  29055   nolabel

exaware# show bgp table ipv4 vpn rd 100:300 labels received 29055
Flags:  s - suppressed, d - damped, h - history, * - valid, > - best
        i - internal, S - stale, = - multipath
Origin:  i - IGP, e - EGP, ? - incomplete
      NEXT      RCVD      LOC
NETWORK      PEER IP  FLAGS  HOP      LABEL  LABEL
-----
150.2.1.0/24  5.5.5.1  *>     5.5.5.1  29055   nolabel
150.3.1.0/24  5.5.5.2  *>     5.5.5.2  29055   nolabel

```

Notes	If VRF is not specified, data is shown for the default VRF. Filtering on RD can only be done under VRF default and SAFI vpn. This command has no detailed view. SAFI labeled-unicast and vpn are supported only in VRF default.
--------------	---

10.2.3.17. show bgp table vsi

Description	Displays BGP routing table for specific VSI.																					
Command Syntax	show bgp table vsi <i>vsi-name</i> [brief detail inconsistent-as] [<i>vsi-prefix</i>]																					
Command Mode	operational																					
Parameters Table	Name	Value		Description																		
	<i>vsi-name</i>	VSI name																				
	<i>vsi-prefix</i>	<2-byte VE>/16																				
Examples	The following examples show sample output from this command:																					
<pre>exaware# show bgp table vsi vpls1 Flags: s - suppressed, d - damped, h - history, * - valid, > - best i - internal, S - stale, = - multipath, a - alternate Origin: i - IGP, e - EGP, ? - incomplete VSI vpls1 :</pre> <table><thead><tr><th>NETWORK</th><th>PEER IP</th><th>FLAGS</th><th>NEXT HOP</th><th>METRIC</th><th>LOC PRF</th><th>WEIGHT</th><th>AS PATH</th><th>ORIGIN</th></tr></thead><tbody><tr><td>1/16</td><td>0.0.0.0</td><td>*></td><td>0.0.0.0</td><td>0</td><td>100</td><td>32768</td><td></td><td>i</td></tr></tbody></table> <pre>exaware# show bgp table vsi vpls1 detail VSI vpls1 : BGP routing table for 1/16 1 path available Local 0.0.0.0 from 0.0.0.0 (192.168.0.82) Origin: IGP, metric: 0, localpref: 100, weight: 32768, valid, sourced, local, best Extended community: RT:1:1 L2-info:VPLS:0x0:1500:100 Route Distinguisher: 35.35.35.35:1 VE block offset: 1, Label-base: 73736, range: 8 Last update: Mon Aug 10 03:15:33 2015</pre>					NETWORK	PEER IP	FLAGS	NEXT HOP	METRIC	LOC PRF	WEIGHT	AS PATH	ORIGIN	1/16	0.0.0.0	*>	0.0.0.0	0	100	32768		i
NETWORK	PEER IP	FLAGS	NEXT HOP	METRIC	LOC PRF	WEIGHT	AS PATH	ORIGIN														
1/16	0.0.0.0	*>	0.0.0.0	0	100	32768		i														
Notes	Default view for this command is brief. When specifying a prefix, default view is detailed.																					

10.3. Common Routing Commands

10.3.1. Common Routing Configuration Commands

10.3.1.1. as-notation

Description	Sets the AS notation as plain or dot.		
Command Syntax	<pre>as-notation {plain dot} no as-notation</pre>		
Command Mode	configuration routing common		
Parameters Table	Name	Value	Description
		plain dot	plain (in the range of 1 - 4294967295) dot (in the range of 1 - 65535.65535)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing common exaware(config-routing-common)# as-notation dot</pre>		

10.3.1.2. maximum-paths

Description	Sets the maximum number of load-balanced paths that may be installed per route in the FIB. If a route in the RIB exceeds this maximum number of paths, all paths over the limit will be inactivated by the RIB. Use the no form of the command to set all Routing Information Base parameters to their default value.		
Command Syntax	<pre>maximum-paths num-paths no maximum-paths</pre>		
Command Mode	configuration routing-common		
Parameters Table	Name	Value	Description
	<i>num-paths</i>	1-30	Maximum number of paths that can be defined to a route in the FIB. (Default = 30)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing common exaware(config-routing-common)# maximum-paths 8</pre>		

10.3.1.3. policy-accounting traffic-counter-description

Description	Sets the description of a specific traffic counter. Use the no form of the command to remove the description.		
Command Syntax	<code>traffic-counter-description id description</code> <code>no traffic-counter-description [id]</code>		
Command Mode	configuration routing common policy-accounting		
Parameters Table	Name	Value	Description
	<i>id</i>	uint32	ID of traffic-counter
	<i>description</i>	String	Description of traffic-counter
Examples	<pre>exaware# config exaware(config)# routing common policy-accounting exaware(config-routing-common-policy-accounting)# traffic-counter-description 5 "NETANYA" exaware# config exaware(config)# routing common policy-accounting exaware(config-routing-common-policy-accounting)# no traffic-counter-description 5</pre>		

10.3.1.4. router-id

Description	Configures the IP address that identifies the router.		
Command Syntax	<code>router-id ip-address</code>		
Command Mode	configuration routing-common		
Parameters Table	Name	Value	Description
	<code>ip-address</code>		IPv4 address of the router.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing common exaware(config-routing-common)# router-id 10.10.10.10</pre>		
Notes	The system automatically selects an IP address to use as the router ID when it is started. The highest local IP address is selected, and loopback interfaces are preferred.		

10.3.1.5. routing common

Description	Enters routing common configuration mode, which is used to configure common Routing Information Base parameters. Use the no form of the command to set all Routing Information Base parameters to their default value.
Command Syntax	routing common no routing common
Command Mode	configuration
Parameters Table	N/A
Examples	This example shows how to enter the routing common configuration mode: Note the change in the command line prompt. exaware# config exaware(config)# routing common exaware(config-routing-common)#

Notes

When the AS notation has been changed:

- Show BGP commands will be updated to display the new notation.
- The regexp will be calculated according to the new notation.
- The bgp local AS will be updated to the new notation (i.e. "routing bgp 1.10 ").
- The notation of as-path in policy will be updated to the new notation.

It will be possible to configure local-AS in both notations – the same instance will be impacted and displayed (according to the selected AS notation).

Policy CLI configuration will support both notations, and the show configuration commands will display values according to the selected notation.

Commands influenced by as-notation are:

- /vrf/rd
- /bgp/vrf/export-rt
- /bgp/vrf/import-rt
- /bgp/vrf/neighbor/afi/unicast/soo (also /bgp/vrf/neighbor-group/afi/unicast/soo)
- All policy extended communities conditions and actions.
- show bgp neighbors advertised/received routes (brief/detail/label view)
- show bgp table (brief/detail/label view)
- show vrf

10.4. DHCP Snooping Commands

10.4.1. DHCP Snooping Configuration Commands

10.4.1.1. dhcp-snooping

Description	Use the dhcp-snooping command to enable/disable DHCP Snooping on a VPLS. Use the no form of this command to restore the default configuration (disable).		
Command Syntax	<code>dhcp-snooping {enable disable}</code> <code>no dhcp-snooping</code>		
Command Mode	configuration l2-services vpls		
Parameters Table	Name	Value	Description
	enable		Enables DHCP Snooping on the VPLS.
	disable		Disables DHCP Snooping on the VPLS (default configuration).
Examples	The following example shows how to use this command: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# dhcp-snooping enable</pre>		

10.4.1.2. dhcp-snooping allow-receiving-information-option

Description	Use the dhcp-snooping allow-receiving-information-option command to allow receiving DHCP packets from DHCP clients with option-82 information. Use the no form of this command to restore the default configuration (enable).		
Command Syntax	<pre>dhcp-snooping allow-receiving-information-option {enable disable} no dhcp-snooping allow-receiving-information-option</pre>		
Command Mode	<pre>configuration l2-services vpls interface configuration l2-services vpls vpws spoke</pre>		
Parameters Table	Name	Value	Description
	enable		Allows receiving DHCP option-82 from DHCP clients.
	disable		Prohibits receiving DHCP option-82 from DHCP clients.
Examples	The following example shows how to use this command: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# interface x-eth0/0/1.5 exaware(config-l2-services-vpls-Service1-eth0/0/1.5)# dhcp-snooping allow-receiving-information-option enable exaware(config-l2-services-vpls-Service1-eth0/0/1.5)# exit exaware(config-l2-services-vpls-Service1)# vpws aaaa spoke exaware(config-l2-services-vpls-vpws-aaaa)# dhcp-snooping allow-receiving-information-option enable</pre>		
Notes	On full mesh interfaces, the option is always enabled.		

10.4.1.3. dhcp-snooping information-option (not supported)

Description	Use the dhcp-snooping information option command to enable DHCP option-82 data insertion. Use the no form of this command to restore the default configuration (enable).		
Command Syntax	<pre>dhcp-snooping information-option {enable disable} no dhcp-snooping information-option</pre>		
Command Mode	<pre>configuration l2-services vpls interface configuration l2-services vpls vpws spoke</pre>		
Parameters Table	Name	Value	Description
	enable		Enables DHCP option-82 data insertion.
	disable		Disables DHCP option-82 data insertion.
Examples	The following example shows how to use this command: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# interface x-eth0/0/1.5 exaware(config-l2-services-vpls-Service1-eth0/0/1.5)# dhcp-snooping information-option enable exaware(config-l2-services-vpls-Service1-eth0/0/1.5)# exit exaware(config-l2-services-vpls-Service1)# vpws aaaa spoke exaware(config-l2-services-vpls-vpws-aaaa)# dhcp-snooping information-option enable</pre>		

10.4.1.4. dhcp-snooping trusted

Description	Use the dhcp-snooping trusted command to configure an interface as trusted/untrusted for DHCP Snooping purpose. Use no form of this command to restore the default configuration (untrusted).		
Command Syntax	<pre>dhcp-snooping trusted {enable disable} no dhcp-snooping trusted</pre>		
Command Mode	<pre>configuration l2-services vpls interface configuration l2-services vpls vpws spoke</pre>		
Parameters Table	Name	Value	Description
	enable		Specifies the interface as trusted.
	disable		Specifies the interface as untrusted (default configuration).
Examples	The following example shows how to use this command: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# interface x-eth0/0/1.5 exaware(config-l2-services-vpls-Service1-eth0/0/1.5)# dhcp-snooping trusted enable exaware(config-l2-services-vpls-Service1-eth0/0/1.5)# exit exaware(config-l2-services-vpls-Service1)# vpws aaaa spoke exaware(config-l2-services-vpls-vpws-aaaa)# dhcp-snooping trusted enable</pre>		
Notes	Full mesh interfaces are always trusted.		

10.4.1.5. dhcpv6-snooping

Description	Use the dhcpv6-snooping command to enable/disable DHCPv6 Snooping on a VPLS. Use the no form of this command to restore the default configuration (disable).		
Command Syntax	<pre>Dhcpv6-snooping {enable disable} no dhcpv6-snooping</pre>		
Command Mode	configuration l2-services vpls		
Parameters Table	Name	Value	Description
	enable		Enables DHCPv6 Snooping on the VPLS.
	disable		Disables DHCPv6 Snooping on the VPLS (default configuration).
Examples	The following example shows how to configure the auto-discovery attributes: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# dhcpv6-snooping enable</pre>		

10.4.1.6. dhcpv6-snooping allow-receiving-information-option

Description	Use the dhcpv6-snooping allow-receiving-information-option command to allow receiving DHCPv6 packets from DHCPv6 clients with interface-ID option. Use the no form of this command to restore the default configuration (enable).		
Command Syntax	<pre>dhcpv6-snooping allow-receiving-information-option {enable disable} no dhcpv6-snooping allow-receiving-information-option</pre>		
Command Mode	<pre>configuration l2-services vpls interface configuration l2-services vpls vpws spoke</pre>		
Parameters Table	Name	Value	Description
	enable		Allows receiving the DHCPv6 Interface-ID option from DHCPv6 client.
	disable		Prohibits receiving the DHCPv6 Interface-ID option from DHCPv6 client.
Examples	The following example shows how to use this command: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# interface x-eth0/0/1.5 exaware(config-l2-services-vpls-Service1-eth0/0/1.5)# dhcpv6-snooping allow-receiving-information-option enable exaware(config-l2-services-vpls-Service1-eth0/0/1.5)# exit exaware(config-l2-services-vpls-Service1)# vpws aaaa spoke exaware(config-l2-services-vpls-vpws-aaaa)# dhcpv6-snooping allow-receiving-information-option enable</pre>		
Notes	On full mesh interfaces, the option is always enabled.		

10.4.1.7. dhcpv6-snooping information-option (not

supported)

Description	Use the dhcpv6-snooping information option command to enable DHCPv6 Interface-ID data insertion. Use the no form of this command to restore the default configuration (enable).		
Command Syntax	<pre>dhcpv6-snooping information-option {enable disable} no dhcpv6-snooping information-option</pre>		
Command Mode	<pre>configuration l2-services vpls interface configuration l2-services vpls vpws spoke</pre>		
Parameters Table	Name	Value	Description
	enable		Enables DHCPv6 Interface-ID data insertion.
	disable		Disables DHCPv6 Interface-ID data insertion.
Examples	The following example shows how to use this command: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# interface x-eth0/0/1.5 exaware(config-l2-services-vpls-Service1-eth0/0/1.5)# dhcpv6-snooping information-option enable exaware(config-l2-services-vpls-Service1-eth0/0/1.5)# exit exaware(config-l2-services-vpls-Service1)# vpws aaaa spoke exaware(config-l2-services-vpls-vpws-aaaa)# dhcpv6-snooping information-option enable</pre>		

10.4.1.8. dhcpv6-snooping trusted

Description	Use the dhcpv6-snooping trust command to configure an interface as trusted/untrusted for DHCPv6 Snooping purpose. Use the no form of this command to restore the default configuration (untrusted).		
Command Syntax	<pre>dhcpv6-snooping trusted {enable disable} no dhcpv6-snooping trusted</pre>		
Command Mode	<pre>configuration l2-services vpls interface configuration l2-services vpls vpws spoke</pre>		
Parameters Table	Name	Value	Description
	enable		Specifies the interface as trusted.
	disable		Specifies the interface as untrusted (default configuration).
Examples	The following example shows how to use this command: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# interface x-eth0/0/1.5 exaware(config-l2-services-vpls-Service1-eth0/0/1.5)# dhcpv6-snooping trusted enable exaware(config-l2-services-vpls-Service1-eth0/0/1.5)# exit exaware(config-l2-services-vpls-Service1)# vpws aaaa spoke exaware(config-l2-services-vpls-vpws-aaaa)# dhcpv6-snooping trusted enable</pre>		
Notes	Full mesh interfaces are always trusted.		

10.4.1.9. polling-interval

Description	Use the polling-interval command to configure the polling interval of the DHCPv4/v6 Snooping Binding tables Use the no form of this command to restore the default configuration (60 min).		
Command Syntax	<code>polling-interval time</code> <code>no polling-interval</code>		
Command Mode	configuration l2-services		
Parameters Table	Name	Value	Description
	<code>time</code>	Integer in range 1..3600	Specifies the polling interval in minutes.
Examples	The following example shows how to use this command: <code>exaware(config)# l2-services</code> <code>exaware(config-l2-services)# polling-interval 60</code>		
Notes	The system polls the DCHP Binding Table and removes from entries if - Lease Time has been sent by server, and it expired. - Lease Time has not been sent by server.		

10.4.2. DHCP Snooping Operational Commands

10.4.2.1. clear dhcp-snooping counters

Description	Use the clear dhcp-snooping counters command to clear the DHCP Snooping counters.		
Command Syntax	clear dhcp-snooping counters [{vpls <i>vpls-name</i> } [{interface <i>interface-name</i> } {vpws <i>vpws-name</i> } {peer <i>ip-address</i> }}]		
Command Mode	operation		
Parameters Table	Name	Value	Description
	vpls-name	string	Clears the DHCP Snooping counters of the specified VPLS. If the argument is skipped, then the counters of all VPLS is cleared.
	interface-name	string	Clears the DHCP Snooping counters of the specified VPLS interface. If the argument is skipped, then the counters of all interfaces of the specified VPLS is cleared.
	vpws-name	string	Clears the DHCP Snooping counters of the specified VPWS interface.
	ip-address	IPv4 address	Clears the DHCP Snooping counters of the PW with specified BGP peer.
Examples	This example shows how to use this command: <pre> exaware# clear dhcp-snooping counters exaware# clear dhcp-snooping counters vpls VPLS1 exaware# clear dhcp-snooping counters vpls VP1 interface x-eth0/0/2.12 exaware# clear dhcp-snooping counters vpls VP1 vpws VPWS-AAA exaware# clear dhcp-snooping counters vpls VP1 peer 10.10.10.1 </pre>		

10.4.2.2. clear dhcpv6-snooping counters

Description	Use the clear dhcpv6-snooping counters command to clear the DHCP Snooping counters.		
Command Syntax	clear dhcpv6-snooping counters [{vpls <i>vpls-name</i> } [interface <i>interface-name</i>] {vpws <i>vpws-name</i> } {peer <i>ip-address</i> }]		
Command Mode	operation		
Parameters Table	Name	Value	Description
	vpls-name	string	Clears the DHCPv6 Snooping counters of the specified VPLS. If the argument is skipped, then the counters of all VPLS is cleared.
	interface <i>if-name</i>	string	Clears the DHCP Snooping counters of the specified VPLS interface. If the argument is skipped, then the counters of all interfaces of the specified VPLS is cleared.
	vpws-name	string	Clears the DHCP Snooping counters of the specified VPWS interface.
	ip-address	IPv4 address	Clears the DHCP Snooping counters of the PW with specified BGP peer.
Examples	This example shows how to use this command: <pre> exaware# clear dhcpv6-snooping counters exaware# clear dhcpv6-snooping counters vpls VPLS1 exaware# clear dhcpv6-snooping counters vpls VPLS1 interface x-eth0/0/2.12 exaware# clear dhcpv6-snooping counters vpls VP1 vpws VPWS-AAA exaware# clear dhcpv6-snooping counters vpls VP1 peer 10.10.10.1 </pre>		

10.4.3. DHCP Snooping Show Commands

10.4.3.1. show dhcp-snooping clients

Description	Use the show dhcp-snooping clients command to display the the DHCP clients and the last passed DHCP Snooping messages.					
Command Syntax	show dhcp-snooping client [vpls vpls-name [interface <i>interface-name</i>]] show dhcp-snooping client detail [vpls vpls-name [interface <i>interface-name</i>] [client <i>mac-addr</i>]]					
Command Mode	operation					
Parameters Table	Name	Value	Description			
	<i>vpls-name</i>	string	Displays the DHCP Snooping messages of the specified VPLS. If the argument is skipped, then the messages of all VPLS is displayed.			
	<i>mac-addr</i>	hh:hh:hh:hh:hh:hh	Displays the DHCP Snooping messages of the specified DHCP clients.			
	<i>interface-name</i>	string	Displays the DHCP Snooping messages on the specified interface.			
	detail		Displays the detailed information.			
Examples	This example shows how to use this command:					
exaware# show dhcp-snooping clients						
Polling Interval: 60 minutes						
VPLS NAME	CLIENT MAC ADDRESS	CLIENT INTERFACE	ALLOCATED IP ADDRESS	SERVER IP ADDRESS	DHCP MESSAGE	RECORD TIME
HRH-EO	00:11:22:33:44:55	x-eth 0/0/7	1.2.3.4/16	42.42.42.42	ACK	2026-02-16T13:25:15+02:00
	00:01:03:04: 44:55	PW-6.6.6.6	1.2.3.5/16	42.42.42.42	OFFER	2026-02-16T13:25:25+02:00
King-P	22:22:22:22:22:22	x-eth 0/0/7			DISCOVER	2026-02-16T13:25:35+02:00
	44:44:44:44:44:44	VPWS-Y	2.3.4.5/16	42.42.42.42	ACK	2026-02-16T13:25:45+02:00

```
exaware# show dhcp-snooping clients detail
```

Polling Interval: 60 minutes

VPLS: HRH-EO

Client MAC Address: 00:11:22:33:44:55

Client Interface: x-eth 0/0/7

Allocated IP Address: 1.2.3.4

Allocated IP Mask: 255.255.0.0

DHCP Server IP Address: 42.42.42.42

DHCP Message: ACK

Record Time: 2026-02-16T13:25:15+02:00

Original Lease Time: 10:00:01

Lease Time Remaining: 20:00:00

Option-82: SubOption=1, Circuit-ID=HostA-x-eth0/0/7

Client MAC Address: 00:11:22:33:44:56

Client Interface: x-eth 0/0/7

Allocated IP Address: 1.2.3.5

Allocated IP Mask: 255.255.0.0

DHCP Server IP Address: 42.42.42.42

DHCP Message: ACK

Record Time: 2026-02-16T13:25:25+02:00

Original Lease Time: 10:00:01

Lease Time Remaining: 20:00:00

Option-82: SubOption=1, Circuit-ID=HostA-x-eth0/0/7

VPLS: APLPHA

Client MAC Address: 00:11:33:33:44:55

Client Interface: x-eth 0/0/7

Allocated IP Address: 2.2.3.5

Allocated IP Mask: 255.0.0.0

DHCP Server IP Address: 42.42.42.42

DHCP Message: ACK

Record Time: 2026-02-16T13:35:15+02:00

Original Lease Time: 10:00:01

Lease Time Remaining: 20:00:00

Option-82: SubOption=1, Circuit-ID=HostA-x-eth0/0/7

Client MAC Address: 00:11:22:33:44:56

Client Interface: x-eth 0/0/7

Allocated IP Address: 2.2.3.4

Allocated IP Mask: 255.255.0.0

DHCP Server IP Address: 42.42.42.42

DHCP Message: ACK

Record Time: 2026-02-16T13:25:45+02:00

Original Lease Time: 10:00:01

Lease Time Remaining: 20:00:00
Option-82: SubOption=1, Circuit-ID=HostA-x-eth0/0/7

10.4.3.2. show dhcp-snooping counters

Description	Use the show dhcp-snooping counters command to display the DHCP Snooping counters.		
Command Syntax	show dhcp-snooping counters [vpls <i>vpls-name</i>] show dhcp-snooping counters detail vpls <i>vpls-name</i> {interface <i>interface-name</i> } {vpws <i>vpws-name</i> } {peer <i>ip-address</i> }		
Command Mode	operation		
Parameters Table	Name	Value	Description
	vpls-name	string	Displays the DHCP Snooping counters of the specified VPLS. If the argument is skipped, then the counters of all VPLS is displayed.
	interface-name	string	Displays the DHCP Snooping counters of the specified VPLS interface.
	vpws-name	string	Displays the DHCP Snooping counters of the specified VPWS interface.
	ip-address	IPv4 address	Displays the DHCP Snooping counters of the PW with specified BGP peer.

Examples

This example shows how to use this command:

```
exaware# show dhcp-snooping counters
```

VPLS NAME	INTERFACE	RX PKTS	TX PKTS	DROPPED PKTS
VPLS1	x-eth0/0/1.3336	13	9	0
	x-eth0/0/2.3336	3	9	0
	10.10.10.1	3	9	0
	VPWS-AAAAA	3	9	0
VPLS2	x-eth0/0/1.54	11	8	0
	x-eth0/0/4.54	5	5	0

```
exaware# show dhcp-snooping counters vpls VPLS1
```

VPLS NAME	INTERFACE	RX PKTS	TX PKTS	DROPPED PKTS
VPLS1	x-eth0/0/1.3336	13	9	0
	x-eth0/0/2.3336	3	9	0
	10.10.10.1	3	9	0
	VPWS-AAAAA	3	9	0

```
exaware# show dhcp-snooping counters detail vpls VPLS1 interface x-eth 0/0/1.3336
```

MESSAGE TYPE	RX PKTS	TX PKTS	DROPPED PKTS
ACK	0	7	0
BOOTP REQUEST	13	0	0
BOOTP REPLY	0	9	0
BOOTP INVALID	0	0	0
DECLINE	0	0	0
DISCOVER	6	0	0
INFORM	6	0	0
NAK	0	0	0
OFFER	0	2	0
REQUEST	1	0	0

10.4.3.3. show dhcpv6-snooping clients

Description	Use the show dhcpv6-snooping clients command to display the DHCPv6 clients and the last passed DHCPv6 Snooping messages.		
Command Syntax	show dhcpv6-snooping client [vpls vpls-name [interface <i>interface-name</i>]] show dhcpv6-snooping client detail [vpls vpls-name [interface <i>interface-name</i>] [client <i>mac-addr</i>]]		
Command Mode	operation		
Parameters Table	Name	Value	Description
	<i>vpls-name</i>	string	Displays the DHCP Snooping messages of the specified VPLS. If the argument is skipped, then the messages of all VPLS is displayed.
	<i>mac-addr</i>	hh:hh:hh:hh:hh:hh	Displays the DHCP Snooping messages of the specified DHCP clients.
	<i>interface-name</i>	string	Displays the DHCP Snooping messages on the specified interface.
	detail		Displays the detailed information.
Examples	This example shows how to use this command:		
Examples	This example shows how to use this command:		

```
exaware# show dhcpv6-snooping clients
```

Polling Interval: 60 minutes

VPLS NAME	CLIENT MAC ADDRESS	CLIENT INTERFACE	ALLOCATED IPV6 ADDRESS	SERVER IPV6 ADDRESS	DHCPV6 MESSAGE	RECORD TIME
HRH-EO	00:11:22:33:44:55	x-eth 0/0/7	2001::1.2.3.4	2001::1.2.2.1	Reply	2026-02-16T13:25:15+02:00
	00:01:03:04:44:55	PW-6.6.6.6	2001::1.2.3.5	2001::1.2.2.1	Advertise	2026-02-16T13:25:25+02:00
King-P	22:22:22:22:22:22	x-eth 0/0/7			Request	2026-02-16T13:25:35+02:00
	44:44:44:44:44:44	VPWS-Y	2001::1.2.3.6	2001::1.2.2.1	Reply	2026-02-16T13:25:45+02:00

```
exaware# show dhcpv6-snooping clients detail
```

Polling Interval: 60 minutes

VPLS: HRH-EO

Client MAC Address: 00:11:22:33:44:55

Client Interface: x-eth 0/0/7

Allocated IP Address: 2001::1.2.3.4

DHCPv6 Server IP Address: 2001::1.2.2.1

DHCPv6 Message: REPLY

Record Time: 2026-02-16T13:25:15+02:00

Original Lease Time: 10:00:01

Lease Time Remaining: 20:00:00

Option-18: SubOption=1, Circuit-ID=HostA-x-eth0/0/7

Client MAC Address: 00:11:22:33:44:56

Client Interface: x-eth 0/0/7

Allocated IP Address: 2001::1.2.3.5

DHCPv6 Server IP Address: 2001::1.2.2.1

DHCPv6 Message: REPLY

Record Time: 10:10 Feb 30, 2029

Original Lease Time: 10:00:01

Lease Time Remaining: 20:00:00

Option-18: SubOption=1, Circuit-ID=HostA-x-eth0/0/7

VPLS: ALHA

Client MAC Address: 00:11:22:33:45:55

Client Interface: x-eth 0/0/7

Allocated IP Address: 2001::1.2.3.4

DHCPv6 Server IP Address: 2001::1.2.2.1

DHCPv6 Message: REPLY

Record Time: 2026-02-16T13:25:25+02:00

Original Lease Time: 10:00:01
Lease Time Remaining: 20:00:00
Option-18: SubOption=1, Circuit-ID=HostA-x-eth0/0/7
Client MAC Address: 00:11:22:33:44:56
Client Interface: x-eth 0/0/7
Allocated IP Address: 2002::1.2.3.5
DHCPv6 Server IP Address: 2001::1.2.2.1
DHCPv6 Message: REPLY
Record Time: 2026-02-16T13:25:35+02:00
Original Lease Time: 10:00:01
Lease Time Remaining: 20:00:00
Option-18: SubOption=1, Circuit-ID=HostA-x-eth0/0/7

10.4.3.4. show dhcpv6-snooping counters

Description	Use the show dhcpv6-snooping counters command to display the DHCPv6 Snooping counters.			
Command Syntax	show dhcpv6-snooping counters [vpls <i>vpls-name</i>] show dhcpv6-snooping counters detail vpls <i>vpls-name</i> {interface <i>interface-name</i> } {vpws <i>vpws-name</i> } {peer <i>ip-address</i> }			
Command Mode	Operation			
Parameters Table	Name	Value	Description	
	vpls-name	string	Displays the DHCPv6 Snooping counters of the specified VPLS. If the argument is skipped, then the counters of all VPLS is displayed.	
	interface-name	string	Displays the DHCP Snooping counters of the specified VPLS interface.	
		vpws-name	string	Displays the DHCP Snooping counters of the specified spoke VPWS interface.
	ip-address	IPv4 address	Displays the DHCP Snooping counters of the PW with specified BGP peer.	

Examples

This example shows how to use this command:

```
exaware# show dhcpv6-snooping counters
```

VPLS NAME	INTERFACE	RX PKTS	TX PKTS	DROPPED PKTS
VPLS1	x-eth0/0/1.3336	13	9	0
	x-eth0/0/2.3336	3	9	0
	10.10.10.1	3	9	0
	VPWS-AAAAA	3	9	0
VPLS2	x-eth0/0/1.54	11	8	0
	x-eth0/0/4.54	5	5	0

```
exaware# show dhcpv6-snooping counters vpls VPLS1
```

VPLS NAME	INTERFACE	RX PKTS	TX PKTS	DROPPED PKTS
VPLS1	x-eth0/0/1.3336	13	9	0
	x-eth0/0/2.3336	3	9	0
	10.10.10.1	3	9	0
	VPWS-AAAAA	3	9	0

```
exaware# show dhcpv6-snooping counters detail vpls VPLS1 interface  
x-eth 0/0/3
```

MESSAGE TYPE	RX PKTS	TX PKTS	DROPPED PKTS
ADVERTISE	0	2	0
CONFIRM	1	0	0
DECLINE	1	0	0
INFO-REQUEST	1	0	0
REBIND	1	0	0
RECONFIGURE	1	0	0
REQUEST	1	0	0
RELAY FORWARD	0	0	0
RELAY REPLY	0	0	0
RELEASE	1	0	0
RENEW	1	0	0
REPLY	1	0	0
SOLICIT	6	0	0

10.5. IGMP Routing Commands

10.5.1. IGMP Routing Configuration Commands

10.5.1.1. last-member-query-timeout

Description	The last member query interval is the max query response interval inserted into group-specific queries sent in response to leave group messages, and is also the amount of time between group-specific query messages. This value may be tuned to modify the leave latency of the network. A reduced value results in reduced time to detect the loss of the last member of a group. Use the no form of the command to remove an interface specific configuration, and the global configuration will be applied.		
Command Syntax	last-member-query-timeout <seconds>		
Command Mode	configuration routing igmp configuration routing igmp interface		
Parameters Table	Name	Value	Description
	seconds	<1..3174>	Last member query timeout. (Default = 1)
Examples	The following example shows how to use this command globally: <pre>exaware# config exaware(config)# routing igmp default exaware(config-routing-igmp-default)# last-member-query-timeout 120</pre> The following example shows how to use this command to configure a specific interface: <pre>exaware# config exaware(config)# routing igmp interface x-eth 0/0/0 exaware(config-routing-igmp-interface-x-eth0/0/0)# last-member-query-timeout 150</pre>		

10.5.1.2. query-interval

Description	The query interval is the interval between General Queries sent by the Querier. Use the no form of the command to remove an interface specific configuration, and the global configuration will be applied.		
Command Syntax	query-interval <seconds>		
Command Mode	configuration routing igmp configuration routing igmp <x-eth agg-eth>		
Parameters Table	Name	Value	Description
	seconds	<1..31744>	The interval between queries sent by the querier. (Default = 60)
Examples	The following example shows how to use this command globally: <pre>exaware# config exaware(config)# routing igmp default exaware(config-routing-igmp-default)# query-interval 150</pre> The following example shows how to use this command to configure a specific interface: <pre>exaware# config exaware(config)# routing igmp interface x-eth 0/0/0 exaware(config-routing-igmp-interface-x-eth0/0/0)# query-interval 125</pre>		

10.5.1.3. query-response-time

Description	The maximum query response interval advertised in IGMPv3 queries on this interface. The query response time used to calculate the Max Resp Code that is inserted into the periodic general queries. The number of seconds represented by the query response time must be less than the query interval value. Use the no form of the command to remove an interface specific configuration, and the global configuration will be applied.		
Command Syntax	query-response-time <seconds>		
Command Mode	configuration routing igmp configuration routing igmp <x-eth agg-eth>		
Parameters Table	Name	Value	Description
	seconds	<0..3174>	The maximum query response interval advertised in MGMDv2 or IGMPv3 queries on this interface. (Default = 10)
Examples	The following example shows how to use this command globally: <pre>exaware# config exaware(config)# routing igmp default exaware(config-routing-igmp-default)# query-response-time 10</pre> The following example shows how to use this command to configure a specific interface: <pre>exaware# config exaware(config)# routing igmp interface x-eth 0/0/0 exaware(config-routing-igmp-interface-x-eth0/0/0)# query-response-time 150</pre>		

10.5.1.4. robustness-factor

Description	The robustness factor allows tuning for the expected packet loss on a network. In the case where packet loss is expected on a network, the robustness factor may be increased. Use the no form of the command to remove an interface specific configuration, and the global configuration will be applied.		
Command Syntax	<code>robustness-factor <number></code>		
Command Mode	configuration routing igmp configuration routing igmp <x-eth agg-eth>		
Parameters Table	Name	Value	Description
	number	<2-255>	Expected packet loss factor. (Default = 2)
Examples	The following example shows how to use this command globally: <pre>exaware# config exaware(config)# routing igmp default exaware(config-routing-igmp-default)# robustness-factor 2</pre> The following example shows how to use this command to configure a specific interface: <pre>exaware# config exaware(config)# routing igmp interface x-eth 0/0/0 exaware(config-routing-igmp-interface-x-eth0/0/0)# robustness-factor 3</pre>		

10.5.1.5. routing igmp default

Description	Enters routing IGMP configuration mode. Use the no form of the command to disable IGMP routing globally.
Command Syntax	routing igmp default no routing igmp
Command Mode	configuration
Parameters Table	N/A
Examples	This example shows how to use this command: Note the change in the command line prompt. exaware# config exaware(config)# routing igmp default exaware(config-routing-igmp-default)#

10.5.1.6. routing igmp interface

Description	Enters routing IGMP configuration mode for a specific interface. Use the no form of the command to disable IGMP routing for the interface.		
Command Syntax	<pre>routing igmp interface <i>interface-name</i> no routing igmp interface <i>interface-name</i></pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>interface-name</i>	{x-eth mgmt} 0/0/ <i>port</i> <i>agg-eth number</i>	Interface name.
Examples	The following example shows how to use this command: Note the change in the command line prompt: <pre>exaware# config exaware(config)# routing igmp interface x-eth 0/0/4 exaware(config-routing-igmp-interface-x-eth-0/0/4) #</pre>		

10.5.1.7. static-group

Description	Specifies the IGMP multicast group address configured statically without IGMP hosts. Used for IGMP joins simulation and for debugging purposes. If the group address is from SSM range – source address needs to be configured. Use the no form of the command to remove the static group configuration from the interface.		
Command Syntax	static-group <group IP address> [source <source IP address>]		
Command Mode	configuration routing igmp interface <x-eth agg-eth>		
Parameters Table	Name	Value	Description
	static-group	Valid multicast IPv4 address	IP address of the multicast group.
	source	Unicast IPv4 address	IP address of the multicast source.
Examples	The following example shows how to use this command globally: <pre>exaware# config exaware(config)# routing igmp default exaware(config-routing-igmp-default)# query-interval 150</pre> The following example shows how to use this command to configure a specific interface: <pre>exaware# config exaware(config)# routing igmp interface x-eth 0/0/0 exaware(config-routing-igmp-interface-x-eth0/0/0)# query-interval 125</pre>		

10.5.2. IGMP Routing Operational Commands

10.5.2.1. clear igmp interface

Description	Clears the IGMP counters at the interface level.		
Command Syntax	<code>clear igmp interface <i>interface-name</i> all</code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>interface-name</i>	<i>x-eth 0/0/port</i> <i>agg-eth number</i>	Interface name.
Examples	The following examples show how to use this command: exaware# clear igmp interface all exaware# clear igmp interface x-eth0/0/0		

10.5.3. IGMP Routing Show Commands

10.5.3.1. show igmp groups

Description	Displays IGMP statistics for groups at the interface level: The options are: - If group is not specified, it shows all groups info. - If group is specified it shows the group info. - If specified interface, it shows all groups under this interface. Displays the following IGMP group membership information: - Group address – group addresses list. - Source address – source addresses list –if not configured will appear as 0.0.0.0. - Interface – Interface list on which this (S,G) pair is configured. - Expire – Indicates the relevance of the source list entry, whereby a non-zero value indicates this is an INCLUDE state value, and a zero value indicates this to be an EXCLUDE state value. - Type – Type of group membership: static or dynamic.			
Command Syntax	show igmp groups [[<group-address>] interface <i>interface-name</i>]			
Command Mode	Operational			
Parameters Table	Name	Value	Description	
	group-address	IP address	IP address of the group.	
	<i>interface-name</i>	x-eth 0/0/ <i>port</i> agg-eth <i>number</i>	Interfaces with configured IGMP.	
Examples	The following examples show sample output from this command: exaware# show igmp groups			
	GROUP ADDRESS	SOURCE ADDRESS	INTERFACE	EXPIRE TYPE
	225.0.0.4	0.0.0.0	x-eth0/0/7	300 dynamic
	232.1.1.1	172.16.1.1	x-eth0/0/1	200 static
			x-eth0/0/3	200 dynamic

232.1.1.2	172.17.1.1	x-eth0/0/3	200	dynamic
	172.17.2.2	x-eth0/3/7	200	static
	172.19.5.1	x-eth0/0/3	200	static
exaware# show igmp groups 232.1.1.1				
GROUP ADDRESS	SOURCE ADDRESS	INTERFACE	EXPIRE	TYPE

232.1.1.1	172.16.1.1	x-eth0/0/1	200	static
		x-eth0/0/3	200	dynamic
	172.17.1.1	x-eth0/0/1	200	dynamic
exaware# show igmp groups interface x-eth0/0/3				
INTERFACE	GROUP ADDRESS	SOURCE ADDRESS	EXPIRE	TYPE

x-eth0/0/3	232.1.1.1	172.16.1.1	200	static
		172.17.1.1	200	dynamic
	232.1.1.2	172.17.1.1	200	static

10.5.3.2. show igmp interface brief

Description	Displays IGMP statistics per interface: - Interface – Interface with IGMP configuration. - Querier - The address of the IGMP or MLD Querier on the IP subnet to which this interface is attached. - Querier Up Time – The time since Querier was last changed. - Reports – The number of reports (joins / leaves) received on this interface. This counter can give an indication of the amount of activity between samples over time. - Groups – The current number of entries for this interface in the Cache Table. - Wrong Version Queries – The number of general queries received whose IGMP or MLD version does not match the equivalent version, over the lifetime of the row entry. Both IGMP and MLD require that all routers on a LAN be configured to run the same version. Thus, if any general queries are received with the wrong version, this indicates a configuration error.				
Command Syntax	show igmp interface brief <i>interface-name</i>				
Command Mode	Operational				
Parameters Table	Name	Value		Description	
	<i>interface-name</i>	<i>x-eth 0/0/port</i> <i>agg-eth number</i>		Interfaces with configured IGMP.	
Examples	The following examples show sample output from this command:				
exaware# show igmp interface brief					
INTERFACE	QUERIER	QUERIER UPTIME	REPORTS	GROUPS	WRONG QUERIES

x-eth0/0/3	192.168.0.1	200	5	2	0
x-eth0/0/4	192.168.5.1	400	2	1	0
x-eth0/0/7	192.168.10.4	300	1	1	0

```
exaware# show igmp interface brief x-eth0/0/3
```

INTERFACE	QUERIER	QUERIER UPTIME	REPORTS	GROUPS	WRONG QUERIES
x-eth0/0/3	192.168.0.1	200	5	2	0

10.5.3.3. show igmp interface detail

Description	Displays IGMP statistics per interface: Configuration parameters: • Query Interval (config) – The interval between General Queries sent by the Querier. • Max Response Time (config) – The maximum query response interval advertised in queries on this interface. • Last Member Query Timeout (config) – How often group-specific query messages are sent. • Robustness factor – Expected packet loss factor. Calculated: • Startup Query Interval – This variable represents the interval between queries sent by a querier on startup. • The value is calculated as $\frac{1}{4}$ of query_interval. • Querier Present Timeout – The time that must pass before a multicast router decides that there is no longer another multicast router which should be the querier. • The value is calculated as $(\text{robustness} * \text{query_interval} + \text{query_response_interval} / 2)$. • Group Membership Timeout – The time without the group queries that must pass before a multicast router decides that there is no memberships on this group. The value is calculated as $(\text{robustness} * \text{query_interval} + \text{query_response_interval})$. Counters and protocol data: • Querier - The address of the IGMP or MLD Querier on the IP subnet to which this interface is attached. • Querier Up Time – The time since Querier was last changed. • Querier Expiry Time – The amount of time remaining before the querier election timeout expires. If the local system is the querier, the value of this object is zero. • Wrong Version Queries – The number of general queries received whose IGMP or MLD version does not match the equivalent version, over the lifetime of the row entry. Both IGMP and MLD require that all routers on a LAN be configured to run the same
--------------------	--

	version. Thus, if any general queries are received with the wrong version, this indicates a configuration error. • Joins – The number of times a group membership has been added on this interface, that is, the number of times an entry for this interface has been added to the Cache Table. This object can give an indication of the amount of activity between samples over time. • Leaves – The number of times a group membership has been removed on this interface, that is, the number of times an entry for this interface has been removed from the IGMP interfaces table. This object can give an indication of the amount of activity between samples over time. • Queries counter – Number of queries sent by the querier. • Groups number – The current number of entries for this interface in the IGMP interfaces table. • Startup Queries – Represents the number of queries sent out on startup, separated by the startup query interval. • Errors (total) ▶ Wrong Version Reports ▶ ASM on SSM range ▶ SSM on ASM range		
Command Syntax	show igmp interface detail [<i>interface-name</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>interface</i> <i>-name</i>	x-eth 0/0/ <i>port</i> agg-eth <i>number</i>	Interfaces with configured IGMP.
Examples	The following example shows sample output from this command:		
<pre>exaware# show igmp interface detail detail interface x-eth 0/0/3 Query interval 60 Max response time 10 Robustness factor 2 Last member query interval 1</pre>			

Startup query interval	15
Querier present timeout	125
Group membership timeout	130
Querier	192.168.0.1
Querier up time	200
Querier expiry time	20
Wrong version queries	0
Joins	3
Leaves	4
Queries	7
Number of groups	2
Startup queries	3
Errors (total)	2
Wrong versions	1
ASM on SSM range	0
SSM on ASM range	1
detail interface x-eth 0/0/4	
Query interval	60
Max response time	10
Robustness factor	3
Last member query interval	1
Startup query interval	15
Querier present timeout	185
Group membership timeout	190
Querier	192.168.5.1
Querier up time	200
Querier expiry time	2
Wrong version queries	0
Joins	3
Leaves	4
Queries	7
Number of groups	2
Startup queries	3
Errors (total)	2
Wrong versions	1
ASM on SSM range	0
SSM on ASM range	1

10.6. IGMP Snooping Commands (Entel)

10.6.1. IGMP Snooping Configuration Commands

10.6.1.1. igmp-snooping

Description	Enable/Disable IGMP Snooping into the given VPLS. Use the no form of this command to restore the default configuration (disable).		
Command Syntax	igmp-snooping disable enable no igmp-snooping		
Command Mode	configuration l2-service vpls		
Parameters Table	Name	Value	Description
	disable		Disable IGMP Snooping.
	enable		Enable IGMP Snooping.
Examples	The following example shows how to use the command: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# igmp-snooping enable</pre>		
Notes	- State disable: - Configuration - All configuration commands igmp-snooping xxx are allowed. - IGMP Snooping dynamic information learned from the IGMP messages is removed. - IP Multicast data including IGMP messages: - IP Multicast data are bridged in accordance with the legacy VPLS rules. - State enable: - IGMP Messages are forwarded in accordance with the IGMP Snooping rules. - IGMP Snooping filtering based on its configuration and its dynamic information learned from IGMP messages is applied to IP Multicast data. - When the state is changed from enable to disable the IGMP configuration is not changed and the IGMP dynamic information is deleted.		

10.6.1.2. igmp-snooping immediate-leave

Description	Enable/Disable the IGMP Snooping Immediate-leave mode. Use the no form of this command to restore the default configuration (disable).		
Command Syntax	igmp-snooping immediate-leave disable enable no igmp-snooping immediate-leave		
Command Mode	configuration l2-service vpls x-eth		
Parameters Table	Name	Value	Description
	disable		Disable IGMP Snooping Immediate-Leave mode.
	enable		Enable IGMP Snooping Immediate-Leave mode.
Examples	The following example shows how to use the command: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# interface x-eth 0/0/3.4 exaware(config-l2-services-vpls-Service1-interface-x-eth-0/0/3.4)# igmp-snooping immediate-leave enable</pre>		
Notes	- o If the option is enabled, then the entry is removed immediately after receiving IGMP leave message. - o The following commands are mutual-exclusive: • igmp-snooping immediate-leave enable • igmp-snooping ipm-subinterface • igmp-snooping router-interface		

10.6.1.3. igmp-snooping ipm-subinterface

Description	Enables/Disables the IGMP Snooping IP Multicast sub-interface mode. Use the no form of this command to restore the default configuration (disable).		
Command Syntax	igmp-snooping ipm-subinterface disable enable no igmp-snooping ipm-sub-interface		
Command Mode	configuration l2-service vpls x-eth-subinterface		
Parameters Table	Name	Value	Description
	disable		Disable IGMP Snooping IP Multicast mode.
	enable		Enable IGMP Snooping IP Multicast mode.
Examples	The following example shows how to use the command: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# interface x-eth 0/0/3.4 exaware(config-l2-services-vpls-Service1-interface-x-eth-0/0/3.4)# igmp-snooping ipm-subinterface enable</pre>		
Notes	- o Only one IP Multicast sub-interface can be configured on the same Ethernet port and the same VPLS. - o The following commands are mutual-exclusive: • igmp-snooping immediate-leave enable • igmp-snooping ipm-subinterface • igmp-snooping router-interface		

10.6.1.4. igmp-snooping query-interval

Description	The query interval is the interval between General Queries sent by the Querier. Use the no form of the command to restore the default value.		
Command Syntax	<pre>igmp-snooping query-interval seconds no igmp-snooping query-interval</pre>		
Command Mode	configuration l2-service vpls		
Parameters Table	Name	Value	Description
	seconds	1..31744	The interval between queries sent by the querier. (Default = 60)
Examples	The following example shows how to use the command: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# igmp-snooping query-interval 150</pre>		

10.6.1.5. igmp-snooping query-response-time

Description	The maximum query response interval advertised in IGMPv3 queries on this interface. Use the no form of the command to restore the default value.		
Command Syntax	<pre>igmp-snooping query-response-time seconds no igmp-snooping query-response-time</pre>		
Command Mode	configuration l2-service vpls		
Parameters Table	Name	Value	Description
	seconds	<0..3174>	The maximum query response interval advertised in IGMPv3 queries on this interface. (Default = 10)
Examples	The following example shows how to use the command: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# igmp-snooping query-response-time 10</pre>		

10.6.1.6. igmp-snooping robustness-factor

Description	The robustness factor allows tuning for the expected packet loss on a network. Use the no form of the command to restore the default value.		
Command Syntax	<code>igmp-snooping robustness-factor <i>number</i></code> <code>no igmp-snooping robustness-factor</code>		
Command Mode	configuration l2-service vpls		
Parameters Table	Name	Value	Description
	number	<2-255>	Expected packet loss factor. (Default = 2)
Examples	The following example shows how to use the command: exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# igmp-snooping robustness-factor 3		

10.6.1.7. igmp-snooping router-interface

Description	Enable/Disable the IGMP Snooping interface as a router's interface. Use the no form of this command to restore the default configuration (disable).		
Command Syntax	igmp-snooping router-interface disable enable no igmp-snooping router-interface		
Command Mode	configuration l2-service vpls x-eth		
Parameters Table	Name	Value	Description
	disable		Specifies the interface as a non-router interface.
	enable		Specifies the interface as a router interface.
Examples	The following example shows how to use the command: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# interface x-eth 0/0/3.4 exaware(config-l2-services-vpls-Service1-interface-x-eth-0/0/3.4)# igmp-snooping router-interface enable</pre>		
Notes	- o The following commands are mutual-exclusive: • igmp-snooping immediate-leave enable • igmp-snooping ipm-subinterface • igmp-snooping router-interface		

10.6.1.8. igmp-snooping static

Description	Create a static IGMP Snooping entry. Use the no form of this command to delete the static entry.		
Command Syntax	<pre>igmp-snooping static group source no igmp-snooping static group source</pre>		
Command Mode	configuration l2-service vpls x-eth		
Parameters Table	Name	Value	Description
	<i>group</i>	IPv4 address	Specify the IPM Destination Group address.
	<i>source</i>	IPv4 address	Specify the IP Source Group address.
Examples	The following example shows how to use the command: <pre>exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# interface x-eth 0/0/3.4 exaware(config-l2-services-vpls-Service1-interface- x-eth-0/0/3.4)# igmp-snooping static 239.1.1.1 10.1.1.0</pre>		
Notes	If the (S,G) entry does not exist, then it is created and the given interface is added.		

10.6.2. IGMP Snooping Operational Commands

10.6.2.1. clear igmp-snooping counters

Description	Clears the IGMP Snooping counters.		
Command Syntax	clear igmp-snooping counters [vpls <i>vpls-name</i> [interface interface-name]]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vpls-name</i>	string	Specify the VPLS name
	interface -name		Specify the interface filter
Examples	The following example shows how to use the command: exaware# clear igmp-snooping counters vpls VPLS1		
Implementation Notes	The filters marked yellow should be implemented in the future.		

10.6.2.2. clear igmp-snooping states

Description	Clears the IGMP Snooping states.		
Command Syntax	clear igmp-snooping states [vpls vpls-name [interface interface-name]]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	vpls-name	string	Specify the VPLS name
	interface -name		Specify the interface filter
Examples	The following example shows how to use the command: exaware# clear igmp-snooping states vpls VPLS1		
Implementation Notes	The filters marked yellow should be implemented in the future.		

10.6.3. IGMP Snooping Show Commands

10.6.3.1. show igmp-snooping counters

Description	Display the IGMP Snooping counters.																														
Command Syntax	show igmp-snooping counters [vpls vpls-name [interface interface-name]]																														
Command Mode	operational																														
Parameters Table	Name	Value	Description																												
	vpls-name	string	Specify the VPLS name																												
	interface -name		Specify the interface filter																												
Examples	The following example shows how to use the command:																														
exaware# show igmp-snooping counters vpls VPLS1																															
<table><tr><td>VPLS NAME</td><td>ATTACHMENT CIRCUIT</td><td>IGMPV3 QUERIES</td><td>IGMPV3 REPORT INCLUDE</td><td>IGMPV3 REPORT EXCLUDE</td><td>IGMPV3 REPORT EXCLUDE X</td><td>IGMPV1&V2 MESSGES</td></tr><tr><td>-----</td><td>-----</td><td>-----</td><td>-----</td><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>VPLS1</td><td>x-eth0/0/1.1</td><td>10</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td></td><td>x-eth0/0/2.1</td><td>0</td><td>10</td><td>6</td><td>0</td><td>0</td></tr></table>				VPLS NAME	ATTACHMENT CIRCUIT	IGMPV3 QUERIES	IGMPV3 REPORT INCLUDE	IGMPV3 REPORT EXCLUDE	IGMPV3 REPORT EXCLUDE X	IGMPV1&V2 MESSGES	-----	-----	-----	-----	-----	-----	-----	VPLS1	x-eth0/0/1.1	10	0	0	0	0		x-eth0/0/2.1	0	10	6	0	0
VPLS NAME	ATTACHMENT CIRCUIT	IGMPV3 QUERIES	IGMPV3 REPORT INCLUDE	IGMPV3 REPORT EXCLUDE	IGMPV3 REPORT EXCLUDE X	IGMPV1&V2 MESSGES																									
-----	-----	-----	-----	-----	-----	-----																									
VPLS1	x-eth0/0/1.1	10	0	0	0	0																									
	x-eth0/0/2.1	0	10	6	0	0																									
Implementation Notes	The filters marked yellow should be implemented in the future.																														

10.6.3.2. show igmp-snooping forwarding-table

Description	Display the IGMP Snooping Forwarding Table.				
Command Syntax	show igmp-snooping forwarding-table [vpls <i>vpls-name</i> [<i>group</i> [<i>source</i>]]]				
Command Mode	operational				
Parameters Table	Name	Value	Description		
	<i>vpls-name</i>	string	Specify the VPLS name		
	<i>group</i>	IP Multicast address	Specified the IP Multicast group		
	<i>source</i>	IP Unicast address	Specify the Source IP address		
Examples	The filters marked yellow should be implemented in the future.				
exaware# show igmp-snooping forwarding-table vpls VPLS1					
Flags: S - Static					
VPLS NAME	MULTICAST GROUP	SOURCES	ATTACHMENT CIRCUITS	FLAGS	IP MULTICAST CIRCUIT
-----	-----	-----	-----	-----	-----
VPLS1	234.1.1.0	3.3.3.3	x-eth0/0/1.2	S	x-eth0/0/1.200
			x-eth0/0/1.3		x-eth0/0/1.200
	234.2.2.2	1.1.1.1	x-eth0/0/1/2		x-eth0/0/1.200
	234.1.1.0	1.1.1.1	x-eth0/0/2.2		
			x-eth0/0/2.3		
	234.2.2.3	1.1.1.1	x-eth0/0/1.2		x-eth0/0/1.200
Implementation Notes	The filters marked by yellow should be implemented in the future.				

10.6.3.3. show igmp-snooping protocol-states

Description	Display the IGMP Snooping Protocol states.		
Command Syntax	show igmp-snooping protocol-states [vpls <i>vpls-name</i> [interface interface-name]]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vpls-name</i>	string	Specify the VPLS name
	<i>interface</i> <i>-name</i>		Specify the interface filter
Examples	The following example shows how to use the command exaware# show igmp-snooping protocol-states vpls VPLS1 Flags: R - Router Interface, M - IP Multicast subinterface , I - Immediate Leave VPLS ATTACHMENT MULTICAST FILTER INCLUDE NAME CIRCUIT GROUP MODE SOURCES FLAGS ----- VPLS1 x-eth0/0/1.1 234.1.1.1 INCLUDE 1.1.1.1 		

10.6.3.4. show igmp-snooping summary

Description	Display the IGMP Snooping summary.																																						
Command Syntax	show igmp-snooping summary [vpls <i>vpls-name</i>]																																						
Command Mode	operational																																						
Parameters Table	Name		Value		Description																																		
	<i>vpls-name</i>		string		Specify the VPLS name																																		
Examples	The following example shows how to use the command																																						
exaware# show igmp-snooping summary vpls VPLS1																																							
FLAGS: R - Router circuit																																							
<table><tr><td>VPLS NAME</td><td>STATUS</td><td>ROBUST VARIABLE</td><td>QUERY INTER</td><td>MAX RES TIME</td><td>GMI</td><td>VPLS CIRCUITS</td><td>FLAGS</td></tr><tr><td>-----</td><td>-----</td><td>-----</td><td>-----</td><td>-----</td><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>VPLS1</td><td>disabled</td><td>2</td><td>125sec</td><td>10sec</td><td>260sec</td><td>x-eth0/0/1.1 x-eth0/0/2.1</td><td>R</td></tr><tr><td>VPLS2</td><td>enabled</td><td>2</td><td>125sec</td><td>10sec</td><td>260sec</td><td>x-eth0/0/1.2 x-eth0/0/3.2</td><td>R</td></tr></table>								VPLS NAME	STATUS	ROBUST VARIABLE	QUERY INTER	MAX RES TIME	GMI	VPLS CIRCUITS	FLAGS	-----	-----	-----	-----	-----	-----	-----	-----	VPLS1	disabled	2	125sec	10sec	260sec	x-eth0/0/1.1 x-eth0/0/2.1	R	VPLS2	enabled	2	125sec	10sec	260sec	x-eth0/0/1.2 x-eth0/0/3.2	R
VPLS NAME	STATUS	ROBUST VARIABLE	QUERY INTER	MAX RES TIME	GMI	VPLS CIRCUITS	FLAGS																																
-----	-----	-----	-----	-----	-----	-----	-----																																
VPLS1	disabled	2	125sec	10sec	260sec	x-eth0/0/1.1 x-eth0/0/2.1	R																																
VPLS2	enabled	2	125sec	10sec	260sec	x-eth0/0/1.2 x-eth0/0/3.2	R																																

10.7. IS-IS Routing Commands

10.7.1. IS-IS Routing Configuration Commands

10.7.1.1. adjacency-check (af-context)

Description	Enables or disables configuring the policy of adjacency based on the protocol-related TLVs in the IS-IS hello packet. Use the no form of the command to set the default (enable).		
Command Syntax	<pre>adjacency-check [enable disable] no adjacency-check</pre>		
Command Mode	configuration routing isis af-ipv4/6 <routing-scheme>		
Parameters Table	Name	Value	Description
		enable	Enable the adjacency-check functionality. (Default)
		disable	Disable the adjacency-check functionality.
Examples	This example shows how to disable adjacency-check functionality: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# af-ipv4 unicast exaware(config-routing-isis-isis1-af-ipv4-unicast)# adjacency-check disable</pre>		
Notes	IS-IS checks adjacency with protocol related TLVs including Protocols Supported TLV or IP Interface Address TLV by default.		

10.7.1.2. af-ipv4

Description	Enter the af-ipv4 configuration sub-mode for IS-IS routing, Use the no form of the command to disable the IS-IS configuration sub-mode.
Command Syntax	<pre>af-ipv4 unicast no af-ipv4 unicast</pre>
Command Mode	<pre>configuration routing isis configuration routing isis interface</pre>
Examples	This example shows how to specify the context for a specific IS-IS instance: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# af-ipv4 unicast exaware(config-routing-isis-isis1-af-ipv4-unicast)#</pre> This example shows how to specify the context for a specific interface for the IS-IS instance: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# af- ipv4 unicast exaware(config-interface-x-eth-0/0/0-af-ipv4-unicast)#</pre>

10.7.1.3. af-ipv6

Description	Enter the af-ipv6 configuration sub-mode for IS-IS routing, Use the no form of the command to disable the IS-IS configuration sub-mode.
Command Syntax	af-ipv6 unicast no af-ipv6 unicast
Command Mode	configuration routing isis configuration routing isis interface
Examples	This example shows how to specify the context for a specific IS-IS instance: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# af-ipv6 unicast exaware(config-routing-isis-isis1-af-ipv6-unicast)#</pre> This example shows how to specify the context for a specific interface for the IS-IS instance: <pre>exaware# config exaware(config)# routing isis isis1 interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# af-ipv6 unicast exaware(config-interface-x-eth-0/0/0-af-ipv6-unicast)#</pre>

10.7.1.4. authentication-key (level 1/2)

Description	Configures the LSP authentication key. Use the no form of the command to disable the authentication password.		
Command Syntax	<code>authentication-key [text hmac-md5] [send-only send-and-receive] snp [send-only send-and-receive]</code>		
Command Mode	configuration routing isis level configuration routing isis interface level		
Parameters Table	Name	Value	Description
	<i>text</i>	String. Use the prefix "\$3\$" to enter an encrypted string	Authentication type is clear text authentication.
	<i>hmac-md5</i>	String. Use the prefix "\$3\$" to enter an encrypted string	Authentication type is HMAC-MD5.
		send-only	Adds authentication to LSP and SNP only when they are sent.
		send-and-receive	Password is authenticated on both received and sent data.
	snp	send-only	Adds authentication to SNP only when they are sent.
	snp	send-and-receive	Password is authenticated on SNP on both received and sent data.
Examples	This example shows how to configure an encrypted password for send only: <pre>exaware# config exaware(config)# routing isis isis1 level-2 exaware(config-routing-isis)# level-2 exaware(config-routing-isis-isis1-level-2)# authentication-key text \$3\$e2fc714c4727ee9395f324cd2e7f331f send-only</pre>		
Notes	The password is always encrypted in the configuration files. The area password must be the same for all the IS-IS routers in one area.		

10.7.1.5. bfd

Description	Enable or disable BFD sessions to the newborns on the given interface. Use the no form of the command to reset to the default state (disable).		
Command Syntax	<pre>bfd enable disable no admin-state</pre>		
Command Mode	<pre>configuration routing isis interface configuration routing isis interface</pre>		
Parameters Table	Name	Value	Description
	enable		A BFD session is used to check the peer state.
	disable		A BFD session is not used to check the peer state.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# bfd enable</pre>		

10.7.1.6. circuit-type

Description	Sets the circuit type for the interface. Use the no form of the command to set the default (level-1-2).		
Command Syntax	<pre>circuit-type level no circuit-type</pre>		
Command Mode	configuration routing isis interface		
Parameters Table	Name	Value	Description
	<i>level</i>	level-1-only	Enables only level-1 adjacencies.
		level-1-2	Enables either level-1 or level-2 adjacencies. (Default)
		level-2-only	Enables only level-2 adjacencies.
Examples	This example shows how to enable level-2 adjacencies: <pre>exaware# config exaware(config)# routing isis isis1 interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# circuit-type level-2-only</pre>		
Notes	If level-1-only or level-2-only is specified in this command, IS-IS sends only the specified level of PDUs. On a point-to point interface, there is only one type of hello packet, so in this case IS-IS hello will be sent regardless of circuit-type. If is-type is configured as either level-1-only or level-2 only, routing for this instance is performed for the specified level only, and only the particular level of PDU is sent on the interface.		

10.7.1.7. csnp-interval

Description	Sets the minimum interval between two consecutive CSNP transmissions. Use the no form of the command to set the default (10 seconds).		
Command Syntax	<code>csnp-interval seconds</code> <code>no csnp-interval</code>		
Command Mode	configuration routing isis interface level		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	The minimum interval between two consecutive CSNP transmissions, in seconds. (Default = 10)
Examples	This example shows how to set the CSNP interval to 60 seconds: <pre>exaware# config exaware(config)# routing isis isis1 interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# network broadcast exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# level-2 exaware(config-interface-x-eth-0/0/0-level-2)# csnp-interval 60</pre>		
Notes	This parameter is only valid on broadcast interfaces (see network), since periodic CSNP is only sent on a broadcast interface. (CSNP on point-to-point interfaces is sent only when adjacency is initiated.)		

10.7.1.8. default-route-originate

Description	Enables distributing reachability information regarding the default destination into LSP. You can also assign a policy to determine default destination distribution. Use the no form of the command to set the default (disable).		
Command Syntax	<pre>default-route-originate [policy <i>policy-name</i>]</pre> <pre>no default-route-originate</pre>		
Command Mode	<pre>configuration routing isis > af-ipv4 unicast</pre> <pre>configuration routing isis > af-ipv6 unicast</pre>		
Parameters Table	Name	Value	Description
	<i>policy-name</i>	existing policy ID	Assigns a policy to determine whether to distribute reachability information regarding the default destination into LSP.
Examples	These examples show how to enable originating the default route and how to enable default route according to the assigned policy. <pre>exaware# config exaware(config)# routing isis isis1 af-ipv4 unicast exaware(config-routing-isis-isis1-af-ipv4-unicast)# default- route-originate</pre> <pre>exaware# config exaware(config)# routing isis isis1 af-ipv4 unicast exaware(config-routing-isis-isis1-af-ipv4-unicast)# default- route-originate policy default-originate-policy1</pre>		
Notes	Configures a default route for the distribution of default information into the respective routing domain. IS-IS creates the default route (0.0.0.0/0) if it does not exist in the IP routing table.		

10.7.1.9. distance

Description	Define the administrative distance. You have the following options: - global: set a global administrative distance - source-sysid: set the administrative distance for routes received from a specific source ID Use the no form of the command to revert to the default settings (115).		
Command Syntax	<pre>distance [global <i>distance1</i>] [source-sysid <i>YYYY.YYYY.YYYY</i> <i>distance2</i>] no distance</pre>		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
	<i>distance1</i>	1-255	Global administrative distance. (Default = 115)
	<i>YYYY.YYYY.YYYY</i>		Specific source ID.
	<i>distance2</i>	1-255	Administrative distance for routes received from the specified source ID. (Default = 115)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# distance global 150</pre>		

10.7.1.10. dynamic-hostname

Description	Enables the dynamic hostname exchange mechanism and system-ID-to-hostname translation. Use the no form of the command to revert to the default setting (global).		
Command Syntax	<pre>dynamic-hostname [global area-tag disable] no dynamic-hostname</pre>		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
		global	Use the global router name as the hostname. (Default)
		area-tag	Use the IS-IS area tag as the hostname.
		disable	Disable dynamic hostname on this IS-IS instance.
Examples	This example shows how to disable the dynamic hostname exchange mechanism and then how to revert to the default, which is enabled and uses the global router name: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# dynamic-hostname disable exaware(config-routing-isis-isis1)# no dynamic-hostname</pre>		

10.7.1.11. fast-reroute

Description	Sets fast-reroute capability for an IGP. Use the no form of the command to set fast-reroute to its default behavior (Disable).		
Command Syntax	fast-reroute {enable disable}		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
		enable	Enable fast-reroute.
		disable	Disable fast-reroute. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing isis default exaware(config-routing-isis-default)# fast-reroute enable exaware(config-routing-isis-default)# no fast-reroute</pre>		

10.7.1.12. graceful-restart grace-interval

Description	Configures the T3 timer interval, which is the length of time that a restarting router retains the forwarding table. Use the no form of the command to configure the default value (600).		
Command Syntax	graceful-restart grace-interval <i>seconds</i>		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
	<i>seconds</i>	0-65535	Length of the grace interval in seconds. (Default = 600)
Examples	This example shows how to set the grace interval to 120 seconds: exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# graceful-restart grace-interval 120		
Notes	The grace interval is the maximum amount of time, in seconds, that the restarting router waits before setting the overload bit to indicate that the LSP database has not been synchronized and the IS-IS graceful restart operation has failed.		

10.7.1.13. graceful-restart helper

Description	Enables or disables the router as a restart helper for a router that is performing a graceful restart. Use the no form of the command to configure the default status (enable).		
Command Syntax	<pre>graceful-restart helper [enable disable] no graceful-restart helper</pre>		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
		<i>enable</i>	The router is a restart helper router. (Default)
		<i>disable</i>	The router is not a restart helper router.
Examples	Both of these commands configure the router to be a restart helper router: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# graceful-restart helper enable exaware(config-routing-isis-isis1)# no graceful-restart helper</pre>		
Notes	A restart helper router for a neighboring router that is performing a graceful restart will maintain the adjacency and send a complete sequence number PDUs (CSNPs) to the restarting router, flooding the entire database.		

10.7.1.14. graceful-restart interface-sync-interval

Description	Configures the interface synchronization interval (T1 timer). Use the no form of the command to set the default (3 seconds).		
Command Syntax	<pre>graceful-restart interface-sync-interval <i>seconds</i> no graceful-restart interface-sync-interval</pre>		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	Interface synchronization interval (T1) in seconds. (Default = 3)
Examples	This example shows how to set the graceful restart interface synchronization interval to 30 seconds: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# graceful-restart interface-sync-interval 30</pre>		

10.7.1.15. graceful-restart interface-sync-retries

Description	Configures the maximum number of synchronization attempts. Use the no form of the command to set the default (1).		
Command Syntax	<pre>graceful-restart interface-sync-retries <i>number</i> no graceful-restart interface-sync-retries</pre>		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
	<i>number</i>	1-3	Maximum number of synchronization attempts. (Default = 1)
Examples	This example shows how to set the number of graceful restart interface synchronization attempts to 3 retries: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# graceful-restart interface-sync-retries 3</pre>		

10.7.1.16. graceful-restart lspdb-level1-sync-interval

Description	Configures the LSP database level 1 synchronization interval (T2 timer). Use the no form of the command to set the default (60 seconds).		
Command Syntax	<pre>graceful-restart lspdb-level1-sync-interval <i>seconds</i> no graceful-restart lspdb-level1-sync-interval</pre>		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
	<i>seconds</i>	0-65535	LSP database synchronization level 1 interval (T2) in seconds. (Default = 60)
Examples	This example shows how to set the graceful restart LSP database synchronization interval to 120 seconds: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# graceful-restart lspdb-level1-sync-interval 120</pre>		

10.7.1.17. graceful-restart lspdb-level2-sync-interval

Description	Configures the LSP database level 2 synchronization interval (T2 timer). Use the no form of the command to set the default (60 seconds).		
Command Syntax	<code>graceful-restart lspdb-level2-sync-interval <i>seconds</i></code> <code>no graceful-restart lspdb-level2-sync-interval</code>		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
	<i>seconds</i>	0-65535	LSP database synchronization level 2 interval (T2) in seconds. (Default = 60)
Examples	This example shows how to set the graceful restart LSP database synchronization interval to 120 seconds: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# graceful-restart lspdb-level2-sync-interval 120</pre>		

10.7.1.18. hello-interval

Description	Sets the time interval between two consecutive hello transmissions. Use the no form of the command to set the default (10 seconds).		
Command Syntax	hello-interval <i>seconds</i> no hello-interval		
Command Mode	configuration routing isis interface level		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	The time interval between two consecutive hello transmissions in seconds. (Default = 10)
Examples	This example shows how to configure a hello interval of 30 seconds: <pre>exaware# config exaware(config)# routing isis isis1 interface x-eth 0/0/0 level-1 exaware(config-interface-x-eth-0/0/0-level-1)# hello-interval 30</pre>		
Notes	DIS sends hello transmissions at three times the rate of non-DIS. If IS-IS is elected as DIS on this interface and the hello interval value is 10 seconds, it sends hello every 3.3 seconds.		

10.7.1.19. hello-multiplier

Description	Sets the multiplier for hello holding time. Use the no form of the command to set the default (3).		
Command Syntax	hello-multiplier <i>value</i> no hello-multiplier		
Command Mode	configuration routing isis <tag> interface <if-name> level		
Parameters Table	Name	Value	Description
	<i>value</i>	2-100	Hello multiplier. (Default = 3)
Examples	This example shows how to configure a hello multiplier of 2: <pre>exaware# config exaware(config)# routing isis isis1 interface loopback 0 level-1 exaware(config-interface-loopback-0-level-1)# hello- multiplier 2</pre>		
Notes	Configures the holding timer in the hello PDU. The holding timer is calculated by multiplying the hello-interval (on page Error! Bookmark not defined.) by this value. After the holding timer has expired without receiving a hello message for the neighbor, the adjacency will break.		

10.7.1.20. hello-padding

Description	Enables or disables padding the IS-IS hello packet. Use the no form of the command to set the default (enable).		
Command Syntax	hello-padding [<i>enable</i> <i>disable</i>] no hello-padding		
Command Mode	configuration routing isis <tag> interface <if-name>		
Parameters Table	Name	Value	Description
		<i>enable</i>	The IS-IS hello packets are padded. (Default)
		<i>disable</i>	The IS-IS hello packets are not padded.
Examples	This example shows how to enable padding: exaware# config exaware(config)# routing isis isis1 interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# hello-padding disable		
Notes	Pads the hello packet to contain the maximum allowed payload.		

10.7.1.21. ignore-attached-bit

Description	Ignores attached bit in L1 LSPs. Use the no form of the command to process attached bit in L1 LSPs.		
Command Syntax	<pre>ignore-attached-bit {enable disable} no ignore-attached-bit</pre>		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
		enable	Ignore attached bit in L1 LSPs.
		disable	Process attached bit in L1 LSPs. (Default = disable)
Examples	This examples shows how to use this command: <pre>exaware# config exaware(config)# routing isis default exaware(config-routing-isis-default)# ignore-attached-bit enable exaware# config exaware(config)# routing isis default exaware(config-routing-isis-default)# no ignore-attached-bit</pre>		

10.7.1.22. interface

Description	Enters routing IS-IS configuration mode for a specific interface for a specified IS-IS instance. You can also configure IS-IS routing for a specific level for a specific interface and IS-IS instance. Use the no form of the command to disable IS-IS routing for the interface or for the level.		
Command Syntax	<pre>interface <i>interface-name</i> no interface <i>interface-name</i></pre>		
Command Mode	configuration isis		
Parameters Table	Name	Value	Description
	<i>interface-name</i>	{x-eth mgmt} 0/0/ <i>port</i> {agg-eth loopback} <i>number</i>	Interface name.
Examples	The following example shows how to use the command: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# interface x-eth 0/0/4</pre>		

10.7.1.23. is-type

Description	Configures the IS-IS adjacency type. The no form of the command disables IS-IS routing on the instance (see Notes.)		
Command Syntax	<code>is-type [level-1-only level-1-2 level-2-only]</code>		
Command Mode	configuration routing isis <tag>		
Parameters Table	Name	Value	Description
	level	<i>level-1-only</i>	Enables only level-1 adjacencies.
		<i>level-1-2</i>	Enables either level-1 or level-2 adjacencies.
		<i>level-2-only</i>	Enables only level-2 adjacencies.
Examples	<pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# is-type level-1-2</pre>		
Notes	There is no default value; this command must be configured in order for the IS-IS instance to be enabled. The no form of the command disables IS-IS routing on the instance, since if no level is configured, IS-IS routing will stop.		

10.7.1.24. level-1

Description	Specifies level 1. Use the no form of the command to disable IS-IS routing for the specific IS-IS instance at level 1.
Command Syntax	level-1 no level-1
Command Mode	configuration routing isis configuration routing isis interface configuration routing isis af-ipv4 unicast configuration routing isis af-ipv6 unicast
Examples	This example shows how to specify level 1 at the IS-IS interface level: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing isis isis1 interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# level-1 exaware(config-interface-x-eth-0/0/0-level-1)#</pre> This example shows how to specify level 1 at the IS-IS context level: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing isis isis1 af-ipv4 unicast exaware(config-routing-isis-isis1-af-ipv4-unicast)# level-1 exaware(config-af-ipv4-unicast-level-1)#</pre>

10.7.1.25. level-2

Description	Specifies level 2. Use the no form of the command to disable IS-IS routing for the specific IS-IS instance at level 2.
Command Syntax	level-2 no level-2
Command Mode	configuration routing isis configuration routing isis interface configuration routing isis af-ipv4 unicast configuration routing isis af-ipv6 unicast
Examples	This example shows how to specify level 2 at the IS-IS interface level: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing isis isis1 interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# level-2 exaware(config-interface-x-eth-0/0/0-level-2)#</pre> This example shows how to specify level 1 at the IS-IS context level: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing isis isis1 af-ipv4 unicast exaware(config-routing-isis-isis1-af-ipv4-unicast)# level-2 exaware(config-af-ipv4-unicast-level-2)#</pre>

10.7.1.26. log-adjacency-changes

Description	Enables or disables writing a message to the log each time an adjacency changes state (up or down). Use the no form of the command to set the default (enable).		
Command Syntax	log-adjacency-changes [<i>enable</i> <i>disable</i>] no log-adjacency-changes		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
		<i>enable</i>	Enable logging adjacency changes. (Default)
		<i>disable</i>	Disable logging adjacency changes.
Examples	This example shows how to use this command: exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# log-adjacency-changes disable		

10.7.1.27. lsp-cs-errors

Description	Configures whether or not the LSP checksum errors should be ignored (permitted). Use the no form of the command to set the default (deny).		
Command Syntax	<pre>lsp-cs-errors [permit deny] no lsp-cs-errors</pre>		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
		permit	Allow checksum errors in the LSP.
		deny	Prohibit checksum errors in the LSP. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# lsp-cs-errors permit</pre>		

10.7.1.28. lsp-fast-flood

Description	Configures the LSP fast flood threshold. Use the no form of the command to set the default (10).		
Command Syntax	<pre>lsp-fast-flood <i>threshold</i> no lsp-fast-flood</pre>		
Command Mode	configuration routing isis interface		
Parameters Table	Name	Value	Description
	<i>threshold</i>	1-4294967295	LSP fast flood threshold. (Default = 10)
Examples	This example shows how to set the threshold to 50: <pre>exaware# config exaware(config)# routing isis isis1 interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# lsp-fast-flood 50</pre>		

10.7.1.29. lsp-gen-interval (level 1/2)

Description	Sets the minimum interval permitted before regenerating the same LSP. Use the no form of the command to set the default (5 seconds).		
Command Syntax	<pre>lsp-gen-interval seconds no lsp-gen-interval</pre>		
Command Mode	configuration routing isis level		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-120	Minimum LSP regeneration interval in seconds. (Default = 5)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing isis isis1 level-2 exaware(config-routing-isis-isis1-level-2)# lsp-gen-interval 60</pre>		
Notes	The smaller the interval, the faster the convergence; however a small interval may cause more frequent flooding.		

10.7.1.30. lsp-interval

Description	Sets the minimum interval between two consecutive LSP transmissions. Use the no command to set the default (33 msec).		
Command Syntax	<pre>lsp-interval <i>milliseconds</i> no lsp-interval</pre>		
Command Mode	configuration routing isis interface		
Parameters Table	Name	Value	Description
	<i>milliseconds</i>	1-4294967295	Minimum LSP transmission interval in milliseconds. (Default = 33)
Examples	This example shows how to set the LSP interval to 80 msec: <pre>exaware# config exaware(config)# routing isis isis1 interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# lsp-interval 80</pre>		
Notes	When flooding or some event triggers transmission of LSP, the LSP will be sent to the interface queue and scheduled to transmit within this interval of time. Two consecutive LSP transmissions are scheduled to have at least this interval between them.		

10.7.1.31. lsp-mtu (level 1/2)

Description	Sets the maximum LSP packet size. Use the no command to set the default (1492 bytes).		
Command Syntax	<code>lsp-mtu bytes</code> <code>no lsp-mtu</code>		
Command Mode	configuration routing isis level		
Parameters Table	Name	Value	Description
	<i>bytes</i>	1492-9175	Maximum LSP packet size in bytes. (Default = 1492)
Examples	This example shows how to set the maximum LSP packet size to 6000 bytes: <pre>exaware# config exaware(config)# routing isis isis1 level-1 exaware(config-routing-isis-isis1-level-1)# lsp-mtu 6000</pre>		

10.7.1.32. lsp-refresh-interval

Description	Sets the LSP refresh interval. Use the no form of the command to set the default value (900 seconds).		
Command Syntax	<pre>lsp-refresh-interval <i>seconds</i> no lsp-refresh-interval</pre>		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	The LSP refresh interval in seconds. (Default = 900)
Examples	This example shows how to set the LSP refresh interval to 1200: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# lsp-refresh-interval 1200</pre>		
Notes	It is recommended to make the LSP refresh interval smaller than the maximum LSP lifetime (see <i>max-lsp-lifetime</i>). Increasing the interval reduces the link utilization caused by flooding of refreshed packets.		

10.7.1.33. max-lsp-lifetime

Description	Sets the maximum LSP lifetime. Use the no form of the command to set the default value (1200 seconds).		
Command Syntax	<code>max-lsp-lifetime <i>seconds</i></code> <code>no max-lsp-lifetime</code>		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
	<i>seconds</i>	350-65535	The maximum LSP lifetime in seconds. (Default = 1200)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# max-lsp-lifetime 2400</pre>		
Notes	It is recommended to make the maximum LSP lifetime larger than the LSP refresh interval (see <i>lsp-refresh-interval</i>).		

10.7.1.34. mesh-group

Description	Does either of the following: • Sets the mesh group ID • Blocks LSPs on the current interface. Use the no form of the command to unblock LSPs.		
Command Syntax	<pre>mesh-group group=id mesh-group blocked no mesh-group</pre>		
Command Mode	configuration routing isis <tag> interface <if-name>		
Parameters Table	Name	Value	Description
	<i>mesh-group</i>	1-4294967295	Mesh group ID.
		blocked	Blocks LSPs on the interface.
Examples	The following example shows how to use this command: <pre>exaware# config exaware(config)# routing isis isis1 interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# mesh-group 12345</pre>		
Notes	If an interface is configured as 'mesh group blocked', the standard LSP database synchronization process is applied when the interface receives CSNP or PSNP.		

10.7.1.35. metric

Description	Sets the default metric. Use the no form of the command to set the default (10).		
Command Syntax	<code>metric value</code> <code>no metric</code>		
Command Mode	configuration routing isis interface level		
Parameters Table	Name	Value	Description
	<code>value</code>	The acceptable range of this value depends on the configured metric-style, as follows: - narrow metric: 1-63 - wide metric: 1-16777214	Metric value. (Default = 10)
Examples	This example shows how to configure a wide metric value: <pre>exaware# config exaware(config)# routing isis isis1 level-2 exaware(config-routing-isis-isis1-level-2)# metric-style wide exaware(config-routing-isis-isis1-level-2)# exit exaware(config-routing-isis-isis1)# interface x-eth 0/0/0 level-2 exaware(config-interface-x-eth-0/0/0-level-2)# metric 478</pre>		
Notes	The interface default metric is put into the following LSPs: - IP reachability information TLVs. - IS reachability information TLVs. The value is used for SPF calculation.		

10.7.1.36. metric-style (level 1/2)

Description	Configures the IS-IS metric style. Use the no form of the command to set the default (narrow).				
Command Syntax	metric-style mode no metric-style				
Command Mode	configuration routing isis <tag> level				
Parameters Table	Name	Value	Description		
	mode	wide	Wide metric style.		
		narrow	Narrow metric style. (Default)		
		wide-transition	Wide metric style with transition to narrow.		
		narrow-transition	Narrow metric style with transition to wide.		
		both	Both wide and narrow metric styles are supported.		
Examples	This example shows how to configure the metric style: exaware# config exaware(config)# routing isis isis1 level-2 exaware(config-routing-isis-isis1-level-2)# metric-style wide				
Notes	The configured metric-style determines the method of TLV encoding and the SPF calculation decision.				
	Metric style	Wide SPF	Wide TLV	Narrow SPF	Narrow TLV
	narrow (default)	off	off	on	on
	narrow- transition	on	off	on	on
	wide	on	on	off	off

	wide- transition	on	on	on	off
	both	on	on	on	on

10.7.1.37. mpls-te

Description	Configures MPLS traffic engineering. You can configure the following parameters: - IS-IS level: specify MPLS traffic engineering at either level 1, level 2, or both - Router ID: Explicitly specify the IP address of the MPLS traffic engineering router Use the no form of the command to disable MPLS traffic engineering.		
Command Syntax	<pre>mpls-te [level {1-only 2-only 1-2}] [router-id <i>ip-address</i>]</pre> <pre>no mpls-te</pre>		
Command Mode	configuration routing isis af-ipv4 unicast		
Parameters Table	Name	Value	Description
		1-only 2-only 1-2	Specify the ISIS level for MPLS traffic engineering.
	<i>ip-address</i>	IPv4 or IPv6 address	MPLS traffic engineering router IP address.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing isis isis1 af-ipv4 unicast exaware(config-routing-isis-isis1-af-ipv4-unicast)# mpls-te level 1-2 router-id 10.10.10.10</pre>		

10.7.1.38. net

Description	Adds a Network Entity Title (NET) for the IS-IS instance. Use the no form of the command to remove a specific NET.		
Command Syntax	<pre>net network-entity-title</pre> <pre>no net network-entity-title</pre>		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
	<i>network-entity-title</i>		The Network Entity Title in the format: xx.xxxx...xxx.YYYY.YYYY.YYYY.00 where: xx.xxxx...xxx = Area Address (The min area is 3 bytes and is constructed of a 1byte AFI and 2byte area ID) YYYY.YYYY.YYYY = System ID (6 bytes) 00 = Selector
Examples	This example shows how to add two net addresses: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# net 10.1010.1111.2222.3333.00</pre>		
Notes	There is no default value; the NET must be configured in order for an IS-IS process to start. The NET defines the ISO address and consists of an area address or ID, a system ID, and a selector. A maximum of three area addresses can be configured. If all the NET addresses are removed, the LSPs will be flushed from the IS-IS database and IS-IS routing will be disabled.		

10.7.1.39. network

Description	Specifies the network type for the interface (not supported on loopback interface). Use the no form of the command to revert to the default network type (broadcast).		
Command Syntax	<pre>network [point-to-point broadcast] no network</pre>		
Command Mode	configuration routing isis interface		
Parameters Table	Name	Value	Description
		<i>point-to-point</i>	Enables IS-IS point-to-point network on the interface.
		<i>broadcast</i>	Enables IS-IS broadcast multi-access network on the interface. (Default)
Examples	This example shows how to set the IS-IS network to point-to-point for an Ethernet interface: <pre>exaware# config exaware(config)# routing isis isis1 interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# network point-to-point</pre>		

10.7.1.40. overload-bit

Description	Sets or unsets the overload-bit in LSPs generated by the router. Use the no form of the command to configure the default (unset).		
Command Syntax	overload-bit [set unset {on-startup <i>delay</i> wait-for-bgp}] no overload-bit		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
		set	Set the overload bit.
		unset	Unset the overload bit. (Default)
		on-startup	Set the overload bit on start up, then clear it after the specified delay interval.
	<i>delay</i>	5-86400	Interval in seconds after which the overload state is exited.
		wait-for-bgp	Wait until converged with BGP to exit overload state.
Examples	This example shows how to set the overload bit on start up and the clear it after 10 minutes: <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# overload-bit on-startup delay 600</pre>		

Notes	If the overload bit is set in the LSPs, the router is not used as a transit router during SPF calculation. The set option causes the router to set the overload bit in its own LSP, so the other routers do not use it as a transit or forwarding router. The router continues to receive LSPs when the overload bit is set. If the on-startup option is specified, the router sets overload bit at startup, but then clears the bit after the specified interval has elapsed.
--------------	--

10.7.1.41. passive

Description	Enables or disables suppressing routing updates on the interface. Use the no form of the command to set the default (disabled).		
Command Syntax	<code>passive [enable disable]</code> <code>no passive</code>		
Command Mode	configuration routing isis interface		
Parameters Table	Name	Value	Description
		<i>enable</i>	Routing updates are suppressed.
		<i>disable</i>	Routing updates are not suppressed. (Default)
Examples	This example shows how to enable padding: <pre>exaware# config exaware(config)# routing isis isis1 interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# passive enable</pre>		

10.7.1.42. pdu-mtu

Description	Configures the MTU size on the interface for hello, PSNP, and CSNP packets. Use the no command to set the default (1492).		
Command Syntax	pdu-mtu <i>bytes</i> no pdu-mtu		
Command Mode	configuration routing isis interface		
Parameters Table	Name	Value	Description
	<i>bytes</i>	1492-9175	Max PDU packet size in bytes. (Default = 1492)
Examples	This example shows how to set the MTU to 5000 bytes: <pre>exaware# config exaware(config)# routing isis isisl interface x-eth 0/0/0 exaware(config-routing-isis-isisl-interface-x-eth-0/0/0)# pdu-mtu 5000</pre>		

10.7.1.43. priority

Description	Sets the priority for LAN DIS election. It sets the priority value in the LAN IS-IS hello PDUs. Use the no form of the command to set the default (64).		
Command Syntax	<pre>priority value no priority</pre>		
Command Mode	configuration routing isis interface level		
Parameters Table	Name	Value	Description
	<i>value</i>	0-127	Priority value. (Default = 64)
Examples	This example shows how to configure a priority of 15: <pre>exaware(config)# routing isis isis1 interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# network broadcast exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# level-2 exaware(config-interface-x-eth-0/0/0-level-2)# priority 15</pre>		
Notes	This command is not valid for point-to-point interfaces (see network) (on page Error! Bookmark not defined.). A higher priority value is preferred in DIS election over lower priority values.		

10.7.1.44. redistribute

Description	Redistributes reachability information from other routing protocols. Use the no form of the command to stop redistribution.		
Command Syntax	<pre>redistribute [bgp ospf connected static] [level-1-only level-2-only level-1-2] [policy <i>policy-name</i>]</pre> <pre>no redistribute</pre>		
Command Mode	<pre>configuration routing isis <tag> af-ipv4 unicast</pre> <pre>configuration routing isis <tag> af-ipv6 unicast</pre>		
Parameters Table	Name	Value	Description
		connected static bgp	Type of protocol to redistribute.
		level-1-only	Redistribute routes into level-1.
		level-1-2	Redistribute routes into level-1 and level 2.
		level-2-only	Redistribute routes into level-2. (Default)
Examples	This example shows how to configure redistribution: <pre>exaware# config exaware(config)# routing isis isis1 af-ipv4 unicast exaware(config-routing-isis-isis1-af-ipv4-unicast)# redistribute static level-1-only</pre>		
Notes	Redistributed IPv4 routes are attached into IP external reachability TLV in LSPs. If <i>metric-style</i> is configured as wide, redistributed routes are attached into Extended IP reachability TLV in LSPs. You can redistribute more than 1 routing type, but only specify 1 routing type per command.		

10.7.1.45. retransmit-interval

Description	Sets the minimum interval between retransmission of LSPs. Use the no form of the command to set the default (5 seconds).		
Command Syntax	<pre>retransmit-interval seconds no retransmit-interval</pre>		
Command Mode	configuration routing isis interface		
Parameters Table	Name	Value	Description
	<i>seconds</i>	0-65535	Interval between retransmission of the same LSP in seconds. (Default = 5)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing isis isis1 interface x-eth 0/0/0 exaware(config-routing-isis-isis1-interface-x-eth-0/0/0)# retransmit-interval 120</pre>		

10.7.1.46. route-leak

Description	Configures the propagation of IS-IS routes between levels. Use the no form of the command to revert to the default behavior (see Note).		
Command Syntax	<pre>route-leak policy <i>policy-name</i> no route-leak</pre>		
Command Mode	<pre>configuration routing isis af-ipv4 unicast level configuration routing isis af-ipv6 unicast level</pre>		
Parameters Table	Name	Value	Description
	<i>policy-name</i>	existing policy name	Policy that determines behavior of propagation of IS-IS routes between levels.
Examples	This example shows how to assign a policy: <pre>exaware# config exaware(config)# routing isis isis1 af-ipv4 unicast level-1 exaware(config-af-ipv4-unicast-level-1)# route-leak policy policy1</pre>		
Notes	If this command is not executed, IS-IS redistributes selected level 1 routes into level 2.		

10.7.1.47. routing isis

Description	Enters routing IS-IS configuration mode. Use the no form of the command to disable IS-IS routing globally or for a specific IS-IS instance.		
Command Syntax	<pre>routing isis [instance-name] no routing isis [instance-name]</pre>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	string - 60 characters or less	Name of the IS-IS instance for which to configure routing. Up to 32 ISIS instances are supported. The default value is string default.
Examples	This example shows how to configure IS-IS routing for IS-IS tag isis1: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)#</pre>		

10.7.1.48. summary-prefix

Description	Configures the summary address to summarize IP reachability information. Use the no form of the command to disable the configured summary address.		
Command Syntax	summary-prefix [<i>address</i> <i>prefix-length</i>] level [<i>level</i>] no summary-prefix		
Command Mode	configuration routing isis af-ipv4 unicast configuration routing isis af-ipv6 unicast		
Parameters Table	Name	Value	Description
	<i>address</i>	IPv4 or IPv6 address	IP address.
	<i>prefix-length</i>	0-32	Prefix length.
	<i>level</i>	1-only 1-2 2-only	Redistribute routes into level-1. Redistribute routes into level-1 and level 2. Redistribute routes into level-2. (Default)
Examples	The following example shows how to use this command: exaware# config exaware(config)# routing isis isis1 af-ipv4 unicast exaware(config-routing-isis-isis1-af-ipv4-unicast)# summary-prefix 10.10.10.10/32 level 1-2		
Notes	If this command is not executed, IS-IS does not configure the summary-address.		

10.7.1.49. suppressed

Description	Suppresses interface address prefixes advertisement. Use the no command to disable interface address prefixes advertisement suppression.		
Command Syntax	<pre>suppressed {enable disable} no suppressed</pre>		
Command Mode	configuration routing isis interface		
Parameters Table	Name	Value	Description
		enable	Enable interface address prefixes advertisement suppression.
		disable	Disable interface address prefixes advertisement suppression. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing isis default interface x-eth 0/0/2 exaware(config-routing-isis-default-interface-x-eth-0/0/2)# suppressed enable</pre>		
Notes	Configuration is available on the following types of interfaces: x-eth, agg-eth, loopback and sub-interface. Default for the command is "disable".		

10.7.1.50. vrf

Description	Specifies VRF for which the IS-IS instance belongs to. Use the no form of the command to restore the default configuration (the default VRF).		
Command Syntax	<code>vrf vrf-name</code> <code>no vrf vrf-name</code>		
Command Mode	configuration routing ospf		
Parameters Table	Name	Value	Description
	<code>vrf-name</code>	Maximum 128 characters, no spaces	Name of the VRF.
Examples	This example shows how to enter configuration mode for the specified VRF: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing isis isis1 exaware(config-routing-isis-isis1)# vrf vrf1 exaware(config-routing-isis-isis1)#</pre>		

10.7.2. IS-IS Routing Operational Commands

10.7.2.1. clear isis database

Description	Clears the IS-IS LSP database either globally or for a specified IS-IS instance.		
Command Syntax	<code>clear isis database [level {1-2,1-only, 2-only}] [tag <i>isis-instance</i>] [purge]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing IS-IS instance name	Specify an IS-IS instance to clear.
	level	1-2 1-only 2-only	Specify an IS-IS level to clear.
		purge	Perform a network-wide purge of LSPs.
Examples	The following example shows how to use this command: <code>exaware# clear isis database level 1-only</code>		

10.7.2.2. clear isis policy default-route-originate

Description	Clears the IS-IS default-route-originate policy counters either globally or for a specified IS-IS instance.		
Command Syntax	<code>clear isis [instance <i>instance-name</i>] policy default-route-originate</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing IS-IS instance name	Specify an IS-IS instance to clear.
Examples	The following example shows how to use this command: <code>exaware# clear isis policy default-route-originate</code>		

10.7.2.3. clear isis policy redistribute

Description	Clears the IS-IS redistribute policy counters either globally or for a specified IS-IS instance.		
Command Syntax	<code>clear isis [instance <i>instance-name</i>] policy redistribute</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing IS-IS instance name	Specify an IS-IS instance to clear.
Examples	The following example shows how to use this command: <code>exaware# clear isis policy redistribute</code>		

10.7.2.4. clear isis policy route-leak

Description	Clears the IS-IS route-leak policy counters either globally or for a specified IS-IS instance.		
Command Syntax	<code>clear isis [instance <i>instance-name</i>] policy route-leak</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing IS-IS instance name	Specify an IS-IS instance to clear.
Examples	The following example shows how to use this command: <code>exaware# clear isis policy route-leak</code>		

10.7.2.5. clear isis process

Description	Clears the IS-IS LSP database and adjacency DB sessions either globally or for a specified IS-IS instance.		
Command Syntax	<code>clear isis [instance <i>instance-name</i>] process</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing IS-IS instance name	Specify an IS-IS instance to clear.
Examples	The following example shows how to use this command: <pre>exaware# clear isis process instance isis1</pre>		

10.7.2.6. clear isis route

Description	Clears the IS-IS routing table either globally or for a specified IS-IS instance.		
Command Syntax	<code>clear isis [instance <i>instance-name</i>] route</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing IS-IS instance name	Specify an IS-IS instance to clear.
Examples	The following example shows how to use this command: <code>exaware# clear isis route instance isis1</code>		

10.7.2.7. clear isis statistics globally

Description	Clears the IS-IS LSP database either globally or for a specified IS-IS instance.		
Command Syntax	<code>clear isis [instance <i>instance-name</i>] statistics global [level-1 level-2]</code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing IS-IS instance name	Specify an IS-IS instance to clear.
		level 1 level 2	IS-IS level-1 global statistics. IS-IS level-2 global statistics. (Default = both)
Examples	The following examples show how to use this command: <code>exaware# clear isis statistics global instance isis1</code> <code>exaware# clear isis statistics interface x-eth 0/0/0</code>		

10.7.2.8. clear isis statistics interface

Description	Clears the IS-IS LSP database either globally or for a specified IS-IS instance.		
Command Syntax	clear isis [instance <i>instance-name</i>] statistics interface [<i>interface-name</i>] [global packets] [level-1 level-2]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing IS-IS instance name	Specify an IS-IS instance to clear.
	level 1 level 2	IS-IS level-1 global statistics. IS-IS level-2 global statistics. (Default = both)	level 1 level 2
Examples	The following examples show how to use this command: <pre>exaware# clear isis statistics global instance isis1</pre> <pre>exaware# clear isis statistics interface x-eth 0/0/0</pre>		

10.7.3. IS-IS Routing Show Commands

10.7.3.1. show isis

Description	Displays IS-IS routing information either globally or for a specified IS-IS instance.		
Command Syntax	show isis [instance <i>instance-name</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing instance name	Instance name
Examples	The following example shows sample output from this command:		
<pre>exaware# show isis instance abc Instance: abc System ID Exaware(0000.0000.0004) IS type: level-1-2 Areas: 47.1234 Interfaces enabled x-eth0/0/1 management0/0/0 Number of active neighbors Broadcast level 1: 0 level 2: 1 Point to point: 0 Number of prefixes IPv4: 2 IPv6: 0 Redistribution enabled IPV4 static routes Administrative distance IPv4: 240 IPv6: 240</pre>			

10.7.3.2. show isis config

Description	Displays IS-IS configuration parameters either globally or for a specified IS-IS instance.		
Command Syntax	show isis [instance <i>instance-name</i>] config [<i>option</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing instance name	Instance name
	<i>option</i>	specific configuration parameter	Query to obtain list of configuration parameter options.
Examples	The following examples show sample output from this command:		
<pre>exaware# show isis instance abc config Instance: abc config system id 0000.0000.0004 (Exaware) is type level-1-2 max lsp lifetime 1200 lsp refresh interval 1112 lsp cs errors deny log adjacency changes enable dynamic hostname global fast reroute enabled ignore attached bit disabled graceful-restart helper enabled grace interval 600 interface sync interval 3 interface sync retries 3 level-1 max lsp sync interval 60 level-2 max lsp sync interval 60 overload-bit status unset level-1 lsp gen interval 5 metric style wide spf-interval initial 50 second 200 max 5000 multiplier 2</pre>			

```
authentication-key enabled
level-2
  lsp gen interval    5
  metric style        wide
  spf-interval initial 50 second 200 max 5000 multiplier 2
  authentication-key enabled
af-ipv4-unicast
  default route originate enabled
  adjacency check      disabled
  level-1 route leak enabled
  level-2 route leak disabled
af-ipv6-unicast
  default route originate disabled
  adjacency check      disabled
  level-1 route leak enabled
  level-2 route leak disabled
```

```
exaware# show isis instance isis1 config ?
```

```
Instance: isis1
Possible completions:
  af-ipv4-unicast
  af-ipv6-unicast
  dynamic-hostname
  graceful-restart
  is-type
  level-1
  level-2
  log-adjacency-changes
  lsp-cs-errors
  lsp-refresh-interval
  max-lsp-lifetime
  overload-bit
  system-id
  |                               Output modifiers
  <cr>
```

```
exaware# show isis instance isis1 config is-type
```

```
Instance: isis1
config is type level-1-only
```

10.7.3.3. show isis database

Description	Displays link state database information either globally or for a specified IS-IS instance. Verbosity level determines the amount of information displayed.				
Command Syntax	show isis [instance <i>instance-name</i>] [level-1 level-2] [brief detail verbose]				
Command Mode	operational				
Parameters Table	Name	Value	Description		
	<i>instance-name</i>	existing instance name	Instance name		
	level	level 1 level 2	Database for level 1. Database for level 2. (Default = both)		
		brief detailed verbose	Database in brief format. Database in detailed format. Database in verbose format. (Default = brief)		
Examples	The following examples show sample output from this command:				
exaware# show isis instance abc database brief					
Instance: abc					
database brief level-1					
LSP ID	SELF	SEQUENCE	CHECKSUM	LIFETIME	ATTRIBUTE

Exaware(0000.0000.0004.00-00)	true	0x000000E8	0x611A	588	[L1 L2]
database brief level-2					
LSP ID	SELF	SEQUENCE	CHECKSUM	LIFETIME	ATTRIBUTE

```
admin(0000.0000.0002.00-00)    false  0x00000003  0x6E11    1194    [ L1 L2 ]
Exaware(0000.0000.0004.00-00)  true   0x000000EA  0x20FE    1192    [ L1 L2 ]
Exaware(0000.0000.0004.01-00)  true   0x00000001  0xB414    1192    [ L1 L2 ]
exaware# show isis instance abc database level-1 detail

Instance: abc
Level 1
LSP ID    Exaware(0000.0000.0004.00-00)
Self      true
Sequence  0x000000E9
Checksum  0x670B
Lifetime  1138 seconds
Attributes [ L1 L2 attached ]
Hostname  Exaware
Area addresses 47.1234
Protocols supported 0xCC(IPv4)
IPv4 interface addresses 10.2.2.4
IP Extended reachability
10.2.2.0/24      25 Up
```

10.7.3.4. show isis interface

Description	Displays IS-IS interface status and configuration either globally or for a specified IS-IS instance. Verbosity level determines the amount of information displayed. (Default = brief)		
Command Syntax	show isis [instance <i>instance-name</i>] interface [brief detail] [<i>interface</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing instance name	Instance name
Examples	The following example shows sample output from this command:		

```
exaware# show isis instance abc interface brief
NETWORK L1 L2 P2P L1 L2 L1 L2
NAME STAT PASSIVE TYPE ADJ ADJ ADJ DIS DIS PTY PTY
-----
loopback1 up enabled - - - - - - - -
loopback2 up enabled - - - - - - - -
loopback5 up enabled - - - - - - - -
loopback3 up enabled - - - - - - - -
loopback0 up enabled - - - - - - - -
x-eth0/1/0.1 up enabled - - - - - - - -
x-eth0/1/0.4 up disabled p2p - - 0 - - - -
x-eth0/1/0.9 up disabled p2p - - 0 - - - -
x-eth0/1/0.7 up disabled p2p - - 0 - - - -
x-eth0/1/0.3 up disabled p2p - - 0 - - - -
x-eth0/1/0.8 up disabled p2p - - 0 - - - -
x-eth0/1/0.6 up disabled p2p - - 0 - - - -
x-eth0/1/0.5 up disabled p2p - - 0 - - - -
x-eth0/1/0.2 up enabled - - - - - - - -
```

```
exaware# show isis interface detail x-eth0/0/1.1001
Instance: abc
interface detail x-eth0/0/1.1001
  BFD                                disabled
  status                            up
  passive                           disabled
  suppressed                         enabled
  network type                       broadcast
  circuit type                       level-1
  local circuit id                   0x05
  extended local circuit id          0x400083E9
  SNPA                               000c.29fd.eab6
  pdu mtu                            1492
  ipv4-address 22.1.1.1/24
  hello padding                      enabled
  retransmit interval                5
  lsp
    interval                         33
    fast flood threshold              10
  level-1
    metric                           10
    wide metric                       10
    priority                          64
    DIS id                           0000.0000.0082.05
    number of active adjacencies      0
    next IIH in                       5s
    hello interval                    10
    hello multiplier                   3
    csnp interval                     10
    restart hello interval             3
    restart hello retries              3
    authentication key                 disabled
    LDP sync status                   enabled
    LDP sync holdtime                  60
```

10.7.3.5. show isis neighbors

Description	Displays IS-IS adjacencies either globally or for a specified IS-IS instance. Verbosity level determines the amount of information displayed. (Default = brief)		
Command Syntax	show isis [instance <i>instance-name</i>] neighbors [brief detail]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing instance name	Instance name
Examples	The following examples show sample outputs from this command:		
<pre>exaware# show isis neighbors brief Instance: abc INTERFACE SYSTEM ID SNPA TYPE STATE HOLDTIME ----- x-eth0/0/0 0000.0000.0002 0019.5b84.7921 L1 up 26s exaware# show isis instance abc neighbors detail Instance: isis1 neighbors detail x-eth0/0/0 admin(0000.0000.0002) SNPA 0019.5b84.7921 adjacency-ID 1 uptime 2m54s ipv4-prefix-count 1 protocol-supported IPv4 info L2 state up holdtime 26s area-addresses 47.5678 ipv4-addresses 10.2.2.2 exaware# show isis instance isis1 neighbors brief Instance: isis1 INTERFACE SYSTEM ID SNPA TYPE STATE HOLDTIME -----</pre>			


```
-----
x-eth0/1/0.102  R7-D-s4 (0010.4255.0007)    ca00.0930.0038  L2    up    8s
x-eth0/1/2.201  IxNet_20_1 (0000.0000.2001) 0000.2354.c4a1  L1    up
22s
                                     L2    up
23s
x-eth0/1/2.202  IxNet_20_2 (0000.0000.2002) 0000.2354.c4a3  L1    up
25s
                                     L2    up
23s
```

10.7.3.6. show isis policy

Description	Displays policies associated with IS-IS configuration either globally or for a specified IS-IS instance. You can show all policies or only the policy assigned to a specific option.		
Command Syntax	<pre>show isis [instance <i>instance-name</i>] policy <i>option</i></pre>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing instance name	Instance name
	<i>option</i>	default-route-originate redistribute route-leak	Show the policy for the specified option only.
Examples	The following example shows sample output from this command: <pre>exaware# show isis policy redistribute Instance: default HIT PROTOCOL POLICY RULE COUNT ----- connected pol rule1 0</pre>		

10.7.3.7. show isis route

Description	Displays IS-IS routing table. Information is displayed either globally or for a specified IS-IS instance.		
Command Syntax	show isis [instance <i>instance-name</i>] route [<i>afi</i>] [all alt]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing instance name	Instance name
	<i>afi</i>	ipv4 ipv6	Show IPv4 routes. Show IPv4 routes.
Examples	The following example shows sample output from this command:		

```
exaware# show isis route ipv4
Flags:  alt - alternate path
```

```
Instance: default
```

PREFIX	TYPE	METRIC	METRIC TYPE	NEXT HOP
1.2.3.4/32	level-2	20	internal	11.1.1.1 x-eth0/0/1.1 11.2.1.1 x-eth0/0/1.2

```
exaware# show isis route ipv4 all
Flags:  alt - alternate path
```

PREFIX	TYPE	METRIC	METRIC TYPE	NEXT HOP
1.2.3.4/32	level-2	20	internal	11.1.1.1 x-eth0/0/1.1 11.2.1.1 x-eth0/0/1.2
1.2.3.4/32	level-2 (alt)	30	internal	12.1.1.1 x-eth0/1/1.1 12.2.1.1 x-eth0/1/1.2

```
exaware# show isis route ipv4 alt
Flags:  alt - alternate path
```

PREFIX	TYPE	METRIC	METRIC TYPE	NEXT HOP
1.2.3.4/32	level-2 (alt)	30	internal	12.1.1.1 x-eth0/1/1.1 12.2.1.1 x-eth0/1/1.2

Notes

Keywords:

all - Shows all routes including alternates.

alt - Shows only alternate routes, if they exist.

10.7.3.8. show isis statistics global

Description	Displays IS-IS global statistics either globally or for a specified IS-IS instance.		
Command Syntax	show isis [instance <i>instance-name</i>] statistics global [level-1 level-2]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing instance name	Instance name
		level 1 level 2	IS-IS level-1 global statistics. IS-IS level-2 global statistics. (Default = both)
Examples	The following example shows sample output from this command:		
<pre>exaware# show isis instance isis1 statistics global Instance: isis1 ISIS Instance 'isis1' Global Statistics Level 1 Link State Packets Purges: 1 Corrupted: 0 Errors Maximum Area Address Mismatches: 0 LSP Database Overflows: 0 Exceed Max LSP Sequence #: 0 General SPF calculation: 9 iSPF calculation: 0 PRC calculation: 0 LSP Sequence number skips: 2 Partition changes: 0 Manual area address delete: 0 Level 2 Link State Packets Purges: 0 Corrupted: 0 Errors Maximum Area Address Mismatches: 0 LSP Database Overflows: 0 Exceed Max LSP Sequence #: 0 General SPF calculation: 16 iSPF calculation: 0 PRC calculation: 0 LSP Sequence number skips: 2 Partition changes: 0 Manual area address delete: 0</pre>			

10.7.3.9. show isis statistics interface

Description	Displays IS-IS interface statistics either globally or for a specified IS-IS instance.		
Command Syntax	show isis [instance <i>instance-name</i>] statistics interface [<i>interface-name</i>] [global packets] [level-1 level-2]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing instance name	Instance name
	<i>interface-name</i>		Interface type and instance. (Default = all)
		global packets	IS-IS interface global statistics. IS-IS interface packets statistics. (Default = both)
		level 1 level 2	IS-IS level-1 global statistics. IS-IS level-2 global statistics. (Default = both)
Examples	The following example shows sample output from this command:		
<pre>exaware# show isis instance isis1 statistics interface Instance: isis1 Interface loopback254 No Global Statistics</pre>			


```

Packets
Level 1
  Rx IIH:      0          Tx IIH:      0
  Rx LSP:      0          Tx LSP:      0
  Rx CSNP:     0          Tx CSNP:     0
  Rx PSNP:     0          Tx PSNP:     0
  Rx Unknown:  0          Tx Unknown:  0
Level 2
  Rx IIH:      0          Tx IIH:      0
  Rx LSP:      0          Tx LSP:      0
  Rx CSNP:     0          Tx CSNP:     0
  Rx PSNP:     0          Tx PSNP:     0
  Rx Unknown:  0          Tx Unknown:  0
Interface x-eth0/1/0.9
Global
Network Type: point-to-point
Level 1
Adjacencies          Total: 0          Changes: 0          Rejected: 0
Errors
  Circuit ID Length Mismatches:      0
  Circuit Maximum Area Address Mismatches: 0
  Invalid Authentication Type:      0
  Invalid Authentication:            0
  Initialization Failures:          0
General
  LAN DIS Changes: 0
Level 2
Adjacencies          Total: 0          Changes: 0          Rejected: 0
Errors
  Circuit ID Length Mismatches:      0
  Circuit Maximum Area Address Mismatches: 0
  Invalid Authentication Type:      0
  Invalid Authentication:            0
  Initialization Failures:          0
General
  LAN DIS Changes:      0
Packets
Level 1
  Rx IIH:      0          Tx IIH:      0
  Rx LSP:      0          Tx LSP:      0
  Rx CSNP:     0          Tx CSNP:     0
  Rx PSNP:     0          Tx PSNP:     0
  Rx Unknown:  0          Tx Unknown:  0
Level 2
  Rx IIH:      0          Tx IIH:     232
  Rx LSP:      0          Tx LSP:      0
  Rx CSNP:     0          Tx CSNP:     0
  Rx PSNP:     0          Tx PSNP:     0
  Rx Unknown:  0          Tx Unknown:  0

```

10.7.3.10. show isis summary

Description	Displays summary IS-IS information either globally or for a specified IS-IS instance.		
Command Syntax	<code>show isis [instance <i>instance-name</i>] summary</code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing instance name	Instance name
Examples	The following example shows sample output from this command: <pre>exaware# show isis summary Instance: default System ID: Exaware(0000.0000.0004) IS type: level-1-2 Areas: 47.1234 Interfaces enabled x-eth0/0/1 management0/0/0 Number of active neighbors Broadcast level 1: 0 level 2: 1 Point to point: 0 Number of prefixes IPv4: 2 IPv6: 0 Redistribution enabled IPV4 static routes Administrative distance IPv4: 240 IPv6: 240</pre>		

10.7.3.11. show isis topology

Description	Displays IS-IS paths to intermediate systems either globally or for a specified IS-IS instance.		
Command Syntax	show isis [instance <i>instance-name</i>] topology [ipv4 ipv6] [level-1 level-2] [<i>system-id</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing instance name	Instance name
	level	level 1 level 2	Database for level 1. Database for level 2. (Default = both)
	<i>system-id</i>	string	System identifier, used to retrieve specific information. (Default = all)
Examples	The following example shows sample output from this command:		
<pre>exaware# show isis instance abc topology Instance: abc Topology ipv4 level-1 SYSTEM ID STATUS METRIC NEXT HOP INTERFACE SNPA ----- - Exaware(0000.0000.0004) self - topology ipv4 level-2 SYSTEM ID STATUS METRIC NEXT HOP INTERFACE SNPA ----- - abc(0000.0000.0002) valid 10 abc(0000.0000.0002) x-eth0/0/0</pre>			

```
0019.5b84.7921  
Exaware(0000.0000.0004) self -
```

10.8. Multicast Routing Commands

10.8.1. Multicast Routing Configuration Commands

10.8.1.1. af-ipv4 (multicast)

Description	Enters the address-family ipv4 mode. Use the no form of the command to remove all af-ipv4 configuration. All relevant parameters will revert to their default values.
Command Syntax	af-ipv4 no af-ipv4
Command Mode	configuration routing-multicast vrf-<vrf-name>
Parameters Table	N/A
Examples	This example shows how to enter the address-family ipv4 mode: exaware# config exaware(config)# routing multicast vrf default exaware(config-routing-multicast-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)#
Notes	The address family is a sub mode for configuration that uses standard IPv4 address prefixes.

10.8.1.2. maximum-routes

Description	Sets the maximum of multicast routes that can be added to a router, and generates an error message when the limit is exceeded. If the threshold parameter is set, a threshold warning message is generated when this threshold is exceeded, and the message continues to occur until the number of multicast routes reaches the limit set by the limit argument. Use the no form of the command to revert to the default maximum value (2147483647).		
Command Syntax	<pre>maximum-routes <i>num-routes</i> [threshold <i>threshold</i>]</pre> <pre>no maximum-routes</pre>		
Command Mode	configuration routing-multicast vrf-< <i>vrf-name</i> > afi-ipv4		
Parameters Table	Name	Value	Description
	<i>num-routes</i>	1- 2147483647	Maximum number of multicast routes that can be added. (Default = 2147483647)
	<i>threshold</i>	percentage of maximum (1-100)	Threshold at which a warning message is generated. (Default = 100)
Examples	This example shows how to configure the maximum and the warning threshold: <pre>exaware# config exaware(config)# routing multicast exaware(config-routing-multicast)# vrf default exaware(config-routing-multicast-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# maximum-routes 10000000 threshold 90</pre>		
Notes	The warning threshold must not exceed the limit.		

10.8.1.3. routing multicast

Description	Enables multicast routing and enters the multicast routing configuration mode. Use the no form of the command to disable multicast routing.
Command Syntax	<code>routing multicast</code> <code>no routing multicast</code>
Command Mode	configuration
Parameters Table	N/A
Examples	This example shows how to enable multicast routing and enter the multicast routing configuration mode: <pre>exaware# config exaware(config)# routing multicast exaware(config-routing-multicast)#</pre>
Notes	This is the main sub-mode for configuring multicast routing.

10.8.1.4. rpf-route

Description	Establishes a static RPF route. Use the no form of the command to clear the specified route.		
Command Syntax	<pre>rpf-route source/prefix-length next-hop ip-address [distance distance]</pre> <pre>no rpf-route source/prefix-length next-hop ip-address</pre>		
Command Mode	configuration routing-multicast vrf-<vrf-name> afi-ipv4		
Parameters Table	Name	Value	Description
	<i>source/prefix-length</i>	IP address / mask	Specifies the multicast source IP address and mask length.
	<i>ip-address</i>	IP address	RPF address for the multicast route. The host IP address can be a directly connected system or a remote system. When it is a remote system, a recursive lookup is done from the unicast routing table to find a directly connected system; the recursive lookup is done up to only one level.

	<i>distance</i>	0-255	Specifies whether a unicast route or multicast static route is used for the RPF lookup. Lower distances have preference. If the multicast static route has the same distance as the other RPF sources, the multicast static route takes precedence. (Default = 0)
Examples	This example shows how to configure a static RPF route: <pre>exaware# config exaware(config)# routing multicast exaware(config-routing-multicast)# vrf default exaware(config-routing-multicast-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# rpf-route 10.10.10.10/32 next-hop 20.20.20.20 distance 150</pre>		

10.8.1.5. ssm-group

Description	To configure a SSM groups prefix, use the ssm-group command. To delete the previous configured SSM groups prefix, use the no form of the command.		
Command Syntax	<code>ssm-group ssm-group-prefix</code>		
Command Mode	configuration routing-multicast vrf <default>		
Parameters Table	Name	Value	Description
	<code>ssm-group-prefix</code>	ipm/len, where ipm – IP Multicast address len – prefix length	SSM groups. Default: IANA prefix 232.0.0.0/8.
Examples	The following example shows how to use this command: <pre>exaware# config exaware(config)# routing multicast exaware(config-routing-multicast)# vrf default exaware(config-routing-multicast-vrf-default)# ssm-group 225.0.0.2/32 exaware(config-routing-multicast-vrf-default)# ssm-group 225.0.0.4/32 exaware(config-routing-multicast-vrf-default)# ssm-group 232.0.0.0/8</pre>		
Notes	• A few ssm-group commands could be configured. • The ssm-group command overrides the default IANA SSM prefix 232.0.0/8. • When the last ssm-group CLI command is deleted, the default configuration is restored.		

10.8.2. Multicast Routing Operational Commands

10.8.2.1. clear multicast routes

Description	Clears entries from the IP multicast routing table.		
Command Syntax	<code>clear multicast routes [af ipv4] [vrf <i>vrf-name</i>] [group {ip <i>ip-address</i> source <i>source-address</i>}]</code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
		af ipv4	Address family.
	<i>vrf-name</i>	default	Only default VRF is currently supported.
	<i>ip-address</i>	IP address	Group IP address.
	<i>source-address</i>	IP address	Source IP address.
Examples	The following example shows how to clear the multicast routing table for the specified address family, vrf, and group: <pre>exaware# clear multicast routes af ipv4 vrf default group {ip 224.1.1.1 source 10.10.10.10}</pre>		
Notes	When this command is used, the MRIB clears the multicast route entries in its multicast route table and removes the entries from the multicast forwarder. The MRIB sends a clear message to the multicast protocols.		

10.8.2.2. clear multicast statistics

Description	Clears multicast-related counters.		
Command Syntax	<pre>clear multicast statistics [af ipv4] [vrf <i>vrf-name</i>] [group {ip <i>ip-address</i> source <i>source-address</i>}]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
		af ipv4	Address family.
	<i>vrf-name</i>	default	Only default VRF is currently supported.
	<i>ip-address</i>	IP address	Group IP address.
	<i>source-address</i>	IP address	Source IP address.
Examples	The following example shows how to clear the multicast-related counters for the specified address family, vrf, and group: <pre>exaware# clear multicast statistics af ipv4 group {ip 224.1.1.1 source 10.10.10.10}</pre>		

10.8.3. Multicast Routing Show Commands

10.8.3.1. show multicast interface

Description	Displays the contents of the MRIB table.																				
Command Syntax	show multicast interface [af ipv4] [vrf <i>vrf-name</i>] [summary]																				
Command Mode	operational																				
Parameters Table	Name	Value	Description																		
		af ipv4	Address family (optional).																		
	<i>vrf-name</i>		Name of existing vrf (optional). (Default = default vrf)																		
		summary	Display only a summary rather than a detailed report.																		
Examples	The following example shows sample output from this command: <pre>exaware# show multicast interface</pre> <table><thead><tr><th>INTERFACE</th><th>UNIQUE ID</th><th>MODULE</th><th>TTL</th><th>LOCAL ADDRESS</th><th>UPTIME</th></tr></thead><tbody><tr><td>x-eth0/0/1</td><td>0</td><td>PIM-SM</td><td>1</td><td>200.1.1.1</td><td>12h56m12s</td></tr><tr><td>x-eth0/0/2</td><td>1</td><td>PIM-SM</td><td>1</td><td>100.1.1.1</td><td>12h56m12s</td></tr></tbody></table>			INTERFACE	UNIQUE ID	MODULE	TTL	LOCAL ADDRESS	UPTIME	x-eth0/0/1	0	PIM-SM	1	200.1.1.1	12h56m12s	x-eth0/0/2	1	PIM-SM	1	100.1.1.1	12h56m12s
INTERFACE	UNIQUE ID	MODULE	TTL	LOCAL ADDRESS	UPTIME																
x-eth0/0/1	0	PIM-SM	1	200.1.1.1	12h56m12s																
x-eth0/0/2	1	PIM-SM	1	100.1.1.1	12h56m12s																

10.8.3.2. show multicast route

Description	Displays the contents of the MRIB table.		
Command Syntax	<pre>show multicast route [af ipv4] [vrf vrf-name] [all summary statistics {global route {ip address [source address]]}]]</pre>		
Command Mode	operational		
Parameters Table	Name	Value	Description
		af ipv4	Address family (optional).
	vrf-name		Name of existing vrf (optional). (Default = default vrf)
		all (default)	Display the contents of the IP multicast routing table, per source and group.
		summary	Display contents of the IP multicast routing table in an abbreviated form, per source and group.
		statistics	Display the route and packet count from the IP multicast routing table. Statistics are displayed either: global for specified route (ip address and optional source address)

Examples

The following examples show sample outputs from this command:

```
exaware# show multicast route
```

GROUP	SOURCE	UPTIME	EXPIRY	MODULE	INTERFACE	FAILURES
233.1.1.1	2.2.2.2	22s	-	PIM-SM	x-eth0/0/1	0
eth0/0/1.3						x-
eth0/0/1.2						x-

```
exaware# show multicast route ipv4 summary
```

GROUP	SOURCE	UPTIME	EXPIRY	MODULE	OUTGOING INTERFACE SIZE
233.1.1.1	2.2.2.2	1m16s	-	PIM-SM	2

```
exaware# show multicast route ipv4 statistics global
```

Total routes	1
Total memory	104
Route limit	2147483647
Route threshold	2147483647

Messages received

No match	483
RPF failure	0
Register	0

Messages sent

No match	483
RPF failure	0
Register	0

Immediate sent update	1
Timed sent update	3
Next poll	39s

```
exaware# show multicast route ipv4 statistics routes 233.1.1.1 2.2.2.2
```

Messages received	
RPF failure	0
Register	0

Messages sent	
RPF failure	0
Register	0
Immediate sent update	1
Timed sent update	3

Register packets	
ACK received	0
NACK received	0
Sent	0

10.8.3.3. show multicast rpf

Description	Displays RPF information for the specified source address.																																														
Command Syntax	show multicast rpf[af ipv4] [vrf <i>vrf-name</i>] [source <i>source-ip</i>] [summary]																																														
Command Mode	operational																																														
Parameters Table	Name	Value	Description																																												
		af ipv4	Address family (optional).																																												
	<i>vrf-name</i>		Name of existing vrf (optional). (Default = default vrf)																																												
	<i>source-ip</i>		Source IP address.																																												
		summary	Summary of multicast RPF table. (Default)																																												
Examples	The following examples show sample outputs from this command:																																														
<pre>exaware# show multicast rpf</pre> <table><thead><tr><th>ROUTE</th><th>NEXT HOP</th><th>INTERFACE</th><th>D/M</th></tr></thead><tbody><tr><td colspan="4">-----</td></tr><tr><td>44.44.44.0/24</td><td>200.1.1.1</td><td>recursive</td><td>0/0</td></tr><tr><td>55.55.55.55/32</td><td>77.77.77.77</td><td>x-eth0/0/1</td><td>0/0</td></tr></tbody></table> <pre>exaware# show multicast rpf af ipv4 vrf default source 44.44.44.1</pre> <table><thead><tr><th>SOURCE IP</th><th>NEXT HOP</th><th>INTERFACE</th><th>ROUTE</th><th>ROUTE ORIGIN</th><th>TYPE</th><th>D/M</th></tr></thead><tbody><tr><td colspan="7">-----</td></tr><tr><td colspan="7">--</td></tr><tr><td>44.44.44.1</td><td>200.1.1.1</td><td>recursive</td><td>44.44.44.0/24</td><td>multicast</td><td>static</td><td>0/0</td></tr></tbody></table>				ROUTE	NEXT HOP	INTERFACE	D/M	-----				44.44.44.0/24	200.1.1.1	recursive	0/0	55.55.55.55/32	77.77.77.77	x-eth0/0/1	0/0	SOURCE IP	NEXT HOP	INTERFACE	ROUTE	ROUTE ORIGIN	TYPE	D/M	-----							--							44.44.44.1	200.1.1.1	recursive	44.44.44.0/24	multicast	static	0/0
ROUTE	NEXT HOP	INTERFACE	D/M																																												

44.44.44.0/24	200.1.1.1	recursive	0/0																																												
55.55.55.55/32	77.77.77.77	x-eth0/0/1	0/0																																												
SOURCE IP	NEXT HOP	INTERFACE	ROUTE	ROUTE ORIGIN	TYPE	D/M																																									

--																																															
44.44.44.1	200.1.1.1	recursive	44.44.44.0/24	multicast	static	0/0																																									

10.9. OSPF Routing Commands

10.9.1. OSPF Routing Configuration Commands

10.9.1.1. area

Description	Specifies an OSPF area. The identifier can be specified as either a decimal value or an IPv4 address.		
Command Syntax	area [<i>decimal</i> <i>ip_address</i>]		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
	<i>decimal</i>	0-4294967295	Area ID as a decimal value.
	<i>ip_address</i>	IPv4 address	Area ID in IPv4 address format.
Examples	This example shows how to use this command: exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10		

10.9.1.1. authentication key

Description	Sets the authentication for the OSPF area/interface or area/virtual-link. Use the no form of the command to disable the authentication password.		
Command Syntax	authentication-key [<i>md5</i> {key-id <i>key password</i> } <i>text</i> <i>null</i>] no authentication-key		
Command Mode	configuration routing ospf vrf area interface configuration routing ospf vrf area virtual-link		
Parameters Table	Name	Value	Description
		<i>md5</i>	Authentication type is MD5.
		<i>text</i>	Authentication type is clear text authentication.
		<i>null</i>	No authentication password.
Examples	This example shows how to configure an encrypted password for send only: <pre> exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf default exaware(config-routing-ospf-ospf1-vrf-default)# area 10.10.10.10 exaware(config-vrf-default-area-10.10.10.10)# interface x-eth 0/0/1 exaware(config-area-10.10.10.10-interface-x-eth-0/0/1)# authentication-key text \$3\$e2fc714c4727ee9395f324cd2e7f331f exaware(config-area-10.10.10.10-interface-x-eth-0/0/1)# exit exaware(config-vrf-default-area-10.10.10.10)# virtual-link 10.3.4.5 exaware(config-vrf-default-area-10.10.10.10-virtual-link- 10.3.4.5)# authentication-key text \$3\$e2fc714c4727ee9395f324cd2e7f44ab </pre>		
Notes	The password is always encrypted in the configuration files. Use the following prefix "\$3\$" in order to enter an encrypted password.		

10.9.1.2. database-filter

Description	Filters outgoing LSAs to an OSPF interface.		
Command Syntax	database-filter [<i>disable</i> <i>enable</i>]		
Command Mode	configuration routing ospf vrf area interface		
Parameters Table	Name	Value	Description
		<i>disable</i>	Do not filter outgoing LSA. (Default)
		<i>enable</i>	Filter outgoing LSA.
	Name	Value	Description
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# database-filter enable </pre>		

10.9.1.3. bfd

Description	Enable or disable BFD sessions to the newborns on the given interface. Use the no form of the command to reset to the default state (disable).		
Command Syntax	<pre>bfd enable disable no admin-state</pre>		
Command Mode	configuration routing ospf vrf interface		
Parameters Table	Name	Value	Description
	enable		A BFD session is used to check the peer state.
	disable		A BFD session is not used to check the peer state.
Examples	This example shows how to use this command: <pre>exaware# config exaware (config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf default exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-routing-ospf-ospf1-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1.6 exaware(config-routing-ospf-ospf1-vrf-vrf1-area-10.10.10.10-interface-x-eth- 0/0/1.6# bfd enable</pre>		

10.9.1.4. dead-interval (interface/virtual-link level)

Description	Sets the time interval, while not receiving packet messages, by which a neighbor is declared down.		
Command Syntax	<code>dead-interval seconds</code>		
Command Mode	configuration routing ospf vrf area interface configuration routing ospf vrf area vitural-link		
Parameters Table	Name	Value	Description
	<code>seconds</code>	1-65535	Dead router detection time in seconds. The default is four times the hello interval, or 40 seconds. As with the hello interval, this value must be the same for all routers and access servers attached to a common network.
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1 exaware(config-area-10.10.10.10-interface-x-eth-0/0/1)# dead- interval 10 exaware(config-area-10.10.10.10-interface-x-eth-0/0/1# exit exaware(config-vrf-default-area-10.10.10.10)# virtual-link 10.3.4.5 exaware(config-vrf-default-area-10.10.10.10-virtual-link- 10.3.4.5)# dead-interval 10 </pre>		

10.9.1.5. default-metric

Description	Specifies a metric for the default summary route sent into a stub or NSSA area.		
Command Syntax	default-metric [<i>value</i>]		
Command Mode	configuration routing ospf vrf area		
Parameters Table	Name	Value	Description
	<i>value</i>	0-16777214	Stub/NSSA advertised default summary metric Metric for the default summary route used for a stub or NSSA. The acceptable value is a 24-bit number. (Default = 1)
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf default exaware(config-routing-ospf-ospf1-vrf-default)# area 10.10.10.10 exaware(config-vrf-default-area-10.10.10.10)# default-metric 20 </pre>		

10.9.1.6. default-route-originate

Description	Used to generate a Type 7 default into the NSSA area. This keyword takes effect only on an NSSA ABR or NSSA Autonomous System Boundary Router (ASBR). If the policy option is used, then the router will advertise an external default route only if the policy condition is met.		
Command Syntax	default-route-originate [[metric <i>metric-value</i>] [metric-type <i>type-value</i>] [always]] [[always] [policy <i>policy-names</i>]]		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
	<i>metric-value</i>	0-16777214	(Optional) OSPF default metric. (Default = 10)
	<i>type-value</i>	1-2	(Optional) OSPF Link State type. External link type associated with the default route advertised in the OSPF routing domain. It can be one of the following values: 1—Type 1 metrics are expressed in the same unit as the OSPF metric. Calculating cost to access external AS includes the distance in the AS. 2—Type 2 metrics are one order of magnitude larger. Calculating cost to access external AS without including the distance in the AS.
	always		Always advertise default-route.

	<i>policy-names</i>	string	Policy name
	Name	Value	Description
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospfl exaware(config-routing-ospf-ospfl)# vrf vrfl exaware(config-routing-ospf-ospfl-vrf-vrfl)# default-route-originate metric 100 metric-type 2</pre>		

Policy Operations:

Command	Tag	prefix	next-hop	rib-metric	rib-route	route-has-mpls-label	Area-id	route-type
default-route-originate	n/a	n/a	n/a	n/a	+	n/a	n/a	n/a

Supported actions per command:

Command	metric	metric-type	tag
default-route-originate	+	+	+

10.9.1.7. distance

Description	Defines OSPF route administrative distances based on route type. Use the no form of the command to set the default.		
Command Syntax	<pre>distance [admin-distance <i>dist</i>] [intra-area <i>dist</i>] [inter-area <i>dist</i>] [external <i>dist</i>]</pre> <pre>no distance</pre>		
Command Mode	configuration ospf vrf		
Parameters Table	Name	Value	Description
	<i>dist</i> (admin-distance)	1-255	OSPF Administrative distance. (Default = 110)
	<i>dist</i> (intra-area)	1-255	Distance for intra-area routes. Sets the distance for all routes within an area. (Default = 110)
	<i>dist</i> (inter-area)	1-255	Distance for inter-area routes. Sets the distance for all routes from one area to another area. (Default = 110)
	<i>dist</i> (external)	1-255	Distance for external routes. Sets the distance for routes from other routing domains, learned by redistribution. (Default = 110)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# distance external 250</pre>		

10.9.1.8. fast-reroute

Description	Sets fast-reroute capability for an IGP.		
	Use the no form of the command to set fast-reroute to its default behavior (Disable).		
Command Syntax	fast-reroute {enable disable}		
Command Mode	configuration routing isis		
Parameters Table	Name	Value	Description
		enable	Enable fast-reroute.
		disable	Disable fast-reroute. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf default exaware(config-routing-ospf-default)# fast-reroute enable exaware(config-routing-ospf-default)# no fast-reroute</pre>		

10.9.1.9. graceful-restart grace-interval

Description	Configures the T3 timer interval, which is the length of time that a restarting router retains the forwarding table. Use the no form of the command to configure the default value (120).		
Command Syntax	graceful-restart grace-interval <i>seconds</i>		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-1800	Length of the grace interval in seconds. (Default = 120)
	Name	Value	Description
Examples	This example shows how to set the grace interval to 10 seconds: exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# graceful-restart grace-interval 10		
Notes	The grace interval is the maximum amount of time, in seconds, that the restarting router waits before setting the overload bit to indicate that the LSP database has not been synchronized and the OSPF graceful restart operation has failed.		

10.9.1.10. graceful-restart helper

Description	Enables or disables the router as a restart helper for a router that is performing a graceful restart. Use the no form of the command to configure the default status (enabled).		
Command Syntax	<pre>graceful-restart helper [<i>enable</i> <i>disable</i>]</pre> <pre>no graceful-restart helper</pre>		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
		<i>enable</i>	The router is a restart helper router. (Default)
		<i>disable</i>	The router is not a restart helper router.
Examples	This command configures the router to be a restart helper router: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# graceful-restart helper disable</pre>		
Notes	A restart helper router for a neighboring router that is performing a graceful restart will maintain the adjacency and send a complete sequence number PDUs (CSNPs) to the restarting router, flooding the entire database.		

10.9.1.11. graceful-restart max-grace-period

Description	Sets the maximum grace period to accept. Use the no form of the command to configure the default status (enabled).		
Command Syntax	graceful-restart max-grace-period <i>seconds</i>		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-1800	The maximum grace period to accept in seconds.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# graceful-restart max-grace-period 200</pre>		

10.9.1.12. graceful-restart state

Description	Enables graceful restart mode. Use the no form of the command to disable graceful restart mode.		
Command Syntax	<pre>graceful-restart [state [enable disable]] no graceful-restart [state]</pre>		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
		<i>enable</i>	Enables graceful restart mode.
		<i>disable</i>	Disables graceful restart mode.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# graceful-restart state enable</pre>		

10.9.1.13. hello-interval (interface/virtual-link level)

Description	Sets the time interval between two consecutive hello transmissions. The value must be the same for all routers and access servers attached to a common network. Use the no form of the command to set the default (10 seconds).		
Command Syntax	<pre>hello-interval <i>seconds</i> no hello-interval</pre>		
Command Mode	<pre>configuration routing ospf vrf area interface configuration routing ospf vrf area virtual-link</pre>		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	The time interval between two consecutive hello transmissions in seconds. (Default = 10)
Examples	This example shows how to configure a hello interval of 30 seconds: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1 exaware(config-area-10.10.10.10-interface-x-eth-0/0/1)# hello-interval 30 exaware(config-area-10.10.10.10-interface-x-eth-0/0/1)# exit exaware(config-vrf-default-area-10.10.10.10)# virtual-link 10.3.4.5 exaware(config-vrf-default-area-10.10.10.10-virtual-link- 10.3.4.5)# hello-interval 30</pre>		
Notes	DIS sends hello transmissions at three times the rate of non-DIS. If OSPF is elected as DIS on this interface and the hello interval value is 10 seconds, it sends hello every 3.3 seconds.		

10.9.1.14. import

Description	Used when the router is an NSSA Area Border Router (ABR) and you want the redistribute command to import routes only into the normal areas, but not into the NSSA area.		
Command Syntax	<code>import [enable disable]</code>		
Command Mode	configuration routing ospf vrf area nssa		
Parameters Table	Name	Value	Description
		<i>enable</i>	Import routes into the normal areas and the NSSA area.
		<i>disable</i>	Import routes only into the normal areas, but not into the NSSA area. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# nssa exaware(config-area-10.10.10.10-nssa)# import disable</pre>		

10.9.1.15. interface

Description	Enters routing OSPF configuration mode for a specific interface for a specified OSPF instance. You can also configure OSPF routing for a specific level for a specific interface and OSPF instance. Use the no form of the command to disable OSPF routing for the interface or for the level.		
Command Syntax	<pre>interface <i>interface-name</i> no interface <i>interface-name</i></pre>		
Command Mode	configuration ospf vrf area interface		
Parameters Table	Name	Value	Description
	<i>interface-name</i>	{x-eth mgmt} 0/0/ <i>port</i> {agg-eth loopback} <i>number</i>	Interface name.
Examples	This example shows how to configure OSPF routing for the following instance: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10 exaware(config-vrf-vrf1-area-0.0.0.10)# interface x-eth 0/0/4 exaware(config-interface-x-eth-0/0/4)#</pre>		

10.9.1.16. interface-bandwidth

Description	This command configures the OSPF interface bandwidth used to calculate the interface default metric. Use the no form of the command to restore the default value. The default value is the actual interface speed.		
Command Syntax	<pre>interface-bandwidth <i>bandwidth</i> [<i>unit</i>]</pre> <pre>no interface-bandwidth</pre>		
Command Mode	configuration routing ospf vrf interface		
Parameters Table	Name	Value	Description
	<i>bandwidth</i>	1kbs-800gbps	Specifies the interface bandwidth in the given units per second.
	<i>unit</i>	bps kbps mbps gbps	Bandwidth unit. (Default = bps)
Examples	This example shows how to use this command: <pre>exaware# config exaware (config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf default exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-routing-ospf-ospf1-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1.6 exaware(config-routing-ospf-ospf1-vrf-vrf1-area-10.10.10.10-interface-x-eth-0/0/1.6)# interface-bandwidth 100 gbps</pre>		
Note	The parameter is used to calculate the interface default metric (see command metric)		

10.9.1.17. log-neighbor-changes

Description	Enables or disables writing a message to the log each time a neighbor changes state (up or down). Use the no form of the command to set the default (enable).		
Command Syntax	<pre>log-neighbor-changes [<i>enable</i> <i>disable</i>]</pre> <pre>no log-neighbor-changes</pre>		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
		<i>enable</i>	Enable logging neighbor changes. (Default)
		<i>disable</i>	Disable logging neighbor changes.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# log-neighbor-changes disable</pre>		

10.9.1.18. max-metric set | unset

Description	Enables OSPF Stub router mode by setting the metric on all links (except of stub links) to maximum value (0xFFFF). OSPF stays in Stub router mode until disabled. Use the no form of the command to disable Stub router mode.		
Command Syntax	<pre>max-metric {set unset} no max-metric</pre>		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
		set unset	Set / unset Stub router mode.
Examples	This example shows how to enable Stub router mode: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf default exaware(config-routing-ospf-ospf1-vrf-default)# max-metric set</pre>		

10.9.1.19. max-metric-on-start

Description	Enables OSPF Stub router mode at system startup by temporarily setting the metric on all links (except of stub links) to maximum value (0xFFFF). OSPF stays in Stub router mode for configured timeout period. When the “wait-for-bgp” option is configured the OSPF Stub router mode will end upon BGP convergence completion event, if such event occurs before the timer expires. Use the no form of the command to disable OSPF Stub router mode at system startup.		
Command Syntax	<pre>max-metric-on-start [timeout] [wait-for-bgp] no max-metric-on-start</pre>		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
	timeout	10-86400	Timeout for Stub router mode. (Default = 600)
	wait-for-bgp		Exit Stub router mode once BGP is converged.
Examples	This example shows how to enable OSPF Stub router mode at system startup for 800 seconds with the wait-for-bgp option: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf default exaware(config-routing-ospf-ospf1-vrf-default)# max-metric-on-start 800 wait-for-bgp</pre>		
Notes	The Stub router mode is not activated upon process restart (only upon system restart).		

10.9.1.20. metric

Description	Specifies the metric of the interface. Use the no form of the command to restore the default value.		
Command Syntax	metric <i>value</i> no metric		
Command Mode	configuration routing ospf vrf area interface configuration routing ospf vrf area virtual-link		
Parameters Table	Name	Value	Description
	<i>value</i>	1-65535	Metric value. Default = ○ Interface: • MAX(1, reference-bandwidth % interface-bandwidth) ○ virtual link: • 1
Examples	This example shows how to configure a wide metric value: <pre> exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# metric 10 exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-routing-ospf-ospf1-vrf-vrf1-area- 10.10.10.10)# virtual-link 18.1.1.1 exaware(config-routing-ospf-ospf1-vrf-vrf1-area-10.10.10.10- virtual-link-18.1.1.1)# metric 100 </pre>		

10.9.1.21. mtu

Description	Configures the OSPF mtu per interface. Use the no form of the command to set the default (1500).		
Command Syntax	mtu <i>value</i> no mtu		
Command Mode	configuration routing ospf vrf area interface		
Parameters Table	Name	Value	Description
	<i>value</i>	An integer into range 594..9200.	The size in bytes of the largest IP datagram that can be sent out the associated interface, without fragmentation.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf default exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10 exaware(config-ospf-ospf1-vrf-vrf1-area-0.0.0.10)# interface x-eth 0/0/4 exaware(config-config-ospf-ospf1-vrf-vrf1-area-0.0.0.10- interface-x-eth-0/0/4)# mtu 1000</pre>		

10.9.1.22. mtu-ignore

Description	Configures prevention of MTU checks on OSPF adjacency establishment. Use the no form of the command to set the default (disable).		
Command Syntax	mtu-ignore enable disable no mtu-ignore		
Command Mode	configuration routing ospf vrf area interface		
Parameters Table	Name	Value	Description
	enable		Disables checking for whether OSPF neighbors are using the MTU on a common interface. The OSPF instance sends the zero value of MTU into the Database Description packet.
	disable		Enables checking for whether OSPF neighbors are using the MTU on a common interface.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf default exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10 exaware(config-ospf-ospf1-vrf-vrf1-area-0.0.0.10)# interface x-eth 0/0/4 exaware(config-config-ospf-ospf1-vrf-vrf1-area-0.0.0.10- interface-x-eth-0/0/4)# mtu-ignore enable</pre>		

10.9.1.23. multi-area-adjacency

Description	Configures an interface to be belong to more than one area.		
Command Syntax	multi-area-adjacency [interface <i>interface-name</i>] [neighbor <i>ip-address</i>]		
Command Mode	configuration ospf vrf area		
Parameters Table	Name	Value	Description
	<i>interface-name</i>	x-eth	Interface name and parameters.
	<i>ip-address</i>	IP address	OSPF neighbor IP address.
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10 exaware(config-vrf-vrf1-area-0.0.0.10)# multi-area-adjacency interface x-eth 0/0/4 neighbor 100.1.1.1 </pre>		

10.9.1.24. network-type

Description	Specifies the network type for the interface (not supported on loopback interface). Use the no form of the command to revert to the default network type (broadcast).		
Command Syntax	<pre>network-type broadcast point-to-point no network-type</pre>		
Command Mode	configuration routing ospf vrf area interface		
Parameters Table	Name	Value	Description
		broadcast	Enables OSPF broadcast multi-access network on the interface. (Default)
		point-to-point	Enables OSPF point-to-point network on the interface.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10 exaware(config-vrf-vrf1-area-0.0.0.10)# interface x-eth 0/0/4 exaware(config-interface-x-eth-0/0/4)# network-type point-to-point</pre>		

10.9.1.25. nssa

Description	Configures an area as a not-so-stubby area (NSSA).
Command Syntax	nssa
Command Mode	configuration routing ospf vrf area
Parameters Table	N/A
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# nssa</pre>
Notes	An NSSA does not flood Type 5 external LSAs from the core into the area, but can import autonomous system external routes in a limited fashion within the area.

10.9.1.26. passive (interface)

Description	Stops sending outgoing hello packets, hence the router cannot form any neighbor relationships via the passive interface. This behavior stops both outgoing and incoming routing updates.		
Command Syntax	<code>passive [<i>disable</i> <i>enable</i>]</code>		
Command Mode	configuration ospf vrf area interface		
Parameters Table	Name	Value	Description
		<i>disable</i> <i>enable</i>	Stops outgoing and incoming routing updates. (Default) Allows sending routing updates.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10 exaware(config-vrf-vrf1-area-0.0.0.10)# interface x-eth 0/0/4 exaware(config-interface-x-eth-0/0/4)# passive enable</pre>		

10.9.1.27. priority (interface level)

Description	Sets the priority for LAN DIS election. It sets the priority value in the LAN OSPF hello PDUs. Use the no form of the command to set the default (1).		
Command Syntax	priority value no priority		
Command Mode	configuration routing ospf vrf [neighbor / area interface]		
Parameters Table	Name	Value	Description
	value	0-255	Priority value. (Default = 1)
Examples	This example shows how to configure a priority of 15: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10 exaware(config-vrf-vrf1-area-0.0.0.10)# interface x-eth 0/0/4 exaware(config-interface-x-eth-0/0/4)# priority 15</pre>		
Notes	This command is not valid for point-to-point interfaces (see network) (on page Error! Bookmark not defined.). A value of 0 will prevent the router from be elected as a DR or BDR. A higher priority value is preferred in DIS election over lower priority values.		

10.9.1.28. range advertise/suppress

Description	Consolidates and summarizes routes at an area boundary by an Area Border Router (ABR).		
Command Syntax	range [<i>prefix/prefix-length</i>] [<i>advertise</i> <i>suppress</i>]		
Command Mode	configuration routing ospf vrf area		
Parameters Table	Name	Value	Description
	<i>prefix</i>	IPv4 address	Area range IPv4 address.
	<i>prefix-length</i>	numeric	IPv4 address mask length.
		<i>advertise</i> <i>suppress</i>	Advertise the range. (Default) Suppress the range.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# range 10.10.10.10/32 suppress</pre>		

10.9.1.29. redistribute

Description	Redistributes reachability information from other routing protocols. If the policy option is used, then the routes from other protocols will be advertised into the OSPF domain only if the policy conditions are met. Use the no form of the command to stop redistribution.		
Command Syntax	<pre>redistribute {<i>bgp</i> <i>connected</i> <i>isis</i> <i>static</i>} [[metric <i>metric-value</i>] [metric-type <i>type-value</i>]] [policy <i>policy-names</i>]</pre> <pre>no redistribute</pre>		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
		bgp connected isis static	Type of protocol to redistribute.
	<i>metric-value</i>	0-16777214	OSPF default metric. (Default = 10)

	<i>type-value</i>	1-2	OSPF Link State type. External link type associated with the default route advertised in the OSPF routing domain. It can be one of the following values: 1—Type 1 metrics are expressed in the same unit as the OSPF metric. Calculating cost to access external AS includes the distance in the AS. 2—Type 2 metrics are one order of magnitude larger. Calculating cost to access external AS without including the distance in the AS.
--	-------------------	-----	---

	<i>policy-names</i>	string	Policy name
Examples	This example shows how to configure redistribution: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# redistribute static policy OSPF-redistribute</pre>		
Notes	You can redistribute more than one routing type, but you can only specify one routing type per command.		

Policy Operations:

Command	Tag	prefix	next-hop	rib-metric	rib-route	route-has-mpls-label	Area-id	route-type
redistribute	+	+	+	+	n/a	+	n/a	+

Supported actions per command:

Command	metric	metric-type	tag
redistribute	+	+	+

10.9.1.30. reference-bandwidth

Description	Configures the OSPF reference bandwidth per interface. Use the no form of the command to set the default.		
Command Syntax	reference-bandwidth <i>bandwidth</i> [<i>unit</i>] no reference-bandwidth		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
	<i>bandwidth</i>	1kbs-800gbps	Bandwidth specified in the given units per second. (Default = 100Mbps)
	<i>unit</i>	bps kbps mbps gbps	Bandwidth unit. (Default = bps)
Examples	This example shows how to use this command: exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1# reference-bandwidth 10		
Note	The parameter is used to calculate the interface default metric (see command metric)		

10.9.1.31. retransmit-interval

Description	Sets the time between link-state advertisement (LSA) retransmissions for adjacencies belonging to the interface. Expect round-trip delay between any two routers on the attached network. The value must be greater than the expected round-trip delay.		
Command Syntax	<code>retransmit-interval <i>seconds</i></code>		
Command Mode	configuration routing ospf vrf area interface configuration routing ospf vrf area virtual-link		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-3600	LSA retransmit interval in seconds. (Default = 5)
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1 exaware(config-area-10.10.10.10-interface-x-eth-0/0/1)# retransmit-interval 100 exaware(config-area-10.10.10.10-interface-x-eth-0/0/1)# exit exaware(config-vrf-default-area-10.10.10.10)# virtual-link 10.3.4.5 exaware(config-vrf-default-area-10.10.10.10-virtual-link- 10.3.4.5)# retransmit-interval 100 </pre>		

10.9.1.32. router-id

Description	Sets a fixed router ID.		
Command Syntax	<code>router-id <i>ip-address</i></code>		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 address	Router-id in IPv4 address format.
Examples	This example shows how to configure a wide metric value: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# router-id 10.10.10.10</pre>		

10.9.1.33. routing ospf

Description	Enters routing OSPF configuration mode. Use the no form of the command to disable OSPF routing globally or for a specific IS-IS instance.		
Command Syntax	<pre>routing ospf [instance-name] no routing ospf [instance-name]</pre>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	string - 60 characters or less	Name of the OSPF instance for which to configure routing. Up to 128 OSPF instances are supported. The default value is string default.
Examples	This example shows how to configure OSPF routing for OSPF tag ospf1: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)#</pre>		

10.9.1.34. spf-interval

Description	Configures the delay time between when OSPF receives a topology change and when it starts a shortest path first (SPF) calculation, and the hold time between two consecutive SPF calculations.		
Command Syntax	spf-interval [min <i>spf-delay-min</i>] [max <i>spf-delay-max</i>]		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
	<i>spf-delay-min</i>	0-2147483647	Min delay time (in milliseconds) between when OSPF receives a topology change and when it starts an SPF calculation. (Default = 50)
	<i>spf-delay-max</i>	0-2147483647	Max delay time (in milliseconds) between when OSPF receives a topology change and when it starts an SPF calculation. (Default = 50000)
Examples	This example shows how to configure a wide metric value: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# spf-interval min 100 max 100000</pre>		

10.9.1.35. stub suppress-summary

Description	Blocks the flooding from AS external advertisements in OSPF and injects a default router into a stub area.		
Command Syntax	stub suppress-summary [<i>enable</i> <i>disable</i>]		
Command Mode	configuration routing ospf vrf area		
Parameters Table	Name	Value	Description
		<i>enable</i>	Advertise the range, and the stub area becomes a totally stub area.
		<i>disable</i>	Suppress the range. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# stub suppress-summary enable</pre>		
Notes	Does not inject inter-area routes into stub. Prevents an Area Border Router (ABR) from sending summary link advertisements into the stub area.		

10.9.1.36. summary-prefix

Description	Configures the summary address to summarize IP reachability information. This command summarizes only routes from other routing protocols that are being redistributed into OSPF. Use the no form of the command to disable the configured summary address.		
Command Syntax	<pre>summary-prefix <i>address/prefix-length</i> [suppress advertise] no summary-prefix</pre>		
Command Mode	configuration routing ospf vrf		
Parameters Table	Name	Value	Description
	<i>address</i>	IPv4 address	Summary address designated for a range of addresses.
	<i>prefix-length</i>	0-32	Prefix length.
		suppress advertise	Suppress routes that match the specified prefix/mask pair. Advertise routes that match the specified prefix/mask pair. (Default)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# summary-prefix 10.10.10.10/32</pre>		
Notes	If this command is not executed, OSPF does not configure the summary-address.		

10.9.1.37. suppress-summary

Description	Sets summary LSA into NSSA.		
Command Syntax	suppress-summary [<i>enable</i> <i>disable</i>]		
Command Mode	configuration routing ospf vrf area nssa		
Parameters Table	Name	Value	Description
		<i>enable</i>	Do not send summary LSA into NSSA.
		<i>disable</i>	Send summary LSA into NSSA.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# nssa exaware(config-area-10.10.10.10-nssa)# suppress-summary disable</pre>		

10.9.1.38. translator-role

Description	Specifies whether a router will perform Type 7-to-5 translation.		
Command Syntax	translator-role [<i>always</i> <i>candidate</i> <i>never</i>]		
Command Mode	configuration routing ospf vrf area nssa		
Parameters Table	Name	Value	Description
		<i>always</i>	NSSA-ABR Translator role.
		<i>candidate</i>	Candidate for translator.
		<i>never</i>	Do not translate.
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# nssa exaware(config-area-10.10.10.10-nssa)# translator-role never </pre>		

10.9.1.39. transmit-delay

Description	Sets the estimated time (in seconds) required to send a link-state update packet on the interface. LSAs in the update packet have their age incremented by this amount before transmission.		
Command Syntax	<code>transmit-delay <i>seconds</i></code>		
Command Mode	configuration routing ospf vrf area interface configuration routing ospf vrf area virtual-link		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-3600	LSA transmission delay in seconds. (Default = 1)
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1 exaware(config-area-10.10.10.10-interface-x-eth-0/0/1)# transmit-delay 100 exaware(config-area-10.10.10.10-interface-x-eth-0/0/1)# exit exaware(config-vrf-default-area-10.10.10.10)# virtual-link 10.3.4.5 exaware(config-vrf-default-area-10.10.10.10-virtual-link- 10.3.4.5)# transmit-delay 100 </pre>		

10.9.1.40.virtual-link

Description	To define an Open Shortest Path First (OSPF) virtual link and enter routing OSPF configuration mode for this virtual link, use the virtual-link command. To remove a virtual link, use the no form of this command.		
Command Syntax	<pre>virtual-link router-id no virtual-link router-id</pre>		
Command Mode	configuration ospf vrf area		
Parameters Table	Name	Value	Description
	router-id	IPv4 address	Router ID associated with the virtual link neighbor. The router ID appears in the show ospf command display.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10 exaware(config-routing-ospf-ospf1-vrf-vrf1-area-0.0.0.10)# virtual-link 10.3.4.5 exaware(config-routing-ospf-ospf1-vrf-vrf1-area-0.0.0.10- virtual-link-10.3.4.5)#</pre>		

10.9.1.41. vrf

Description	Enters routing OSPF configuration mode for a specified VRF. Use the no form of the command to disable OSPF routing for the specified VRF.		
Command Syntax	<pre>vrf vrf-name no vrf vrf-name</pre>		
Command Mode	configuration routing ospf		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Maximum 128 characters, no spaces	Name of the VRF for which to configure OSPF routing. A default VRF is pre-configured in the system. To configure a user-defined VRF for OSPF routing, you must first define it. See vrf (on page 1494).
Examples	This example shows how to enter configuration mode for the specified VRF: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)#</pre>		

10.9.2. OSPF Operational Commands

10.9.2.1. clear ospf process

Description	Clears the OSPF LSP database and adjacency DB sessions either globally or for a specified OSPF tag.		
Command Syntax	<code>clear ospf [instance <i>instance-name</i>] process</code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing OSPF instance name	Specify an OSPF instance to clear.
Examples	The following example shows how to use this command: <code>exaware# clear ospf instance ospf1 process</code>		

10.9.2.2. clear ospf route

Description	Clears the OSPF routing table. You have the following options: • Clear the LSP database for a specified OSPF tag • Clear the LSP database for a specified address family and routing scheme		
Command Syntax	<code>clear ospf [instance <i>instance-name</i>] route</code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing OSPF instance name	Specify an OSPF instance to clear.
Examples	The following example shows how to use this command: <code>exaware# clear ospf instance ospf1 route</code>		

10.9.2.3. clear ospf statistics (interface)

Description	Clears the OSPF LSP database.		
Command Syntax	<pre>clear ospf[instance <i>instance-name</i>] statistics interface [<i>interface-name parameter</i>]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing OSPF instance name	Specify an OSPF instance to clear.
	<i>interface-name</i>	agg-eth x-eth all	Specify an OSPF interface port for which to clear statistics.
	<i>parameter</i>	x/y/z	
Examples	The following examples show how to use this command: <pre>exaware# clear ospf instance ospf1 statistics interface x-eth 0/0/1</pre>		

10.9.2.4. clear ospf statistics (virtual-link)

Description	Clears the OSPF LSP database.		
Command Syntax	<pre>clear ospf statistics virtual-link [{name <i>virtual-link-name</i>} {transit-area <i>area</i> remote-id <i>router-id</i>}]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>name</i>	string	The virtual link name assigned by the system (see command show ospf virtual-link).
	<i>area</i>	IPv4 Address or integer	The transit area
	<i>remote-id</i>	IPv4 Address	The router-id of the remote router.
Examples	The following examples show how to use this command: <pre>exaware# clear ospf statistics virtual-link name VLINK2 exaware# clear ospf statistics virtual-link transit-area 0.0.02. router-id 10.1.1.1</pre>		

10.9.3. OSPF Show Commands

10.9.3.1. show ospf border-routers

Description	Displays OSPF routing table entries to an Area Border Router (ABR) and Autonomous System Boundary Router (ASBR). Include the tag ID parameter to view data about specified instances.																																																						
Command Syntax	show ospf [tag tag-id] border-routers																																																						
Command Mode	Operational																																																						
Parameters Table	Name	Value	Description																																																				
	tag-id	<0-65535>	Tag ID. If this argument is included, only information for the specified routing tag is displayed.																																																				
Examples	The following example shows sample output from this command:																																																						
<pre>exaware# show ospf tag 1 border-routers</pre> <table><thead><tr><th>TYPE</th><th>ROUTER ID</th><th>METRIC</th><th>NEXTHOP</th><th>INTERFACE</th><th>ROUTER TYPE</th><th>AREA ID</th></tr></thead><tbody><tr><td>INTRA</td><td>10.10.10.1</td><td>50</td><td>106.1.1.1</td><td>VLINK1</td><td>ABR</td><td>0.0.0.0</td></tr><tr><td>INTRA</td><td>1.2.3.4</td><td>1</td><td>101.1.2.3</td><td>x-eth0/0/1.10</td><td>ABR</td><td>0.0.0.0</td></tr><tr><td>INTRA</td><td>52.52.52.52</td><td>1</td><td>102.1.1.1</td><td>x-eth0/0/0.10</td><td>ABR, ASBR</td><td>0.0.0.1</td></tr><tr><td></td><td></td><td></td><td>103.1.1.17</td><td>x-eth0/0/0.11</td><td>ABR, ASBR</td><td>0.0.0.1</td></tr><tr><td></td><td></td><td></td><td>104.1.1.1</td><td>x-eth0/0/0.12</td><td>ABR, ASBR</td><td>0.0.0.1</td></tr><tr><td></td><td></td><td></td><td>105.1.1.1</td><td>x-eth0/0/0.13</td><td>ABR, ASBR</td><td>0.0.0.1</td></tr></tbody></table>							TYPE	ROUTER ID	METRIC	NEXTHOP	INTERFACE	ROUTER TYPE	AREA ID	INTRA	10.10.10.1	50	106.1.1.1	VLINK1	ABR	0.0.0.0	INTRA	1.2.3.4	1	101.1.2.3	x-eth0/0/1.10	ABR	0.0.0.0	INTRA	52.52.52.52	1	102.1.1.1	x-eth0/0/0.10	ABR, ASBR	0.0.0.1				103.1.1.17	x-eth0/0/0.11	ABR, ASBR	0.0.0.1				104.1.1.1	x-eth0/0/0.12	ABR, ASBR	0.0.0.1				105.1.1.1	x-eth0/0/0.13	ABR, ASBR	0.0.0.1
TYPE	ROUTER ID	METRIC	NEXTHOP	INTERFACE	ROUTER TYPE	AREA ID																																																	
INTRA	10.10.10.1	50	106.1.1.1	VLINK1	ABR	0.0.0.0																																																	
INTRA	1.2.3.4	1	101.1.2.3	x-eth0/0/1.10	ABR	0.0.0.0																																																	
INTRA	52.52.52.52	1	102.1.1.1	x-eth0/0/0.10	ABR, ASBR	0.0.0.1																																																	
			103.1.1.17	x-eth0/0/0.11	ABR, ASBR	0.0.0.1																																																	
			104.1.1.1	x-eth0/0/0.12	ABR, ASBR	0.0.0.1																																																	
			105.1.1.1	x-eth0/0/0.13	ABR, ASBR	0.0.0.1																																																	
Name			Value		Description																																																		
TYPE			INTRA/INTER		INTRA - Intra-area route INTER - Inter-area route																																																		
ROUTER ID			A.B.C.D		Destination router ID																																																		
METRIC					Destination router metric																																																		

NEXTHOP		Next hop IPv6 address toward destination
INTERFACE		Next hop interface name or Virtual Link name assigned by the system.
ROUTER TYPE	ABR/ASBR	Destination router type
AREA ID	A.B.C.D	Decimal IP address

10.9.3.2. show ospf database

Description	Displays link state database information either globally or for a specified OSPF instance.							
Command Syntax	show ospf [instance <i>instance-name</i>] database [adv-router self-originate] [<i>brief</i> <i>detail</i>]							
Command Mode	operational							
Parameters Table	Name	Value	Description					
	<i>instance-name</i>	existing ospf instance	Specify an OSPF instance to display.					
	self-originate		Database for level 1. Database for level 2. (Default = both)					
		<i>brief</i> <i>detail</i>	Database in brief format. (Default) Database in detailed format.					
Examples	The following example shows sample output from this command:							
exaware# show ospf instance 1 database brief								
Instance: 1 VFR: default OSPF link state database (Area 0.0.0.1)								
Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	OPT	LEN
Router	10.1.70.1/0	false	10.255.70.103	0x80000002	0x4112	215	0x20	48
Router	10.1.71.2/0	true	10.255.71.242	0x80000002	0x11b1	214	0x20	48
Summary	23.1.1.0/0	true	10.255.71.242	0x80000002	0x6d72	172	0x20	28
Summary	24.1.1.0/0	false	10.255.71.242	0x80000002	0x607e	177	0x20	28
NSSA	33.1.1.1/0	false	10.255.71.242	0x80000002	0x73bd	217	0x28	36

10.9.3.3. show ospf database asbr-summary

Description	Displays information about the Autonomous System Boundary Router (ASBR) summary LSAs. Generated by Area Border Routers (ABRs) either globally or for a specified OSPF instance. The link-state ID is the router ID of the described ASBR.		
Command Syntax	show ospf [instance <i>instance-name</i>] database asbr-summary [adv-router brief detail self-originate]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing ospf instance	Specify an OSPF instance to display.
Examples	The following example shows sample output from this command:		
exaware# show ospf instance 1 database asbr-summary			
Instance: 1 VFR: default			
Summary ASBR Link States (Area 0.0.0.0)			
Type	LINK ID	SELF	ADV ROUTER
ASBR-summ	33.1.1.1/0	false	10.255.71.242

10.9.3.4. show ospf database external

Description	Displays information only about the external LSAs. Generated by the Autonomous System Border Routers (ASBRs) either globally or for a specified OSPF instance. The link-state ID is the external network number.							
Command Syntax	show ospf [instance <i>instance-name</i>] database external [adv-router brief detail self-originate]							
Command Mode	operational							
Parameters Table	Name	Value	Description					
	<i>tag-name</i>	existing ospf tag	Specify an OSPF tag to display.					
Examples	The following example shows sample output from this command:							
exaware# show ospf instance 1 database external								
Instance: 1 VFR: default AS External Link States								
Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	OPT	LEN
External	33.1.1.1	false	10.255.71.242	0x80000002	0x73bd	217	0x28	36
	Metric Type		2					
	Metric		0					
	Forward Address		0.0.0.0					
	External Route Tag		0					

10.9.3.5. show ospf database network

Description	Displays information only about the network LSAs. Generated by Designated Routers (DR) and describes the set of routers attached to a particular network either globally or for a specified OSPF instance. They are flooded in the area that contains the network. The link-state ID is the IP interface address of the DR.								
Command Syntax	show ospf [instance <i>instance-name</i>] database network [adv-router brief detail self-originate]								
Command Mode	operational								
Parameters Table	Name	Value	Description						
	<i>instance-name</i>	existing ospf instance	Specify an OSPF instance to display.						
Examples	The following example shows sample output from this command:								
exaware# show ospf instance 1 database network									
Instance: 1 VFR: default Net Link States (Area 0.0.0.0)									
Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	OPT	LEN	
Network	33.1.1.1	false	10.255.71.242	0x80000002	0x73bd	217	0x28	36	
	Attached Router		20.0.4.101/24						
	Attached Router		20.0.3.100/24						

10.9.3.6. show ospf database nssa-external

Description	Displays information only about the not so stubby area (NSSA) external LSAs either globally or for a specified OSPF instance. Generated by Area Border Routers (ABRs).							
Command Syntax	show ospf [instance <i>instance-name</i>] database nssa-external [adv-router brief detail self-originate]							
Command Mode	operational							
Parameters Table	Name	Value	Description					
	tag-name	existing ospf tag	Specify an OSPF tag to display.					
Examples	The following example shows sample output from this command:							
exaware# show ospf instance 1 database nssa-external								
Instance: 1 VFR: default								
Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	OPT	LEN

NSSA	10.0.0.0/32	false	4.4.4.4	0x80000002	0x73bd	217	0x28	36
	Metric Type		2					
	Metric		10					
	Forward Address		0.0.0.0					
	Admin Tag		1000					

10.9.3.7. show ospf database opaque-area

Description	Displays information about the area-local (link state type 10) scope LSAs either globally or for a specified OSPF instance. Type-10 Opaque LSAs are not flooded beyond the borders of their associated area.							
Command Syntax	show ospf [instance <i>instance-name</i>] database opaque-area [adv-router brief detail self-originate]							
Command Mode	operational							
Parameters Table	Name	Value	Description					
	<i>instance-name</i>	existing ospf instance	Specify an OSPF instance to display.					
Examples	The following example shows sample output from this command:							
exaware# show ospf instance 1 database opaque-area								
Area-Local Opaque-LSA (Area 0.0.0.0)								
Instance: 1 VFR: default								
Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	OPT	LEN

opaque-area	1.0.0.1/0	false	20.0.4.101	0x800000c7	0xf597	5	0x28	36
Opaque-ID		0						
Opaque-Type		1	(Traffic Engineering LSA)					
MPLS TE router ID		172.20.1.12						
Link ID		20.0.3.200						
Interface Address		20.0.3.101						
Neighbor Address		20.0.3.20						
Admin-Group		10						
Maximum Link Bandwidth		10.00 Gbps						
Reservable Bandwidth		10.00 Gbps						
Unreserved Bandwidth		10.00 Gbps						
Priority 0		: 10.00 Gbps						
Priority 1		: 10.00 Gbps						
Priority 2		: 10.00 Gbps						
Priority 3		: 10.00 Gbps						
Priority 4		: 10.00 Gbps						
Priority 5		: 10.00 Gbps						

Priority 6	:	10.00 Gbps	
Priority 7	:	10.00 Gbps	
TE Metric			0

10.9.3.8. show ospf database opaque-as

Description	Displays information about the opaque Type 11 either globally or for a specified OSPF instance. LSAs. Type 11 denotes that the LSA is flooded throughout the autonomous system.		
Command Syntax	show ospf [instance <i>instance-name</i>] database opaque-as [adv-router brief detail self-originate]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing ospf instance	Specify an OSPF instance to display.
Examples	The following example shows sample output from this command:		
exaware# show ospf instance 1 database opaque-as			
Instance: 1 VFR: default AS-Global Opaque-LSA			
Type	LINK ID	SELF	ADV ROUTER
opaque-as	1.0.0.1/0	false	20.0.4.101

10.9.3.9. show ospf database opaque-link

Description	Displays information about the opaque Type 9 LSAs either globally or for a specified OSPF instance. Type 9 denotes a link-local scope. The LSAs are not flooded beyond the local network.							
Command Syntax	show ospf [instance <i>instance-name</i>] database opaque-link [adv-router brief detail self-originate]							
Command Mode	operational							
Parameters Table	Name	Value	Description					
	tag-name	existing ospf tag	Specify an OSPF tag to display.					
Examples	The following example shows sample output from this command:							
exaware# show ospf instance 1 database opaque-link								
Instance: 1 VFR: default								
Link-Local Opaque-LSA (Link hme0:10.10.10.50)								
Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	OPT	LEN

opaque-as	1.0.0.1/0	false	20.0.4.101	0x80000001	0xf597	325	0x28	36
Opaque ID		657687						
Opaque Type		Unassigned						

10.9.3.10. show ospf database router

Description	Displays information only about the router LSAs either globally or for a specified OSPF instance. Generated by each router for each area to which it belongs. These packets describe the states of the router’s links to the area and are only flooded within a particular area. The link ID is the originating router’s ID.								
Command Syntax	show ospf [instance <i>instance-name</i>] database router [adv-router brief detail self-originate]								
Command Mode	operational								
Parameters Table	Name	Value	Description						
	<i>instance-name</i>	existing ospf instance	Specify an OSPF instance to display.						
Examples	The following example shows sample output from this command:								
exaware# show ospf instance 1 database router									
Instance: 1 VFR: default									
Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	OPT	LEN	
Router	10.1.70.1/0	false	10.255.70.103	0x80000002	0x4112	215	0x20	48	
Router	10.1.71.2/0	true	10.255.71.242	0x80000002	0x11b1	214	0x20	48	

10.9.3.11. show ospf database summary

Description	Displays information only about the summary LSAs either globally or for a specified OSPF instance. Generated by Area Border Routers (ABRs) and describe inter-area routes to various networks. They can also be used for aggregating routes. The link-state ID is the destination network number.							
Command Syntax	show ospf [instance <i>instance-name</i>] database summary [adv-router brief detail self-originate]							
Command Mode	operational							
Parameters Table	Name	Value	Description					
	tag-name	existing ospf tag	Specify an OSPF tag to display.					
Examples	The following example shows sample output from this command:							
exaware# show ospf instance 1 database summary detail								
Instance: 1 VFR: default								
Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	OPT	LEN

Summary	33.1.1.1/0	false	10.255.71.242	0x80000002	0x73bd	217	0x28	36
Metric		10						

10.9.3.12. show ospf flood-list

Description	Displays a list of link-state advertisements (LSAs) waiting to be flooded over an interface either globally or for a specified OSPF instance.		
Command Syntax	show ospf [instance <i>instance-name</i>] flood-list [area <i>area-id</i>] [<i>detail</i> <i>brief</i>] [interface <i>interface-name</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing ospf instance	Specify an OSPF instance to display.
	<i>area-id</i>	0-4294967295	Name of the OSPF VRF. The vrf-name argument can be specified as an arbitrary string.
		or IP address	Area number/ip used to define the particular area.
		<i>detail</i>	Database in detail format.
		<i>brief</i>	Database in brief format.
<i>interface-name</i>		Interface name.	
Examples	The following example shows sample output from this command:		
<pre>exaware# show ospf instance 1 flood-list interface x-eth 0/0/1 Instance: 1 VFR: default Interface x-eth0/0/1 Queue length 20 LS flooding 12 msec Type LINK ID ADV ROUTER SEQUENCE CHECKSUM AGE ----- External 10.2.195.0 192.168.0.163 0x80000009 0xFB61 0 External 10.1.192.0 192.168.0.163 0x80000009 0x2938 0 External 10.2.194.0 192.168.0.163 0x80000009 0x757 0</pre>			

Router	10.1.71.2	10.255.71.242	0x80000002	0x11b1	214
Summary	23.1.1.0	10.255.71.242	0x80000002	0x6d72	172
Summary	24.1.1.0	10.255.71.242	0x80000002	0x607e	177
NSSA	33.1.1.1	10.255.71.242	0x80000002	0x73bd	217

10.9.3.13. show ospf interface

Description	Displays OSPF-related interface information either globally or for a specified OSPF instance.						
Command Syntax	show ospf [instance <i>instance-name</i>] interface [area <i>area-id</i>] [<i>detail</i> <i>brief</i>]						
Command Mode	Operational						
Parameters Table	Name	Value	Description				
	<i>instance-name</i>	existing ospf instance	Specify an OSPF instance to display.				
	<i>area-id</i>	0-4294967295 or IP address	Name of the OSPF VRF. The vrf-name argument can be specified as an arbitrary string. Area number/ip used to define the particular area.				
		<i>detail</i> <i>brief</i>	Database in detail format. Database in brief format.				
Examples	The following examples show sample output from this command:						
exaware# show ospf instance 1 interface brief							
Instance: 1 VFR: default							
NAME	STATE	PASSIVE	TYPE	NETWORK AREA	DR ID	BDR ID	NBRS

loopback0	Loopback	enabled	-	0.0.0.10			
x-eth0/1/1	Down	disabled	-	0.0.0.10			0
x-eth0/1/2	BDR	disabled	broadcast	0.0.0.11	10.10.10.4	10.10.10.5	1
x-eth0/1/3	DR	disabled	broadcast	0.0.0.11	20.20.20.4	20.20.20.5	1

```
exaware# show ospf interface detail
```

NAME	STATE	PASSIVE	NETWORK TYPE	AREA	ID	DR ID	BDR NBR
x-eth0/0/0.10	point-to-point	disable	point-to-point	0.0.0.1	-	-	1

```

Tag                v2
Status             up
MTU                1500
MTU Ignore         Disable
Metric             1
Adjacent neighbors  1
Priority
Authentication key disabled
BFD                disabled
Local Address       102.1.1.1/24
Timer intervals
  Hello interval    10 seconds
  Dead interval     40 seconds
  Wait              40 seconds
  Retransmit        5 seconds
  Transmit Delay    1 seconds
  LDP Sync status   Enabled
  LDP Sync holdtime 60 seconds

```

10.9.3.14. show ospf neighbors

Description	Displays OSPF neighbors operational data on an individual interface basis either globally or for a specified OSPF instance.				
Command Syntax	show ospf [instance <i>instance-name</i>] neighbors [area <i>area-id</i>] [<i>detail</i> <i>brief</i>] [interface <i>interface-name</i>]				
Command Mode	Operational				
Parameters Table	Name	Value	Description		
	<i>instance-name</i>	existing ospf instance	Specify an OSPF instance to display.		
	<i>area-id</i>	0-4294967295 or IP address	Name of the OSPF VRF. The vrf-name argument can be specified as an arbitrary string. Area number/ip used to define the particular area.		
		<i>detail</i> <i>brief</i>	Database in detail format. Database in brief format.		
	<i>interface-name</i>		Interface name.		
Examples	The following example shows sample output from this command:				
exaware# show ospf instance 1 neighbors					
Instance: 1 VFR: default					
NEIGHBOR	STATE	INTERFACE	ADDRESS	PRIORITY	DEAD TIME

20.0.3.200	Full	x-eth0/0/1	192.168.254.225	2	34
20.0.4.200	Full	x-eth0/0/2	10.1.1.129	2	37
20.0.5.200	Full	VLINK1	101.1.1.1	-	40

10.9.3.15. show ospf request-list

Description	Displays request-list kept during synchronization of adjacent routers either globally or for a specified OSPF instance.		
Command Syntax	show ospf [instance <i>instance-name</i>] request-list [area <i>area-id</i>] [<i>detail</i> <i>brief</i>] [interface <i>interface-name</i>] [neighbor <i>neighbor-id</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>tag-name</i>	existing ospf tag	Specify an OSPF tag to display.
	<i>area-id</i>	0-4294967295 or IP address	Name of the OSPF VRF. The vrf-name argument can be specified as an arbitrary string. Area number/ip used to define the particular area.
		<i>detail</i> <i>brief</i>	Database in detail format. Database in brief format.
	<i>interface-name</i>		Interface name.
	<i>neighbor-id</i>	IPv4 address	IPv4 address of the OSPF neighbor.
	Examples	The following example shows sample output from this command:	
<pre>exaware# show ospf instance 1 request-list neighbor 10.0.124.4 interface x-eth 0/0/1 Instance: 1 VRF: default OSPF Router with ID (192.168.1.11) (Tag 1) Neighbor 10.0.124.4</pre>			

```
interface x-eth0/0/1
```

Type	LINK ID	ADV ROUTER	SEQUENCE	CHECKSUM	AGE
External	10.2.195.0	192.168.0.163	0x80000009	0xFB61	0
External	10.1.192.0	192.168.0.163	0x80000009	0x2938	0
External	10.2.194.0	192.168.0.163	0x80000009	0x757	0
Router	10.1.71.2	10.255.71.242	0x80000002	0x11b1	214
Summary	23.1.1.0	10.255.71.242	0x80000002	0x6d72	172
Summary	24.1.1.0	10.255.71.242	0x80000002	0x607e	177
NSSA	33.1.1.1	10.255.71.242	0x80000002	0x73bd	217

10.9.3.16. show ospf route

Description	Displays the current state of the routing table either globally or for a specified OSPF instance.																				
Command Syntax	show ospf [instance <i>instance-name</i>] route [area <i>area-id</i>] [<i>detail</i> <i>brief</i>]																				
Command Mode	operational																				
Parameters Table	Name	Value	Description																		
	<i>instance-name</i>	existing ospf instance	Specify an OSPF instance to display.																		
	<i>area-id</i>	0-4294967295 or IP address	Name of the OSPF VRF. The vrf-name argument can be specified as an arbitrary string. Area number/ip used to define the particular area.																		
		<i>detail</i> <i>brief</i>	Database in detail format. Database in brief format.																		
Examples	The following example shows sample output from this command:																				
<pre>exaware# show ospf instance 1 ospf route Flags: O - Intra area, IA - Inter area E1 - External type 1, E2 - External type 2 N1 - NSSA external type 1, N2 - NSSA external type 2 C - Directly connected Instance: 1 VFR: default</pre> <table><thead><tr><th>PREFIX</th><th>FLAGS</th><th>METRIC</th><th>ADMIN TAG</th><th>NEXTHOP ADDRESS</th><th>NEXTHOP INTERFACE</th></tr></thead><tbody><tr><td>10.10.10.4/32</td><td>E2</td><td>10</td><td>0</td><td>20.0.3.200</td><td>etr0/0/1.1</td></tr><tr><td>10.10.10.5/32</td><td>E2</td><td>10</td><td>100</td><td>20.0.4.200</td><td>etr0/0/1</td></tr></tbody></table>				PREFIX	FLAGS	METRIC	ADMIN TAG	NEXTHOP ADDRESS	NEXTHOP INTERFACE	10.10.10.4/32	E2	10	0	20.0.3.200	etr0/0/1.1	10.10.10.5/32	E2	10	100	20.0.4.200	etr0/0/1
PREFIX	FLAGS	METRIC	ADMIN TAG	NEXTHOP ADDRESS	NEXTHOP INTERFACE																
10.10.10.4/32	E2	10	0	20.0.3.200	etr0/0/1.1																
10.10.10.5/32	E2	10	100	20.0.4.200	etr0/0/1																

20.0.3.0/24	C	10	0	etr0/0/1.2
-------------	---	----	---	------------

10.9.3.17. show ospf statistics (interface)

Description	Displays OSPF traffic statistics information per interface either globally or for a specified OSPF instance.		
Command Syntax	show ospf [instance <i>instance-name</i>] statistics [interface { <i>interface-name</i> }] [{ <i>detail</i> <i>brief</i> }]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>tag-name</i>	existing ospf tag	Specify an OSPF tag for which to show statistics.
	<i>interface-name</i>		Interface name.
		<i>detail</i>	Database in detail format.
		<i>brief</i>	Database in brief format.
Examples	The following examples show sample output from this command:		
<pre>exaware# show ospf instance 1 statistics brief Instance: 1 VFR: default TYPE SENT RECEIVED ----- Hello 51 0 DB Desc 0 0 LS Req 0 0 LS Upd 0 0 LS Ack 0 0 exaware# show ospf statistics detail Instance: 1 VFR: default Hello Sent: 1233 Received: 158955 Network Mask mismatch: 1 Hello Interval mismatch: 2 Dead Interval mismatch: 1</pre>			

Options mismatch:	3	
DB Description		
Sent: 59569	Received: 56597	
Unknown Neighbor ID:	1	
MTU mismatch:	14	
Negotiation Failed:	5	
MS-bit mismatch:	2	
I-bit mismatch:	2	
Options mismatch:	1	
Sequence Number mismatch:	0	
Neighbor state mismatch:	12	
LS Update		
Sent: 1289	Received: 5534	
Unknown Neighbor ID:	0	
LSA checksum error:	0	
Unknown LS type:	2	
LS age is more than MaxAge:	0	
LSA data mismatch:	1	
Neighbor state mismatch:	0	
LS Request		
Sent: 1173628	Received: 1140318	
Unknown Neighbor ID:	1	
Neighbor state mismatch:	7	
LS Ack		
Sent: 429727	Received: 1145790	
Unknown Neighbor ID:	3	
Neighbor state mismatch:	0	
Header Errors		
Invalid packet type:	4	
Version mismatch:	1	
Area mismatch:	30	
Network address mismatch:	0	
Instance ID mismatch:	0	
Authentication type mismatch:	2	
Authentication error:	5	
Checksum error:	2	
Total		
Sent: 1665446	Received: 2507194	
Failed: 23	Discarded: 52	

10.9.3.18. show ospf statistics virtual-link

Description	Displays OSPF traffic statistics information per interface.		
Command Syntax	show ospf [tag <i>tag-name</i>] statistics virtual-link [{name <i>virtual-link-name</i> } {transit-area <i>area</i> remote-id <i>router-id</i> }]] [{ <i>detail</i> <i>brief</i> }]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>tag-name</i>	existing ospf tag	Specify an OSPF tag for which to show statistics.
	<i>name</i>	string	The virtual link name assigned by the system (see command show ospf virtual-link).
	<i>area</i>	IPv4 Address or integer	The transit area
	<i>remote-id</i>	IPv4 Address	The router-id of the remote router.
		<i>detail</i> <i>brief</i>	Database in detail format. Database in brief format.
Examples	The following examples show sample output from this command:		
exaware# show ospf statistics virtual-link brief TYPE SENT RECEIVED ----- Hello 51 0 DB Desc 0 0 LS Req 0 0 LS Upd 0 0 LS Ack 0 0			

```
exaware# show ospf statistics virtual-link detail
Hello
  Sent: 1233                      Received: 158955
  Network Mask mismatch:          1
  Hello Interval mismatch:        2
  Dead Interval mismatch:         1
  Options mismatch:               3

DB Description
  Sent: 59569                     Received: 56597
  Unknown Neighbor ID:            1
  MTU mismatch:                   14
  Negotiation Failed:             5
  MS-bit mismatch:                2
  I-bit mismatch:                 2
  Options mismatch:               1
  Sequence Number mismatch:       0
  Neighbor state mismatch:        12

LS Update
  Sent: 1289                      Received: 5534
  Unknown Neighbor ID:            0
  LSA checksum error:             0
  Unknown LS type:                2
  LS age is more than MaxAge:     0
  LSA data mismatch:              1
  Neighbor state mismatch:        0

LS Request
  Sent: 1173628                   Received: 1140318
  Unknown Neighbor ID:            1
  Neighbor state mismatch:        7

LS Ack
  Sent: 429727                    Received: 1145790
  Unknown Neighbor ID:            3
  Neighbor state mismatch:        0

Header Errors
  Invalid packet type:            4
  Version mismatch:               1
  Area mismatch:                  30
  Network address mismatch:       0
  Instance ID mismatch:           0
  Authentication type mismatch:   2
  Authentication error:           5
  Checksum error:                 2

Total
  Sent: 1665446                   Received: 2507194
```


Failed: 23

Discarded: 52

10.9.3.19. show ospf summary

Description	Displays summary OSPF information either globally or for a specified OSPF instance.		
Command Syntax	show ospf [instance <i>instance-name</i>] summary		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>instance-name</i>	existing ospf instance	Specify an OSPF instance to display.
Examples	The following example shows sample output from this command: <pre> exaware# show ospf instance 1 summary Instance: 1 VFR: default VRF default Router-ID 20.0.4.101 Number of area 2 Number of Interface 12 Number of Virtual-links 1 Type Area Border Router \ AS Border Router Opaque LSA enabled Graceful restart State enabled Interval 600 seconds Helper enabled Max grace period 600 seconds LSA refresh time 50 minutes SPF Calc Timers Min hold time 5 seconds Maximum wait time 5 seconds Area 0.0.0.0 Interfaces 2 Full state neighbors 1 Authentication key disable SPF calculation 3 Number of LSA 4 Checksum Sum 0x2A6AC </pre>		

10.9.3.20. show ospf virtual-links

Description	Displays parameters and the current state of Open Shortest Path First (OSPF) virtual links.
Command Syntax	show ospf virtual-links
Command Mode	Operational
Examples	The following example shows sample output from this command: exaware# show ospf virtual-links Virtual Link: VLINK1 Router-ID 10.10.10.1 Transit Area: 0.0.0.1 Operational Status: UP Metric 1 Adjacency State: Full Local IPv4 Address: 145.1.1.1 Remote IPv4 Address: 12.1.1.1 Authentication Key: disable Timer intervals HELLO Interval: 30sec Dead Interval: 10sec Retransmission Interval: 100sec Transmission Delay: 10sec

10.10. OSPFv3 Routing Commands

10.10.1. OSPFv3 Routing Configuration Commands

10.10.1.1. area

Description	Specifies an OSPFv3 area. The identifier can be specified as either a decimal value or an IPv4 address.		
Command Syntax	area [<i>decimal</i> <i>ip_address</i>]		
Command Mode	configuration routing ospfv3 vrf		
Parameters Table	Name	Value	Description
	<i>decimal</i>	0-4294967295	Area ID as a decimal value.
	<i>ip_address</i>	IPv4 address	Area ID in IPv4 address format.
Examples	This example shows how to use this command: exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# area 10.10.10.10		

10.10.1.2. bfd

Description	Enable or disable BFD sessions to the newborns on the given interface. Use the no form of the command to reset to the default state (disable).		
Command Syntax	<pre>bfd enable disable no admin-state</pre>		
Command Mode	configuration routing ospfv3 vrf interface		
Parameters Table	Name	Value	Description
	enable		A BFD session is used to check the peer state.
	disable		A BFD session is not used to check the peer state.
Examples	This example shows how to use this command: <pre>exaware# config exaware (config)# routing ospfv3 ospf3 exaware(config-routing-ospfv3-ospf3)# vrf vrf1 exaware(config-routing-ospfv3-ospf3-vrf-vrf1)# area 10.10.10.10 exaware(config-routing-ospfv3-ospf3-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1.6 exaware(config-routing-ospfv3-ospf3-vrf-vrf1-area-10.10.10.10-interface-x-eth- 0/0/1.6)# bfd enable</pre>		

10.10.1.3. dead-interval (interface/virtual-link level)

Description	Sets the time interval, while not receiving packet messages, by which a neighbor is declared down.		
Command Syntax	<code>dead-interval seconds</code>		
Command Mode	configuration routing ospfv3 vrf area interface configuration routing ospfv3 vrf area virtual-link		
Parameters Table	Name	Value	Description
	<code>seconds</code>	1-65535	Dead router detection time in seconds. The default is four times the hello interval, or 40 seconds. As with the hello interval, this value must be the same for all routers and access servers attached to a common network.
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1 exaware(config-area-10.10.10.10-interface-x-eth-0/0/1)# dead- interval 10 exaware(config-routing-ospfv3-ospfv3-vrf-vrf1-area- 10.10.10.10-interface-x-eth-0/0/1.6)# exit exaware(config-routing-ospfv3-ospfv3-vrf-vrf1-area- 10.10.10.10)# virtual-link 10.3.4.5 exaware(config-routing-ospfv3-ospfv3-vrf-vrf1-area- 10.10.10.10-virtual-link-10.3.4.5)# dead-interval 10 </pre>		

10.10.1.4. default-metric

Description	Specifies a metric for the default summary route sent into a stub or NSSA area.		
Command Syntax	default-metric [<i>value</i>]		
Command Mode	configuration routing ospfv3 vrf area		
Parameters Table	Name	Value	Description
	<i>value</i>	0-16777214	Stub/NSSA advertised default summary metric Metric for the default summary route used for a stub or NSSA. The acceptable value is a 24-bit number. (Default = 1)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf default exaware(config-routing-ospfv3-ospfv31-vrf-default)# area 10.10.10.10 exaware(config-vrf-default-area-10.10.10.10)# default-metric 20</pre>		

10.10.1.5. default-route-originate

Description	Used to generate a Type 7 default into the NSSA area. This keyword takes effect only on an NSSA ABR or NSSA Autonomous System Boundary Router (ASBR). If the policy option is used, then the router will advertise an external default route only if the policy condition is met.		
Command Syntax	default-route-originate [[metric <i>metric-value</i>] [metric-type <i>type-value</i>] [always]]		
Command Mode	configuration routing ospfv3 vrf		
Parameters Table	Name	Value	Description
	<i>metric-value</i>	0-16777214	(Optional) OSPF default metric. (Default = 10)
	<i>type-value</i>	1-2	(Optional) OSPF Link State type. External link type associated with the default route advertised in the OSPF routing domain. It can be one of the following values: 1—Type 1 metrics are expressed in the same unit as the OSPF metric. Calculating cost to access external AS includes the distance in the AS. 2—Type 2 metrics are one order of magnitude larger. Calculating cost to access external AS without including the distance in the AS.
	always		Always advertise default-route.

Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# default-route-originate metric 100 metric-type 2</pre>
-----------------	--

Policy Operations:

Command	Tag	prefix	next-hop	rib-metric	rib-route	route-has-mpls-label	Area-id	route-type
default-route-originate	n/a	n/a	n/a	n/a	+	n/a	n/a	n/a

Supported actions per command:

Command	metric	metric-type	tag
default-route-originate	+	+	+

10.10.1.6. graceful-restart grace-interval

Description	Configures the T3 timer interval, which is the length of time that a restarting router retains the forwarding table. Use the no form of the command to configure the default value (120).		
Command Syntax	graceful-restart grace-interval <i>seconds</i>		
Command Mode	configuration routing ospfv3 vrf		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-1800	Length of the grace interval in seconds. (Default = 120)
Examples	This example shows how to set the grace interval to 10 seconds: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# graceful- restart grace-interval 10</pre>		
Notes	The grace interval is the maximum amount of time, in seconds, that the restarting router waits before setting the overload bit to indicate that the LSP database has not been synchronized and the OSPFv3 graceful restart operation has failed.		

10.10.1.7. graceful-restart helper

Description	Enables or disables the router as a restart helper for a router that is performing a graceful restart. Use the no form of the command to configure the default status (enabled).		
Command Syntax	<pre>graceful-restart helper [enable disable] no graceful-restart helper</pre>		
Command Mode	configuration routing ospfv3 vrf		
Parameters Table	Name	Value	Description
		<i>enable</i>	The router is a restart helper router. (Default)
		<i>disable</i>	The router is not a restart helper router.
Examples	This command configures the router to be a restart helper router: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# graceful- restart helper disable</pre>		
Notes	A restart helper router for a neighboring router that is performing a graceful restart will maintain the adjacency and send a complete sequence number PDUs (CSNPs) to the restarting router, flooding the entire database.		

10.10.1.8. graceful-restart max-grace-period

Description	Sets the maximum grace period to accept.		
	Use the no form of the command to configure the default status (enabled).		
Command Syntax	graceful-restart max-grace-period <i>seconds</i>		
Command Mode	configuration routing ospfv3 vrf		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-1800	The maximum grace period to accept in seconds.
Examples	This example shows how to use this command: exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# graceful-restart max-grace-period 200		

10.10.1.9. graceful-restart state

Description	Enables graceful restart mode. Use the no form of the command to disable graceful restart mode.		
Command Syntax	<pre>graceful-restart [state [enable disable]] no graceful-restart [state]</pre>		
Command Mode	configuration routing ospfv3 vrf		
Parameters Table	Name	Value	Description
		<i>enable</i>	Enables graceful restart mode.
		<i>disable</i>	Disables graceful restart mode.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# graceful- restart state enable</pre>		

10.10.1.10. hello-interval (interface/virtual-link level)

Description	Sets the time interval between two consecutive hello transmissions. The value must be the same for all routers and access servers attached to a common network. Use the no form of the command to set the default (10 seconds).		
Command Syntax	<pre>hello-interval <i>seconds</i> no hello-interval</pre>		
Command Mode	<pre>configuration routing ospfv3 vrf area interface configuration routing ospfv3 vrf area virtual-link</pre>		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	The time interval between two consecutive hello transmissions in seconds. (Default = 10)
Examples	This example shows how to configure a hello interval of 30 seconds: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1 exaware(config-area-10.10.10.10-interface-x-eth-0/0/1)# hello-interval 30 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1-area- 10.10.10.10-interface-x-eth-0/0/1.6)# exit exaware(config-routing-ospfv3-ospfv31-vrf-vrf1-area- 10.10.10.10)# virtual-link 10.3.4.5 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1-area- 10.10.10.10-virtual-link-10.3.4.5)# hello-interval 30</pre>		
Notes	DIS sends hello transmissions at three times the rate of non-DIS. If OSPFv3 is elected as DIS on this interface and the hello interval value is 10 seconds, it sends hello every 3.3 seconds.		

10.10.1.11. interface

Description	Enters routing OSPFv3 configuration mode for a specific interface for a specified OSPFv3 instance. You can also configure OSPFv3 routing for a specific level for a specific interface and OSPFv3 instance. Use the no form of the command to disable OSPFv3 routing for the interface or for the level.		
Command Syntax	<pre>interface <i>interface-name</i> no interface <i>interface-name</i></pre>		
Command Mode	configuration ospfv3 vrf area interface		
Parameters Table	Name	Value	Description
	<i>interface-name</i>	{x-eth mgmt} 0/0/ <i>port</i> {agg-eth loopback} <i>number</i>	Interface name.
Examples	This example shows how to configure OSPFv3 routing for the following instance: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# area 10 exaware(config-vrf-vrf1-area-0.0.0.10)# interface x-eth 0/0/4 exaware(config-interface-x-eth-0/0/4)#</pre>		

10.10.1.12. interface-bandwidth

Description	This command configures the OSPFv3 interface bandwidth used to calculate the interface default metric. Use the no form of the command to restore the default value. The default value is the actual interface speed.		
Command Syntax	<pre>interface-bandwidth <i>bandwidth</i> [<i>unit</i>]</pre> <pre>no interface-bandwidth</pre>		
Command Mode	configuration routing ospfv3 vrf interface		
Parameters Table	Name	Value	Description
	<i>bandwidth</i>	1kbs-800gbps	Specifies the interface bandwidth in the given units per second.
	<i>unit</i>	bps kbps mbps gbps	Bandwidth unit. (Default = bps)
Examples	This example shows how to use this command: <pre>exaware# config exaware (config)# routing ospfv3 ospf3 exaware(config-routing-ospfv3-ospf3)# vrf vrf1 exaware(config-routing-ospfv3-ospf3-vrf-vrf1)# area 10.10.10.10 exaware(config-routing-ospfv3-ospf3-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1.6 exaware(config-routing-ospfv3-ospf3-vrf-vrf1-area-10.10.10.10-interface-x-eth-0/0/1.6)# interface-bandwidth 100 gbps</pre>		
Note	The parameter is used to calculate the interface default metric (see command metric)		

10.10.1.13. log-neighbor-changes

Description	Enables or disables writing a message to the log each time a neighbor changes state (up or down). Use the no form of the command to set the default (enable).		
Command Syntax	<pre>log-neighbor-changes [<i>enable</i> <i>disable</i>]</pre> <pre>no log-neighbor-changes</pre>		
Command Mode	configuration routing ospfv3 vrf		
Parameters Table	Name	Value	Description
		<i>enable</i>	Enable logging neighbor changes. (Default)
		<i>disable</i>	Disable logging neighbor changes.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# log-neighbor-changes disable</pre>		

10.10.1.14. max-metric set | unset

Description	Enables OSPFv3 Stub router mode by setting the metric on all links (except of stub links) to maximum value (0xFFFF). OSPFv3 stays in Stub router mode until disabled. Use the no form of the command to disable Stub router mode.		
Command Syntax	<pre>max-metric {set unset} no max-metric</pre>		
Command Mode	configuration routing ospfv3 vrf		
Parameters Table	Name	Value	Description
		set unset	Set / unset Stub router mode.
Examples	This example shows how to enable Stub router mode: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf default exaware(config-routing-ospfv3-ospfv31-vrf-default)# max-metric set</pre>		

10.10.1.15. max-metric-on-start

Description	Enables OSPFv3 Stub router mode at system startup by temporarily setting the metric on all links (except of stub links) to maximum value (0xFFFF). OSPFv3 stays in Stub router mode for configured timeout period. When the “wait-for-bgp” option is configured the OSPFv3 Stub router mode will end upon BGP convergence completion event, if such event occurs before the timer expires. Use the no form of the command to disable OSPFv3 Stub router mode at system startup.		
Command Syntax	<pre>max-metric-on-start [timeout] [wait-for-bgp] no max-metric-on-start</pre>		
Command Mode	configuration routing ospfv3 vrf		
Parameters Table	Name	Value	Description
	timeout	10-86400	Timeout for Stub router mode. (Default = 600)
	wait-for-bgp		Exit Stub router mode once BGP is converged.
Examples	This example shows how to enable OSPFv3 Stub router mode at system startup for 800 seconds with the wait-for-bgp option: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf default exaware(config-routing-ospfv3-ospfv31-vrf-default)# max-metric-on-start 800 wait-for-bgp</pre>		
Notes	The Stub router mode is not activated upon process restart (only upon system restart).		

10.10.1.16. metric

Description	Specifies the metric of the interface. Use the no form of the command to restore the default value.		
Command Syntax	metric <i>value</i> no metric		
Command Mode	configuration routing ospf vrf area interface configuration routing ospf vrf virtual-link		
Parameters Table	Name	Value	Description
	value	1-65535	Metric value. Default = ○ Interface: • MAX(1, reference-bandwidth % interface-bandwidth) ○ virtual link: • 1
Examples	This example shows how to configure a wide metric value: <pre> exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# metric 10 exaware# config exaware(config)# routing ospfv3 ospf3 exaware(config-routing-ospfv3-ospf3)# vrf vrf1 exaware(config-routing-ospfv3-ospf3-vrf-vrf1)# area 10.10.10.10 exaware(config-routing-ospfv3-ospf3-vrf-vrf1-area- 10.10.10.10)# virtual-link 18.1.1.1 exaware(config-routing-ospfv3-ospf3-vrf-vrf1-area- 10.10.10.10-virtual-link-18.1.1)# metric 100 </pre>		

10.10.1.17. mtu

Description	Configures the OSPFv3 mtu per interface. Use the no form of the command to set the default (1500).		
Command Syntax	<pre>mtu value</pre> <pre>no mtu</pre>		
Command Mode	configuration routing ospfv3 vrf area interface		
Parameters Table	Name	Value	Description
	<i>value</i>	An integer into range 594..9200.	The size in bytes of the largest IP datagram that can be sent out the associated interface, without fragmentation.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospfv3 ospf1 exaware(config-routing-ospfv3-ospf1)# vrf default exaware(config-routing-ospfv3-ospf1-vrf-vrf1)# area 10 exaware(config-ospfv3-ospf1-vrf-vrf1-area-0.0.0.10)# interface x-eth 0/0/4 exaware(config-config-ospfv3-ospf1-vrf-vrf1-area-0.0.0.10- interface-x-eth-0/0/4)# mtu 1000</pre>		

10.10.1.18. mtu-ignore

Description	Configures prevention of MTU checks on OSPFv3 adjacency establishment. Use the no form of the command to set the default (disable).		
Command Syntax	mtu-ignore enable disable no mtu-ignore		
Command Mode	configuration routing ospfv3 vrf area interface		
Parameters Table	Name	Value	Description
	enable		Disables checking for whether OSPFv3 neighbors are using the MTU on a common interface. The OSPF instance sends the zero value of MTU into the Database Description packet.
	disable		Enables checking for whether OSPFv3 neighbors are using the MTU on a common interface.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospfv3 ospf1 exaware(config-routing-ospfv3-ospf1)# vrf default exaware(config-routing-ospfv3-ospf1-vrf-vrf1)# area 10 exaware(config-ospfv3-ospf1-vrf-vrf1-area-0.0.0.10)# interface x-eth 0/0/4 exaware(config-config-ospfv3-ospf1-vrf-vrf1-area-0.0.0.10- interface-x-eth-0/0/4)# mtu-ignore enable</pre>		

10.10.1.19. network-type

Description	Specifies the network type for the interface (not supported on loopback interface). Use the no form of the command to revert to the default network type (broadcast).		
Command Syntax	<pre>network-type broadcast point-to-point no network-type</pre>		
Command Mode	configuration routing ospfv3 vrf area interface		
Parameters Table	Name	Value	Description
		broadcast	Enables OSPFv3 broadcast multi-access network on the interface. (Default)
		point-to-point	Enables OSPFv3 point-to-point network on the interface.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrfl exaware(config-routing-ospfv3-ospfv31-vrf-vrfl)# area 10 exaware(config-vrf-vrfl-area-0.0.0.10)# interface x-eth 0/0/4 exaware(config-interface-x-eth-0/0/4)# network-type point-to-point</pre>		

10.10.1.20. passive (interface)

Description	Stops sending outgoing hello packets, hence the router cannot form any neighbor relationships via the passive interface. This behavior stops both outgoing and incoming routing updates.		
Command Syntax	<code>passive [<i>disable</i> <i>enable</i>]</code>		
Command Mode	configuration ospfv3 vrf area interface		
Parameters Table	Name	Value	Description
		<i>disable</i> <i>enable</i>	Stops outgoing and incoming routing updates. (Default) Allows sending routing updates.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# area 10 exaware(config-vrf-vrf1-area-0.0.0.10)# interface x-eth 0/0/4 exaware(config-interface-x-eth-0/0/4)# passive enable</pre>		

10.10.1.21. priority (interface level)

Description	Sets the priority for LAN DIS election. It sets the priority value in the LAN OSPFv3 hello PDUs. Use the no form of the command to set the default (1).		
Command Syntax	priority value no priority		
Command Mode	configuration routing ospfv3 vrf [neighbor / area interface]		
Parameters Table	Name	Value	Description
	value	0-255	Priority value. (Default = 1)
Examples	This example shows how to configure a priority of 15: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# area 10 exaware(config-vrf-vrf1-area-0.0.0.10)# interface x-eth 0/0/4 exaware(config-interface-x-eth-0/0/4)# priority 15</pre>		
Notes	This command is not valid for point-to-point interfaces (see network) (on page c). A value of 0 will prevent the router from be elected as a DR or BDR. A higher priority value is preferred in DIS election over lower priority values.		

10.10.1.22. range advertise/suppress

Description	Consolidates and summarizes routes at an area boundary by an Area Border Router (ABR).		
Command Syntax	range [<i>prefix/prefix-length</i>] [<i>advertise</i> <i>suppress</i>]		
Command Mode	configuration routing ospf vrf area		
Parameters Table	Name	Value	Description
	<i>prefix</i>	IPv6 address	Area range IPv6 address.
	<i>prefix-length</i>	numeric	IPv6 address mask length.
		<i>advertise</i> <i>suppress</i>	Advertise the range. (Default) Suppress the range.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospf ospf1 exaware(config-routing-ospf-ospf1)# vrf vrf1 exaware(config-routing-ospf-ospf1-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# range 2001:0eb8:1234::/64 suppress</pre>		

10.10.1.23. redistribute

Description	Redistributes reachability information from other routing protocols. If the policy option is used, then the routes from other protocols will be advertised into the OSPFv3 domain only if the policy conditions are met. Use the no form of the command to stop redistribution.		
Command Syntax	redistribute { <i>bgp</i> <i>connected</i> <i>isis</i> <i>static</i> } [[metric <i>metric-value</i>] [metric-type <i>type-value</i>]] [policy <i>policy-names</i>] no redistribute		
Command Mode	configuration routing ospfv3 vrf		
Parameters Table	Name	Value	Description
		bgp	Type of protocol to redistribute.
		connected	
		isis	
		static	
Examples	This example shows how to configure redistribution: exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# redistribute static		
Notes	You can redistribute more than one routing type, but you can only specify one routing type per command.		

Policy Operations:

Command	Tag	prefix	next-hop	rib-metric	rib-route	route-has- mpls-label	Area-id	route-type
redistribute	+	+	+	+	n/a	+	n/a	+

Supported actions per command:

Command	metric	metric-type	tag
redistribute	+	+	+

10.10.1.24. reference-bandwidth

Description	Configures the OSPFv3 reference bandwidth per interface. Use the <code>no</code> form of the command to set the default.		
Command Syntax	<pre>reference-bandwidth <i>bandwidth</i> [<i>unit</i>]</pre> <pre>no reference-bandwidth</pre>		
Command Mode	configuration ospfv3 vrf		
Parameters Table	Name	Value	Description
	<i>bandwidth</i>	1kbs-800gbps	Bandwidth specified in given units per second. (Default = 100Mbps)
	<i>unit</i>	bps kbps mbps gbps	Bandwidth unit. (Default = bps)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# reference-bandwidth 1 gbps</pre>		
Note	The parameter is used to calculate the interface default metric (see command <code>metric</code>)		

10.10.1.25. retransmit-interval

Description	Sets the time between link-state advertisement (LSA) retransmissions for adjacencies belonging to the interface. Expect round-trip delay between any two routers on the attached network. The value must be greater than the expected round-trip delay.		
Command Syntax	retransmit-interval <i>seconds</i>		
Command Mode	configuration routing ospfv3 vrf area interface configuration routing ospfv3 vrf area virtual-link		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-1800	LSA retransmit interval in seconds. (Default = 5)
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1 exaware(config-area-10.10.10.10-interface-x-eth-0/0/1)# retransmit-interval 100 exaware(config-routing-ospfv3-ospfv3-vrf-vrf1-area- 10.10.10.10-interface-x-eth-0/0/1.6)# exit exaware(config-routing-ospfv3-ospfv3-vrf-vrf1-area- 10.10.10.10)# virtual-link 10.3.4.5 exaware(config-routing-ospfv3-ospfv3-vrf-vrf1-area- 10.10.10.10-virtual-link-10.3.4.5)# retransmit-interval 100 </pre>		

10.10.1.26. router-id

Description	Sets a fixed router ID.		
Command Syntax	<code>router-id <i>ip-address</i></code>		
Command Mode	configuration routing ospfv3 vrf		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 address	Router-id in IPv4 address format.
Examples	This example shows how to configure a wide metric value: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# router-id 10.10.10.10</pre>		

10.10.1.27. routing ospfv3

Description	Enters routing OSPFv3 configuration mode. This mode configures OSPFv3 routing for a specified OSPFv3 instance tag at a global level for that tag. You can also configure OSPFv3 routing for the following in relation to the specified OSPFv3 instance: • A specific management or loopback interface (see routing ospf interface (on page Error! Bookmark not defined.)). • A specific level (see routing ospf level (on page Error! Bookmark not defined.)). • A specific context (IP address family - routing scheme or IP address family - routing scheme-level) (see routing ospfv3 <context> (on page Error! Bookmark not defined.)). • A specific level for a specific management or loopback interface (see routing ospfv3 interface (on page Error! Bookmark not defined.)). Use the no form of the command to disable OSPFv3 routing globally or for a specific OSPFv3 instance.		
Command Syntax	<pre>routing ospfv3 tag no routing ospfv3 no routing ospfv3 tag</pre>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	tag	string - 60 characters or less	ID of the OSPFv3 instance for which to configure routing.
Examples	This example shows how to configure OSPFv3 routing for OSPFv3 tag ospfv31: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)#</pre>		

10.10.1.28. spf-interval

Description	Configures the delay time between when OSPFv3 receives a topology change and when it starts a shortest path first (SPF) calculation, and the hold time between two consecutive SPF calculations.		
Command Syntax	spf-interval [min <i>spf-delay-min</i>] [max <i>spf-delay-max</i>]		
Command Mode	configuration routing ospfv3 vrf		
Parameters Table	Name	Value	Description
	<i>spf-delay-min</i>	0-2147483647	Min delay time (in milliseconds) between when OSPFv3 receives a topology change and when it starts an SPF calculation. (Default = 50)
	<i>spf-delay-max</i>	0-2147483647	Max delay time (in milliseconds) between when OSPFv3 receives a topology change and when it starts an SPF calculation. (Default = 50000)
Examples	This example shows how to configure a wide metric value: <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# spf-interval min 100 max 100000</pre>		

10.10.1.29. stub suppress-summary

Description	Blocks the flooding from AS external advertisements in OSPFv3 and injects a default router into a stub area.		
Command Syntax	stub suppress-summary [<i>enable</i> <i>disable</i>]		
Command Mode	configuration routing ospfv3 vrf area		
Parameters Table	Name	Value	Description
		<i>enable</i>	Advertise the range, and the stub area becomes a totally stub area.
		<i>disable</i>	Suppress the range. (Default)
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# stub suppress- summary enable </pre>		
Notes	Does not inject inter-area routes into stub. Prevents an Area Border Router (ABR) from sending summary link advertisements into the stub area.		

10.10.1.30. transmit-delay

Description	Sets the estimated time (in seconds) required to send a link-state update packet on the interface. LSAs in the update packet have their age incremented by this amount before transmission.		
Command Syntax	transmit-delay <i>seconds</i>		
Command Mode	configuration routing ospfv3 vrf area interface configuration routing ospfv3 vrf area virtual-link		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-1800	LSA transmission delay in seconds. (Default = 1)
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)# area 10.10.10.10 exaware(config-vrf-vrf1-area-10.10.10.10)# interface x-eth 0/0/1 exaware(config-area-10.10.10.10-interface-x-eth-0/0/1)# transmit-delay 100 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1-area- 10.10.10.10-interface-x-eth-0/0/1.6)# exit exaware(config-routing-ospfv3-ospfv31-vrf-vrf1-area- 10.10.10.10)# virtual-link 10.3.4.5 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1-area- 10.10.10.10-virtual-link-10.3.4.5)# transmit-delay 100 </pre>		

10.10.1.31. virtual-link

Description	To define an Open Shortest Path First v3 (OSPFv3) virtual link and enter routing OSPF configuration mode for this virtual link, use the virtual-link command. To remove a virtual link, use the no form of this command.		
Command Syntax	<pre>virtual-link <i>router-id</i> no virtual-link <i>router-id</i></pre>		
Command Mode	configuration ospfv3 vrf area		
Parameters Table	Name	Value	Description
	<i>router-id</i>	IPv4 address	Router ID associated with the virtual link neighbor. The router ID appears in the show ospf command display.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# routing ospfv3 ospf31 exaware(config-routing-ospfv3-ospf31)# vrf vrf1 exaware(config-routing-ospfv3-ospf31-vrf-vrf1)# area 10 exaware(config-routing-ospfv3-ospf31-vrf-vrf1-area-0.0.0.10)# virtual-link 10.3.4.5 exaware(config-routing-ospfv3-ospf31-vrf-vrf1-area-0.0.0.10- virtual-link-10.3.4.5)#</pre>		

10.10.1.32. vrf

Description	Enters routing OSPFv3 configuration mode for a specified VRF. Use the no form of the command to disable OSPFv3 routing for the specified VRF.		
Command Syntax	<pre>vrf vrf-name no vrf vrf-name</pre>		
Command Mode	configuration routing ospfv3		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Maximum 128 characters, no spaces	Name of the VRF for which to configure OSPFv3 routing. A default VRF is pre-configured in the system. To configure a user-defined VRF for OSPFv3 routing, you must first define it. See vrf (on page 1494).
Examples	This example shows how to enter configuration mode for the specified VRF: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing ospfv3 ospfv31 exaware(config-routing-ospfv3-ospfv31)# vrf vrf1 exaware(config-routing-ospfv3-ospfv31-vrf-vrf1)#</pre>		

10.10.2. OSPFv3 Routing Operational Commands

10.10.2.1. clear ospfv3 process

Description	Clears the OSPFv3 LSP database and adjacency DB sessions either globally or for a specified OSPFv3 tag.		
Command Syntax	<code>clear ospf process [tag <i>ospf-instance</i>]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>ospfv3-instance</i>	existing ospfv3 tag	Specify an OSPFv3 tag to clear.
Examples	The following example shows how to use this command: <code>exaware# clear ospfv3 process tag ospfv31</code>		

10.10.2.2. clear ospfv3 route

Description	Clears the OSPFv3 routing table. You have the following options: • Clear the LSP database for a specified OSPFv3 tag • Clear the LSP database for a specified address family and routing scheme		
Command Syntax	<code>clear ospfv3 route [tag <i>tag-name</i>]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>tag name</i>	existing ospfv3 tag	Specify an OSPFv3 tag to clear.
Examples	The following example shows how to use this command: <code>exaware# clear ospfv3 route tag ospfv31</code>		

10.10.2.3. clear ospfv3 statistics (interface)

Description	Clears the OSPFv3 LSP database.		
Command Syntax	<code>clear ospfv3 statistics interface [<i>interface-name</i>]</code>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>interface-name</i>	{x-eth mgmt} 0/0/ <i>port</i> {agg-eth loopback} <i>number</i>	Specify an OSPFv3 interface for which to clear statistics.
Examples	The following examples show how to use this command: <pre>exaware# clear ospfv3 statistics interface x-eth 0/0/1</pre>		

10.10.2.4. clear ospfv3 statistics (virtual-link)

Description	Clears the OSPF LSP database.		
Command Syntax	<pre>clear ospfv3 statistics virtual-link [{name <i>virtual-link-name</i>} {transit-area <i>area</i> remote-id <i>router-id</i>}]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>name</i>	string	The virtual link name assigned by the system (see command show ospf virtual-link).
	<i>area</i>	IPv4 Address or integer	The transit area
	<i>remote-id</i>	IPv4 Address	The router-id of the remote router.
Examples	The following examples show how to use this command: <pre>exaware# clear ospfv3 statistics virtual-link name VLINK2 exaware# clear ospfv3 statistics virtual-link transit-area 0.0.02. router-id 10.1.1.1</pre>		

10.10.3. OSPFv3 Routing Show Commands

10.10.3.1. show ospfv3

Description	Displays information per OSPFv3 instance.		
Command Syntax	show ospfv3 [tag <i>process-id</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>process-id</i>	0-65535	Process ID. If this argument is included, only information for the specified routing process is included.
Examples	The following example shows sample output from this command: <pre>exaware# show ospfv3 Process 0 VRF default Router-ID 20.0.4.101 Number of area 1 Number of Interface 10 Number of Virtual-links 3 Type Area Border Router Graceful restart enabled LSA refresh time 50 minutes SPF Calc Timers Min hold time 5 second Maximum wait time 5 second Area 0.0.0.100 Interfaces 10 Full state neighbors 40 Authentication disable SPF calculation 3 Number of LSA 4 Checksum Sum 0x2A6AC</pre>		

10.10.3.2. show ospfv3 border-routers

Description	Displays OSPFv3 routing table entries to an Area Border Router (ABR) and Autonomous System Boundary Router (ASBR). Include the tag ID parameter to view data about specified instances.																																																			
Command Syntax	show ospfv3 [tag tag-id] border-routers																																																			
Command Mode	Operational																																																			
Parameters Table	Name	Value	Description																																																	
	tag-id	<0-65535>	Tag ID. If this argument is included, only information for the specified routing tag is displayed.																																																	
Examples	The following example shows sample output from this command:																																																			
exaware# show ospfv3 tag 1 border-routers																																																				
<table><thead><tr><th>TYPE</th><th>ROUTER ID</th><th>METRIC</th><th>NEXTHOP</th><th>INTERFACE</th><th>ROUTER TYPE</th><th>AREA</th></tr></thead><tbody><tr><td>INTRA</td><td>1.2.3.40</td><td>1</td><td>2000::20c:29ff::ff45</td><td>VLINK1</td><td>10</td><td>ABR</td></tr><tr><td>INTRA</td><td>1.2.3.4</td><td>1</td><td>fe80::20c:29ff:fea8:ff45</td><td>x-eth0/0/1.10</td><td>ABR</td><td>.0.0.0</td></tr><tr><td>INTRA</td><td>52.52.52.52</td><td>1</td><td>fe80::20c:29ff:fea8:ff46</td><td>x-eth0/0/0.10</td><td>ABR, ASBR</td><td></td></tr><tr><td></td><td></td><td></td><td>fe80::20c:29ff:fea8:ff47</td><td>x-eth0/0/0.11</td><td>ABR, ASBR</td><td></td></tr><tr><td></td><td></td><td></td><td>fe80::c801:9ff:fe23:1c</td><td>x-eth0/0/0.12</td><td>ABR, ASBR</td><td></td></tr><tr><td></td><td></td><td></td><td>fe80::200:4ff:fec0:8e72</td><td>x-eth0/0/0.13</td><td>ABR, ASBR</td><td></td></tr></tbody></table>				TYPE	ROUTER ID	METRIC	NEXTHOP	INTERFACE	ROUTER TYPE	AREA	INTRA	1.2.3.40	1	2000::20c:29ff::ff45	VLINK1	10	ABR	INTRA	1.2.3.4	1	fe80::20c:29ff:fea8:ff45	x-eth0/0/1.10	ABR	.0.0.0	INTRA	52.52.52.52	1	fe80::20c:29ff:fea8:ff46	x-eth0/0/0.10	ABR, ASBR					fe80::20c:29ff:fea8:ff47	x-eth0/0/0.11	ABR, ASBR					fe80::c801:9ff:fe23:1c	x-eth0/0/0.12	ABR, ASBR					fe80::200:4ff:fec0:8e72	x-eth0/0/0.13	ABR, ASBR	
TYPE	ROUTER ID	METRIC	NEXTHOP	INTERFACE	ROUTER TYPE	AREA																																														
INTRA	1.2.3.40	1	2000::20c:29ff::ff45	VLINK1	10	ABR																																														
INTRA	1.2.3.4	1	fe80::20c:29ff:fea8:ff45	x-eth0/0/1.10	ABR	.0.0.0																																														
INTRA	52.52.52.52	1	fe80::20c:29ff:fea8:ff46	x-eth0/0/0.10	ABR, ASBR																																															
			fe80::20c:29ff:fea8:ff47	x-eth0/0/0.11	ABR, ASBR																																															
			fe80::c801:9ff:fe23:1c	x-eth0/0/0.12	ABR, ASBR																																															
			fe80::200:4ff:fec0:8e72	x-eth0/0/0.13	ABR, ASBR																																															
Name		Value	Description																																																	
TYPE		INTRA/INTER	INTRA - Intra-area route INTER - Inter-area route																																																	

ROUTER ID	A.B.C.D	Destination router ID
METRIC		Destination router metric
NEXTHOP		Next hop IPv6 address toward destination
INTERFACE		Next hop interface name or Virtual Link name assigned by the system.
ROUTER TYPE	ABR/ASBR	Destination router type
AREA ID	A.B.C.D	Decimal IP address
TYPE	INTRA/INTER	INTRA - Intra-area route INTER - Inter-area route

10.10.3.3. show ospfv3 database

Description	Displays link state database information for the specified OSPF instance.						
Command Syntax	show ospfv3 [<i>process-id</i>] database [adv-router <i>ip-address</i>] [self-originate] [brief detail]						
Command Mode	operational						
Parameters Table	Name	Value	Description				
	<i>process-id</i>	0-65535	Process ID. If this argument is included, only information for the specified routing process is included.				
	<i>ip-address</i>	IP address	Displays all LSAs of the specified router.				
	self-originate		Displays only self-originated LSAs (from the local router).				
		brief detail	Displays the database in brief format. Displays the database in detailed format.				
Examples	The following example shows sample output from this command:						
exaware# show ospfv3 database brief							
OSPF link state database (Area 0.0.0.1)							
Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	LEN

Router	0.0.0.1	false	10.255.71.242	0x80000002	0x4112	215	48
Inter-Area Prefix	23.1.1.0	true	10.255.71.242	0x80000002	0x6d72	172	28
Inter-Area Router	24.1.1.0	false	10.255.71.242	0x80000002	0x607e	177	28

10.10.3.4. show ospfv3 database external

Description	Displays information only about the external LSAs. Generated by the Autonomous System Border Routers (ASBRs). The link-state ID is the external network number.
Command Syntax	show ospfv3 database external [adv-router brief detail self-originate]
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command:
<pre>exaware# show ospfv3 database external detail AS External Link States Type LINK ID SELF ADV ROUTER SEQUENCE CHECKSUM AGE OPT LEN ----- External 33.1.1.1 false 10.255.71.242 0x800000002 0x73bd 217 0x28 36 Metric Type 2 Metric 0 Prefix 2222:: Prefix Length 56 External Route Tag 0</pre>	

10.10.3.5. show ospfv3 database grace

Description	Displays information about the state for the graceful restart link.																
Command Syntax	show ospfv3 database grace [adv-router brief detail self-originate]																
Command Mode	operational																
Parameters Table	N/A																
Examples	The following example shows sample output from this command:																
<pre>exaware# show ospf database external</pre> <table><tr><th>Type</th><th>LINK ID</th><th>SELF</th><th>ADV ROUTER</th><th>SEQUENCE</th><th>CHECKSUM</th><th>AGE</th><th>LEN</th></tr><tr><td>Grace</td><td>10.0.0.0</td><td>false</td><td>4.4.4.4</td><td>0x80000002</td><td>0x73bd</td><td>217</td><td>36</td></tr></table> <pre>Interface x-eth 0/0/1 Grace Period: 320 Restart Reason: Software Restart</pre>		Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	LEN	Grace	10.0.0.0	false	4.4.4.4	0x80000002	0x73bd	217	36
Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	LEN										
Grace	10.0.0.0	false	4.4.4.4	0x80000002	0x73bd	217	36										

10.10.3.6. show ospfv3 database inter-prefix

Description	Displays information about Inter-Area-Prefix LSAs.
Command Syntax	show ospfv3 database inter-prefix [adv-router brief detail self-originate]
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command:
<pre>exaware# show ospfv3 database inter-prefix Inter Area Prefix Link States (Area 0) Type LINK ID SELF ADV ROUTER SEQUENCE CHECKSUM AGE LEN ----- Inter-Prefix-Links 1.0.0.1 false 20.0.4.101 0x80000001 0xf597 325 36 Metric 1 Prefix 3002:: Prefix Length 56</pre>	

10.10.3.7. show ospfv3 database inter-router

Description	Displays information about Inter-Router LSAs.
Command Syntax	show ospfv3 database inter-router [adv-router brief detail self-originate]
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command: exaware# show ospfv3 database inter-router Type LINK ID SELF ADV ROUTER SEQUENCE CHECKSUM AGE LEN ----- Inter-Prefix-Links 1.0.0.1 false 20.0.4.101 0x80000001 0xf597 325 36 Dest-router-id 10.255.4.97 Options 0x19 Metric 10

10.10.3.8. show ospfv3 database intra-prefix

Description	Displays information about the Intra-Prefix LSAs.																								
Command Syntax	show ospfv3 database intra-prefix [adv-router brief detail self-originate]																								
Command Mode	operational																								
Parameters Table	N/A																								
Examples	The following example shows sample output from this command:																								
<pre>exaware# show ospfv3 database intra-prefix</pre> Intra Area Prefix Link States (Area 1) <table><tr><th>Type</th><th>LINK ID</th><th>SELF</th><th>ADV ROUTER</th><th>SEQUENCE</th><th>CHECKSUM</th><th>AGE</th><th>LEN</th></tr><tr><td colspan="8">-----</td></tr><tr><td>Intra-Prefix</td><td>10.1.70.1</td><td>false</td><td>10.255.70.103</td><td>0x80000002</td><td>0x4112</td><td>215</td><td>48</td></tr></table> <pre>Referenced LSA Type 2001 Referenced Link State ID 0 Referenced Advertising Router 10.0.0.206 Prefixes Num 1 Prefix 8006:: Prefix Length 56 Prefix-options 0x2 Metric 0</pre>		Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	LEN	-----								Intra-Prefix	10.1.70.1	false	10.255.70.103	0x80000002	0x4112	215	48
Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	LEN																		

Intra-Prefix	10.1.70.1	false	10.255.70.103	0x80000002	0x4112	215	48																		

10.10.3.9. show ospfv3 database link

Description	Displays information about the link LSAs.
Command Syntax	show ospfv3 database link [adv-router brief detail self-originate]
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command:
<pre>exaware# show ospfv3 database link Link (Type-8) Link States (Area 0) Type LINK ID SELF ADV ROUTER SEQUENCE CHECKSUM AGE LEN ----- Link 1.0.0.1 false 20.0.4.101 0x80000001 0xf597 325 36 Local Address fe80::204:c0ff:fe22:73fe Router Priority 1 Number of Prefixes 1 Options 0x11 Prefix 7002:: Prefix Length 56 Prefix-options 0x0</pre>	

10.10.3.10. show ospfv3 database network

Description	Displays information only about the network LSAs. Generated by Designated Routers (DR) and describes the set of routers attached to a particular network. They are flooded in the area that contains the network. The link-state ID is the IP interface address of the DR.
Command Syntax	show ospfv3 database network [adv-router brief detail self-originate]
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command: <pre>exaware# show ospfv3 database network detail Net Link States (Area 0.0.0.0) Type LINK ID SELF ADV ROUTER SEQUENCE CHECKSUM AGE LEN ----- Network 0.0.0.1 false 10.255.71.242 0x80000002 0x73bd 217 36 Attached Router Num 2 Attached Router 20.0.4.101 Attached Router 20.0.3.100</pre>

10.10.3.11. show ospfv3 database router

Description	Displays information only about the router LSAs. Generated by each router for each area to which it belongs. These packets describe the states of the router’s links to the area and are only flooded within a particular area. The link ID is the originating router’s ID.
Command Syntax	show ospfv3 database router [adv-router brief detail self-originate]
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command:

```
exaware# show ospfv3 database router detail
```

Type	LINK ID	SELF	ADV ROUTER	SEQUENCE	CHECKSUM	AGE	LEN

Router	0.0.0.1	false	10.255.71.242	0x80000002	0x73bd	217	36

Links Num

2

Type

Metric

Local Interface ID

Neighbor Interface ID

Neighbor Router ID

Transit Network

10

11

1

10.0.0.207

Type

Metric

Local Interface ID

Neighbor Interface ID

Neighbor Router ID

Transit Network

10

2

1

10.0.0.0

10.10.3.12. show ospfv3 flood-list

Description	Displays a list of link-state advertisements (LSAs) waiting to be flooded over an interface.				
Command Syntax	show ospfv3 flood-list [area <i>area-id</i>] [<i>detail</i> <i>brief</i>] [interface <i>interface-name</i>]				
Command Mode	operational				
Parameters Table	Name	Value	Description		
	<i>area-id</i>	0-4294967295 or IP address	Name of the OSPFv3 VRF. The vrf-name argument can be specified as an arbitrary string. Area number/ip used to define the particular area.		
		<i>detail</i> <i>brief</i>	Database in detail format. Database in brief format.		
	<i>interface-name</i>		Interface name.		
Examples	The following example shows sample output from this command:				
exaware# show ospfv3 flood-list interface x-eth 0/0/1 Interface x-eth0/0/1 Queue length 20 LS flooding 12 msec					
Type	LINK ID	ADV ROUTER	SEQUENCE	CHECKSUM	AGE
Router	0.0.0.199	10.0.0.207	0x80000009	0xFB61	0
Inter-Area Prefix	0.0.0.200	10.0.0.207	0x80000009	0x757	0
Inter-Area Router	10.0.0.206	10.0.0.207	0x80000002	0x11b1	214

10.10.3.13. show ospfv3 interface

Description	Displays OSPFv3-related interface information.						
Command Syntax	show ospfv3 interface [area <i>area-id</i>] [<i>detail</i> <i>brief</i>]						
Command Mode	Operational						
Parameters Table	Name	Value	Description				
	<i>area-id</i>	0-4294967295 or IP address	Name of the OSPFv3 VRF. The vrf-name argument can be specified as an arbitrary string. Area number/ip used to define the particular area.				
		<i>detail</i>	Database in detail format.				
		<i>brief</i>	Database in brief format.				
Examples	The following example shows sample output from this command:						
exaware# show ospfv3 interface brief							
NETWORK							
NAME	STATE	PASSIVE	TYPE	AREA	DR ID	BDR ID	NBRS

loopback0	Loopback	enabled	-	0.0.0.10			
x-eth0/1/1	Down	disabled	-	0.0.0.10			0
x-eth0/1/2	BDR	disabled	broadcast	0.0.0.11	10.10.10.4	10.10.10.5	1
x-eth0/1/3	DR	disabled	broadcast	0.0.0.11	20.20.20.4	20.20.20.5	1

10.10.3.14. show ospfv3 neighbors

Description	Displays OSPFv3 neighbors operational data on an individual interface basis. Note: If vrf is not specified, data is shown for default vrf.				
Command Syntax	show ospfv3 neighbors [area <i>area-id</i>] [<i>detail</i> <i>brief</i>] [interface <i>interface-name</i>]				
Command Mode	Operational				
Parameters Table	Name	Value	Description		
	<i>area-id</i>	0-4294967295	Name of the OSPFv3 VRF. The vrf-name argument can be specified as an arbitrary string.		
		or IP address	Area number/ip used to define the particular area.		
		<i>detail</i>	Database in detail format.		
		<i>brief</i>	Database in brief format.		
	<i>interface-name</i>		Interface name.		
Examples	The following example shows sample output from this command:				
exaware# show ospfv3 neighbors OSPFv3 process 0					
NEIGHBOR	STATE	INTERFACE	ADDRESS	PRIORITY	DEAD TIME
192.168.254.225	Full	x-eth0/0/1	fe80::290:69ff:fe9b:e002	2	34
10.1.1.129	Full	x-eth0/0/2	fe80::290:69ff:fe9b:e001	2	37
10.1.1.1	Full	VLINK1	2000::290:69ff:13af:e001	-	40

10.10.3.15. show ospfv3 request-list

Description	Displays request-list kept during synchronization of adjacent routers.				
Command Syntax	show ospfv3 request-list [area <i>area-id</i>] [<i>detail</i> <i>brief</i>] [interface <i>interface-name</i>] [neighbor <i>neighbor-id</i>]				
Command Mode	operational				
Parameters Table	Name	Value	Description		
	<i>area-id</i>	0-4294967295 or IP address	Name of the OSPFv3 VRF. The vrf-name argument can be specified as an arbitrary string. Area number/ip used to define the particular area.		
		<i>detail</i> <i>brief</i>	Database in detail format. Database in brief format.		
	<i>interface-name</i>		Interface name.		
	<i>neighbor-id</i>	IPv4 address	IPv4 address of the OSPFv3 neighbor.		
Examples	The following example shows sample output from this command:				
<pre>exaware# show ospfv3 request-list neighbor 10.0.124.4 interface x-eth 0/0/1 OSPF Router with ID (192.168.1.11) (Process ID 1) Neighbor 10.0.0.207 interface x-eth0/0/1 prefix fe80::3034:30ff:fe33:3742</pre>					
Type	LINK ID	ADV ROUTER	SEQUENCE	CHECKSUM	AGE
Router	10.2.195.0	192.168.0.163	0x80000009	0xFB61	0
Network	10.1.192.0	192.168.0.163	0x80000009	0x2938	0
Inter-Area	10.2.194.0	192.168.0.163	0x80000009	0x757	0

As-external	10.1.71.2	10.255.71.242	0x80000002	0x11b1	214
-------------	-----------	---------------	------------	--------	-----

10.10.3.16. show ospfv3 route

Description	Displays the current state of the routing table.																	
Command Syntax	show ospfv3 route [area <i>area-id</i>] [<i>detail</i> <i>brief</i>]																	
Command Mode	operational																	
Parameters Table	Name	Value	Description															
	<i>area-id</i>	0-4294967295 or IP address	Name of the OSPFv3 VRF. The vrf-name argument can be specified as an arbitrary string. Area number/ip used to define the particular area.															
		<i>detail</i> <i>brief</i>	Database in detail format. Database in brief format.															
Examples	The following example shows sample output from this command:																	
<pre>exaware# show ospfv3 route Flags: O - Intra area, IA - Inter area E1 - External type 1, E2 - External type 2 C - Directly connected, D - Discard</pre> <table><thead><tr><th>PREFIX</th><th>FLAGS</th><th>METRIC</th><th>NEXTHOP ADDRESS</th><th>NEXTHOP INTERFACE</th></tr></thead><tbody><tr><td>10.10.10.4/32</td><td>E2</td><td>10</td><td>fe80::290:69ff:fe9b:e002</td><td>etr0/0/1.1</td></tr><tr><td>10.10.10.5/32</td><td>E2</td><td>10</td><td>fe80::290:69ff:fe9b:e002</td><td>etr0/0/1.1</td></tr></tbody></table>				PREFIX	FLAGS	METRIC	NEXTHOP ADDRESS	NEXTHOP INTERFACE	10.10.10.4/32	E2	10	fe80::290:69ff:fe9b:e002	etr0/0/1.1	10.10.10.5/32	E2	10	fe80::290:69ff:fe9b:e002	etr0/0/1.1
PREFIX	FLAGS	METRIC	NEXTHOP ADDRESS	NEXTHOP INTERFACE														
10.10.10.4/32	E2	10	fe80::290:69ff:fe9b:e002	etr0/0/1.1														
10.10.10.5/32	E2	10	fe80::290:69ff:fe9b:e002	etr0/0/1.1														

10.10.3.17. show ospfv3 (statistics)

Description	Displays OSPFv3 traffic statistics information per interface.		
Command Syntax	show ospfv3 [tag <i>tag-name</i>] statistics [interface { <i>interface-name</i> }] [{ <i>detail</i> <i>brief</i> }]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>tag-name</i>	existing ospfv3 tag	Specify an OSPFv3 tag for which to show statistics.
	<i>interface-name</i>		Interface name.
		<i>detail</i> <i>brief</i>	Database in detail format. Database in brief format.

Examples

The following examples show sample output from this command:

```
exaware# show ospfv3 statistics brief
TYPE      SENT    RECEIVED
-----
Hello      51      0
DB Desc    0        0
LS Req     0        0
LS Upd     0        0
LS Ack     0        0

exaware# show ospfv3 statistics detail
Hello
  Sent: 1233                      Received: 158955
  Network Mask mismatch:          1
  Hello Interval mismatch:        2
  Dead Interval mismatch:         1
  Options mismatch:               3

DB Description
  Sent: 59569                      Received: 56597
  Unknown Neighbor ID:            1
  MTU mismatch:                   14
```


Negotiation Failed:	5	
MS-bit mismatch:	2	
I-bit mismatch:	2	
Options mismatch:	1	
Sequence Number mismatch:	0	
Neighbor state mismatch:	12	
LS Update		
Sent: 1289	Received: 5534	
Unknown Neighbor ID:	0	
LSA checksum error:	0	
Unknown LS type:	2	
LS age is more than MaxAge:	0	
LSA data mismatch:	1	
Neighbor state mismatch:	0	
LS Request		
Sent: 1173628	Received: 1140318	
Unknown Neighbor ID:	1	
Neighbor state mismatch:	7	
LS Ack		
Sent: 429727	Received: 1145790	
Unknown Neighbor ID:	3	
Neighbor state mismatch:	0	
Header Errors		
Invalid packet type:	4	
Version mismatch:	1	
Area mismatch:	30	
Network address mismatch:	0	
Instance ID mismatch:	0	
Authentication type mismatch:	2	
Authentication error:	5	
Checksum error:	2	
Total		
Sent: 1665446	Received: 2507194	
Failed: 23	Discarded: 52	

10.10.3.18. show ospfv3 statistics (virtual-link)

Description	Displays OSPFv3 traffic statistics information per interface.		
Command Syntax	show ospfv3 statistics virtual-link [{name <i>virtual-link-name</i> } {transit-area <i>area</i> remote-id <i>router-id</i> }] [{detail brief}]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>name</i>	string	The virtual link name assigned by the system (see command show ospf virtual-link).
	<i>area</i>	IPv4 Address or integer	The transit area
	<i>remote-id</i>	IPv4 Address	The router-id of the remote router.
		<i>detail</i> <i>brief</i>	Database in detail format. Database in brief format.

Examples The following examples show sample output from this command:

```
exaware# show ospfv3 statistics virtual-link brief
TYPE      SENT    RECEIVED
-----
Hello      51      0
DB Desc    0        0
LS Req     0        0
LS Upd     0        0
LS Ack     0        0

exaware# show ospfv3 statistics virtual-link detail
Hello
  Sent: 1233                      Received: 158955
  Network Mask mismatch:          1
```

Hello Interval mismatch:	2	
Dead Interval mismatch:	1	
Options mismatch:	3	
DB Description		
Sent: 59569	Received: 56597	
Unknown Neighbor ID:	1	
MTU mismatch:	14	
Negotiation Failed:	5	
MS-bit mismatch:	2	
I-bit mismatch:	2	
Options mismatch:	1	
Sequence Number mismatch:	0	
Neighbor state mismatch:	12	
LS Update		
Sent: 1289	Received: 5534	
Unknown Neighbor ID:	0	
LSA checksum error:	0	
Unknown LS type:	2	
LS age is more than MaxAge:	0	
LSA data mismatch:	1	
Neighbor state mismatch:	0	
LS Request		
Sent: 1173628	Received: 1140318	
Unknown Neighbor ID:	1	
Neighbor state mismatch:	7	
LS Ack		
Sent: 429727	Received: 1145790	
Unknown Neighbor ID:	3	
Neighbor state mismatch:	0	
Header Errors		
Invalid packet type:	4	
Version mismatch:	1	
Area mismatch:	30	
Network address mismatch:	0	
Instance ID mismatch:	0	
Authentication type mismatch:	2	
Authentication error:	5	
Checksum error:	2	
Total		
Sent: 1665446	Received: 2507194	
Failed: 23	Discarded: 52	

10.10.3.19. show ospfv3 virtual-links

Description	Displays parameters and the current state of Open Shortest Path First v3 (OSPFv3) virtual links.
Command Syntax	show ospfv3 virtual-links
Command Mode	Operational
Examples	The following example shows sample output from this command: <pre>exaware# show ospfv3 virtual-links Virtual Link: VLINK1 Router-ID 10.10.10.1 Transit Area: 0.0.0.1 Operational Status: UP Metric 1 Adjacency State: Full Local IPv6 Address: 2000::20c:29ff:fea8:ff45 Remote IPv6 Address: 2000::20c:29ff:fea8:ff46` Cost: 102 Timer intervals HELLO Interval: 30sec Dead Interval: 10sec Retransmission Interval: 100sec Transmission Delay: 10sec</pre>

10.11. PIM Commands

10.11.1. PIM Configuration Commands

10.11.1.1. af-ipv4

Description	Enters the address-family ipv4 configuration sub-mode for PIM routing, as follows: • IP address family for VRF default (only VRF currently supported) • IP address family for the specified interface Use the no form of the command to remove all ipv4-related configuration. All relevant parameters will revert to their default values.
Command Syntax	<pre>af-ipv4 no af-ipv4</pre>
Command Mode	<pre>configuration routing-pim-<instance> vrf-<vrf-name> configuration routing-pim-<instance> interface-<interface-instance></pre>
Parameters Table	N/A
Examples	The following example shows how to enter ipv4 configuration sub-mode for a specified VRF: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)#</pre> The following example shows how to enter ipv4 configuration sub-mode for a specified interface: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# interface x-eth 0/0/2 exaware(config-routing-pim-default-interface-x-eth- 0/0/2)# af-ipv4 exaware(config-interface-x-eth-0/0/2-af-ipv4)#</pre>

10.11.1.2. anycast-rp

Description	Configures anycast RP in the RP set. Use the no form of the command to remove the anycast configuration.		
Command Syntax	<pre>anycast-rp ip-address [member-addr member-address] no anycast-rp</pre>		
Command Mode	configuration routing-pim-<instance> vrf-<vrf-name> af-ipv4/6		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IP address	Unicast IP address of the anycast RP set. An anycast RP set is a collection of RPs in the same domain.
	<i>member-address</i>	IP address	Destination IP address where Register messages are copied and sent. A Member RP is an individual RP member in the anycast RP set.
Examples	The following example shows how to configure anycast RP: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# anycast-rp 10.10.10.32 member-addr 10.10.10.10</pre>		

10.11.1.3. bsr candidate-bsr

Description	Sets the router to announce its candidacy as a bootstrap router using the specified IP address of the interface. Use the no form of the command to remove candidate BSR status.		
Command Syntax	<pre>bsr candidate-bsr {<i>interface interface-instance</i>} [hash-mask-length <i>mask-length</i>] [<i>priority priority</i>]</pre> <pre>no bsr candidate-bsr</pre>		
Command Mode	configuration routing-pim-< <i>instance</i> > vrf-< <i>vrf-name</i> > af-ipv4/6		
Parameters Table	Name	Value	Description
	<i>interface interface-instance</i>	x-eth, agg-eth, x/y/z, loopback integer ID	Incoming interface type and instance.
	<i>mask-length</i>	0-32	Hash mask length for RP selection. (Default = 30)
	<i>priority</i>	0-255	Priority for the BSR candidate. (Default = 64)
Examples	The following example shows how to configure the candidate BSR: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# bsr candidate-bsr x-eth 0/0/0 priority 100</pre>		

10.11.1.4. bsr candidate-rp

Description	Sets the router to advertise itself to the bootstrap router as a candidate rendezvous point using the specified IP address of the interface. Use the no form of the command to remove candidate RP status.		
Command Syntax	<pre>bsr candidate-rp {<i>interface interface-instance</i>} [<i>interval interval</i>] [<i>priority priority</i>]</pre> <pre>no bsr candidate-rp</pre>		
Command Mode	configuration routing-pim-< <i>instance</i> > vrf-< <i>vrf-name</i> > af-ipv4/6		
Parameters Table	Name	Value	Description
	<i>interface interface-instance</i>	x-eth, agg-eth, x/y/z loopback integer ID	Incoming interface type and instance. Can only be specified for non-broadcast interfaces.
	<i>interval</i>	0-16383	Advertisement interval in seconds. (Default = 60)
	<i>priority</i>	0-255	Priority for the BSR candidate. (Default = 10)
Examples	The following example shows how to give a router the candidate RP status: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# bsr candidate-rp x-eth 0/0/0 interval 45 priority 100</pre>		

10.11.1.5. bsr-border

Description	Enables or disables sending and receiving BSR messages on the specified interface. Use the no form of the command to disable sending and receiving BSR messages.		
Command Syntax	<pre>bsr-border {enable disable} no bsr-border</pre>		
Command Mode	configuration routing-pim- <i><instance></i> interface- <i><interface></i> af-ipv4/6		
Parameters Table	Name	Value	Description
		enable	Enable sending and receiving BSR messages.
		disable	Disable sending and receiving BSR messages. (Default)
Examples	The following example shows how to enable sending and receiving BSR messages: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# interface x-eth 0/0/2 exaware(config-routing-pim-default-interface-x-eth- 0/0/2)# af-ipv4 exaware(config-interface-x-eth-0/0/2-af-ipv4)# bsr-border enable</pre>		

10.11.1.6. bsr-unicast-message

Description	Enables or disables sending and receiving unicast BSR messages on the interface. Use the no form of the command to disable sending and receiving unicast BSR messages.		
Command Syntax	<pre>bsr-unicast-message {enable disable} no bsr-unicast-message</pre>		
Command Mode	configuration routing-pim- <i><instance></i> interface- <i><interface></i> af-ipv4/6		
Parameters Table	Name	Value	Description
		enable	Enable unicast BSR messages functionality.
		disable	Disable unicast BSR messages functionality. (Default)
Examples	This example shows how to enable sending and receiving unicast BSR messages. <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# interface x-eth 0/0/2 exaware(config-routing-pim-default-interface-x-eth- 0/0/2)# af-ipv4 exaware(config-interface-x-eth-0/0/2-af-ipv4)# bsr- unicast-message enable</pre>		

10.11.1.7. dr-priority

Description	Configures the router priority to be elected as the designated router over the LAN segment . Use the no form of the command to revert to the default router priority (1).		
Command Syntax	<pre>dr-priority <i>priority</i> no dr-priority</pre>		
Command Mode	configuration routing-pim-< <i>instance</i> > interface-< <i>interface</i> > af-ipv4 af-ipv6		
Parameters Table	Name	Value	Description
	<i>priority</i>	1- 4294967294	Router priority for LAN election/ (Default = 1)
Examples	The following example shows how to set the router priority to 5: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# interface x-eth 0/0/2 exaware(config-routing-pim-default-interface-x-eth- 0/0/2)# af-ipv4 exaware(config-interface-x-eth-0/0/2-af-ipv4)# dr- priority 5</pre>		

10.11.1.8. hello holdtime

Description	Configures PIM hello packets holdtime in seconds. Use the no form of the command to revert to the default hello packets holdtime.		
Command Syntax	<pre>hello holdtime <i>seconds</i> no hello holdtime</pre>		
Command Mode	configuration routing-pim- <i><instance></i> interface- <i><interface></i> af-ipv4/6		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	Hello packets holdtime in seconds.
Examples	The following example shows how to set the hello packets holdtime to 90 seconds: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# interface x-eth 0/0/2 exaware(config-routing-pim-default-interface-x-eth- 0/0/2)# af-ipv4 exaware(config-interface-x-eth-0/0/2-af-ipv4)# hello holdtime 90</pre>		

10.11.1.9. hello interval

Description	Configures PIM hello packets interval for the DR in seconds. Use the no form of the command to revert to the default hello packets interval (30 seconds).		
Command Syntax	<pre>hello interval <i>seconds</i> no hello interval</pre>		
Command Mode	configuration routing-pim-< <i>instance</i> > interface-< <i>interface</i> > af-ipv4/6		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-18724	Hello packets interval for the DR in seconds. (Default = 30)
Examples	The following example shows how to set the hello packets interval to 90 seconds: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# interface x-eth 0/0/2 exaware(config-routing-pim-default-interface-x-eth- 0/0/2)# af-ipv4 exaware(config-interface-x-eth-0/0/2-af-ipv4)# hello interval 90</pre>		

10.11.1.10. interface

Description	Enters the interface configuration mode for PIM routing, for the specified interface. Only the 'default' PIM instance is currently supported. Use the no form of the command to remove all interface-related configuration. All relevant parameters will revert to their default values.		
Command Syntax	<pre>interface <i>interface-name</i> no interface <i>interface-name</i></pre>		
Command Mode	<pre>configuration routing-pim-<i><instance></i></pre>		
Parameters Table	Name	Value	Description
	<i>interface-name</i>	<pre>{x-eth mgmt} 0/0/<i>port</i> {agg-eth loopback} <i>number</i></pre>	Interface name.
Examples	The following example shows how to enter interface configuration sub-mode for a specified interface: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# interface x-eth 0/0/2 exaware(config-routing-pim-default-interface-x-eth- 0/0/2) #</pre>		

10.11.1.11. join-prune-interval

Description	Configures the join/prune timer. Use the no form of the command to revert to the default value (60 seconds).		
Command Syntax	<pre>join-prune-interval seconds no join-prune-interval</pre>		
Command Mode	configuration routing-pim-<instance> vrf-<vrf-name> af-ipv4/6		
Parameters Table	Name	Value	Description
	seconds	1-65535	Join/prune timer interval in seconds. (Default = 60)
Examples	The following example shows how to set the join/prune timer interval to 120 seconds: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# join-prune-interval 120</pre>		

10.11.1.12. passive

Description	Enables suppressing routing updates on the interface. Use the no form of the command to disable suppressing routing updates.		
Command Syntax	<pre>passive {enable disable} no passive</pre>		
Command Mode	configuration routing-pim- <i><instance></i> interface- <i><interface></i> af-ipv4/6		
Parameters Table	Name	Value	Description
		enable	Enable passive mode.
		disable	Disable passive mode. (Default)
Examples	The following example shows how to enable passive mode: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# interface x-eth 0/0/2 exaware(config-routing-pim-default-interface-x-eth- 0/0/2)# af-ipv4 exaware(config-interface-x-eth-0/0/2-af-ipv4)# passive enable</pre>		

10.11.1.13. register checksum

Description	Enables register packets checksum calculation on the whole packet. Use the no form of the command to disable register packets checksum calculation.		
Command Syntax	<pre>register checksum {enable disable} no register checksum</pre>		
Command Mode	configuration routing-pim-<instance> vrf-<vrf-name> af-ipv4/6		
Parameters Table	Name	Value	Description
		enable	Enable register checksum functionality.
		disable	Disable register checksum functionality. (Default)
Examples	The following example shows how to enable checksum functionality: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# register checksum enable</pre>		

10.11.1.14. register rp-reachability

Description	Enables RP reachability check-in registering for the DR. Use the no form of the command to disable RP reachability check-in registering.		
Command Syntax	<pre>register rp-reachability {enable disable} no register rp-reachability</pre>		
Command Mode	configuration routing-pim-<instance> vrf-<vrf-name> af-ipv4/6		
Parameters Table	Name	Value	Description
		enable	Enable RP reachability usage.
		disable	Disable RP reachability usage. (Default)
Examples	The following example shows how to enable RP reachability usage: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# register rp-reachability enable</pre>		

10.11.1.15. register source

Description	Configures the source address of register packets sent by the DR. Use the no form of the command to remove the source address configuration.		
Command Syntax	<pre>register source {ip-address interface interface-instance} no register source</pre>		
Command Mode	configuration routing-pim-<instance> vrf-<vrf-name> af-ipv4/6		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 address	Source address.
	<i>interface interface-instance</i>	x-eth, agg-eth, x/y/z loopback integer ID	Incoming interface type and interface name.
Examples	The following example shows how to configure the source of the register packets: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# register source loopback 15</pre>		
Notes	The configured address must be reachable to be used by the RP to send corresponding Register-Stop messages in response. It is normally the loopback interface address, but can also be other physical addresses. This address must be advertised by unicast routing protocols on the DR. The configured interface is not required to be PIM enabled.		

10.11.1.16. register timers keepalive-interval

Description	Configures the register packets keepalive interval in seconds. Use the no form of the command to revert to the default value.		
Command Syntax	<pre>register timers keepalive-interval <i>seconds</i> no register timers keepalive-interval</pre>		
Command Mode	configuration routing-pim-< <i>instance</i> > vrf-< <i>vrf-name</i> > af-ipv4/6		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	Register packets keepalive interval in seconds.
Examples	The following example shows how to configure the keepalive interval to 300: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# register timers keepalive-interval 300</pre>		

10.11.1.17. register timers rate-limit

Description	Configures rate of register packets sent by the DR per second. Use the no form of the command to revert to the default value.		
Command Syntax	<pre>register timers rate-limit <i>seconds</i> no register timers rate-limit</pre>		
Command Mode	configuration routing-pim-< <i>instance</i> > vrf-< <i>vrf-name</i> > af-ipv4/6		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-65535	Rate of register packets sent per second.
Examples	The following example shows how to configure the rate of sending register packets to 120: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# register timers rate-limit 120</pre>		

10.11.1.18. register timers suppress-interval

Description	Configures the register packets suppression interval for the DR in seconds. Use the no form of the command to revert to the default value (60 seconds).		
Command Syntax	<pre>register timers suppress-interval seconds no register timers suppress-interval</pre>		
Command Mode	configuration routing-pim-<instance> vrf-<vrf-name> af-ipv4/6		
Parameters Table	Name	Value	Description
	seconds	12-65535	Register packets suppression interval in seconds. (Default = 60)
Examples	The following example shows how to configure the suppression interval to 300: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# register timers suppress-interval 300</pre>		

10.11.1.19. routing pim

Description	Enables the default PIM routing instance and enters the PIM routing configuration mode. Use the no form of the command to disable PIM routing (default).		
Command Syntax	<pre>routing pim default no routing pim</pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
		default	Only 'default' PIM instance is supported.
Examples	The following example shows how to configure the maximum and the warning threshold: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)#</pre>		
Notes	This is the main sub mode for configuring PIM routing.		

10.11.1.20. rp-address

Description	Statically configures a Rendezvous Point (RP) for a range of multicast groups. The configuration will associate a single multicast group, or multicast groups list with the IP address of a Rendezvous Point (RP). The RP can also be configured with the override parameter whose default value is TRUE. When FALSE, the router will use dynamic mapping of the group to the RP. Use the no form of the command to remove the static RP address configuration. Only vrf default is supported.		
Command Syntax	<pre>rp-address <ip-address> group {<multicast-group-name> <ip-prefix group>} override {enable disable} no rp-address</pre>		
Command Mode	configuration routing-pim-<instance> vrf-<vrf-name> af-ipv4/6		
Parameters Table	Name	Value	Description
	<i>ip-address</i>	IPv4 address	The address of the Rendezvous Point.
	group	Multicast IP-address or, Set of multicast IP-addresses or, IP-prefix group	Allow you to specify a single or a range of multicast group(s).

	override	enable disable	Enable/disable static RP override of dynamically learned RP. (Default = enable)
Examples	The following example shows how to set the static RP address: <pre> exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af- ipv4 exaware(config-routing-pim-default-vrf-default-af- ipv4)# rp-address 192.168.5.1 group MG override enable exaware(config-routing-pim-default-vrf-default-af- ipv4)# rp-address 192.168.5.2 group 232.2.2.0/30 override enable exaware(config-routing-pim-default-vrf-default-af- ipv4)# rp-address 192.168.5.3 group (232.2.2. 0/32, 232.2.3. 0/32) </pre>		
Notes	Unicast IP addresses, or IPv6 addresses that are configured for group will be ignored. This behavior is similar to other routing protocols.		

10.11.1.21. rp-use-priority

Description	Enables ignoring the RP-SET priority value, and using only the hashing mechanism for RP selection. Use the no form of the command to disable using the RP-SET priority value. Only vrf default is supported.		
Command Syntax	<pre>rp-use-priority {enable disable} no rp-use-priority</pre>		
Command Mode	configuration routing-pim-<instance> vrf-<vrf-name> af-ipv4/6		
Parameters Table	Name	Value	Description
		enable	Enable using RP set priority. (Default)
		disable	Disable using RP set priority.
Examples	The following example shows how to disable using RP set priority and using only the hashing mechanism for RP selection: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# rp-use-priority disable</pre>		

10.11.1.22. spt-switchover

Description	Enables PIM last hop router switchover to shortest path tree. Use the no form of the command to disable spt switchover.		
Command Syntax	<pre>spt-switchover {enable disable} no spt-switchover</pre>		
Command Mode	configuration routing-pim-<instance> vrf-<vrf-name> af-ipv4/6		
Parameters Table	Name	Value	Description
		enable	Enable spt switchover. (Default)
		disable	Disable spt switchover.
Examples	The following example shows how to disable checksum functionality: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# spt-switchover disable</pre>		

10.11.1.23. ssm

Description	Enables PIM source-specific multicast. Use the no form of the command to disable source-specific multicast.		
Command Syntax	<pre>ssm {enable disable} no ssm</pre>		
Command Mode	configuration routing-pim-<instance> vrf-<vrf-name> af-ipv4/6		
Parameters Table	Name	Value	Description
		enable	Enable SSM.
		disable	Disable SSM. (Default)
Examples	The following example shows how to enable SSM: <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)# af-ipv4 exaware(config-vrf-default-af-ipv4)# ssm enable</pre>		

10.11.1.24. vrf

Description	Enters the vrf routing configuration mode for the specified vrf. Use the no form of the command to remove all vrf-related configuration. All relevant parameters will revert to their default values. Only vrf default is supported.		
Command Syntax	<pre>vrf vrf-name</pre> <pre>no vrf vrf-name</pre>		
Command Mode	configuration routing-pim-<instance>		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	String, maximum 32 characters	Name of the VRF.
Examples	The following example shows how to enter configuration mode for the specified VRF: Note the change in the command line prompt. <pre>exaware# config exaware(config)# routing pim default exaware(config-routing-pim-default)# vrf default exaware(config-routing-pim-default-vrf-default)#</pre>		

10.11.2. PIM Operational Commands

10.11.2.1. clear pim bsr

Description	Clears RP learned through the BSR.		
Command Syntax	<code>clear pim bsr [af ipv4] [vrf <i>vrf-name</i>]</code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
		af ipv4	Address family.
	<i>vrf-name</i>		VRF name.
Examples	The following example shows how to use this command: exaware# clear pim bsr		

10.11.2.2. clear pim table

Description	Clears PIM topology table and multicast forwarding table.		
Command Syntax	<code>clear pim table [af ipv4] [vrf <i>vrf-name</i>] [group { ip <i>ip-address</i> source <i>source-address</i> }]</code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
		<i>af ipv4</i>	Address family.
	<i>vrf-name</i>		VRF name.
	<i>ip-address</i>	IP address	Group IP address.
	<i>source-address</i>	IP address	Source IP address.
Examples	The following example shows how to use this command: <pre>exaware# clear pim table af ipv4 group { ip 22.1.1.1 source 10.10.10.10 }</pre>		

10.11.3. PIM Show Commands

10.11.3.1. show pim bsr

Description	Displays information about the bootstrap router.		
Command Syntax	show pim bsr [ipv4] [vrf <i>vrf-name</i>] [detail]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
		ipv4	Address family to display (optional).
	<i>vrf-name</i>		VRF to display (optional).
		detail	Show detailed bootstrap router information. (Default)
Examples	The following example shows sample output from this command: <pre> exaware# show pim bsr detail address 200.1.1.50 elected disable uptime 0s priority 64 hash mask length 30 state accept-preferred BSR candidate disable candidate-RP 200.1.1.2 name x-eth0/0/1 advertisement interval 60 next advertisement 3s backoff counter 2 </pre>		

10.11.3.2. show pim interface

Description	Displays information about the PIM interface.					
Command Syntax	show pim interface [ipv4] [vrf <i>vrf-name</i>] [brief detail]					
Command Mode	operational					
Parameters Table	Name	Value	Description			
		ipv4	Address family to display (optional).			
	<i>vrf-name</i>		VRF to display (optional).			
		brief (Default)				
		detail				
Examples	The following examples show sample output from this command:					
exaware# show pim interface ipv4 brief						
INTERFACE	ADDRESS	INDEX	NEIGHBOR COUNT	DR PRIORITY	DR ADDRESS	
x-eth0/0/1	200.1.1.1	0	2	1	200.1.1.50	
x-eth0/0/2	100.1.1.1	2	1	1	100.1.1.102	
exaware# show pim interface ipv4 detail						
detail x-eth0/0/1						
address		200.1.1.1				
index		0				
passive		disable				
dr address		200.1.1.50				
bsr border		disable				
hello period s		30				
next hello s		8				
triggered hello period s		5				
hello rx count		50				
hello tx count		54				

```
propagation delay ms      1000
unicast bsr msg support   disable
neighbors 200.1.1.2
neighbors 200.1.1.50
detail x-eth0/0/2
  address                  100.1.1.1
  index                    2
  passive                  disable
  dr address               100.1.1.102
  bsr border               disable
  hello period s           30
  next hello s             3
  triggered hello period s 5
  hello rx count           51
  hello tx count           55
  propagation delay ms     1000
  unicast bsr msg support  disable
  neighbors 100.1.1.102
```

10.11.3.3. show pim join

Description	Displays contents of the MRIB table.		
Command Syntax	show pim join [ipv4] [vrf <i>vrf-name</i>] [all summary detail]		
Command Mode	operational		
Parameters Table	Name	Value	Description
		ipv4	Address family to display (optional).
	<i>vrf-name</i>		VRF to display (optional).
		all (Default)	
		summary	
		detail	
Examples	The following examples show sample output from this command:		

```
exaware# show pim join ipv4 summary
```

```
(*, *, RP)          0
(*, G)              3
(S, G)              3
(S, G, RPT)         3
Forwarding Cache Records 2
```

```
exaware# show pim join ipv4 all
```

```
pim join ipv4 all XG
```

```
(*, G) Joins
```

GROUP	UPTIME	RP	RPF NEIGHBOR	RPF INTERFACE	UPSTREAM STATE	OUTGOING
INTERFACE						

231.1.1.1	1m8s	1.1.1.100	-	-	joined	x-
eth0/0/1						
232.1.1.1	1m12s	1.1.1.100	-	-	joined	x-
eth0/0/1.1						

```

234.1.1.1 1m8s 1.1.1.100 - - joined x-
eth0/0/1

eth0/0/1.3 x-

pim join ipv4 all SG
(S, G) Joins

                                RPF          RPF          UPSTREAM
OUTGOING
SOURCE      GROUP      UPTIME NEIGHBOR      INTERFACE      STATE      SPT
BIT      INTERFACE
-----
201.1.1.100 231.1.1.1 1m12s 101.1.1.100 x-eth0/0/1.1 joined
disabled x-eth0/0/1
202.1.1.100 232.1.1.1 1m12s 102.1.1.100 x-eth0/0/1.2 joined
disabled x-eth0/0/1.1
204.1.1.100 234.1.1.1 1m12s 104.1.1.100 x-eth0/0/1.4 joined
disabled x-eth0/0/1

x-eth0/0/1.3

pim join ipv4 all SGRPT
(S, G, RPT) Joins

                                RPF          RPF
UPSTREAM
SOURCE      GROUP      UPTIME  RP          NEIGHBOR  INTERFACE  STATE
-----
201.1.1.100 231.1.1.1 1m13s 1.1.1.100 - - not-
pruned
202.1.1.100 232.1.1.1 1m13s 1.1.1.100 - - not-
pruned
204.1.1.100 234.1.1.1 1m13s 1.1.1.100 - - not-
pruned

exaware# show pim join ipv4 detail

pim join ipv4 detail XG 231.1.1.1
  uptime          1m13s
  RP              1.1.1.100
  upstream state   joined
  SPT switchover   disabled
  join prune timer off
  state            "Join Desired"
  downstream x-eth0/0/1

```

```
state                joined
expiry timer         107s
prune pending timer  off
macro state          "Could Assert, Assert Track"
assert
  state no-info
  timer off
  winner
    address          0.0.0.0
    metric            4294967295
    metric preference 4294967295
    RPT bit           enabled
joined-olist x-eth0/0/1
FCR 201.1.1.100
  keepalive timer 169s
  packet counter  0
  inherited-olist x-eth0/0/1
pim join ipv4 detail SG 201.1.1.100 231.1.1.1
  uptime          1m17s
  RPF neighbor     101.1.1.100
  RPF interface    x-eth0/0/1.1
  upstream state   joined
  SPT bit          disabled
  join prune timer 5s
  keepalive timer  205s
  state            "Join Desired"
downstream x-eth0/0/1
  state                joined
  expiry timer         171s
  prune pending timer  off
  macro state          "Assert Track"
  assert
    state no-info
    timer off
    winner
      address          0.0.0.0
      metric            4294967295
      metric preference 4294967295
      RPT bit           enabled
joined-olist x-eth0/0/1
inherited-olist x-eth0/0/1
pim join ipv4 detail SGRPT 201.1.1.100 231.1.1.1
  uptime          1m19s
  RP              1.1.1.100
  upstream state   not-pruned
  override timer  off
```

```
state          "RPT Join"  
inherited-olist x-eth0/0/1
```


10.11.3.4. show pim neighbors

Description	Displays information about the PIM neighbors.				
Command Syntax	show pim neighbor [ipv4] [vrf <i>vrf-name</i>] [brief detail]				
Command Mode	operational				
Parameters Table	Name	Value	Description		
		ipv4	Address family to display (optional).		
		<i>vrf-name</i>	VRF to display (optional).		
		brief (Default)			
		detail			
Examples	The following examples show sample output from this command:				

```
exaware# show pim neighbors ipv4 brief
                                     DR
INTERFACE      ADDRESS        UPTIME        EXPIRES        PRIORITY    DR
-----
x-eth0/0/1     200.1.1.2      2D1h24m28s   1m42s          1           false
x-eth0/0/1     200.1.1.50     2D1h24m33s   1m19s          10          true
x-eth0/0/2     100.1.1.102    2D1h24m34s   1m40s          1           true
exaware# show pim neighbors ipv4 detail
detail x-eth0/0/1 200.1.1.2
  uptime                2D1h24m31s
  expires                1m39s
  dr priority            1
  dr                     false
  hello holdtime         105
  tracking support        false
  lan delay s             1
  override interval s    3
  gen id                 159152341
detail x-eth0/0/1 200.1.1.50
  uptime                2D1h24m36s
  expires                1m16s
  dr priority            10
  dr                     true
```

```
hello holdtime      105
```

```
tracking support    false
lan delay s         0
  override interval s 2
  gen id             1523682350
detail x-eth0/0/2 100.1.1.102
  uptime             2D1h24m37s
  expires             1m37s
  dr priority        1
  dr                  true
  hello holdtime     105
  gen id             3962646278
```

10.11.3.5. show pim next-hop

Description	Displays PIM next-hop information.				
Command Syntax	show pim next-hop [ipv4] [vrf <i>vrf-name</i>] [brief]				
Command Mode	operational				
Parameters Table	Name	Value	Description		
		ipv4	Address family to display (optional).		
	<i>vrf-name</i>		VRF to display (optional).		
		brief (Default)			
Examples	The following example shows sample output from this command:				
exaware# show pim next-hop					
Flags: N = New, R = RP, S = Source, U = Unreachable					
DESTINATION	TYPE	NEXTHOP NUM	NEXTHOP ADDRESS	INTERFACE	METRIC
PREFERENCE					

--					
201.1.1.100	S	1	101.1.1.100	x-eth0/0/1.1	10 33
202.1.1.100	S	1	102.1.1.100	x-eth0/0/1.2	10 33
204.1.1.100	S	1	104.1.1.100	x-eth0/0/1.4	10 33

10.11.3.6. show pim rp

Description	Displays PIM RP information, for either BSR or a static defined RP.		
Command Syntax	show pim rp [ipv4] [vrf <i>vrf-name</i>] [all [bsr static]] [group <i>ip-address</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
		ipv4	Address family to display (optional).
	<i>vrf-name</i>		VRF to display (optional).
		all (Default)	One of the following: - All PIM RP information (default) - All BSR information All static RP information
	<i>ip-address</i>		IP address of a specific group for which to display RP information.
Examples	The following examples show sample output from this command:		
exaware# show pim rp ipv4 all static			
GROUP	ADDRESS	UPTIME	OVERVERRIDE DYNAMIC INACTIVE SSM

224.0.0.0/4	200.1.1.2	2D20h11m29s	false false false
exaware## show pim rp ipv4 all bsr			
GROUP	GROUP		
GROUP	EXPIRES	ADDRESS	BIDIR SOURCE PRIORITY UPTIME
EXPIRES			

224.0.0.0/4	-	200.1.1.2	false 100.1.1.102 10 2m37s
1m51s		100.1.1.1	false 100.1.1.102 10 2m36s
1m51s			
exaware# show pim rp ipv4 group 224.0.0.0			

```
pim rp ipv4 group 224.0.0.0
```

10.12. Routing and Forwarding Information Base Commands

10.12.1. L2 Services Configuration Commands

10.12.1.1. admin-state

Description	Configures admin-state of the VPLS service. Use the no form of the command to remove the admin-state configured and use the default value.		
Command Syntax	admin-state {up down} no admin-state		
Command Mode	configuration l2-services vpls		
Parameters Table	Name	Value	Description
		up down	Set VPLS admin state to up (default). Set VPLS admin state to down.
Examples	The following example shows how to configure the admin-state of the VPLS service: exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# admin-state up exaware(config-l2-services-vpls-Service1)# no admin-state		

10.12.1.2. auto-discovery

Description	Configures auto-discovery attributes. Use the no form of the command to remove everything in auto-discovery.
Command Syntax	auto-discovery no auto-discovery
Command Mode	configuration l2-services vpls
Parameters Table	N/A
Examples	The following example shows how to configure the auto-discovery attributes: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# auto-discovery exaware(config-l2-service-vpls-Service1s)# no auto-discovery</pre>

10.12.1.3. block-size

Description	Configures the label block size. Use the no form of the command to remove the block-size and use its default value.		
Command Syntax	<pre>block-size size no block-size</pre>		
Command Mode	configuration l2-services vpls auto-discovery		
Parameters Table	Name	Value	Description
	size	2, 4, 8, 16	Block size. (Default = 8)
Examples	The following example shows how to configure the block size: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# auto-discovery exaware(config-vpls-Service1-auto-discovery)# block-size 2 exaware(config-vpls-Service1-auto-discovery)# no block-size</pre>		

10.12.1.4. control-word (vpls)

Description	To enable Control Word in VPLS packets, use the control-word CLI command. Use the no form of this command, to disable (default).														
Command Syntax	control-word enable disable no control-word														
Command Mode	configuration l2-services vpls auto-discovery														
Parameters Table	Name	Value	Description												
	enable		A Control Word must be present in VPLS packets,												
	disable		A Control Word must not be present in VPLS packets,												
Examples	This example shows how to use this command: exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# auto-discovery exaware(config-vpls-Service1-auto-discovery)# control-word enable														
Notes	This following table specifies the result depended on configuration on both sides: <table><tr><td>local side</td><td>Enable</td><td>disable</td></tr><tr><td>remote side</td><td></td><td></td></tr><tr><td>enable</td><td>3</td><td>1</td></tr><tr><td>disable</td><td>1</td><td>2</td></tr></table> 1: PW is not set 2: PW is set without Control Word. 3: PW is set with Control Word.			local side	Enable	disable	remote side			enable	3	1	disable	1	2
local side	Enable	disable													
remote side															
enable	3	1													
disable	1	2													

10.12.1.5.export-rt

Description	Configures a list of route-target values to be exported. Use the no form of the command to remove the auto-discovery export-rt configured route-target values.		
Command Syntax	<pre>export-rt route-target no export-rt route-target</pre>		
Command Mode	configuration l2-services vpls auto-discovery		
Parameters Table	Name	Value	Description
	route-target	<2-byte ASN>:<4-byte number> <4-byte ASN>L:<2-byte number> <IPv4 address>:<2-byte number>	2-byte ASN notation. 4-byte ASN notation. IPv4 address notation.
Examples	The following example shows how to configure the route-target values to be exported: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# auto- discovery exaware(config-vpls-Service1-auto-discovery)# export-rt 100:200 exaware(config-vpls-Service1-auto-discovery)# export-rt 100:201 exaware(config-vpls-Service1-auto-discovery)# no export-rt 100:200 exaware(config-vpls-Service1-auto-discovery)# no export-rt</pre>		

10.12.1.6. import-rt

Description	Configures a list of route-target values to be imported. Use the no form of the command to remove the auto-discovery import-rt configured route-target values.		
Command Syntax	<pre>import-rt route-target no import-rt route-target</pre>		
Command Mode	configuration l2-services vpls auto-discovery		
Parameters Table	Name	Value	Description
	<i>route-target</i>	<2-byte ASN>:<4-byte number>	2-byte ASN notation.
		<4-byte ASN>L:<2-byte number>	4-byte ASN notation.
		<IPv4 address>:<2-byte number>	IPv4 address notation.
Examples	The following example shows how to configure the route-target values to be exported: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# auto- discovery exaware(config-vpls-Service1-auto-discovery)# import-rt 100:200 exaware(config-vpls-Service1-auto-discovery)# import-rt 100:201 exaware(config-vpls-Service1-auto-discovery)# no import-rt 100:200 exaware(config-vpls-Service1-auto-discovery)# no import-rt</pre>		

10.12.1.7. interface

Description	Configures an interface (attachment circuit) under the VPLS service. Use the no form of the command to remove the interface configuration under the VPLS service.		
Command Syntax	<pre>interface <i>if-name</i> no interface <i>if-name</i></pre>		
Command Mode	configuration l2-services vpls		
Parameters Table	Name	Value	Description
	<i>if-name</i>	interface	VPLS service interface (attachment circuit).
Examples	The following example shows how to configure an attachment circuit under the VPLS service: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# interface x-eth 0/0/3 exaware(config-l2-services-vpls-Service1)# interface x-eth 0/0/4 exaware(config-l2-services-vpls-Service1)# no interface x-eth 0/0/3 exaware(config-l2-services-vpls-Service1)# no interface</pre>		
Notes	Only interfaces with l2-transport set to 'enable' can be configured as an attachment circuit. An attachment circuit can be configured on interface types x-eth, agg-eth, and on sub-interfaces of all these types.		

10.12.1.8. l2-services

Description	To enter L2 service configuration mode, use the l2-service command. To remove the L2 service configuration, use the no form of this command.		
Command Syntax	<pre>l2-services no l2-services</pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>vpls-name</i>	string	Virtual Private LAN service name.
Examples	The following example shows how to configure the VPLS service name: <pre>exaware(config)# l2-services exaware(config-l2-services)#</pre>		

10.12.1.9. mac-address-static (not supported in v8.0.0)

Description	Use the mac-address-static command to configure a static MAC address. Use the no form of the command to remove the static MAC address.		
Command Syntax	<pre>mac-address-static mac-address {interface interface peer ip-address vpws vpws-name} no mac-address-static [mac-address]</pre>		
Command Mode	configuration l2-services vpls		
Parameters Table	Name	Value	Description
	<i>mac-address</i>	xx:xx:xx:xx:xx:xx	MAC address.
	<i>interface</i>	Interface name	AC interface name.
	<i>ip-address</i>	A.B.C.D	PW peer IP address.
	<i>vpws-name</i>	string	VPWS name.
Examples	The following example shows how to use this command: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# mac- address-static 00:01:01:02:03:04 interface x-eth 0/0/12.4 exaware(config-l2-services-vpls-Service1)# mac- address-static 00:01:01:12:13:04 peer 10.23.1.2 exaware(config-l2-services-vpls-Service1)# mac- address-static 00:01:01:aa:bb:04 vpls VPWS1 exaware(config-l2-services-vpls-Service1)# no mac- address-static 00:11:11:12:13:04</pre>		

10.12.1.10. mac-aging-time

Description	Use the mac-aging-time command to configure the VPLS MAC address table aging time. Use the no form of the command to restore the default configuration.		
Command Syntax	<code>mac-aging-time seconds</code> <code>no mac-aging-time</code>		
Command Mode	configuration l2-services vpls		
Parameters Table	Name	Value	Description
	seconds	0, 40-2400	Specifies the aging time in seconds. A value of 0 means that aging time is disabled. The default value is 300 second.
Examples	The following example shows how to use this command: <code>exaware(config)# l2-services</code> <code>exaware(config-l2-services)# vpls service1</code> <code>exaware(config-l2-servicesvpls-service1)# mac-aging-time 400</code>		
Notes	Devices with Jerico1 support only the following subset of the range defined above: 0, 100-600.		

10.12.1.11. mac-limit

Description	Sets the maximum number of MAC addresses learned for the entire VPLS service. Use the no form of the command to remove the mac-limit configured.		
Command Syntax	<pre>mac-limit limit</pre> <pre>no mac-limit limit</pre>		
Command Mode	configuration l2-services vpls		
Parameters Table	Name	Value	Description
	<i>limit</i>	1-250000	Maximum MAC addresses learned. (Default = 65520)
Examples	The following examples show how to set the maximum number of MAC addresses learned: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# mac-limit 5000</pre> <pre>exaware(config-l2-services-vpls-Service1)# no mac-limit</pre>		
Notes	Mac-limit under the VPLS service limits the number of MAC addresses learned for the entire VPLS service.		

10.12.1.12. mtu

Description	Configures a VPLS service signaled mtu. Use the no form of the command to remove the VPLS service signaled mtu configuration and use the default value.		
Command Syntax	<pre>mtu <i>number</i> no mtu</pre>		
Command Mode	configuration l2-services vpls		
Parameters Table	Name	Value	Description
	<i>number</i>	594-9216	MTU size in bytes of the VPLS. (Default = 1500)
Examples	The following example shows how to use this command: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# mtu 2500 exaware(config-l2-services-vpls-Service1)# no mtu</pre>		

10.12.1.13. rd

Description	Configures the route-distinguisher value. Use the no form of the command to remove everything in auto-discovery except the rd.		
Command Syntax	<pre>rd route-distinguisher no rd</pre>		
Command Mode	configuration l2-services vpls auto-discovery		
Parameters Table	Name	Value	Description
	<i>route-distinguisher</i>	<2-byte ASN>:<4-byte number>	2-byte ASN notation.
		<4-byte ASN>L:<2-byte number>	4-byte ASN notation.
		<IPv4 address>:<2-byte number>	IPv4 address notation.

10.12.1.14. split-horizon-group

Description	Sets the split-horizon-group the VPLS interface should belong to. The VPLS interface discards all frames arriving from other VPLS interfaces in the group. Use the no form of the command to remove the interface from the split horizon group.
Command Syntax	<pre>split-horizon-group 1 2 no split-horizon-group</pre>
Command Mode	<pre>configuration l2-services vpls interface configuration l2-services vpls vpws spoke</pre>
Examples	The following example shows how to use this command: <pre>exaware(config)#l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# interface x-eth 0/0/3 exaware(config-vpls-Service1-interface-x-eth-0/0/3)# split- horizon-group 1 exaware(config-vpls-Service1-interface-x-eth-0/0/3)# exit exaware(config-l2-services-vpls-Service1)# vpws aaaa spoke exaware(config-vpls-Service1-vpws-aaaa)# split-horizon- group 1 exaware(config-vpls-Service1-vpws-aaaa)# exit</pre>

10.12.1.15. ve-id

Description	Configures the VPLS edge identifier.		
Command Syntax	<code>ve-id <i>number</i></code>		
Command Mode	configuration l2-services vpls		
Parameters Table	Name	Value	Description
	<i>number</i>	1- 65535	VPLS edge identifier.
Examples	The following example shows how to configure the VPLS edge identifier: <pre>exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# ve-id 53</pre>		
Notes	VE-ID is mandatory in VPLS and cannot be deleted.		

10.12.1.16. vpls

Description	Configures VPLS service name. Use the no form of the command to remove the VPLS service configuration.		
Command Syntax	<pre>vpls vpls-name</pre> <pre>no vpls</pre>		
Command Mode	configuration l2-services		
Parameters Table	Name	Value	Description
	<i>vpls-name</i>	string	Virtual Private LAN service name.
Examples	The following example shows how to configure the VPLS service name: <pre>exaware(config)# l2-services</pre> <pre>exaware(config-l2-services)# vpls Service1</pre> <pre>exaware(config-l2-services)# no vpls Service1</pre>		
Notes	VPLS service must be configured with VLAN edge identifier (ve-id) and route-distinguisher (auto-discovery/rd) to pass commit validation.		

10.12.1.17. vpws(vpls)

Description	Adds a VPWS endpoint of the VPLS service. Use the no form of the command to remove the endpoint.		
Command Syntax	vpws vpws-name [spoke mesh] no vpws		
Command Mode	configuration l2-services vpls		
Parameters Table	Name	Value	Description
	vpws-name	string	VPWS name.
	spoke		The attached VPWS is a spoke link. It is the default.
	mesh		The attached VPWS is a mesh link.
Examples	The following example shows how to define VPLS endpoints: <pre> exaware# config exaware(config)# l2-services exaware(config-l2-services)# vpls Service1 exaware(config-l2-services-vpls-Service1)# interface x-eth 0/0/3 exaware(config-l2-services-vpls-Service1)# interface x-eth 0/0/4 exaware(config-l2-services-vpls-Service1)# vpws aaaa spoke exaware(config-l2-services-vpls-Service1)# vpws bbbb spoke exaware(config-l2-services-vpls-Service1)# vpws cccc mesh exaware(config-l2-services-vpls-Service1)# vpws dddd mesh </pre>		

10.12.2. L2 Services Operational Commands

10.12.2.1.clear vpls mac address-table

Description	Clears VPLS mac address-table.		
Command Syntax	clear vpls mac address-table name <i>vpls-name</i> [source { <i>interface</i> peer-ip <i>ip-address</i> vpws <i>vpws-name</i> mac <i>mac-address</i> }		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>vpls-name</i>	String	Name of VPLS service.
	<i>interface</i>	Interface name	AC interface name.
	<i>ip-address</i>	A.B.C.D	PW peer IP address.
	<i>vpws-name</i>	String	VPWS name.
	<i>mac-address</i>	xx:xx:xx:xx:xx:xx	MAC address.
Examples	The following examples show how to use this command: The following command removes all dynamic MAC addresses from VPLS vplsService1. <pre>exaware# clear vpls mac address-table name vplsService1</pre> The following command removes all dynamic MAC addresses from VPLS vplsService1 defined on x-eth 0/1/2.6. <pre>exaware# clear vpls mac address-table name vplsService1 source x-eth 0/1/2.6</pre> The following command removes all dynamic MAC addresses from VPLS vplsService1 defined on PW specified by peer IP address 10.1.12.4. <pre>exaware# clear vpls mac address-table name vplsService1 source peer-ip 10.1.12.4</pre> The following command removes all dynamic MAC addresses from VPLS vplsService1 defined on VPWS VPWS1. <pre>exaware# clear vpls mac address-table name vplsService1 source vpws VPWS1</pre> The following command removes dynamic MAC address 00:43:54:43:54:43 from VPLS vplsService1. <pre>exaware# clear vpls mac address-table name vplsService1 mac 00:43:54:43:54:43</pre>		

10.12.2.2. clear vpws secondary

Description	To reroute traffic from the primary PW to the secondary one or revert vice versa, use the clear vpws secondary command.		
Command Syntax	clear vpws secondary name <i>vpws-name</i> {reroute-secondary revert-primary}		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>vpws-name</i>	String	Name of VPWS service.
	reroute-secondary		Reroutes traffic from the primary PW to the secondary one.
	revert-primary		Reverts traffic from the secondary PW to the primary one.
Examples	The following examples show how to use this command: <pre>exaware# clear vpws secondary mane VPWS-A reroute-secondary</pre>		

10.12.3. L2 Services Show Commands

10.12.3.1. show fib mpls vpws

Description	Displays FIB MPLS VPWS information.
Command Syntax	show fib mpls vpws
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command: <pre>exaware# show fib mpls vpws Legend for VLAN-OP: -o: outer tag -i: inner tag CM - ingress vlan-op configuration mismatch IN LABEL NEIGHBOR PW-ID AC ----- 65536 3.3.3.3 1 agg-eth1.1 push(0x8100:1) 65537 3.3.3.4 1 agg-eth1.2 push-i(0x8100:3),push-o(0x88a8:4) [CM]</pre>
Name	Value
VLAN-OP	pop
	transparent
	swap(vlan tag)
	push(ethertype:vlan tag)
	push-i(ethertype:vlan tag),push-o(ethertype:vlan tag)
	swap-i(ethertype:vlan tag),push-o(ethertype:vlan tag)
	none

10.12.3.2. show fib vpls-ac

Description	Displays fib vpls-ac information.																								
Command Syntax	show fib vpls-ac																								
Command Mode	operational																								
Parameters Table	N/A																								
Examples	The following example shows sample output from this command:																								
<pre>exaware# show fib vpls-ac Legend for VLAN-OP: -o: outer tag -i: inner tag Legend for FLAGS: CM - ingress vlan-op configuration mismatch</pre> <table><tr><th>VPLS</th><th>AC INTERFACE</th><th>INGRESS VLAP-OP</th><th>EGRESS VLAN-OP</th><th>FLAGS</th></tr><tr><td>vplsA</td><td>x-eth0/1/10.202</td><td>swap (300)</td><td>swap (202)</td><td></td></tr><tr><td></td><td>x-eth0/1/5.4</td><td>pop-pop</td><td>push-i (0x8100:3),push-o (0x88a8:4)</td><td>CM</td></tr><tr><td></td><td>x-eth0/1/5.5</td><td>pop-swap (10030)</td><td>swap-i (0x8100:400),push-o (0x9100:401)</td><td></td></tr></table>						VPLS	AC INTERFACE	INGRESS VLAP-OP	EGRESS VLAN-OP	FLAGS	vplsA	x-eth0/1/10.202	swap (300)	swap (202)			x-eth0/1/5.4	pop-pop	push-i (0x8100:3),push-o (0x88a8:4)	CM		x-eth0/1/5.5	pop-swap (10030)	swap-i (0x8100:400),push-o (0x9100:401)	
VPLS	AC INTERFACE	INGRESS VLAP-OP	EGRESS VLAN-OP	FLAGS																					
vplsA	x-eth0/1/10.202	swap (300)	swap (202)																						
	x-eth0/1/5.4	pop-pop	push-i (0x8100:3),push-o (0x88a8:4)	CM																					
	x-eth0/1/5.5	pop-swap (10030)	swap-i (0x8100:400),push-o (0x9100:401)																						
Name		Value																							
INGRESS VLAN-OP		pop																							
		pop-o,swap-i(vlan tag)																							
		pop-pop																							
		transparent																							
		swap (vlan tag)																							
EGRESS VLAN-OP		pop																							
		transparent																							
		swap (vlan tag)																							

	push(ethertype:vlan tag)
	push-i(ethertype:vlan tag),push-o(ethertype:vlan tag)
	swap-i(ethertype:vlan tag),push-o(ethertype:vlan tag)
	none

10.12.3.3. show fib vpls-pw

Description	Displays fib vpls-pw information.																																																																																								
Command Syntax	show fib vpls-pw																																																																																								
Command Mode	operational																																																																																								
Parameters Table	N/A																																																																																								
Examples	The following example shows sample output from this command:																																																																																								
<pre>exaware# show fib vpls-pw</pre> PW Type: B - BGP Mesh PW; L - LDP Mesh PW(VPWS); S - Spoke PW(VPWS). <table><thead><tr><th></th><th>VPLS</th><th>NEIGHBOR</th><th>LOCAL LABEL</th><th>VPN</th><th>EGRESS INTERFACE</th><th>NH IP</th><th>L2 ADDRESS</th><th>MPLS LABELS</th><th>TUNNEL NAME</th><th>PW TYPE</th></tr></thead><tbody><tr><td>--</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>vplsA</td><td>3.3.3.3</td><td>65534</td><td>65583</td><td>x-eth0/1/9.1</td><td>1.1.1.2</td><td>00:00:a3:ae:e8:5a</td><td>20</td><td></td><td></td><td>B</td></tr><tr><td></td><td>3.3.3.4</td><td>65538</td><td>27346</td><td>x-eth0/1/9.1</td><td>1.1.1.3</td><td>00:00:a3:be:e8:5a</td><td>30</td><td></td><td></td><td>B</td></tr><tr><td></td><td>3.4.3.6</td><td>65540</td><td>27356</td><td>x-eth0/1/9.1</td><td>1.1.1.3</td><td>00:00:a3:aa:e8:5a</td><td>40</td><td></td><td></td><td>S</td></tr><tr><td></td><td>3.4.3.7</td><td>65540</td><td>27356</td><td>x-eth0/1/9.1</td><td>1.1.1.3</td><td>00:00:a3:aa:e8:5a</td><td>50</td><td></td><td></td><td>S</td></tr><tr><td></td><td>3.5.3.8</td><td>65540</td><td>27356</td><td>x-eth0/1/9.2</td><td>1.1.1.3</td><td>00:00:a3:bb:e8:5a</td><td>60</td><td></td><td></td><td>L</td></tr><tr><td></td><td>3.5.3.9</td><td>65540</td><td>27356</td><td>x-eth0/1/9.2</td><td>1.1.1.3</td><td>00:00:a3:cc:e8:5a</td><td>70</td><td></td><td></td><td>L</td></tr></tbody></table>			VPLS	NEIGHBOR	LOCAL LABEL	VPN	EGRESS INTERFACE	NH IP	L2 ADDRESS	MPLS LABELS	TUNNEL NAME	PW TYPE	--											vplsA	3.3.3.3	65534	65583	x-eth0/1/9.1	1.1.1.2	00:00:a3:ae:e8:5a	20			B		3.3.3.4	65538	27346	x-eth0/1/9.1	1.1.1.3	00:00:a3:be:e8:5a	30			B		3.4.3.6	65540	27356	x-eth0/1/9.1	1.1.1.3	00:00:a3:aa:e8:5a	40			S		3.4.3.7	65540	27356	x-eth0/1/9.1	1.1.1.3	00:00:a3:aa:e8:5a	50			S		3.5.3.8	65540	27356	x-eth0/1/9.2	1.1.1.3	00:00:a3:bb:e8:5a	60			L		3.5.3.9	65540	27356	x-eth0/1/9.2	1.1.1.3	00:00:a3:cc:e8:5a	70			L
	VPLS	NEIGHBOR	LOCAL LABEL	VPN	EGRESS INTERFACE	NH IP	L2 ADDRESS	MPLS LABELS	TUNNEL NAME	PW TYPE																																																																															
--																																																																																									
vplsA	3.3.3.3	65534	65583	x-eth0/1/9.1	1.1.1.2	00:00:a3:ae:e8:5a	20			B																																																																															
	3.3.3.4	65538	27346	x-eth0/1/9.1	1.1.1.3	00:00:a3:be:e8:5a	30			B																																																																															
	3.4.3.6	65540	27356	x-eth0/1/9.1	1.1.1.3	00:00:a3:aa:e8:5a	40			S																																																																															
	3.4.3.7	65540	27356	x-eth0/1/9.1	1.1.1.3	00:00:a3:aa:e8:5a	50			S																																																																															
	3.5.3.8	65540	27356	x-eth0/1/9.2	1.1.1.3	00:00:a3:bb:e8:5a	60			L																																																																															
	3.5.3.9	65540	27356	x-eth0/1/9.2	1.1.1.3	00:00:a3:cc:e8:5a	70			L																																																																															

10.12.3.4. show fib vpws

Description	Displays FIB VPWS service information.																																																		
Command Syntax	show fib vpws																																																		
Command Mode	operational																																																		
Parameters Table	N/A																																																		
Examples	The following example shows sample output from this command:																																																		
<pre>exaware# show fib vpws Legend for VLAN-OP: -o: outer tag -i: inner tag CM - ingress vlan-op configuration mismatch</pre>																																																			
<table><tr><th>AC</th><th>INTERFACE</th><th>NEIGHBOR</th><th>PW-ID</th><th>VLAN-OP</th><th>VPN LABEL</th><th>INTERFACE</th><th>NH IP</th><th>L2 ADDRESS</th><th>MPLS LABELS</th></tr><tr><td colspan="11">-----</td></tr><tr><td></td><td>agg-eth1.1</td><td>3.3.3.3</td><td>1</td><td>pop[CM]</td><td>16</td><td>x-eth0/1/9.1</td><td>1.1.1.2</td><td>00:00:a3:ae:e8:5a</td><td>20</td></tr><tr><td></td><td>agg-eth1.2</td><td>2.3.3.3</td><td>2</td><td>swap(3)</td><td>12</td><td>x-eth0/1/9.1</td><td>1.1.1.3</td><td>00:00:a3:be:e8:5a</td><td>30</td></tr></table>											AC	INTERFACE	NEIGHBOR	PW-ID	VLAN-OP	VPN LABEL	INTERFACE	NH IP	L2 ADDRESS	MPLS LABELS	-----												agg-eth1.1	3.3.3.3	1	pop[CM]	16	x-eth0/1/9.1	1.1.1.2	00:00:a3:ae:e8:5a	20		agg-eth1.2	2.3.3.3	2	swap(3)	12	x-eth0/1/9.1	1.1.1.3	00:00:a3:be:e8:5a	30
AC	INTERFACE	NEIGHBOR	PW-ID	VLAN-OP	VPN LABEL	INTERFACE	NH IP	L2 ADDRESS	MPLS LABELS																																										

	agg-eth1.1	3.3.3.3	1	pop[CM]	16	x-eth0/1/9.1	1.1.1.2	00:00:a3:ae:e8:5a	20																																										
	agg-eth1.2	2.3.3.3	2	swap(3)	12	x-eth0/1/9.1	1.1.1.3	00:00:a3:be:e8:5a	30																																										
Name		Value																																																	
VLAN-OP		pop																																																	
		pop-o,swap-i(vlan tag)																																																	
		pop-pop																																																	
		transparent																																																	
		swap(vlan tag)																																																	
		none																																																	

10.12.3.5. show fib xconnect

Description	Displays fib xconnect information.										
Command Syntax	show fib xconnect										
Command Mode	operational										
Parameters Table	N/A										
Examples	The following example shows sample output from this command: <pre>exaware# show fib xconnect Legend for VLAN-OP: -o: outer tag -i: inner tag CM - ingress vlan-op configuration mismatch</pre> <table> <thead> <tr> <th>INCOMING INTERFACE</th><th>OUTGOING INTERFACE</th><th>VLAN-OP</th></tr> </thead> <tbody> <tr> <td>x-eth0/1/10.200</td><td>x-eth0/1/10.202</td><td>swap (300) [CM]</td></tr> <tr> <td>x-eth0/1/10.202</td><td>x-eth0/1/10.200</td><td>swap (100)</td></tr> </tbody> </table>		INCOMING INTERFACE	OUTGOING INTERFACE	VLAN-OP	x-eth0/1/10.200	x-eth0/1/10.202	swap (300) [CM]	x-eth0/1/10.202	x-eth0/1/10.200	swap (100)
INCOMING INTERFACE	OUTGOING INTERFACE	VLAN-OP									
x-eth0/1/10.200	x-eth0/1/10.202	swap (300) [CM]									
x-eth0/1/10.202	x-eth0/1/10.200	swap (100)									
	Name	Value									
	VLAN-OP	pop									
		pop-o,swap-i(vlan tag)									
		pop-pop									
		transparent									
		swap(vlan tag)									
		push(ethertype:vlan tag)									
		push-i(ethertype:vlan tag),push-o(ethertype:vlan tag)									
		swap-i(ethertype:vlan tag),push-o(ethertype:vlan tag)									

	swap-i(ethertype:vlan tag),swap-o(ethertype:vlan tag)
	none

10.12.3.6. show route mpls vpws

Description	Displays MPLS VPWS label information.															
Command Syntax	show route mpls vpws															
Command Mode	operational															
Parameters Table	N/A															
Examples	The following example shows sample output from this command:															
exaware# show route mpls vpws Flags: 1 - ldp																
<table><tr><th>FLAGS</th><th>IN LABEL</th><th>NEIGHBOR</th><th>PW-ID</th><th>AC INTERFACE</th></tr><tr><td colspan="5">-----</td></tr><tr><td>1</td><td>1000</td><td>1.1.1.1</td><td>64</td><td>x-eth 0/1/0</td></tr></table>		FLAGS	IN LABEL	NEIGHBOR	PW-ID	AC INTERFACE	-----					1	1000	1.1.1.1	64	x-eth 0/1/0
FLAGS	IN LABEL	NEIGHBOR	PW-ID	AC INTERFACE												

1	1000	1.1.1.1	64	x-eth 0/1/0												

10.12.3.7. show route vpws

Description	Displays VPWS route information.															
Command Syntax	show route vpws															
Command Mode	operational															
Parameters Table	N/A															
Examples	The following example shows sample output from this command:															
exaware# show route vpws Flags: 1 - ldp																
<table><tr><td>FLAGS</td><td>NEIGHBOR</td><td>PW-ID</td><td>OUT LABEL</td><td>INTERFACE</td></tr><tr><td colspan="5">-----</td></tr><tr><td>1</td><td>1.1.1.1</td><td>64</td><td>2000</td><td>recursive</td></tr></table>		FLAGS	NEIGHBOR	PW-ID	OUT LABEL	INTERFACE	-----					1	1.1.1.1	64	2000	recursive
FLAGS	NEIGHBOR	PW-ID	OUT LABEL	INTERFACE												

1	1.1.1.1	64	2000	recursive												

10.12.3.8. show vpls detail

Description	Displays VPLS detailed service information.																																																																																
Command Syntax	show vpls detail [<i>vpls-name</i>]																																																																																
Command Mode	operational																																																																																
Parameters Table	Name	Value	Description																																																																														
	<i>vpls-name</i>	String	Name of VPLS service.																																																																														
Examples	The following examples show sample output from this command:																																																																																
<pre>exaware# show vpls detail vplsService1 PW flags: c - control-word, s - sequence, f - flush, d - down bit State: CM - Control word mismatch, DB - Down bit, ES - Exceeded maximum scale PW Type: B - BGP MESH PW; L - LDP Mesh PW(VPWS); S - Spoke PW(VPWS). VPWS flags: MM - MTU mismatch, CW - Control word mismatch, EM - Encapsulation mismatch OL - no outgoing label, NT - No MPLS tunnel VPLS State: Up, Name: vplsService1 VE-ID: 4, RD: 1.2.3.4:100 MAC limit: 42000, MAC lock: disable Block size: 16, MTU: 1500 DHCP Snooping: v4,v6 IGMP Snooping: Status: Enabled Robustness Variable: 2 Query Interval: 125 seconds MAX Query Response Time: 10 seconds Group Membership Interval(GMI): 260 seconds AC interfaces: IGMP Snooping Flags: M-Multicast Sub-interface, I-Immediate Leave is Enabled, R-Router Interface</pre>																																																																																	
<table><tr><th>INTERFACE</th><th>STATE</th><th>SPLIT HORIZON GROUP</th><th colspan="3">DHCP SNOOPIONG INFORM. OPTION</th><th>IGMP SNOOPING FLAGS</th></tr><tr><th></th><th></th><th></th><th>TRUSTED</th><th>RECEIVE</th><th>INSERT</th><th></th></tr><tr><td colspan="7">-----</td></tr><tr><td>x-eth0/1/3.4</td><td>Up</td><td>-</td><td>v4</td><td>v4, v6</td><td>v4, v6</td><td>I</td></tr><tr><td>x-eth0/2/1.8</td><td>Down(ES)</td><td>1</td><td>v4, v6</td><td>v4</td><td>v4, v6</td><td>R</td></tr><tr><td>x-eth0/2/1.9</td><td>Down</td><td>1</td><td>v6</td><td>v6</td><td>v4, v6</td><td>I</td></tr><tr><td>x-eth0/2/1.10</td><td>Up</td><td>1</td><td>N</td><td>N</td><td>v4, v6</td><td>-</td></tr><tr><td>x-eth0/2/1.11</td><td>Invalid</td><td>-</td><td>N</td><td>N</td><td>v4, v6</td><td>-</td></tr><tr><td>x-eth0/0/1.99</td><td>Up</td><td>N</td><td>N</td><td>N</td><td>-</td><td>M</td></tr></table>				INTERFACE	STATE	SPLIT HORIZON GROUP	DHCP SNOOPIONG INFORM. OPTION			IGMP SNOOPING FLAGS				TRUSTED	RECEIVE	INSERT		-----							x-eth0/1/3.4	Up	-	v4	v4, v6	v4, v6	I	x-eth0/2/1.8	Down(ES)	1	v4, v6	v4	v4, v6	R	x-eth0/2/1.9	Down	1	v6	v6	v4, v6	I	x-eth0/2/1.10	Up	1	N	N	v4, v6	-	x-eth0/2/1.11	Invalid	-	N	N	v4, v6	-	x-eth0/0/1.99	Up	N	N	N	-	M															
INTERFACE	STATE	SPLIT HORIZON GROUP	DHCP SNOOPIONG INFORM. OPTION			IGMP SNOOPING FLAGS																																																																											
			TRUSTED	RECEIVE	INSERT																																																																												

x-eth0/1/3.4	Up	-	v4	v4, v6	v4, v6	I																																																																											
x-eth0/2/1.8	Down(ES)	1	v4, v6	v4	v4, v6	R																																																																											
x-eth0/2/1.9	Down	1	v6	v6	v4, v6	I																																																																											
x-eth0/2/1.10	Up	1	N	N	v4, v6	-																																																																											
x-eth0/2/1.11	Invalid	-	N	N	v4, v6	-																																																																											
x-eth0/0/1.99	Up	N	N	N	-	M																																																																											
PWs: <table><tr><th rowspan="2">NEIGHBOR</th><th rowspan="2">STATE</th><th rowspan="2">RD</th><th colspan="3">VE-ID/ VPWS</th><th rowspan="2">PW TYPE</th><th rowspan="2">SPLIT HORIZON GROUP</th><th rowspan="2">PW FLAGS</th><th colspan="3">DHCP SNOOPIONG INFORM. OPTION</th></tr><tr><th>LOCAL NAME</th><th>REMOTE LABEL</th><th>MTU</th><th>TRUSTED</th><th>RECEIVE</th><th>INSERT</th></tr><tr><td colspan="12">-----</td></tr><tr><td>10.1.2.3</td><td>Up</td><td>1.2.3.4:1</td><td>3</td><td>1234</td><td>2345</td><td>1500 B</td><td>mesh</td><td></td><td>v4, v6</td><td>v4, v6</td><td>v4, v6</td></tr><tr><td>10.1.2.4</td><td>Down(CM)</td><td>1.2.3.5:1</td><td>4</td><td>1235</td><td>2346</td><td>1500 B</td><td>mesh</td><td>c, s, f</td><td>v4, v6</td><td>v4, v6</td><td>v4, v6</td></tr><tr><td>10.1.5.9</td><td>Down(DB)</td><td>1.2.3.6:1</td><td>5</td><td>1236</td><td>2347</td><td>1500 B</td><td>mesh</td><td>s, d</td><td>v4, v6</td><td>v4, v6</td><td>v4, v6</td></tr><tr><td>10.1.6.9</td><td>Down(ES)</td><td>1.2.3.6:1</td><td>5</td><td>1236</td><td>2347</td><td>1500 B</td><td>mesh</td><td>s, d</td><td>v4, v6</td><td>v4, v6</td><td>v4, v6</td></tr></table>				NEIGHBOR	STATE	RD	VE-ID/ VPWS			PW TYPE	SPLIT HORIZON GROUP	PW FLAGS	DHCP SNOOPIONG INFORM. OPTION			LOCAL NAME	REMOTE LABEL	MTU	TRUSTED	RECEIVE	INSERT	-----												10.1.2.3	Up	1.2.3.4:1	3	1234	2345	1500 B	mesh		v4, v6	v4, v6	v4, v6	10.1.2.4	Down(CM)	1.2.3.5:1	4	1235	2346	1500 B	mesh	c, s, f	v4, v6	v4, v6	v4, v6	10.1.5.9	Down(DB)	1.2.3.6:1	5	1236	2347	1500 B	mesh	s, d	v4, v6	v4, v6	v4, v6	10.1.6.9	Down(ES)	1.2.3.6:1	5	1236	2347	1500 B	mesh	s, d	v4, v6	v4, v6	v4, v6
NEIGHBOR	STATE	RD	VE-ID/ VPWS				PW TYPE	SPLIT HORIZON GROUP	PW FLAGS				DHCP SNOOPIONG INFORM. OPTION																																																																				
			LOCAL NAME	REMOTE LABEL	MTU	TRUSTED				RECEIVE	INSERT																																																																						

10.1.2.3	Up	1.2.3.4:1	3	1234	2345	1500 B	mesh		v4, v6	v4, v6	v4, v6																																																																						
10.1.2.4	Down(CM)	1.2.3.5:1	4	1235	2346	1500 B	mesh	c, s, f	v4, v6	v4, v6	v4, v6																																																																						
10.1.5.9	Down(DB)	1.2.3.6:1	5	1236	2347	1500 B	mesh	s, d	v4, v6	v4, v6	v4, v6																																																																						
10.1.6.9	Down(ES)	1.2.3.6:1	5	1236	2347	1500 B	mesh	s, d	v4, v6	v4, v6	v4, v6																																																																						

```

10.1.4.3 Up      -      PWa  1244  2348  1500 S  -      v4,v6  v4,v6  v4,v6
10.1.4.3 Up      -      PWb  1245  2349  1500 S  -      v4      v4      v4
10.1.4.4 Down    -      PWc  1246  2348  1500 S  -      NT      v4,v6  v4,v6  v4,v6
10.1.4.5 Down    -      PWd  1256  -      1500 S  -      OL      v4,v6  v4,v6  v4,v6
10.1.4.6 Down    -      PWe  1257  -      1500 S  -      d      v4,v6  v4      v4,v6

exaware# show vpls detail vplsService2
PW flags: c - control-word, s - sequence, f - flush, d - down bit
State:      CM - Control word mismatch, DB - Down bit,
           ES - Exceeded maximum scale
PW Type: B - BGP MESH PW; L - LDP Mesh PW(VPWS); S - Spoke PW(VPWS).
VPWS flags: MM - MTU mismatch, CW - Control word mismatch, EM - Encapsulation mismatch
OL - no outgoing label, NT - No MPLS tunnel

VPLS State: Up, Name: vplsService2
  VE-ID: 6, RD: 1.2.3.4:200
  MAC limit: 42000, MAC lock: disable
  Block size: 16, MTU: 1500
  DHCP Snooping: v4,v6

AC interfaces:
INTERFACE      STATE      PLIT      DHCP SNOOPING
                HORIZON    GROUP      TRUSTED RECEIVE INSERT
-----
x-eth0/1/3.4   Up         -          v4         v4,v6      v4,v6
x-eth0/2/1.8   Down(ES)  1          v4,v6      v4         v4,v6
x-eth0/2/1.9   Down       1          v6         v6         v4,v6
x-eth0/2/1.10 Up         1          N          N          v4,v6
x-eth0/2/1.11 Invalid -          N          N          v4,v6

PWs:
NEIGHBOR STATE  RD      VE-ID/
VPWS  LOCAL REMOTE  PW  SPLIT  DHCP SNOOPING
NAME  LABEL LABEL MTU  TYPE HORIZON PW  INFORM. OPTION
-----
10.1.2.3 Up      -      PW1  1234  2345  1500 L  mesh  v4,v6  v4,v6  v4,v6
10.1.2.4 Down(CM) -      PW2  1235  2346  1500 L  mesh  c,s,f v4,v6  v4,v6  v4,v6
10.1.5.9 Down(DB) -      PW3  1236  2347  1500 L  mesh  s,d   v4,v6  v4,v6  v4,v6
10.1.6.9 Down(ES) -      PW4  1236  2347  1500 L  mesh  s,d   v4,v6  v4,v6  v4,v6
10.1.4.3 Up      -      PWa  1244  2348  1500 S  -      v4,v6  v4,v6  v4,v6
10.1.4.3 Up      -      PWb  1245  2349  1500 S  -      v4      v4      v4
10.1.4.4 Down    -      PWc  1246  2348  1500 S  -      NT      v4,v6  v4,v6  v4,v6
10.1.4.5 Down    -      PWd  1256  -      1500 S  -      OL      v4,v6  v4,v6  v4,v6
10.1.4.6 Down    -      PWe  1257  -      1500 S  -      d      v4,v6  v4      v4,v6

```

10.12.3.9. show vpls label-blocks

Description	Displays VPLS label blocks.		
Command Syntax	show vpls label-blocks [<i>vpls-name</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vpls-name</i>	String	Name of VPLS service.
Examples	The following examples show sample output from this command:		
<pre>exaware# show vpls label-blocks name vplsService1 Legend for remote label-blocks status: BO - block out of range ND - remote site is not designated VPLS vplsService1 VE-ID: 4, RD: 1.2.3.4:100 Local label blocks: LABEL BLOCK LABEL OFFSET SIZE BASE ----- 0 8 81920 8 8 82736 Remote label blocks: RD VE-ID LABEL BLOCK LABEL NEIGHBOR STATUS OFFSET SIZE BASE ----- 1.2.3.4:100 3 4 4 8200 10.1.2.3 Active 1.2.3.4:100 3 8 4 8148 10.1.2.3 Inactive(BO) 1.2.3.4:100 6 8 4 8128 10.1.2.3 Inactive(ND) 1.2.3.4:100 6 12 4 8000 10.1.2.3 Inactive(ND) 2.3.4.5:210 5 0 16 1600 10.1.2.3 Active 2.3.4.6:310 3 0 16 12440 10.1.2.3 Inactive(ND)</pre>			
Notes	Remote blocks status: Active: Remote blocks may be active if they cover the local VE-ID. Inactive: Remote blocks may be inactive in the following cases: - Multiple-VE(ND): Several VE-IDs signaled from the same RD. The lowest VE-ID is selected as active. Other higher VE-ID values are inactive. - Block out of range(BO):received block offset doesn't cover local VE-ID.		

10.12.3.10. show vpls mac address-table

Description	Displays VPLS mac address-table.		
Command Syntax	show vpls mac address-table [name <i>vpls-name</i>] [source { <i>interface</i> <i>peer-ip</i> <i>vpws-name</i> } mac <i>mac-address</i>] [node <i>location</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vpls-name</i>	String	Name of VPLS service.
	<i>interface</i>	Interface name	AC interface name.
	<i>peer-ip</i>	A.B.C.D	PW peer IP address.
	<i>vpws-name</i>	string	Attached VPWS name.
	<i>mac-address</i>	xx:xx:xx:xx:xx:xx	MAC address.
	<i>location</i>	0/lc0	CPU location in chassis/card notation.
Examples	The following examples show sample output from this command:		
exaware# show vpls mac address-table name vplsService1 Flags values: D - dynamic, S - static			
VPLS NAME	MAC ADDRESS	SOURCE	FLAGS
a1	00:34:29:a2:93:23	x-eth0/0/1.1	D
	00:54:8a:04:01:b4	10.1.12.4	D
	00:25:22:04:01:b4	VPWS-AAAAA	D

10.12.3.11. show vpls mac summary

Description	Displays VPLS mac address-table summary.		
Command Syntax	show vpls mac summary [name <i>vpls-name</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vpls-name</i>	String	Name of VPLS service.
Examples	The following examples show sample output from this command:		
exaware# show vpls mac summary			
VPLS NAME Locked SOURCE MAC COUNT			

vplsService1	No	x-eth0/1/2.6	103
		10.1.12.4	23
		VPWS-AAAAA	6
vplsService2	Yes	x-eth0/1/2.7	103
		10.1.12.14	23
		VPWS-BBBB	6

10.12.3.12. show vpls summary

Description	Displays VPLS summary information.
Command Syntax	<code>show vpls summary</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following examples show sample output from this command: <code>exaware# show vpls summary</code> Total vpls: 5, Up: 3, Down: 2

10.12.3.13. show vpws

Description	Displays VPWS service information.								
Command Syntax	show vpws [detail brief] [<i>vpws-name</i> <i>vpws-state</i> <i>interface</i> <i>neighbor</i> <i>pw-id</i>]								
Command Mode	Operational								
Parameters Table	Name	Value			Description				
	vpws-name	String			VPWS service name.				
	vpws-state	up/down/invalid			VPWS state.				
	interface	interface name			Interface name.				
	neighbor	IP address			Neighbor IP address.				
	pw-id	1-4294967295			Pseudo-wire ID.				
Examples	The following examples show sample output from this command:								
exaware# show vpws									
VPWS NAME	STATE	ST	LOCAL AC/ VPLS NAME	AC STATE	PW NEIGHBOR	PW ID	PW STATE	VPWS STATE	
VPWS1	A	PRI	x-eth0/0/1.1	Up	10.2.2.2	1	Up	Up	
TowardsIxia	A	PRI	a1	-	10.3.3.3	2	Up	Up	
exaware# show vpws detail									
PW flags: MM - MTU mismatch, CW - Control word mismatch, EM - Encapsulation mismatch									
OL - No outgoing label, NT - No MPLS tunnel									
VPWS: State: Down, Name: VPWS1, Role: PRI, State: Active									
AC: State: Up, INTERFACE MTU: 9182									
PW: State: Down: (OL), Neighbor 10.2.2.2, PW-ID: 3									
Control Word: not present									
Protocol: LDP, Encapsulation: MPLS, Uptime:0									
ITEM	Local			Remote					
-----	-----			-----					
Label:	65536			N/A					
Interface:	x-eth0/0/1.1			N/A					
VLAN:	N/A			N/A					

```

MTU: 1500 N/A
Control word: Preferred Disabled
Eth type: TAGGED N/A
VCCV CV types: N/A N/A
VCCV CC types: N/A N/A
Status TLV Faults: N/A

```

VPWS: State: Down, Name: TowardsIxia, Role: PRI, State: Active

VPLS: a1

PW: State: Down: (OL), Neighbor 10.3.3.3, PW-ID: 3

Protocol: LDP, Encapsulation: MPLS, Uptime:0

ITEM	Local	Remote
-----	-----	-----
Label:	65539	N/A
Interface:	a1, SPOKE	N/A
VLAN:	N/A	N/A
MTU:	1500	N/A
Control word:	Enabled	N/A
Eth type:	TAGGED	N/A
VCCV CV types:	N/A	N/A
VCCV CC types:	N/A	N/A
Status TLV Faults:	N/A	N/A

10.12.3.14. show vpws summary

Description	Displays VPWS service summary information.
Command Syntax	<code>show vpws summary</code>
Command Mode	operational
Parameters Table	N/A
Examples	The following example shows sample output from this command: <pre>exaware# show vpws summary</pre> Total vpws: 2, Up: 0, Down: 2, Invalid: 0

10.12.3.15. show xconnect

Description	Displays xconnect service information.						
Command Syntax	show xconnect [admin-state <i>state</i>] [interface <i>if-name</i> name <i>xc-name</i> state <i>xc-state</i> summary]						
Command Mode	Operational						
Parameters Table	Name	Value			Description		
	<i>admin-state</i>	up/down			Filter xconnect services that match admin-state value.		
	<i>if-name</i>	Interface			Filter xconnect services that contain interface.		
	<i>xc-name</i>	String			Filter xconnect services that match service name.		
	<i>xc-state</i>	up/down/invalid			Filter xconnect services that match state value.		
Examples	The following examples show sample output from this command:						
<pre>exaware# show xconnect XC XC XC IF-1 IF-2 NAME ADMIN STATE INTERFACE1 STATE INTERFACE2 STATE ----- xc-64 Up Down x-eth 0/1/0 Up x-eth 0/1/7.3 down xc-65 Up Invalid agg-eth 10.9 Up agg-eth 8 up exaware# show xconnect summary</pre>							
Total xconnect: 4, Up: 2, Down: 1, Invalid: 1							

10.12.4. L3 Services Operational Commands

10.12.4.1. clear forwarding

Description	Clears forwarding counters for a specified module/engine or for all modules/engines. You can clear the counters for either drops or exceptions.		
Command Syntax	clear forwarding [drops exceptions hw-status] [module <i>module</i>] [engine <i>engine</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>module</i>	#/lc# or #/cpm# or all	Module on which to clear the forwarding counters.
	<i>engine</i>	integer or all	Engine on which to clear the forwarding counters.
Examples	The following example shows how to use this command: <pre>exaware# clear forwarding exceptions module 0/lc0 engine all</pre>		

10.12.4.2. clear forwarding mtu-discards

Description	Clears the mtu-discards counters and Lists.
Command Syntax	<code>clear forwarding mtu-discards</code>
Command Mode	operational
Examples	The following example shows how to use this command: <pre>exaware# clear forwarding mtu-discards</pre>

10.12.5. L3 Services Show Commands

10.12.5.1. show fib adjacency all

Description	Displays all adjacencies on the specified location CPU (LC/CPM). If no location is specified, the default location is selected.		
Command Syntax	show fib adjacency all [detail] [node <i>location</i>]		
Command Mode	Operational		
Parameters Table	Name	Value	Description
		detail	Display detailed information.
	<i>location</i>	#/lc# or #/cpm#	CPU location in chassis/card notation.
Examples	The following examples show sample output from this command:		
<pre>exaware# show fib adjacency all Headers: F - Flags Flags: P - Protection installed EGRESS INTERFACE F NH IP L2 ADDRESS MPLS LABELS TUNNEL NAME ----- - x-eth0/1/19.316 P 34.30.16.2 00:00:23:57:a8:01 1274 toTR319_1032 x-eth0/1/19.316 34.30.16.2 00:00:23:57:a8:01 1389 toTR319_1064 exaware# show fib adjacency all detail Interface: x-eth0/1/19.316 Nexthop: 34.30.16.2 NH L2 Addr: 00:00:23:57:a8:01 Labels: 1274 Tunnel: ID: 340 Name: toTR319_1032 Counter id: 550 Protection: Type: bypass Interface: x-eth0/0/1 Nexthop: 4.5.6.7 NH L2 Addr: 00:75:75:75:75:75 Labels: 800,1274 Bypass Tunnel: ID: 550 Name: Bypass>10.1.1.1>34.30.16.2 Counter id: 555 HW info:</pre>			

HW index:	30559
Ref count:	2

10.12.5.2. show fib adjacency interface

Description	Displays all adjacencies to a specific egress interface on the specified location CPU (LC/CPM). If no location is specified, the default location is selected.		
Command Syntax	<pre>show fib adjacency interface <i>if-name</i> [detail] [node <i>location</i>]</pre>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>if-name</i>	string	Egress-interface name.
		detail	Show detailed information.
	<i>location</i>	#/lc# or #/cpm#	CPU location in chassis/card notation.
Examples	The following example shows sample output from this command: <pre>exaware# show fib adjacency interface mgmt0/0/0 node 0/cpm0</pre> <pre>adjacency TYPE EGRESS IFACE NH IP ----- ipv4 mgmt0/0/0 20.1.1.6 ipv4 mgmt0/0/0 20.1.1.7 glean mgmt0/0/0 0.0.0.0</pre>		

10.12.5.3. show fib adjacency interface glean

Description	Displays glean adjacency (with unknown IP address) to a specific egress interface on the specified location CPU (LC/CPM). If no location is specified, default location is selected.		
Command Syntax	<pre>show fib adjacency interface <i>if-name</i> glean [detail] [node <i>location</i>]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	<i>if-name</i>	string	Egress-interface name.
		detail	Show detailed information.
	<i>location</i>	#/lc# or #/cpm#	CPU location in chassis/card notation.
Examples	The following example shows sample output from this command: <pre>exaware# show fib adjacency interface mgmt0/0/1 glean node 0/cpm0</pre> <pre>EGRESS NH L2 TUNNEL INTERFACE IP ADDRESS MPLS LABELS NAME -----</pre> <pre>mgmt0/0/1</pre>		

10.12.5.4. show fib adjacency interface nh-ip

Description	Displays adjacency to specific next hop (with specific IP address) to a specific egress interface on the specified location CPU (LC/CPM). If no location is specified, the default location is selected.														
Command Syntax	<pre>show fib adjacency interface <i>if-name</i> [ipv4 ipv6] nh-ip <i>ip-address</i> [detail] [node <i>location</i>]</pre>														
Command Mode	operational														
Parameters Table	Name	Value	Description												
	<i>if-name</i>	string	Egress-interface name.												
	<i>ip-address</i>	IPv4 address	IP address of next-hop.												
		detail	Show detailed information.												
	<i>location</i>	#/lc# or #/cpm#	CPU location in chassis/card notation.												
Examples	The following example shows sample output from this command: <pre>exaware# show fib adjacency interface mgmt0/0/0 ipv4 nh-ip 20.1.1.7 node 0/cpm0</pre> <table> <thead> <tr> <th>TYPE</th> <th>EGRESS IFACE</th> <th>NH IP</th> <th>REF COUNT</th> </tr> </thead> <tbody> <tr> <td>-----</td> <td></td> <td></td> <td></td> </tr> <tr> <td>ipv4</td> <td>management0/0/0</td> <td>20.1.1.7</td> <td>1</td> </tr> </tbody> </table>			TYPE	EGRESS IFACE	NH IP	REF COUNT	-----				ipv4	management0/0/0	20.1.1.7	1
TYPE	EGRESS IFACE	NH IP	REF COUNT												

ipv4	management0/0/0	20.1.1.7	1												

10.12.5.5. show fib all

Description	Displays all routes for the default VRF. If no location is specified, the default location is selected.		
Command Syntax	show fib [ipv4 ipv6] all [hierarchy] [node <i>location</i>]		
Command Mode	operational		
Parameters Table	Name	Value	Description
		hierarchy	Display the FIB hierarchy information.
	<i>location</i>	#/lc# or #/cpm#	CPU location in chassis/card notation.
Examples	The following examples show sample output from this command:		
<pre>exaware# show fib ipv4 all node 0/lc0 Headers: F - flags Flags: I - internal FIB route, A - ARP route EGRESS MPLS TUNNEL PREFIX F VPN INTERFACE NH IP L2 ADDRESS LABELS NAME ----- 0.0.0.0/0 I null0 - - 11.22.33.0/24 x-eth0/0/2 - - 11.22.33.44/32 44 local - - 11.22.33.255/32 I null0 - - 25.1.19.0/24 x-eth0/0/1 - - 25.1.19.79/32 79 local - - 25.1.19.255/32 I null0 - - 99.99.97.0/24 x- eth0/0/0 99.99.99.43 00:0c:29:bf:b6:ee 7894 99.99.98.0/24 x- eth0/0/0 99.99.99.43 00:0c:29:bf:b6:ee x- eth0/0/0 99.99.99.42 unresolved exaware# show fib ipv4 all hierarchy node 0/lc0 Headers: F - flags</pre>			

Flags: I - internal FIB route,
A - ARP route

PREFIX	F	VPN	HIERARCHY TREE
0.0.0.0/0	I		drop
2.2.2.0/24			x-eth0/0/0
2.2.2.2/32			locally received
2.2.2.3/32	A		2.2.2.3 (00:11:22:33:44:55) via x-eth0/0/0
2.2.2.255/32	I		drop
4.4.4.0/28			2.2.2.3 (00:11:22:33:44:55) via x-eth0/0/0
4.4.5.0/28			mpath(0x008) -> 2.2.2.3 (00:11:22:33:44:55) via x-
eth0/0/0			2.2.2.4 (unresolved) via x-
eth0/0/0			
4.4.6.0/28			2.2.2.3 (00:11:22:33:44:55) via x-eth0/0/0
5.5.5.0/28			indir(0x00a) -> 2.2.2.3 (00:11:22:33:44:55) via x-eth0/0/0

10.12.5.6. show fib mpls

Description	Displays the (incoming label match) FIB routes.						
Command Syntax	show fib mpls [label <i>value</i>] [label-space <i>space</i>] [node <i>location</i>]						
Command Mode	Operational						
Parameters Table	Name	Value		Description			
	<i>value</i>	unsignedInt		Specific label to display.			
	<i>space</i>	numeric		MPLS label space.			
	<i>location</i>	#/lc#		LC location in chassis/card notation.			
Examples	The following example shows sample output from this command:						
exaware# show fib mpls label 54497							
IN	EGRESS				MPLS	TUNNEL	
LABEL	OPCODE	VPN	INTERFACE	NH IP	L2 ADDRESS	LABELS	NAME

54497	swap		x-eth0/0/19.3	34.30.16.2	00:00:23:57:a8:01	1145	PC34_TR114_TR319
	push	1040	x-eth0/0/9	41.68.0.1	68:21:5f:00:e1:ef	3	PC34_TR114_TR320
10	pop & lookup VRF DZ						

10.12.5.7. show fib multicast

Description	Displays multicast FIB routes. If no location is specified, the default location is selected.																																
Command Syntax	show fib multicast [ipv4 ipv6] [vrf <i>vrf-name</i>] {all group <i>addr</i> } [detail] [node <i>location</i>]																																
Command Mode	Operational																																
Parameters Table	Name	Value	Description																														
	<i>vrf-name</i>	Existing vrf name	VRF for which to show routes.																														
		all	Show all groups.																														
	<i>addr</i>	Multicast IP address	IP group address for which to show routes.																														
		detail	Show detailed information.																														
	<i>location</i>	#/lc#	LC location in chassis/card notation.																														
Examples	The following example shows sample output from this command:																																
exaware# show fib multicast ipv4 all node 0/lc0																																	
Flags: R - punt for pim Register, L - punt for Local listener, F - Forward																																	
<table><tr><th>GROUP</th><th>SOURCE</th><th>RPF INTERFACE</th><th>FLAGS</th><th>OUTPUT LIST</th></tr><tr><td colspan="5">-----</td></tr><tr><td>0.0.0.0/0</td><td>0.0.0.0/0</td><td>any</td><td></td><td></td></tr><tr><td>224.0.0.0/24</td><td>0.0.0.0/0</td><td>any</td><td>L</td><td></td></tr><tr><td>229.1.2.3/32</td><td>2.2.2.2/32</td><td>x-eth0/0/0</td><td>F</td><td>x-eth0/0/2.3 x-eth0/0/5.2 x-eth0/0/11</td></tr><tr><td>255.255.255.255/32</td><td>0.0.0.0/0</td><td>any</td><td>L</td><td></td></tr></table>				GROUP	SOURCE	RPF INTERFACE	FLAGS	OUTPUT LIST	-----					0.0.0.0/0	0.0.0.0/0	any			224.0.0.0/24	0.0.0.0/0	any	L		229.1.2.3/32	2.2.2.2/32	x-eth0/0/0	F	x-eth0/0/2.3 x-eth0/0/5.2 x-eth0/0/11	255.255.255.255/32	0.0.0.0/0	any	L	
GROUP	SOURCE	RPF INTERFACE	FLAGS	OUTPUT LIST																													

0.0.0.0/0	0.0.0.0/0	any																															
224.0.0.0/24	0.0.0.0/0	any	L																														
229.1.2.3/32	2.2.2.2/32	x-eth0/0/0	F	x-eth0/0/2.3 x-eth0/0/5.2 x-eth0/0/11																													
255.255.255.255/32	0.0.0.0/0	any	L																														

10.12.5.8. show fib prefix

Description	Displays the route for a specific prefix in the default VRF. If no location is specified, the default location is selected.						
Command Syntax	show fib [ipv4 ipv6] prefix <i>ip-prefix</i> [hierarchy] [node <i>location</i>]						
Command Mode	operational						
Parameters Table	Name	Value			Description		
	<i>ip-prefix</i>	IPv4/IPv6 prefix			IP prefix for lookup.		
		hierarchy			Display FIB hierarchy information.		
	<i>location</i>	#/lc# or #/cpm#			CPU location in chassis/card notation.		
Examples	The following example shows sample output from this command:						
exaware# show fib ipv4 prefix 34.44.16.64/32							
EGRESS							
PREFIX	F	VPN	INTERFACE	NH IP	L2 ADDRESS	MPLS LABELS	TUNNEL NAME

34.44.16.64/32			x-eth0/1/19.316	34.30.16.2	00:00:23:57:a8:01	1274	toTR319_10
			x-eth0/1/19.316	34.30.16.2	00:00:23:57:a8:01	1389	toTR319_20

10.12.5.9. show fib summary

Description	Displays the FIB prefix count per protocol/vrf.																							
Command Syntax	show fib summary [node <i>location</i>]																							
Command Mode	operational																							
Parameters Table	Name	Value	Description																					
	<i>location</i>	<i>#/lc#</i>	LC location in chassis/card notation.																					
Examples	The following example shows sample output from this command: <pre>exaware# show fib summary</pre> <table><thead><tr><th>TYPE</th><th>VRF NAME</th><th>PREFIX COUNT</th></tr></thead><tbody><tr><td>-----</td><td></td><td></td></tr><tr><td>ipv4uc</td><td>default</td><td>9</td></tr><tr><td>ipv4mc</td><td>default</td><td>3</td></tr><tr><td>ipv6uc</td><td>default</td><td>2</td></tr><tr><td>ipv6mc</td><td>default</td><td>4</td></tr><tr><td>mpls</td><td>-</td><td>0</td></tr></tbody></table>			TYPE	VRF NAME	PREFIX COUNT	-----			ipv4uc	default	9	ipv4mc	default	3	ipv6uc	default	2	ipv6mc	default	4	mpls	-	0
TYPE	VRF NAME	PREFIX COUNT																						

ipv4uc	default	9																						
ipv4mc	default	3																						
ipv6uc	default	2																						
ipv6mc	default	4																						
mpls	-	0																						

10.12.5.10. show fib vrf

Description	Displays FIB information for a specific VRF.		
Command Syntax	<pre>show fib vrf <i>vrf-name</i> [ipv4 ipv6] show fib vrf <i>vrf-name</i> [ipv4 ipv6] all [[hierarchy] [node node]] show fib vrf <i>vrf-name</i> [ipv4 ipv6] prefix <i>ip-prefix</i> [[hierarchy] [node <i>node</i>]]</pre>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Existing vrf name	VRF for which to show routes.
		all	List information for all table entries.
	<i>ip-prefix</i>	IP address/prefix	List information for only the specified IP addresses.
		hierarchy	Show hierarchy information.
	<i>node</i>	#/lc# or #/cpm#	Location for which to show the VRF information.

Examples	The following examples show sample output from this command:
----------	--

```
exaware# show fib vrf default ipv4 all node 0/cpm0
```

Headers: F - flags
Flags: I - internal FIB route,
A - ARP route

PREFIX	F	VPN	EGRESS INTERFACE	NH IP	L2 ADDRESS	MPLS LABELS	TUNNEL NAME
0.0.0.0/0	I		null0				

```
exaware# show fib vrf default ipv4 all hierarchy node 0/cpm0
```

Headers: F - flags
Flags: I - internal FIB route,
A - ARP route

PREFIX	F	VPN	HIERARCHY TREE
0.0.0.0/0	I		drop

10.12.5.11. show forwarding mtu-discards

Description	Displays the number of packets discarded due to MTU violation.														
Command Syntax	<code>show forwarding mtu-discards</code>														
Command Mode	operational														
Examples	The following examples show sample output from this command: <pre>exaware# show forwarding mtu-discards</pre> Time since last clear statistics: 1hour 14min 54sec Local Discards: 85 Remote Discards: 12 Last Local Discards List <table> <thead> <tr> <th>SIP</th><th>DIP</th></tr> </thead> <tbody> <tr><td>100.100.100.1</td><td>100.100.101.1</td></tr> <tr><td>200.200.200.1</td><td>100.100.101.1</td></tr> <tr><td>200.200.200.1</td><td>100.100.101.6</td></tr> </tbody> </table> Last Remote Discards List <table> <thead> <tr> <th>SIP</th><th>DIP</th></tr> </thead> <tbody> <tr><td>10.10.10.1</td><td>10.10.101.1</td></tr> <tr><td>20.20.20.1</td><td>100.100.101.1</td></tr> </tbody> </table>	SIP	DIP	100.100.100.1	100.100.101.1	200.200.200.1	100.100.101.1	200.200.200.1	100.100.101.6	SIP	DIP	10.10.10.1	10.10.101.1	20.20.20.1	100.100.101.1
SIP	DIP														
100.100.100.1	100.100.101.1														
200.200.200.1	100.100.101.1														
200.200.200.1	100.100.101.6														
SIP	DIP														
10.10.10.1	10.10.101.1														
20.20.20.1	100.100.101.1														
Notes	<i>Time since last clear statistics</i> is the time passed since the mtu-discards were last cleared. <i>Local Discards counts packets discarded locally.</i> The <i>Local Discards</i> counter is affected by low filter and is unreliable. Remote Discards counts packets discarded by a remote router. Last packets Lists display maximum of 10 packets each.														

10.12.5.12. show route

Description	Displays RIB information for a specific IP address. Only selected routes are shown.					
Command Syntax	show route [<i>ip-address</i> <i>ip-prefix</i> <i>ip-address/prefix</i>] [ipv4 ipv6] [all] [unselected]					
Command Mode	Operational					
Parameters Table	Name	Value	Description			
	ip-address	IP address	IP address for which to show the routes			
	ip-prefix	IP/length	IP prefix for which to show the routes.			
	ip-address/prefix	IP address/length	IP address and prefix length for which to show the routes.			
		all	Both selected and unselected routes.			
		unselected	Unselected routes only.			
Examples	The following example shows sample output from this command:					
exaware# show route ipv4 all						
Headers: D - distance, M - metric						
Flags: s - static, c - connected, L - leaked, Lc - leaked-connected						
b - bgp (V - MPLS VPN route)						
i - isis (L1 - level1, L2 - level2, ia - inter-area)						
o - ospf (IA- inter, E1- ext1, E2- ext2, N1- nssa1, N2- nssa2)						
r - RSVP-TE tunnel, > - selected						
FLAGS	PREFIX	D/M	NEXT HOP	GW-VRF	OUT LABEL	INTERFACE
>s	50.50.50.0/24	1/0	192.0.0.2	vrfA		agg-eth0
>s	60.60.60.0/24	1/0	192.0.0.2			agg-eth0
>L	60.60.60.0/24	3/0	192.0.0.2	vrfA		agg-eth0
>L	60.60.60.0/24	4/0	192.0.0.2	vrfB		agg-eth0
>c	192.0.0.0/8	0/0				agg-eth0
exaware# show route 34.44.16.64						
FLAGS	PREFIX	D/M	NEXT HOP	GW-VRF	OUT LABEL	INTERFACE
r	34.44.16.64/32	5/160	34.30.16.2		1274	x-eth0/1/19.316 toTR319 1032

34.30.16.2	1189	x-eth0/1/19.316	toTR319_1064
------------	------	-----------------	--------------

10.12.5.13. show route all

Description	Displays RIB information for a specific VRF. Both selected routes and unselected routes (not downloaded to the FIB) are shown. If no VRF is specified, the default VRF is displayed.																				
Command Syntax	show route [vrf <i>vrf-name</i>] { <i>afi</i> [<i>prefix</i>] <i>prefix</i> } all																				
Command Mode	operational																				
Parameters Table	Name	Value	Description																		
	<i>vrf-name</i>	Name, *	Name of VRF instance. (Default = default)																		
	<i>afi</i>	ipv4 / ipv6	AFI name. (Default = all)																		
	<i>prefix</i>	prefix	Prefix. (Default = all)																		
Examples	The following example shows sample output from this command:																				
<pre>exaware# show route 50.0.0.0/30 all Headers: D - distance, M - metric Flags: s - static, c - connected, L - leaked, Lc - leaked-connected b - bgp (LU - label-unicast) i - isis (L1 - level1, L2 - level2, ia - inter-area) o - ospf (IA - inter, E1 - ext1, E2 - ext2, N1 - nssa1, N2 - nssa2) r - RSVP-TE tunnel, > - selected S - stale route, B - blocked by max prefix Label flags: V - VPN label</pre> <table><thead><tr><th>FLAGS</th><th>PREFIX</th><th>D/M</th><th>NEXT HOP</th><th>OUT GW-VRF LABEL</th><th>INTERFACE</th></tr></thead><tbody><tr><td>>b</td><td>50.0.0.0/30</td><td>20/0</td><td>10.41.0.1</td><td></td><td>x-eth0/1/10.1</td></tr><tr><td></td><td></td><td></td><td>10.41.0.3</td><td></td><td>x-eth0/1/10.2 alternative-path</td></tr></tbody></table> <pre>exaware# show route ipv4 all Headers: D - distance, M - metric</pre>				FLAGS	PREFIX	D/M	NEXT HOP	OUT GW-VRF LABEL	INTERFACE	>b	50.0.0.0/30	20/0	10.41.0.1		x-eth0/1/10.1				10.41.0.3		x-eth0/1/10.2 alternative-path
FLAGS	PREFIX	D/M	NEXT HOP	OUT GW-VRF LABEL	INTERFACE																
>b	50.0.0.0/30	20/0	10.41.0.1		x-eth0/1/10.1																
			10.41.0.3		x-eth0/1/10.2 alternative-path																

```
Flags:  s - static, c - connected, L - route leaking
        b - bgp (LU - label-unicast)
        i - isis (L1 - level1, L2 - level2, ia - inter-area)
        o - ospf (IA - inter, E1 - ext1, E2 - ext2, N1 - nssa1, N2 - nssa2)
        r - RSVP-TE tunnel, > - selected
        S - stale route, B - blocked by max prefix
Label flags: V - VPN label
```

FLAGS	PREFIX	D/M	NEXT HOP	GW-VRF	OUT LABEL	INTERFACE
>b	50.0.0.0/30	20/0	10.41.0.1			x-eth0/1/10.1
			10.41.0.3			x-eth0/1/10.2 alternative-path

10.12.5.14. show route ip-prefix

Description	Displays RIB information for the specified prefix for a specific VRF. If the prefix does not exist, the longest prefix match is displayed. If no VRF is specified, the default VRF is displayed. Use the all keyword to display results that were not downloaded to the FIB (not selected).		
Command Syntax	show route [vrf <i>vrf-name</i>] [ipv4 ipv6] <i>ip-prefix</i> [all]		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	string	Name of VRF instance. (Default = default)
	<i>ip-prefix</i>	IP address/prefix	IP prefix for longest prefix match lookup.
Examples	The following example shows sample output from this command: <pre>exaware# show route vrf default ipv4 192.0.0.1/32</pre> Headers: D - distance, M - metric Flags: s - static, c - connected, L - leaked, Lc - leaked-connected b - bgp (V - MPLS VPN route) i - isis (L1 - level1, L2 - level2, ia - inter-area) o - ospf (IA - inter, E1 - ext 1, E2 - ext2, N1 - nssa 1, N2 - nssa2) r - RSVP-TE tunnel <pre> FLAGS PREFIX D/M NEXT HOP GW-VRF OUT LABEL INTERFACE ----- c 192.0.0.0/8 0/0 </pre>		

10.12.5.15. show route protocol

Description	Displays summary of RIB information for the specified protocol for a specific VRF. If no VRF is specified, the default VRF is displayed. Use the all keyword to display results that were not downloaded to the FIB (not selected).		
Command Syntax	<pre>show route [vrf <i>vrf-name</i>] [ipv4 ipv6] [<i>select-status</i>] [ip- prefix <i>prefix</i>] [protocol <i>protocol</i>]</pre>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	string	Name of VRF instance. (Default = default)
	<i>prefix</i>	IP address/prefix	Prefix.
	<i>select-status</i>	all unselected	Show all or only unselected routes. (Default = selected)
	<i>protocol</i>	static connected bgp isis ospf	Route origin protocol.

Examples	The following example shows sample output from this command:																												
<pre>exaware# show route vrf default ipv4 protocol static</pre>																													
Headers: D - distance, M - metric																													
Flags: s - static, c - connected, L - leaked, Lc - leaked-connected																													
b - bgp (V - MPLS VPN route)																													
i - isis (L1 - level1, L2 - level2, ia - inter-area)																													
o - ospf (IA - inter, E1 - ext 1, E2 - ext2, N1 - nssa 1, N2 -																													
nssa2)																													
r - RSVP-TE tunnel																													
<table><tr><td></td><td></td><td></td><td></td><td>OUT</td><td></td><td></td></tr><tr><td>FLAGS</td><td>PREFIX</td><td>D/M</td><td>NEXT HOP</td><td>GW-VRF</td><td>LABEL</td><td>INTERFACE</td></tr><tr><td colspan="7">-----</td></tr><tr><td>s</td><td>50.50.50.0/24</td><td>1/0</td><td>192.0.0.2</td><td></td><td></td><td>agg-eth0</td></tr></table>						OUT			FLAGS	PREFIX	D/M	NEXT HOP	GW-VRF	LABEL	INTERFACE	-----							s	50.50.50.0/24	1/0	192.0.0.2			agg-eth0
				OUT																									
FLAGS	PREFIX	D/M	NEXT HOP	GW-VRF	LABEL	INTERFACE																							

s	50.50.50.0/24	1/0	192.0.0.2			agg-eth0																							

10.12.5.16. show route mpls

Description	Displays MPLS route information.
Command Syntax	<code>show route mpls [label]</code>
Command Mode	Operational
Parameters Table	N/A
Examples	The following example shows sample output from this command: exaware# show route mpls 53120 FLAGS IN LABEL FEC NEXT HOP OUT LABEL INTERFACE ----- r 54497 6.0.0.0/24 201.0.0.2 1145 x-eth0/1/19.316 PC34_TR114_TR319

10.12.5.17. show route summary

Description	Displays summary of RIB information per routing protocol for a specific VRF If no VRF is specified, the default VRF is displayed.		
Command Syntax	show route [vrf <i>vrf-name</i>] [ipv4 ipv6] summary		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	string	Name of VRF instance. (Default = default)
Examples	The following example shows sample output from this command: <pre>exaware# show route vrf default ipv4 summary Router ID: 192.0.0.1 Maximum-path: 30 PROTOCOL COUNT ----- connected 1 static 1 total 2</pre>		

10.12.5.18. show route vrf

Description	Displays RIB information for a specific VRF.		
Command Syntax	<pre>show route vrf <i>vrf-name</i> {ipv4 ipv6} [IPv4/6- address/prefix-length] [all] [protocol <i>protocol</i>] [summary] [unselected]</pre>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Existing vrf name	VRF for which to show routes.
		IPv4/6-address/prefix-length	Default prefix is 32 for IPv4 and 128 for IPv6.
		all	Both selected and unselected routes.
	<i>protocol</i>	static connected bgp isis ospf	Filter by protocol.
		summary	Show route summary.
		unselected	Unselected routes only.

Examples	The following example shows sample output from this command:					
<pre>exaware# show route vrf default ipv4 protocol connected all Headers: D - distance, M - metric Flags: s - static, c - connected, b - bgp (V - MPLS VPN route) i - isis (L1 - level1, L2 - level2, ia - inter-area) o - ospf (IA- inter, E1- ext1, E2- ext2, N1- nssa1, N2- nssa2) r - RSVP-TE tunnel, > - selected, RL - recursive-loop</pre>						
FLAGS	PREFIX	D/M	GW-VRF	NEXT HOP	OUT LABEL	INTERFACE
>c	10.0.0.1/32	0/0				loopback5
>c	10.0.0.30/32	0/0				loopback0
>c	10.0.0.31/32	0/0				loopback1
>c	10.0.0.254/32	0/0				loopback254
>c	22.22.22.1/32	0/0				loopback2
>c	110.110.110.110/32	0/0				loopback110
>c	172.16.110.0/30	0/0				x-eth0/1/1.110
>c	172.16.111.0/29	0/0	VRFA			x-eth0/1/1.111
>c	172.16.113.0/30	0/0				x-eth0/1/1.113
>c	172.172.17.17/32	0/0				loopback3
>c	199.199.199.199/32	0/0				loopback199

10.13. Routing Leak Commands

10.13.1. Leak Configuration Commands

10.13.1.1. route

Description	Defines a route imported from another VRF to the relevant VRF. Use the no form of the command to clear a specific route.		
Command Syntax	<pre>route destination/prefix-length gw-vrf vrf-name distance distance no route destination/prefix-length</pre>		
Command Mode	configuration routing leak		
Parameters Table	Name	Value	Description
	<i>destination</i>	IP address	Destination prefix of the route.
	<i>prefix-length</i>	IPv4: 0-32 IPv6: 0-128	Destination prefix length.
	<i>vrf-name</i>	Maximum of 128 characters, no spaces	Name of the VRF where the route is imported from.
	<i>distance</i>	1-254	Distance value. (Default = 3)
Examples	This example shows how to configure routing leak: <pre>exaware# config exaware(config)# routing leak vrf vrf1 af-ipv4 unicast exaware(config-routing-leak-vrf-vrf1-af-ipv4-unicast)# route 102.10.10.00/24 gw-vrf vrf2 exaware(config-routing-leak-vrf-vrf1-af-ipv4-unicast)# route 102.10.11.00/24 gw-vrf vrf2 exaware(config-routing-leak-vrf-vrf1-af-ipv4-unicast)# route 102.11.11.00/24 gw-vrf vrf3 distance 10</pre>		

10.13.1.2. routing leak

Description	Enters routing leak configuration mode. This mode configures routing leak for a specified VRF. Use the no form of the command to disable routing leak.		
Command Syntax	<pre>routing leak vrf <i>vrf-name</i> {af-ipv4 ipv6} unicast no routing leak [vrf <i>vrf-name</i> [{af-ipv4 af-ipv6} unicast]]</pre>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Maximum of 128 characters, no spaces	Name of the VRF for which to configure static routing. A default VRF is pre-configured in the system. To configure a user-defined VRF for static routing, you must first define it. See vrf (on page 1494). For each VRF, there are two possible IP address families and two possible routing schemes, for a total of four possible contexts per VRF.
		af-ipv4 af-ipv6	Address family of the specified VRF. For each address family, there are two possible routing schemes.
		unicast	Type of routing scheme.

Examples

This example shows how to enter the routing leak configuration mode for the following context:

```
exaware# config
exaware(config)# routing leak vrf vrfl af-ipv4 unicast
exaware(config-routing-leak-vrf-vrfl-af-ipv4-unicast) #
```

10.14. Static Routing Commands

10.14.1. Static Configuration Commands

10.14.1.1. route

Description	Defines the static route for the relevant VRF and context. Use the no form of the command to clear a specific route.		
Command Syntax	<pre>route destination/prefix-length {{interface null} {tunnel-te tunnel-te-name} {{gateway ip-address [gw-vrf gw-vrf-name]} {direct-gateway ip-address [gw-vrf gw-vrf-name]}} [distance distance][description description] [tag tag] no route destination/prefix-length</pre>		
Command Mode	configuration routing static vrf af		
Parameters Table	Name	Value	Description
	destination	IPv4 address or global IPv6 address	Destination prefix of the route.
	prefix-length	IPv4: 0-32 IPv6: 0-128	Destination prefix length.
	null		Traffic is discarded
	tunnel-te	existing RSVP tunnel name	RSVP tunnel-te name.

	<i>gw-vrf-name</i>	Maximum of 128 characters, no spaces	Name of the VRF where the gateway is located. If the parameter is omitted the vrf-name configured into the routing static command is applied. The following vrf-names cannot be configured: • management The parameter cannot be configured in the context of the following VRFs; • management The parameter is allowed only for the following address families: • af-ipv4 unicast af-ipv6 unicast
--	--------------------	--------------------------------------	---

	<i>ip-address</i>	IP address	Gateway prefix of the next hop.
	<i>description</i>	String up to 255 characters	Default: empty string
	<i>tag</i>	1-4294967295	Default: 1
	<i>distance</i>	1-254	Distance metric. (Default = 1)
Examples	This example shows how to configure a static route, specifying the interface: <pre>exaware# config exaware(config)# routing static vrf vrfl af-ipv4 unicast exaware(config-vrf-vrfl-af-ipv4-unicast)# route 102.10.10.0/24 gateway 10.10.11.11 exaware(config-vrf-vrfl-af-ipv4-unicast)# route 102.10.20.8/25 gateway 10.10.11.11 exaware(config-vrf-vrfl-af-ipv4-unicast)# route 110.1.1.12/32 gw-vrf vrfl2 gateway 10.10.11.11</pre>		
Notes	• If interface null is configured, the traffic for the specified destination will be discarded. • The host part (the last bits after the mask) of <i>destination</i> must be zero.		

10.14.1.2. routing static

Description	Enters routing static configuration mode. This mode configures static routing for a specified VRF in four different contexts, as indicated in the command syntax. Use the no form of the command to disable static routing. You can disable static routing at any level: globally, for a specific VRF, for a specific IP address family, or for a specific context.		
Command Syntax	<pre>routing static vrf <i>vrf-name</i> af-ipv4 unicast routing static vrf <i>vrf-name</i> af-ipv6 unicast no routing static no routing static vrf <i>vrf-name</i> no routing static vrf <i>vrf-name</i> (af-ipv4 af-ipv6) no routing static vrf <i>vrf-name</i> (af-ipv4 af-ipv6) unicast</pre>		
Command Mode	Configuration		
Parameters Table	Name	Value	Description
	<i>vrf-name</i>	Maximum of 128 characters, no spaces	Name of the VRF for which to configure static routing. A default VRF is pre-configured in the system. To configure a user-defined VRF for static routing, you must first define it. See vrf (on page 1494). For each VRF, there are two possible IP address families and two possible routing schemes, for a total of four possible contexts per VRF.
		af-ipv4 / af-ipv6	Address family of the specified VRF. For each address family, there are two possible routing schemes.
		unicast	Type of routing scheme.

Examples

This example shows how to enter the static route configuration mode for the following context:

- VRF tag: VRF1
- IP address family: IPv4
- routing scheme: unicast

Note the change in the command line prompt.

```
exaware# config
exaware(config)# routing static vrf vrf1 af-ipv4 unicast
exaware(config-vrf-vrf1-af-ipv4-unicast)#
```

10.15. VRF Commands

10.15.1. VRF Configuration Commands

10.15.1.1. description

Description	Provides a user-defined description of the VRF. Use the no form of the command to remove the description.		
Command Syntax	description <i>description</i> no description		
Command Mode	configuration vrf < <i>vrf-name</i> >		
Parameters Table	Name	Value	Description
	<i>description</i>		Text describing the VRF. (No spaces)
Examples	This example shows how to use this command: exaware# config exaware(config)# vrf vrf1 exaware(config-vrf-vrf1)# description NewVRF		

10.15.1.2. maximum-prefix

Description	Configures maximum prefixes for VRF installed to FIB. Use the no form of the command to restore the default maximum (524288).		
Command Syntax	<pre>maximum-prefix [number <i>number</i>] [threshold <i>threshold</i>] [action <i>action</i>]</pre> <pre>no maximum-prefix</pre>		
Command Mode	<pre>configuration vrf af-ipv4 unicast</pre> <pre>configuration vrf af-ipv6 unicast</pre>		
Parameters Table	Name	Value	Description
	<i>number</i>	1-4294967295	Maximum number of prefixes installed to FIB. (Default = 524288)
	<i>threshold</i>	1-100	Threshold (percentage of <i>number</i>) at which warnings will be generated. (Default = 75)
	<i>action</i>	limit warn	Action taken when reaching the maximum. (Default = warn)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# vrf vrf1 exaware(config-vrf-vrf1)# af-ipv4 unicast exaware(config-vrf-vrf1-af-ipv4-unicast)# maximum-prefix number 200 threshold 75 action warning-only</pre>		

10.15.1.3. rd

Description	Configures the route-distinguisher value for the VRF. Use the no form of the command to delete the RD.		
Command Syntax	<pre>rd route-distinguisher no rd</pre>		
Command Mode	configuration vrf <vrf-name>		
Parameters Table	Name	Value	Description
	route-distinguisher	<2-byte ASN>:<4-byte number>	2-byte ASN notation.
		<4-byte ASN>L:<2-byte number>	4-byte ASN notation.
		<IPv4 address>:<2-byte number>	IPv4 address notation.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# vrf vrf1 exaware(config-vrf-vrf1)# rd 100:200</pre>		
Notes	You cannot delete the RD value of a VRF attached to BGP instance.		

10.15.1.4. vrf

Description	Creates a VRF.		
	Use the no form of the command to disable the specified VRF.		
Command Syntax	<code>vrf vrf-name</code> <code>no vrf vrf-name</code>		
Command Mode	configuration		
Parameters Table	Name	Value	Description
	<code>vrf-name</code>	string, maximum 128 characters	User-defined name of the VRF.
Examples	This example shows how to create a VRF named vrf1: exaware# config exaware(config)# vrf vrf1		
Notes	A default VRF is pre-configured in the system. Until user-defined VRFs are added, this is the only VRF.		

10.15.2. VRF Show Commands

10.15.2.1. show vrf

Description	Displays VRF information.								
Command Syntax	show vrf <i>vrf-name</i>								
Command Mode	operational								
Parameters Table	Name			Value			Description		
	<i>vrf-name</i>			Name, *			VRF name.		
Examples	The following example shows sample output from this command:								
exaware# show vrf *									
VRF NAME	RD	DEFAULT VRF LOOKUP	AFI	PRFFIX COUNT	BLOCKED COUNT	PREFIX LIMIT	LIMIT ACTION	THRESHOLD	ACTIVE ALARMS

default			ipv4-uc	1111	0				
			ipv6-uc	111	0				
VRF1	1:1111	enabled	ipv4-uc	1205	0				
			Ipv6-uc	11	0				
VRF2	1:1112	disabled	ipv4-uc	111	0				
			Ipv6-uc	1	0				
ipv6-uc	1	0							

10.16. VRRP Commands

10.16.1. VRRP Configuration Commands

10.16.1.1. admin-state

Description	This command configures the administrative state of a VRRP router. Use the no form of the command to restore the default interval.		
Command Syntax	admin-status state enabled disabled no admin-state		
Command Mode	configuration in-band interface vrrp-ipv4		
Parameters Table	Name	Value	Description
	enabled		Specifies the VRRP router can run. The default value is enabled.
	disabled		Specifies the VRRP router cannot run.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv4 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# admin-state disabled exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)#</pre>		

10.16.1.2. advertise-interval-v2

Description	This command configures the advertise interval of VRRPv2. Use the no form of the command to restore the default interval.		
Command Syntax	<pre>advertise-interval-v2 seconds no advertise-interval-v2</pre>		
Command Mode	configuration\n in-band interface vrrp-ipv4		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-255	Specifies the advertise interval in seconds. The default value is 1.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv4 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# version vrrp-v2 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# advertise-interval-v2 5 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)#</pre>		
Note	A value configured by this command is only applied when the vrrp-v2 version is configured.		

10.16.1.3. advertise-interval-v2-and-v3

Description	This command configures the advertise interval of VRRPv2-and-v3. Use the no form of the command to restore the default interval.		
Command Syntax	<pre>advertise-interval-v2-and-v3 seconds no advertise-interval-v2-and-v3</pre>		
Command Mode	configuration in-band interface vrrp-ipv4		
Parameters Table	Name	Value	Description
	<i>seconds</i>	1-255	Specifies the advertise interval in seconds. The default value is 1.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv4 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# version vrrp-v2-and-v3 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# advertise-interval-v2-and-v3 5 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)#</pre>		
Note	A value configured by this command is only applied when the vrrp-v2-and-v3 version is configured.		

10.16.1.4. advertise-interval-v3

Description	This command configures the advertise interval of VRRPv3. Use the no form of the command to restore the default interval.		
Command Syntax	<pre>advertise-interval-v3 centi-secs no advertise-interval-v3</pre>		
Command Mode	configuration in-band interface vrrp-ipv4 configuration in-band interface vrrp-ipv6		
Parameters Table	Name	Value	Description
	centi-secs	10-4095	Specifies the advertise interval in centi-seconds. The default value is 100.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv4 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# version vrrp-v3 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# advertise-interval-v3 50 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)#</pre>		
Note	A value configured by this command is only applied when the vrrp-v3 version is configured.		

10.16.1.5.hold-time

Description	This configures the hold time (delay before replacing of the current Master). Use the no form of the command to reset the default hold time.		
Command Syntax	<pre>hold-time seconds no preempt</pre>		
Command Mode	configuration in-band interface vrrp-ipv4 configuration in-band interface vrrp-ipv6		
Parameters Table	Name	Value	Description
	<i>seconds</i>	Integer in range 0..3600	Specifies the delay before the new virtual router with a higher priority replaces the current Master. The default value is 0.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv4 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# hold-time 4 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)#</pre>		

10.16.1.6. log-state-change

Description	This command allows sending syslog messages and notifications when the protocol state is changed. By default, the messages are not sent. Use the no form of the command to restore the default.
Command Syntax	<code>log-state-change</code> <code>no log-state-change</code>
Command Mode	configuration in-band interface vrrp-ipv4 configuration in-band interface vrrp-ipv6
Examples	This example shows how to use this command: exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv4 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# log-state-change exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)#

10.16.1.7.preempt

Description	This command allows replacing the current Master with lower priority. By default, the preemption is allowed. Use the no form of the command to prohibit preemption the default.		
Command Syntax	preempt enabled disabled no preempt		
Command Mode	configuration in-band interface vrrp-ipv4 configuration in-band interface vrrp-ipv6		
Parameters Table	Name	Value	Description
	enabled		The current VRRP router can replace the Master with lower priority.
	disabled		The current VRRP router cannot replace the Master with lower priority.
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv4 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# preempt disabled exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# </pre>		

10.16.1.8.priority

Description	This command configures the virtual router priority. Use the no form of the command to restore the default priority.		
Command Syntax	priority <i>value</i> no priority		
Command Mode	configuration in-band interface vrrp-ipv4 configuration in-band interface vrrp-ipv6		
Parameters Table	Name	Value	Description
	<i>value</i>	1-254	Specifies the virtual router priority. The default value is 100.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv4 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# priority 120 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)#</pre>		

10.16.1.9.version

Description	This command configures the VRRP protocol version for the IPv4 virtual router. Use the no form of the command to restore the default.		
Command Syntax	version vrrp-v2 vrrp-v3 vrrp-v2-and-v3 no version		
Command Mode	configuration in-band interface vrrp-ipv4		
Parameters Table	Name	Value	Description
	vrrp-v2		VRRPv2 protocol.
	vrrp-v3		VRRPv3 protocol. The default value is vrrp-v3.
	vrrp-v2-and-v3		VRRPv2-and-v3 protocol.
Examples	This example shows how to use this command: <pre> exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv4 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# version 2 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# </pre>		

10.16.1.10. virtual-ipv4-address

Description	This command configures an IPv4 address of the virtual address. Use the no form of the command to remove the IPv4 address.		
Command Syntax	<pre>virtual-ipv4-address address no virtual-ipv4-address</pre>		
Command Mode	configuration in-band interface vrrp-ipv4		
Parameters Table	Name	Value	Description
	<i>address</i>	IPv4 address	Specifies the virtual IPv4 address of a virtual VRRP router.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv4 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)# virtual-ipv4-address 133.12.34.5 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)#</pre>		
Note	Only one virtual IPv4 addresses can be configured for one virtual router.		

10.16.1.11. virtual-ipv6-address

Description	This command configures an IPv6 global address of the virtual address. Use the no form of the command to remove the IPv6 address.		
Command Syntax	<pre>virtual-ipv6-address address no virtual-ipv6-address eui-64 address</pre>		
Command Mode	configuration in-band interface vrrp-ipv6		
Parameters Table	Name	Value	Description
	<i>address</i>	IPv6 global address	Specifies the virtual IPv6 global address of a virtual VRRP router.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv6 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv6-10)# virtual-ipv6-address 2000:0df8:1234::1 exaware(config-interface-x-eth-0/0/1-vrrp-ipv6-10)#</pre>		
Note	Only one global virtual IPv6 addresses can be configured for one virtual router.		

10.16.1.12. virtual-ipv6-address link-local

Description	This command configures an IPv6 link-local address of the virtual address. Use the no form of the command to restore the default IPv6 link local address.		
Command Syntax	<pre>virtual-ipv6-address link-local address no virtual-ipv6-address link-local</pre>		
Command Mode	configuration in-band interface vrrp-ipv6		
Parameters Table	Name	Value	Description
	<i>address</i>	IPv6 link-local address	Specifies the virtual IPv6 link-local address of a virtual VRRP router.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv6 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv6-10)# virtual-ipv6-address link-local fe80::260:3eff exaware(config-interface-x-eth-0/0/1-vrrp-ipv6-10)#</pre>		

10.16.1.13. vrrp-ipv4

Description	This command enters VRRP IPv4 Command Mode and creates a new VRRP virtual router, if does not exist. Use the no form of the command to remove the VRRP virtual router.		
Command Syntax	<pre>vrrp-ipv4 value no vrrp-ipv4</pre>		
Command Mode	configuration in-band interface (ethernet port, sub-interface, or LAG)		
Parameters Table	Name	Value	Description
	<i>value</i>	1-255	VRRP IPv4 virtual router identifier. It uniquely identifies VRRP IPv4 virtual router on an interface.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv4 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv4-10)#</pre>		

10.16.1.14. vrrp-ipv6

Description	This command enters VRRP IPv6 Command Mode and creates a new VRRP virtual router, if does not exist. Use the no form of the command to remove the VRRP virtual router.		
Command Syntax	vrrp-ipv6 <i>value</i> no vrrp-ipv6		
Command Mode	configuration in-band interface (ethernet port, sub-interface, or LAG)		
Parameters Table	Name	Value	Description
	<i>value</i>	1-255	VRRP IPv6 virtual router identifier. It uniquely identifies VRRP IPv6 virtual router on an interface.
Examples	This example shows how to use this command: exaware# config exaware(config)# interface x-eth 0/0/1 exaware(config-interface-x-eth-0/0/1)# vrrp-ipv6 vrid 10 exaware(config-interface-x-eth-0/0/1-vrrp-ipv6-10)#		

10.16.2. VRRP Operational Commands

10.16.2.1.clear vrrp counters

Description	This command clears the VRRP counters.		
Command Syntax	<pre>clear vrrp counters total-errors clear vrrp counters all clear vrrp counters interface [x-eth eth-if} {agg-eth agg-id}] clear vrrp counters {ipv4 IPv6} [interface [x-eth eth- if} {agg-eth agg-id}] clear vrrp counters {ipv4 ipv6} interface x-eth eth- if} {agg-eth agg-id}} vrid vrid-value</pre>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	all		Clear all the VRRP counters.
	total-errors		Clear only the total VRRP error counters.
	interface		Clear only the interface VRRP counters.
	ipv4		Clear only the interface VRRPv4 counters.
	ipv6		Clear only the interface VRRPv6 counters.
	eth-if	0/0/port[subinterface]	Clear the counters of VRRP routers running on this interface.

	<i>agg-id</i>	0-79	Clear the counters of VRRP routers running on this interface.
	<i>vrid-value</i>	1-255	Clear only counters of the specified VRRP router.
	Name	Value	Description
Examples	This example shows how to use this command: <pre>exaware# clear vrrp all exaware#</pre>		

10.16.3. VRRP Show Commands

10.16.3.1.show vrrp counters

Description	This command displays the VRRP counters.		
Command Syntax	<pre>show vrrp counters total-errors show vrrp counters all [detail] show vrrp counters {ipv4 ipv6} [interface [{x-eth eth-if} {agg-eth agg-id}]] [detail] show vrrp counters {ipv4 ipv6} interface {x-eth eth-if} {agg-eth agg-id}} vrid vrid-value [detail]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	all		Display all the VRRP counters.
	total-errors		Display only the total VRRP error counters.
	interface		Display only interface VRRP counters.
	ipv4		Display only the interface VRRPv4 counters.
	ipv6		Display only the interface VRRPv6 counters.
	eth-if	0/0/port[.subinterface]	Display only counters of VRRP routers running on this interface.
	agg-id	0-79	Display only counters of VRRP routers running on this interface.
	vrid-value	1-255	Display only counters of the specified VRRP router.
Examples			

```
exaware# show vrrp counters total-errors
```

Interface	IpVer	Interval Errors	Checksum Errors	Length Errors	Type Errors	Version Errors	TTL Errors	VRIDs Unknown
x-eth0/0/4	IPv4	2	1	0	0	0	0	0
x-eth0/0/5	IPv4	2	1	0	0	0	0	0

```
exaware# show vrrp counters all
```

Interface	IpVer	Vrid	MasterTrans	AdvSent	AdvRecv	PriZeroPktsSnt	PriZeroPktsRcv	InvalidPktsRcvd
-----------	-------	------	-------------	---------	---------	----------------	----------------	-----------------

--								
x-eth0/0/4	Ipv4	2	0	1	100	0	0	0
x-eth0/0/4	Ipv6	2	0	1	10	0	0	0

```
exaware# show vrrp counters all detail
```

```
interface x-eth 0/0/1
```

```
  vrrp-ipv4
```

```
    vrid: 1
```

master-transitions	2
advertisement-rcvd	3
advertisement-sent	105
interval-errors	1
priority-zero-pkts-rcvd	1
priority-zero-pkts-sent	0
invalid-type-pkts-rcvd	0
invalid-version	0
invalid-ttl	0
address-list-errors	0
packet-length-errors	0

checksum-error	0
vrid: 2	
master-transitions	1
advertisement-rcvd	0
advertisement-sent	225
interval-errors	0
priority-zero-pkts-rcvd	0
priority-zero-pkts-sent	0
invalid-type-pkts-rcvd	0
invalid-version	0
invalid-ttl	0
address-list-errors	0
packet-length-errors	0
vrrp-ipv6	
vrid: 1	
master-transitions	2
advertisement-rcvd	3
advertisement-sent	105
interval-errors	1
priority-zero-pkts-rcvd	1
priority-zero-pkts-sent	0
invalid-type-pkts-rcvd	0
invalid-version	0
invalid-ttl	0
address-list-errors	0
packet-length-errors	0
checksum-error	0
interface x-eth 0/0/2	
vrrp-ipv4	
vrid: 1	
master-transitions	0

advertisement-rcvd	0
advertisement-sent	225
interval-errors	0
priority-zero-pkts-rcvd	0
priority-zero-pkts-sent	0
invalid-type-pkts-rcvd	0
invalid-version	0
invalid-ttl	0
address-list-errors	0
packet-length-errors	0

10.16.3.2.show vrrp status

Description	This command displays the VRRP status.		
Command Syntax	<pre>show vrrp status all [detail] show vrrp status {ipv4 ipv6} [interface [{x-eth eth-if} {agg-eth agg-id}]] [detail] show vrrp status {ipv4 ipv6} interface {x-eth eth-if} {agg-eth agg-id}} vrid vrid-value [detail]</pre>		
Command Mode	Operational		
Parameters Table	Name	Value	Description
	all		Display statuses of all the VRRP routers.
	ipv4		Display statuses of only the VRRPv4 routers.
	ipv6		Display statuses of only the VRRPv4 routers.
	interface		Display statuses of only the VRRP routers running on this interface.
	x-eth eth-if	0/0/port[.subinterface]	Display statuses only of the VRRP routers running on this interface.
	agg-eth agg-id	0-79	Display statuses only of the VRRP routers running on this LAG.
	vrid-value	1-255	Display statuses only of the specified VRRP router.

Examples

This example shows how to use this command:

```
exaware# show vrrp status all
```

interface	IpVer	Vrid	Version	State	Oper. Prior.	AdvInterval (by Master)	Preemp/ Hold-time	VirtualIP	MasterIP
x-eth0/0/1	IPv4	5	vrrp-v3	Master	255	155 csec	Ena/0	1.1.1.2	1.1.1.2
x-eth0/0/1	IPv4	6	vrrp-v3	Slave	100	155 csec	Ena/10	1.1.1.3	1.2.2.2
x-eth0/0/2	IPv4	1	vrrp-v2	Slave	100	15 sec	Ena/0	1.3.1.1	1.3.2.2

```
exaware# show vrrp status all detail
```

```
interface x-eth 0/0/1
```

```
  vrrp-ipv4
```

```
    vrid: 1
```

```
      Version: vrrp-v3
```

```
      Configured Priority: 200
```

```
      Operational Priority: 255
```

```
      Advertisement Interval: 30 csec
```

```
      Preemption: enabled
```

```
      Hold-time: 4 sec
```

```
      Skew Time: 1 msec
```

```
      Virtual IP Address: 10.2.1.10
```

```
      Virtual MAC Address: 00:00:5e:00:01:01
```

```
      State: Master
```

```
      Last Event Type: owner-preempt
```

```
      Last Event Time: 04-Jan-2020, 15:13:14
```

```
    vrid: 2
```

```
      Version: vrrp-v2
```

```
      Configured Priority: 100
```

```
      Operational Priority: 100
```

```
      Advertisement Interval: 3 sec
```

```
      Preemption: disabled
```

```
      Hold-time: 0
```

```
      Skew Time: 609 msec
```

Virtual IP Address: 10.2.2.100
Virtual MAC Address: 00:00:5e:00:01:02
State: Backup
Last Event Type: owner-preempt
Last Event Time: 04-Jan-2020, 15:13:14
Master Router: 10.2.2.20
Master Priority: 200
Master Advertisement Interval: 3 sec
Master Down Interval: 11000 msec

vrrp-ipv6

vrid: 1

Version: vrrp-v3
Configured Priority: 200
Operational Priority: 200
Advertisement Interval: 30 csec
Preemption: enabled
Hold-time: 4 sec
Skew Time: 65 msec
Virtual IP Address: fe80::fe00:0201 2002:10::2.1.10
Virtual MAC Address: 00:00:5e:00:02:01
State: Master
Last Event Type: event-higher-priority-backup
Last Event Time: 04-Jan-2020, 15:15:01

11. Timing Commands

11.1. PTP Commands

11.1.1. PTP Configuration Commands

11.1.1.1. **announce-receipt-timeout**

Description	Use this command to set announce-receipt-timeout. Use the no form of the command to restore the default value (3).		
Command Syntax	<code>announce-receipt-timeout seconds</code> <code>no announce-receipt-timeout</code>		
Command Mode	configuration timing ptp clock		
Parameters Table	Name	Value	Description
	<i>seconds</i>	Integer into Range 1-10	Administrative state. g8571 Default = 3
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timeout ptp exaware(config-timeout-ptp)# clock g8275.1 exaware(config-timeout-ptp-clock-g8275.1)# announce- receipt-timeout 5</pre>		

11.1.1.2. clock

Description	Enter ptp clock configuration mode. Use the no form of the command to delete the clock.
Command Syntax	<code>clock g8575.1</code> <code>no clock g8575.1</code>
Command Mode	configuration timing ptp
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timeout ptp exaware(config-timeout-ptp)# clock g8275.1 exaware(config-timeout-ptp-clock-g8275.1)#</pre>

11.1.1.3. clock-identity

Description	Enter ptp clock configuration mode. Use the no form of the command to delete the clock.
Command Syntax	<pre>clock-identity clockID no clock g8575.1</pre>
Command Mode	configuration timing ptp clock
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timeout ptp exaware(config-timeout-ptp)# clock g8275.1 exaware(config-timeout-ptp-clock-g8275.1)# clock-identity 01;02;03;04;05:06:07:08</pre>
Notes	The default clock identity is by the following the below steps: 1. Take the first 3 bytes of the chassis' base MAC address (OUI). 2. Follow with the 0xFFFE. 3. Append the last 3 bytes of the MAC address.

11.1.1.4. clock-step

Description	Use this command to set clock timestamping mode. Use the no form of the command to restore the default value (one-step).		
Command Syntax	<pre>clock-step step {one-step two-step} no clock-step</pre>		
Command Mode	configuration timing ptp clock		
Parameters Table	Name	Value	Description
	step	one-step	The timestamp is inserted into the sync message.
		two-step	The timestamp is inserted into a follow-up message that follows the sync message.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timeout ptp exaware(config-timeout-ptp)# clock g8275.1 exaware(config-timeout-ptp-clock-g8275.1)# clock-step two-step</pre>		

11.1.1.5. clock-type

Description	Use this command to set clock type. Use the no form of the command to restore the default value (boundary-clock).
Command Syntax	<code>clock-type boundary-clock</code> <code>no clock-type</code>
Command Mode	configuration timing ptp clock
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timeout ptp exaware(config-timeout-ptp)# clock g8275.1 exaware(config-timeout-ptp-clock-g8275.1)# clock-type boundary-clock</pre>

11.1.1.6. domain

Description	Use this command to set the current synchronization domain. Use the no form of the command to restore the default value.		
Command Syntax	<code>domain domain</code> <code>no domain</code>		
Command Mode	configuration timing ptp clock		
Parameters Table	Name	Value	Description
	<code>domain</code>	24-43	Default = 25
Examples	This example shows how to use this command: <code>exaware# config</code> <code>exaware(config)# timeout ptp</code> <code>exaware(config-timeout-ptp)# clock g8275.1</code> <code>exaware(config-timeout-ptp-clock-g8275.1)# domain 30</code>		

11.1.1.7. interface

Description	Creates a new clock interface if it is not created already. Use the no form of the command to delete the interface.		
Command Syntax	Interface x-eth <i>instance</i> no domain		
Command Mode	configuration timing ptp clock		
Parameters Table	Name	Value	Description
	<i>instance</i>	x/x/x	Physical interface only.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timeout ptp exaware(config-timing-ptp)# clock g8275.1 exaware(config-timing-ptp-clock-g8275.1)# interface 0/0/2</pre>		

11.1.1.8. holdover

Description	Use this command to set clock holdover duration. Use the no form of the command to restore the default value.		
Command Syntax	<pre>holdover holdover no holdover</pre>		
Command Mode	configuration timing ptp clock		
Parameters Table	Name	Value	Description
	holdover	0-1440	0 means no holdover. Default = 120.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timeout ptp exaware(config-timeout-ptp)# clock g8275.1 exaware(config-timeout-ptp-clock-g8275.1)# holdover 130</pre>		

11.1.1.9. local-priority

Description	Use this command to specify the local attribute of the local clock. Use the no form of the command to restore the default value.		
Command Syntax	<pre>local-priority npri no local-priority</pre>		
Command Mode	configuration timing ptp clock		
Parameters Table	Name	Value	Description
	<i>pri</i>	1-255	Default = 128
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timeout ptp exaware(config-timeout-ptp)# clock g8275.1 exaware(config-timeout-ptp-clock-g8275.1)# local-priority 130</pre>		

11.1.1.10. master-only

Description	Use this command to set the interface to a master-only port. Use the no form of the command to restore the default value (disable).		
Command Syntax	<pre>master-only {enable disable} no master-only</pre>		
Command Mode	configuration timing ptp clock interface		
Parameters Table	Name	Value	Description
	enable		The interface may serve as master only by the BC.
	disable		
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timeout ptp exaware(config-timeout-ptp)# clock g8275.1 exaware(config-timeout-ptp-clock-g8275.1)# interface x-eth 0/0/4 exaware(config-timeout-ptp-clock-g8275.1-interface-x-eth- 0/0/4)# master-only enable</pre>		

11.1.1.11. non-forwardable-mac

Description	Use this command to set the non-forwardable multicast address as destination mac address. Use the no form of the command to restore the default value (disable).		
Command Syntax	<pre>non-forwardable-mac {enable disable} no non-forwardable-mac</pre>		
Command Mode	configuration timing ptp clock		
Parameters Table	Name	Value	Description
	enable		Use non-forwardable DA MAC address to ensure the packet is not forwarded beyond the LAN.
	disable		
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timeout ptp exaware(config-timing-ptp)# clock g8275.1 exaware(config-timing-ptp-clock-g8275.1)# non-forwardable-mac enable</pre>		

11.1.1.12. priority2

Description	Use this command to set the Priority2 attribute of the local clock. Use the no form of the command to restore the default value.		
Command Syntax	<pre>priority2 pri no priority2</pre>		
Command Mode	configuration timing ptp clock		
Parameters Table	Name	Value	Description
	<i>pri</i>	1-255	Default = 128
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timeout ptp exaware(config-timeout-ptp)# clock g8275.1 exaware(config-timeout-ptp-clock-g8275.1)# priority2 130</pre>		

11.1.1.13. servo-alg

Description	Use this command to configure the ptp clock servo algorithm. Use the no form of the command to restore the default value (full-on-path-synce).		
Command Syntax	<pre>servo-alg alg no clock-step</pre>		
Command Mode	configuration timing ptp clock		
Parameters Table	Name	Value	Description
	<i>alg</i>	full-on-path	The timestamp is inserted into the sync message.
		full-on-path-synce	The timestamp is inserted into a follow-up message that follows the sync message.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timeout ptp exaware(config-timeout-ptp)# clock g8275.1 exaware(config-timeout-ptp-clock-g8275.1)# servo-alg full-on-path</pre>		

11.1.1.14. squelch

Description	Use this command to configure the ptp clock output squelch mode. Use the no form of the command to restore the default value (disable).		
Command Syntax	<pre>squelch {enable disable} no squelch</pre>		
Command Mode	configuration timing ptp clock		
Parameters Table	Name	Value	Description
	enable		Output clocks are squelched upon the clock state changes to Holdover.
	disable		
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timeout ptp exaware(config-timeout-ptp)# clock g8275.1 exaware(config-timeout-ptp-clock-g8275.1)# squelch enable</pre>		

11.1.2. PTP Operational Commands

11.1.2.1. clear ptp statistics

Description	Clear the PTP statistics.		
Command Syntax	<code>clear ptp statistics</code>		
Command Mode	operational		
Parameters Table	Name	Value	Description
	<i>steps</i>	1-255	Default = 255
Examples	This example shows how to use this command: <code>exaware# clear ptp statistics</code>		

11.1.3. PTP Show Commands

11.1.3.1. show ptp clock dataset parent

Description	Show the data set of the master clock this clock is synchronized to.
Command Syntax	<code>show ptp clock dataset parent</code>
Command Mode	operational
Examples	This example shows how to use this command: <pre>exaware# show ptp clock dataset parent</pre> Parent Port ID: 00:80:40:FF:FE:04:05:08-01 Observed Parent Offset: 65535 Observed Parent Clock Phase Change Rate: 2417483647 Grandmaster Identity: 00:80:40:FF:FE:04:05:08 Grandmaster Priority2: 128 Grandmaster Clock Class: 6 Grandmaster Clock Accuracy: 33

11.1.3.2. show ptp clock summary

Description	Show the info of the PTP clock.
Command Syntax	<code>show ptp clock summary</code>
Command Mode	operational
Examples	This example shows how to use this command: <pre>exaware# show ptp clock summary Clock Identity: 00:80:40:FF:FE:01:01:12 Profile: G8275.1 Clock Type: boundary-clock Domain: 24 Servo Alg: full-on-path-synce Num Of Interfaces: 2 Clock Class: 6 Clock Accuracy: 33 Scaled Log Variance: 20061 Priority2: 128 Clock Step: one-step Current UTC Offset: 37 Time Traceable: Yes Frequency Traceable: Yes Time Source: GPS</pre>

11.1.3.3. show ptp interface summary

Description	Show the info of the PTP clock interface.
Command Syntax	<code>show ptp interface summary</code>
Command Mode	operational
Examples	This example shows how to use this command: <pre>exaware# show ptp interface parent Interface Name: x-eth 0/0/0 State: Slave Log Announce Interval: -3 Announce Receipt Timeout: 3 Log Sync Interval: -4 Log Min Delay Req Interval: -4 Peer Mean Path Delay: 13ns Delay Mechanism: End To End PTP Version: 2 Local Priority: 128 Master-Only: No Port Identity: 00:80:40:FF:FE:01:01:12-01</pre>

11.1.3.4. show ptp servo

Description	Show the info of the PTP servo.
Command Syntax	show ptp servo
Command Mode	operational
Examples	This example shows how to use this command: <pre>exaware# show ptp servo</pre> Servo Alg: full-on-path-synce Servo State: Phase Locked Servo State Duration: 300 secs Frequency Correction: -119523.000 Phase Correction: 0 ns Offset From Master: 15 ns Mean Path Delay: 45 ns Sync Packet Rate: 16 Delay Packet Rate: 16

11.1.3.5. show ptp statistics brief

Description	Show PTP statistics per PTP clock.
Command Syntax	<code>show ptp statistics brief</code>
Command Mode	operational
Examples	This example shows how to use this command: <pre>exaware# show ptp ptp statistics brief</pre> Time since last clear statistics: 1hour 14min 54sec TX Packets: 1580 RX Packets: 6580 RX Packets Discarded: 0

11.1.3.6. show ptp statistics detail

Description	Show PTP statistics per PTP interface.																												
Command Syntax	show ptp statistics detail																												
Command Mode	operational																												
Examples	This example shows how to use this command: <pre>exaware# show ptp ptp statistics detail</pre> <table><tr><th></th><th colspan="3">PTP Interface x-eth 0/0/1</th></tr><tr><th></th><th>Sent</th><th>Received</th><th>Dropped</th></tr><tr><td>Announce</td><td>3</td><td>82</td><td>0</td></tr><tr><td>Sync</td><td>0</td><td>660</td><td>0</td></tr><tr><td>Follow-up</td><td>0</td><td>660</td><td>0</td></tr><tr><td>Delay-Req</td><td>559</td><td>0</td><td>0</td></tr><tr><td>Delay-Resp</td><td>0</td><td>559</td><td>0</td></tr></table>		PTP Interface x-eth 0/0/1				Sent	Received	Dropped	Announce	3	82	0	Sync	0	660	0	Follow-up	0	660	0	Delay-Req	559	0	0	Delay-Resp	0	559	0
	PTP Interface x-eth 0/0/1																												
	Sent	Received	Dropped																										
Announce	3	82	0																										
Sync	0	660	0																										
Follow-up	0	660	0																										
Delay-Req	559	0	0																										
Delay-Resp	0	559	0																										

11.2. Synchronization Commands

11.2.1. Synchronization Configuration Commands

11.2.1.1. **eeec**

Description	Use this command to configure Synchronous G.8262.1 eEEC mode. Use the no form of the command to set to the default (disable).
Command Syntax	<code>eeec enable disable</code> <code>no eeec</code>
Command Mode	configuration timing synchronization
Examples	This example shows how to use this command: <code>exaware# config</code> <code>exaware(config)# timing synchronization</code> <code>exaware(config-timing-synchronization)# eeec enable</code>

11.2.1.2. interface

Description	Use this command to configure a synchronization interface. Enters synchronization interface configuration mode. Use the no form of the command to delete the interface.		
Command Syntax	<pre>interface if-name no interface if-name</pre>		
Command Mode	configuration timing synchronization		
Parameters Table	Name	Value	Description
	<i>if-name</i>	x-eth 0/0/ <i>n</i>	SyncE Physical interface,ysical interface.
		ptp	PTP interface.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timing synchronization exaware(config-timing-synchronization)# interface x-eth 0/0/2</pre>		
Notes	Configuring x-eth SyncE will automatically configure the ESMC mode of the physical interface.		

11.2.1.3. ql-mode

Description	Configure the clock selection mode. Use the no form of the command to set to the default (ql-enabled).		
Command Syntax	ql-mode enable disable no ql-mode		
Command Mode	configuration timing synchronization		
Parameters Table	Name	Value	Description
	Enable		Compare clock sources by QL.
	disable		Don't compare clock sources by QL.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timing synchronization exaware(config-timing-synchronization)# ql-mode enable</pre>		

11.2.1.4. **quality-level**

Description	Use this command to force the quality level (QL) for the timing source. Use the no form of this command to unconfigure quality-level.		
Command Syntax	<pre>quality-level ql no quality-level</pre>		
Command Mode	configuration timing synchronization interface		
Parameters Table	Name	Value	Description
	<i>ql</i>	ql-prc	Primary Reference Clock (option1).
		ql-ssu-a	Types I or V slave clock (option1)
		ql-ssu-b	Type VI slave clock (option1)
		ql-sec	Synchronous Equipment Clock (option1)
		ql-dnu	This signal should not be used for synchronization (option1)
		ql-stu	Synchronized – traceability unknown (option2)
		ql-prs	PRS traceable (option2)
		ql-st2	Traceable to stratum 2 (option2)
		ql-st3e	Traceable to stratum 3E (option2)
		ql-smc	Traceable to SONET clock self timed (option2)
		ql-prov	Provisionable by the network operator (option2)
		ql-tnc	Traceable to transit node clock (option2)

	ql-dus	This signal should not be used for synchronization (option2)
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timing synchronization exaware(config-timing-synchronization)# interface x-eth 0/0/2 exaware(config-timing-synchronization-x-eth-0/0/2)# quality-level prc</pre>	
Notes	ql-prc, ql-ssu-a, ql-ssu-b, ql-sec and ql-dnu should be available for ssm-option option1 only. ql-prs, ql-st2, ql-st3e, ql-smc, ql-prov, ql-tnc and ql-dus should be available for ssm-option option2 only.	

11.2.1.5. output-source

Description	Use this command to use a SyncE interface as a timing output source. Use the no form of the command to set to the default (disable).		
Command Syntax	<pre>output-source {enable disable} no output-source</pre>		
Command Mode	configuration timing synchronization interface		
Parameters Table	Name	Value	Description
	<i>minutes</i>	0-12	The wait-to-restore timer in minutes.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timing synchronization exaware(config-timing-synchronization)# interface x-eth 0/0/2 exaware(config-timing-synchronization-x-eth-0/0/2)# output-source enable</pre>		

11.2.1.6. priority

Description	Use this command to set an input timing source. Synchronization packets are received from this source and sent to the clock selection algorithm. Use the no form of this command to delete an input source.		
Command Syntax	<code>priority pri</code>		
Command Mode	configuration timing synchronization interface		
Parameters Table	Name	Value	Description
	<code>pri</code>	1-255	The input priority. SyncE Physical interface.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timing synchronization exaware(config-timing-synchronization)# interface x-eth 0/0/2 exaware(config-timing-synchronization-x-eth-0/0/2)# priority 1</pre>		
Notes	1 is the highest, 255 is the lowest.		

11.2.1.7. source-id

Description	Use this command to set the clock source identifier for a SyncE interface. Use the no form of this command to unset this value.		
Command Syntax	<pre>source-id source-id no source-id</pre>		
Command Mode	configuration timing synchronization interface		
Parameters Table	Name	Value	Description
	<i>source-id</i>	1-255	The source identifier number for the interface.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timing synchronization exaware(config-timing-synchronization)# interface x-eth 0/0/2 exaware(config-timing-synchronization-x-eth-0/0/2)# source-id 1</pre>		
Notes	Assign the same clock source identifier if interfaces are connected to same clock source.		

11.2.1.8. ssm-option

Description	Use this command to set the synchronization option for the internal clock that is locked in frequency to an incoming signal. Use the no form of the command to set to the default (<i>option1</i>).
Command Syntax	<code>ssm-option {option1 option2}</code> <code>no ssm-option</code>
Command Mode	configuration timing synchronization
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timing synchronization exaware(config-timing- synchronization)# ssm-option option2</pre>
Notes	The command fails to change the type if any of the sync-references is configured with quality-level.

11.2.1.9. wait-to-restore

Description	Use this command to set the wait-to-restore timer in minutes. Use the no form of this command to reset the wait-to-restore time to default value 5.		
Command Syntax	<pre>wait-to-restore <i>minutes</i> no source-id</pre>		
Command Mode	configuration timing synchronization interface		
Parameters Table	Name	Value	Description
	<i>minutes</i>	0-12	The wait-to-restore timer in minutes.
Examples	This example shows how to use this command: <pre>exaware# config exaware(config)# timing synchronization exaware(config-timing-synchronization)# interface x-eth 0/0/2 exaware(config-timing-synchronization-x-eth-0/0/2)# wait-to-restore 10</pre>		

11.2.2. Synchronization Operational Commands

11.2.2.1. **clear synchronization statistics esmc**

Description	Clear the PTP statistics.
Command Syntax	<code>clear synchronization statistics esmc</code>
Command Mode	operational
Examples	This example shows how to use this command: <pre>exaware# clear synchronization statistics esmc</pre>

11.2.2.2. clear synchronization wait-to-restore

Description	Clear the PTP statistics.
Command Syntax	<code>clear synchronization wait-to-restore interface <i>if-name</i></code>
Command Mode	operational
Examples	This example shows how to use this command: <pre>exaware# clear interface wait-to-restore x-eth 0/0/2</pre>

11.2.3. Synchronization Show Commands

11.2.3.1. show synchronization interface detail

Description	Show all synchronization interfaces one after the other.
Command Syntax	<code>show synchronization interface detail</code>
Command Mode	operational
Examples	This example shows how to use this command: <pre>exaware# show synchronization interface detail</pre> <pre>Interface x-eth 0/0/0 Link State: Up In / Out Source: Input ESMC Status: OK Is Selected: Yes Configured QL: QL-PRC Received QL: QL-PRC Operational: QL QL-PRC Priority: 1 Clock Source ID: 1 Configured Wait-to-restore (min): 5 WTR Remaining [min]: 0 Hold-off Timer: Not Running Timing Loop Detected: No</pre> <pre>Interface x-eth 0/0/9 Link State: Up In / Out Source: Output ESMC Status: OK Configured QL: QL-PRC Operational QL: QL-PRC Clock Source ID: 1</pre>

11.2.3.2. show synchronization interface input

Description	Use this command to display status and configuration of all synchronization interfaces.							
Command Syntax	show synchronization interface input							
Command Mode	operational							
Examples	This example shows how to use this command:							
exaware# show synchronization interface input								
If Name	Is Selected	Pri	Rcv QL	Oper QL	Link State	WTR Reamining (min)	Hold-off Timer	
x-eth 0/0/1	Yes	2	QL-PRC	QL-PRC	UP	0	Not Running	
x-eth 0/0/6	No	6	NA	QL-Fail	DOWN	0	Not Running	
ptp G8275.1	No	9	QL-PRC	QL-PRC	UP	4	Not Running	

11.2.3.3. show synchronization interface output

Description	Use this command to display status and configuration of all synchronization interfaces.
Command Syntax	show synchronization interface output
Command Mode	operational
Examples	This example shows how to use this command:

```
exaware# show synchronization interface output
```

If Name	Link State	Configured QL	Oper QL
x-eth 0/0/9	UP	NA	QL-PRC

11.2.3.4. show synchronization statistics esmc

Description	Use this command to display Ethernet Synchronization statistics.																
Command Syntax	show synchronization statistics esmc																
Command Mode	operational																
Examples	This example shows how to use this command: <pre>exaware# show synchronization statistics esmc</pre> Time since last clear statistics: 1hour 14min 54sec <table><tr><th>If Name</th><th>ESMC Status</th><th>ECMS Received</th><th>ESMC Sent</th></tr><tr><td>x-eth 0/0/1</td><td>OK</td><td>1015</td><td>405</td></tr><tr><td>x-eth 0/0/6</td><td>OK</td><td>0</td><td>165852</td></tr><tr><td>x-eth 0/0/9</td><td>OK</td><td>950</td><td>950</td></tr></table>	If Name	ESMC Status	ECMS Received	ESMC Sent	x-eth 0/0/1	OK	1015	405	x-eth 0/0/6	OK	0	165852	x-eth 0/0/9	OK	950	950
If Name	ESMC Status	ECMS Received	ESMC Sent														
x-eth 0/0/1	OK	1015	405														
x-eth 0/0/6	OK	0	165852														
x-eth 0/0/9	OK	950	950														

11.2.3.5. show synchronization summary

Description	Show the details of the clock.
Command Syntax	<code>show synchronization summary</code>
Command Mode	operational
Examples	This example shows how to use this command: <pre>exaware# show synchronization summary</pre> Synchronization summary Clock-selection Mode: QL-Enable Network Clock Type: Option1 eEEEC Mode: Enabled DPLL Clock State: Locked Clock State: Locked State Duration: 3600 Seconds

Index

A

- aaa accounting event/destination • 148
- aaa authentication-order • 151, 160, 180
- aaa authorization-order • 154
- AAA Commands • 148
- AAA Configuration Commands • 148
- AAA Operational Commands • 184
- aaa radius • 157
- aaa radius-retry • 161
- aaa role • 162, 182
- AAA Show Commands • 185
- aaa tacacs • 177
- aaa user • 181
- abort • 252
- accounting policy • 705
- ACL Actions • 574
- ACL Conditions • 572
- ACL Groups • 572
- ACL Policies • 572
- ACL Policy Configuration Commands
 - Actions • 625
 - Traffic Conditions • 624
- ACL Policy Operational Commands • 634
- ACL Policy Show Commands • 637
- adjacency-check (af-context) • 1125
- admin-state • 52, 406, 457, 868, 871, 873, 876, 879, 882, 883, 886, 890, 891, 904, 925, 936, 954, 955, 960, 963, 965, 970, 972, 976, 1037, 1038, 1041, 1130, 1226, 1306, 1410, 1521, 1522, 1523, 1524, 1525, 1526, 1527, 1528, 1529, 1530, 1531, 1532, 1533, 1534, 1535, 1536, 1537, 1538, 1539, 1540, 1541, 1542, 1543, 1544, 1545, 1548, 1549, 1550, 1551, 1552, 1553, 1554, 1555, 1557, 1559, 1561, 1563
- admin-state (xconnect) • 143
- affinity (Interface) • 458
- affinity (Tunnel) • 459
- affinity-map • 460
- af-ipv4 • 407, 892, 1126, 1369
- af-ipv4 (multicast) • 1209
- af-ipv6 • 895, 1127
- af-l2vpn vpls • 898
- aggregate-address • 706, 899
- aggregator-id • 53
- alias • 253
- allocate-label • 900
- allow-as-in • 901
- alternate-path • 902
- annotate • 254
- anycast-rp • 1371
- area • 1223, 1305
- arp ageing-timeout • 54
- ARP Commands • 17
- ARP Configuration Commands • 17, 42, 50
- arp interface • 17, 43, 44, 45, 46, 47, 48, 50
- ARP Operational Commands • 19
- as-override • 903
- as-path • 648, 676, 678
- as-path-length • 651
- as-path-unique-length • 652
- Attach Point for QoS Actions • 578
- Attach points for conditions • 576
- authentication key • 1224
- authentication-key (level 1/2) • 1128
- auto-bandwidth adjust-interval • 461
- auto-bandwidth adjust-threshold • 463
- auto-bandwidth max-bandwidth • 464
- auto-bandwidth min-bandwidth • 466
- auto-bandwidth monitor • 467
- auto-bandwidth overflow-limit • 468
- auto-bandwidth sample-interval • 469

auto-discovery • 1068, 1069, 1070, 1071, 1072,
1073, 1074, 1075, 1076, 1077, 1078, 1079,
1082, 1085, 1089, 1411

B

bandwidth • 471

BGP Attach Points • 583

BGP Configuration Commands • 865, 868, 883,
886, 890, 1271, 1479, 1521, 1535, 1536,
1542, 1553, 1555

bgp extended-asn-capability • 906

bgp first-as-enforcement • 905

bgp graceful-restart • 907

bgp local-preference • 909

BGP Operational Commands • 980

bgp path-selection • 910

BGP Policy Configuration Commands

 Actions • 678

 Attachment Points • 704

 Routing Conditions • 648

BGP Routing Actions • 582

BGP Routing Commands • 865, 890, 1521, 1542

BGP Routing Conditions • 581

BGP Show Commands • 1004

bgp timers • 916

block-size • 1412

bsr candidate-bsr • 1372

bsr candidate-rp • 1373

bsr-border • 1374

bsr-unicast-message • 1375

C

cancel • 255

capability • 917

circuit-type • 1131

clear aaa • 184

clear acl counters • 634

clear acl forwarding counters • 635

clear arp • 19

clear bgp all • 980

clear bgp groups • 984

clear bgp neighbors • 988

clear bgp policy accounting • 993, 997

clear bgp policy aggregate-address • 994

clear bgp policy allocate-label • 995

clear bgp policy default-route-originate • 996

clear bgp policy neighbors • 998

clear bgp policy network • 1001

clear bgp policy redistribute • 1002

clear ctrl-plane acl counters • 636

clear flow-table counters flowspec • 38

clear forwarding • 1451, 1452

clear history • 292

clear igmp interface • 1100

clear interface counters • 98, 99, 378

clear interface routing-policy-accounting • 100

clear isis database • 1181

clear isis process • 1182, 1183, 1184, 1185

clear isis route • 1186

clear isis statistics • 1187, 1188

clear lacp counters • 101

clear ldp • 439

clear ldp policy label-allocation • 742

clear ldp statistics • 440

clear multicast routes • 1216

clear multicast statistics • 1217

clear ndp • 20

clear ospf policy default-route-originate • 747

clear ospf policy inter-area • 748

clear ospf policy redistribute • 749

clear ospf policy rib-filter • 750

clear ospf process • 1271

clear ospf route • 1272

clear ospf statistics • 1273, 1274, 1345

clear ospfv3 policy inter-area • 755

clear ospfv3 policy redistribute • 756

clear ospfv3 policy rib-filter • 757

clear ospfv3 process • 1342
clear ospfv3 route • 1343
clear ospfv3 statistics • 1344
clear pim bsr • 1396
clear pim table • 1397
clear qos counters • 843
clear rsvp auto-bandwidth • 530
clear rsvp bypass-tunnel • 531
clear rsvp bypass-tunnels • 532
clear rsvp statistics • 533
clear rsvp statistics neighbor • 534
clear rsvp statistics neighbors • 535
clear rsvp tunnel • 536
clear rsvp tunnel optimize • 537
clear rsvp tunnel revert-primary • 538
clear rsvp tunnels • 539
clear rsvp tunnels optimize • 540
clear rsvp tunnels revert-primary • 541
clear system alarm history • 377
clear vpls mac address-table • 1428, 1430
CLI Command Reference • 15
client-to-client-reflection • 920
cluster-id • 921
commit • 256
Common Routing Commands • 1062
community • 653, 683
community [not supported in release 1.1] • 335, 336, 339, 347, 348, 349, 357
compare • 293
confederation • 922
Configuration Mode • 252
configure • 294
csnp-interval • 1132
cspf • 473
ctrl-plane policy acl • 633

D

damping • 687

database-filter • 1225
dead-interval (interface level) • 1227, 1307
debug • 219
default-deny • 645
default-forward-to • 625
default-metric • 1228, 1308
default-permit • 646
default-route-originate • 709, 734, 923, 1133, 1229, 1309
deny • 628, 644
description • 55, 924, 1491, 1496, 1497, 1498, 1499, 1501, 1502, 1505, 1506, 1507
destination ip • 624, 791, 1110, 1111, 1113, 1117, 1118, 1119, 1120, 1121, 1122, 1123, 1124
destination port • 624, 794
discovery-hello-holdtime (timers) • 408, 412, 438
discovery-hello-interval (timers) • 410, 411, 438
discovery-hellos • 413
distance • 926, 1134, 1232
distance source-ip • 927
dns • 259
Documentation Conventions • 15
dr-priority • 1376
dscp • 624, 759, 802
DSCP values • 777, 802
dynamic-hostname • 1135

E

ebgp-multihop • 928
edit • 261
end • 262
ether-type • 56, 83, 624, 804
exec-policy • 566, 585
exit • 263, 295
explicit-null • 415, 475, 929
export-policy • 710, 930
export-rt • 931, 1415
extcommunity • 656, 690

F

facility-override (log remote-server) • 222
fast-external-turnover • 932
fast-reroute • 1136, 1233
File Commands • 202
file copy • 202
file delete • 210
file edit • 211
file list • 212
file monitor start • 213
file monitor stop • 214
file rename • 215
file show • 218
filter • 225
flap-damping • 711, 933
Flow Commands • 37
Flowspec Commands • 37
Flowspec Operational Commands • 38
Flowspec Show Commands • 39
forward-to • 629
frag-non-initial • 624, 805

G

General Routing Policy Actions • 644
graceful-restart grace-interval • 1137, 1234, 1312
graceful-restart helper • 1138, 1235, 1313
graceful-restart interface-sync-interval • 1139
graceful-restart interface-sync-retries • 1140
graceful-restart lspdb-level1-sync-interval • 1141
graceful-restart lspdb-level2-sync-interval • 1142
graceful-restart max-grace-period • 1236, 1314
graceful-restart state • 1237, 1315

H

hello holdtime • 1377
hello interval • 1378
hello-holdtime • 416

hello-interval • 417, 1143, 1144
hello-interval (interface level) • 1238, 1316
hello-messages interval • 476
hello-messages neighbor • 479
hello-messages receipt • 480
hello-messages timeout • 478, 481
hello-multiplier • 1144
hello-padding • 1145
help • 296
hop-limit • 484

I

icmpv4 • 60
icmpv4-code • 624, 810
icmpv4-type • 624, 807
icmpv6 • 61
icmpv6-code • 624, 817
icmpv6-type • 624, 815
ieee-802.1p • 624, 820
IGMP Routing Commands • 1092
IGMP Routing Configuration Commands • 1092
IGMP Routing Operational Commands • 1100
IGMP Routing Show Commands • 1101
ignore-attached-bit • 1146
igp-metric • 487
igp-shortcut • 488
igp-synchronization-delay • 418
import • 1239
import-policy • 712, 937
import-rt • 938, 1416
inbound-soft-reconfiguration • 939
Intended Audience • 16
inter-area policy • 744, 752
interface • 419, 489, 872, 1147, 1240, 1318, 1340, 1379, 1417
interface (vpws) • 129
interface agg-eth • 62
Interface Commands • 52

Interface Configuration Commands • 52
interface loopback • 63
interface mgmt • 64
Interface Operational Commands • 98
Interface Show Commands • 102
interface x-eth • 65
interface1 (xconnect) • 144
interface2 (xconnect) • 145
ip-source • 490
ipv4-address • 66, 67
ipv6-address • 68
ISIS Attach Points • 584
ISIS Policy Configuration Commands
 Actions • 730
 Attachment Points • 733
 Routing Conditions • 720
ISIS Routing Actions • 584
IS-IS Routing Commands • 1125
ISIS Routing Conditions • 583
IS-IS Routing Configuration Commands • 1062,
 1125
IS-IS Routing Operational Commands • 1181
IS-IS Routing Show Commands • 1189
isis-metric • 730
is-type • 1131, 1148

J

join-prune-interval • 1380

K

keep-multiplier • 489, 491, 516

L

L2 Services Configuration Commands • 129,
 136, 142, 1068, 1077, 1078, 1275, 1410,
 1451, 1453
L2 Services Operational Commands • 1428
L2 Services Show Commands • 1431
l2-transport • 59, 71, 87, 92
label-allocation • 420, 741

lACP • 72
lACP-interval • 73
lACP-rx-interval • 74
lACP-tx-interval • 75
last-member-query-timeout • 1092
LDP Commands • 406
LDP Configuration Commands • 406
LDP Operational Commands • 439
LDP Policy Configuration Commands
 Attachment Points • 740
 Routing Conditions • 737
LDP Policy Operational Commands • 742
LDP Policy Show Commands • 743
LDP Show Commands • 441
ldp-synchronization • 421
ldp-synchronization-holdtime • 422
ldp-tunneling • 494
level • 720
level-1 • 1149
level-2 • 1150
load • 264
local-address • 424, 940
local-gen-pkt • 624
local-preference • 659, 694
local-protection (interface) • 495
local-protection (tunnel) • 496
log copy • 244
log delete • 246
log monitor start • 242
log monitor stop • 243
log output console • 231
log output file • 232
log output remote-server • 233
log output user • 237
log-adjacency-changes • 1151
Logging and Debugging Commands • 219
Logging and Debugging Configuration
 Commands • 219

Logging Operational Commands • 242
log-neighbor-changes • 941, 1242, 1320
logout • 297
lsp-cs-errors • 1152
lsp-fast-flood • 1153
lsp-gen-interval (level 1/2) • 1154
lsp-interval • 1155
lsp-mtu (level 1/2) • 1156
lsp-refresh-interval • 1157
lsp-style • 497

M

mac-limit • 1421
mac-lock • 1419, 1420
Management Commands • 148, 1110, 1119, 1121
maximum-paths • 1063
maximum-prefix • 945, 1492
maximum-routes • 1210
max-lsp-lifetime • 1158
max-metric set | unset • 1243, 1321
max-metric-on-start • 1244, 1322
max-reservable-bw • 489, 498
med • 660, 695
mesh-group • 1159
metric • 1160
metric (interface level) • 1245, 1323
metric-style (level 1/2) • 1161
metric-type • 731
minimum bandwidth • 761, 780
minimum-interval • 946
min-links • 77, 78
mpls • 79
MPLS Commands • 406
mpls exp • 764
mpls exp topmost • 624, 821
mpls ldp • 425
mpls-te • 1164

mtu • 57, 58, 80, 380, 396, 1422
mtu (pw-profile) • 139
mtu-ipv4 • 81
multi-area-adjacency • 1249
Multicast Routing Commands • 1209
Multicast Routing Configuration Commands • 1209
Multicast Routing Operational Commands • 1216
Multicast Routing Show Commands • 1218
multipath • 947
multipath ebgp | ibgp • 948

N

ndp ageing-timeout • 82
ndp interface • 18
neighbor • 949, 1247, 1248, 1325, 1326
neighbor (vpws) • 130
neighbor-group • 950
neighbor-in • 713
neighbor-out • 714
net • 1165
Netflow Commands • 42
network • 717, 951, 1167, 1173, 1253, 1329
network-type • 1250, 1327
nexthop • 501
next-hop • 661
next-hop • 697
next-hop-self • 952
nssa • 1251
ntp • 248
NTP Commands • 248
NTP Configuration Commands • 248
NTP Operational Commands • 251

O

Operational Mode • 292
optimization-timer • 502
orf • 718, 953
orf-prefix • 664

origin • 667, 698
OSPF Policy Configuration Commands • 744
OSPF Policy Operational Commands • 747
OSPF Policy Show Commands • 751
OSPF Routing Commands • 1223
OSPF Routing Configuration Commands • 1223
OSPFv3 Policy Configuration Commands • 752
OSPFv3 Policy Operational Commands • 755
OSPFv3 Policy Show Commands • 758
OSPFv3 Routing Commands • 1305
OSPFv3 Routing Configuration Commands • 1305
OSPFv3 Routing Operational Commands • 1342
OSPFv3 Routing Show Commands • 1346
os-shell • 298
overload-bit • 1168

P

passive • 1171, 1381
passive (interface) • 1252, 1328
passive-session • 956
password • 426, 957, 958, 978
path • 506
path (command mode) • 503, 508
pdu-mtu • 1172
peer • 668
permit • 632, 647
PIM Commands • 1369
PIM Configuration Commands • 1369
PIM Operational Commands • 1396, 1510
PIM Show Commands • 1398, 1513
ping • 299
police • 767
Policing • 580
policy • 84, 959
policy (attach QoS and ACL) • 565, 607
Policy Actions • 569
Policy Commands • 562

Policy Conditions • 568
Policy Configuration Commands • 585
Policy Configuration Show Commands • 611
policy global-parameters • 564, 566, 589
policy group acl-forwarding • 590
policy group bgp-as-path • 571, 592, 650, 682
policy group bgp-community • 571, 595, 655, 686
policy group bgp-extcommunity • 571, 598, 658, 693
policy group ip-prefix • 564, 571, 602, 663, 666, 670, 672, 675, 723, 727, 739, 793, 831
policy group mac-address • 606
Policy Groups • 571
policy named condition • 564, 588
Policy Overview • 562
policy-accounting traffic-counter-description • 1064
precedence • 624, 784, 785, 823, 824
prefix • 671, 721, 737
preserve-next-hop • 961
priority • 509, 1173
priority (interface level) • 1253, 1329
priority-queue • 781
private-as • 962
protocol-ip • 624, 825
pwd • 273
pw-profile • 136, 140, 414, 474, 1413
pw-profile (vpws) • 131

Q

QoS Actions • 577
QoS Groups • 575
QoS Marking • 580
QoS Policies • 575
QoS Policy Configuration Commands
 Actions • 759
 Traffic Conditions • 791
QoS Policy Operational Commands • 843

QoS Policy Show Commands • 844

QoS/Traffic Conditions • 575

qos-tag • 828

query-interval • 1094, 1114

query-response-time • 1095, 1115

queue max size • 786

R

range advertise/suppress • 1254, 1330

rd • 1423, 1493

record • 513

redistribute • 719, 735, 964, 1174, 1255, 1331

reference-bandwidth • 1241, 1259, 1319, 1333

refresh-time • 489, 493, 514

register checksum • 1382

register rp-reachability • 1383

register source • 1384

register timers keepalive-interval • 1385

register timers rate-limit • 1386

register timers suppress-interval • 1387

remote-as-number • 966

resolved • 274

retransmit-interval • 1175, 1260, 1334

retry-limit • 517

retry-limit-cspf • 518

retry-timer • 519

reuse • 88, 89

revert-timer • 521

rib-filter policy • 746, 754

rib-metric • 724

rib-route • 673, 725

robustness-factor • 1096, 1116

rollback • 275

rotate (log file) • 238

route • 1479, 1484

route-has-mpls-label • 728

route-has-no-mpls-label • 729

route-leak • 736, 1176

Router Administration Commands • 252

route-reflector-client • 967

router-id • 428, 968, 1065, 1261, 1335

Routing and Forwarding Information Base
Commands • 31, 32, 1068, 1410

routing bgp • 969

Routing Commands • 865, 1521

routing common • 1066

routing common as-notation • 1062

Routing Groups • 581

routing igmp default • 1097

routing igmp interface • 1098

routing isis • 1177

routing isis <context> • 1336

routing isis interface • 1336

routing isis level • 1336

routing multicast • 1211

routing ospf • 1262

routing ospfv3 • 1336

routing pim • 1388

Routing Policies • 581

routing static • 1481, 1488

routing-policy-accounting • 89

rp-address • 1389

rp-route • 1212

rp-use-priority • 1392

rsvp • 522

RSVP Attach Points • 584

RSVP Commands • 457

RSVP Configuration Commands • 457

RSVP Operational Commands • 530

RSVP Routing Actions • 584

RSVP Routing Conditions • 584

RSVP Show Commands • 542

rule-default (QoS and ACL) • 566, 610

run • 276

S

- save • 277
- secondary • 523
- send • 303
- send-community • 971
- Services Commands • 129
- session • 284
- session autowizard* • 284, 304
- session complete-on-space* • 284, 305
- session configure-prompt • 306
- session display-level* • 284, 285
- session editor • 261, 284, 286
- session history* • 284, 307
- session idle-timeout* • 284, 308
- session ignore-leading-spaces* • 284, 309
- session paginate* • 284, 310
- session prompt-timestamp • 287
- session record start / stop • 311
- session screen-length • 312
- session screen-width • 313
- session show-defaults* • 284, 314
- session terminal-type • 315
- session-keepalive-interval (timers) • 429, 438
- session-keepalive-timeout (timers) • 431, 438
- shape • 760, 787, 789
- show (output filters) • 316
- show acl forwarding interface • 637
- show acl forwarding policy • 639
- show acl policy • 640
- show arp/ndp • 21
- show bgp auto-groups • 1005
- show bgp damping • 1008
- show bgp groups • 1012
- show bgp neighbors advertised/received routes** • 37, 1015
- show bgp neighbors advertised/received routes label view • 1023
- show bgp neighbors brief/detail • 1027
- show bgp neighbors received-prefix-filter • 1030
- show bgp policy • 1031
- show bgp policy allocate-label • 1036
- show bgp summary • 1042
- show bgp table brief/detail** • 37, 1048, 1056
- show bgp table label view • 1058
- show bgp table vsi • 1061
- show candidate • 278
- show configuration (operational) • 319
- show configuration aaa • 164, 183, 185
- show configuration commit • 279
- show configuration policy acl • 641
- show configuration policy condition • 611
- show configuration policy group • 612
- show configuration policy qos • 844
- show configuration rollback • 280
- show configuration running • 281
- show ctrl-plane acl • 643
- show fib adjacency all • 1453
- show fib adjacency interface • 1455
- show fib adjacency interface glean • 1456
- show fib adjacency interface nh-ip • 1457
- show fib all • 1458
- show fib mpls • 1460
- show fib mpls vpws • 1431
- show fib multicast • 1461
- show fib prefix • 1462
- show fib summary • 1463
- show fib vpls-ac • 1432
- show fib vpls-pw • 1434
- show fib vpws • 1435
- show fib vrf • 1464, 1466
- show fib xconnect • 1436
- show flow-table flowspec • 39
- show history • 283, 323
- show history (configuration) • 283
- show id • 324
- show igmp groups • 1101

show igmp interface brief • 1104	show ospf border-routers • 1275
show igmp interface detail • 1106	show ospf database • 1277
show interface • 102	show ospf database asbr-summary • 1278
show interface average-rate • 106	show ospf database external • 1279
show interface brief • 108	show ospf database network • 1280
show interface description • 116	show ospf database nssa-external • 1281
show interface detail • 109	show ospf database opaque-area • 1282
show interface ip • 117	show ospf database opaque-as • 1284
show interface optical-monitoring • 120, 122	show ospf database opaque-link • 1285
show interface routing-policy-accounting • 123, 125, 128	show ospf database router • 1286
show isis interface • 1194	show ospf database summary • 1287
show isis neighbors • 1196	show ospf flood-list • 1288
show isis policy • 1198	show ospf interface • 1290
show isis route • 1199	show ospf neighbors • 1292, 1304, 1368
show isis summary • 1206	show ospf policy • 751
show isis tag • 1189	show ospf request-list • 1293
show isis tag config • 1190	show ospf route • 1295
show isis tag database • 1192	show ospf statistics • 1297, 1299
show isis tag statistics global • 1202	show ospf summary • 1302
show isis tag statistics interface • 1204	show ospfv3 • 1346
show isis tag topology • 1207	show ospfv3 border-routers • 1347
show ldp • 441	show ospfv3 database • 1349
show ldp adjacency • 443	show ospfv3 database external • 1350
show ldp interfaces • 444	show ospfv3 database grace • 1351
show ldp local-bindings • 445	show ospfv3 database inter-prefix • 1352
show ldp local-bindings vpws • 447	show ospfv3 database inter-router • 1353
show ldp neighbors • 448	show ospfv3 database intra-prefix • 1354
show ldp policy label-allocation • 743	show ospfv3 database link • 1355
show ldp statistics messages • 453	show ospfv3 database network • 1356
show ldp table • 454	show ospfv3 database router • 1357
show ldp table vpws • 455	show ospfv3 flood-list • 1358
show ldp tunnels • 456	show ospfv3 interface • 1359
show log • 247	show ospfv3 neighbors • 1360
show multicast interface • 1218	show ospfv3 policy • 758
show multicast route • 1219	show ospfv3 request-list • 1361
show multicast rpf • 1222	show ospfv3 route • 1363
show ntp associations • 251	show ospfv3 statistics • 1364, 1366
	show pim bsr • 1398

show pim interface • 1399	show system module • 404
show pim join • 1401	show system processes • 397
show pim neighbors • 1405	show system snapshot • 399, 400, 401
show pim next-hop • 1407	show system software images • 200, 201, 369
show pim rp • 1408	show system software version • 370
show policy • 613	show system uptime • 403
show qos interface agg-verbose • 847	show version • 370
show qos interface brief • 850, 861	show vpls detail • 1441
show qos interface detail • 857	show vpls label-blocks • 1443
show qos policy • 864	show vpls mac address-table • 1444
show route • 1468	show vpls mac summary • 1445
show route all • 1470	show vpls summary • 1446
show route ip-prefix • 1472	show vpws • 1447
show route mpls • 1475	show vpws summary • 1449
show route mpls vpws • 1439	show vrf • 1495
show route protocol • 1473	show who • 186
show route summary • 1476	show xconnect • 1450
show route vpws • 1440	snmp • 44, 46, 47, 350
show route vrf • 1477	snmp [ip-address] • 333, 342, 345, 353
show rsvp bypass • 542	SNMP Commands • 333
show rsvp interface • 544	SNMP Configuration Commands • 333, 358
show rsvp lsp • 546	Software Download Commands • 187, 361
show rsvp neighbors • 552	Software Download Operational Commands • 187, 202, 361
show rsvp paths • 553	Software Download Show Commands • 200, 369
show rsvp statistics • 554	soo • 974
show rsvp statistics neighbor • 555	source ip • 624, 829
show rsvp summary • 556	source port • 624, 832
show rsvp tunnels • 557	spf-interval • 1263, 1337
show running-config • 281	spt-switchover • 1393
show session • 325	ssh • 326
show system alarm • 386	ssm • 1394
show system alarm configuration-mismatch • 387	ssm-group • 1215
show system alarm history • 388	standby • 524
show system build-parameters • 389	Static Routing Commands • 1479, 1484
show system date • 390, 391	static-group • 1099
show system hardware • 392	status • 351
show system memory • 395	strict-capability-match • 973

stub suppress-summary • 1264, 1338
summary-prefix • 1178, 1265
suppressed • 1179
suppress-specific-route • 699
suppress-summary • 1266
system banner announcement • 371
system banner login • 372
System Commands • 371
System Configuration Commands • 371
system date • 379
system default-management-source • 373
system hostname • 374
System Module Commands • 404
System Module Show Commands • 404
System Operational Commands • 377
system recovery reset factory-default • 381
system recovery reset snapshot • 382
system restart • 383
System Show Commands • 386
system snapshot create • 384
system software download • 187, 189, 192, 194,
197, 361, 369
system software install • 366
system software switch • 367
system software verify • 368
system tech-support • 385
system time-zone • 375, 376

T

tag • 288, 677, 700, 732
targeted-hello-holdtime • 433
targeted-hello-interval • 434
targeted-hellos • 435
targeted-peer • 436
tcp • 289
tcp-flags • 624, 839
Technical Support • 16
telnet-server • 290

timers • 437, 975
top • 291
traceroute • 329
traffic-counter • 701
translator-role • 1267
transmit-delay • 1268, 1339
transport (log remote-server) • 241
ttl • 624, 842
tunnel-bypass • 525
tunnel-destination • 527
tunnel-source • 528
tunnel-te • 529
type (pw-profile) • 141

U

unsuppress-specific-routes • 702

V

ve-id • 1425
vlan-id • 93, 94
vlan-op-ingress • 95
vlan-tags • 96
vpls • 76, 1418, 1426, 1427
vpws • 135
vrf • 42, 97, 337, 338, 344, 346, 352, 354, 355,
356, 358, 360, 977, 1180, 1270, 1341, 1395,
1482, 1489, 1494
VRF Commands • 1491, 1496
VRF Configuration Commands • 1491, 1496
VRF Show Commands • 1495

W

weight • 703, 979

X

xconnect • 147

